CHAPTER 21

AIR CONDITIONING

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21-EFFECTIVE PAGES		
1 thru 9	Jun 15/2026	
10	BLANK	
21-HOW TO USE THE FIM		
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21-FAULT CODE INDEX		
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21-MAINT MSG INDEX		
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21-00 TASKS		
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21-00 TASKS (cont)		
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21-00 TASKS (cont)		
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254	Feb 15/2024	
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258	BLANK	
21-00 TASK SUPPORT		
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318	Jun 15/2025	
21-22 TASKS		
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203	Oct 15/2023	
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A 206	Jun 15/2026	
21-25 TASKS		
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21-27 TASKS		
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204	Feb 15/2026	
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206	Feb 15/2019	
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212	Jun 15/2019	
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214	Feb 15/2026	

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217	Jun 15/2021	
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220	Feb 15/2026	
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256	BLANK	
21-27 TASK SUPPORT		
301	Feb 15/2019	
302	BLANK	
21-31 TASKS		
201	Feb 15/2026	
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203	Oct 15/2022	
204	Feb 15/2023	
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206	Oct 15/2025	
207	Oct 15/2022	
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221	Jun 15/2020	
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224	Feb 15/2020	
225	Feb 15/2020	
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227	Feb 15/2020	

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262	Sep 15/2023	
263	Sep 15/2023	
21-31 TASKS (cont)		
264	Sep 15/2023	
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21-33 TASKS		
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21-33 TASKS (cont)		
203	Feb 15/2026	
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215	Feb 15/2026	
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21-45 TASKS		
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207	Feb 15/2026	
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214	Feb 15/2026	
215	Feb 15/2026	
216	Feb 15/2026	
217	Feb 15/2026	
218	Feb 15/2026	
21-51 TASKS		
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203	Feb 15/2026	

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237	Feb 15/2026	
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239	Oct 15/2025	
21-51 TASKS (cont)		
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268	Feb 15/2026	
269	Feb 15/2026	
270	BLANK	
21-52 TASKS		
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203	Jun 15/2015	
204	Jun 15/2015	
21-52 TASKS (cont)		
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O 205	Jun 15/2026	
O 206	Jun 15/2026	
R 207	Jun 15/2026	
O 208	Jun 15/2026	
A 209	Jun 15/2026	
A 210	BLANK	
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21-61 TASKS		
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295	Jun 15/2025	
296	BLANK	
21-62 TASKS		
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298.18	Feb 15/2026	
298.19	Feb 15/2026	
298.20	Feb 15/2026	
298.21	Feb 15/2026	
298.22	Feb 15/2026	
298.23	Feb 15/2026	
298.24	Feb 15/2026	
298.25	Feb 15/2026	
21-62 TASKS (cont)		
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298.28	Feb 15/2026	
298.29	Feb 15/2026	
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298.58	Feb 15/2026	
298.59	Feb 15/2026	
298.60	Feb 15/2026	
298.61	Feb 15/2026	
21-62 TASKS (cont)		
298.62	Feb 15/2026	
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298.69	Feb 15/2026	
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298.95	Feb 15/2026	
298.96	Feb 15/2026	
298.97	Feb 15/2026	

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21-EFFECTIVE PAGES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 21
AIR CONDITIONING

Subject/Page	Date	COC
21-62 TASKS (cont)		
298.98	Feb 15/2026	
298.99	Feb 15/2026	
298.100	Feb 15/2026	
298.101	Feb 15/2026	
298.102	Feb 15/2026	
298.103	Feb 15/2026	
298.104	Feb 15/2026	
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298.201	Feb 15/2026	
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298.203	Feb 15/2026	
298.204	BLANK	

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21-EFFECTIVE PAGES

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CHAPTER 21
AIR CONDITIONING

Subject/Page	Date	COC
21-62 TASK SUPPORT		
301	Oct 15/2024	
302	Feb 15/2026	
303	Feb 15/2026	
304	Oct 15/2016	
305	Oct 15/2016	
306	Oct 15/2016	
307	Jun 15/2022	
308	Jun 15/2022	
309	Jun 15/2022	
310	Jun 15/2022	
311	Jun 15/2022	
312	Jun 15/2022	
313	Jun 15/2022	
314	Jun 15/2022	
315	Feb 15/2026	
316	Feb 15/2026	
317	Feb 15/2026	
318	Jun 15/2022	
319	Jun 15/2022	
320	Jun 15/2022	
321	Jun 15/2022	
322	Feb 15/2026	
21-75 TASKS		
201	Jun 15/2015	
202	Jun 15/2015	
203	Jun 15/2015	
204	Jun 15/2016	
205	Jun 15/2016	
206	BLANK	

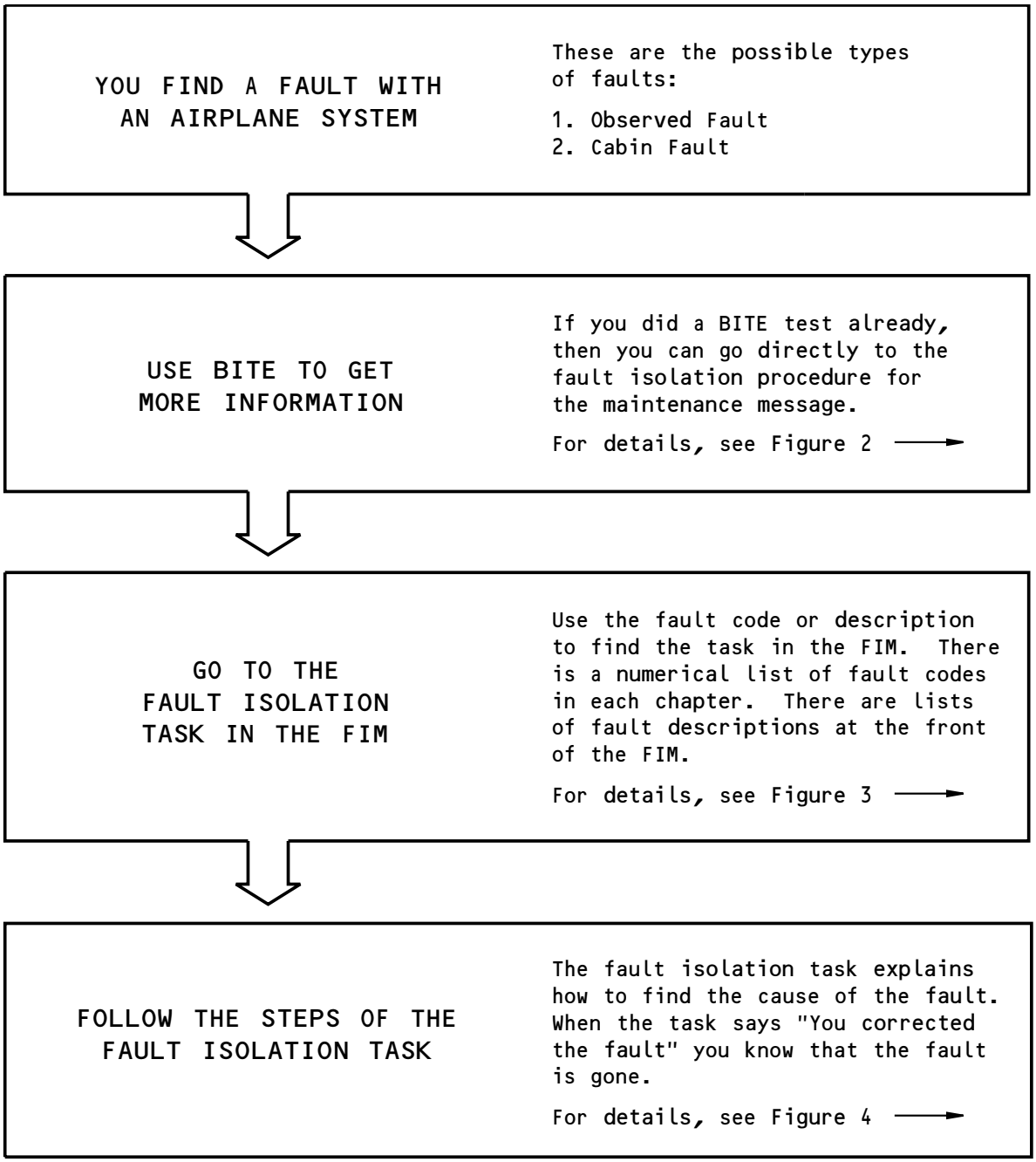
Subject/Page	Date	COC

Subject/Page	Date	COC

A = Added, R = Revised, D = Deleted, O = Overflow, C = Customer Originated Change

21-EFFECTIVE PAGES

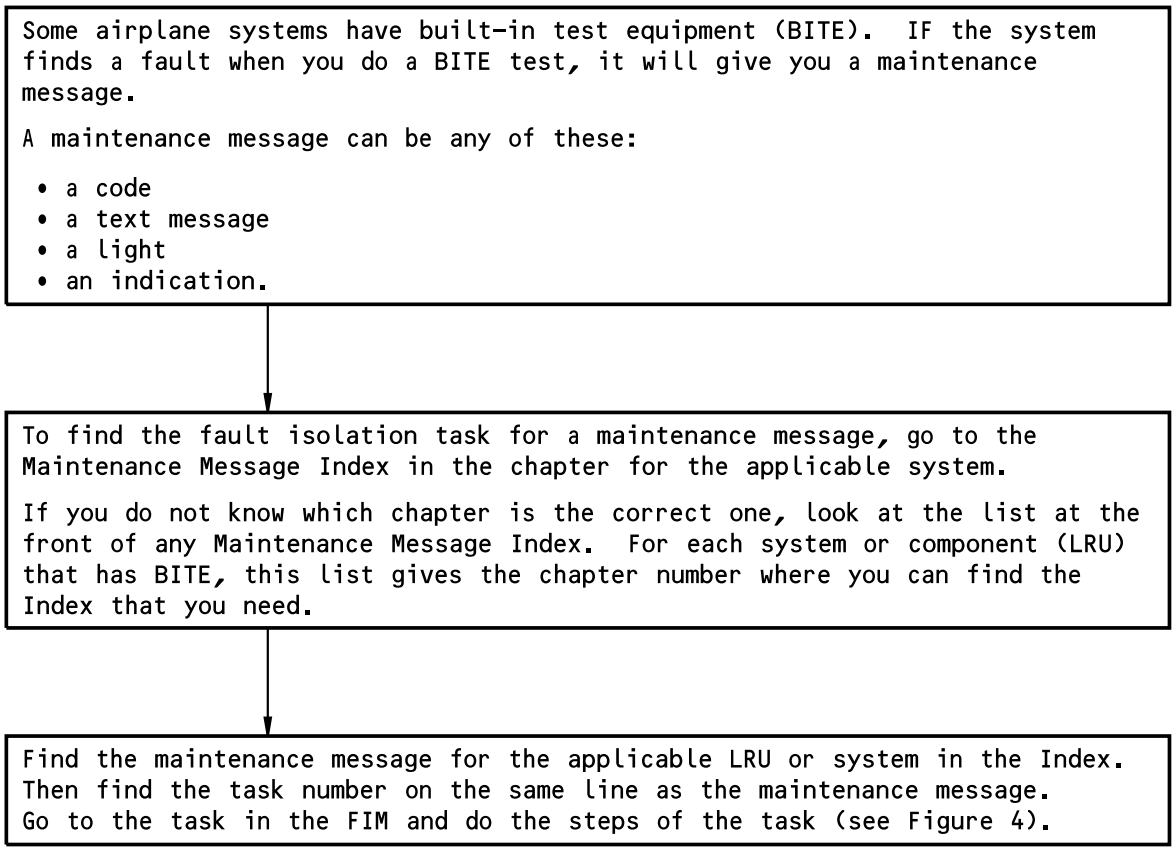
ECCN 9E991 BOEING PROPRIETARY - See title page for details



G04902 S0000148576_V1

Basic Fault Isolation Process
Figure 1

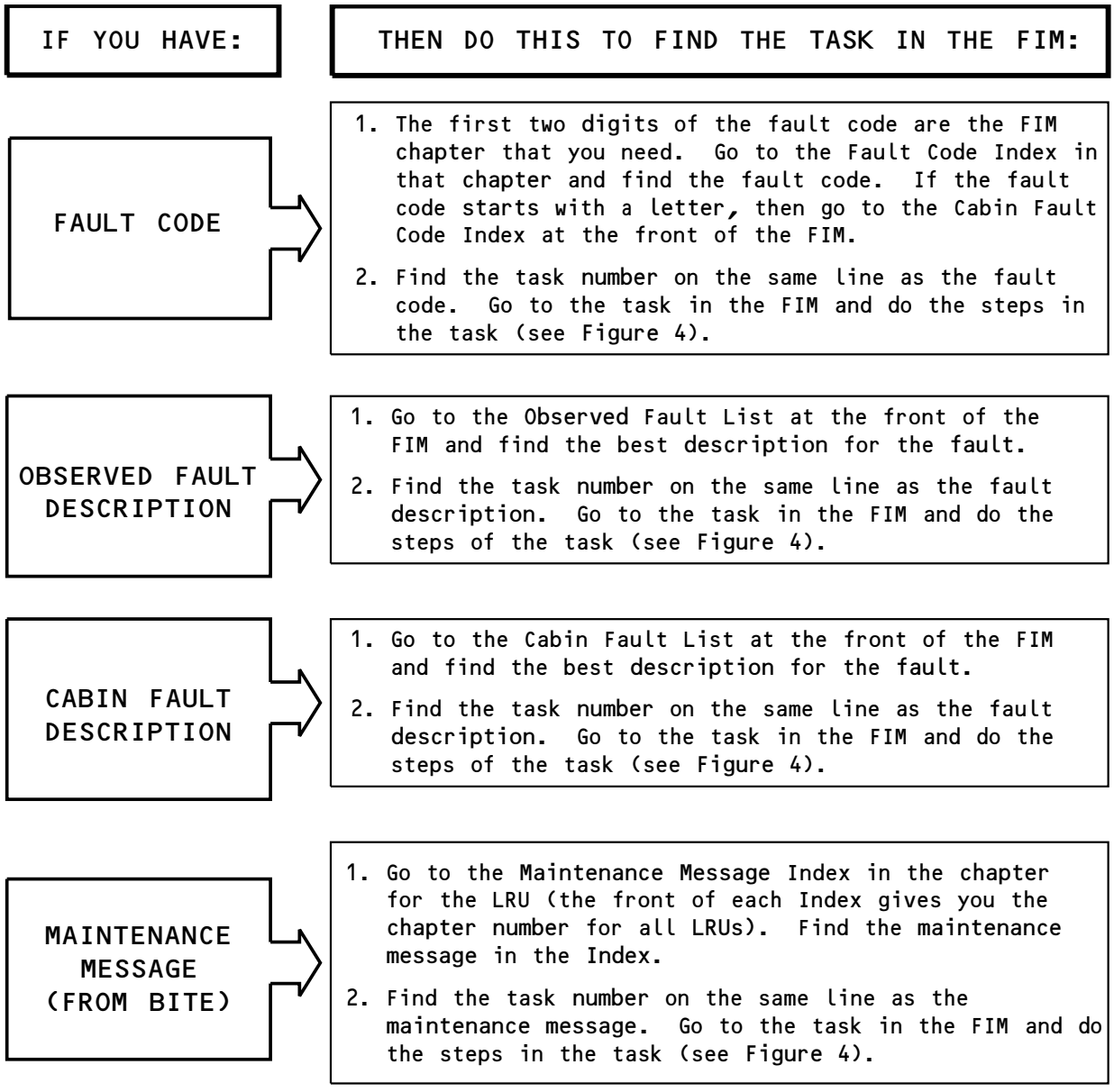
ECCN 9E991 BOEING PROPRIETARY - See title page for details



G04950 S0000148578_V1

Getting Fault Information from BITE
Figure 2

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G04979 S0000148579_V2

Finding the Fault Isolation Task in the FIM
Figure 3

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ASSUMED CONDITIONS AT START OF TASK

- External electrical power is ON
- Hydraulic power and pneumatic power are OFF
- Engines are shut down
- No equipment in the system is deactivated

POSSIBLE CAUSES

- The list of possible causes has the most likely cause first and the least likely cause last.
- You can use the maintenance records of your airline to determine if the fault occurred before. Compare the list of possible causes to the past maintenance actions. This will help prevent repetition of the same maintenance actions.

INITIAL EVALUATION PARAGRAPH

- The primary purpose of the Initial Evaluation paragraph at the start of the task is to help you find out if you can detect the fault right now:
  - If you cannot detect the fault right now, then the task cannot isolate the fault and the Initial Evaluation paragraph will say that there was an intermittent fault.
  - If you have an intermittent fault, you must use your judgement (and follow your airline's policy) to decide which maintenance action to take. Then monitor the airplane to see if the fault happens again on subsequent flights.
- The Initial Evaluation paragraph can also help you find out which Fault Isolation Procedure to use to isolate and correct the fault.

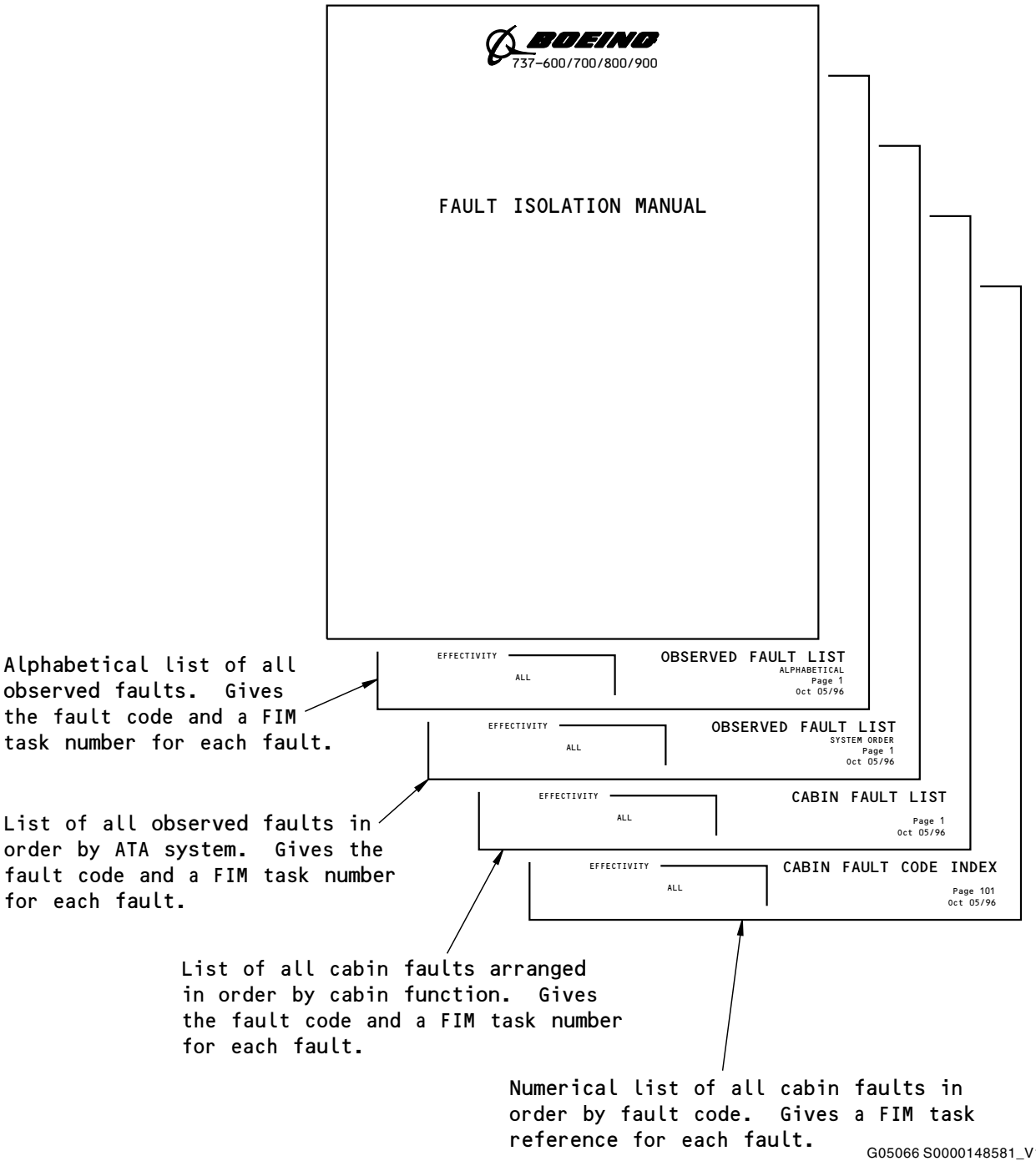
FAULT ISOLATION STEPS

- The FIM task steps are presented in a specified order. The "If... then" statements will guide you along a logical path. But if you do not plan to follow the FIM task exactly, make sure that you read it before you start to isolate the fault. Some FIM procedures start with important steps that have an effect on the other steps in the procedure.
- When you are at the endpoint of the path, the step says "...you corrected the fault." Complete the step and exit the procedure.

G05009 S0000148580_V3

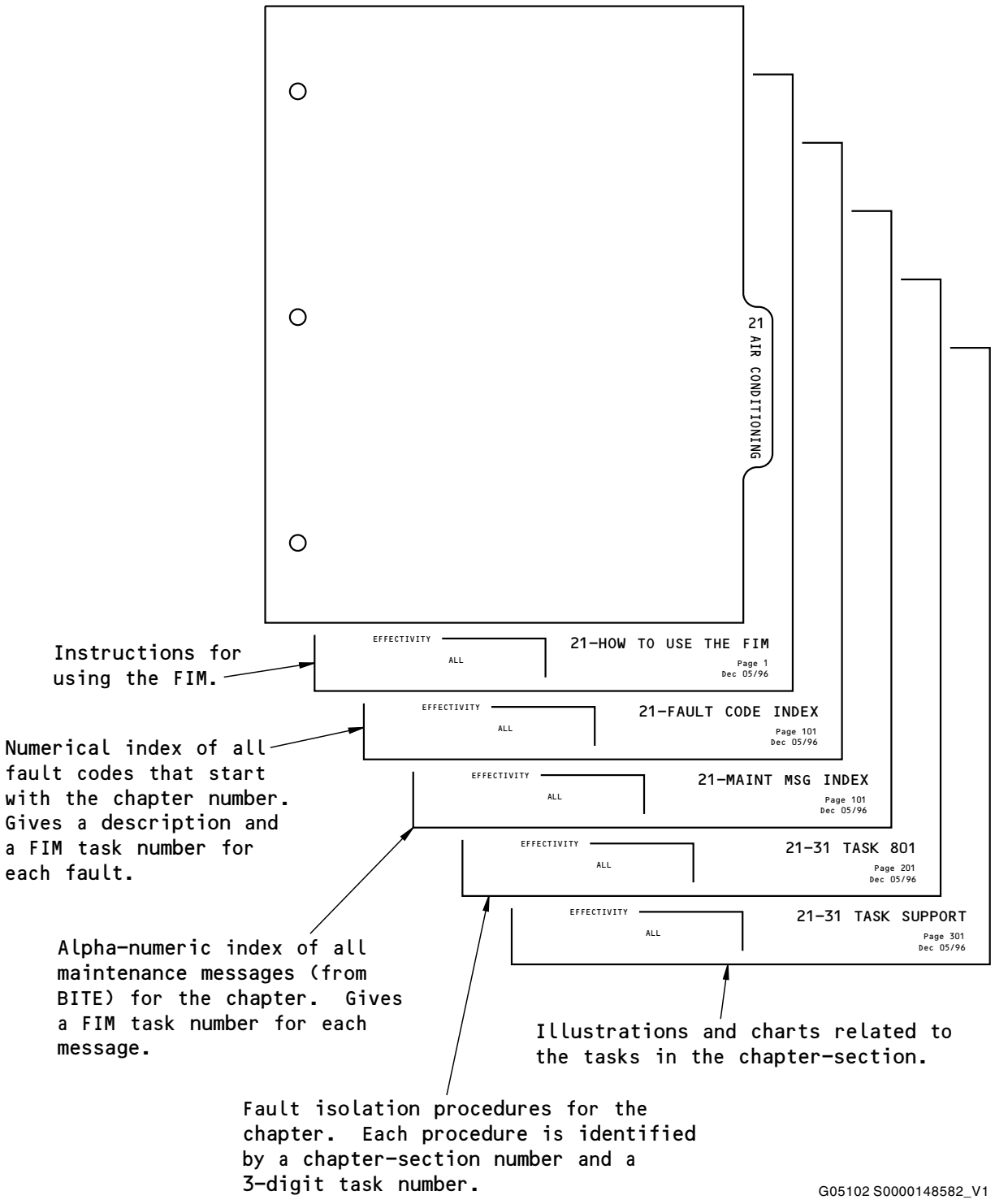
Doing the Fault Isolation Task
Figure 4

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Subjects at Front of FIM
Figure 5

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Subjects in Each FIM Chapter
Figure 6

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FAULT CODE	FAULT DESCRIPTION	GO TO FIM TASK
210 020 00	Smoke: Stops with recirculation fan(s) off, isolation valve closed, and right pack switch at off - flight compartment and passenger cabin.	21-00 TASK 801
210 030 00	Smoke: Stops with recirculation fan(s) off, isolation valve closed, and right pack switch at off - passenger cabin.	21-00 TASK 801
210 040 00	Smoke: Stops with recirculation fan(s) off, isolation valve closed, right pack switch in auto, and left pack switch off - flight compartment.	21-00 TASK 802
210 050 00	Smoke: Stops with recirculation fan(s) off, isolation valve closed, right pack switch in auto, and left pack switch off - flight compartment and passenger cabin.	21-00 TASK 802
210 060 00	Smoke: Stops with recirculation fan(s) off, isolation valve closed, right pack switch in auto, and left pack switch off - passenger cabin.	21-00 TASK 802
210 070 00	Smoke: does not stop with recirculation fans OFF, isolation valve closed, and both pack switches at OFF - flight compartment.	21-00 TASK 803
210 080 00	Smoke: does not stop with recirculation fans OFF, isolation valve closed, and both pack switches at OFF - flight compartment and passenger cabin.	21-00 TASK 803
210 090 00	Smoke: does not stop with recirculation fans OFF, isolation valve closed, and both pack switches at OFF - passenger cabin.	21-00 TASK 803
210 671 00	Smoke or fumes in cabin: Source unknown.	21-00 TASK 806
212 010 31	Windshield: Air outlet does not operate - captain's.	21-22 TASK 801
212 010 32	Windshield: Air outlet does not operate - first officer's.	21-22 TASK 801
212 020 31	Foot air outlet: does not operate - captain's.	21-22 TASK 801
212 020 32	Foot air outlet: does not operate - first officer's.	21-22 TASK 801
212 030 00	Main Deck Air Distribution System Shutoff Valves Do Not Operate.	21-23 TASK 801
212 040 00	Recirculation fan (single-fan system): does not operate.	21-25 TASK 801
212 040 41	Recirculation fan (dual-fan system): does not operate - left.	21-25 TASK 802
212 040 42	Recirculation fan (dual-fan system): does not operate - right.	21-25 TASK 802
212 050 00	Smoke: Stops with recirculation fan(s) off - flight compartment.	21-00 TASK 804
212 060 00	Smoke: Stops with recirculation fan(s) off - flight compartment and passenger cabin.	21-00 TASK 804
212 070 00	Smoke: Stops with recirculation fan(s) off - passenger cabin.	21-00 TASK 804
212 080 00	EQUIP COOLING EXHAUST OFF light: light on with switch at ALTERNATE.	21-27 TASK 805
212 090 00	EQUIP COOLING EXHAUST OFF light: light on with switch at NORMAL.	21-27 TASK 806

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FAULT CODE	FAULT DESCRIPTION	GO TO FIM TASK
212 100 00	EQUIP COOLING SUPPLY OFF light: light on with switch at ALTERNATE.	21-27 TASK 807
212 110 00	EQUIP COOLING SUPPLY OFF light: light on with switch at NORMAL.	21-27 TASK 808
212 115 00	Equipment Cooling Supply Fans Do Not Shut Off When Commanded During a Cargo Compartment Smoke Alert.	21-27 TASK 810
212 120 00	Equipment Cooling Exhaust Fans Do Not Shut Off When Commanded During a Cargo Compartment Smoke Alert.	21-27 TASK 811
212 201 00	Conditioned Air Distribution: Low flow to the Flight Deck.	21-22 TASK 802
213 010 00	AUTO FAIL light: light on and ALTN light on, AUTO FAIL light goes off when pressurization mode selector switch is at ALTN.	21-31 TASK 801
213 020 00	AUTO FAIL light: light on, ALTN light not on.	21-31 TASK 801
213 030 00	Manual mode control of aft outflow valve: does not operate.	21-31 TASK 811
213 040 00	CABIN ALT indication: fluctuates.	21-31 TASK 801
213 050 00	Pressurization: cannot keep the correct cabin pressure.	21-31 TASK 801
213 060 00	Cabin altitude warning system: does not operate.	21-33 TASK 801
213 080 00	CABIN CLIMB Indicator: does not operate, pointer does not move or fluctuates.	21-33 TASK 802
213 090 00	FLT ALT display: indication blank.	21-31 TASK 803
213 100 00	LAND ALT display: indication blank.	21-31 TASK 803
213 110 00	MANUAL VALVE indicator: does not operate, pointer does not move or fluctuates.	21-31 TASK 803
213 121 00	Cabin altitude and differential pressure gauge: indicates high cabin pressure differential and low cabin altitude.	21-33 TASK 803
215 010 00	Fog, high humidity, or ice in cabin.	21-51 TASK 801
215 015 00	Noise, pounding heard through air distribution system.	21-51 TASK 802
215 020 41	PACK TRIP OFF light: light on at cruise, light stays off after reset - left.	21-52 TASK 824
215 020 42	PACK TRIP OFF light: light on at cruise, light stays off after reset - right.	21-52 TASK 825
215 030 41	PACK TRIP OFF light: light on at cruise, light comes on again after reset - left.	21-52 TASK 830
215 030 42	PACK TRIP OFF light: light on at cruise, light comes on again after reset - right.	21-52 TASK 831
215 040 41	PACK TRIP OFF light: light on, cannot reset - left.	21-52 TASK 816
215 040 42	PACK TRIP OFF light: light on, cannot reset - right.	21-52 TASK 817
215 050 41	PACK TRIP OFF light: light on during approach, light stays off after reset - left.	21-52 TASK 822
215 050 42	PACK TRIP OFF light: light on during approach, light stays off after reset - right.	21-52 TASK 823

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FAULT CODE	FAULT DESCRIPTION	GO TO FIM TASK
215 060 00	Temperature indication problem - air temperature selector at PACK L.	21-61 TASK 854
215 060 41	PACK TRIP OFF light: light on during approach, light comes on again after reset - left.	21-52 TASK 832
215 060 42	PACK TRIP OFF light: light on during approach, light comes on again after reset - right.	21-52 TASK 833
215 070 00	Temperature indication problem - air temperature selector at PACK R.	21-61 TASK 855
215 080 41	PACK TRIP OFF light: light on during climb, light stays off after reset - left.	21-52 TASK 820
215 080 42	PACK TRIP OFF light: light on during climb, light stays off after reset - right.	21-52 TASK 821
215 090 41	PACK TRIP OFF light: light on during climb, light comes on again after reset - left.	21-52 TASK 834
215 090 42	PACK TRIP OFF light: light on during climb, light comes on again after reset - right.	21-52 TASK 835
215 110 41	PACK TRIP OFF light: light on on ground, light stays off after reset - left.	21-52 TASK 818
215 110 42	PACK TRIP OFF light: light on on ground, light stays off after reset - right.	21-52 TASK 819
215 120 41	PACK TRIP OFF light: light on on ground, light comes on again after reset - left.	21-52 TASK 828
215 120 42	PACK TRIP OFF light: light on on ground, light comes on again after reset - right.	21-52 TASK 829
215 140 00	RAM DOOR FULL OPEN Light: light is not on, airplane on the ground.	21-51 TASK 804
215 160 00	RAM DOOR FULL OPEN Light: light stays on, airplane in the air.	21-51 TASK 803
215 201 41	PACK Light: is ON without MASTER CAUTION Recall - L PACK.	21-53 TASK 820
215 201 42	PACK Light: is ON without MASTER CAUTION Recall - R PACK.	21-53 TASK 820
215 202 41	PACK Light: comes ON During MASTER CAUTION Recall - L PACK.	21-53 TASK 821
215 202 42	PACK Light: comes ON During MASTER CAUTION Recall - R PACK.	21-53 TASK 821
215 203 00	Packs, Divergent Pack Outlet Temperatures Noted Between Left and Right Packs.	21-53 TASK 819
215 350 00	Pack airflow/temperature diagnostics tool: Tool shows pack airflow is too low or too high.	21-52 TASK 857
215 351 00	Pack airflow/temperature diagnostics tool: Tool shows pack outlet temperature is too cold or too hot.	21-52 TASK 856

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FAULT CODE	FAULT DESCRIPTION	GO TO FIM TASK
216 010 00	Air mix valve indicator: does not operate, pointer does not move, temperature is OK - CONT CABIN.	21-61 TASK 841
216 020 00	Air mix valve indicator: does not operate, pointer does not move, temperature is OK - PASS CABIN.	21-61 TASK 841
216 040 00	AUTO temperature control: does not operate - CONT CAB.	21-62 TASK 905
216 050 00	AUTO temperature control: does not operate - AFT CAB.	21-62 TASK 905
216 060 00	AUTO temperature control: does not operate - FWD CAB.	21-62 TASK 905
216 070 00	AUTO temperature control: does not operate - PASS CABIN.	21-61 TASK 862
216 080 00	DUCT OVERHEAT light: light on, stays off after reset - CONT CABIN.	21-61 TASK 839
216 085 00	DUCT OVERHEAT light: light on, stays off after reset - PASS CABIN.	21-61 TASK 840
216 090 00	DUCT OVERHEAT light: light on, comes on again after reset - CONT CABIN.	21-61 TASK 848
216 095 00	DUCT OVERHEAT light: light on, comes on again after reset - PASS CABIN.	21-61 TASK 849
216 100 00	DUCT OVERHEAT light: light on, cannot reset - CONT CABIN.	21-61 TASK 837
216 105 00	DUCT OVERHEAT light: light on, cannot reset - PASS CABIN.	21-61 TASK 838
216 120 00	Temperature indication problem - air temperature selector at AFT ZONE CABIN.	21-61 TASK 857
216 130 00	Temperature indication problem - air temperature selector at FWD ZONE CABIN.	21-61 TASK 856
216 140 00	Temperature indication problem - air temperature selector at CONT CAB SUPPLY DUCT.	21-61 TASK 858
216 160 00	Temperature indication problem - air temperature selector at SUPPLY DUCT.	21-61 TASK 842
216 170 00	Temperature indication problem - air temperature selector at AFT SUPPLY DUCT.	21-61 TASK 860
216 180 00	Temperature indication problem - air temperature selector at FWD SUPPLY DUCT.	21-61 TASK 859
216 190 00	Temperature indication problem - air temperature selector at PASS CABIN.	21-61 TASK 843
216 201 00	ZONE TEMP light: light on, stays off after reset - aft cabin.	21-62 TASK 903
216 202 00	ZONE TEMP light: light on, cannot reset - aft cabin.	21-62 TASK 904
216 203 00	ZONE TEMP light: light on, comes on again after reset - aft cabin.	21-62 TASK 903
216 211 00	ZONE TEMP light: light on, stays off after reset - control cabin.	21-62 TASK 903
216 212 00	ZONE TEMP light: light on, cannot reset - control cabin.	21-62 TASK 904
216 213 00	ZONE TEMP light: light on, comes on again after reset - control cabin.	21-62 TASK 903

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FAULT CODE	FAULT DESCRIPTION	GO TO FIM TASK
216 221 00	ZONE TEMP light: light on, stays off after reset - forward cabin.	21-62 TASK 903
216 222 00	ZONE TEMP light: light on, cannot reset - forward cabin.	21-62 TASK 904
216 223 00	ZONE TEMP light: light on, comes on again after reset - forward cabin.	21-62 TASK 903
216 224 00	Zone Temperature Control: ZONE TEMP Light Comes ON unexpectedly during MASTER CAUTION Recall - CONT CAB.	21-62 TASK 903
216 224 44	Zone Temperature Control: ZONE TEMP Light Comes ON unexpectedly during MASTER CAUTION Recall - FWD CAB.	21-62 TASK 903
216 224 45	Zone Temperature Control: ZONE TEMP Light Comes ON unexpectedly during MASTER CAUTION Recall - AFT CAB.	21-62 TASK 903
216 224 48	Zone Temperature Control: ZONE TEMP Light Comes ON unexpectedly during MASTER CAUTION Recall - FWD CAB and AFT CAB.	21-62 TASK 903
216 224 99	Zone Temperature Control: ZONE TEMP Light Comes ON.	21-62 TASK 904
216 310 00	Trim air valve does not operate.	21-62 TASK 838
216 400 00	Temperature: Too hot, cannot make cooler in AUTO mode, OK in MANUAL mode - CONT CABIN.	21-61 TASK 801
216 405 00	AUTO temperature control: Too cold, no temperature control in AUTO - PASS CABIN.	21-62 TASK 905
216 410 00	Temperature: Too hot, cannot make cooler in AUTO mode, OK in MANUAL mode - PASS CABIN.	21-61 TASK 801
216 420 00	Temperature: Too cold, cannot make warmer in AUTO mode, OK in MANUAL mode - CONT CABIN.	21-61 TASK 801
216 430 00	Temperature: Too cold, cannot make warmer in AUTO mode, OK in MANUAL mode - PASS CABIN.	21-61 TASK 801
216 435 00	AUTO temperature control: Too cold, no temperature control in AUTO - CONT CABIN.	21-62 TASK 905
216 440 00	Temperature: Too hot, cannot make cooler in AUTO and MANUAL mode, air mix valve does not move to COLD - CONT CABIN.	21-61 TASK 844
216 450 00	Temperature: Too hot, cannot make cooler in AUTO and MANUAL mode, air mix valve does not move to COLD - PASS CABIN.	21-61 TASK 846
216 460 00	Temperature: Too cold, cannot make warmer in AUTO and MANUAL mode, air mix valve does not move to HOT - CONT CABIN.	21-61 TASK 845
216 470 00	Temperature: Too cold, cannot make warmer in AUTO and MANUAL mode, air mix valve does not move to HOT - PASS CABIN.	21-61 TASK 847
216 510 00	Temperature control: Cabin temperatures are uncontrollable.	21-62 TASK 905
216 612 00	Temperature control: cannot cool cabins, all temperature selectors set to full COLD.	21-62 TASK 849

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FAULT CODE	FAULT DESCRIPTION	GO TO FIM TASK
217 821 41	Purge valve for air conditioning pack air supply cleaner: does not operate correctly during system test - left pack.	21-75 TASK 801
217 821 42	Purge valve for air conditioning pack air supply cleaner: does not operate correctly during system test - right pack.	21-75 TASK 801

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LRU/SYSTEM	SHORT NAME	CHAPTER
Air Data Inertial Reference System	ADIRS	34
Air Traffic Controller Transponder - 1 (Left)	ATC XPDR - 1 (L)	34
Air Traffic Controller Transponder - 2 (Right)	ATC XPDR - 2 (R)	34
Airborne Vibration Monitor System Signal Conditioner	AVM SIG COND	77
Antiskid Control Unit	ANTISKID	32
Attendant Control Panel	ACP	23
Automatic Direction Finder Receiver - 1	ADF RECVR - 1	34
Automatic Direction Finder Receiver - 2	ADF RECVR - 2	34
Autothrottle Computer	A/T COMPUTER	22
Auxiliary Power Unit	APU	49
Auxiliary Power Unit Generator Control Unit	APU GCU	24
Bus Power Control Unit	BPCU	24
Cabin Pressure Controller	CAB PRESS CON	21
Cabin Temperature Controller	CAB TEMP CONT	21
Cargo Electronic Unit - Lower Aft	CEU - LWR AFT	26
Cargo Electronic Unit - Lower Forward	CEU - LWR FWD	26
Cargo Electronic Unit - Main Aft	CEU - MAIN AFT	26
Cargo Electronic Unit - Main Forward	CEU - MAIN FWD	26
Cargo Fire Control Panel	CFCP	26
Common Display System	CDS	31
Compartment Overheat Detection Control Module	WING/BODY OHT	26
Digital Flight Control System	DFCS	22
Distance Measurement Equipment Interrogator	DME INTRROGTR	34
Electrical Meters, Battery, and Galley Power Module	P5-13	24
Electronic Engine Controller - 1	ENGINE - 1	73
Electronic Engine Controller - 2	ENGINE - 2	73
Emergency Locator Transmitter	ELT	23
Engine Accessory Unit	EAU	78
Engine Accessory Unit/TR DEPLOY ENG 1	EAU/TR DPLOY-ENG 1	78
Engine Accessory Unit/TR DEPLOY ENG 2	EAU/TR DPLOY-ENG 2	78
Engine Accessory Unit/TR STOW ENG 1	EAU/TR STOW-ENG 1	78
Engine Accessory Unit/TR STOW ENG 2	EAU/TR STOW-ENG 2	78
Engine and Auxiliary Power Unit Fire Detection Control Module	ENG/APU FIRE	26
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LRU/SYSTEM	SHORT NAME	CHAPTER
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Pack/Zone Temperature Controller - Left	PACK/ZN CON - L	21
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Proximity Switch Electronics Unit	PSEU	32
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Window Heat Control Unit - Right Side	WHCU - R SIDE	30
Window Heat Control Unit 1 - Left Forward and Right Side	WHCU1 - L FWD/R SIDE	30
Window Heat Control Unit 2 - Right Forward and Left Side	WHCU2 - R FWD/L SIDE	30

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LRU/SYSTEM	MAINTENANCE MESSAGE	GO TO FIM TASK
35 DEG CONT L	POSN 1 - NO GO	21-52 TASK 802
35 DEG CONT L	POSN 2 - GO, 35° F CONTROL VALVE NOT OPEN	21-52 TASK 812
35 DEG CONT L	POSN 2 - NO GO	21-52 TASK 802
35 DEG CONT L	POSN 3 - NO GO	21-52 TASK 802
35 DEG CONT L	POSN 4 - GO, 35° F CONTROL VALVE NOT CLOSED	21-52 TASK 813
35 DEG CONT L	POSN 4 - NO GO	21-52 TASK 802
35 DEG CONT L	POSN 5 - NO GO	21-52 TASK 810
35 DEG CONT R	POSN 1 - NO GO	21-52 TASK 802
35 DEG CONT R	POSN 2 - GO, 35° F CONTROL VALVE NOT OPEN	21-52 TASK 812
35 DEG CONT R	POSN 2 - NO GO	21-52 TASK 802
35 DEG CONT R	POSN 3 - NO GO	21-52 TASK 802
35 DEG CONT R	POSN 4 - GO, 35° F CONTROL VALVE NOT CLOSED	21-52 TASK 813
35 DEG CONT R	POSN 4 - NO GO	21-52 TASK 802
35 DEG CONT R	POSN 5 - NO GO	21-52 TASK 810
CAB PRESS CON	ACFT RATE HI	21-31 TASK 815
CAB PRESS CON	AUTO/MAN ERROR	21-31 TASK 821
CAB PRESS CON	CAB DIFF PRESS HI	21-31 TASK 816
CAB PRESS CON	CAB PRESS SW ACTIV	21-31 TASK 820
CAB PRESS CON	CAB RATE HI	21-31 TASK 818
CAB PRESS CON	CONTRL LRU FAIL	21-31 TASK 801
CAB PRESS CON	DADC #1 INVALID	21-31 TASK 806
CAB PRESS CON	DADC #2 INVALID	21-31 TASK 807
CAB PRESS CON	DRV ENBL ERROR	21-31 TASK 819
CAB PRESS CON	HI CAB ALT	21-31 TASK 817
CAB PRESS CON	LDG GEAR DISAGREE	21-31 TASK 822
CAB PRESS CON	LO INFL/HI LEAKG	21-31 TASK 823
CAB PRESS CON	N1/N2 LOWRU FAIL	21-31 TASK 810
CAB PRESS CON	OFV LRU FAIL	21-31 TASK 802
CAB PRESS CON	OFV OR WIRING	21-31 TASK 805
CAB PRESS CON	SEL PNL LRU FAIL	21-31 TASK 803
CAB PRESS CON	SMC #1 INVALID	21-31 TASK 808
CAB PRESS CON	SMC #2 INVALID	21-31 TASK 809
CAB PRESS CON	SYSTEM WIRING	21-31 TASK 804
CAB TEMP CONT	ANTICIPATOR SENSOR - LEFT PACK - NO GO (Position 7)	21-61 TASK 831
CAB TEMP CONT	ANTICIPATOR SENSOR - RIGHT PACK - NO GO (Position 7)	21-61 TASK 832

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LRU/SYSTEM	MAINTENANCE MESSAGE	GO TO FIM TASK
CAB TEMP CONT	CABIN SENSOR - LEFT PACK - NO GO (Position 6)	21-61 TASK 833
CAB TEMP CONT	CABIN SENSOR - RIGHT PACK - NO GO (Position 6)	21-61 TASK 834
CAB TEMP CONT	CLOSE VALVE DRIVE CHECK - LEFT PACK - NO GO (Position 5)	21-61 TASK 865
CAB TEMP CONT	CLOSE VALVE DRIVE CHECK - RIGHT PACK - NO GO (Position 5)	21-61 TASK 865
CAB TEMP CONT	DEADBAND CHECK - LEFT PACK - NO GO (Position 4)	21-61 TASK 865
CAB TEMP CONT	DEADBAND CHECK - RIGHT PACK - NO GO (Position 4)	21-61 TASK 865
CAB TEMP CONT	DUCT LIMIT SENSOR - LEFT PACK - NO GO (Position 8)	21-61 TASK 829
CAB TEMP CONT	DUCT LIMIT SENSOR - RIGHT PACK - NO GO (Position 8)	21-61 TASK 830
CAB TEMP CONT	OPEN VALVE DRIVE CHECK - LEFT PACK - NO GO (Position 3)	21-61 TASK 865
CAB TEMP CONT	OPEN VALVE DRIVE CHECK - RIGHT PACK - NO GO (Position 3)	21-61 TASK 865
CAB TEMP CONT	POWER CHECK - LEFT PACK - NO GO (Position 2)	21-61 TASK 865
CAB TEMP CONT	POWER CHECK - RIGHT PACK - NO GO (Position 2)	21-61 TASK 865
CAB TEMP CONT	TEMP SELECTOR - LEFT PACK - NO GO	21-61 TASK 835
CAB TEMP CONT	TEMP SELECTOR - RIGHT PACK - NO GO	21-61 TASK 836
PACK/ZN CON - L	BKUP ZONE CABIN SENSOR	21-62 TASK 815
PACK/ZN CON - L	BKUP ZONE SELECTOR	21-62 TASK 809
PACK/ZN CON - L	CONTROLLER FAULT - PACK/ZONE	21-62 TASK 802
PACK/ZN CON - L	CONTROLLER FAULT - RAM AIR	21-62 TASK 802
PACK/ZN CON - L	CONTROLLER FAULT - STANDBY	21-62 TASK 802
PACK/ZN CON - L	FLT DECK CABIN SENSOR	21-62 TASK 811
PACK/ZN CON - L	FLT DECK DUCT SENSOR	21-62 TASK 813
PACK/ZN CON - L	FLT DECK SELECTOR	21-62 TASK 805
PACK/ZN CON - L	FLT DECK TRIM VALVE	21-62 TASK 833
PACK/ZN CON - L	RAM AIR ACT	21-62 TASK 837
PACK/ZN CON - L	RAM AIR SNSR	21-62 TASK 821
PACK/ZN CON - L	SRADA EXIT	21-62 TASK 853
PACK/ZN CON - L	SRADA INLET	21-62 TASK 851
PACK/ZN CON - L	STBY OTHER PACK - PACK SNSR	21-62 TASK 824
PACK/ZN CON - L	STBY OTHER PACK - TCV	21-62 TASK 829
PACK/ZN CON - L	THIS PACK - MIX MANF SENSOR	21-62 TASK 827
PACK/ZN CON - L	THIS PACK - PACK SNSR	21-62 TASK 825
PACK/ZN CON - L	THIS PACK - TCV	21-62 TASK 830

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LRU/SYSTEM	MAINTENANCE MESSAGE	GO TO FIM TASK
PACK/ZN CON - L	ZONE CABIN SENSOR	21-62 TASK 817
PACK/ZN CON - L	ZONE DUCT SENSOR	21-62 TASK 819
PACK/ZN CON - L	ZONE SELECTOR	21-62 TASK 808
PACK/ZN CON - L	ZONE TRIM VALVE	21-62 TASK 835
PACK/ZN CON - R	BKUP ZONE CABIN SENSOR	21-62 TASK 816
PACK/ZN CON - R	BKUP ZONE SELECTOR	21-62 TASK 807
PACK/ZN CON - R	CONTROLLER FAULT - PACK/ZONE	21-62 TASK 803
PACK/ZN CON - R	CONTROLLER FAULT - RAM AIR	21-62 TASK 803
PACK/ZN CON - R	CONTROLLER FAULT - STANDBY	21-62 TASK 803
PACK/ZN CON - R	FLT DECK CABIN SENSOR	21-62 TASK 810
PACK/ZN CON - R	FLT DECK DUCT SENSOR	21-62 TASK 812
PACK/ZN CON - R	FLT DECK SELECTOR	21-62 TASK 804
PACK/ZN CON - R	FLT DECK TRIM VALVE	21-62 TASK 832
PACK/ZN CON - R	RAM AIR ACT	21-62 TASK 836
PACK/ZN CON - R	RAM AIR SNSR	21-62 TASK 820
PACK/ZN CON - R	SRADA EXIT	21-62 TASK 854
PACK/ZN CON - R	SRADA INLET	21-62 TASK 852
PACK/ZN CON - R	STBY OTHER PACK - PACK SNSR	21-62 TASK 823
PACK/ZN CON - R	STBY OTHER PACK - TCV	21-62 TASK 828
PACK/ZN CON - R	THIS PACK - MIX MANF SENSOR	21-62 TASK 826
PACK/ZN CON - R	THIS PACK - PACK SNSR	21-62 TASK 822
PACK/ZN CON - R	THIS PACK - TCV	21-62 TASK 831
PACK/ZN CON - R	ZONE CABIN SENSOR	21-62 TASK 814
PACK/ZN CON - R	ZONE DUCT SENSOR	21-62 TASK 818
PACK/ZN CON - R	ZONE SELECTOR	21-62 TASK 806
PACK/ZN CON - R	ZONE TRIM VALVE	21-62 TASK 834
PFTC	21-51001 PFTC CNTRL INOP	21-62 TASK 858
PFTC	21-51002 PFTC CNTRL INOP	21-62 TASK 858
PFTC	21-51003 PFTC CNTRL INOP	21-62 TASK 858
PFTC	21-51004 PFTC CNTRL INOP	21-62 TASK 858
PFTC	21-51005 PFTC CNTRL INOP	21-62 TASK 858
PFTC	21-51006 PFTC CNTRL INOP	21-62 TASK 858
PFTC	21-51007 PFTC CNTRL INOP	21-62 TASK 858
PFTC	21-51008 PFTC CNTRL INOP	21-62 TASK 858
PFTC	21-51009 PFTC CNTRL INOP	21-62 TASK 858

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LRU/SYSTEM	MAINTENANCE MESSAGE	GO TO FIM TASK
PFTC	21-51010 PFTC CNTRL INOP	21-62 TASK 858
PFTC	21-51017 PFTC CNTRL INOP	21-62 TASK 858
PFTC	21-51018 PFTC CNTRL INOP	21-62 TASK 858
PFTC	21-51019 PFTC CNTRL INOP	21-62 TASK 858
PFTC	21-51030 DISPLAY	21-62 TASK 858
PFTC	21-51034 PFTC CNTRL INOP	21-62 TASK 858
PFTC	21-51035 PFTC CNTRL INOP	21-62 TASK 858
PFTC	21-51036 PFTC CNTRL INOP	21-62 TASK 858
PFTC	21-51037 PFTC CNTRL INOP	21-62 TASK 858
PFTC	21-51038 PFTC CNTRL INOP	21-62 TASK 858
PFTC	21-51041 PFTC CNTRL INOP	21-62 TASK 858
PFTC	21-51042 PFTC CNTRL INOP	21-62 TASK 858
PFTC	21-51043 PFTC CNTRL INOP	21-62 TASK 858
PFTC	21-51044 PFTC CNTRL INOP	21-62 TASK 858
PFTC	21-51045 PFTC CNTRL INOP	21-62 TASK 858
PFTC	21-51046 STBY 28V PWR OFF	21-62 TASK 858
PFTC	21-51053 PFTC CNTRL INOP	21-62 TASK 862
PFTC	21-51056 PFTC CNTRL INOP	21-62 TASK 863
PFTC	21-51061 PFTC CNTRL INOP	21-62 TASK 859
PFTC	21-51066 PFTC CNTRL INOP	21-62 TASK 860
PFTC	21-51067 PFTC CNTRL INOP	21-62 TASK 860
PFTC	21-51068 PFTC CNTRL INOP	21-62 TASK 860
PFTC	21-51069 PFTC CNTRL INOP	21-62 TASK 857
PFTC	21-51070 PFTC CNTRL INOP	21-62 TASK 857
PFTC	21-51073 PFTC CNTRL INOP	21-62 TASK 876
PFTC	21-51074 PFTC CNTRL INOP	21-62 TASK 876
PFTC	21-51075 PFTC CNTRL INOP	21-62 TASK 876
PFTC	21-51078 PFTC CNTRL INOP	21-62 TASK 877
PFTC	21-51079 PFTC CNTRL INOP	21-62 TASK 877
PFTC	21-51080 PFTC CNTRL INOP	21-62 TASK 877
PFTC	21-51083 PFTC CNTRL INOP	21-62 TASK 874
PFTC	21-51084 PFTC CNTRL INOP	21-62 TASK 874
PFTC	21-51085 PFTC CNTRL INOP	21-62 TASK 874
PFTC	21-51086 PFTC STBY CNTRL	21-62 TASK 875
PFTC	21-51087 PFTC STBY CNTRL	21-62 TASK 875

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LRU/SYSTEM	MAINTENANCE MESSAGE	GO TO FIM TASK
PFTC	21-51090 PFTC STBY CNTRL	21-62 TASK 858
PFTC	21-51091 PFTC STBY CNTRL	21-62 TASK 858
PFTC	21-51092 PFTC STBY CNTRL	21-62 TASK 858
PFTC	21-51093 PFTC STBY CNTRL	21-62 TASK 858
PFTC	21-51097 PFTC STBY CNTRL	21-62 TASK 864
PFTC	21-51100 PFTC STBY CNTRL	21-62 TASK 868
PFTC	21-51101 PFTC STBY CNTRL	21-62 TASK 868
PFTC	21-51110 PFTC CNTRL INOP	21-62 TASK 887
PFTC	21-51111 PFTC CNTRL INOP	21-62 TASK 887
PFTC	21-51112 PFTC CNTRL INOP	21-62 TASK 888
PFTC	21-51113 PFTC CNTRL INOP	21-62 TASK 888
PFTC	21-51114 PFTC CNTRL INOP	21-62 TASK 889
PFTC	21-51115 PFTC CNTRL INOP	21-62 TASK 889
PFTC	21-51116 PFTC CNTRL INOP	21-62 TASK 890
PFTC	21-51117 PFTC CNTRL INOP	21-62 TASK 890
PFTC	21-51118 PFTC CNTRL INOP	21-62 TASK 891
PFTC	21-51119 PFTC CNTRL INOP	21-62 TASK 891
PFTC	21-51120 PFTC CNTRL INOP	21-62 TASK 892
PFTC	21-51121 PFTC CNTRL INOP	21-62 TASK 892
PFTC	21-51122 PFTC CNTRL INOP	21-62 TASK 893
PFTC	21-51123 PFTC CNTRL INOP	21-62 TASK 893
PFTC	21-51124 PFTC CNTRL INOP	21-62 TASK 894
PFTC	21-51125 PFTC CNTRL INOP	21-62 TASK 894
PFTC	21-51126 PFTC CNTRL INOP	21-62 TASK 895
PFTC	21-51127 PFTC CNTRL INOP	21-62 TASK 895
PFTC	21-51128 PFTC CNTRL INOP	21-62 TASK 896
PFTC	21-51129 PFTC CNTRL INOP	21-62 TASK 896
PFTC	21-51131 ARINC 429 INVLD	21-62 TASK 883
PFTC	21-51132 ARINC 429 INVLD	21-62 TASK 884
PFTC	21-51133 ARINC 429 INVLD	21-62 TASK 881
PFTC	21-51134 ARINC 429 INVLD	21-62 TASK 882
PFTC	21-51135 ARINC 429 INVLD	21-62 TASK 885
PFTC	21-51136 ARINC 429 INVLD	21-62 TASK 886
PFTC	21-51143 ARINC 429 INVLD	21-62 TASK 883
PFTC	21-51144 ARINC 429 INVLD	21-62 TASK 883

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LRU/SYSTEM	MAINTENANCE MESSAGE	GO TO FIM TASK
PFTC	21-51145 ARINC 429 INVLD	21-62 TASK 884
PFTC	21-51146 ARINC 429 INVLD	21-62 TASK 884
PFTC	21-51147 ARINC 429 INVLD	21-62 TASK 881
PFTC	21-51148 ARINC 429 INVLD	21-62 TASK 881
PFTC	21-51149 ARINC 429 INVLD	21-62 TASK 882
PFTC	21-51150 ARINC 429 INVLD	21-62 TASK 882
PFTC	21-51151 ARINC 429 INVLD	21-62 TASK 885
PFTC	21-51152 ARINC 429 INVLD	21-62 TASK 885
PFTC	21-51153 ARINC 429 INVLD	21-62 TASK 886
PFTC	21-51154 ARINC 429 INVLD	21-62 TASK 886
PFTC	21-51190 F/D TEMP SENSOR	21-62 TASK 865
PFTC	21-51191 F/D TEMP SENSOR	21-62 TASK 865
PFTC	21-51193 ZONE TEMP SENSOR	21-62 TASK 867
PFTC	21-51194 ZONE TEMP SENSOR	21-62 TASK 867
PFTC	21-51196 STBY ZONE SENSOR	21-62 TASK 868
PFTC	21-51197 STBY ZONE SENSOR	21-62 TASK 868
PFTC	21-51199 F/D DUCT SENSOR	21-62 TASK 866
PFTC	21-51200 F/D DUCT SENSOR	21-62 TASK 866
PFTC	21-51202 ZONE DUCT SENSOR	21-62 TASK 869
PFTC	21-51203 ZONE DUCT SENSOR	21-62 TASK 869
PFTC	21-51205 PACK TEMP SENSOR	21-62 TASK 871
PFTC	21-51206 PACK TEMP SENSOR	21-62 TASK 871
PFTC	21-51208 STBY PACK TEMP	21-62 TASK 872
PFTC	21-51209 STBY PACK TEMP	21-62 TASK 872
PFTC	21-51211 MIXMAN TEMP SNSR	21-62 TASK 873
PFTC	21-51212 MIXMAN TEMP SNSR	21-62 TASK 873
PFTC	21-51217 F/D SELECTOR	21-62 TASK 862
PFTC	21-51218 F/D SELECTOR	21-62 TASK 862
PFTC	21-51220 ZONE SELECTOR	21-62 TASK 863
PFTC	21-51221 ZONE SELECTOR	21-62 TASK 863
PFTC	21-51224 STBY SELECTOR	21-62 TASK 864
PFTC	21-51225 STBY SELECTOR	21-62 TASK 864
PFTC	21-51230 PACK IN PRESSURE	21-62 TASK 859
PFTC	21-51231 PACK IN PRESSURE	21-62 TASK 859
PFTC	21-51232 PACK IN PRESSURE	21-62 TASK 859

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LRU/SYSTEM	MAINTENANCE MESSAGE	GO TO FIM TASK
PFTC	21-51233 PACK IN PRESSURE	21-62 TASK 859
PFTC	21-51239 PACK FLOW SENSOR	21-62 TASK 860
PFTC	21-51240 PACK FLOW SENSOR	21-62 TASK 860
PFTC	21-51246 PACK FLOW SENSOR	21-62 TASK 860
PFTC	21-51247 PACK FLOW SENSOR	21-62 TASK 860
PFTC	21-51248 PACK FLOW SENSOR	21-62 TASK 860
PFTC	21-51249 PACK FLOW SENSOR	21-62 TASK 860
PFTC	21-51250 FLOW CTRL VALVE	21-62 TASK 857
PFTC	21-51251 FLOW CTRL VALVE	21-62 TASK 857
PFTC	21-51252 FLOW CTRL VALVE	21-62 TASK 857
PFTC	21-51253 FLOW CTRL VALVE	21-62 TASK 857
PFTC	21-51254 FLOW CTRL VALVE	21-62 TASK 857
PFTC	21-51255 FLOW CTRL VALVE	21-62 TASK 902
PFTC	21-51256 FLOW CTRL VALVE	21-62 TASK 861
PFTC	21-51257 FLOW CTRL VALVE	21-62 TASK 861
PFTC	21-51260 TEMP CTRL VALVE	21-62 TASK 874
PFTC	21-51261 TEMP CTRL VALVE	21-62 TASK 874
PFTC	21-51263 TEMP CTRL VALVE	21-62 TASK 874
PFTC	21-51264 TEMP CTRL VALVE	21-62 TASK 874
PFTC	21-51265 TEMP CTRL VALVE	21-62 TASK 874
PFTC	21-51266 TEMP CTRL VALVE	21-62 TASK 874
PFTC	21-51267 TEMP CTRL VALVE	21-62 TASK 874
PFTC	21-51268 F/D TRIM VALVE	21-62 TASK 876
PFTC	21-51269 F/D TRIM VALVE	21-62 TASK 876
PFTC	21-51271 F/D TRIM VALVE	21-62 TASK 876
PFTC	21-51272 F/D TRIM VALVE	21-62 TASK 876
PFTC	21-51273 F/D TRIM VALVE	21-62 TASK 876
PFTC	21-51274 F/D TRIM VALVE	21-62 TASK 876
PFTC	21-51275 F/D TRIM VALVE	21-62 TASK 876
PFTC	21-51276 ZONE TRIM VALVE	21-62 TASK 877
PFTC	21-51277 ZONE TRIM VALVE	21-62 TASK 877
PFTC	21-51279 ZONE TRIM VALVE	21-62 TASK 877
PFTC	21-51280 ZONE TRIM VALVE	21-62 TASK 877
PFTC	21-51281 ZONE TRIM VALVE	21-62 TASK 877
PFTC	21-51282 ZONE TRIM VALVE	21-62 TASK 877

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LRU/SYSTEM	MAINTENANCE MESSAGE	GO TO FIM TASK
PFTC	21-51283 ZONE TRIM VALVE	21-62 TASK 877
PFTC	21-51300 STBY TCV VALVE	21-62 TASK 875
PFTC	21-51301 STBY TCV VALVE	21-62 TASK 875
PFTC	21-51310 F/D TEMP SENSOR	21-62 TASK 865
PFTC	21-51311 F/D SELECTOR	21-62 TASK 862
PFTC	21-51312 F/D DUCT SENSOR	21-62 TASK 866
PFTC	21-51313 PACK TEMP SENSOR	21-62 TASK 871
PFTC	21-51314 MIXMAN TEMP SNSR	21-62 TASK 873
PFTC	21-51331 DISPLAY	21-62 TASK 858
PFTC	21-51332 PFTC CNTRL INOP	21-62 TASK 858
PFTC	21-51333 PFTC CNTRL INOP	21-62 TASK 858
PFTC	21-51400 SRADA CNTRL	21-62 TASK 878
PFTC	21-51401 SRADA INLET	21-62 TASK 878
PFTC	21-51402 SRADA EXIT	21-62 TASK 879
PFTC	21-51403 SRADA RAM SNSR	21-62 TASK 870
PFTC	21-51405 SRADA INLET	21-62 TASK 878
PFTC	21-51406 SRADA EXIT	21-62 TASK 879
PFTC	21-51407 SRADA RAM SNSR	21-62 TASK 870
PFTC	21-51408 SRADA CNTRL	21-62 TASK 880
PFTC	21-51409 SRADA INLET	21-62 TASK 878
PFTC	21-51410 SRADA EXIT	21-62 TASK 879
PFTC	21-51411 SRADA RAM SNSR	21-62 TASK 870
PFTC	21-51500 PFTC CNTRL INOP	21-62 TASK 860
PFTC	21-51501 PFTC CNTRL INOP	21-62 TASK 857
PFTC	21-51502 PFTC CNTRL INOP	21-62 TASK 860
PFTC	21-51503 PFTC CNTRL INOP	21-62 TASK 857
PFTC	21-51504 PFTC CNTRL INOP	21-62 TASK 857
PFTC	21-51505 PFTC CNTRL INOP	21-62 TASK 876
PFTC	21-51506 PFTC CNTRL INOP	21-62 TASK 877
PFTC	21-51507 PFTC CNTRL INOP	21-62 TASK 874
PFTC	21-51508 F/D TRIM VALVE	21-62 TASK 876
PFTC	21-51509 ZONE TRIM VALVE	21-62 TASK 877
PFTC	21-51510 TEMP CTRL VALVE	21-62 TASK 874

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801. Smoke or Fumes Reported During Flight in the Passenger Cabin or Flight Compartment; Smoke Stops With ISOLATION VALVE Closed and R PACK Switch OFF - Fault Isolation

A. Description

(1) Smoke or fumes flow into the passenger cabin or the flight compartment through the Air Distribution System during flight.

(2) This problem is usually caused by contamination of the air by oil, glycol, fuel, or hydraulic fluid.

(3) In flight, contamination can be caused by leakage of these fluids in one of the engines.

(4) Smoke or fumes can also be caused by oil, glycol, hydraulic fluid, fuel fumes or exhaust fumes being ingested into the inlet of one of the engines.

(5) From the inlet, the contaminant can go into the Pneumatic and/or the Air Conditioning System and vaporize at higher temperatures.

(6) Fumes can also be caused by the use of petroleum jelly during maintenance of the Pneumatic Ducts or Air Distribution Ducts.

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(7) Make sure that the fault report shows the smoke or fumes were stopped with the ISOLATION VALVE closed, the R PACK Switch OFF, and the Recirculation Fan(s) OFF.

NOTE: 737-600 and 737-700 Airplanes have ONE Recirculation Fan and Filter.

737-800 and 737-900 Airplanes have TWO Recirculation Fans and Filters. Make sure you do a check of both Recirculation Fans and Filters on 737-800 and 737-900 Airplanes.

(a) This means that Engine 1, the Pneumatic Air Supply from the Engine 1, the Left Pack, and the related Air Distribution Ducts were not the source of the smoke or fumes.

(b) The Fault Isolation Procedure below tells you how to find the cause of smoke in the Engine 2, the APU Pneumatic System, the Pneumatic Air Supply from the Engine 2, the Right Pack, the Recirculation System, and the related Air Distribution Ducts.

OMR 807

(8) Make sure that the fault report shows the smoke or fumes were stopped with the ISOLATION VALVE closed and the R PACK Switch OFF.

(a) This means that Engine 1, the Pneumatic Air Supply from Engine 1, the Right Pack, and the related Air Distribution Ducts were not the source of the smoke or fumes.

(b) The Fault Isolation Procedure below tells you how to find the cause of smoke in the Engine 2, the APU Pneumatic System, the Pneumatic Air Supply from the Engine 2, the Right Pack, the Recirculation System, and the related Air Distribution Ducts.

OMR ALL

(9) Do not do this procedure with the passengers or the crew on the airplane since it can cause smoke or fumes in the passenger cabin.

(10) (SDS SUBJECT 21-20-00)

(11) (SDS SUBJECT 36-00-00)

B. Possible Causes

(1) Contamination in the Engine 2 Pneumatic System

(2) Hydraulic Fluid Leakage

(3) Contamination in the Air Distribution System

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(4) Recirculation Fan(s) Overheating

OMR ALL

C. Related Data

(1) WDM 21-25-11

(2) SSM 21-25-11

D. Fault Isolation Procedure

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(1) Do these checks of the Air Recirculation System:

NOTE: Examine the Air Recirculation Fan(s) and Filter(s) only if there is smoke or fumes in the passenger compartment.

NOTE: 737-600 and 737-700 Airplanes have ONE Recirculation Fan and Filter.

737-800 and 737-900 Airplanes have TWO Recirculation Fans and Filters. Make sure you do a check of both Recirculation Fans and Filters on 737-800 and 737-900 Airplanes.

(a) Do this task: Recirculation Air Filter - Removal, AMM TASK 21-25-01-000-801.

(b) Examine the Air Recirculation Filter(s) for:

- Contaminants: dust, dirt, lint, and oil
- Rips in the filter
- Breaks in the Filter Frame

1) If necessary, install new Air Recirculation Filter(s). This is the task: Recirculation Air Filter - Installation, AMM TASK 21-25-01-400-801.

(c) Do the Recirculation System - Operational Test, AMM TASK 21-25-00-710-801-001 or Recirculation System - Operational Test, AMM TASK 21-25-00-710-802-002 to see if the Air Recirculation Fan(s) or the related wiring generates smoke when it operates.

1) If there is smoke from the Air Recirculation Fan(s) or the related wiring, then do the applicable steps below:

a) Replace the Air Recirculation Fan(s). These are the tasks:

- Recirculation Fan Removal, AMM TASK 21-25-02-000-801
- Recirculation Fan Installation, AMM TASK 21-25-02-400-801

b) Repair the applicable wiring.

c) Do the Repair Confirmation at the end of this task.

2) If there is no smoke or fumes from the Air Recirculation Fan(s) or the related wiring, and the Operational Test is satisfactory, then continue.

OMR ALL

(2) Do these checks for contamination in the Engine 2 Pneumatic System:

(a) Look for leakage of oil in Engine No. 2. This is the task: External Oil System Inspection, AMM TASK 79-00-00-200-802-F00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Do an inspection of the vents and drains for indications of leakage of oil, hydraulic fluid, or fuel in Engine 2. This is the task: Engine Vents and Drains Inspection, AMM TASK 71-71-00-200-801-F00.

NOTE: It is not necessary to do this task again if you did it during the external oil system inspection.

(c) If you find indications of leakage in Engine 2, then do these steps:

1) Do the corrective action shown in the inspection procedures or repair the leakage problems that you find.

2) Do the Repair Confirmation at the end of this task.

(d) If you do not find indications of leakage, then look for indications of oil leakage internal to the engine. To do this, do this task: Internal Oil System Inspection, AMM TASK 79-00-00-200-803-F00.

1) If you find indications of leakage, then do these steps:

a) Repair the leakage.

b) Do the Repair Confirmation at the end of this task.

2) If you do not find indications of leakage, then continue:

(3) Do this check for Hydraulic Fluid Leakage:

(a) Do this task: Hydraulic Fluid Leakage Problem - Fault Isolation, 29-10 TASK 812.

(b) If you find hydraulic fluid leakage, then do the corrective action indicated in the procedure.

1) Do the Repair Confirmation at the end of this task.

(c) If you do not find hydraulic fluid leakage, then continue.

(4) Do this check for contamination in the Air Distribution System:

(a) Do the Oil Contamination Removal Procedure with Engine 2 and the Right Pack ON until smoke or fumes appear in the cabin. This is the task: Oil Contamination Removal from Air Conditioning and Pneumatic Systems, AMM TASK 21-00-01-100-801.

(b) If smoke or fumes appear, then do these steps:

1) Do a check of these areas in the Air Distribution System for the source of the contamination:

a) Each Pack Component

b) Main Distribution Manifold

c) Air Distribution Ducts where the smoke or fumes first appear

2) Replace the component or clean the duct that is the source of the smoke.

3) Do the Repair Confirmation at the end of this task.

(c) If smoke or fumes do not appear, then do the Full Oil Contamination Removal Procedure with the TWO engines, the APU, and the TWO Packs ON. This is the task: Oil Contamination Removal from Air Conditioning and Pneumatic Systems, AMM TASK 21-00-01-100-801.

(d) If smoke or fumes appear, then do these steps:

1) Do a check of these areas in the Air Distribution System for the source of the contamination:

a) Each Pack Component

b) Main Distribution Manifold

ECCN 9E991 BOEING PROPRIETARY - See title page for details

c) Air Distribution Ducts where the smoke or fumes first appear

2) Replace the component or clean the duct that is the source of the smoke.

3) Do the Repair Confirmation at the end of this task.

(e) If smoke or fumes do not appear, and if the problem does not occur on the subsequent flight, then there was an intermittent problem.

E. Repair Confirmation

(1) Do this task: Oil Contamination Removal from Air Conditioning and Pneumatic Systems, AMM TASK 21-00-01-100-801.

(a) If the Oil Contamination Removal Procedure is done until there is no smoke or fumes, and the problem does not occur on the subsequent flight, then you corrected the problem.

——— END OF TASK ———

802. Smoke or Fumes Reported During Flight in the Passenger Cabin or Flight Compartment; Smoke Stops With ISOLATION VALVE Closed and L PACK Switch OFF - Fault Isolation

A. Description

(1) Smoke or fumes flow into the passenger cabin or the flight compartment through the Air Distribution System during flight.

(2) This problem is usually caused by contamination of the air by oil, glycol, fuel, or hydraulic fluid.

(3) In flight, contamination can be caused by leakage of these fluids in one of the engines.

(4) Smoke or fumes can also be caused by oil, glycol, hydraulic fluid, fuel fumes or exhaust fumes being ingested into the inlet of one of the engines.

(5) From the inlet, the contaminant can go into the Pneumatic and/or the Air Conditioning System and vaporize at higher temperatures.

(6) Fumes can also be caused by the use of petroleum jelly during maintenance of the Pneumatic Ducts or Air Distribution Ducts.

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(7) Make sure that the fault report shows the smoke or fumes were stopped with the ISOLATION VALVE closed, the L PACK Switch OFF, and the Recirculation Fan(s) OFF.

NOTE: 737-600 and 737-700 Airplanes have ONE Recirculation Fan and Filter.

737-800 and 737-900 Airplanes have TWO Recirculation Fans and Filters. Make sure you do a check of both Recirculation Fans and Filters on 737-800 and 737-900 Airplanes.

(a) This means that Engine 2, the Pneumatic Air Supply from the Engine 2, the Right Pack, and the related Air Distribution Ducts were not the source of the smoke or fumes.

(b) The Fault Isolation Procedure below tells you how to find the cause of smoke in the Engine 1, the APU Pneumatic System, the Pneumatic Air Supply from the Engine 1, the Left Pack, the Recirculation System, and the related Air Distribution Ducts.

OMR 807

(8) Make sure that the fault report shows the smoke or fumes were stopped with the ISOLATION VALVE closed, and the Left PACK Switch OFF.

(a) This means that Engine 2, the Pneumatic Air Supply from the Engine 2, the Right Pack, and the related Air Distribution Ducts were not the source of the smoke or fumes.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 807 (Continued)

(b) The Fault Isolation Procedure below tells you how to find the cause of smoke in the Engine 1, the APU Pneumatic System, the Pneumatic Air Supply from the Engine 1, the Left Pack, the Recirculation System, and the related Air Distribution Ducts.

OMR ALL

(9) Do not do this procedure with the passengers or the crew on the airplane since it can cause smoke or fumes in the passenger cabin.

(10) (SDS SUBJECT 21-20-00)

(11) (SDS SUBJECT 36-00-00)

B. Possible Causes

(1) Contamination in the Engine 1 Pneumatic System

(2) Contamination in the APU Pneumatic System

(3) Hydraulic Fluid Leakage

(4) Contamination in the Air Distribution System

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(5) Recirculation Fan(s) Overheating

OMR ALL

C. Related Data

(1) WDM 21-25-11

(2) SSM 21-25-11

D. Fault Isolation Procedure

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(1) Do these checks of the Air Recirculation System:

NOTE: Examine the Air Recirculation Fan(s) and Filter(s) only if there is smoke or fumes in the passenger compartment.

NOTE: 737-600 and 737-700 Airplanes have ONE Recirculation Fan and Filter.

737-800 and 737-900 Airplanes have TWO Recirculation Fans and Filters. Make sure you do a check of both Recirculation Fans and Filters on 737-800 and 737-900 Airplanes.

(a) Do this task: Recirculation Air Filter - Removal, AMM TASK 21-25-01-000-801.

(b) Examine the Air Recirculation Filter(s) for:

- Contaminants: dust, dirt, lint, and oil
- Rips in the filter
- Breaks in the Filter Frame

1) If necessary, install new Air Recirculation Filter(s). This is the task: Recirculation Air Filter - Installation, AMM TASK 21-25-01-400-801.

(c) Do the Recirculation System - Operational Test, AMM TASK 21-25-00-710-801-001 or Recirculation System - Operational Test, AMM TASK 21-25-00-710-802-002 to see if the Air Recirculation Fan(s) or the related wiring generates smoke when it operates.

1) If there is smoke from the Air Recirculation Fan(s) or the related wiring, then do the applicable steps below:

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a) Replace the Air Recirculation Fan(s). These are the tasks:
- Recirculation Fan Removal, AMM TASK 21-25-02-000-801
- Recirculation Fan Installation, AMM TASK 21-25-02-400-801

b) Repair the applicable wiring.

c) Do the Repair Confirmation at the end of this task.

2) If there is no smoke or fumes from the Air Recirculation Fan(s) or the related wiring, and the Operational Test is satisfactory, then continue.

OMR ALL

(2) Do these checks for contamination in the APU Pneumatic System:

(a) Do this task:Fumes or Smoke in Cabin During APU Operation - Fault Isolation, 49-55 TASK 802.

1) If you do not find an indication of contamination, then continue.

(3) Do these checks for contamination in the Engine 1 Pneumatic System:

(a) Look for leakage of oil in Engine 1. This is the task: External Oil System Inspection, AMM TASK 79-00-00-200-802-F00.

(b) Do an inspection of the vents and drains for indications of leakage of oil, hydraulic fluid, or fuel in Engine 1. This is the task: Engine Vents and Drains Inspection, AMM TASK 71-71-00-200-801-F00.

NOTE: It is not necessary to do this task again if you did it during the external oil system inspection.

(c) If you find indications of leakage in Engine 1, then do these steps:

1) Do the corrective action shown in the inspection procedures or repair the leakage problems that you find.

2) Do the Repair Confirmation at the end of this task.

(d) If you do not find indications of leakage, then look for indications of oil leakage internal to the engine. This is the task: Internal Oil System Inspection, AMM TASK 79-00-00-200-803-F00.

1) If you find indications of leakage, then do these steps:

a) Repair the leakage.

b) Do the Repair Confirmation at the end of this task.

2) If you do not find indications of leakage, then continue.

(4) Do this check for Hydraulic Fluid Leakage:

(a) Do this task: Hydraulic Fluid Leakage Problem - Fault Isolation, 29-10 TASK 812.

(b) If you find hydraulic fluid leakage, then do the corrective action indicated in the procedure.

1) Do the Repair Confirmation at the end of this task.

(c) If you do not find hydraulic fluid leakage, then continue.

(5) Do this check for contamination in the Air Distribution System:

(a) Do the Oil Contamination Removal Procedure with Engine 1 and the Left Pack ON until smoke or fumes appear in the cabin. This is the task: Oil Contamination Removal from Air Conditioning and Pneumatic Systems, AMM TASK 21-00-01-100-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) If smoke or fumes appear, then do these steps:

1) Do a check of these areas in the Air Distribution System for the source of the contamination:

a) Each Pack Component

b) Main Distribution Manifold

c) Air Distribution Ducts where the smoke or fumes first appear

2) Replace the component or clean the duct that is the source of the smoke.

3) Do the Repair Confirmation at the end of this task.

(c) If smoke or fumes do not appear, then do the Full Oil Contamination Removal Procedure with the two engines, the APU, and the two Packs ON. This is the task: Oil Contamination Removal from Air Conditioning and Pneumatic Systems, AMM TASK 21-00-01-100-801.

(d) If smoke or fumes appear, then do these steps:

1) Do a check of these areas in the Air Distribution System for the source of the contamination:

a) Each Pack Component

b) Main Distribution Manifold

c) Air Distribution Ducts where the smoke or fumes first appear

2) Replace the component or clean the duct that is the source of the smoke.

3) Do the Repair Confirmation at the end of this task.

(e) If smoke or fumes do not appear, and if the problem does not occur on the subsequent flight, then there was an intermittent problem.

E. Repair Confirmation

(1) Do this task: Oil Contamination Removal from Air Conditioning and Pneumatic Systems, AMM TASK 21-00-01-100-801.

(a) If the Oil Contamination Removal Procedure is done until there is no smoke or fumes, and the problem does not occur on the subsequent flight, then you corrected the problem.

——— END OF TASK ———

803. Smoke or Fumes in the Passenger Cabin or Flight Compartment, Reported During Flight, Smoke Does Not Stop With Isolation Valve Closed, Each of the Two Pack Switches Off (In Turn) - Fault Isolation

A. Description

(1) Smoke or fumes flow into the passenger cabin or the flight compartment through the air distribution system during flight. This problem is usually caused by contamination of the air by oil, glycol, fuel, or hydraulic fluid. Fumes can also be caused by the use of petroleum jelly during maintenance of the pneumatic ducts or air distribution ducts.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799

(2) Make sure the fault report shows the smoke or fumes were not stopped with the isolation valve closed, the recirculation fan off, the right pack switch set at OFF and then the left pack switch at OFF. Usually during the flight operations procedure, the right pack switch and the left pack switch are turned off in sequence, not at the same time. This usually means that the source of the smoke or fumes is the main air distribution manifold or the air duct system downstream. It is unlikely to be in the pneumatic air supply or the engine since each pack and each engine was used independently with the isolation valve closed. Thus, the source of the smoke or fumes must be in the main air distribution manifold or the air duct system downstream.

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(3) Make sure the fault report shows the smoke or fumes were not stopped with the isolation valve closed, the two recirculation fans off, the right pack switch set at OFF and then the left pack switch at OFF. Usually during the flight operations procedure, the right pack switch and the left pack switch are turned off in sequence, not at the same time. This usually means that the source of the smoke or fumes is the main air distribution manifold or the air duct system downstream. It is unlikely to be in the pneumatic air supply or the engine since each pack and each engine was used independently with the isolation valve closed. Thus, the source of the smoke or fumes must be in the main air distribution manifold or the air duct system downstream.

OMR 807

(4) Make sure the fault report shows the smoke or fumes were not stopped with the isolation valve closed and the right pack switch set at OFF and then the left pack switch at OFF. Usually during the flight operations procedure, the right pack switch and the left pack switch are turned off in sequence, not at the same time. This usually means that the source of the smoke or fumes is the main air distribution manifold or the air duct system downstream. It is unlikely to be in the pneumatic air supply or the engine since each pack and each engine was used independently with the isolation valve closed. Thus, the source of the smoke or fumes must be in the main air distribution manifold or the air duct system downstream.

OMR ALL

(5) Do not do this procedure with the passengers or the crew on the airplane since it can cause smoke or fumes in the passenger cabin.

(6) (SDS SUBJECT 21-20-00)

(7) (SDS SUBJECT 36-00-00)

B. Possible Causes

(1) Hydraulic fluid leakage

(2) Contamination in the air distribution system

C. Fault Isolation Procedure

(1) Do this check for hydraulic fluid leakage:

(a) Do this task: Hydraulic Fluid Leakage Problem - Fault Isolation, 29-10 TASK 812.

(b) If you find hydraulic fluid leakage, then do the corrective action indicated in the procedure.

1) Do the Repair Confirmation procedure at the end of this task.

(c) If you do not find hydraulic fluid leakage, then continue.

(2) Do this check of the air distribution system:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Do this task: Oil Contamination Removal from Air Conditioning and Pneumatic Systems, AMM TASK 21-00-01-100-801.

(b) If smoke or fumes appear, then do these steps:

1) Do a check of these areas in the air distribution system for the source of the contamination:

a) The main distribution manifold

b) The air distribution ducts where the smoke first appears

2) Replace the component or clean the duct that is the source of the smoke.

3) Do the Repair Confirmation at the end of this task.

(c) If smoke or fumes do not appear, then there was an intermittent fault.

D. Repair Confirmation

(1) Do this task: Oil Contamination Removal from Air Conditioning and Pneumatic Systems, AMM TASK 21-00-01-100-801.

(a) If the oil contamination removal procedure is done until there is no smoke or fumes, and if the problem does not occur on the subsequent flight, then you corrected the fault.

——— END OF TASK ———

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

804. Smoke or Fumes in the Passenger Cabin - Stops With Recirculation Fan Off - Fault Isolation

A. Description

(1) Smoke or fumes flow into the passenger cabin or the flight compartment through the air distribution system. This problem is usually caused by contamination of the air by oil, glycol, fuel, or hydraulic fluid. If you think the problem is caused by one of these types of contamination, then do the applicable fault isolation procedure below:.

(a) Do this task: Smoke or Fumes Reported During Flight in the Passenger Cabin or Flight Compartment; Smoke Stops With ISOLATION VALVE Closed and R PACK Switch OFF - Fault Isolation, 21-00 TASK 801.

(b) Do this task: Smoke or Fumes Reported During Flight in the Passenger Cabin or Flight Compartment; Smoke Stops With ISOLATION VALVE Closed and L PACK Switch OFF - Fault Isolation, 21-00 TASK 802.

OMR 701-799

(2) But if the problem was corrected when the recirculation fan was turned off and the air conditioning system was in its normal configuration with the two packs in operation and the isolation valve open, then the only possible cause is the air recirculation system. Make sure the fault report shows the smoke or fumes were stopped with the recirculation fan off. If this was the only measure taken to stop the smoke and the smoke was stopped, this means that the recirculation system was the cause of the smoke or fumes.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(3) But if the problem was corrected when either one or both of the recirculation fans were turned off and the air conditioning system was in its normal configuration with the two packs in operation and the isolation valve open, then the only possible cause is the air recirculation system. Make sure the fault report shows the smoke or fumes were stopped with the recirculation fans off. If this was the only measure taken to stop the smoke and the smoke was stopped, this means that the recirculation system was the cause of the smoke or fumes.

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(4) Do not do this procedure with the passengers or the crew on the airplane since it can cause smoke or fumes in the passenger cabin.

(5) (SDS SUBJECT 21-20-00)

(6) (SDS SUBJECT 36-00-00)

B. Possible Causes

(1) Recirculation Fan Overheating

(2) Wiring

C. Related Data

(1) WDM 21-25-11

(2) SSM 21-25-11

D. Fault Isolation Procedure

(1) Do these checks of the Air Recirculation System:

NOTE: Examine the Air Recirculation Fans and filters only if there is smoke or fumes in the passenger compartment.

(a) Do this task: Recirculation Air Filter - Removal, AMM TASK 21-25-01-000-801.

OMR 701-799

(b) Examine the Air Recirculation Filter for:

- Contaminants: dust, dirt, lint, and oil
- Rips in the Filter
- Breaks in the Filter Frame

1) If necessary, install a new Air Recirculation Filter. This is the task: Recirculation Air Filter - Installation, AMM TASK 21-25-01-400-801.

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(c) Examine the Air Recirculation Filters for:

- Contaminants: dust, dirt, lint, and oil
- Rips in the Filter
- Breaks in the Filter Frame

1) If necessary, install new Air Recirculation Filters. This is the task: Recirculation Air Filter - Installation, AMM TASK 21-25-01-400-801.

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OMR 701-799

(d) Do the Recirculation System - Operational Test, AMM TASK 21-25-00-710-801-001 to see if the Air Recirculation Fan or the related wiring generates smoke when it operates.

1) If there is smoke from the Air Recirculation Fan or the related wiring, then do the applicable steps below:

a) Replace the Air Recirculation Fan. These are the tasks:

- Recirculation Fan Removal, AMM TASK 21-25-02-000-801
- Recirculation Fan Installation, AMM TASK 21-25-02-400-801

b) Repair the applicable wiring.

c) Do this task: Recirculation System - Operational Test, AMM TASK 21-25-00-710-801-001.

<1> If the Operational Test is satisfactory and the problem does not occur on the subsequent flight, then you corrected the problem..

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(e) Do the Recirculation System - Operational Test, AMM TASK 21-25-00-710-802-002 to see if the Air Recirculation Fans or the related wiring generate smoke when they operate.

1) If there is smoke from the Air Recirculation Fans or the related wiring, then do the applicable steps below:

a) Replace the applicable Air Recirculation Fan(s). These are the tasks:

- Recirculation Fan Removal, AMM TASK 21-25-02-000-801
- Recirculation Fan Installation, AMM TASK 21-25-02-400-801

b) Repair the applicable wiring.

c) Do this task: Recirculation System - Operational Test, AMM TASK 21-25-00-710-802-002.

<1> If the Operational Test is satisfactory and the problem does not occur on the subsequent flight, then you corrected the problem..

OMR ALL

——— END OF TASK ———

806. Smoke or Fumes in the Cabin, Source Unknown - Fault Isolation

A. General

CAUTION DO NOT REPLACE PARTS UNTIL YOU IDENTIFY, AND REMOVE THE SOURCE OF THE FUMES. THE FUMES CAN CAUSE THE CONTAMINATION OF A NEW PART. THEN THE NEW PART WILL BE THE SOURCE OF THE FUMES.

(1) To troubleshoot for cabin fumes or smoke, look for obvious sources inside the cabin and then work to the outside. Start with newly leaking fluids from the engine cowling, APU compressors, and APU drain masts.

(2) Smoke or fumes can be caused by an internal APU/engine failures which can release oil into the pneumatic and/or air conditioning systems. Another source of smoke or fumes is oil, glycol or hydraulic fluid being ingested into the inlet of the APU or engines. Contamination that is ingested by the APU or engines can enter the pneumatic and/or air conditioning systems.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(3) Additional sources of smoke or fumes can be the recirculation fans, gasper fans (if installed), equipment cooling fans, cabin temperature sensor fan, compressor assembly, IFE Equipment (if installed), materials in the cargo compartments, ground carts, fuel leaks that have entered the ducts of the air conditioning system, and hydraulic fluid from the reservoir pressurization module tubing.

OMR 807

(4) Additional sources of smoke or fumes can be the gasper fans (if installed), equipment cooling fans, cabin temperature sensor fan, compressor assembly, IFE Equipment (if installed), materials in the cargo compartments, ground carts, fuel leaks that have entered the ducts of the air conditioning system, and hydraulic fluid from the reservoir pressurization module tubing.

OMR ALL

(5) Odors in the cabin after an engine has been overhauled or the compressor has been washed are not uncommon. These conditions are not considered a fault. Put the airplane "on watch" as necessary.

(6) Odors in the cabin can occur again after the correction of a problem with a bleed air source (APU, engines). Additional flights or the replacement of ozone converters (if installed), trim air mufflers, heat exchangers, or cooling pack components may be necessary to remove all the contamination.

(7) Electrical burning odors in the captains side of the cabin only can be a display unit or other electrical equipment in the forward part of the flight deck. These conditions may not cause a fault. Inspect the flight deck equipment and display units as necessary.

(8) Faults in different systems make fumes with different smells. The table that follows has descriptions of the fumes from different systems to help you isolate the fault.

FUMES	DESCRIPTION	FAULT ISOLATION TASK(S)
OIL	SWEAT, LOCKER ROOM, DIRTY SOCK, RANCID CHEESE, WET DOG, BURNING RUBBER, SOUR MILK, (FRESH OIL) SWEET, MILD IRRITATION TO EYES	Check the APU, Left and Right Engine for Oil Contamination, Check the Cooling Packs for Oil Contamination, Check the Ozone Converter for odor, If present, work upstream to find the source of the contamination
HYDRAULIC FLUID	STRONG IRRITATION TO EYES, PUNGENT, ACRID	Check the Hydraulic Reservoir Pressurization Module tubing and Check Valves for Hydraulic Oil Residue. Check hydraulic system for overservicing condition.
FUEL	(BURNED FUEL) KEROSENE, (UNBURNED FUEL) ACRID, BITTER	Check the engines as cause of Oil/Fuel Fumes/Smoke in Cabin (FIM 71-05)
OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999		
ELECTRICAL FAULTS	ACRID, BURNING RUBBER, SULFUR	Check the Cabin Equipment, Check Recirculation Fans and Cargo Compartments, Check Light Ballasts and Galley Equipment
OMR 807		
ELECTRICAL FAULTS	ACRID, BURNING RUBBER, SULFUR	Check the Cabin Equipment, Check Cargo Compartments, Check Light Ballasts and Galley Equipment
OMR ALL		

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(Continued)

FUMES	DESCRIPTION	FAULT ISOLATION TASK(S)
OTHER	COOKED CHICKEN (BIRD STRIKE), BURNING RESIN, ACRID ODOR THAT IRRITATES NOSE AND THROAT, HOT MUSTY SMELL, HAZE IN FLIGHT WITH NO SMELL	Check the Engines for evidence of Bird Strike, Check Trim Valve operation for zone that is source of complaint. Check under Aft Cargo Floor for dripped fluids or dust around the APU Bleed Duct, Check that fumes are not being ingested into bleed air from an external source. Check cargo floors above and below and behind cargo sidewalls for dust or lint and for unwanted materials spilled or left behind

(9) Do not do this procedure with the crew or passengers on the airplane. This procedure can generate smoke in the airplane.

B. Evaluation With the use of Bleed Air Sampling Tool Kit

NOTE: This procedure should be performed only if you are using the air sampler, SPL-11153. The air sampler, SPL-11153 was designed to provide access to the high pressure bleed air via the airplane pneumatic ground port. Smoke and/or Burning Order (SBO) events may not be confirmed to bleed system sources. The air sampler, SPL-11153 is designed specifically to assist with the isolation of engine and APU oil odor sources.

(1) Install the air sampler, SPL-11153, to the Pneumatic Ground Air Connector.

(2) Make sure that electrical power is supplied to the airplane Supply Electrical Power, AMM TASK 24-22-00-860-811.

(3) Make sure these switches on the pilot's overhead panel, P5, which follow are set to the indicated positions:

(a) On the AIR CONDITIONING and BLEED AIR Module:

1) APU, BLEED 1, and BLEED 2 Switches are set to OFF.

2) ISOLATION VALVE Switch is set to CLOSED.

3) L PACK and R PACK Switches are set to OFF.

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

4) RECIRC FAN Switch set to OFF.

OMR ALL

(b) On the ANTI-ICE module:

1) WING, ENG 1, and ENG 2 Switches are set to OFF.

(4) To supply bleed air using the APU, do these steps:

(a) Start the APU and let the APU operation become stable (APU Starting and Operation, AMM TASK 49-11-00-860-801).

(b) On the BLEED AIR Module of the Overhead Panel, P5, set switches that follow to the indicated positions:

1) Put the ISOLATION Valve Switch to the AUTO position.

2) Put the APU switch to the ON position.

(5) To supply bleed air using the engines, do these steps:

(a) Do this task to start the engines: Start the Engine Procedure (Selection), AMM TASK 71-00-00-800-807-F00.

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(b) On the BLEED AIR Module of the Overhead Panel, P5, set switches that follow to the indicated positions:

1) Put the ISOLATION Valve Switch to the AUTO position.

2) Put the 1, and 2 BLEED Switches to the ON position.

(6) Allow bleed air to run through theair sampler, SPL-11153 , for 10 minutes.

(7) Do these steps to remove bleed air:

(a) If you provided bleed air using the APU, do these steps:

1) Put the APU switch, located on the BLEED AIR module of the Overhead Panel, P5, to OFF.

2) Put the ISOLATION VALVE switch, located on the BLEED AIR module of the Overhead Panel, P5, to the CLOSE position.

3) Stop the operation of the APU (APU Usual Shutdown, AMM TASK 49-11-00-860-802).

(b) If you provided bleed air using the engines, do these steps:

1) Put the ISOLATION VALVE switch, located on the BLEED AIR module of the Overhead Panel, P5, to the CLOSE position.

2) Put the L PACK, and R PACK switches, located on the BLEED AIR module of the Overhead Panel, P5, to the OFF position.

3) Stop the operation of the engines (Stop the Engine Procedure (Usual Engine Stop), AMM TASK 71-00-00-700-819-F00).

(8) Remove the air sampling pad from the air sampler, SPL-11153.

(9) Remove the air sampler, SPL-11153, from the airplane.

(10) Extract data from the sample using a spectrometer per supplier instructions.

(11) Use the data to determine the source of the of the smoke or fumes.

(12) Based on the results of the spectrometer, do repairs as needed.

(13) Do the Repair Confirmation at the end of this task.

C. Initial Evaluation

(1) Do a check of the Low Pressure Ground Connector.

(a) Make sure that the check is done less than an hour after the engines are stopped.

(b) To get access to the Ground Air Check Valve for the Cooling Packs, open the ECS low pressure connection door, 191E, do this task: Finding an Access Door or Panel on the Lower Half of the Fuselage, AMM TASK 06-41-00-800-801.

(c) Hold the low pressure flapper valve open.

(d) Use a mirror to look inside the low pressure ground connector for signs of oil contamination.

(e) Feel the surfaces inside the low pressure ground connector to see if fresh oil is present.

(f) If oil contamination is found, the probable source is the left/right engine or APU. The pack must be cleaned. This would rule out E/E cooling and recirculation.

(2) Do a visual inspection of the drain mast.

(a) If nothing is found, then follow the detailed instructions.

(3) Check hydraulic system for overservicing condition.

(a) If necessary, do this task, Hydraulic Reservoir Servicing, AMM TASK 12-12-00-610-801.

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(4) Do a check for existing faults for the cabin and IFE Equipment.

NOTE: Faults for the cabin and IFE Equipment may be related to fumes in the cabin or flight deck.

(5) Start the APU. This is the task: APU Starting and Operation, AMM TASK 49-11-00-860-801.

(6) Start the left and right engines. This is the task: Start the Engine Procedure (Selection), AMM TASK 71-00-00-800-807-F00.

WARNING OBSERVE PROPER SAFETY PRECAUTIONS AROUND RUNNING ENGINE. WEAR EAR PROTECTORS AND STAY CLEAR OF ENGINE HAZARD AREAS. SEE "ENGINE GROUND SAFETY PRECAUTIONS" FOR ENGINE HAZARD DESCRIPTION.

(7) Make sure that you obey all safety precautions when you are near an engine that is in operation. To obey the safety precautions, comply with the directions in this task:Engine Ground Safety Precautions, AMM TASK 71-00-00-800-805-F00.

(8) On the BLEED AIR module of the Overhead Panel, P5, set switches that follow to the indicated positions:

(a) Put the ISOLATION VALVE Switch to the OPEN position.

(b) Put the ENGINE 1, and ENGINE 2 Switches to the ON position.

(c) Put the APU Switch to the ON position.

(9) Make sure the hydraulic reservoirs are pressurized. This is the task: Hydraulic Reservoirs Pressurization, AMM TASK 29-09-00-860-801 or Hydraulic Reservoirs Pressurization, AMM TASK 29-09-00-860-807.

(10) Supply conditioned air with the left and right cooling packs. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(11) Make sure the switches that follow are set to the indicated positions:

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(a) On the P5-10 air conditioning panel:

OMR 701-799

1) RECIRC FAN Switch is set to OFF.

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

2) L RECIRC FAN and R RECIRC FAN Switches are set to OFF.

OMR ALL

(b) On the P5-11 thermal anti-ice panel:

1) WING, ENGINE 1, and ENGINE 2, ANTI-ICE switches are set to OFF.

(12) Operate the cooling packs, APU and engines in each of the configurations shown in the table below.

(a) Operate the cooling packs with the trim air ON and the cabin temperatures selected to full Warm (W).

(b) Repeat the steps with the trim air OFF and the cabin temperatures selected to full Cold (C).

(c) Use the REFERENCE NUMBER table to find the applicable fault isolation procedure to perform when fumes are present.

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APU	L ENG	R ENG	L PACK	R PACK	FUMES	REFERENCE NUMBER(S)
ON	OFF	OFF	ON	OFF	YES	4,1
ON	OFF	OFF	OFF	ON	YES	4,1
OFF	ON	OFF	ON	OFF	YES	4,2
OFF	OFF	ON	OFF	ON	YES	4,3

REFERENCE NUMBER	FAULT ISOLATION TASK TITLE
1	Check the APU Bleed Air for Oil Contamination
2	Check the Left Engine Bleed Air for Oil Contamination
3	Check the Right Engine Bleed Air for Oil Contamination
4	Check the Cooling Packs for Oil Contamination

(13) If no fumes or smoke appear, then there was an intermittent fault.

NOTE: Occurrences of smoke or fumes in flight are not always repeatable on the ground.

(a) If no fumes or smoke appear, you may be able to determine the source of the smoke occurrence from the description of the smell and its location from the crew members on the flight.

(b) If a description of the smell of the fumes is available use the fumes description table in the General Section to determine which Fault Isolation Procedure is applicable.

(c) If the type of fumes can not be identified, then do the Fault Isolation Task from the beginning.

(14) Stop the Cooling Packs. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(15) Stop the Left and Right engines. This is the task: Stop the Engine Procedure (Usual Engine Stop), AMM TASK 71-00-00-700-819-F00.

(16) Stop the APU. This is the task: APU Usual Shutdown, AMM TASK 49-11-00-860-802.

D. Fault Isolation Procedure - Check the APU Bleed Air for Oil Contamination

(1) Do the steps that follow to check the APU for the correct quantity of oil:

(a) Get access to the APU Oil Level Sight Gage:

Open this access panel:

Number	Name/Location
315A	APU Cowl Door

1) To open the APU Access Doors, do these steps:

a) Open the three latches.

b) Open the APU Cowl Door.

c) Remove the retainer pin from the rod end of the forward hold-open rod on the APU Cowl Door.

d) Remove the retainer pin from the spring clip on the aft hold-open rod.

e) Disconnect the two hold-open rods from the two spring clips.

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f) Connect the two rod ends of the two hold-open rods to the two brackets in the APU Compartment.

g) Install the two retainer pins in the two rod ends.

(b) Make sure the APU Oil Level Sight Gage shows the correct quantity of oil.

(c) If the APU oil level is less than FULL, do a check of the maintenance records to see if there is a change in oil consumption averages. Do this task: APU Oil Level Inspection, AMM TASK 12-13-31-200-801.

NOTE: The oil consumption for a new or newly overhauled APU is 0.008 quarts (8 cc) per hour. Oil consumption more than this rate does not necessarily indicate a problem with the APU. If the oil consumption increases rapidly, then there could be a problem with the APU.

(2) Check the APU Magnetic Chip Detectors for debris:

(a) Do this task: Magnetic Drain Plug Inspection, AMM TASK 49-91-81-200-801

NOTE: Debris in the oil can cause damage to bearings and seals. Damage to seals can cause oil loss and contamination of the pneumatic system. Oil contamination of the pneumatic system can cause smoke and fumes.

(3) Inspect the APU Inlet area and Inlet Duct:

(a) Examine the fuselage exterior near the APU Inlet Door, and the inside of the APU Inlet Duct for signs that fluid went into the APU Inlet.

1) If fluid went into the APU inlet, find the cause of the fluid and correct the problem before you continue.

(4) Do the steps that follow to check the APU:

(a) Remove the APU Bleed Air Valve. This is the task: Bleed Air Valve Removal, AMM TASK 49-52-11-000-801.

(b) Remove the APU Surge Control Valve. This is the task: Surge Control Valve Removal, AMM TASK 49-52-41-000-801.

(c) Examine the APU Bleed Air Valve and the upstream APU Bleed Duct for wet oil.

(d) Examine the APU Surge Control Valve and the upstream duct for wet oil.

(e) Re-install the APU Bleed Air Valve. This is the task: Bleed Air Valve Installation, AMM TASK 49-52-11-400-801.

(f) Re-install the APU Surge Control Valve. This is the task: Surge Control Valve Installation, AMM TASK 49-52-41-400-801.

(g) Inspect the APU Load Compressor for damage. do this task: APU Engine Inspection, AMM TASK 49-21-00-200-801.

(h) Inspect the APU Power Section Compressor for damage. This is the task: APU Engine Inspection, AMM TASK 49-21-00-200-801.

(5) Supply electrical power with a ground cart. This is the task: Supply External Power, AMM TASK 24-22-00-860-813.

NOTE: Do not operate the APU or the engines to supply electrical power to the airplane.

(6) To get access to the cooling packs, open these access panels, in this sequence:

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

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WARNING PUT ON GLOVES FOR PROTECTION FROM HOT SURFACES WHEN YOU CONNECT OR DISCONNECT PNEUMATIC FITTINGS. PACK OPERATION CAN MAKE THE GROUND AIR CONNECTOR VERY HOT. A HOT GROUND AIR CONNECTOR WILL BURN YOUR BARE HANDS.

(7) Install the deactivator, SPL-2462, in the ground air connector as follows:

(a) Put the deactivator, SPL-2462, in the ground air connector so that the check valve flappers are open.

(b) Install the coupling on the supply hose from the ground air cart onto the nipple of the ground air connector check valve assembly so that the deactivator, SPL-2462 is held in place.

NOTE: The supply hose should not be connected to the ground cart as the exhaust from the hose should be vented to ambient.

(8) Make sure these switches on the P5 Overhead Panel, are set to the indicated positions:

(a) On the P5-10 Air Conditioning Panel:

1) L PACK and R PACK Switches are set to OFF.

OMR 701-799

2) RECIRC FAN Switch is set to OFF.

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

3) L RECIRC FAN and R RECIRC FAN Switches are set to OFF.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

4) TRIM AIR Switch is set to OFF.

OMR ALL

(b) On the P5-11 thermal anti-ice panel:

1) WING, ENGINE 1, and ENGINE 2, ANTI-ICE switches are set to OFF.

WARNING DO NOT PUT YOUR HAND OR BODY IN THE AIR THAT COMES OUT OF THE PNEUMATIC GROUND CONNECTOR. THE HOT AIR WILL CAUSE INJURIES.

(9) Do the steps which follow to do a check for oil contamination from the APU:

NOTE: The APU bleed air will go out through the check valve in the pneumatic ground air connector.

(a) Start the APU and let the APU operation become stable. To start and operate the APU, do this task: APU Starting and Operation, AMM TASK 49-11-00-860-801.

(b) Make sure the ISOLATION VALVE switch on the P5-10 air conditioning panel is set to OPEN.

(c) Make sure the APU BLEED switch, on the P5-10 air conditioning panel, is set to ON.

(d) Operate the APU for 5 minutes.

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WARNING DO NOT PUT YOUR HAND OR BODY IN THE AIR THAT COMES OUT OF THE PNEUMATIC GROUND CONNECTOR. THE HOT AIR WILL CAUSE INJURIES.

(e) Stand near the pneumatic ground air connector and smell if there is contaminated air.

NOTE: You can do a smell check of the bleed air from a different APU if it is necessary to compare the bleed air fumes.

(f) Look for smoke in the APU Bleed Air.

(g) If there is contaminated Bleed Air from the APU, do the steps that follow:

1) Replace the APU. These are the tasks:

- APU Power Plant Removal, AMM TASK 49-11-00-000-801, APU Power Plant Removal, AMM TASK 49-11-00-000-801, APU Power Plant Removal (Fishpole Hoist Procedure), AMM TASK 49-11-00-000-802, APU Power Plant Removal (Hydraulic Jack Procedure), AMM TASK 49-11-00-000-803
- APU Power Plant Installation, AMM TASK 49-11-00-400-801, APU Power Plant Installation (Fishpole Hoist Procedure), AMM TASK 49-11-00-400-802, APU Power Plant Installation (Hydraulic Jack Procedure), AMM TASK 49-11-00-400-803

2) Remove the oil contamination. This is the task: Oil Contamination Removal from Air Conditioning and Pneumatic Systems, AMM TASK 21-00-01-100-801.

3) If the oil contamination removal task does not remove the fumes, it may be necessary to replace components of the affected cooling pack. To replace the components, do the steps that follow:

a) Replace the Heat Exchanger for the oil contaminated Cooling Pack. These are the tasks:

- Heat Exchanger and Plenum/Diffuser Assembly - Removal, AMM TASK 21-51-03-000-802-001 or Heat Exchanger and Plenum/Diffuser Assembly - Removal, AMM TASK 21-51-03-000-805-002
- Heat Exchanger and Plenum/Diffuser Assembly - Installation, AMM TASK 21-51-03-400-801-001 or Heat Exchanger and Plenum/Diffuser Assembly - Installation, AMM TASK 21-51-03-400-804-002

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

b) Replace the condenser for the oil contaminated Cooling Pack. These are the tasks:

- Condenser Removal, AMM TASK 21-51-12-000-801
- Condenser Installation, AMM TASK 21-51-12-400-802

c) Replace the Reheater for the oil contaminated Cooling Pack. These are the tasks:

- Reheater Removal, AMM TASK 21-51-13-000-801
- Reheater Installation, AMM TASK 21-51-13-400-801

OMR ALL

d) If there is smoke or fumes from the Left Cooling Pack, replace the Flight Deck Mufflers. These are the tasks:

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- Flight Deck Air Distribution System Muffler/Duct Assembly Removal, AMM TASK 21-22-05-000-801
- Flight Deck Air Distribution System Muffler/Duct Assembly Installation, AMM TASK 21-22-05-400-801

4) Do the Repair Confirmation at the end of this task.

(h) If there is no contaminated Bleed Air from the APU, then continue.

(i) Set the ISOLATION VALVE Switch on the P5-10 Air Conditioning Panel to CLOSE.

(j) Stop the operation of the APU. This is the task: APU Usual Shutdown, AMM TASK 49-11-00-860-802.

WARNING PUT ON GLOVES FOR PROTECTION FROM HOT SURFACES WHEN YOU CONNECT OR DISCONNECT PNEUMATIC FITTINGS. PACK OPERATION CAN MAKE THE GROUND AIR CONNECTOR VERY HOT. A HOT GROUND AIR CONNECTOR WILL BURN YOUR BARE HANDS.

(10) Remove the Ground Cart Supply Hose and deactivator, SPL-2462, from the Pneumatic Ground connector.

(11) Close these access panels in this sequence:

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

E. Fault Isolation Procedure - Check the Left Engine Bleed Air for Oil Contamination

(1) Do the steps that follow to check the left engine for the correct quantity of oil:

(a) Do the applicable steps from this task:

Replenish the Engine Oil, AMM TASK 12-13-11-600-801

(b) Make sure the engine has the correct quantity of oil.

(c) If the oil level is near or over the top of the sight glass, correct the oil level before you continue. Do the task to drain the engine oil (Drain the Engine Oil, AMM TASK 12-13-11-600-803).

NOTE: Too much oil in the engine(s) can cause smoke or fumes in the pneumatic system.

(d) If the oil level is less than FULL, do a check of the maintenance records to see if there is a change in oil consumption averages.

NOTE: An increase in oil consumption can be an indication of oil loss and possible contamination of the pneumatic system. This can cause smoke or fumes in the pneumatic system.

OMR ALL; ENGINES WITH DEBRIS MONITORING SYSTEM

(2) Examine the Debris Monitoring System for debris:

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OMR ALL; ENGINES WITH DEBRIS MONITORING SYSTEM (Continued)

(a) Do this task:CDU Interrogation of the DMS Detectors, AMM TASK 79-00-00-710-801-F00.

NOTE: Debris in the oil can cause damage to bearings and seals. Damage to seals can cause oil loss and contamination of the pneumatic system. Oil contamination of the pneumatic system can cause smoke and fumes.

OMR ALL

(3) Do the steps that follow to do a check of the left engine for damage:

(a) Examine the area behind the fan blades for large oil stains or puddles of oil.

(b) Examine the area behind the fan blades for signs of a bird strike.

(4) Supply electrical power with a Ground Cart. This is the task: Supply External Power, AMM TASK 24-22-00-860-813.

NOTE: Do not operate the APU or the engines to supply electrical power to the airplane.

(5) Make sure these switches on the P5 overhead panel, are set to the indicated positions:

(a) On the P5-10 air conditioning panel:

1) APU, ENGINE 1, and ENGINE 2 Switches are set to OFF

2) ISOLATION VALVE Switch is set to CLOSE

3) L PACK and R PACK Switches are set to OFF.

OMR 701-799

4) RECIRC FAN Switch is set to OFF.

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

5) L RECIRC FAN and R RECIRC FAN Switches are set to OFF.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

6) TRIM AIR Switch is set to OFF.

OMR ALL

(b) On the P5-11 Thermal Anti-Ice Panel:

(c) WING, ENGINE 1, and ENGINE 2, ANTI-ICE Switches are set to OFF.

(6) To get access to the cooling packs, open these access panels, in this sequence:

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

WARNING PUT ON GLOVES FOR PROTECTION FROM HOT SURFACES WHEN YOU CONNECT OR DISCONNECT PNEUMATIC FITTINGS. PACK OPERATION CAN MAKE THE GROUND AIR CONNECTOR VERY HOT. A HOT GROUND AIR CONNECTOR WILL BURN YOUR BARE HANDS.

(7) Install the deactivator, SPL-2462, in the Ground Air Connector as follows:

(a) Put the deactivator, SPL-2462, in the ground air connector so that the check valve flappers are open.

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(b) Install the coupling on the supply hose from the ground air cart onto the nipple of the ground air connector check valve assembly so that the deactivator, SPL-2462 is held in place.

NOTE: The supply hose should not be connected to the ground cart as the exhaust from the hose should be vented to ambient.

WARNING

DO NOT PUT YOUR HAND OR BODY IN THE AIR THAT COMES OUT OF THE PNEUMATIC GROUND CONNECTOR. THE HOT AIR WILL CAUSE INJURIES.

(8) Do these steps which follow to check for oil contamination from the left engine:

NOTE: The left engine bleed air will go out through the pneumatic ground air connector duct opening.

(a) Start the left engine and let the engine operate at ground idle speed. To start and operate the left engine, do this task: Start the Engine Procedure (Selection), AMM TASK 71-00-00-800-807-F00.

1) Use the APU to start the engine as it is still in operation.

2) Set the APU BLEED switch, on the P5-10 air conditioning panel, to OFF.

3) Set the ISOLATION VALVE switch on the P5-10 air conditioning panel to OPEN.

(b) Set the engine pneumatic BLEED 1 switch, on the P5-10 air conditioning panel, to ON.

WARNING

OBSERVE PROPER SAFETY PRECAUTIONS AROUND RUNNING ENGINE. WEAR EAR PROTECTORS AND STAY CLEAR OF ENGINE HAZARD AREAS. SEE "ENGINE GROUND SAFETY PRECAUTIONS" FOR ENGINE HAZARD DESCRIPTION.

(c) Make sure that you obey all safety precautions when you are near an engine that is in operation. To obey the safety precautions, comply with the directions in this task:Engine Ground Safety Precautions, AMM TASK 71-00-00-800-805-F00.

(d) Let the engine operate for 2 minutes.

WARNING

DO NOT PUT YOUR HAND OR BODY IN THE AIR THAT COMES OUT OF THE PNEUMATIC GROUND CONNECTOR. THE HOT AIR WILL CAUSE INJURIES.

(e) Stand near the pneumatic ground air connector and smell if there is contaminated air.

NOTE: You can do a smell check of the bleed air from the right engine if it is necessary to compare the bleed air fumes.

(f) If there is contaminated bleed air from the left engine, do the steps that follow:

1) Check the left engine as cause of oil/fuel fumes/smoke in the cabin Fumes or Smoke in Cabin; Pneumatic Power Supplied by Engine - Fault Isolation, 71-05 TASK 807.

2) Do the oil contamination removal task, To remove the oil contamination, do this task: Oil Contamination Removal from Air Conditioning and Pneumatic Systems, AMM TASK 21-00-01-100-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

3) If the oil contamination removal task does not remove the fumes it may be necessary to replace components of the affected cooling pack. To replace the components, do the tasks that follow.

a) Replace the heat exchanger for the oil contaminated cooling pack. These are the tasks:

- Heat Exchanger and Plenum/Diffuser Assembly - Removal, AMM TASK 21-51-03-000-802-001 or Heat Exchanger and Plenum/Diffuser Assembly - Removal, AMM TASK 21-51-03-000-805-002
- Heat Exchanger and Plenum/Diffuser Assembly - Installation, AMM TASK 21-51-03-400-801-001 or Heat Exchanger and Plenum/Diffuser Assembly - Installation, AMM TASK 21-51-03-400-804-002

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

b) Replace the condenser for the oil contaminated cooling pack. These are the tasks:

- Condenser Removal, AMM TASK 21-51-12-000-801
- Condenser Installation, AMM TASK 21-51-12-400-802

c) Replace the reheater for the oil contaminated cooling pack. These are the tasks:

- Reheater Removal, AMM TASK 21-51-13-000-801
- Reheater Installation, AMM TASK 21-51-13-400-801

OMR ALL

d) If there is smoke or fumes from the left cooling pack replace the flight deck mufflers. These are the tasks:

- Flight Deck Air Distribution System Muffler/Duct Assembly Removal, AMM TASK 21-22-05-000-801
- Flight Deck Air Distribution System Muffler/Duct Assembly Installation, AMM TASK 21-22-05-400-801

4) Do the Repair Confirmation at the end of this task.

(g) If there is not contaminated bleed air from the left engine, then continue.

(h) Set the engine pneumatic BLEED 1 switch to OFF.

(i) Stop the operation of the left engine, do this task: Stop the Engine Procedure (Usual Engine Stop), AMM TASK 71-00-00-700-819-F00.



WARNING PUT ON GLOVES FOR PROTECTION FROM HOT SURFACES WHEN YOU CONNECT OR DISCONNECT PNEUMATIC FITTINGS. PACK OPERATION CAN MAKE THE GROUND AIR CONNECTOR VERY HOT. A HOT GROUND AIR CONNECTOR WILL BURN YOUR BARE HANDS.

(9) Remove the ground cart supply hose and deactivator, SPL-2462, from the Pneumatic Ground Connector.

(10) Close these access panels in this sequence:

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
192DR	ECS High Pressure Access Door

F. Fault Isolation Procedure - Check the Right Engine Bleed Air for Oil Contamination

(1) Do the steps that follow to check the right engine for the correct quantity of oil:

(a) Do this task:

Replenish the Engine Oil, AMM TASK 12-13-11-600-801.

(b) Make sure the engine has the correct quantity of oil.

(c) If the oil level is near or over the top of the sight glass, correct the oil level before you continue. Do this task to drain the engine oil: Drain the Engine Oil, AMM TASK 12-13-11-600-803.

NOTE: Too much oil in the engine(s) can cause smoke or fumes in the pneumatic system.

(d) If the oil level is less than FULL, do a check of the maintenance records to see if there is a change in oil consumption averages.

NOTE: An increase in oil consumption can be an indication of oil loss and possible contamination of the pneumatic system. This can cause smoke or fumes in the pneumatic system.

OMR ALL; ENGINES WITH DEBRIS MONITORING SYSTEM

(2) Examine the Debris Monitoring System for debris:

(a) Do this task:CDU Interrogation of the DMS Detectors, AMM TASK 79-00-00-710-801-F00.

NOTE: Debris in the oil can cause damage to bearings and seals. Damage to seals can cause oil loss and contamination of the pneumatic system. Oil contamination of the pneumatic system can cause smoke and fumes.

OMR ALL

(3) Do the steps that follow to do a check of the right engine for damage:

(a) Examine the area behind the fan blades for large oil stains or puddles of oil.

(b) Examine the area behind the fan blades for signs of a bird strike.

(4) Supply electrical power with a ground cart. This is the task: Supply External Power, AMM TASK 24-22-00-860-813.

NOTE: Do not operate the APU or the engines to supply electrical power to the airplane.

(5) Make sure these switches on the P5 overhead panel, are set to the indicated positions:

(a) On the P5-10 Air Conditioning Panel:

1) APU, ENGINE 1, and ENGINE 2 Switches are set to OFF

2) ISOLATION VALVE Switch is set to CLOSE

3) L PACK and R PACK Switches are set to OFF.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799

4) RECIRC FAN Switch is set to OFF.

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

5) L RECIRC FAN and R RECIRC FAN Switches are set to OFF.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

6) TRIM AIR Switch is set to OFF.

OMR ALL

(b) On the P5-11 thermal anti-ice panel:

(c) WING, ENGINE 1, and ENGINE 2, ANTI-ICE switches are set to OFF.

(6) To get access to the cooling packs, open these access panels, in this sequence:

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

WARNING

PUT ON GLOVES FOR PROTECTION FROM HOT SURFACES WHEN YOU CONNECT OR DISCONNECT PNEUMATIC FITTINGS. PACK OPERATION CAN MAKE THE GROUND AIR CONNECTOR VERY HOT. A HOT GROUND AIR CONNECTOR WILL BURN YOUR BARE HANDS.

(7) Install the deactivator, SPL-2462, in the ground air connector as follows:

(a) Put the deactivator, SPL-2462, in the ground air connector so that the check valve flappers are open.

(b) Install the coupling on the supply hose from the ground air cart onto the nipple of the ground air connector check valve assembly so that the deactivator, SPL-2462 is held in place.

NOTE: The supply hose should not be connected to the ground cart as the exhaust from the hose should be vented to ambient.

WARNING

DO NOT PUT YOUR HAND OR BODY IN THE AIR THAT COMES OUT OF THE PNEUMATIC GROUND CONNECTOR. THE HOT AIR WILL CAUSE INJURIES.

(8) Do the steps which follow to check for oil contamination from the right engine:

NOTE: The right engine bleed air will go out through the pneumatic ground air connector duct opening.

(a) Start the right engine and let the engine operate at ground idle speed. To start and operate the right engine, do this task: Start the Engine Procedure (Selection), AMM TASK 71-00-00-800-807-F00.

1) Use the APU to start the engine as it is still in operation.

2) Set the APU BLEED switch, on the P5-10 air conditioning panel, to OFF.

3) Set the ISOLATION VALVE switch on the P5-10 air conditioning panel to OPEN.

(b) Set the engine pneumatic BLEED 1 switch, on the P5-10 air conditioning panel, to ON.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING OBSERVE PROPER SAFETY PRECAUTIONS AROUND RUNNING ENGINE. WEAR EAR PROTECTORS AND STAY CLEAR OF ENGINE HAZARD AREAS. SEE "ENGINE GROUND SAFETY PRECAUTIONS" FOR ENGINE HAZARD DESCRIPTION.

(c) Make sure that you obey all safety precautions when you are near an engine that is in operation. To obey the safety precautions, comply with the directions in this task: Engine Ground Safety Precautions, AMM TASK 71-00-00-800-805-F00.

(d) Let the engine operate for 2 minutes.

WARNING DO NOT PUT YOUR HAND OR BODY IN THE AIR THAT COMES OUT OF THE PNEUMATIC GROUND CONNECTOR. THE HOT AIR WILL CAUSE INJURIES.

(e) Stand near the pneumatic ground air connector and smell if there is contaminated air.

NOTE: You can do a smell check of the bleed air from the left engine if it is necessary to compare the bleed air fumes.

(f) If there is contaminated bleed air from the right engine, do the steps that follow:

1) Check the right engine as cause of oil/fuel fumes/smoke in the cabin Fumes or Smoke in Cabin; Pneumatic Power Supplied by Engine - Fault Isolation, 71-05 TASK 807.

2) Remove the oil contamination. This is the task: Oil Contamination Removal from Air Conditioning and Pneumatic Systems, AMM TASK 21-00-01-100-801.

3) If the oil contamination removal task does not remove the fumes it may be necessary to replace components of the affected cooling pack. To replace the components, do the tasks that follow.

a) Replace the heat exchanger for the oil contaminated cooling pack. These are the tasks:

- Heat Exchanger and Plenum/Diffuser Assembly - Removal, AMM TASK 21-51-03-000-802-001 or Heat Exchanger and Plenum/Diffuser Assembly - Removal, AMM TASK 21-51-03-000-805-002
- Heat Exchanger and Plenum/Diffuser Assembly - Installation, AMM TASK 21-51-03-400-801-001 or Heat Exchanger and Plenum/Diffuser Assembly - Installation, AMM TASK 21-51-03-400-804-002

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

b) Replace the condenser for the oil contaminated cooling pack. These are the tasks:

- Condenser Removal, AMM TASK 21-51-12-000-801
- Condenser Installation, AMM TASK 21-51-12-400-802

c) Replace the reheater for the oil contaminated cooling pack. These are the tasks:

- Reheater Removal, AMM TASK 21-51-13-000-801
- Reheater Installation, AMM TASK 21-51-13-400-801

OMR ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

d) If there is smoke or fumes from the left cooling pack replace the flight deck mufflers. These are the tasks:

- Flight Deck Air Distribution System Muffler/Duct Assembly Removal, AMM TASK 21-22-05-000-801
- Flight Deck Air Distribution System Muffler/Duct Assembly Installation, AMM TASK 21-22-05-400-801

4) Do the Repair Confirmation at the end of this task.

(g) If there is not contaminated bleed air from the right engine, then continue.

(h) Set the engine pneumatic BLEED 1 Switch to OFF.

(i) Stop the operation of the right engine. This is the task: Stop the Engine Procedure (Usual Engine Stop), AMM TASK 71-00-00-700-819-F00.

WARNING PUT ON GLOVES FOR PROTECTION FROM HOT SURFACES WHEN YOU CONNECT OR DISCONNECT PNEUMATIC FITTINGS. PACK OPERATION CAN MAKE THE GROUND AIR CONNECTOR VERY HOT. A HOT GROUND AIR CONNECTOR WILL BURN YOUR BARE HANDS.

(9) Remove the ground cart supply hose and deactivator, SPL-2462, from the Pneumatic Ground Connector.

(10) Close these access panels in this sequence:

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

G. Fault Isolation Procedure - Check the Cooling Packs for Oil Contamination

(1) Do these steps to check for oil contamination from the conditioned air supplied by the left and right cooling packs:

(a) Supply pressure to the pneumatic system. To do this, do this task: Supply Pressure to the Pneumatic System (Selection), AMM TASK 36-00-00-860-801.

NOTE: Do not use a contaminated source to pressurize the pneumatic system.

(b) Supply conditioned air with the Left and Right Cooling Packs. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

NOTE: Operate the Left and Right Cooling Pack independently of the other to find which pack has the contamination.

OMR 701-799

(c) Turn the CONT CABIN and PASS CABIN Temperature Selectors, on the P5-17 Cabin Temperature Control Panel, to MANUAL WARM.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(d) Put the TRIM AIR Switch to the ON position.

(e) Turn the CONT CAB, FWD CAB, AND AFT CAB temperature selectors, on the P5-17 cabin temperature control panel, to the full WARM position.

OMR ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(f) Let the packs operate and smell the air that goes into the control cabin and the passenger cabin for contaminated air.

NOTE: You can do a smell check of the conditioned air from a different airplane if it is necessary to compare the conditioned air fumes.

(g) If you smell contaminated air from a cooling pack, then do the steps that follow to the applicable pack:

1) Do the oil contamination removal task, To remove the oil contamination, do this task: Oil Contamination Removal from Air Conditioning and Pneumatic Systems, AMM TASK 21-00-01-100-801.

2) If the oil contamination removal task does not remove the fumes do the steps that follow:

a) Replace the heat exchanger for the oil contaminated cooling pack. These are the tasks:

- Heat Exchanger and Plenum/Diffuser Assembly - Removal, AMM TASK 21-51-03-000-802-001 or Heat Exchanger and Plenum/Diffuser Assembly - Removal, AMM TASK 21-51-03-000-805-002
- Heat Exchanger and Plenum/Diffuser Assembly - Installation, AMM TASK 21-51-03-400-801-001 or Heat Exchanger and Plenum/Diffuser Assembly - Installation, AMM TASK 21-51-03-400-804-002

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

b) Replace the condenser for the oil contaminated cooling pack. These are the tasks:

- Condenser Removal, AMM TASK 21-51-12-000-801
- Condenser Installation, AMM TASK 21-51-12-400-802

c) Replace the reheater for the oil contaminated cooling pack. These are the tasks:

- Reheater Removal, AMM TASK 21-51-13-000-801
- Reheater Installation, AMM TASK 21-51-13-400-801

OMR ALL

d) If there is smoke or fumes from the left cooling pack replace the flight deck mufflers. These are the tasks:

- Flight Deck Air Distribution System Muffler/Duct Assembly Removal, AMM TASK 21-22-05-000-801
- Flight Deck Air Distribution System Muffler/Duct Assembly Installation, AMM TASK 21-22-05-400-801

3) Do the Repair Confirmation at the end of this task.

(h) If you do not smell contaminated air from a cooling pack, then do the step that follows and continue.

1) Shut down the Cooling Packs. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804

(i) Remove pressure from the Pneumatic System. This is the task: Remove Pressure from the Pneumatic System, AMM TASK 36-00-00-860-806.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799, 802 PRE SB 737-29-1106

H. Fault Isolation Procedure - Check the Hydraulic Reservoir Pressurization Modules

(1) Make sure that system A and B Hydraulic Reservoirs are depressurized. This is the task: Hydraulic Reservoirs Depressurization, AMM TASK 29-09-00-860-808.

(2) Get access to the flight deck or passenger compartment to smell the conditioned air for contamination.

(3) Do these steps to make sure the hydraulic reservoir pressurization module is not the source of the smoke or fumes:

(a) Start the APU and let the APU operation become stable. This is the task: APU Starting and Operation, AMM TASK 49-11-00-860-801.

(b) Supply conditioned air to the airplane with the left and right cooling packs. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(c) Pressurize the hydraulic system reservoir. This is the task: Hydraulic Reservoirs Pressurization, AMM TASK 29-09-00-860-807

(d) Operate the APU and cooling packs for 5 minutes.

(e) Stop the cooling packs. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(f) Set the APU BLEED Switch, on the P5-10 Air Conditioning Panel, to OFF.

1) Wait for approximately 5 minutes.

NOTE: This will permit hydraulic fluid to leak past the check valve if there is a failure.

(g) Set the APU BLEED Switch, on the P5-10 air conditioning panel, to ON.

(h) Supply conditioned air to the airplane with the left and right cooling packs, do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(i) If the fumes are present, then there is a problem with the hydraulic reservoir pressurization module. Replace the Hydraulic Reservoir Pressurization Module. These are the tasks:

- Hydraulic Reservoir Pressurization Module - Removal, AMM TASK 29-09-06-000-805
- Hydraulic Reservoir Pressurization Module - Installation, AMM TASK 29-09-06-400-805

1) Do the Repair Confirmation at the end of this task.

(j) If no fumes are present, then continue.

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 802 POST SB 737-29-1106

I. Fault Isolation Procedure - Check the Air Charging Valve Manifold Assemblies

(1) Make sure that system A and B hydraulic reservoirs are depressurized. To depressurize the hydraulic reservoirs, do this task: Hydraulic Reservoirs Depressurization, AMM TASK 29-09-00-860-802.

(2) Get access to the flight deck or passenger compartment to smell the conditioned air for contamination.

(3) Do these steps to make sure the air charging valve manifold assembly is not the source of the smoke or fumes:

(a) Start the APU and let the APU operation become stable. To start and operate the APU, do this task: APU Starting and Operation, AMM TASK 49-11-00-860-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 802 POST SB 737-29-1106 (Continued)

(b) Supply conditioned air to the airplane with the left and right cooling packs, do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(c) Pressurize the hydraulic system reservoir. To pressurize the hydraulic reservoir, do this task: Hydraulic Reservoirs Pressurization, AMM TASK 29-09-00-860-801

(d) Operate the APU and cooling packs for 5 minutes.

(e) Stop the cooling packs, do this task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(f) Set the APU BLEED switch, on the P5-10 air conditioning panel, to OFF.

1) Wait for approximately 5 minutes.

NOTE: This will permit hydraulic fluid to leak past the check valve if there is a failure.

(g) Set the APU BLEED switch, on the P5-10 air conditioning panel, to ON.

(h) Supply conditioned air to the airplane with the left and right cooling packs, do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(i) If the fumes are present, then there is a problem with the air charging valve manifold assembly. Replace the air charging valve manifold assembly. These are the tasks:

- Air Charging Valve Manifold Assembly - Removal, AMM TASK 29-09-02-000-801
- Air Charging Valve Manifold Assembly - Installation, AMM TASK 29-09-02-400-801

1) Do the Repair Confirmation at the end of this task.

(j) If no fumes are present, then continue.

OMR ALL

J. Fault Isolation Procedure - Check the Cabin Equipment

(1) Do these steps to make sure the cabin equipment is not the source of the smoke or fumes:

(a) Operate the cabin equipment, galley equipment, IFE equipment, and light ballasts.

(b) If there is smoke or fumes from the equipment, then do the steps below:

1) Replace the applicable equipment.

(c) Operate the applicable equipment for approximately 3 minutes.

(d) If the fumes go away, then you corrected the fault.

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

K. Fault Isolation Procedure - Check Recirculation Fans

(1) Do these checks of the Air Recirculation System:

NOTE: Examine the Air Recirculation Fans and filters only if there is smoke or fumes in the passenger compartment.

(a) Do this task: Recirculation Air Filter - Removal, AMM TASK 21-25-01-000-801.

OMR 701-799

(b) Examine the Air Recirculation Filter for:

- Contaminants: dust, dirt, lint, and oil
- Rips in the filter

ECCN 9E991 BOEING PROPRIETARY - See title page for details

- Breaks in the Filter Frame

1) If necessary, install a new Air Recirculation Filter. This is the task: Recirculation Air Filter - Installation, AMM TASK 21-25-01-400-801.

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(c) Examine the Air Recirculation Filters for:

- Contaminants: dust, dirt, lint, and oil
- Rips in the filter
- Breaks in the Filter Frame

1) If necessary, install new Air Recirculation Filters. This is the task: Recirculation Air Filter - Installation, AMM TASK 21-25-01-400-801.

OMR 701-799

(d) Do the Recirculation System - Operational Test, AMM TASK 21-25-00-710-801-001 to see if the Air Recirculation Fan or the related wiring generates smoke when it operates.

1) If there is smoke from the Air Recirculation Fan or the related wiring, then do the applicable steps below:

a) Replace the Air Recirculation Fan. These are the tasks:

- Recirculation Fan Removal, AMM TASK 21-25-02-000-801
- Recirculation Fan Installation, AMM TASK 21-25-02-400-801

b) Repair the applicable wiring.

c) Do the Repair Confirmation at the end of this task.

2) If there is no smoke or fumes from the Air Recirculation Fan or the related wiring, and the Operational Test is satisfactory, then continue.

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(e) Do the Recirculation System - Operational Test, AMM TASK 21-25-00-710-802-002 to see if the Air Recirculation Fans or the related wiring generate smoke when they operate.

1) If there is smoke from the Air Recirculation Fans or the related wiring, then do the applicable steps below:

a) Replace the applicable Air Recirculation Fan(s). These are the tasks:

- Recirculation Fan Removal, AMM TASK 21-25-02-000-801
- Recirculation Fan Installation, AMM TASK 21-25-02-400-801

b) Repair the applicable wiring.

c) Do the Repair Confirmation at the end of this task.

2) If there is no smoke or fumes from the Air Recirculation Fans or the related wiring, and the Operational Test is satisfactory, then continue.

OMR ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

L. Repair Confirmation

(1) Start the Left and Right Engines. This is the task: Start the Engine Procedure (Selection), AMM TASK 71-00-00-800-807-F00.

WARNING OBSERVE PROPER SAFETY PRECAUTIONS AROUND RUNNING ENGINE. WEAR EAR PROTECTORS AND STAY CLEAR OF ENGINE HAZARD AREAS. SEE "ENGINE GROUND SAFETY PRECAUTIONS" FOR ENGINE HAZARD DESCRIPTION.

(2) Make sure that you obey all safety precautions when you are near an engine that is in operation. To obey the safety precautions, comply with the directions in this task:Engine Ground Safety Precautions, AMM TASK 71-00-00-800-805-F00.

(3) Start the APU. This is the task: APU Starting and Operation, AMM TASK 49-11-00-860-801.

(4) Set the BLEED AIR Switches, on the P5 panel, to the positions shown:

(a) APU, ENGINE 1, and ENGINE 2 Switches are set to ON

(b) Put the ISOLATION VALVE Switch to the OPEN position.

(5) Make sure that the Hydraulic Reservoirs are pressurized. To pressurize the Hydraulic Reservoirs, do this task: Hydraulic Reservoirs Pressurization, AMM TASK 29-09-00-860-801 or Hydraulic Reservoirs Pressurization, AMM TASK 29-09-00-860-807.

(a) To pressurize the Hydraulic Reservoirs, do this task: Hydraulic Reservoirs Pressurization, AMM TASK 29-09-00-860-801 or Hydraulic Reservoirs Pressurization, AMM TASK 29-09-00-860-807.

OMR 701-799

(6) Set the RECIRC FAN switch to the applicable position as follows:

(a) If the fault is electrical, set the RECIRC FAN switch to the AUTO position.

(b) If the fault is oil, fuel or hydraulic fluid, set the RECIRC FAN switch to the OFF position.

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(7) Set the L RECIRC FAN and R RECIRC FAN switches to the applicable position as follows:

(a) If the fault is electrical, set the L RECIRC FAN and R RECIRC FAN switches to the AUTO position.

(b) If the fault is oil, fuel or hydraulic fluid, set the L RECIRC FAN and R RECIRC FAN switches to the OFF position.

OMR ALL

(8) Supply conditioned air with the applicable left or right cooling packs. To supply conditioned air, do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(9) Let the systems operate for approximately 5 minutes.

(a) If no fumes or smoke appear, then you have corrected the fault.

1) Put the airplane back to its usual condition:

a) Stop the operation of the left and right cooling packs, do this task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

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OMR 701-799

b) Set the RECIRC FAN Switch to the AUTO position.

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

c) Set the L RECIRC FAN and R RECIRC FAN switches to the AUTO position.

OMR ALL

d) Stop the operation of the left and right engines. This is the task: Stop the Engine Procedure (Usual Engine Stop), AMM TASK 71-00-00-700-819-F00.

e) Stop the operation of the APU. This is the task: APU Usual Shutdown, AMM TASK 49-11-00-860-802.

f) If it is not necessary, remove the Electrical Power. This is the task: Remove External Power, AMM TASK 24-22-00-860-814.

(b) If there are still fumes or smoke, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

807. Airplane Cannot Maintain Cabin Pressure On a Single Pack Within Specified Limits - Fault Isolation

A. Description

Pressurization requires proper operation of the pneumatic (engine bleed) system, the air conditioning packs, and the cabin pressure control system. The pneumatic system supplies air to the air conditioning packs. The air conditioning packs regulate the air flow into the fuselage. The cabin pressure control system regulates air escaping from the fuselage to control cabin pressure. Air leaking from the fuselage must be within specified limits in order for the packs to operate efficiently.

Unusual pressurization occurrences are typically associated with low air inflow, high fuselage leakage, or a combination of the two. Examples of theses occurrences are:

- Cabin altitude is above the pressurization schedule during cruise
- Cabin altitude climbs during cruise
- Cabin altitude climbs during initial descent

The inflow of air depends on the bleed air source (engines, APU) and air conditioning pack flow. Bleed air pressure is one method for determining if sufficient air is available to the packs. Low bleed pressures may indicate poor bleed performance even if the pressures are not zero.

NOTE: Bleed air pressure must exceed cabin pressure by at least 10 psi to ensure air flow into the cabin through an air conditioning pack.

Airplanes have controlled air leakage from the fuselage. Air exits the fuselage from galley ventilation and drains, aft outflow valves, forward outflow valves (where applicable), and certain bulkhead penetrations such as control cables. Air also leaks from the fuselage in uncontrolled ways. These paths can include worn door seals, bilge drain valves, bulkhead penetrations, and APU bleed air duct bulkhead seals.

NOTE: Fault Code 30 on the Cabin Pressure Controller indicates low inflow or high fuselage leakage and not a problem with the Cabin Pressure Controller.

The fault isolation procedure below will help determine if the inability of the airplane to maintain cabin pressure within the specified limits is caused by problems with the bleed system, the air conditioning packs, a leaky fuselage, or a combination of the three.

B. Possible Causes

(1) Air conditioning packs cannot pressurize the airplane to specification.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799

(a) Water separator

(b) 35° system

OMR ALL

(c) Secondary heat exchanger

(2) Fuselage leaks excessively.

(a) Air leaking around cabin doors and over-wing hatches.

(b) Air leaking around service door seals.

(c) Air leaking from bilge drains (they should close at about 2.0 psid).

(d) Air leaking from APU duct to fuselage seals at station 540, station 727 and station 1016.

(e) Air leaking from nose gear view port

(f) Air leaking from forward outflow valve/EE cooling overboard vent

(3) Bleed system does not supply enough pressure to the packs to allow the packs to operate efficiently.

(a) Precooler control valve not modulating correctly or stuck closed

(b) Precooler control valve sensor (390° F sensor) is out-of-tolerance, stuck closed or plugged

(c) Pressure regulator and shutoff valve (PRSOV) has a butterfly valve that is sticking

(d) 450° F thermostat has failed open

(e) Bleed air regulator (M1180) regulates control pressure too low

(f) High stage valve has a butterfly valve that is sticking

(g) High stage regulator not regulating properly

(h) Sense lines have pressure leaks

(i) Sense lines are obstructed

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	5	C00259	AIR CONDITIONING BLEED AIR VALVE ISLN
A	6	C01470	AIR CONDITIONING BLEED AIR XDCR LEFT
A	7	C00796	AIR CONDITIONING BLEED AIR VALVES L
B	5	C00077	AIR CONDITIONING BLEED AIR PRESS IND
B	6	C01469	AIR CONDITIONING BLEED AIR XDCR RIGHT
B	7	C00797	AIR CONDITIONING BLEED AIR VALVES R

D. Related Data

(1) Cabin Pressure Leak Test, AMM TASK 05-51-91-790-801

(2) Airplane Pack Operation - Confidence Check, AMM TASK 21-00-05-780-801

(3) Equipment Cooling Overboard Exhaust Valve Removal, AMM TASK 21-27-05-000-801Equipment Cooling Overboard Exhaust Valve Installation, AMM TASK 21-27-05-400-801

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(4) Cabin Pressure Controller (CPC) Removal, AMM TASK 21-31-01-000-801, Cabin Pressure Controller (CPC) Installation, AMM TASK 21-31-01-400-801

(5) Cabin Pressure Control Module - Removal, AMM TASK 21-31-02-000-801, Cabin Pressure Control Module - Installation, AMM TASK 21-31-02-400-801

(6) Aft Outflow Valve Assembly Removal, AMM TASK 21-31-03-000-801, Aft Outflow Valve Assembly Installation, AMM TASK 21-31-03-400-801

(7) Heat Exchanger and Plenum/Diffuser Assembly - Removal, AMM TASK 21-51-03-000-802-001 or Heat Exchanger and Plenum/Diffuser Assembly - Removal, AMM TASK 21-51-03-000-805-002Heat Exchanger and Plenum/Diffuser Assembly - Installation, AMM TASK 21-51-03-400-801-001 or Heat Exchanger and Plenum/Diffuser Assembly - Installation, AMM TASK 21-51-03-400-804-002

(8) Engine Bleed Air System Leak Check Using the APU, AMM TASK 36-11-00-700-802

(9) Engine Bleed Air Crossover Operational Test, AMM TASK 36-11-00-710-801

(10) 737–SL-36–013

(11) 737–SL-21-045

(12) 737–SL-36–068

(13) 737–SL-36–073

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(14) Low Limit (35 Degree F) Controller Removal, AMM TASK 21-51-31-000-801, Low Limit (35 Degree F) Controller Installation, AMM TASK 21-51-31-400-801

(15) Water Separator Removal, AMM TASK 21-51-05-000-801, Water Separator Installation, AMM TASK 21-51-05-400-802

(16) Low Limit (35 Degree Fahrenheit) Temperature Sensor Removal, AMM TASK 21-51-32-000-801, Low Limit (35 Degree Fahrenheit) Temperature Sensor Installation, AMM TASK 21-51-32-400-801

OMR ALL

E. Initial Evaluation of the Packs Ability to Pressurize the Cabin

(1) Install the Downlock Pins to lock the Landing Gear. This is the task: Landing Gear Downlock Pins Installation, AMM TASK 32-00-01-480-801.

(2) Open these access panels:

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

(3) Do these steps on the Flight Management Computer (FMC) to display the FMCS ANALOG page 1/4:

(a) Press the INIT REF function key on the CDU front panel.

(b) Push the LSK adjacent to <INDEX (LSK 6L) to show the INIT REF INDEX 1/1 page.

(c) Push the LSK adjacent to MAINT> (LSK 6R) to show the MAINT BITE INDEX page.

(d) Push the LSK adjacent to <FMCS (LSK 1L) to show the FMCS BITE 1/1 page.

(e) Push the LSK adjacent to <FMC LEFT or <FMC RIGHT.

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(f) Push the LSK adjacent to <DISCRETES (LSK 4L) to show the FMCS ANALOG DISC 1/4 page.

(g) Use the ECS PACK H/L (high flow, low flow) line entry on the ANALOG DISC 1/4 page.

(h) The following information can be displayed on the FMC:

	LEFT	RIGHT
ECS PACK	ON	OFF
ECS PACK H/L	LO	HI
ISOLATION VALVE	OPEN	OPEN

NOTE: ECS PACK indicates if the left or right pack is on or off. ECS PACK H/L indicates the pack is in the low flow mode or if it has been commanded to the high flow mode. A pack can be commanded to the high flow mode even if it is off. ISOLATION VALVE indicates if the valve is open or closed.

OMR 701-799

(4) Set the RECIRC FAN switch on the air conditioning panel to AUTO.

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(5) Set the L and R RECIRC FAN switches on the air conditioning panel to AUTO.

OMR ALL

(6) Start the APU by doing this task: APU Starting and Operation, AMM TASK 49-11-00-860-801.

(a) Let the APU become stable at the governed speed.

(7) Set the BLEED 1 and BLEED 2 switches on the P5-10 air conditioning panel to the ON position.

NOTE: The cabin is not being pressurized at this point.

(8) Do these steps to make sure the pack flow control and shutoff valves operate correctly:

(a) Pressurize the pneumatic system with the APU by doing this task: Supply Pressure to the Pneumatic System with the APU, AMM TASK 36-00-00-860-803.

(b) Set the APU BLEED switch on the air conditioning panel to ON.

(c) Set the L and R PACK switches on the air conditioning panel to AUTO.

(d) Make sure the ECS PACK H/L indication on the FMC display shows LO for both packs.

(e) Set the L PACK switch to HIGH.

(f) Make sure the ECS PACK H/L indication on the FMC display shows HI for the left pack and LO for the right pack.

(g) Set the R PACK switch to HIGH.

(h) Make sure the ECS PACK H/L indication on the FMC display shows HI for both packs.

(i) Put the L and R PACK switches on the air conditioning panel to AUTO.

(j) Make sure the ECS PACK H/L indication on the FMC display shows LO for both packs.

(k) Put the L and R PACK switches to OFF.

(9) Pressurize the airplane with both packs as follows:

(a) Pressurize the airplane to 4.0 psid as follows:

NOTE: Follow the instructions in this procedure when you pressurize the airplane: (AMM TASK 05-51-91-790-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Start the right engine using the APU bleed air.
   a) APU BLEED switch is set to ON.
2) Set the ISOLATION VALVE switch to CLOSE.
3) Supply pneumatic pressure to the right pack with the right engine at ground idle. To supply pneumatic pressure with an engine, do this task: Supply Pressure to the Pneumatic System with One or Both Engines, AMM TASK 36-00-00-860-804.
   a) BLEED 2 switch is set to ON.
   b) BLEED 1 switch is set to OFF.
   c) R PACK switch is set to AUTO.
4) Supply pneumatic pressure to the left pack with the APU. To supply pneumatic pressure with the APU, do this task: Supply Pressure to the Pneumatic System with the APU, AMM TASK 36-00-00-860-803.
   a) APU BLEED switch is set to ON.
   b) L PACK switch is set AUTO.

(b) Manually operate the outflow valve MANUAL VALVE CLOSE/OPEN switch to maintain the cabin pressure of 4.0 psid as follows:

NOTE: Pressurizing the airplane to 4.0 psid for the short duration of the test will not harm the airplane.

1) Make sure the pressurization mode selector is set to MAN.
2) Move the MANUAL VALVE CLOSE/OPEN toggle switch as necessary to maintain 4.0 psid on the DIFF PRESS gauge.

(c) If a differential pressure of 4.0 psid cannot be maintained, either a pack supply or airplane leakage problem exists. Do the applicable fault isolation tasks.

(10) Do this check of the right pack's capability to pressurize the passenger cabin in AUTO mode (low flow) with the right engine running:

(a) Make sure the pressure differential on the DIFF PRESS/CABIN ALT indicator on the P5-17 panel indicates a steady 4.0 psid.
   1) If a differential pressure of 4.0 psid cannot be maintained, either a pack supply or airplane leakage problem exists. Do the applicable fault isolation tasks.

OMR 701-799

(b) Make sure the RECIRC FAN switch on the air conditioning panel is set to AUTO.

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(c) Make sure the L and R RECIRC FAN switches on the air conditioning panel is set to AUTO.

OMR ALL

(d) Set the ISOLATION VALVE switch is in the CLOSE position.

(e) Set the L PACK switch on the P5-10 air conditioning panel to OFF.

(f) Set the APU BLEED switch on the P5-10 air conditioning panel to OFF.

(g) Set the BLEED 1 switch to OFF.

(h) Make sure the BLEED 2 switch is set to ON.

(i) Make sure the R PACK switch is set to AUTO (low flow).

(j) Set the temperature selector for the right pack to COLD.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(k) Slowly move the right engine throttle until N2% increases to 72% (+0%, -2%).

(l) Manually operate the outflow valve MANUAL VALVE CLOSE/OPEN switch as necessary to maintain 4.0 psid on the DIFF PRESS gauge.

(m) Make a record of the cabin rate of change, right engine bleed duct pressure, and right engine N2% information in the troubleshooting data sheet fields of 21-00 TASK SUPPORT Figure 302.

(n) Monitor the ANALOG DISC 1/4 page on the FMC CDU to make sure the pack and isolation valve operation is as commanded.

(o) Move the right engine throttle to the ground idle position.

(11) Do this task to check the right pack's capability to pressurize the passenger cabin in the HIGH mode (high flow).

OMR 701-799

(a) Set the RECIRC FAN switch on the air conditioning panel to AUTO.

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(b) Set the R RECIRC FAN switch on the air conditioning panel to AUTO.

OMR ALL

(c) Make sure the ISOLATION VALVE switch in the CLOSE position.

(d) Make sure the L PACK switch is OFF.

(e) Make sure the APU BLEED switch is OFF.

(f) Make sure the engine BLEED 1 switch is OFF.

(g) Make sure the BLEED 2 switch is ON.

(h) Set the R PACK switch to HIGH.

(i) Make sure the temperature selector for the right pack is set to COLD.

(j) Slowly move the right engine throttle until N2% increases to 72% (+0%, -2%).

(k) Manually operate the outflow valve MANUAL VALVE CLOSE/OPEN switch as necessary to maintain 4.0 psid on the DIFF PRESS gauge.

(l) Make sure the ECS PACK H/L indication on the FMC Maintenance Discrete page shows HI for the right pack.

(m) Make a record of the cabin rate of change, right engine bleed duct pressure, and right engine N2% information in the troubleshooting data sheet fields of 21-00 TASK SUPPORT Figure 302 once HIGH mode is achieve and indications are stable. A cabin rate of -500 fpm is normal.

(n) Move the right engine throttle to the ground idle position.

(o) Put the R PACK switch to AUTO.

(p) Manually operate the outflow valve MANUAL VALVE CLOSE/OPEN switch as necessary to achieve a steady 4.0 psid cabin pressure on the DIFF PRESS indicator.

(12) Do this check of the left pack capability to pressurize the passenger cabin in AUTO mode (low flow) with the left engine running.

NOTE: Cabin pressure will fluctuate during the switch over to Bleed 1, L PACK.

(a) Set the APU BLEED switch to ON.

(b) Start the left engine with the APU bleed air. Do this task:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Start the Engine Procedure (Normal Start), AMM TASK 71-00-00-800-808-F00

OMR 701-799

(c) Set the RECIRC FAN switch on the air conditioning panel to AUTO.

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(d) Set the R RECIRC FAN switch on the air conditioning panel to AUTO.

OMR ALL

(e) Make sure the ISOLATION VALVE switch is in the CLOSE position.

(f) Make sure the APU BLEED switch is set to OFF.

(g) Set the BLEED 1 switch to ON.

NOTE: A dual bleed light should illuminate. Ignore this indication.

(h) Make sure the L PACK switch is set to AUTO.

(i) Set the R PACK switch to OFF.

(j) Put the BLEED 2 switch to OFF.

(k) Set the temperature selector for the left pack to cold.

(l) Shut down the right engine: AMM TASK 71-00-00-700-819-F00.

(m) Slowly move the left engine throttle until N2% increases to 72% (+0%, -2%).

(n) Manually operate the outflow valve MANUAL VALVE CLOSE/OPEN switch as necessary to maintain 4.0 psid on the DIFF PRESS gauge.

(o) Make a record of the cabin rate of change, left engine bleed duct pressure, and left engine N2% information in the troubleshooting data sheet fields of 21-00 TASK SUPPORT Figure 302.

(p) Monitor the ANALOG DISC 1/4 page on the FMC CDU to make sure the pack and isolation valve operation is as commanded.

(q) Move the left engine throttle to the ground idle position.

(13) Do this check of the left pack capability to maintain pressure to the passenger cabin with the pack in the HIGH Mode (high flow).

OMR 701-799

(a) Set the RECIRC FAN switch on the air conditioning panel to AUTO.

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(b) Set the R RECIRC FAN switch on the air conditioning panel to AUTO.

OMR ALL

(c) Make sure the ISOLATION VALVE switch is set to CLOSE.

(d) Make sure the APU BLEED switch is OFF.

(e) Make sure the BLEED 1 switch is ON.

(f) Make sure the R PACK switch is OFF.

(g) Make sure the BLEED 2 switch is OFF.

(h) Put the L PACK switch to HIGH.

(i) Make sure the temperature selector for the left pack is set to cold.

(j) Slowly move the left engine throttle until N2% increases to 72% (+0%, -2%).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(k) Stabilize the cabin pressure to 4.0 psid by adjusting the outflow valve.

(l) Make sure the ECS PACK H/L indication on the FMC DISCRETE page shows HI for the left pack.

(m) Make a record of the cabin rate of change, left engine bleed duct pressure, and left engine N2% information in the troubleshooting data sheet fields of 21-00 TASK SUPPORT Figure 302 once HIGH Mode is achieved and indications are stable. A cabin rate of -500 fpm is normal.

(n) Move the left engine throttle to the ground idle position.

(o) Return the LPACK switch to AUTO.

(p) Manually operate the outflow valve MANUAL VALVE CLOSE/OPEN switch as necessary to achieve a steady 4.0 psid cabin pressure on the DIFF PRESS indicator.

F. Dispatchability Troubleshooting Table

(1) The following is a Cabin Pressurization Confidence Dispatch Guide. Print out the table and use it as a quick reference guide when determining the dispatchability of an airplane.

Cabin Pressurization Confidence Dispatch Guide

SYSTEM	MINIMUM REQUIREMENT	REQUIREMENT SATISFIED	
		Yes	No
Single Pack Low Flow Mode	Able to maintain cabin differential pressure of 3.9 - 4.1 psid and cabin rate of change is no more than ±100 feet per minute.	Confidence is high that airplane will maintain pressure with single pack in high flow mode while at cruise.	Confidence is low that airplane will maintain pressure on single pack at cruise. Emphasis needs to be placed on the following: Outflow: Was airplane leak down less than 100 seconds? If so, check for cabin leaks and run Phase III test again. Inflow: If the cabin leak down rate was acceptable, then the pack performance needs to be investigated. Perform various pack system health checks per AMM 21-51-00/500 series pages.
Single Pack High Flow Mode	Cabin rate of change is -500 feet per minute or greater.	Confidence is high that airplane will maintain pressure with single pack in high flow mode while at cruise	Confidence is low that airplane will maintain pressure on single pack at cruise. Emphasis needs to be placed on the following: Outflow: Was airplane leak down less than 100 seconds? If so, check for cabin leaks and run Phase III test again. Inflow: If the cabin leak down rate was acceptable, then the pack performance needs to be investigated. Perform various pack system health checks per AMM 21-51-00/500 series pages.

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Cabin Pressurization Confidence Dispatch Guide (Continued)

SYSTEM	MINIMUM REQUIREMENT	REQUIREMENT SATISFIED	
		Yes	No
Airplane/Structure Air Leakage	Leak down rate from 4.0 to 2.5 psid is greater than or equal to a MINIMUM of 100 seconds (110 to 120 seconds is the target).	Cabin leakage is acceptable. Operators should consider conducting AMM 05-51-91 leakage test on C-check basis and monitor performance.	Inspect and repair any air leaks. Specifically: Doors and hatches - missing or torn seals, poor contact between door seals and airplane due to dirt, etc., inspect sealing compound areas and repair door seals per AMM 52-09-10. Aft outflow valve - Test per AMM 21-31-00/501. Bilge drains - Should close at approximately 2.0 psid. Cabin pressure safety relief valve, outflow valve seal, negative pressure relief valve seal, water service panel, flight deck windows, pressure bulkhead penetrations should all have minimal leakage. Significant leaks should be resolved as required.
Engine Bleed Duct Pressure - Ground Idle	With N1 at approximately 20%, duct pressure should be 15 to 20 psi. Reference AMM 36-11-00, page 516.	Continue to maintain bleed system components on 1C-check intervals as specified in MPD Section 1.1.	Consider accomplishing the engine bleed air distribution system adjustment/test AMM 36-11-00, 500 series pages to help improve system performance.
Engine Bleed Duct Pressure - Flight Idle	With N1 at approximately 33%, duct pressure should be 26 to 38 psi. Reference AMM 36-11-00, page 516.	Continue to maintain bleed system components on 1C-check intervals as specified in MPD Section 1.1.	Consider accomplishing the engine bleed air distribution system adjustment/test AMM 36-11-00, 500 series pages to help improve system performance.
Cabin Pressurization Control System	Successful completion of the Phase I, II and III tests and cabin pressure can be maintained.	Confidence is high that cabin pressure can be maintained.	Consider accomplishing pressurization control system adjustment/test per AMM 21-31-00, 500 series pages.

G. Fault Isolation of Low Pack Air Flow

OMR 701-799

(1) Contaminants mixed with moisture and ice in the water separator can cause low air flow.

(2) Do the following to check the coalescer bag pressure indicator: Coalescer Bag Pressure Indicator Inspection, AMM TASK 21-51-05-400-803.

(3) If the pressure indicator is red, do the following to clean the coalescer bag in the water separator.

(a) Remove the water separator by doing the following: Water Separator Removal, AMM TASK 21-51-05-000-801

(b) Remove the coalescer bag by doing the following: Coalescer Bag Removal, AMM TASK 21-51-05-020-801

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OMR 701-799 (Continued)

(c) Clean the coalescer bag by doing the following: Coalescer Bag Cleaning, AMM TASK 21-51-05-100-801

(d) Install the coalescer bag after cleaning by doing the following: Coalescer Bag Installation, AMM TASK 21-51-05-400-801

(e) Install the water separator by doing the following: Water Separator Installation, AMM TASK 21-51-05-400-802

(4) The 35° system can cause ice to form in the water separator when not working properly. The BITE procedure monitors faults within the controller and also malfunctions caused by a defective 35° sensor and valve of the same subsystem. The cover of the 35° controller has a chart describing the BITE instructions and the GO/NO GO criteria. The BITE procedure checks the following:

NOTE: The FLIGHT position of the BITE switch is the default position.

BITE CHECKS	
SWITCH S1 POSITION	OPERATION CHECKED
1	POWER: 15.0 AND ±1.5 VDC
2	FULL-ON HEAT - OPEN DRIVE
3	DEADBAND
4	FULL-ON COOL - CLOSE DRIVE
5	SENSOR OPEN OR SHORT

(a) Perform the 35° controller BITE test.

(b) Replace faulty components determined by the BITE test.

OMR ALL

(5) Over a period of time, moisture mixed with contaminants in the air can form sludge in the secondary heat exchanger. This can cause low air flow.

(a) Perform the following procedure to clean the secondary heat exchanger: Heat Exchanger and Plenum/Diffuser Assembly Cleaning Using Backflush Equipment, AMM TASK 21-51-03-160-807

OMR 701-799

(6) If you have an air pack - air flow and temperature diagnostic, SPL-6191 tool, follow the procedure after the fuselage leak check to evaluate the pack flow.

OMR ALL

H. Repair Confirmation for Low Pack Air Flow

OMR 701-799

(1) Perform the procedure to evaluate the packs again to see if the problem has been solved. If the water separator, coalescer bag, and 35° system are not causing low pack air flow, evaluate the leakage in the fuselage and the bleed air pressure to determine if sufficient air is available to the packs. Low bleed pressures may indicate poor bleed performance even if the pressures are not zero.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(2) Perform the procedure to evaluate the packs again to see if the problem has been solved. If pressurization problems still exist, evaluate the leakage in the fuselage and the bleed air pressure to determine if sufficient air is available to the packs. Low bleed pressures may indicate poor bleed performance even if the pressures are not zero.

OMR ALL

I. Fault Isolation of Fuselage Leakage

(1) Check the airplane for sources of leakage in the fuselage (21-00 TASK SUPPORT Figure 301).

(a) Pressurize the airplane with the APU by following this task: AMM TASK 36-00-00-860-803.

(b) Examine these areas from inside the airplane:

1) Main cabin doors

NOTE: There should not be any significant areas where the air jets out the door seals.

2) Over-wing hatches

(c) Examine these areas from outside the airplane:

1) Cargo door seals

2) Bilge drains

NOTE: Bilge drains should close at about 2.0 psid. All bilge drains open create the equivalent of a 4.75 diameter hole (approx.). As few as four or five open drains can cause a noticeable increase in leakage.

3) E/E bay doors

4) Main cabin doors

5) Over-wing hatches

6) APU duct to fuselage seal at station 661.34 (Open access panel 192K forward of the MLG forward bulkhead.)

7) APU duct to fuselage seal at station 727 (Open the access panel at the aft end of the keel beam to access the APU duct at station 727.)

8) APU duct to fuselage seal at station 1016 (Open the door into the tail of the airplane and look at the aft pressure bulkhead for APU duct at station 1016.)

9) Nose gear view port (Feel for air flow from the site glass window.) (Reference IPC 56-41-31-01).

10) Forward outflow valve/E/E cooling overboard vent

(2) Repair all pressure leaks that are found. Perform the airplane leak down check in the Repair Confirmation section to make sure the pressure leakage is within acceptable limits.

J. Repair Confirmation for Fuselage Leakage

(1) Perform the following airplane Leak Down Check to confirm the airplane can maintain pressure within acceptable limits.

(a) Pressurize the airplane with the APU by following this task: AMM PAGEBLOCK 05-51-91/201.

(b) When 4.0 psid on the DIF PRESS gauge has been achieved and stabilized, fully close (manually) the aft outflow valve.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Make sure that the indication on the gauge above the CLOSED/OPEN switch on the P5-17 pressurization control panel shows that the valve is fully closed.

(c) Set the L and R PACK switches to OFF.

(d) Set the BLEED 1 and 2 switches to OFF.

(e) Continue to operate the APU.

NOTE: The APU is kept running in the event a bleed source is needed in case of a very high rapid cabin depressurization below 2.5 psi and for electrical power if a ground source is not used.

(f) Start the check at 4.0 psid (time zero).

(g) Make a record of the pressure differential indications on the 21-00 TASK SUPPORT Figure 302 at 10 second intervals until the cabin pressure reduces to 2.5 psi.

NOTE: After completing the bleed down check, a pack may be switched ON to reduce cabin rate of depressurization.

(h) Continue to depressurize the airplane and put it back to its unpressurized condition.

(i) Shut down the APU per APU Usual Shutdown, AMM TASK 49-11-00-860-802

(2) Best performance of the air conditioning packs and airplane structure is as follows:

Table 201/21-00-00-993-801 Best Performance Data

SYSTEM	BEST PERFORMANCE
Single Pack - Low Flow Mode	Able to maintain or increase cabin differential at 4.0 PSID and cabin rate of change of 0 or negative value (cabin pressurizing).
Single Pack - High Flow Mode	Able to maintain or increase cabin differential at 4.0 PSID and cabin rate of change of -500 feet/minute or greater (cabin pressurizing).
Cabin Leakage Rate from 4.0 to 2.5 psi	Greater than 110 seconds is excellent; 100 -110 seconds is good; less than 100 seconds is unacceptable and requires service..

(3) If the test data agrees with the best performance data for the packs (inflow) and the airplane structure (outflow), you can be very sure that cabin pressure can be maintained at cruise (FL250) on a single pack.

OMR 701-799

K. Fault Isolation Procedure - Diagnostic Check of 737-600s and 737-700s Air Conditioning Packs Using an Air Flow and Temperature Diagnostic Tool

NOTE: This procedure requires the use of the specially designed air pack - air flow and temperature diagnostic, SPL-6191 tool to measure the performance level of the air conditioning packs on 737-600s and 737-700s. Skip this procedure if you do not have an air flow and temperature diagnostic tool or have 737-600s and 737-700s in your fleet.

(1) The air pack - air flow and temperature diagnostic, SPL-6191 is designed to measure the temperature and flow coming out of the air conditioning packs in various modes of operation on 737-600s and 737-700s. The data collected can be used to isolate problems with the pack flow control valve, 35° system, coalescer bag, and heat exchangers.

(2) Open these access panels:

Number	Name/Location
192CL	ECS Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799 (Continued)

(Continued)

Number	Name/Location
192CR	ECS Access Door

(3) Remove the center panel on the aft bulkhead in the forward cargo compartment.

(4) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	12	C00132	MASTER CAUTION ANNUNCIATOR BUS 1
B	13	C00131	MASTER CAUTION ANNUNCIATOR BAT
C	11	C01275	MASTER CAUTION ANNUNCIATOR CONT 1
C	12	C01276	MASTER CAUTION ANNUNCIATOR CONT 2
C	13	C01277	MASTER CAUTION ANNUNCIATOR CONT 3
C	14	C01278	MASTER CAUTION ANNUNCIATOR CONT 4
D	11	C00133	INDICATOR MASTER DIM DIM/TST CONT
D	13	C00311	INDICATOR MASTER DIM BUS 1
D	14	C00312	INDICATOR MASTER DIM BUS 2
D	15	C01401	LANDING GEAR AIR/GND RELAY
D	17	C01027	LANDING GEAR NOSE GEAR STEER

(5) Put the LIGHTS switch on the P2-1 Panel to TEST and make sure that these lights come on and then go off:

(a) MASTER CAUTION and AIR COND annunciator lights on the P7 Panel

(b) PACK TRIP OFF lights on the P5-10 Panel

(6) Do these steps on the Flight Management Computer (FMC) to display the FMCS ANALOG page 1/4.

(a) Press the INIT REF function key on the CDU front panel.

(b) Push the LSK adjacent to <INDEX (LSK 6L) to show the INIT REF INDEX 1/1 page.

(c) Push the LSK adjacent to MAINT> (LSK 6R) to show the MAINT BITE INDEX page.

(d) Push the LSK adjacent to <FMCS (LSK 1L) to show the FMCS BITE 1/1 page.

(e) Push the LSK adjacent to either Left or Right Prompts.

(f) Push the LSK adjacent to <DISCRETES (LSK 4L) to show the FMCS ANALOG DISC 1/4 page.

(g) Use the ECS PACK H/L (high flow, low flow) line entry on the ANALOG DISC 1/4 page.

(h) The following information can be displayed on the FMC:

LEFT RIGHT

ECS PACK ON OFF

ECS PACK H/L LO HI

ISOLATION VALVE OPEN (CLOSED)

(7) Set the positions of these switches on the P5-10 Panel.

(a) Set the engine BLEED 1 switch to OFF.

(b) Set the engine BLEED 2 switch to OFF.

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(c) Set the APU BLEED switch to OFF.

(d) Set the L PACK and R PACK switches to OFF.

(e) Set the ISOLATION VALVE switch to CLOSE.

(f) Set the RECIRC FAN switch to OFF.

(8) Remove the right cold-air check valve (Conditioned Air Check Valve Removal, AMM TASK 21-51-07-000-801-001).

(a) Loosen the clamps that secure the check valve to the ducts.

(b) Move the clamps away from the check valve.

(c) Remove the check valve.

(d) Install the air pack - air flow and temperature diagnostic, SPL-6191.

(e) Install clamps onto the air pack - air flow and temperature diagnostic, SPL-6191.

(f) Tighten clamps to 50 pounds-inches (5.65 Newton-meters).

(9) Start the APU (APU Starting and Operation, AMM TASK 49-11-00-860-801).

NOTE: The electrical load should be less than 25 Amperes.

(a) Record the outside ambient temperature and dew point on the 21-00 TASK SUPPORT Figure 303.

(b) Set the positions of these switches on the P5-10 Panel:

1) Set the ISOLATION VALVE switch to AUTO.

2) Set the APU BLEED switch to ON.

(c) Allow the duct pressure to stabilize and record the right duct pressure (psi) _____.

(d) Record the flow and temperature from the measuring tool:

FLOW_____ lbs/min, TEMP_____ °F

(10) Set the positions of these switches on the P5-10 panel for high flow as follows:

NOTE: This step checks that solenoid “C” is energized. Airflow is expected to be zero.

(a) Set the R PACK switch to AUTO.

(b) Make sure the right pack temperature selector is set to MANUAL FULL COOL.

(11) Check the flow measuring tool connections for leaks and correct any that are found.

(12) Measure and record the right pack flow and temperature for HIGH FLOW.

(a) Verify correct configuration with aircraft FMC and allow readings to stabilize on the air pack - air flow and temperature diagnostic, SPL-6191.

(b) Place the AIR TEMP selector on the P5-10 panel, to SUPPLY DUCT position. When the supply duct temperature is stable, record the supply duct pressure and temperature on the data collection sheet.

(c) Record HIGH FLOW right pack flow and temperature from measuring tool on the data collection sheet.

(d) Record the position of the following component indicators on the data collection sheet.

1) 35°F Control Valve

2) Flow Control Valve

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3) Mix Valve
4) Water Separator Pressure Indicator

(13) Do these steps and record the RIGHT Pack Readings for AUTO FLOW.
(a) Put the R PACK Switch to AUTO.
(b) Put the BLEED 1 or BLEED 2 Switch to ON.
(c) On the Flight Management Computer (FMC) make sure that the PACK configuration is correct and allow readings to stabilize on the flow measuring tool.
(d) On the P5-10 Panel, put the AIR TEMP Selector to the SUPPLY DUCT position.
(e) When the Supply Duct Temperature is stable, record the Supply Duct Pressure and Temperature on the Data Collection sheet.
(f) Record from the Measuring Tool the AUTO FLOW RIGHT Pack Flow and Temperature on the Data Collection Sheet.
(g) Record the position of these component indicators on the Data Collection Sheet:
1) 35°F control valve
2) Flow Control Valve
3) Mix Valve
4) Water Separator Pressure Indicator

(14) Do these steps and record the Right Pack Readings for APU HIGH FLOW:
(a) Put the R PACK switch to HIGH
(b) Put the BLEED 1 and 2 switches to OFF
(c) On the FMC make sure that the PACK configuration is correct and allow readings to stabilize on the flow measuring tool.
(d) On the P5-10 Panel, put the AIR TEMP Selector to the SUPPLY DUCT position.
(e) When the Supply Duct Temperature is stable, record the Supply Duct Pressure and Temperature on the Data Collection Sheet.
(f) Record from the Measuring Tool the APU HIGH FLOW RIGHT Pack Flow and Temperature on the Data Collection Sheet.
(g) Record the position of the following component indicators on the data collection sheet.
1) 35°F Control Valve
2) Flow Control Valve
3) Mix Valve
4) Water Separator Pressure Indicator

(15) Put these switches on the P5-10 Panel:
(a) Put the R PACK Switch to OFF
(b) Put the APU BLEED switch to OFF.
(c) Make sure that the BLEED 1 and 2 Switches are OFF.
(d) Make sure that the Right Pack indicated Duct Pressure is Zero.
(e) On the FMC, make sure that the PACK Configuration is correct.

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(16) Remove Flow Measuring Tool and reinstall Right Cold-Air Check Valve.

NOTE: Make sure the orientation of the valve is correct before tightening the clamps.

(17) Torque the clamps to 50 pound-inches (5.65 Newton-meters).

(18) Remove the Left Cold-Air Check Valve (Conditioned Air Check Valve Removal, AMM TASK 21-51-07-000-801-001).

(a) Loosen the clamps that secure the Check Valve to the ducts.

(b) Move the clamps away from the check valve.

(c) Remove the Check Valve.

(d) Install the air pack - air flow and temperature diagnostic, SPL-6191.

(e) Install clamps onto the air pack - air flow and temperature diagnostic, SPL-6191.

(f) Torque clamps to 50 pounds-inches (5.65 Newton-meters).

(19) Set the positions of these switches on the P5-10 Panel as follows:

(a) Set the ISOLATION VALVE Switch to CLOSED.

(b) Set the APU BLEED Switch to ON.

(c) Allow the Duct Pressure to stabilize and record the Left Duct Pressure (psi) _____.

(d) Record the Flow and Temperature from the Measuring Tool:

FLOW_____ lbs/min, TEMP_____ °F

(20) Set the positions of these switches on the P5-10 panel for high flow as follows:

NOTE: This step checks that solenoid “C” is energized. Airflow is expected to be zero.

(a) Set the L PACK Switch to AUTO.

(b) Make sure the left pack temperature selector is set to MANUAL FULL COOL.

(21) Measure and record the left pack flow and temperature for HIGH FLOW:

(a) Verify correct configuration with aircraft FMC and allow readings to stabilize on the air pack - air flow and temperature diagnostic, SPL-6191.

(b) Place the AIR TEMP selector on the P5-10 panel, to SUPPLY DUCT position. When the supply duct temperature is stable, record the supply duct pressure and temperature on the data collection sheet.

(c) Record HIGH FLOW left pack flow and temperature from measuring tool on the data collection sheet.

(d) Record the position of the following component indicators on the data collection sheet.

1) 35°F Control Valve

2) Flow Control Valve

3) Mix Valve

4) Water Separator Pressure Indicator

(22) Set the positions of these switches on the P5-10 panel for AUTO FLOW as follows:

(a) Make sure that the L PACK Switch is in AUTO.

(b) Put the BLEED 1 or BLEED 2 S to ON.

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(c) On the FMC, make sure that the PACK Configuration is correct and allow readings to stabilize on the flow measuring tool.

(d) Place the AIR TEMP selector on the P5-10 panel to the SUPPLY DUCT position.

(e) When the supply duct temperature is stable, record the supply duct pressure and temperature on the data collection sheet.

(f) Record from the measuring tool the AUTO FLOW left pack flow and temperature on the data collection sheet.

(g) Record the position of the following component indicators on the data collection sheet.

1) 35°F Control Valve
2) the flow Vontrol Valve
3) Mix Valve
4) Water Separator Pressure Indicator

(23) Set the positions of these switches on the P5-10 panel for APU HIGH FLOW as follows:

(a) Put the L PACK Switch to HIGH.

(b) Put the ENGINE BLEED switch to OFF.

(c) Put the AIR TEMP selector on the P5-10 panel the to SUPPLY DUCT position.

(d) When the Supply Duct Temperature is stable, record the Supply Duct Pressure and Temperature on the Data Collection Sheet.

(e) Record from Measuring Tool APU HIGH FLOW LEFT Pack Flow and Temperature on the Data Collection Sheet.

(f) Record the position of these Component Indicators on the Data Collection Sheet.

1) 35°F Control Valve
2) Flow Control Valve
3) Mix Valve
4) Water Separator Pressure Indicator

(24) Set the positions of these switches on the P5-10 Panel:

(a) Put the L PACK Switch to OFF.

(b) Put the APU BLEED Switch to OFF.

(c) Make sure that the Left Pack indicated Duct Pressure is Zero.

(d) On the FMC, make sure that the PACK Configuration is correct.

(25) Remove flow measuring tool and reinstall left cold-air check valve.

NOTE: Make sure the orientation of the valve is correct before tightening the clamps.

(26) Tighten the clamps to 50 pound-inches (5.65 Newton-meters).

(27) Set the ISOLATION VALVE switch to OPEN.

(28) Set the APU BLEED switch to ON.

(29) Check the right cold-air check valve for leaks and correct as necessary.

(30) Move the ISOLATION VALVE switch to CLOSED.

(31) Check the left cold-air check valve for leaks and correct as necessary.

(32) Set the APU BLEED switch to OFF.

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(33) Shut down the APU (APU Usual Shutdown, AMM TASK 49-11-00-860-802).

(34) Compare the values in the 21-00 TASK SUPPORT Figure 303 with the flow curves in 21-00 TASK SUPPORT Figure 304. Refer to the21-00 TASK SUPPORT Figure 305 to troubleshoot the system if the values collected do not fall within the parameters of the flow curves.

L. Repair Confirmation After Use of the Air Flow and Temperature Diagnostic Tool

(1) Do the procedure to evaluate the packs again to see if the problem has been solved.

(2) Continue to evaluate the leakage in the fuselage and the Bleed Air Pressure to determine if sufficient air is available to the packs.

(a) Low Bleed Pressures can indicate poor Bleed Performance even if the pressures are not zero.

OMR ALL

M. Initial Evaluation of Engine Bleed System Performance

NOTE: This procedure is given for Engine-1. Engine-2 is similar unless otherwise stated.

(1) Make sure that the applicable Engine Start Lever is in the CUTOFF position.

(a) Install a DO-NOT-OPERATE tag on the Engine Start Lever.

(2) Make sure that the L and R PACK Switches on the P5-10 Panel are in the OFF position.

(3) Make sure that the APU BLEED Switch on the P5-10 Panel is in the OFF position.

(4) Make sure that the ISOLATION VALVE Switch on the P5-10 Panel is in the CLOSE position.

NOTE: For Engine-2, the ISOLATION VALVE Switch should be in the OPEN position.

(5) Open the Thrust Reverser. This is the task: Open the Thrust Reverser (Selection), AMM TASK 78-31-00-010-801-F00

(6) Use a dental mirror, STD-3907 to look at the Position Indicator/Manual Override Nut [5] on the Precooler Control Valve [4] to make sure it is in the OPEN position (21-00 TASK SUPPORT Figure 306).

(a) Use a 3/4-inch wrench on the Manual Override Nut [5] to close the Precooler Control Valve [4] to make sure it moves smoothly.

(b) Remove the wrench and allow the valve to return to the OPEN position by spring force only.

(c) If the Precooler Control Valve [4] does not close smoothly or return to the OPEN position, replace the Precooler Control Valve. These are the tasks:

- Precooler Control Valve Removal, AMM TASK 36-12-02-000-801
- Precooler Control Valve Installation, AMM TASK 36-12-02-400-801

(7) Look at the Position Indicator/Manual Override Nut [2] on the PRSOV [1] to make sure it is in the CLOSED position (21-00 TASK SUPPORT Figure 306).

(a) Use a 3/8-inch wrench on the Manual Override Nut [2] to open the PRSOV [1] to make sure it moves smoothly.

(b) Remove the wrench and allow the valve to return to the CLOSED position by spring force only.

(c) If the PRSOV [1] does not open smoothly or return to the CLOSED position, replace the PRSOV. These are the tasks:

- PRSOV Removal, AMM TASK 36-11-04-000-801

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- PRSOV Installation, AMM TASK 36-11-04-400-801

(8) Look at the Position Indicator/Manual Override Nut [10] on the High Stage Valve [9] to make sure it is in the CLOSED position (21-00 TASK SUPPORT Figure 306).

(a) Use a 3/8-inch wrench on the Manual Override Nut [10] to open the High Stage Valve [9] to make sure it moves smoothly.

(b) Remove the wrench and allow the valve to return to the CLOSED position by spring force only.

(c) If the High Stage Valve does not open smoothly or return to the CLOSED position, replace the High Stage Valve. These are the tasks:

- High Stage Valve Removal, AMM TASK 36-11-06-000-801
- High Stage Valve - Installation, AMM TASK 36-11-06-400-801

N. Fault Isolation Procedure - Do the Bleed Air System Leak Check

(1) Use the APU to pressurize the Bleed Air System. This is the task: Supply Pressure to the Pneumatic System with the APU, AMM TASK 36-00-00-860-803.

NOTE: The PRSOV [1] must be closed when you pressurize the Bleed Air System.

(2) Put the applicable Engine BLEED Switch in the ON position.

WARNING

USE A RATCHET-TYPE WRENCH TO OPEN THE PRSOV. PRESSURE IN THE SYSTEM CAN CAUSE THE PRSOV TO OPEN QUICKLY. THIS CAN PULL THE WRENCH FROM YOUR HANDS. INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Use a 3/8-inch socket on a ratchet-type wrench to turn the Manual Override Nut [2] on the PRSOV [1]. Once you begin turning the Manual Override Nut [2], the air pressure should move the PRSOV to the fully open position.

WARNING

REMOVE THE PRESSURE FROM THE PNEUMATIC DUCTS BEFORE YOU REMOVE A PNEUMATIC SYSTEM COMPONENT. HOT HIGH PRESSURE AIR CAN CAUSE INJURIES TO PERSONNEL OR DAMAGE TO EQUIPMENT.

(4) If the PRSOV did not move fully open, test the Relief Valve Assembly [6] on the Bleed Air Regulator [3] for major air leakage.

NOTE: A small amount of air leakage from the Relief Valve Assembly [6] is normal.

(a) Press down on the top of the Relief Valve Assembly [6] to stop the air flow.

(b) If the PRSOV [1] moves fully open when the air flow stops, there is too much leakage from the Relief Valve Assembly [6]. Replace the Bleed Air Regulator [3] by doing these tasks.

1) Bleed Air Regulator Removal, AMM TASK 36-11-03-000-801
2) Bleed Air Regulator Installation, AMM TASK 36-11-03-400-801

(5) Apply leak detector, G50135 to the sense lines and fittings to determine if air leakage exists.

(a) Replace leaking sense lines and fittings as necessary.

NOTE: Apply Pure Nickel Special compound, D00006 to all male threads before connecting the pneumatic sense lines.

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(6) Replace the PRSOV [1] if it does not go fully open when the system is pressurized and there is no major leakage from the Relief Valve Assembly [6] on the Bleed Air Regulator [3] or leaks in the sense lines. Do these tasks:

(a) PRSOV Removal, AMM TASK 36-11-04-000-801

(b) PRSOV Installation, AMM TASK 36-11-04-400-801

(7) Make sure the Precooler Control Valve [4] moves fully closed when the PRSOV is fully open.

(8) If the Precooler Control Valve [4] does not move fully closed, inspect for air leaks by doing the following.

(a) Apply leak detector, G50135 to the supply and control pressure sense lines, fittings, and the 390° sensor.

(b) Replace leaking sense lines, fittings, and the 390° sensor as necessary.

NOTE: Apply Pure Nickel Special compound, D00006 to all male threads before connecting the pneumatic sense lines.

(9) If the Precooler Control Valve [4] does not move fully closed and no air leaks were found, do the following.

(a) Remove the test cap [12] on the control pressure sense line from the Precooler Control Valve [4] to the 390° sensor and make sure air flows out of the test port.

(b) If air does not flow out of the test port, do the following.

1) Reconnect the test cap [12].

2) Disconnect the sense line from the Precooler Control Valve [4] to make sure there is air flow out of the valve's control pressure port.

3) If air does not flow out of the control pressure port, replace the Precooler Control Valve [4].

a) Precooler Control Valve Removal, AMM TASK 36-12-02-000-801

b) Precooler Control Valve Installation, AMM TASK 36-12-02-400-801

4) If there is air flow out of the control pressure port on the Precooler Control Valve [4], there is blockage in the sense line between the test port and the valve.

a) Replace the sense line.

(c) If air flows out of the test port when the test cap is removed, do the following.

1) Reconnect the test cap [12].

2) Disconnect the sense line at the 390° sensor to make sure there is no blockage in the sense line between the test port and the sensor.

3) Replace the sense line if there is no air flow out of the end.

(10) Listen for air leakage in the engine.

NOTE: Air leakage in the engine is an indication that the High Stage Valve [9] is allowing air to backflow into the engine.

(a) Replace the High Stage Valve [9] if air can be heard in the engine when the system is pressurized. Do these tasks:

1) High Stage Valve Removal, AMM TASK 36-11-06-000-801

2) High Stage Valve - Installation, AMM TASK 36-11-06-400-801

(11) Disconnect the Supply Pressure Sense Line [8] from the High Stage Regulator [7].

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(12) Do a check for air leakage at the supply pressure inlet on the High Stage Regulator [7].

NOTE: Air leaking from the supply pressure inlet when the Bleed Air System is pressurized with the APU indicates the reverse flow diaphragm inside the High Stage Regulator [7] is damaged.

(a) Replace the High Stage Regulator [7] if air leakage is detected from the supply pressure inlet. Do these tasks.

1) High Stage Regulator Removal, AMM TASK 36-11-07-000-801

2) High Stage Regulator Installation, AMM TASK 36-11-07-400-801

(b) If no air leakage is detected, reconnect the Supply Pressure Sense Line [8] to the High Stage Regulator [7].

NOTE: Apply Pure Nickel Special compound, D00006 to all male threads before connecting the pneumatic sense lines.

O. Put the Airplane Back To Its Usual Condition

(1) Remove pneumatic pressure from the Bleed Air System. Do this task:

(a) Remove Pressure from the Pneumatic System, AMM TASK 36-00-00-860-806

(2) Put the ISOLATION VALVE switch on the P5-10 Air Conditioning Module in the AUTO position.

(3) Put the L and R BLEED switches on the P5-10 panel in the OFF position.

(4) Close the thrust reverser. This is the task: Close the Thrust Reverser (Selection), AMM TASK 78-31-00-010-804-F00

(5) Remove the electrical power from the airplane. This is the task: Remove Electrical Power, AMM TASK 24-22-00-860-812.

P. Engine Bleed System Troubleshooting Table

(1) The following is an Engine Bleed System Quick Reference Troubleshooting Table. Print out the table and use it as a quick reference troubleshooting guide along with 21-00 TASK SUPPORT Figure 307 when working on the bleed system of an airplane.

Engine Bleed System Quick Reference Troubleshooting Table

STEP	737-600/700/800/900	TROUBLESHOOTING STEP	EXPECTATION - NORMAL OPERATION	REMEDY FOR ABNORMAL OPERATION
1	AMM 36-11-00/500 Task 36-11-00-700-802 Para 6.F.(6) FIM 36-10-804, F.(6)	Open the thrust reverser.		
2.A	AMM 36-11-00/500 Task 36-11-00-700-802 Para 6.F.(7) FIM 36-10-804, F.(7)	Look at the position indicator on the precooler control valve.	Should be open.	
2.B	AMM 36-11-00/500 Task 36-11-00-700-802 Para 6.F.(7)(a),(b) FIM 36-10-804, F.(7)(a), (b)	Manually wrench CLOSED the precooler control valve and allow it to move OPEN by spring force only.	Should move smoothly to the CLOSED position and return to the OPEN position by spring force only.	If the valve does not move smoothly and return to OPEN, replace per the R/I in AMM 36-12-02/400 series pages.

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Engine Bleed System Quick Reference Troubleshooting Table (Continued)

STEP	737-600/700/800/900	TROUBLESHOOTING STEP	EXPECTATION - NORMAL OPERATION	REMEDY FOR ABNORMAL OPERATION
3.A	AMM 36-11-00/500 Task 36-11-00-700-802 Para 6.F.(8) FIM 36-10-804, F.(9)	Look at the position indicator on the pressure regulating shutoff valve (PRSOV).	Should be CLOSED.	
3.B	AMM 36-11-00/500 Task 36-11-00-700-802 Para 6.F.(8)(a), (b) FIM 36–10–804, F.(9)(a), (b)	Manually wrench OPEN the PRSOV and allow it to move CLOSED by spring force only.	Should move smoothly to the OPEN position and return to the CLOSED position by spring force only.	If the valve does not move smoothly and return to CLOSED, replace per the R/I in AMM 36-11-04/400 series pages.
4.A	AMM 36-11-00/500 Task 36-11-00-700-802 Para 6.F.(9) FIM 36-10-804, F.(10)	Look at the position indicator on the high stage valve.	Should be CLOSED.	
4.B	AMM 36-11-00/500 Task 36-11-00-700-802 Para 6.F.(9)(a), (b) FIM 36-10-804, F.(10)(a), (b)	Manually wrench OPEN the high stage valve and allow it to move CLOSED by spring force only.	Should move smoothly to the OPEN position and return to the CLOSED position by spring force only.	If the valve does not move smoothly and return to CLOSED, replace per the R/I in AMM 36-11-06/400 series pages.
5A	AMM 36-11-00/500 Task 36-11-00-700-802 Para 6.G.(1)(a) FIM 36-10-804, F.(11)	Supply pneumatic pressure from the APU or ground supply (pressure should be less than 40 psig).		
5.B	AMM 36-11-00/500 Task 36-11-00-700-802 Para 6.G.(3), (4) FIM 36-10-804, F.(13)	Manually OPEN the PRSOV with a wrench if it does not OPEN when the APU bleed is selected ON. WARNING: Use a ratchet-type wrench when opening the PRSOV to avoid injury.	PRSOV will move to and stay in the FULL OPEN position. Low control pressure from the BAR or leaking control pressure lines can result in low duct pressure when the PRSOV is regulating the pneumatic pressure. Check for leaking relief valve on the bleed air regulator, leaking flex line to the PRSOV, leaking control pressure line from the PRSOV to the 450° F sensor, excessive venting of the 450° F sensor.	Check for leaks on: 1. the supply and control pressure lines. 2. relief valve on the regulator. If the relief valve is leaking, replace the bleed air regulator per the R/I in AMM 36-11-03/400 series pages.

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Engine Bleed System Quick Reference Troubleshooting Table (Continued)

STEP	737-600/700/800/900	TROUBLESHOOTING STEP	EXPECTATION - NORMAL OPERATION	REMEDY FOR ABNORMAL OPERATION
5.C	AMM 36-11-00/500 Task 36-11-00-700-802 Para 6.G.(7) FIM 36-10-804, F.(20)	When the PRSOV moves OPEN, the precooler control valve should move CLOSED.	The precooler control valve moves to the CLOSED position.	If the valve does not move CLOSED, check for leaks on the supply and control pressure lines. If no leaks are found, replace the precooler control valve per the R/I in AMM 36-12-02/400 series pages.
5.D	AMM 36-11-00/500 Task 36-11-00-700-802 Para 6.G.(9) FIM 36-10-804, F.(21)(a)	Disconnect the test cap on the precooler control valve control pressure line.	The precooler control valve should move to the OPEN position.	If the valve does not move OPEN; check for blockage in the sense line.
5.E	AMM 36-11-00/500 Task 36-11-00-700-802 Para 6.G.(10)	Listen for air leakage in the engine.	Air leakage in the engine indicates the high stage valve is allowing air to backflow into the engine.	If you can hear air leakage in the engine, replace the high stage valve per the R/I in the AMM: 36–11–06/400 series pages.
5.F	AMM 36-11-00/500 Task 36-11-00-700-802 Para 6.H.(11), (12) FIM 36-10-804, F.(15)	Disconnect the supply line connection from the high stage regulator port.	No air should flow from the supply port.	If air flows from the supply port, the reverse flow diaphragm is ruptured. Replace the high stage regulator per the R/I in AMM 36-11-07/400 series pages
6.A	AMM 36-11-00/500 Task 36-11-00-700-802 Para 6.H.(1)-(3)	Return pneumatic system to normal configuration.		
6.B	AMM 36-11-00/500 Task 36-11-00-700-802 Para 6.H.(4), (5)	Return airplane to normal configuration.		

Q. Repair Confirmation

(1) Do this task: Start the Engine Procedure (Selection), AMM TASK 71-00-00-800-807-F00

(2) Put these switches, on the P5-10 panel, to the positions that follow:

(a) ISOLATION VALVE - CLOSED

(b) ISOLATION VALVE - CLOSED

(c) 2 BLEED - OFF

(d) APU BLEED - OFF

(e) L PACK - OFF

(f) R PACK - OFF

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(3) Examine the dual duct pressure indicator on the P5-10 panel.

NOTE: Ignore the fluctuation in the dual duct pressure indicator with these conditions: you move the BLEED switch, the isolation valve is closed, or the PACK switch is in the OFF position.

(a) Make sure the dual duct pressure indicator shows 10.0 psig or less.

(4) Make sure that engine is at idle.

(5) Put the engine 1 BLEED switch to the ON position.

(a) Permit the pressure in the pneumatic ducts to become stable.

(b) Make sure the dual duct pressure indicator shows 10.0 - 25.0 psig.

NOTE: Pneumatic duct pressure must be a minimum of 18 psig to supply sufficient air for cabin pressurization.

(6) Put the L PACK switch to the AUTO position.

(7) Slowly increase the engine power in 10% increments when following these steps and log the pressure values in the DUCT PRESSURE VALUES AS N1% INCREASES table below.

NOTE: The duct pressures should follow the graphs in 21-00 TASK SUPPORT Figure 308. The minimum serviceable pneumatic duct pressure under in-flight airplane configuration and operating conditions are shown in 21-00 TASK SUPPORT Figure 309.

(a) As N1 increases, make sure the duct pressure is at 32 ± 6 psig and becomes stable when N1 is at 40%.

(b) Continue to slowly increase the engine power to 80% N1.

(c) Make sure the duct pressure goes to 42 ± 8 psig and becomes stable.

INCREASE N1%

N1%	IDLE	20%	30%	40%
LEFT ENGINE				
RIGHT ENGINE				
PSIG RANGE	18–20	18–20	26–38	26–38

INCREASE N1%

N1%	50%	60%	70%	80%
LEFT ENGINE				
RIGHT ENGINE				
PSIG RANGE	26–38	34–50	34–50	34–50

DUCT PRESSURE VALUES AS N1% INCREASES

(8) Slowly decrease the engine power.

(a) Log the pressure values at every 10% N1 interval in the DUCT PRESSURE VALUES AS N1% DECREASES table below.

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(b) Make sure the duct pressure goes to 32 ± 6 psig and becomes stable as shown in the graphs in 21-00 TASK SUPPORT Figure 308.

NOTE: When the engine bleed system goes from the 5th-stage bleed to the 9th-stage bleed, the duct pressure can decrease to approximately 20.0 psig. This can occur before the high- stage valve opens and regulates the bleed air.

DECREASE N1%

N1%	80%	70%	60%	50%
LEFT ENGINE				
RIGHT ENGINE				
PSIG RANGE	34–50	34–50	34–50	26–38

DECREASE N1%

N1%	40%	30%	20%	IDLE
LEFT ENGINE				
RIGHT ENGINE				
PSIG RANGE	26–38	26–38	18–20	18–20

DUCT PRESSURE VALUES AS N1% DECREASES

(9) Put the engine 1 BLEED switch to the OFF position.

(10) Put the L PACK switch to the OFF position.

(11) After the duct pressure is stable.

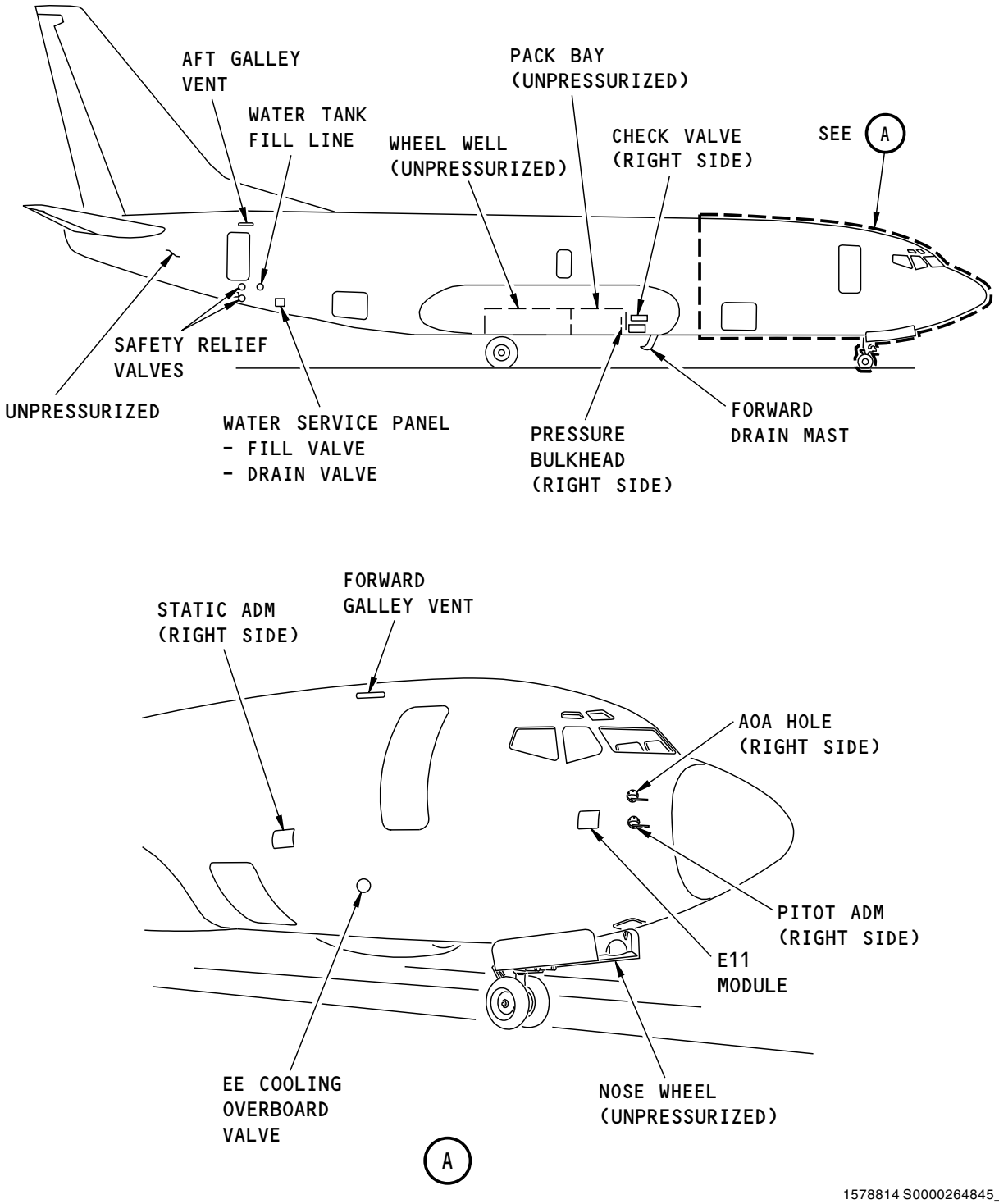
(a) Make sure the duct pressure is at 10.0 psig or less.

(12) To stop the engine, do this task: Stop the Engine Procedure (Usual Engine Stop), AMM TASK 71-00-00-700-819-F00

(13) Do this procedure again for the engine 2 bleed system, if it is necessary.

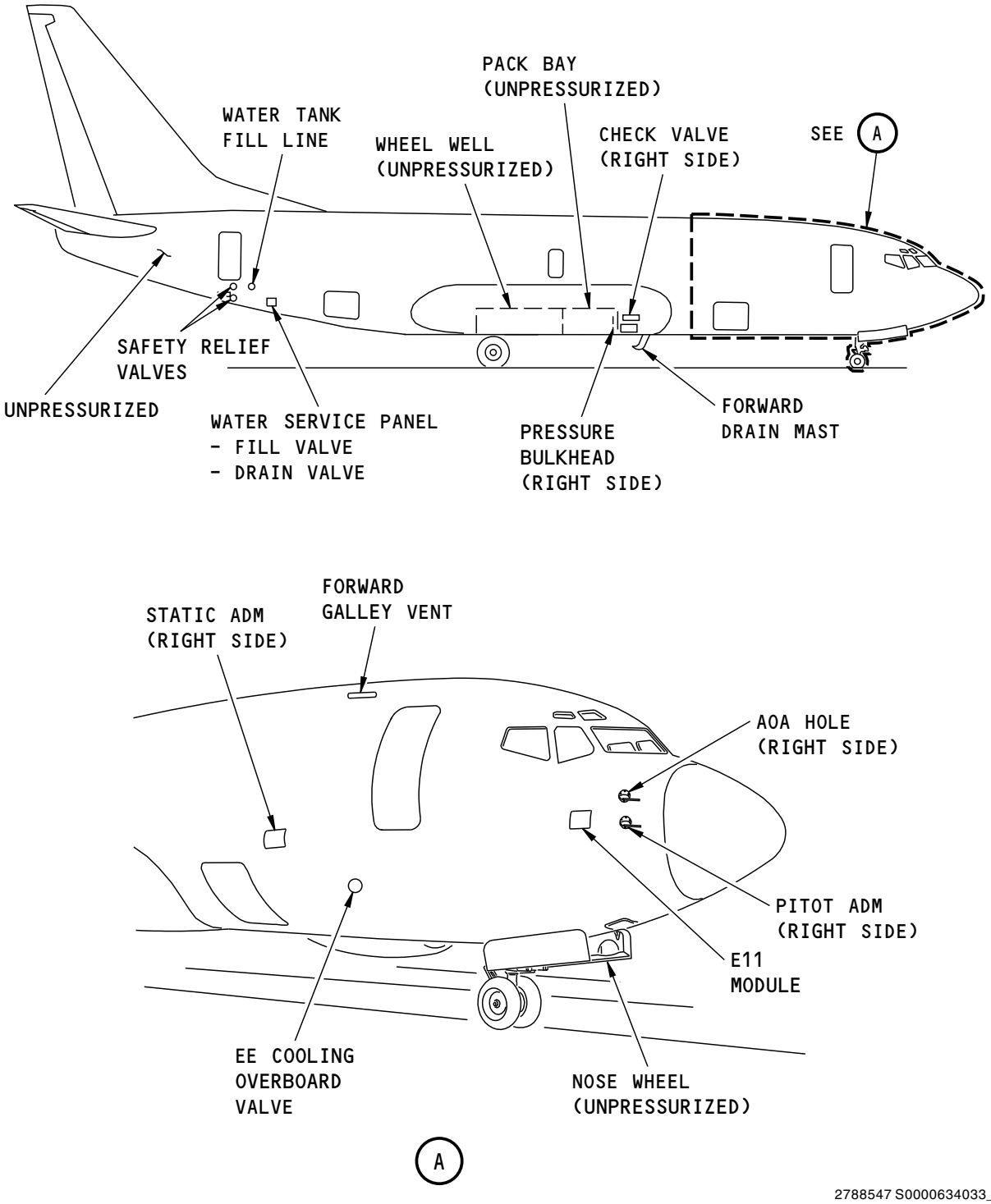
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuselage Leakage Sources
Figure 301/21-00-00-990-810 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuselage Leakage Sources
Figure 301/21-00-00-990-810 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AIRPLANE _____ HOURS _____ CYCLES _____ DATE _____

INITIAL CABIN PRESSURIZATION (0.0 TO 4.0 PSID)

CONDITION OF CARGO AND MAIN ENTRY DOOR SEALS _____

FLOW CONTROL VALVE INDICATOR MODULATION LEFT VALVE _____ RIGHT VALVE _____

ANY OBVIOUS AIR LEAK NOISE FROM INSIDE CABIN _____

ANY OBVIOUS LEAKS OUTSIDE AIRPLANE _____

ESTIMATED TIME TO PRESSURIZE FROM 0.0 TO 4.0 PSID _____

BLEED SOURCE(S) USED TO ACHIEVE 4.0 PSID LEFT ENGINE RIGHT ENGINE APU

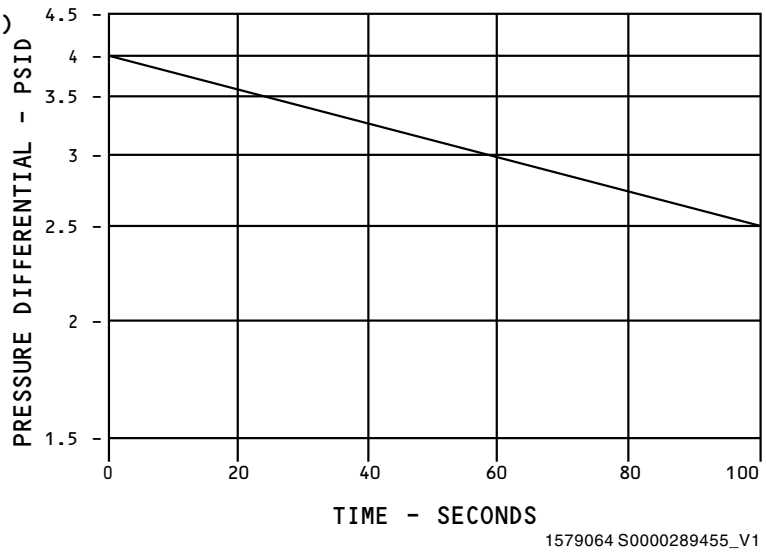
SINGLE PACK ABILITY TO PRESSURIZE THE CABIN (LOW AND HIGH FLOW MODE)

	CABIN RATE OF CHANGE	BLEED SOURCE & DUCT PRESSURE	ENGINE N2 %
LEFT PACK LOW FLOW MODE	_____	_____	_____
LEFT PACK HIGH FLOW MODE	_____	_____	_____
RIGHT PACK LOW FLOW MODE	_____	_____	_____
RIGHT PACK HIGH FLOW MODE	_____	_____	_____

AIRPLANE LEAK DOWN (NO PACK OPERATION)

CABIN TEMPERATURE _____

TIME (SECONDS)	CABIN (PSID)	CABIN (RATE OF CHANGE)
10	_____	_____
20	_____	_____
30	_____	_____
40	_____	_____
50	_____	_____
60	_____	_____
70	_____	_____
80	_____	_____
90	_____	_____
100	_____	_____
110	_____	_____
120	_____	_____



1579064 S0000289455_V1

Single Pack Operation Troubleshooting Data Sheet
Figure 302/21-00-00-990-811

ECCN 9E991 BOEING PROPRIETARY - See title page for details

DATA COLLECTION SHEET

Airplane _____ Pack: Left or Right Date: _____

Ambient Temperature: _____ °C or °F Dew Point: _____

Flow Mode: HIGH

Pack Flow: _____ lbs/min Pack Temperature: _____ °F

Duct Pressure _____ PSI Duct Temperature: _____ °C or F

35 °F Valve Position: Circle one Full Close Half Closed _____ %Closed

Flow Control Valve: Indicate percent open _____ % Open

Water Separator Indicator Position: Red Black

Flow Mode: AUTO

Pack Flow: _____ lbs/min Pack Temperature: _____ °F

Duct Pressure _____ PSI Duct Temperature: _____ °C or F

35 °F Valve Position: Circle one Full Close Half Closed _____ %Closed

Flow Control Valve: Indicate percent open _____ % Open

Water Separator Indicator Position: Red Black

Flow Mode: APU HIGH

Pack Flow: _____ lbs/min Pack Temperature: _____ °F

Duct Pressure _____ PSI Duct Temperature: _____ °C or F

35 °F Valve Position: Circle one Full Close Half Closed _____ %Closed

Flow Control Valve: Indicate percent open _____ % Open

Water Separator Indicator Position: Red Black

Any other observations:

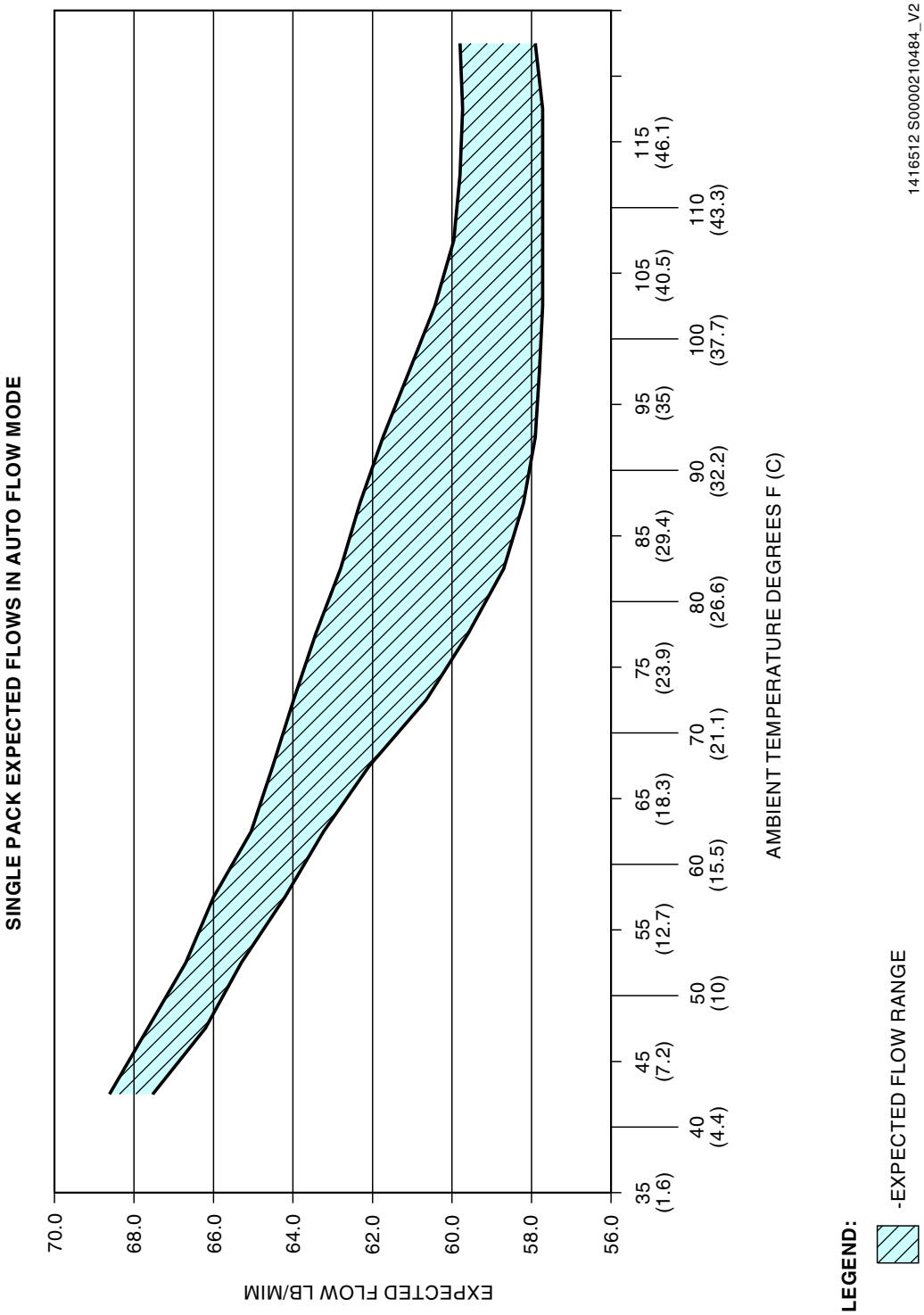
_____

_____

1579082 S0000296382_V1

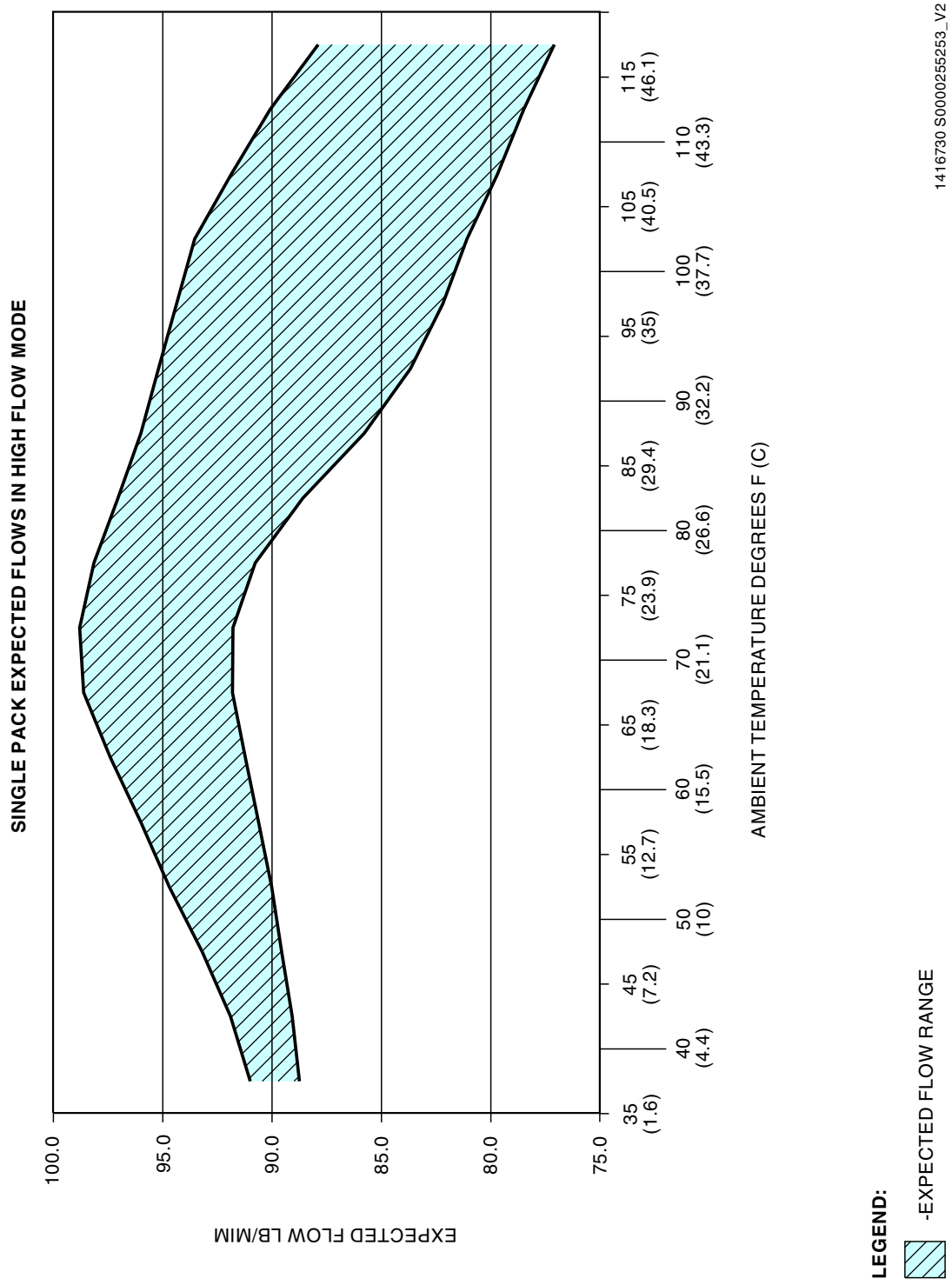
Pack Flow Control and Pack Cooling System Data Collection Sheet
Figure 303/21-00-00-990-812

ECCN 9E991 BOEING PROPRIETARY - See title page for details



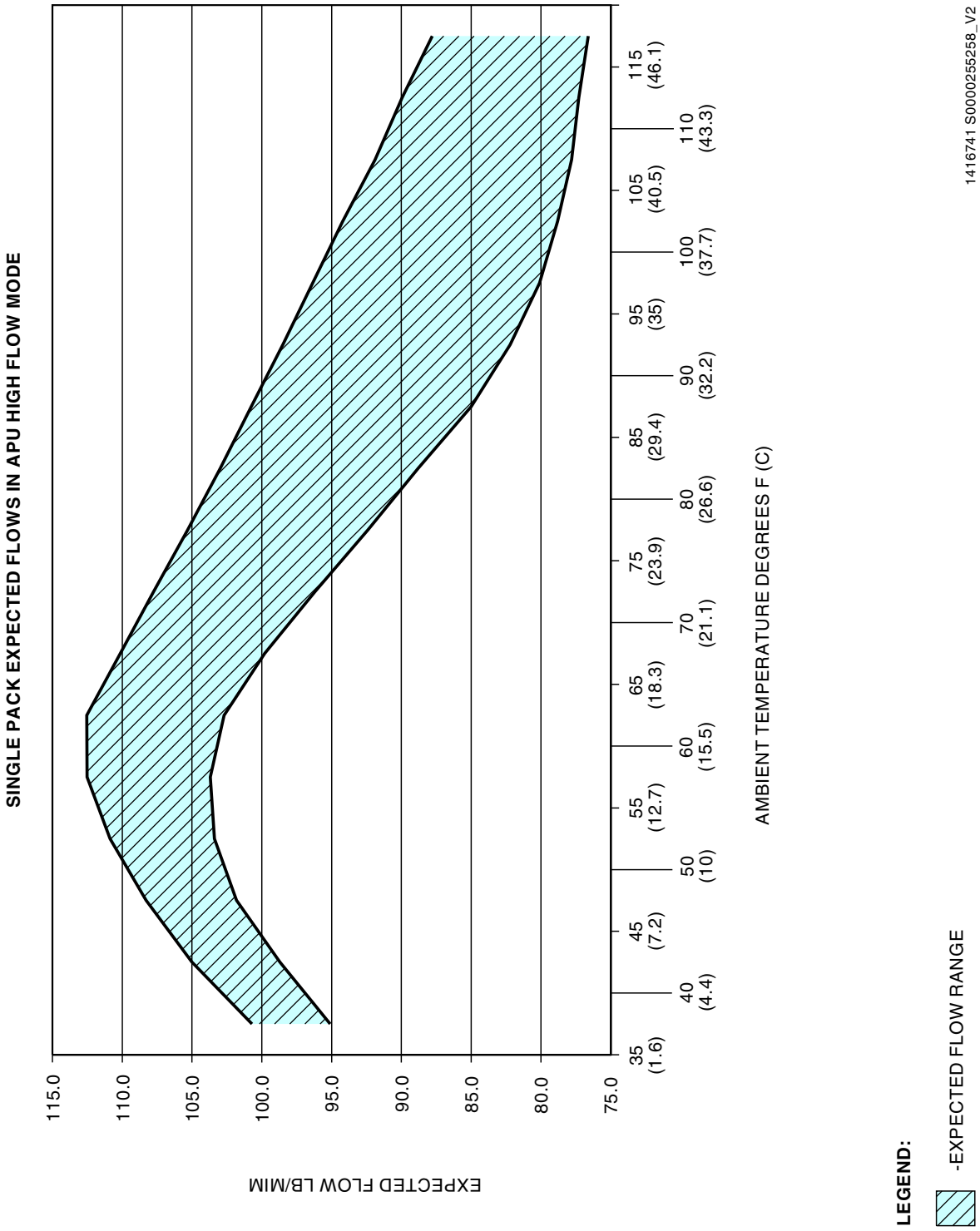
Single Pack Expected Flows
Figure 304/21-00-00-990-813 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



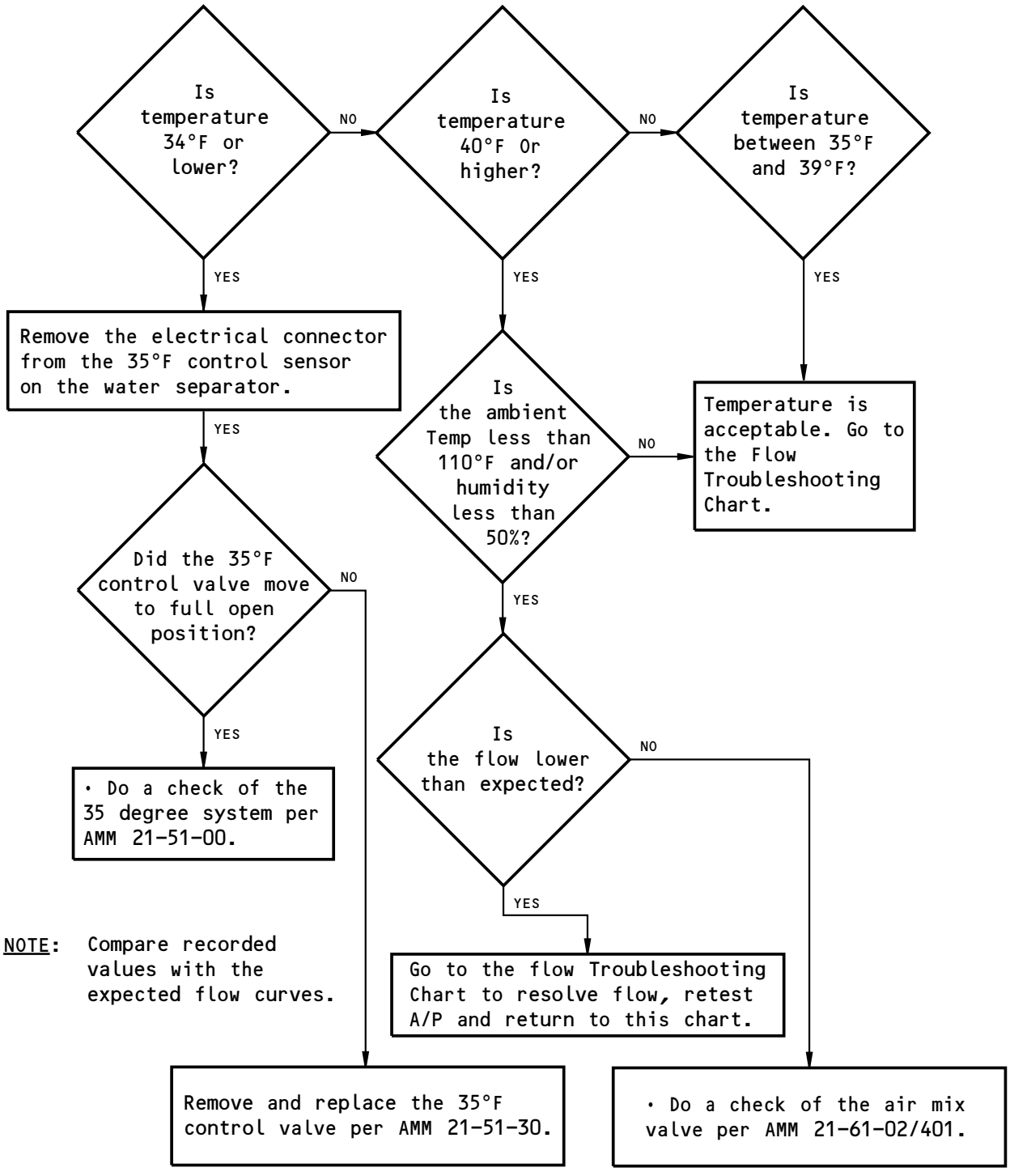
Single Pack Expected Flows
Figure 304/21-00-00-990-813 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



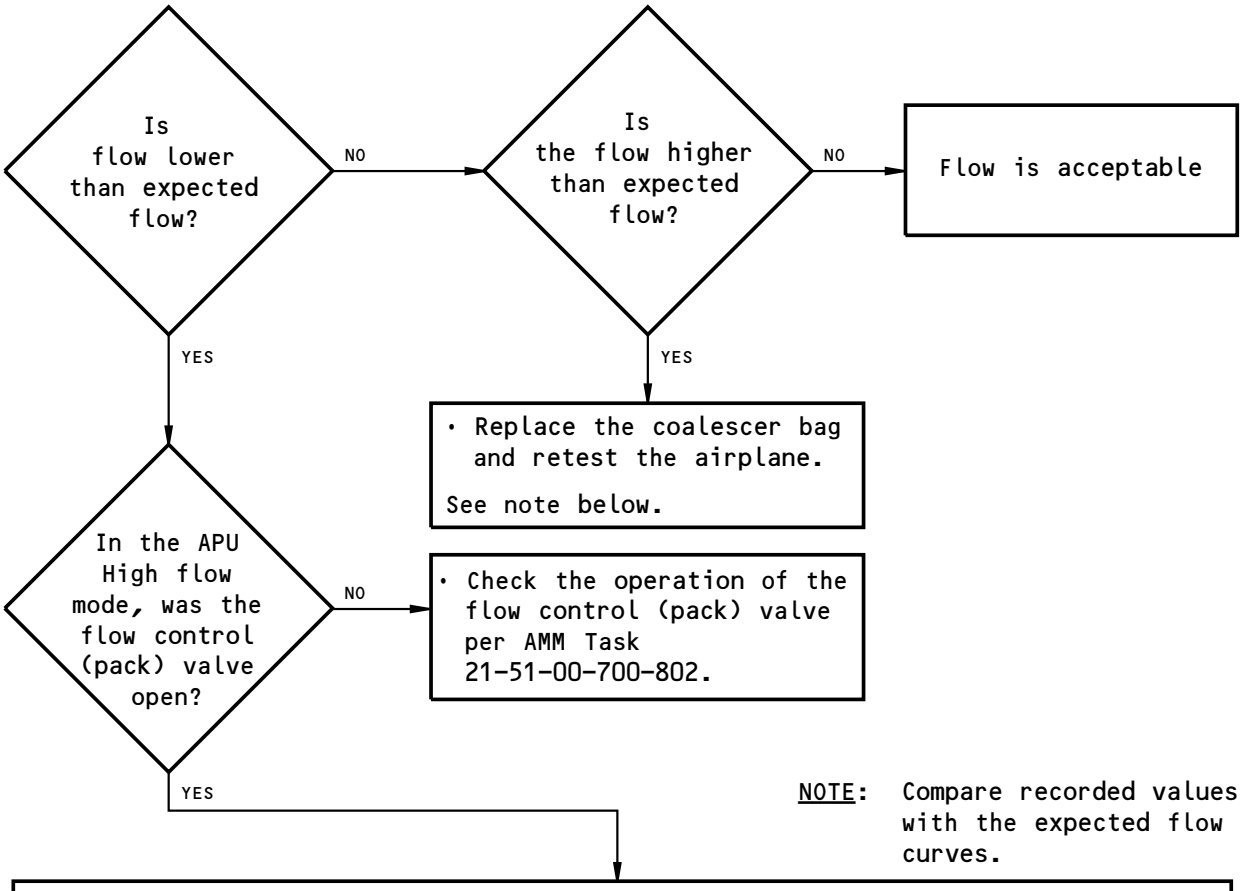
Single Pack Expected Flows
Figure 304/21-00-00-990-813 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Troubleshooting Charts
Figure 305/21-00-00-990-814 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

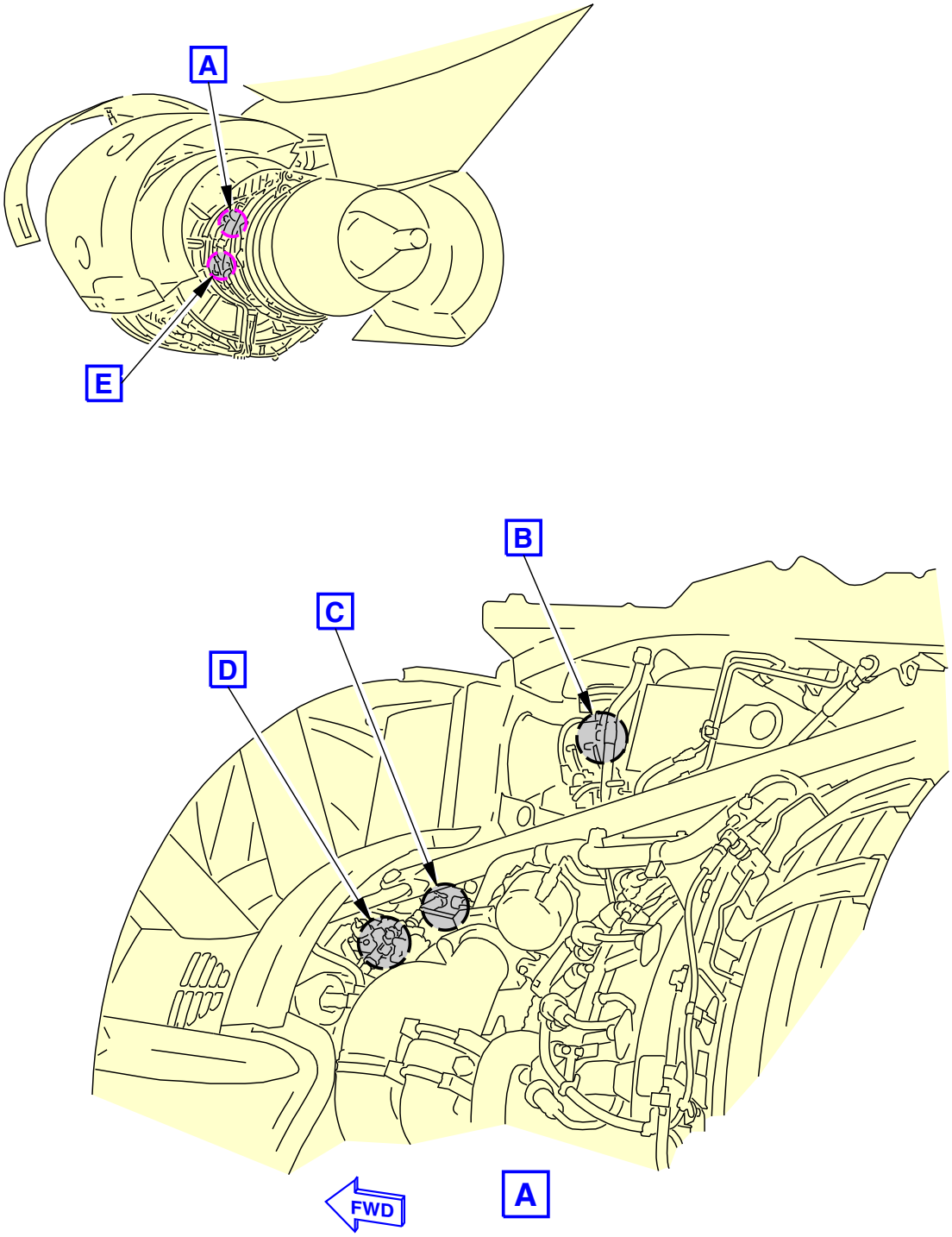


NOTE: High flow readings are an indication of water droplets in the pack flow which causes the FCI tool to indicate high flow readings. Water droplets are an indication that the coalescer bag is not adequately removing moisture.

1650543 S0000297528_V1

Troubleshooting Charts
Figure 305/21-00-00-990-814 (Sheet 2 of 2)

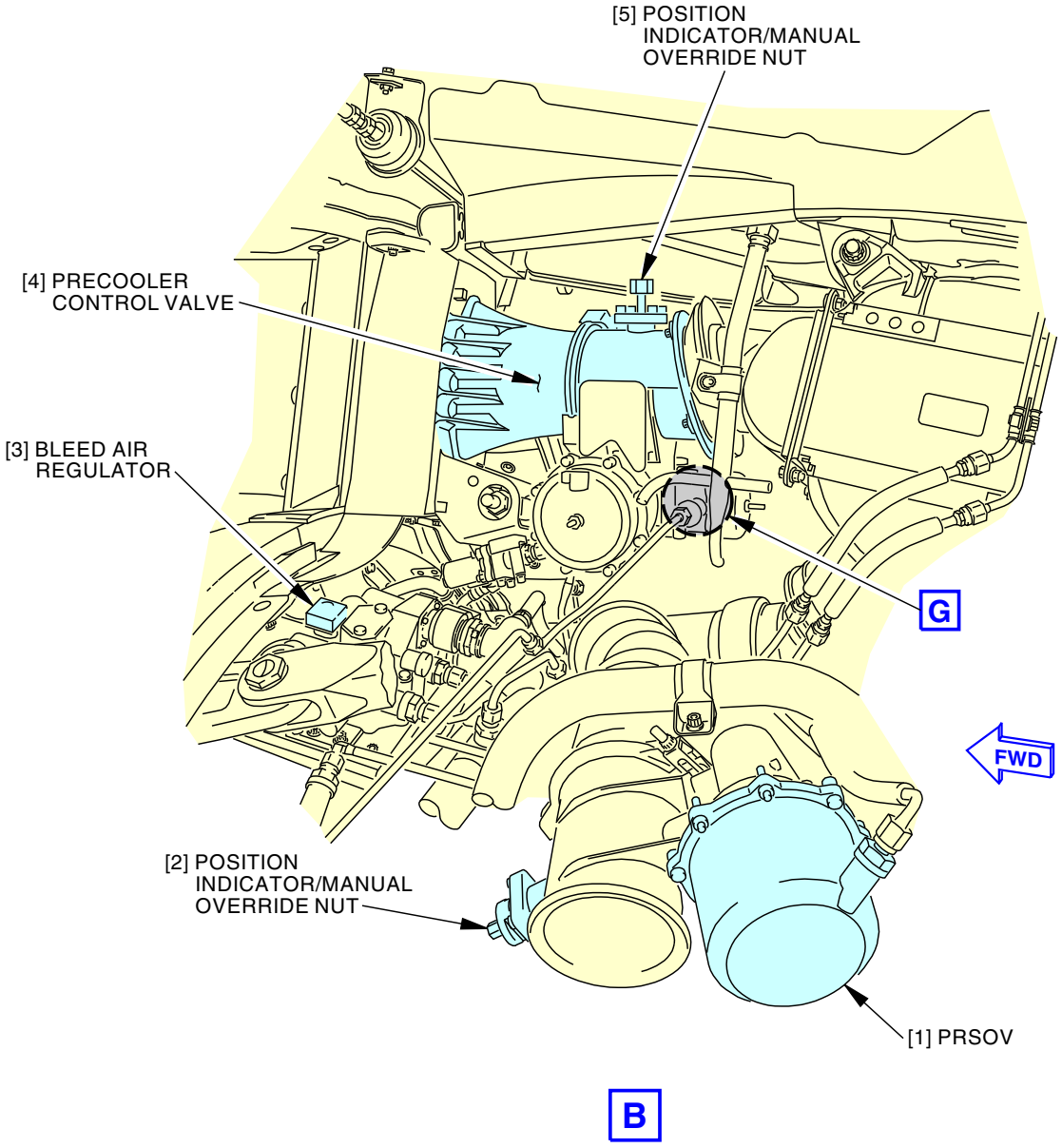
ECCN 9E991 BOEING PROPRIETARY - See title page for details



U71200 S0000214518_V3

Engine Bleed Air System Leakage Check
Figure 306/21-00-00-990-815 (Sheet 1 of 6)

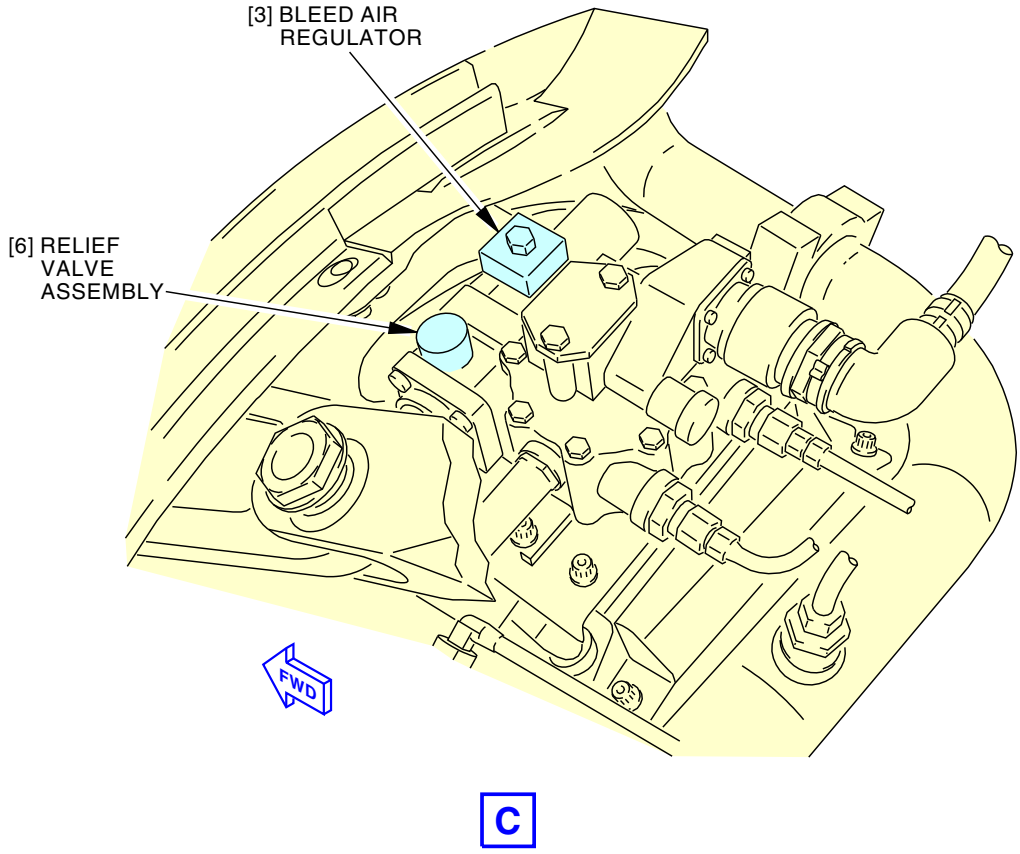
ECCN 9E991 BOEING PROPRIETARY - See title page for details



U71886 S0000214519_V3

Engine Bleed Air System Leakage Check
Figure 306/21-00-00-990-815 (Sheet 2 of 6)

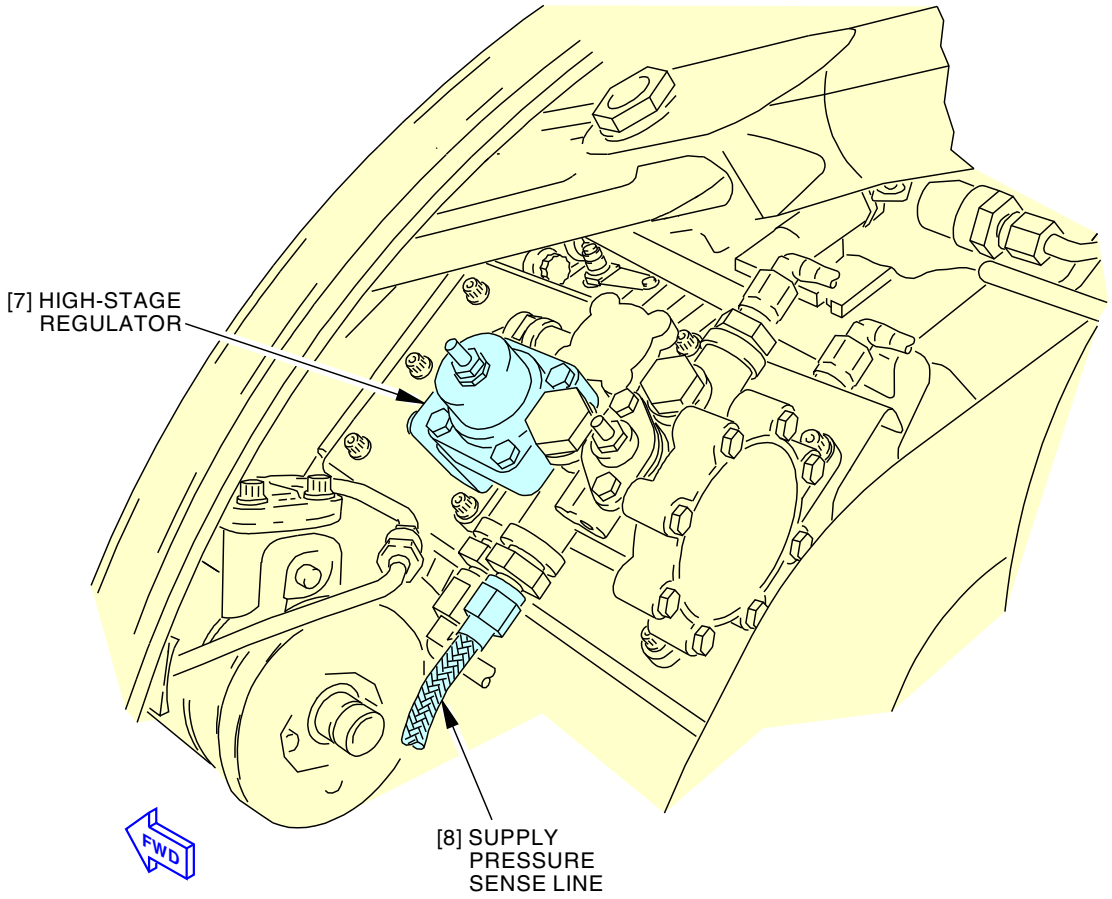
ECCN 9E991 BOEING PROPRIETARY - See title page for details



U71900 S0000214520_V2

Engine Bleed Air System Leakage Check
Figure 306/21-00-00-990-815 (Sheet 3 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

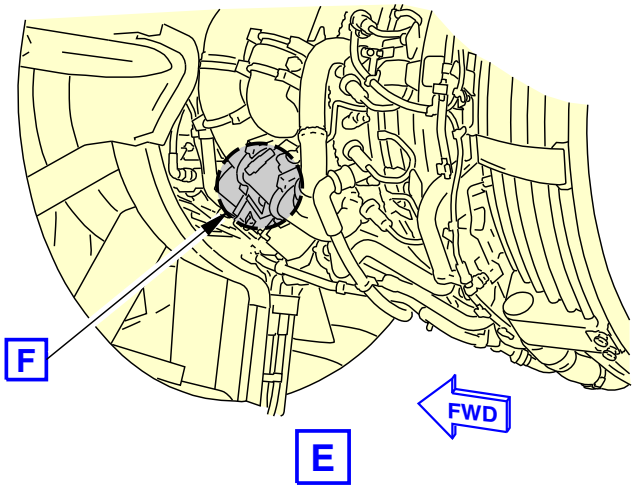


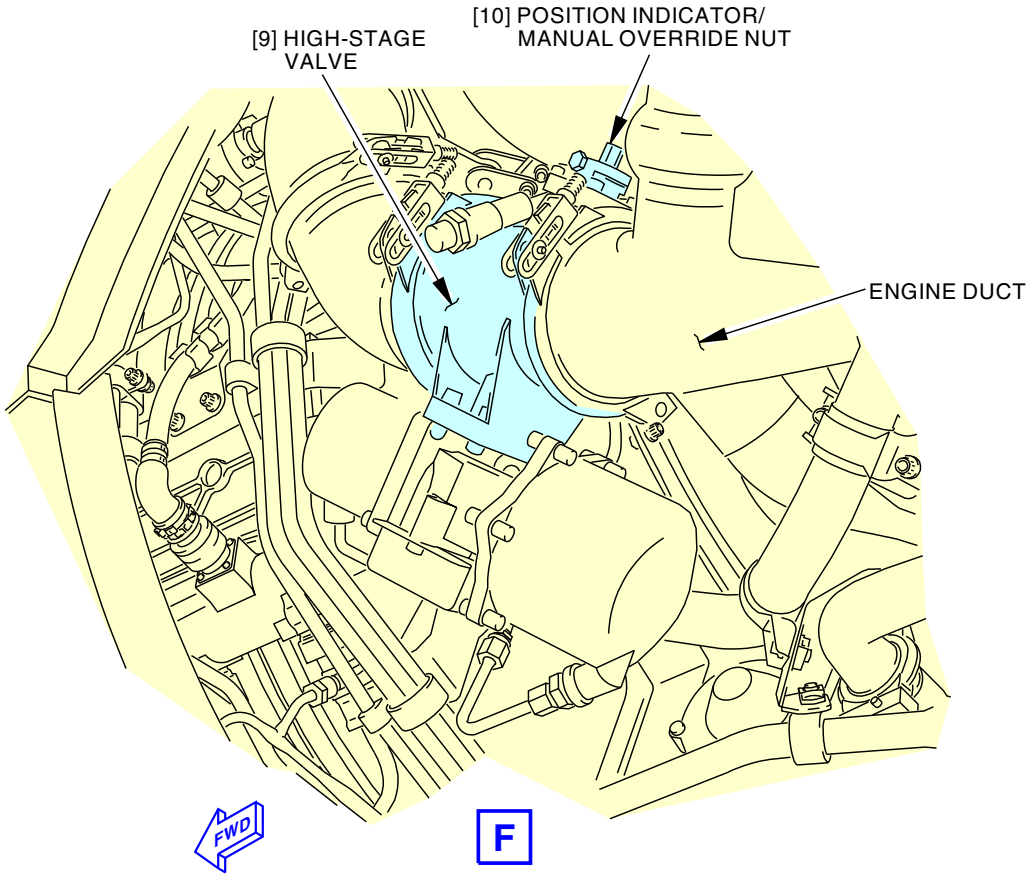
D

U71901 S0000214521_V3

Engine Bleed Air System Leakage Check
Figure 306/21-00-00-990-815 (Sheet 4 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

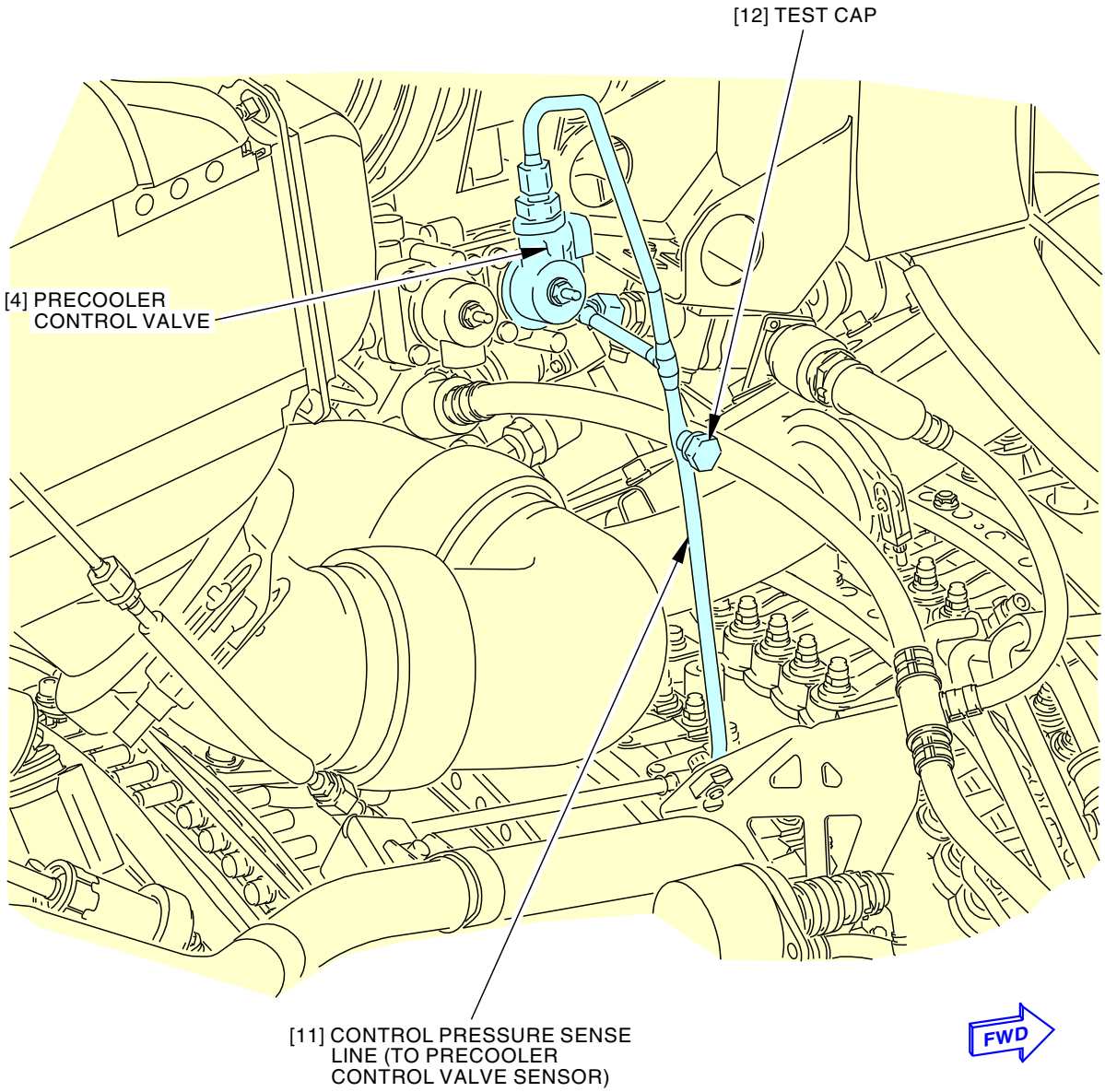




U71902 S0000214522_V2

Engine Bleed Air System Leakage Check
Figure 306/21-00-00-990-815 (Sheet 5 of 6)

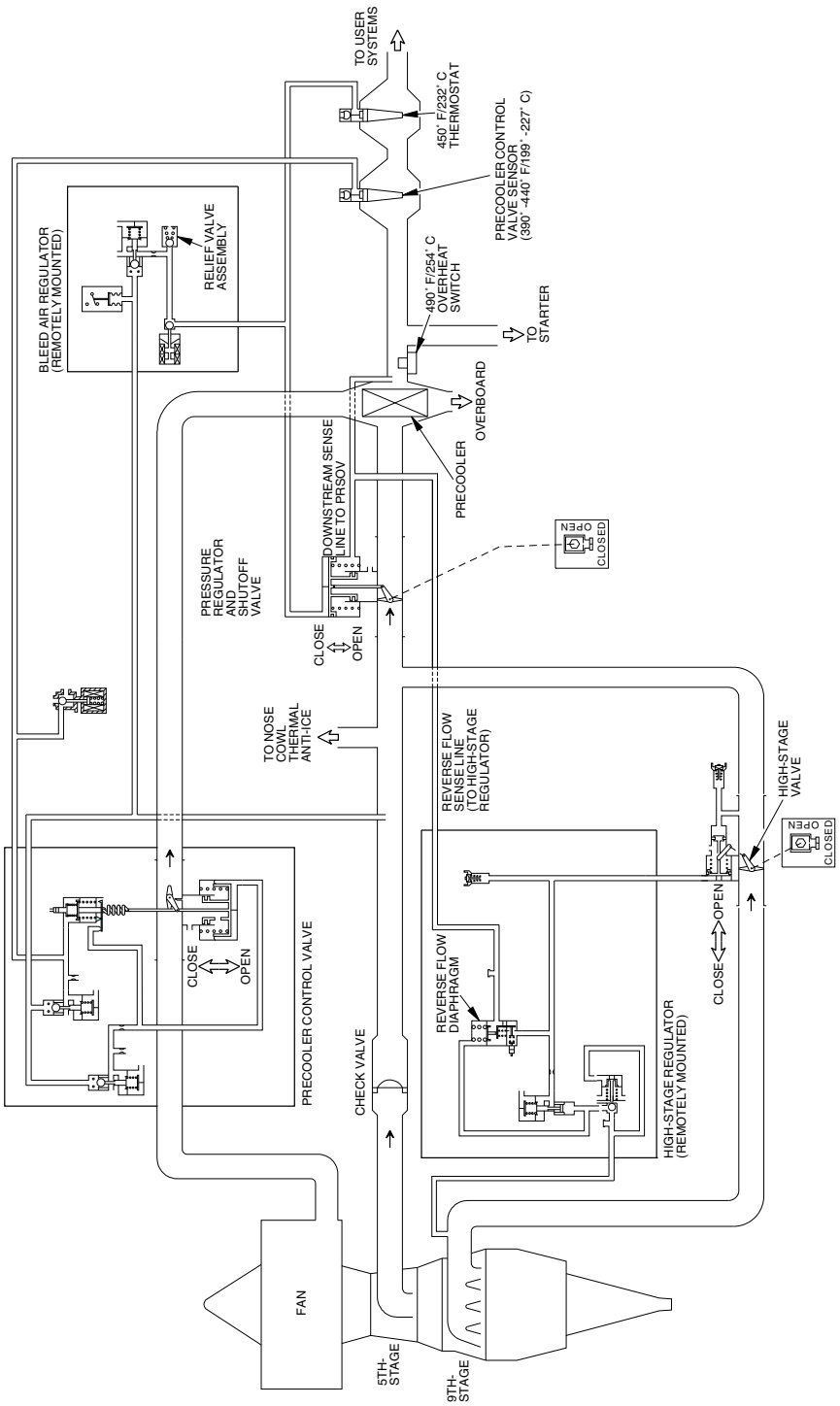
ECCN 9E991 BOEING PROPRIETARY - See title page for details



1337969 S0000236688_V3

Engine Bleed Air System Leakage Check
Figure 306/21-00-00-990-815 (Sheet 6 of 6)

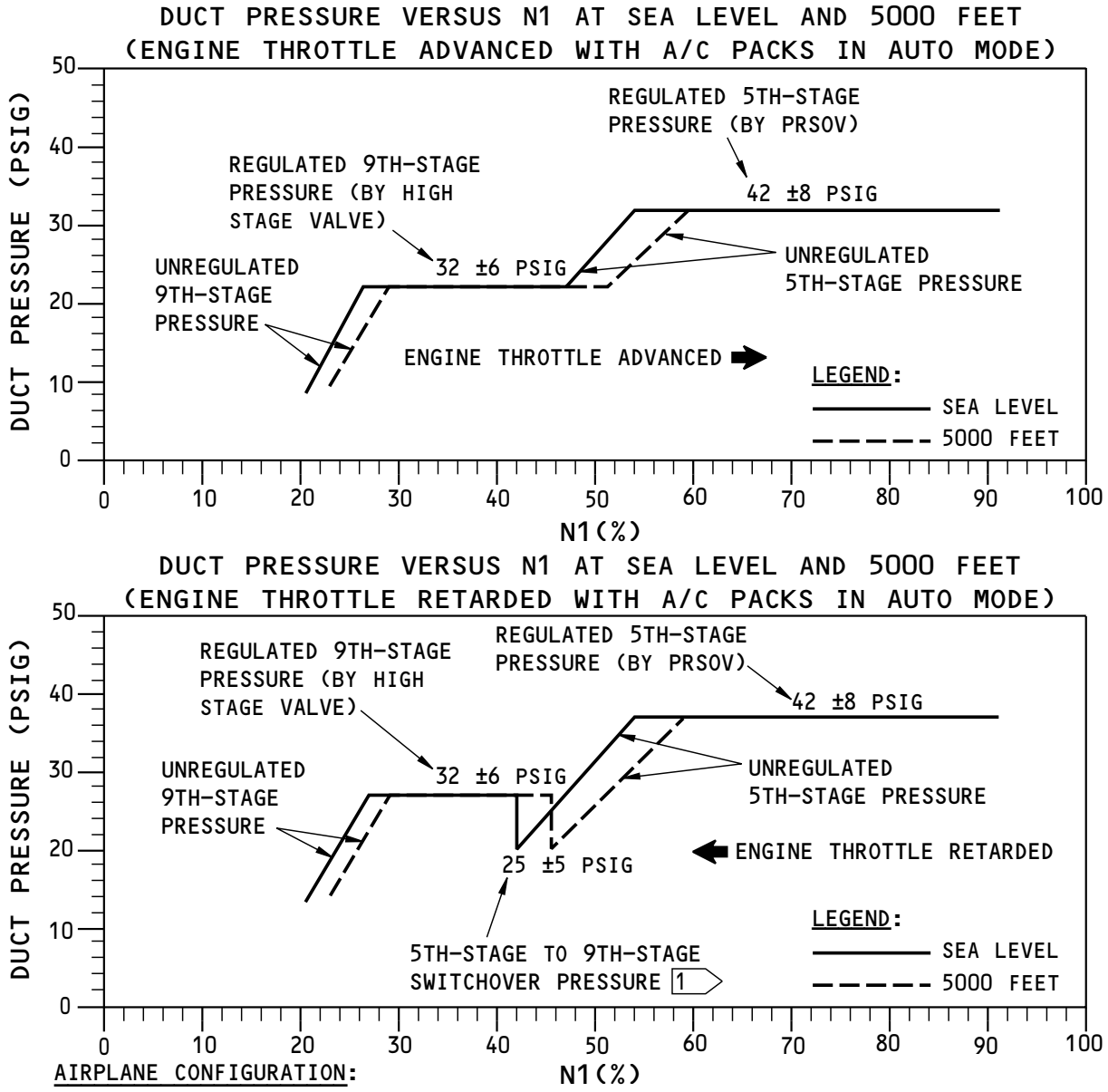
ECCN 9E991 BOEING PROPRIETARY - See title page for details



U71199 S0000214494_V2

Engine Bleed Air Distribution System Control Schematic
Figure 307/21-00-00-990-816

ECCN 9E991 BOEING PROPRIETARY - See title page for details



AIRPLANE CONFIGURATION:
ASSOCIATED PACK: AUTO
ASSOCIATED BLEED: ON
ASSOCIATED CTAI: OFF
ISOLATION VALVE: CLOSED
WTAI: OFF

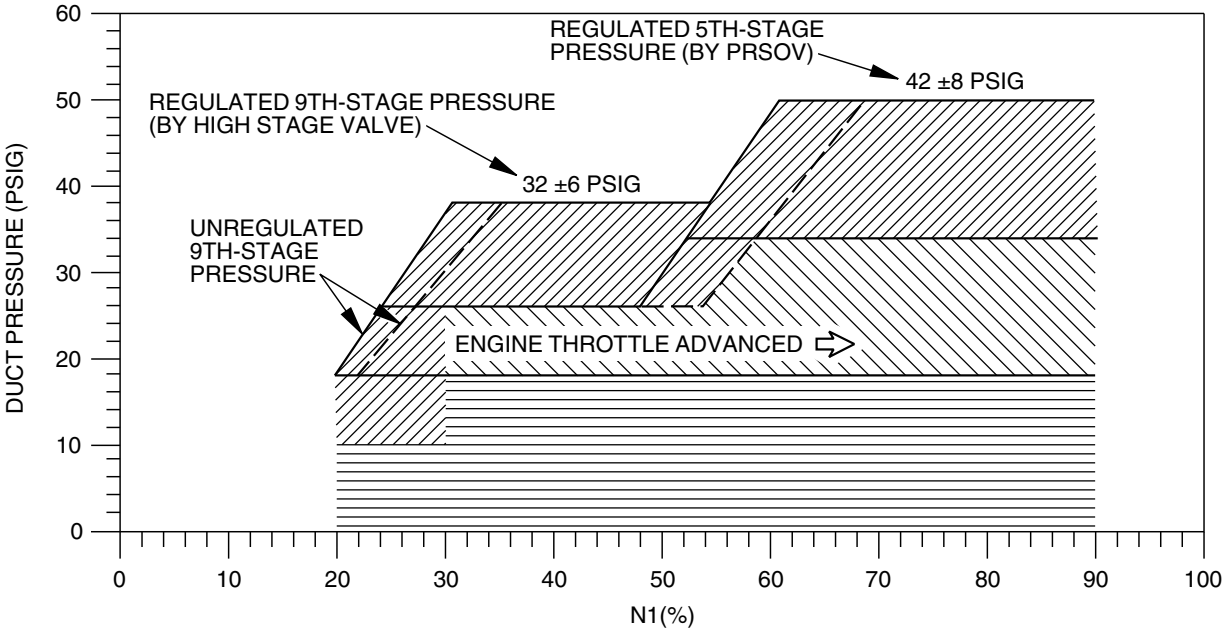
1 WHEN THE ENGINE THROTTLE IS RETARDED AND THE ENGINE BLEED SYSTEM SWITCHOVER OCCUR FROM 5TH-STAGE PRESSURE TO 9TH-STAGE PRESSURE, DUCT PRESSURE CAN DECAY TO AS LOW AS 20 PSIG BEFORE THE HIGH STAGE VALVE OPENS AND REGULATES THE DUCT PRESSURE TO NOMINAL 32 PSIG.

G50747 S0006744860_V2

Duct Pressure Versus N1 at Sea Level and 5000 Feet
Figure 308/21-00-00-990-817

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MINIMUM SERVICEABLE PNEUMATIC DUCT PRESSURE



AIRPLANE CONFIGURATION:
ASSOCIATED PACK: AUTO
WING ANTI-ICE: OFF
COWL ANTI-ICE: OFF

NOTE: MINIMUM POWER IN FLIGHT IS GREATER THAN MINIMUM POWER ON THE GROUND.

LEGEND:

- SEA LEVEL
- 5000 FEET
- BLEED SYSTEM OPERATING NORMALLY.
- BLEED SYSTEM PERFORMANCE DRIFTING. AIRPLANE CAN BE OPERATED NORMALLY BUT ACTION TO RESTORE BLEED SYSTEM TO OPTIMUM OPERATION SHOULD BE TAKEN AT A CONVENIENT OPPORTUNITY.
- BLEED SYSTEM INOPERATIVE. RESTORE TO NORMAL OPERATION BEFORE FLIGHT OR IMPOSE THE MEL RESTRICTION FOR CONTINUED OPERATION.

1418175 S0000255925_V5

Minimum Serviceable Pneumatic Duct Pressure
Figure 309/21-00-00-990-818

ECCN 9E991 BOEING PROPRIETARY - See title page for details

801. Windshield or Foot Air Outlet Does Not Operate - Fault Isolation

A. Description

(1) The captain's or the first officer's windshield or foot air outlet has no air flow or less air flow than expected.

B. Possible Causes

(1) Windshield and foot air outlet valve

(2) The cables for the windshield and foot air outlet valve

(3) The air distribution ducts for the flight compartment

(4) Defective muffler/duct assembly

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(5) Left Standby Temperature Control Valve, V140

(6) Right Standby Temperature Control Valve, V142

OMR ALL

C. Related Data

(1) SDS SUBJECT 21-22-00

D. Initial Evaluation

(1) Supply conditioned air to the flight compartment with the left air conditioning pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) Make sure that the left air conditioning pack is on.

(3) Pull the applicable knob on the P1 panel (captain's instrument panel) or the P3 panel (first officer's instrument panel) for the applicable outlet (FOOT AIR or WINDSHIELD AIR).

(4) If the air flow from the applicable air outlet is satisfactory, then there was an intermittent fault.

(5) If the air flow from the applicable air outlet is not satisfactory, then do the Fault Isolation Procedure below.

E. Fault Isolation Procedure

(1) Do these steps to do a check of the other air distribution outlets in the flight compartment:

(a) Examine all of the other air distribution outlets in the flight compartment for satisfactory airflow:

(b) If you find that all of the other air distribution outlets in the flight compartment do not have satisfactory air flow, then do this task: Low Conditioned Air Flow to the Flight Deck - Fault Isolation, 21-22 TASK 802.

(c) If all of the other air distribution outlets in the flight compartment have satisfactory air flow, then continue.

(2) Do these steps to do a check of the cables for the applicable windshield and foot air outlet valve:

(a) To get access to the captain's windshield and foot air outlet valve and cables, remove the display units on the captain's instrument panel (P1). This is the task: Display Unit Removal, AMM TASK 31-62-11-000-801.

(b) To get access to the first officer's windshield and foot air outlet valve and cables, remove the display units on the first officer's instrument panel (P3). This is the task: Display Unit Removal, AMM TASK 31-62-11-000-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Do a check to see if the cables for the applicable air outlet are connected and correctly adjusted. This is the task: Air Outlet Valve Control Cables Adjustment and Test, AMM TASK 21-22-02-820-801.

1) If there is a problem with the cables, then do these steps:

a) If it is necessary, adjust the cables. This is the task: Air Outlet Valve Control Cables Adjustment and Test, AMM TASK 21-22-02-820-801.

b) If it is necessary, replace the cables. These are the tasks:

- Air Outlet Valve Control Cables Removal, AMM TASK 21-22-02-000-801
- Air Outlet Valve Control Cables Installation, AMM TASK 21-22-02-400-801

c) Do the Repair Confirmation at the end of this task.

2) If the cables for the applicable air outlet are connected and correctly adjusted, then continue.

(3) Do these steps to do a check of the applicable windshield and foot air outlet valve:

(a) Remove the applicable windshield and foot air outlet valve. This is the task: Windshield and Foot Air Outlet Valve Removal, AMM TASK 21-22-01-000-801.

(b) Do a check to see if the air flow from the open air distribution duct where you removed the valve is satisfactory.

1) If the air flow from the duct is satisfactory, then do these steps:

a) Replace the windshield and foot air outlet valve with a new valve. This is the task: Windshield and Foot Air Outlet Installation, AMM TASK 21-22-01-400-801.

b) Do the Repair Confirmation at the end of this task.

2) If the air flow from the duct is not satisfactory, then continue.

(4) Examine the air distribution ducts for the applicable outlet.

(a) Look for leakage or blockage in the air distribution ducts between the connection to the valve and applicable air outlet.

(b) Repair the air distribution ducts or remove the blockage.

(c) Install the applicable windshield and foot air outlet valve. This is the task: Air Outlet Valve Control Cables Installation, AMM TASK 21-22-02-400-801.

(d) Do the Repair Confirmation at the end of this task.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(5) Replace the Left Standby Temperature Control Valve, V140. These are the tasks:

- Standby Temperature Control Valve Removal, AMM TASK 21-51-11-000-801
- Standby Temperature Control Valve Installation, AMM TASK 21-51-11-400-801

(a) Do the Repair Confirmation at the end of this task.

(6) Replace the Right Standby Temperature Control Valve, V142. These are the tasks:

- Standby Temperature Control Valve Removal, AMM TASK 21-51-11-000-801
- Standby Temperature Control Valve Installation, AMM TASK 21-51-11-400-801

(a) Do the Repair Confirmation at the end of this task.

OMR ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Repair Confirmation

(1) Do these steps to make sure you corrected the problem:

(a) If conditioned air is not supplied to the flight compartment, do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(b) Make sure the left air conditioning pack is on.

(c) Pull the applicable FOOT AIR or WINDSHIELD AIR knob on the P1 or P3 panel to open to applicable valve.

(d) If the air flow at the applicable outlet is satisfactory, then you corrected the problem.

(e) If the air flow at the applicable outlet is not satisfactory, then continue the Fault Isolation Procedure at the subsequent step.

(2) Do these steps to put the airplane back to its usual condition:

(a) Make sure the applicable knobs on the P1 and P3 panel, WINDSHIELD AIR and FOOT AIR are pushed in to the closed position.

(b) Do this task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

——— END OF TASK ———

802. Low Conditioned Air Flow to the Flight Deck - Fault Isolation

A. Description

(1) The flow of conditioned air from the flight deck conditioned air outlets is low and seems to be decreasing with continued use.

B. Possible Causes

(1) Disconnected, crushed or leaking flight deck distribution duct

(2) Defective muffler/duct assembly - collapsed lining material in the muffler

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(3) Duct assembly with blocked wire mesh screen

NOTE: This duct assembly is roughly shaped like a “Y”.

(4) Left Standby Temperature Control Valve, V140

(5) Right Standby Temperature Control Valve, V142

OMR ALL

C. Related Data

(1) SDS SUBJECT 21-22-00

D. Initial Evaluation

(1) Supply conditioned air to the flight compartment with the left air conditioning pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(a) Make sure that the L PACK switch on the P5 air conditioning module is set to ON.

(2) Open or operate the conditioned air outlets in the flight compartment.

(3) If the air flow from the air outlets appears to be satisfactory, then there was an intermittent fault.

(4) If the conditioned air flow from all the air outlets listed below appears to be unsatisfactory, then do the Fault Isolation Procedure.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Overhead outlets and gasper

(b) Underseat diffusers

(c) Foot air diffusers

(d) Windshield air diffusers

(e) Sidewall outlets (shoulder warmers)

(f) Individual panel gaspers.

E. Fault Isolation Procedure

(1) Make sure that conditioned air is supplied to the flight compartment.

(a) Examine the left flow control and shutoff valve to make sure that it is open.

(2) Do these steps to do a check of the flight compartment air distribution ducts:

(a) Do a check of the air distribution ducts between the mix manifold outlet and the flight compartment outlets for air leakage, an icing in the ducting, or blockage.

NOTE: The air distribution ducts between the mix manifold outlet and the flight compartment outlets are located on the left side of the forward cargo compartment and the Electronics/Equipment bay.

(b) If you find air leakage, damaged ducts, icing in the ducting or blocked ducts, then do these steps:

1) Repair the leaking or damaged/blocked duct.

a) If you determine that the blockage is at the muffler/duct assembly in the mix bay, then replace the muffler/duct assembly. These are the tasks:

- Flight Deck Air Distribution System Muffler/Duct Assembly Removal, AMM TASK 21-22-05-000-801
- Flight Deck Air Distribution System Muffler/Duct Assembly Installation, AMM TASK 21-22-05-400-801

2) Do the Repair Confirmation at the end of this task.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(3) Clean the flight deck duct assembly with the screen. This is the task: Flight Deck Duct Assembly Screen Cleaning, AMM TASK 21-22-07-100-801.

NOTE: Muffler debris can build up on the wire mesh screen located inside of the “Y” duct assembly and reduce the airflow.

(a) Do the Repair Confirmation at the end of this task.

(4) Replace the Left Standby Temperature Control Valve, V140. These are the tasks:

- Standby Temperature Control Valve Removal, AMM TASK 21-51-11-000-801
- Standby Temperature Control Valve Installation, AMM TASK 21-51-11-400-801

(a) Do the Repair Confirmation at the end of this task.

(5) Replace the Right Standby Temperature Control Valve, V142. These are the tasks:

- Standby Temperature Control Valve Removal, AMM TASK 21-51-11-000-801
- Standby Temperature Control Valve Installation, AMM TASK 21-51-11-400-801

(a) Do the Repair Confirmation at the end of this task.

OMR ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Repair Confirmation

(1) Do these steps to make sure you corrected the problem.

(a) If conditioned air is not supplied to the flight compartment, do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(b) Make sure that the L PACK switch on the P5-10 air conditioning panel is set to ON.

(c) Make sure there is satisfactory airflow from these outlets:

1) Overhead outlets and gasper

2) Underseat diffusers

3) Foot air diffusers

4) Windshield air diffusers

5) Sidewall outlets (shoulder warmers)

6) Individual panel gaspers.

(d) If the conditioned air flow from the outlets is satisfactory, then you corrected the problem.

(e) If the conditioned air flow from the outlets is not satisfactory, then continue the Fault Isolation Procedure at the subsequent step.

(2) Do these steps to put the airplane back to its usual condition:

(a) Do this task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

——— END OF TASK ———

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801. Main Deck Air Distribution System Shutoff Valves Do Not Operate - Fault Isolation

A. Description

(1) This task is for these Observed Faults:

(a) The Shutoff Valve that supplies conditioned air to the Main Deck Forward Cargo area does not shut OFF when commanded.

(b) The Shutoff Valve that supplies conditioned air to the Main Deck Aft Cargo area does not shut OFF when commanded.

(2) For a detailed description of the Main Deck Air Distribution System refer to SDS SUBJECT 21-23-00.

B. Possible Causes

(1) Main Deck FWD (AFT) Shutoff Valve, V165 (V166)

(2) Main Deck FWD (AFT) Shutoff Valve Control Relay, R1212 (R1211)

(3) Wiring

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-11

Row	Col	Number	Name
C	1	C01705	CARGO FIRE VLV POWER
C	2	C01569	CARGO FIRE MAIN SYS EXT
C	3	C01570	CARGO FIRE MAIN SYS SOV

D. Related Data

(1) WDM 21-23-11

(2) WDM 26-23-21

(3) SSM 21-23-11

(4) SSM 26-23-21

E. Initial Evaluation

(1) Do this task: Operational Test of Shutoff Valves for the Main Deck Air Distribution System, AMM TASK 21-23-05-710-802.

(a) If the Operational Test is satisfactory, then there was an intermittent fault.

(b) If the Operational Test is not satisfactory for one of the valves, then do the “Fault Isolation Procedure - One Shutoff Valve Does Not Operate”.

(c) If the Operational Test is not satisfactory for the FWD and AFT Valves, then do the “Fault Isolation Procedure - FWD and AFT Shutoff Valves Do Not Operate”.

F. Fault Isolation Procedure - One Shutoff Valve Does Not Operate

(1) Do this voltage check at the applicable FWD (AFT) Shutoff Valve (WDM 21-23-11):

(a) Remove the applicable Aft Bulkhead Liner(s) in the Forward Cargo Compartment. This is the task: Forward Cargo Compartment Aft Bulkhead Liner - Removal, AMM TASK 25-52-17-000-801.

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(b) Disconnect connector D13234 (D13236) from the Main Deck FWD (AFT) Shutoff Valve, V165 (V166), as applicable.

NOTE: The data in parenthesis applies to the Aft Shutoff Valve. Use the data for the valve that does not operate.

(c) On the Cargo Smoke Detection and Fire Suppression Module, push the DEPR Switch

(d) Do a check for 28V DC between pins 3 and 4 of connector D13234 (D13236).

(e) If you find 115V AC between the pins, then do these steps:

1) Replace the applicable FWD (AFT) Shutoff Valve, V165 (V166). These are the tasks

- Main Deck Air Distribution System Shutoff Valve - Removal, AMM TASK 21-23-05-000-802
- Main Deck Air Distribution Shutoff Valve - Installation, AMM TASK 21-23-05-400-802

2) Do the Repair Confirmation at the end of this task.

(f) If you do not find 115V AC between the pins, then continue.

(2) Do this check of the FWD (AFT) Shutoff Valve Ground Wiring (WDM 21-23-11):

FWD VLV, V165
D13234
pin 4 . . . . . GD650-AC

AFT VLV, V166
D13236
pin 4 . . . . . GD678-AC

(a) If you find a problem with the wiring, then do these steps:

1) Repair the problems that you find.

2) Re-connect the applicable connector D13234 (D13236).

3) Close this circuit breaker:

F/O Electrical System Panel, P6-11

Row	Col	Number	Name
C	3	C01570	CARGO FIRE MAIN SYS SOV

4) Do the Repair Confirmation at the end of this task.

(b) If you do not find a problem with the wiring, then do this step and continue:

1) Re-connect the applicable connector D13234 (D13236).

(3) Do a check for 28V DC to the applicable Main Deck FWD (AFT) Shutoff Valve Control Relay, R1212 (R1211) (WDM 21-23-11):

(a) In the Equipment Bay, get access to the E5-2 Shelf.

(b) Remove the applicable FWD (AFT) Shutoff Valve Control Relay, R1212 (R1211).

(c) Do a check for 28V DC between pins Y1 and Y2 of connector D16766 (D16764).

1) If you find 28V DC between pins Y1 and Y2, then do these steps:

a) Examine the wiring between the applicable FWD (AFT) SOV Control Relay, R1212 (R1211) and FWD (AFT) Shutoff Valve, V165 (V166) as follows:

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MAIN DECK FWD SUTOFF VALVE

RLY R1212	**V165**
D16766	**D13234**
pin A3	pin 3

MAIN DECK AFT SHUTOFF VALVE

RLY R881	**V166**
D16764	**D13236**
pin A3	pin 3

<1> If the wiring is OK, replace the applicable FWD (AFT) SOV Control Relay, R1212 (R1211).

<a> Do the Repair Confirmation at the end of this task.

<2> If you find problems in the wiring, do these steps:

<a> Repair the problems that you find.

<b> Re-install the applicable FWD (AFT) SOV Control Relay, R1212 (R1211).

<c> Do the Repair Confirmation at the end of this task.

2) If you do not find 28V DC between pins Y1 and Y2, then you must fault isolate the Fire Suppression System.

(4) Re-install the applicable Aft Bulkhead Liner in the Forward Cargo Compartment. This is the task: Forward Cargo Compartment Aft Bulkhead Liner - Installation, AMM TASK 25-52-17-400-801.

G. Fault Isolation Procedure - FWD and AFT Shutoff Valves Do Not Operate

(1) In the Equipment Bay, get access to Terminal Board TB530 in the E5-2 Shelf.

(2) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-11

Row	Col	Number	Name
C	3	C01570	CARGO FIRE MAIN SYS SOV

(3) Examine the wiring between the Load Control Center - Right (LCC-R) and Terminal Block TB530 in the E5-2 Shelf as follows (WDM 21-23-11):

LCC-R	**TERM BLOCK**
D40800P	**TB530**
pin 20	term YA3

(a) If you find a problem in the wiring, do these steps:

1) Repair the problems that you find.

2) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-11

Row	Col	Number	Name
C	3	C01570	CARGO FIRE MAIN SYS SOV

3) Do the Repair Confirmation at the end of this task.

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(b) If the wiring is OK, you must fault isolate the Fire Suppression System.

H. Repair Confirmation

(1) Do this task: Operational Test of Shutoff Valves for the Main Deck Air Distribution System, AMM TASK 21-23-05-710-802.

(a) If the Operational Test is satisfactory, then you corrected the problem.

(b) If the Operational Test is not satisfactory, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

803. Main Deck Overboard Exhaust Valve Does Not Operate - Fault Isolation

A. Description

(1) This task is for the overboard exhaust valve that does not open when commanded.

(2) SDS 21-23-00

B. Possible Causes

(1) Mix Manifold Dump Valve (Overboard Exhaust Valve), V198

(2) Main Deck Fwd SOV Cont Relay, R1212

(3) Wiring

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-11

Row	Col	Number	Name
C	1	C01705	CARGO FIRE VLV POWER
C	2	C01569	CARGO FIRE MAIN SYS EXT
C	3	C01570	CARGO FIRE MAIN SYS SOV

D. Related Data

(1) SSM 26-23-21

(2) SSM 21-23-11

(3) WDM 26-23-21

(4) WDM 21-23-11

E. Initial Evaluation

(1) Do this task: Operational Test of Shutoff Valves for the Main Deck Air Distribution System, AMM TASK 21-23-05-710-802.

(a) If the operational test is not satisfactory for the valve, then do the Fault Isolation Procedure below.

(b) If the operational test is satisfactory for the valve, then there was an intermittent fault.

F. Fault Isolation Procedure

(1) Do this check for 28V DC at the overboard exhaust valve:

(a) Remove the center aft bulkhead liner(s) in the forward cargo compartment. To do this, do this task: Forward Cargo Compartment Aft Bulkhead Liner - Removal, AMM TASK 25-52-17-000-801

(b) Disconnect connector D17533 from the overboard exhaust valve, V198.

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CAUTION DISCONNECT ALL ELECTRICAL CONNECTORS FROM THE FIRE EXTINGUISHER BOTTLE BEFORE YOU PUSH THE DISCHARGE SWITCH. IF CONNECTED, THE BOTTLE WILL DISCHARGE.

(c) If the electrical connectors are not disconnected from the fire extinguisher bottles, disconnect the electrical connectors per applicable steps of the procedure that follows:

1) Do this task: Cargo Fire Extinguisher Bottle Removal, AMM TASK 26-23-01-000-801-001.

(d) Press the DEPR/DISCH guarded switch on the P8-76 cargo fire control panel.

NOTE: The main deck system must be armed per the adjustment/test procedure for the valve.

(e) Do a check for 28V DC between pins 5 and 4 of connector D17533.

(f) If there is 28V DC between pins 5 and 4, then do these steps:

1) Replace the overboard exhaust valve (V198). These are the tasks:

- Main Deck Air Distribution System Shutoff Valve - Removal, AMM TASK 21-23-05-000-802
- Main Deck Air Distribution Shutoff Valve - Installation, AMM TASK 21-23-05-400-802

2) Do this task: Operational Test of Shutoff Valves for the Main Deck Air Distribution System, AMM TASK 21-23-05-710-802.

3) If the operational test is satisfactory, then you corrected the fault.

(g) If there is not 28V DC between pins 5 and 4, then continue.

(2) Do this check of the ground circuit wiring for the overboard exhaust valve:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-11

Row	Col	Number	Name
C	1	C01705	CARGO FIRE VLV POWER

(b) Do a check of the wiring between connector D17533, pin 4 and the ground GD3254-AC.

Valve Connector D17533		Ground GD3254-AC
pin 4	. . . . .	gnd

(c) If you find a problem with the wiring, then do these steps:

1) Repair any problems that you find.

2) Re-connect connector D17533.

3) Close this circuit breaker:

F/O Electrical System Panel, P6-11

Row	Col	Number	Name
C	1	C01705	CARGO FIRE VLV POWER

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4) Do this task: Operational Test of Shutoff Valves for the Main Deck Air Distribution System, AMM TASK 21-23-05-710-802.

5) If the operational test is satisfactory, then you corrected the fault.

(d) If you do not find a problem with the wiring, then do this step and continue:

1) Re-connect connector D17533.

(3) Do a check for 28V DC to main deck fwd shutoff valve cont relay, R1212:

(a) Get access to the E5-2 shelf in the equipment bay.

(b) Remove the shutoff valve control relay, R1212.

(c) Do a check for 28V DC between pins Y1 and Y2 of connector D16766.

(d) If there is 28V DC between pins Y1 and Y2, then do these steps:

1) Do a continuity check between connector D16766, pin B3 and connector D17533, pin 5.

RELAY CONNECTOR D16766		VALVE CONNECTOR D17533
pin B3	. . . . .	pin 5

2) If there is continuity between D16766, pin B3 and connector D17533, pin 5, do these steps:

a) Replace the shutoff valve relay, R1212.

b) Do this task: Operational Test of Shutoff Valves for the Main Deck Air Distribution System, AMM TASK 21-23-05-710-802.

c) If the operational test was satisfactory, then you corrected the fault.

3) If there is not continuity between D16766, pin B3 and connector D17533, pin 5, do these steps:

a) Repair any problem that you find.

b) Re-install the shutoff valve control relay R1212.

c) Do this task: Operational Test of Shutoff Valves for the Main Deck Air Distribution System, AMM TASK 21-23-05-710-802.

d) If the operational test was satisfactory, then you corrected the fault.

(e) If there is not 28V DC between pins Y1 and Y2, then you must fault isolate the fire suppression system.

(4) Re-install the center aft bulkhead liner in the forward cargo compartment. To do this, do this task: Forward Cargo Compartment Aft Bulkhead Liner - Installation, AMM TASK 25-52-17-400-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799

801. Recirculation Fan Does Not Operate - Fault Isolation

A. Description

(1) You set the RECIRC FAN switch on the air conditioning panel to AUTO and the recirculation fan does not come on as shown in the table below.

Table 201

L PACK Switch Position	R PACK Switch Position	Fan Condition
OFF	OFF	ON
OFF	AUTO	ON
OFF	HIGH	ON
AUTO	OFF	ON
AUTO	AUTO	ON
AUTO	HIGH	OFF
HIGH	OFF	ON
HIGH	AUTO	OFF
HIGH	HIGH	OFF

B. Possible Causes

(1) Recirculation fan, M1148

(2) Cabin air recirculation fan relay, R331

(3) Air conditioning accessory unit, M324

(4) Switch for the recirculation fan, S3

(5) Wiring

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
E	1	C01015	AIR CONDITIONING RECIRC FAN CONT
E	4	C00884	AC RECIRC FAN RIGHT CABIN AIR

D. Related Data

(1) (SSM 21-25-11)

(2) (WDM 21-25-11)

E. Initial Evaluation

(1) AIRPLANES WITH CAB/UTIL SWITCH ON THE P5-13 PANEL;

Make sure that the CAB/UTIL switch on the P5-13 panel is set to the ON position.

(2) Do this task: Recirculation System - Operational Test, AMM TASK 21-25-00-710-801-001.

(3) If the operational test shows that the recirculation fan does not operate with any setting of the L PACK switch or the R PACK switch, then do the Fault Isolation Procedure - Recirculation Fan Does Not Come On At Any Setting of L PACK and R PACK Switches.

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OMR 701-799 (Continued)

(4) If the operational test shows that the recirculation fan operates but does not come on as specified in the operational test, then do the Fault Isolation Procedure - Recirculation Fan Does Not Come On as Specified With the L PACK and R PACK Switches.

(5) If the operational test does not show a problem, then there was an intermittent fault.

F. Fault Isolation Procedure - Recirculation Fan Does Not Come On At Any Setting of L PACK and R PACK Switches

NOTE: You must do the steps in the Initial Evaluation before you do these steps.

NOTE: This procedure isolates the fault to one of these possible causes:

- recirculation fan, M1148
- recirculation fan relay, R331
- recirculation fan switch, S3
- wiring
- air conditioning accessory unit, M324

(1) Do this check of the recirculation fan, M1148:

(a) Make sure the RECIRC FAN switch on the P5-10 air conditioning panel is in the AUTO position.

(b) Disconnect connector D3020 from the recirculation fan, M1148.

(c) Do a continuity check between these pins of the recirculation fan, M1148:

1) Pin 5 and pin 6
2) Pin 2 and pin 3
3) Pin 3 and pin 4
4) Pin 2 and pin 4.

(d) If there is an open circuit between any pair of pins, then do these steps:

1) Replace the recirculation fan, M1148.

These are the tasks:

Recirculation Fan Removal, AMM TASK 21-25-02-000-801,

Recirculation Fan Installation, AMM TASK 21-25-02-400-801.

2) If you did not do it as part of the replacement procedure, then, do this task: Recirculation System - Operational Test, AMM TASK 21-25-00-710-801-001

3) If the operational test is satisfactory, then you corrected the fault.

(e) If there is continuity between each pair of pins, then continue.

(2) Do these checks of the recirculation fan circuit:

(a) Do a check for an open circuit between pin 5 of connector D3020 and structure ground.

(b) If there is an open circuit, then do these steps:

1) Repair the wiring between pin 5 of D3020 and structure ground.

2) Do this task: Recirculation System - Operational Test, AMM TASK 21-25-00-710-801-001.

3) If the operational test is satisfactory, then you corrected the fault.

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OMR 701-799 (Continued)

(c) If there is continuity between pin 5 of connector D3020 and structure ground, then continue.

(d) Do these checks for the correct voltage at connector D3020:

1) Do a check for 28 VDC between pins 6 and 5 (ground) of connector D3020.

2) Install a jumper wire between pins 5 and 6 of connector D3020.

3) Do a check for 3-phase 115 VAC from pins 2, 3, and 4 of connector D3020 and structure ground.

4) If there is not 3-phase 115 VAC at pins 2, 3, 4, or 28 VDC at pin 6 of connector D3020, then do these steps:

a) Replace the recirculation fan relay, R331.

b) Do this task: Recirculation System - Operational Test, AMM TASK 21-25-00-710-801-001.

c) If the operational test is satisfactory, then you corrected the fault.

d) If the operational test is not satisfactory, then continue.

e) Repair the wiring between these pins of connector D3020 and the P6-4 circuit breaker (P6-4E4 C884):

D3020		C884
pin 2	. . . . .	term A1
pin 3	. . . . .	term B1
pin 4	. . . . .	term C1

f) Do this task: Recirculation System - Operational Test, AMM TASK 21-25-00-710-801-001.

g) If the operational test is satisfactory, then you corrected the fault.

5) If there is 3-phase 115 VAC at pins 2, 3, 4 and 28 VDC at pin 6, then continue.

(3) Replace the RECIRC FAN switch, S3 on the air conditioning panel.

(a) Do this task: Recirculation System - Operational Test, AMM TASK 21-25-00-710-801-001.

(b) If the operational test is satisfactory, then you corrected the fault.

(c) If the operational test is not satisfactory, then continue.

(4) Replace the air conditioning accessory unit, M324.

These are the tasks:

Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801,

Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

(a) If you did not do it after installation, then, do this task: Air Conditioning Accessory Unit - Operational Test, AMM TASK 21-51-02-710-801-001.

(b) If the operational test of the air conditioning accessory unit, M324 is satisfactory, then you corrected the fault.

(c) If the operational test of the air conditioning accessory unit, M324 is not satisfactory, then continue.

(5) Do this check of the wiring:

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OMR 701-799 (Continued)

(a) Do a check for an open circuit between these pins of connector D458A on the air conditioning accessory unit and the circuit breaker panel (P6-4E1, C1015):

D458A		C1015
pin 1	. . . . .	term L

(b) Do a check for an open circuit between these pins of connector D458B on the air conditioning accessory unit and the circuit breaker panel (P6-4E1, C1015):

D458B		C1015
pin 2	. . . . .	term L

(c) Do a check for an open circuit between these pins on connector D3020 for the recirculation fan and connector D458B at the air conditioning relay module, M324:

D3020		D458B
pin 6	. . . . .	pin 1
pin 6	. . . . .	pin 2

(d) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.
2) Do this task: Recirculation System - Operational Test, AMM TASK 21-25-00-710-801-001.
3) If the operational test is satisfactory, then you corrected the fault.

G. Fault Isolation Procedure - Recirculation Fan Does Not Come On As Specified With the L PACK and R PACK Switches

NOTE: You must do the steps in the Initial Evaluation before you do these steps.

NOTE: This procedure isolates the fault to this cause:

- air conditioning accessory unit, M324

(1) Do these steps to replace the air conditioning accessory unit, M324 and make sure you corrected the fault:

(a) Replace the air conditioning accessory unit, M324.

These are the tasks:

Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801,

Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

(b) If you did not do it after installation, do this task: Air Conditioning Accessory Unit - Operational Test, AMM TASK 21-51-02-710-801-001

(c) If the operational test of the air conditioning accessory unit, M324 is satisfactory, then you corrected the fault.

——— END OF TASK ———

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OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

802. Left or Right Recirculation Fan Does Not Operate - Fault Isolation

A. Description

(1) You set the L and R RECIRC FAN switches on the air conditioning panel to AUTO and the recirculation fans do not come on as shown in the table below.

Table 202

L PACK Switch Position	R PACK Switch Position	L Fan Condition	R Fan Condition
OFF	OFF	ON	ON
OFF	AUTO	ON	ON
OFF	HIGH	ON	ON
AUTO	OFF	ON	ON
AUTO	AUTO	ON	ON
AUTO	HIGH	ON	ON
HIGH	OFF	ON	ON
HIGH	AUTO	ON	ON
HIGH	HIGH	OFF	ON

B. Possible Causes

(1) Left recirculation fan, M1440
(2) Right recirculation fan, M1441
(3) Left cabin air recirculation fan relay, R429
(4) Right cabin air recirculation fan relay, R331
(5) Left air conditioning accessory unit, M324
(6) Right air conditioning accessory unit, M1455
(7) Switch for the left recirculation fan, S11
(8) Switch for the right recirculation fan, S3
(9) Wiring

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
E	7	C00911	A/C RECIRC FAN LEFT CABIN AIR
E	9	C01156	A/C RECIRC FAN LEFT CONT

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
E	2	C01023	A/C RECIRC FAN RIGHT CONT.
E	4	C00884	AC RECIRC FAN RIGHT CABIN AIR

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D. Related Data

(1) (SSM 21-25-11)

(2) (WDM 21-25-11)

E. Initial Evaluation

(1) AIRPLANES WITH CAB/UTIL SWITCH ON THE P5-13 PANEL;

Make sure that the CAB/UTIL switch on the P5-13 panel is set to the ON position.

(2) Do this task: Recirculation System - Operational Test, AMM TASK 21-25-00-710-802-002.

(3) If the operational test shows that the left recirculation fan does not operate with any setting of the L PACK switch or the R PACK switch, then do the Fault Isolation Procedure - Left Recirculation Fan Does Not Come On At Any Setting of L PACK and R PACK Switches.

(4) If the operational test shows that the right recirculation fan does not operate with any setting of the L PACK switch or the R PACK switch, then do the Fault Isolation Procedure - Right Recirculation Fan Does Not Come On At Any Setting of L PACK and R PACK Switches.

(5) If the operational test shows that the left or right recirculation fan operates but does not come on as specified in the operational test, then do the Fault Isolation Procedure - Left or Right Recirculation Fan Does Not Come On as Specified With the L PACK and R PACK Switches.

(6) If the operational test does not show a problem, then there was an intermittent fault.

F. Fault Isolation Procedure - Right Recirculation Fan Does Not Come On At Any Setting of L PACK and R PACK Switches

NOTE: You must do the steps in the Initial Evaluation before you do these steps.

NOTE: This procedure isolates the fault to one of these possible causes:

- recirculation fan, M1441
- recirculation fan relay, R331
- recirculation fan switch, S3
- wiring

(1) Do this check of the right recirculation fan, M1441:

(a) Disconnect connector D3854 from the right recirculation fan.

(b) Do a continuity check between these pins of the connector on the recirculation fan:

FAN CONNECTOR		FAN CONNECTOR
pin 5	. . . . .	pin 6
pin 2	. . . . .	pin 3
pin 3	. . . . .	pin 4

(c) If there is not continuity between a pair of pins, then do these steps:

1) Replace the right recirculation fan, M1441.

These are the tasks:

Recirculation Fan Removal, AMM TASK 21-25-02-000-801,

Recirculation Fan Installation, AMM TASK 21-25-02-400-801.

2) If you did not do it as part of the replacement procedure, do this task: Recirculation System - Operational Test, AMM TASK 21-25-00-710-802-002

ECCN 9E991 BOEING PROPRIETARY - See title page for details

3) If the operational test is satisfactory, then you corrected the fault.

(d) If there is continuity between all the pairs of pins, then continue.

(2) Do this check for a path to ground for the right recirculation fan:

(a) Do a continuity check between pin 5 of the connector D3854 and structure ground.

(b) If there is not continuity between pin 5 and the structure ground, then do these steps:

1) Repair the wiring.

2) Do this task: Recirculation System - Operational Test, AMM TASK 21-25-00-710-802-002.

3) If the operational test is satisfactory, then you corrected the fault.

(c) If there is continuity between pin 5 and the structure ground, then continue.

(3) Do this voltage check at the right recirculation fan, M1441:

(a) Make sure the R RECIRC FAN switch on the P5-10 air conditioning panel is in the AUTO position.

(b) Disconnect connector D3854 from the recirculation fan, M1441.

(c) Do a check for 28 VDC between pin 6 of connector D3854 and structure ground.

(d) If there is not 28 VDC at pin 6 of D3854, then do these steps:

1) Do a check for 28 VDC between terminal X2 of right cabin air recirculation fan relay, R331 and structure ground.

2) If there is not 28 VDC at terminal X2, then replace right cabin air recirculation fan relay, R331.

a) Do this task: Recirculation System - Operational Test, AMM TASK 21-25-00-710-802-002.

b) If the operational test is satisfactory, then you corrected the fault.

c) If the operational test is not satisfactory, then continue.

d) Do a check of the wiring between these pins of circuit breaker C1023 on the P6-4 panel and relay R331 behind the P6 panel:

C1023		R331
term L	. . . . .	term X1

e) Repair the wiring.

f) Do this task: Recirculation System - Operational Test, AMM TASK 21-25-00-710-802-002.

g) If the operational test is satisfactory, then you corrected the fault.

3) If there is 28 VDC at terminal X2 of recirculation fan relay, R331, then replace right recirculation fan switch, S3.

a) Do this task: Recirculation System - Operational Test, AMM TASK 21-25-00-710-802-002.

b) If the operational test is satisfactory, then you corrected the fault.

c) If the operational test is not satisfactory, then continue.

d) Do a check for an open circuit between these pins of the R331 relay and D680 connector on the P5-10 panel:

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R331	D680
term X2	pin 46

e) Do a check for an open circuit between these pins of the D680 connector on the P5-10 panel and the D40562P connector on the E4-1 electronic equipment rack:

D680	D40562P
pin 47	pin A13

f) Do a check for an open circuit between these pins of the D40562P connector on the E4-1 electronic equipment rack and connector D3854 for the recirculation fan:

D40330P	D3854
pin 3	pin 6

g) If you find an open circuit, then repair the wiring.

h) Do this task: Recirculation System - Operational Test, AMM TASK 21-25-00-710-802-002.

i) If the operational test is satisfactory, then you corrected the fault.

(e) If there is 28 VDC at pin 6 of D3854, then do these steps:

1) Install a jumper wire between pin 5 and pin 6 of connector D3854.

2) Do a check for 200 VAC 3-phase power between pins 2, 3, and 4 of connector D3854.

3) If there is not 200 VAC 3-phase power between pins 2, 3, and 4 of connector D3854, then replace right recirculation fan relay, R331.

a) Do this task: Recirculation System - Operational Test, AMM TASK 21-25-00-710-802-002.

b) If the operational test is satisfactory, then you corrected the fault.

c) If the operational test is not satisfactory, then continue.

d) Do a check of the wiring between these pins of circuit breaker C884 on the P6-4 panel and connector D3854 for the right recirculation fan:

C884	D3854
term A1	pin 2
term B1	pin 3
term C1	pin 4

e) If you find a problem with the wiring, then repair the wiring.

f) Do this task: Recirculation System - Operational Test, AMM TASK 21-25-00-710-802-002.

g) If the operational test is satisfactory, then you corrected the fault.

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G. Fault Isolation Procedure - Left Recirculation Fan Does Not Come On At Any Setting of L PACK and R PACK Switches

NOTE: You must do the steps in the Initial Evaluation before you do these steps.

NOTE: This procedure isolates the fault to one of these possible causes:

- recirculation fan, M1440
- recirculation fan relay, R429
- recirculation fan switch, S11
- wiring

(1) Do this check of the left recirculation fan, M1440:

(a) Disconnect connector D3856 from the left recirculation fan, M1440.

(b) Do a continuity check between these pins on the connector of the fan:

FAN		CONNECTOR
pin 5	. . . . .	pin 6
pin 2	. . . . .	pin 3
pin 3	. . . . .	pin 4

(c) If there is not continuity between a pair of pins, then do these steps:

1) Replace the left recirculation fan, M1440.

These are the tasks:

Recirculation Fan Removal, AMM TASK 21-25-02-000-801,

Recirculation Fan Installation, AMM TASK 21-25-02-400-801.

2) If you did not do it as part of the replacement procedure, then, do this task: Recirculation System - Operational Test, AMM TASK 21-25-00-710-802-002

3) If the operational test is satisfactory, then you corrected the fault.

(d) If there is continuity between all the pairs of pins, then continue.

(2) Do this check for a path to ground for the left recirculation fan:

(a) Do a continuity check between pin 5 of the connector D3856 and structure ground.

(b) If there is not continuity between pin 5 and structure ground, then do these steps:

1) Repair the wiring.

2) Do this task: Recirculation System - Operational Test, AMM TASK 21-25-00-710-802-002.

3) If the operational test is satisfactory, then you corrected the fault.

(c) If there is continuity between pin 5 and structure ground, then continue.

(3) Do this check for 28 VDC at the left recirculation fan, M1440:

(a) Make sure the L RECIRC FAN switch on the P5-10 air conditioning panel is in the AUTO position.

(b) Disconnect connector D3856 from the recirculation fan, M1440.

(c) Do a check for 28 VDC between pin 6 of connector D3856 and structure ground.

(d) If there is not 28 VDC at pin 6 of connector D3856, then do these steps:

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OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999 (Continued)

1) Do a check for 28 VDC between terminal X2 of left cabin air recirculation fan relay, R429 and structure ground.
2) If there is not 28 VDC at terminal X2, then replace left cabin air recirculation fan relay, R429.
   a) Do this task: Recirculation System - Operational Test, AMM TASK 21-25-00-710-802-002.
   b) If the operational test is satisfactory, then you corrected the fault.
   c) If the operational test is not satisfactory, then continue.
   d) Do a check of the wiring between these pins of circuit breaker C1156 on the P18-3 panel and relay R429 of the J22 junction box (located in left side of the nose wheel well):

C1156		R429
term L	. . . . .	term X1

   e) Repair the wiring.
   f) Do this task: Recirculation System - Operational Test, AMM TASK 21-25-00-710-802-002.
   g) If the operational test is satisfactory, then you corrected the fault.
3) If there is 28 VDC at terminal X2 of recirculation fan relay, R429, then replace left recirculation fan switch, S11.
   a) Do this task: Recirculation System - Operational Test, AMM TASK 21-25-00-710-802-002.
   b) If the operational test is satisfactory, then you corrected the fault.
   c) If the operational test is not satisfactory, then continue.
   d) Do a check for an open circuit between these pins of relay R429 in the J22 junction box and connector D646 on the P5-10 air conditioning panel:

R429		D646
term X2	. . . . .	pin 53

   e) Do a check for an open circuit between these pins of connector D646 on the P5-10 panel and connector D40554P on the E4-1 electronic equipment rack:

D646		D40554P
pin 54	. . . . .	pin B6

   f) Do a check for an open circuit between these pins of connector D40244P on the E4-1 rack and connector D3856 for recirculation fan:

D40244P		D3856
pin B4	. . . . .	pin 6

   g) If you find a problem with the wiring, then repair the wiring.
   h) Do this task: Recirculation System - Operational Test, AMM TASK 21-25-00-710-802-002.

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i) If the operational test is satisfactory, then you corrected the fault.

(e) If there is 28 VDC at pin 6 of connector D3856, then do these steps:

1) Install a jumper wire between pin 5 and pin 6 of connector D3856.

2) Do a check for 200 VAC 3-phase power between pins 2, 3, and 4 of connector D3856.

3) If there is not 200 VAC 3-phase power between pins 2, 3, and 4 of connector D3856, then replace right recirculation fan relay, R429.

a) Do this task: Recirculation System - Operational Test, AMM TASK 21-25-00-710-802-002.

b) If the operational test is satisfactory, then you corrected the fault.

c) If the operational test is not satisfactory, then continue.

d) Do a check of the wiring between these pins of circuit breaker C911 on the P18-3 panel and connector D3856 for the recirculation fan:

C911		D3856
term A1	. . . . .	pin 2
term B1	. . . . .	pin 3
term C1	. . . . .	pin 4

e) If you find a problem with the wiring, then repair the wiring.

f) Do this task: Recirculation System - Operational Test, AMM TASK 21-25-00-710-802-002.

g) If the operational test is satisfactory, then you corrected the fault.

H. Fault Isolation Procedure - Left or Right Recirculation Fan Does Not Come On As Specified With the L PACK and R PACK Switches

NOTE: You must do the steps in the Initial Evaluation before you do these steps.

NOTE: This procedure isolates the fault to this cause:

- air conditioning accessory unit, M324
- air conditioning accessory unit, M1455

(1) Replace the two air conditioning accessory units, M324 and M1455.

These are the tasks:

Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801,

Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

(a) If you did not do it after installation, then, do this task: Air Conditioning Accessory Unit - Operational Test, AMM TASK 21-51-02-710-802-002.

(b) If the operational test of the air conditioning accessory units, M324 and M1455 is satisfactory, then you corrected the fault.

END OF TASK

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805. EQUIP COOLING EXHAUST OFF Light ON With the Switch at ALTN - Fault Isolation

A. Description

(1) The EQUIP COOLING EXHAUST OFF Light on the equipment cooling panel is ON with the EQUIP COOLING EXHAUST Switch at ALTN.

(2) The EQUIP COOLING EXHAUST OFF Light comes ON when the Alternate Exhaust Fan does not pull enough cooling air through the Equipment Cooling Ducts. The monitoring circuit in the Exhaust Low Flow Detector then detects an inadequate air flow condition and provides a ground to allow the light to come ON.

(3) If the aircraft is parked in a very cold location for many hours with no power on the airplane, the temperature in the aircraft can decrease low enough that this fault can occur when power is applied to the aircraft and the equipment cooling system is first started. If the OFF light comes on, examine the fans for operation. If the fans are in operation, let the temperature increase in the Equipment Cooling Bay to more than 32°F (0°C). Then cycle the circuit breakers for the flow detectors. With the fans in operation and the OFF Light extinguished, then the aircraft can be safely be dispatched.

B. Possible Causes

(1) Alternate Exhaust Fan, M99

(2) Low Flow Detector, M100 (dirty or defective)

(3) Alternate Exhaust Fan Control Relay, R30

(4) Equipment Cooling Switch, S59

(5) Wiring

(6) Duct Blockage or Leakage

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	17	C01433	EQPT COOLING LOW FLOW DETECT SUPPLY
A	18	C01434	EQPT COOLING LOW FLOW DETECT EXHAUST

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	14	C01435	EQPT COOLING EXHAUST FAN CONT NORMAL
C	15	C01436	EQPT COOLING EXHAUST FAN CONT ALTN

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
E	1	C00836	EQPT CLG EXH FAN PWR-NORM

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
OMR 701-799, 802			
A	10	C00837	EQPT CLG EXH FAN PWR-ALTN
OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
D	12	C00837	EQPT CLG EXH FAN PWR-ALTN

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OMR ALL

D. Related Data

(1) WDM 21-27-21

(2) SSM 21-27-21

E. Initial Evaluation

(1) Cycle this circuit breaker to reset the system:

(a) Open and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	18	C01434	EQPT COOLING LOW FLOW DETECT EXHAUST

1) If the EQUIPMENT COOLING EXHAUST OFF and EQUIPMENT COOLING SUPPLY OFF Light is not ON after the system has been powered for more than 10 minutes, then you corrected the problem.

2) If the EQUIPMENT COOLING EXHAUST OFF and EQUIPMENT COOLING SUPPLY OFF Light stays ON after the system has been powered for more than 10 minutes, then continue.

(2) Put the EQUIP COOLING EXHAUST Switch to ALTN.

(a) If the EQUIP COOLING EXHAUST OFF Light is OFF or goes OFF in approximately 20 seconds, then there was an intermittent fault.

(b) If the EQUIP COOLING EXHAUST OFF Light is ON, then continue.

(3) Put the EQUIP COOLING EXHAUST Switch to NORM.

(a) If the EQUIP COOLING EXHAUST OFF Light goes OFF in 20 seconds, then do the “Fault Isolation Procedure - Fan Circuit Problem” below.

(b) If the EQUIP COOLING EXHAUST OFF Light does not go OFF in 20 seconds, then do the “Fault Isolation Procedure - Airflow or Indication Circuit Problem” below.

F. Fault Isolation Procedure - Fan Circuit Problem

(1) Do this check of the Alternate Exhaust Fan.

(a) Do these steps to get access to the Alternate Exhaust Fan:

1) Open this access panel to enter the EE Compartment:

Number	Name/Location
117A	Electronic Equipment Access Door

2) Remove the access panel on the top of the raised platform that is just aft and outboard of this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

(b) Disconnect connector D162 from the Alternate Exhaust Fan (WDM 21-27-21).

NOTE: The Alternate Exhaust Fan is the more aft of the two Exhaust Fans.

(c) Do a continuity check at the Alternate Exhaust Fan Receptacle as follows (WDM 21-27-21) :

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FAN RECEPTACLE		FAN RECEPTACLE
pin 5	. . . . .	pin 6
pin 1	. . . . .	pin 2
pin 2	. . . . .	pin 3
pin 1	. . . . .	pin 3

1) If you do not find continuity between one or more set of pins, replace the Alternate Exhaust Fan. These are the tasks:
   - Equipment Cooling Exhaust Fan Removal, AMM TASK 21-27-04-000-801
   - Equipment Cooling Exhaust Fan Installation, AMM TASK 21-27-04-400-801

   a) Do the Repair Confirmation at the end of this task.
2) If there is continuity between all the pins, then continue.

(2) Do this check for a path to Ground for the Alternate Exhaust Fan (WDM 21-27-21):
   (a) Put the EQUIP COOLING EXHAUST Switch to ALTN.
   (b) Do a check for continuity between pin 6 of connector D162 and the Structure Ground.
      1) If you do not find continuity, do these steps:
         a) Lower the P5 Forward Overhead Panel to get access to the electrical connections on the Equipment Cooling Switch.
         b) Do a continuity check between terminals 4 and 5 of the Equipment Cooling Switch.
            <1> If you do not find continuity between terminals 4 and 5, then replace the Equipment Cooling Switch (WDM 21-27-21).
               <a> Put the switch in the ALTN position.
               <b> Re-install connector D162 to the Alternate Exhaust Fan.
               <c> Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	15	C01436	EQPT COOLING EXHAUST FAN CONT ALTN

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
OMR 701-799, 802			
A	10	C00837	EQPT CLG EXH FAN PWR-ALTN
OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
D	12	C00837	EQPT CLG EXH FAN PWR-ALTN

OMR ALL

               <d> Do the Repair Confirmation at the end of this task.
            <2> If you find continuity between terminals 4 and 5, then do these steps:
               <a> Repair the wiring between pin 6 of connector D162 and the Structure Ground.
               <b> Re-install connector D162 to the Alternate Exhaust Fan.

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<c> Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	15	C01436	EQPT COOLING EXHAUST FAN CONT ALTN

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
OMR 701-799, 802			
A	10	C00837	EQPT CLG EXH FAN PWR-ALTN
OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
D	12	C00837	EQPT CLG EXH FAN PWR-ALTN

OMR ALL

<d> Do the Repair Confirmation at the end of this task.

2) If you find continuity between pin 6 of connector D162 and the Structure Ground, then continue.

(3) Do a check for 28V DC Power at the Alternate Exhaust Fan connector as follows (WDM 21-27-21):

(a) Put the EQUIP COOLING EXHAUST Switch to ALTN.

(b) Make sure that this circuit breaker is closed:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	15	C01436	EQPT COOLING EXHAUST FAN CONT ALTN

(c) Put the EQUIP COOLING EXHAUST Switch to ALTN.

(d) Do a check for 28V DC at pin 5 of connector D162.

1) If you do not find 28V DC, then do a check for 28V DC at pin X1 of the Alternate Exhaust Fan Control Relay connector, D1626.

a) If you find 28V DC at pin X1 of connector D1626, then examine and repair the wiring between pin X2 of the Alternate Exhaust Fan Control Relay connector, D1626 and pin 5 of the Alternate Exhaust Fan.

<1> If the problem continues, replace the Alternate Exhaust Fan Control Relay, R30.

<a> Do the Repair Confirmation at the end of this task.

b) If you do not find 28V DC at pin X1 of connector D1626, then examine/repair the wiring as follows:

LOADER CTR-R	**ALTN EXCH FAN CTRL RLY**
D40396P	**D1626**
pin 11	pin X1

2) If you find 28V DC, at pin 5 of the Alternate Exhaust Fan connector, D162, then continue.

(4) Do a check for 3-Phase 115V AC Power to Alternate Exhaust Fan as follows:

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(a) Install a jumper wire between pins 5 and 6 on connector D162.

(b) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	15	C01436	EQPT COOLING EXHAUST FAN CONT ALTN

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
OMR 701-799, 802			
A	10	C00837	EQPT CLG EXH FAN PWR-ALTN
OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
D	12	C00837	EQPT CLG EXH FAN PWR-ALTN

OMR ALL

(c) Do a check for 115V AC at pins 1, 2 and 3 of the Alternate Exhaust Fan connector, D162.

1) If you find 115 VAC, then replace the Alternate Exhaust Fan, M99.

a) Do the Repair Confirmation at the end of this task.

2) If you do not find 115V AC, then examine the wiring as follows:

P92 PNL	**ALTN EXCH FAN**
C837	**D162**
term A	pin 1
term B	pin 2
term C	pin 3

a) Repair the wiring as necessary.

<1> Remove the jumper wire from connector D162.

<2> Install connector D162 on the Alternate Exhaust Fan.

<3> Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	15	C01436	EQPT COOLING EXHAUST FAN CONT ALTN

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
OMR 701-799, 802			
A	10	C00837	EQPT CLG EXH FAN PWR-ALTN
OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
D	12	C00837	EQPT CLG EXH FAN PWR-ALTN

OMR ALL

<4> Do the Repair Confirmation at the end of this task.

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G. Fault Isolation Procedure - Airflow or Indication Circuit Problem

(1) Do these steps to clean and test the Low Flow Detector:

(a) Remove the Low Flow Detector. This is the task: Equipment Cooling Low Flow Sensors Removal, AMM TASK 21-27-03-000-802.

(b) Clean the Low Flow Detector. This is the task: Equipment Cooling Low Flow Sensors Cleaning, AMM TASK 21-27-03-400-803.

(c) Install the Low Flow Detector. This is the task: Equipment Cooling Low Flow Sensors Installation, AMM TASK 21-27-03-400-802.

(d) Do the Repair Confirmation at the end of this task.

(2) Replace the Equipment Cooling Exhaust Low Flow Detector. These are the tasks:

- Equipment Cooling Low Flow Sensors Removal, AMM TASK 21-27-03-000-802
- Equipment Cooling Low Flow Sensors Installation, AMM TASK 21-27-03-400-802

(a) Do the Repair Confirmation at the end of this task.

(3) Do this check of the wiring (WDM 21-27-21):

(a) Disconnect connector D164 from the Exhaust Low Flow Detector.

(b) Disconnect connector D11418 from the P91 Power Distribution Panel 1 in the EE Compartment.

(c) Lower the P5 Forward Overhead Panel to get access to the electrical connections on the EQUIP COOLING EXHAUST OFF Light, L47.

(d) Do a check of the wiring between the Exhaust Low Flow Detector and the EQPT LOW FLOW DETECT EXHAUST Circuit Breaker, the P91 Panel and the EQUIP COOLING EXHAUST OFF Light as follows (WDM 21-27-21):

D164		**C1434**
pin 1	. . . . .	term L
D164		**D11418**
pin 9	. . . . .	pin 11
D164		**L47**
pin 9	. . . . .	term 2

(e) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-install the P5 Forward Overhead Panel.

3) Re-connect connector D11418 to the P91 Power Distribution Panel 1 in the EE Compartment.

4) Re-connect connector D164 to the Exhaust Low Flow Detector.

5) Do the Repair Confirmation at the end of this task.

(4) Examine the Equipment Cooling Exhaust System Ducts for blockage or leakage.

(a) Repair the leakage or blockage as necessary.

(b) Do the Repair Confirmation at the end of this task.

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H. Repair Confirmation

(1) Put the EQUIP COOLING EXHAUST Switch to ALTN.

(2) Make sure that the Alternate Exhaust Fan Operates.

(a) If the EQUIP COOLING EXHAUST OFF Light is OFF or goes OFF in 20 seconds, then you corrected the problem.

(b) If the EQUIP COOLING EXHAUST OFF Light does not go OFF in 20 seconds then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

806. EQUIP COOLING EXHAUST OFF Light ON With the Switch at NORMAL - Fault Isolation

A. Description

(1) The EQUIP COOLING EXHAUST OFF Light ON the equipment cooling panel is ON with the EQUIP COOLING EXHAUST Switch at NORM.

(2) The EQUIP COOLING EXHAUST OFF Light comes ON when the Normal Exhaust Fan does not pull enough cooling air through the Equipment Cooling Ducts. The monitoring circuit in the Exhaust Low Flow Detector then detects an inadequate air flow condition and provides a Ground to allow the light to come ON.

(3) If the aircraft is parked in a very cold location for many hours with no power on the airplane, the temperature in the aircraft can decrease low enough that this fault can occur when power is applied to the aircraft and the equipment cooling system is first started. If the OFF light comes on, examine the fans for operation. If the fans are in operation, let the temperature increase in the Equipment Cooling Bay to more than 32°F (0°C). Then cycle the circuit breakers for the flow detectors. With the fans in operation and the OFF Light extinguished, then the aircraft can be safely be dispatched.

B. Possible Causes

(1) Normal Exhaust Fan, M98

(2) Low Flow Detector, M100 (dirty or defective)

(3) Normal Exhaust Fan Control Relay, R29

(4) Equipment Cooling Switch, S59

(5) Wiring

(6) Duct Blockage or Leakage

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	18	C01434	EQPT COOLING LOW FLOW DETECT EXHAUST

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	14	C01435	EQPT COOLING EXHAUST FAN CONT NORMAL
C	15	C01436	EQPT COOLING EXHAUST FAN CONT ALTN

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Power Distribution Panel Number 1, P91

Row	Col	Number	Name
E	1	C00836	EQPT CLG EXH FAN PWR-NORM

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
OMR 701-799, 802			
A	10	C00837	EQPT CLG EXH FAN PWR-ALTN
OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
D	12	C00837	EQPT CLG EXH FAN PWR-ALTN

OMR ALL

D. Related Data

(1) WDM 21-27-21

(2) SSM 21-27-21

E. Initial Evaluation

(1) Cycle this circuit breaker to reset the system:

(a) Open and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	18	C01434	EQPT COOLING LOW FLOW DETECT EXHAUST

1) If the EQUIPMENT COOLING EXHAUST OFF and EQUIPMENT COOLING SUPPLY OFF Light is not ON after the system has been powered for more than 10 minutes, then you corrected the fault.

2) If the EQUIPMENT COOLING EXHAUST OFF and EQUIPMENT COOLING SUPPLY OFF Light stays ON after the system has been powered for more than 10 minutes, then continue.

(2) Put the EQUIP COOLING EXHAUST Switch to NORM (21-27 TASK SUPPORT Figure 301).

(a) If the EQUIP COOLING EXHAUST OFF Light is OFF or goes OFF in approximately 20 seconds, then there was an intermittent fault.

(b) If the EQUIP COOLING EXHAUST OFF Light is ON, then continue.

(3) Put the EQUIP COOLING EXHAUST Switch to ALTN.

(a) If the EQUIP COOLING EXHAUST OFF Light goes OFF in 20 seconds, then do the “Fault Isolation Procedure - Fan Circuit Problem” below.

(b) If the EQUIP COOLING EXHAUST OFF Light does not go OFF in 20 seconds, then do the “Fault Isolation Procedure - Airflow or Indication Circuit Problem” below.

F. Fault Isolation Procedure - Fan Circuit Problem

(1) Do this check of the Normal Exhaust Fan:

(a) Do these steps to get access to the Normal Exhaust Fan:

1) Open this access panel to enter the EE Compartment:

Number	Name/Location
117A	Electronic Equipment Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Remove the access panel on the top of the raised platform that is just aft and right of this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

(b) Disconnect connector D160 from the Normal Exhaust Fan (WDM 21-27-21).

NOTE: The Normal Exhaust Fan is the more forward of the two Exhaust Fans.

(c) Do a continuity check between these pins on the connector of the fan (WDM 21-27-21)

FAN RECEPTACLE D160		FAN RECEPTACLE D160
pin 5	. . . . .	pin 6
pin 1	. . . . .	pin 2
pin 2	. . . . .	pin 3
pin 1	. . . . .	pin 3

1) If there is no continuity between one or more sets of pins, replace the Normal Exhaust Fan. These are the tasks:
   - Equipment Cooling Exhaust Fan Removal, AMM TASK 21-27-04-000-801,
   - Equipment Cooling Exhaust Fan Installation, AMM TASK 21-27-04-400-801
   a) Do the Repair Confirmation at the end of this task.
2) If there is continuity between all the pins, then continue.

(2) Do this check for a path to Structure Ground for the Normal Exhaust Fan (WDM 21-27-21):

(a) Put the EQUIP COOLING EXHAUST Switch to NORM (21-27 TASK SUPPORT Figure 301).

(b) Do a continuity check between pin 6 of connector D160 and Structure Ground:

NORM EXH FAN D160		Structure Ground
pin 6	. . . . .	GD516-DC
pin 6	. . . . .	GD534-DC

(c) If you do not find continuity to Structure Ground, do these steps:

1) Lower the P5 Forward Overhead Panel to get access to the electrical connections on the Equipment Cooling Switch.
2) Do a continuity check between terminals 2 and 3 of the Equipment Cooling Switch.
3) If you do not find continuity between terminals 2 and 3, then do these steps:
   a) Replace the Equipment Cooling Switch.

   NOTE: Put the switch in the NORM position.

   b) Re-install connector D160 on the Normal Exhaust Fan.

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c) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	14	C01435	EQPT COOLING EXHAUST FAN CONT NORMAL

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
E	1	C00836	EQPT CLG EXH FAN PWR-NORM

d) Do the Repair Confirmation at the end of this task.

4) If you find continuity between terminals 2 and 3, then do these steps:

a) Repair the wiring between pin 6 of connector D160 and the Structure Ground.

b) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	14	C01435	EQPT COOLING EXHAUST FAN CONT NORMAL

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
E	1	C00836	EQPT CLG EXH FAN PWR-NORM

c) Re-install connector D160 on the Normal Exhaust Fan.

d) Do the Repair Confirmation at the end of this task.

(d) If you find continuity between pin 6 of connector D160 and the Structure Ground, then continue.

(3) Do this check for power to the Normal Exhaust Fan (WDM 21-27-21):

(a) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	14	C01435	EQPT COOLING EXHAUST FAN CONT NORMAL

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
E	1	C00836	EQPT CLG EXH FAN PWR-NORM

(b) Put the EQUIP COOLING EXHAUST Switch to NORM (21-27 TASK SUPPORT Figure 301).

(c) Do a check for 28V DC at pin 5 of connector D160.

1) If you do not find 28V DC, then do a check for 28V DC at pin X1 of the Normal Exhaust Fan Control Relay connector, D1624.

a) If you find 28V DC at pin X1 of connector D1624, then examine and repair the wiring between pin X2 of the Normal Exhaust Fan Control Relay connector, D1624 and pin 5 of the Normal Exhaust Fan.

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<1> If the problem continues, replace the Normal Exhaust Fan Control Relay, R29.

<a> Do the Repair Confirmation at the end of this task.

b) If you do not find 28V DC at pin X1 of connector D1624, then examine/repair the wiring as follows:

LOADER CTR-R	NORM EXH FAN CTRL RLY
D40812P	D1624
pin 19	pin X1

2) If you find 28V DC at pin 5 of the Normal Exhaust Fan connector D160, then continue.

(4) Do a check for 3-Phase 115V AC Power to Normal Exhaust Fan as follows:

(a) Install a jumper wire between pins 5 and 6 on connector D160.

(b) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	14	C01435	EQPT COOLING EXHAUST FAN CONT NORMAL

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
E	1	C00836	EQPT CLG EXH FAN PWR-NORM

(c) Do a check for 115V AC at pins 1, 2 and 3 of the Normal Exhaust Fan connector, D160.

1) If you find 115 VAC, then replace the Normal Exhaust Fan, M98.

a) Do the Repair Confirmation at the end of this task.

2) If you do not find 115V AC, then examine the wiring as follows:

P91 PNL C836	NORM EXH FAN D160
term A	pin 1
term B	pin 2
term C	pin 3

a) Repair the wiring as necessary.

<1> Remove the jumper wire from connector D160.

<2> Install connector D160 on the Normal Exhaust Fan.

<3> Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	14	C01435	EQPT COOLING EXHAUST FAN CONT NORMAL

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Power Distribution Panel Number 1, P91

Row	Col	Number	Name
E	1	C00836	EQPT CLG EXH FAN PWR-NORM

<4> Do the Repair Confirmation at the end of this task.

G. Fault Isolation Procedure - Airflow or Indication Circuit Problem

(1) Do these steps to clean and test the Equipment Cooling Low Flow Detector:

(a) Do this task to remove the Low Flow Detector: Equipment Cooling Low Flow Sensors Removal, AMM TASK 21-27-03-000-802.

(b) Do this task to clean the Low Flow Detector: Equipment Cooling Low Flow Sensors Cleaning, AMM TASK 21-27-03-400-803.

(c) Do this task to install the Low Flow Detector: Equipment Cooling Low Flow Sensors Installation, AMM TASK 21-27-03-400-802.

(d) Do the Repair Confirmation at the end of this task.

(2) Replace the Equipment Cooling Exhaust Low Flow Detector. These are the tasks:

- Equipment Cooling Low Flow Sensors Removal, AMM TASK 21-27-03-000-802
- Equipment Cooling Low Flow Sensors Installation, AMM TASK 21-27-03-400-802

(a) Do the Repair Confirmation at the end of this task.

(3) Do this check of the wiring (WDM 21-27-21):

(a) Disconnect connector D164 from the Exhaust Low Flow Detector.

(b) Disconnect connector D11418 from the P91 Power Distribution Panel 1 in the EE Compartment.

(c) Lower the P5 Forward Overhead Panel to get access to the electrical connections on the EQUIP COOLING EXHAUST OFF Light, L47.

(d) Do a check of the wiring between the Exhaust Low Flow Detector and the EQPT LOW COOLING EXHAUST OFF Circuit Breaker, the P91 Panel and the EQUIP COOLING EXHAUST OFF Light as follows (WDM 21-27-21):

D164		**C1434**
pin 1	. . . . .	term L
D164		**D11418**
Pin 9	. . . . .	Pin 11
D164		**L47**
pin 9	. . . . .	term 2

(e) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-install the P5 Forward Overhead Panel.

3) Re-connect connector D11418 to the P91 Power Distribution Panel No. 1 in the EE Compartment.

4) Re-connect connector D164 to the Exhaust Low Flow Detector.

5) Do the Repair Confirmation at the end of this task.

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(4) Examine the Equipment Cooling Exhaust System Ducts for blockage or leakage.

(a) Repair the leakage or blockage.

(b) Do the Repair Confirmation at the end of this task.

H. Repair Confirmation

(1) Put the EQUIP COOLING EXHAUST Switch to NORM (21-27 TASK SUPPORT Figure 301).

(2) Make sure that the Normal Exhaust Fan Operates.

(a) If the EQUIP COOLING EXHAUST OFF Light is OFF or goes OFF in 20 seconds, then you corrected the problem.

(b) If the EQUIP COOLING EXHAUST OFF Light does not go OFF in 20 seconds then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

807. EQUIP COOLING SUPPLY OFF Light On With the Switch at ALTN - Fault Isolation

A. Description

(1) The EQUIP COOLING SUPPLY OFF Light on the Equipment Cooling Panel is ON with the EQUIP COOLING SUPPLY Switch to ALTN.

(2) The EQUIP COOLING SUPPLY OFF Light comes ON when the Alternate Supply Fan does not blow enough cooling air through the Equipment Cooling Duct. The monitoring circuit in the Supply Low Flow Detector then detects an inadequate air flow condition and provides a ground to allow the light to come ON.

(3) If the aircraft is parked in a very cold location for many hours with no power on the airplane, the temperature in the aircraft can decrease low enough that this fault can occur when power is applied to the aircraft and the equipment cooling system is first started. If the OFF light comes on, examine the fans for operation. If the fans are in operation, let the temperature increase in the Equipment Cooling Bay to more than 32°F (0°C). Then cycle the circuit breakers for the flow detectors. With the fans in operation and the OFF Light extinguished, then the aircraft can be safely be dispatched.

B. Possible Causes

(1) Alternate Supply Fan, M1323

(2) Supply Low Flow Detector, M1915 (dirty or defective)

(3) Alternate Supply Fan Control Relay, R373

(4) EQUIP COOLING SUPPLY Switch, S870

(5) Equipment Cooling Air Filter

(6) Wiring

(7) Duct Blockage or Leakage

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	17	C01433	EQPT COOLING LOW FLOW DETECT SUPPLY

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F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	12	C01116	EQPT COOLING SUPPLY FAN CONT-NORMAL
C	13	C01117	EQPT COOLING SUPPLY FAN CONTROL-ALTN

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
A	8	C00935	EQPT CLG SPLY FAN PWR-ALTN

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
OMR 701-799, 802			
A	8	C00934	EQPT CLG SPLY FAN PWR-NORM
OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
D	10	C00934	EQPT CLG SPLY FAN PWR-NORM

OMR ALL

D. Related Data

(1) WDM 21-27-31

(2) SSM 21-27-31

E. Initial Evaluation

(1) Cycle this circuit breaker to reset the system:

(a) Open and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	17	C01433	EQPT COOLING LOW FLOW DETECT SUPPLY

1) If the EQUIPMENT COOLING EXHAUST OFF and EQUIPMENT COOLING SUPPLY OFF Light is not ON after the system has been powered for more than 10 minutes, then you corrected the problem.

2) If the EQUIPMENT COOLING EXHAUST OFF and EQUIPMENT COOLING SUPPLY OFF Light stays ON after the system has been powered for more than 10 minutes, then continue.

(2) Put the EQUIP COOLING SUPPLY Switch to ALTN (21-27 TASK SUPPORT Figure 301).

(a) If the EQUIP COOLING SUPPLY Light is OFF or goes OFF in 20 seconds, then there was an intermittent fault.

(b) If the EQUIP COOLING SUPPLY Light is ON, then continue.

(3) Put the EQUIP COOLING SUPPLY Switch to NORM.

(a) If the EQUIP COOLING SUPPLY Light goes OFF in 20 seconds, then do the Fault Isolation Procedure - Fan Circuit Problem below.

(b) If the EQUIP COOLING SUPPLY Light does not go OFF within 20 seconds, then do the Fault Isolation Procedure - Airflow or Indication Circuit Problem below.

F. Fault Isolation Procedure - Fan Circuit Problem

(1) Do these steps check the Alternate Supply Fan:

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(a) Open this access panel to enter the EE Compartment:

Number	Name/Location
117A	Electronic Equipment Access Door

(b) Disconnect connector D3472 from the Alternate Supply Fan (WDM 21-27-31).

NOTE: The Alternate Supply Fan is the lower of the two Supply Fans.

(c) Do a continuity check at the Alternate Supply Fan Connector as follows:

FAN CONNECTOR		FAN CONNECTOR
pin 5	. . . . .	pin 6
pin 1	. . . . .	pin 2
pin 2	. . . . .	pin 3
pin 1	. . . . .	pin 3

1) If you do not find continuity between any two pins, replace the Alternate Supply Fan. These are the tasks:
   - Equipment Cooling Supply Fan Removal, AMM TASK 21-27-02-000-801
   - Equipment Cooling Supply Fan Installation, AMM TASK 21-27-02-400-801

   a) Do the Repair Confirmation at the end of this task.

2) If you find continuity between all the pins, then continue.

(2) Do this check for a path to Ground for the Alternate Supply Fan (WDM 21-27-31):

(a) Put the EQUIP COOLING SUPPLY Switch to ALTN (21-27 TASK SUPPORT Figure 301).

(b) Do a continuity check between pin 6 of connector D3472 and the Structure Ground.

1) If you do not find continuity, do these steps:

a) Lower the P5 Forward Overhead Panel to get access to the electrical connections on the Equipment Cooling Switch, S870.

b) Do a continuity check between terminals 4 and 5, of the Equipment Cooling Switch.

<1> If you do not find continuity between terminals 4 and 5, then replace the Equipment Cooling Switch (WDM 21-27-31).

<a> Put the switch in the ALTN position.

<b> Re-install connector D3472 to the Alternate Supply Fan.

<c> Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	13	C01117	EQPT COOLING SUPPLY FAN CONTROL-ALTN

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
A	8	C00935	EQPT CLG SPLY FAN PWR-ALTN

<d> Do the Repair Confirmation at then end of this task.

<2> If you find continuity between terminals 4 and 5, then do these steps.

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<a> Repair the wiring between pin 6 of connector D3472 and the Structure Ground.

<b> Re-install connector D3472 to the Alternate Supply Fan, M1323.

<c> Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	13	C01117	EQPT COOLING SUPPLY FAN CONTROL-ALTN

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
A	8	C00935	EQPT CLG SPLY FAN PWR-ALTN

<d> Do the Repair Confirmation at the end of this task.

(c) If you do not find continuity between pin 6 of connector D3472 and the Structure Ground, do these steps:

1) Lower the P5 Forward Overhead Panel to get access to the electrical connections on the Equipment Cooling Switch, S870.

2) Do a continuity check between terminals 4 and 5 of the Equipment Cooling Supply Switch, S870.

3) If you do not find continuity between terminals 4 and 5, then do these steps:

a) Replace the Equipment Cooling Supply Switch, S870.

NOTE: Put the switch in the ALTN position.

b) Re-install connector D3472 to the Alternate Supply Fan, M1323.

c) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	13	C01117	EQPT COOLING SUPPLY FAN CONTROL-ALTN

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
A	8	C00935	EQPT CLG SPLY FAN PWR-ALTN

d) Do the Repair Confirmation at the end of this task.

4) If you find continuity between terminals 4 and 5, then do these steps:

a) Repair the wiring between pin 6 of connector D3472 and the Structure Ground.

b) Re-install connector D3472 on the Alternate Supply Fan, M1323.

c) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	13	C01117	EQPT COOLING SUPPLY FAN CONTROL-ALTN

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
A	8	C00935	EQPT CLG SPLY FAN PWR-ALTN

d) Do the Repair Confirmation at the end of this task.

(d) If there is continuity check between pin 6 of connector D3472 and the Structure Ground, then continue.

(3) Do this check for power to the Alternate Supply Fan:

(a) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	14	C01435	EQPT COOLING EXHAUST FAN CONT NORMAL

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
E	1	C00836	EQPT CLG EXH FAN PWR-NORM

(b) Put the EQUIP COOLING SUPPLY Switch to ALTN (21-27 TASK SUPPORT Figure 301).

(c) Do a check for 28V DC at pin 5 of connector D3472.

(d) If you do not find 28V DC at pin 5, do these steps:

1) Do a check for 28V DC at pin X1 of connector D3468 of the Alternate Supply Fan Control Relay R373.

2) If you find 28V DC at pin X1 of connector D3468, then replace the Alternate Exhaust Fan Control Relay R373.

3) If you do not find 28V DC at pin X1 of connector D3468, then repair the wiring between these pins:

C935		D3472
term L	. . . . .	pin 5

4) Re-connect connector D3472 to the Alternate Supply Fan.

5) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	13	C01117	EQPT COOLING SUPPLY FAN CONTROL-ALTN

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
A	8	C00935	EQPT CLG SPLY FAN PWR-ALTN

6) Do the Repair Confirmation at the end of this task.

(e) If you find 28V DC at pin 5, then continue.

(f) Install a jumper wire between pin 5 and pin 6 on D3472.

(g) Do a check for 3-phase 115V AC at pins 1, 2 and 3 of connector D3472.

(h) If you do not find 115V AC at pins 1, 2 and 3 of D3472, then do these steps:

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1) Repair the wiring between these connectors (WDM 21-27-31):

C935		D3472
term A	. . . . .	pin 1
term B	. . . . .	pin 2
term C	. . . . .	pin 3

2) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	13	C01117	EQPT COOLING SUPPLY FAN CONTROL-ALTN

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
A	8	C00935	EQPT CLG SPLY FAN PWR-ALTN

3) Remove the jumper wire from connector D3472.

4) Re-connect connector D3472 to the Alternate Supply Fan.

5) Do the Repair Confirmation at the end of this task.

G. Fault Isolation Procedure - Airflow or Indication Circuit Problem

(1) Do these steps to clean and test the Supply Low Flow Detector, M1915:

(a) Remove the Low Flow Detector, Equipment Cooling Low Flow Sensors Removal, AMM TASK 21-27-03-000-802.

(b) Clean the Low Flow Detector, Equipment Cooling Low Flow Sensors Cleaning, AMM TASK 21-27-03-400-803.

(c) Install the Low Flow Detector, Equipment Cooling Low Flow Sensors Installation, AMM TASK 21-27-03-400-802.

(d) Do the Repair Confirmation at the end of this task.

(2) Replace the Equipment Cooling Air Filter. These are the tasks:

- Equipment Cooling Air Filter Removal, AMM TASK 21-27-01-000-801
- Equipment Cooling Air Filter Installation, AMM TASK 21-27-01-400-801

(a) Do the Repair Confirmation at the end of this task.

(3) Replace the Equipment Cooling Supply Low Flow Detector. These are the task:

- Equipment Cooling Low Flow Sensors Removal, AMM TASK 21-27-03-000-802
- Equipment Cooling Low Flow Sensors Installation, AMM TASK 21-27-03-400-802

(a) Do the Repair Confirmation at the end of this task.

(4) Do this check of the wiring (WDM 21-27-31):

(a) Disconnect connector D3464 from the Supply Low Flow Detector.

(b) Disconnect connector D11452 from the P92 Power Distribution Panel 2.

(c) Lower the P5 Forward Overhead Panel to get access to the electrical connections on the EQUIP COOLING SUPPLY OFF Light, L1141.

(d) Do a check of the wiring between these pins of connector D3464 and connector D11452.

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D3464	D11452
pin 9	pin 7

(e) Do a check of the wiring between these pins of connector D3464 and L1141:

D3464	L1141
pin 9	term 2

(f) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.
2) Raise and close P5 Forward Overhead Panel.
3) Re-connect connector D11452 to the P92 Power Distribution Panel No. 2.
4) Re-connect connector D3464 to the Supply Low Flow connector, M1915.
5) Do the Repair Confirmation at the end of this task.

(g) If you do not find a problem with the wiring, then do these steps and continue:

1) Raise and close the P5 Forward Overhead Panel.
2) Re-connect connector D11452 to the P92 Power Distribution Panel 2 in the EE Compartment.
3) Re-connect connector D3464 to the Supply Low Flow Detector, M1915.

(5) Examine all of the ducts of the Equipment Cooling Supply System for Leakage or Blockage.

(a) Repair the Leakage or Blockage.
(b) Do the Repair Confirmation at the end of this task.

H. Repair Confirmation

(1) Put the EQUIP COOLING SUPPLY Switch to ALTN.
(2) Make sure that the Alternate Supply Fan Operates.

(a) If the EQUIP COOLING SUPPLY OFF Light goes OFF in 20 seconds, then you corrected the problem.
(b) If the EQUIP COOLING SUPPLY OFF Light goes does not go OFF in 20 seconds, then continue the Fault Isolation Procedure at the subsequent step.

END OF TASK

808. EQUIP COOLING SUPPLY OFF Light is ON With the Switch at NORM - Fault Isolation

A. Description

(1) The EQUIP COOLING SUPPLY OFF Light on the EQUIP COOLING Panel is ON with the EQUIP COOLING SUPPLY Switch at NORM.
(2) The EQUIP COOLING SUPPLY OFF Light comes ON when the Normal Supply Fan does not blow enough cooling air through the Equipment Cooling Duct. The monitoring circuit in the Supply Low Flow Detector then detects an inadequate air flow condition and provides a ground which causes the light to come ON.

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(3) If the aircraft is parked in a very cold location for many hours with no power on the airplane, the temperature in the aircraft can decrease low enough that this fault can occur when power is applied to the aircraft and the equipment cooling system is first started. If the OFF light comes on, examine the fans for operation. If the fans are in operation, let the temperature increase in the Equipment Cooling Bay to more than 32°F (0°C). Then cycle the circuit breakers for the flow detectors. With the fans in operation and the OFF Light extinguished, then the aircraft can be safely be dispatched.

B. Possible Causes

(1) Normal Supply Fan, M1322

(2) Supply Low Flow Detector, M1915 (dirty or defective)

(3) Normal Supply Fan Control Relay, R347

(4) EQUIP COOLING SUPPLY Switch, S870

(5) Equipment Cooling Air Filter

(6) Wiring

(7) Duct Blockage or Leakage

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	17	C01433	EQPT COOLING LOW FLOW DETECT SUPPLY

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	12	C01116	EQPT COOLING SUPPLY FAN CONT-NORMAL
C	13	C01117	EQPT COOLING SUPPLY FAN CONTROL-ALTN

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
A	8	C00935	EQPT CLG SPLY FAN PWR-ALTN

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
OMR 701-799, 802			
A	8	C00934	EQPT CLG SPLY FAN PWR-NORM
OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
D	10	C00934	EQPT CLG SPLY FAN PWR-NORM

OMR ALL

D. Related Data

(1) WDM 21-27-31

(2) SSM 21-27-31

E. Initial Evaluation

(1) Cycle this circuit breaker to reset the system:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Open and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	17	C01433	EQPT COOLING LOW FLOW DETECT SUPPLY

1) If the EQUIPMENT COOLING EXHAUST OFF and EQUIPMENT COOLING SUPPLY OFF Light is not ON after the system has been powered for more than 10 minutes, then you corrected the fault.

2) If the EQUIPMENT COOLING EXHAUST OFF and EQUIPMENT COOLING SUPPLY OFF Light stays ON after the system has been powered for more than 10 minutes, then continue.

(2) Put the EQUIP COOLING SUPPLY Switch to NORM (21-27 TASK SUPPORT Figure 301).

(a) If the EQUIP COOLING SUPPLY OFF Light is OFF or goes OFF in approximately 20 seconds, then there was an intermittent fault.

(b) If the EQUIP COOLING SUPPLY OFF Light is ON, then continue.

(3) Put the EQUIP COOLING SUPPLY Switch to ALTN.

(a) If the EQUIP COOLING SUPPLY OFF Light goes OFF in 20 seconds, then do the “Fault Isolation Procedure - Fan Circuit Problem” below.

(b) If the EQUIP COOLING SUPPLY OFF Light does not go OFF in 20 seconds, then do the “Fault Isolation Procedure - Airflow or Indication Circuit Problem” below.

F. Fault Isolation Procedure - Fan Circuit Problem

(1) Do these steps check the Normal Supply Fan:

(a) Open this access panel to enter the EE Compartment:

Number	Name/Location
117A	Electronic Equipment Access Door

(b) Disconnect connector D3470 from the Normal Supply Fan (WDM 21-27-31).

NOTE: The Normal Supply Fan is the higher of the two Supply Fans on the right side of the EE Compartment.

(c) Do a continuity check between these pins on the connector of the fan:

FAN CONNECTOR		FAN CONNECTOR
pin 5	. . . . .	pin 6
pin 1	. . . . .	pin 2
pin 2	. . . . .	pin 3
pin 1	. . . . .	pin 3

(d) If you do not find continuity between one or more set of pins, then do these steps:

1) Replace the Normal Supply Fan. These are the tasks:

- Equipment Cooling Supply Fan Removal, AMM TASK 21-27-02-000-801
- Equipment Cooling Supply Fan Installation, AMM TASK 21-27-02-400-801

2) Do the Repair Confirmation at the end of this task

(e) If you find continuity between all the pins, then continue.

(2) Do this check for a path to Ground for the Normal Supply Fan (WDM 21-27-31):

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(a) Put the EQUIP COOLING SUPPLY Switch to NORM (21-27 TASK SUPPORT Figure 301).

(b) Do a continuity check between pin 6 of connector D3470 and the Structure Ground.

(c) If you do not find continuity check between pin 6 of connector D3470 and the Structure Ground, do these steps:

1) Lower the P5 Forward Overhead Panel to get access to the electrical connections on the Equipment Cooling Switch, S870.

2) Do a continuity check between terminals 2 and 3 of the Equipment Cooling Supply Switch, S870.

3) If you do not find continuity between terminals 2 and 3, then do these steps:

a) Replace the Equipment Cooling Supply Switch, S870.

NOTE: Put the Switch in the NORM position.

b) Re-install connector D3470 to the Normal Supply Fan, M1322.

c) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	12	C01116	EQPT COOLING SUPPLY FAN CONT-NORMAL

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
OMR 701-799, 802			
A	8	C00934	EQPT CLG SPLY FAN PWR-NORM
OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
D	10	C00934	EQPT CLG SPLY FAN PWR-NORM

OMR ALL

d) Do the Repair Confirmation at the end of this task.

4) If you find continuity between terminals 2 and 3, then do these steps:

a) Repair the wiring between pin 6 of connector D3470 and the Structure Ground.

b) Re-install connector D3470 on the Normal Supply Fan, M1322.

c) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	12	C01116	EQPT COOLING SUPPLY FAN CONT-NORMAL

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
OMR 701-799, 802			
A	8	C00934	EQPT CLG SPLY FAN PWR-NORM
OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
D	10	C00934	EQPT CLG SPLY FAN PWR-NORM

OMR ALL

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d) Do the Repair Confirmation at the end of this task.

(d) If you find continuity between pin 6 of connector D3470 and the Structure Ground, then continue.

(3) Do this check for 28V DC power to the Normal Supply Fan (WDM 21-27-31):

(a) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	12	C01116	EQPT COOLING SUPPLY FAN CONT-NORMAL

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
OMR 701-799, 802			
A	8	C00934	EQPT CLG SPLY FAN PWR-NORM
OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
D	10	C00934	EQPT CLG SPLY FAN PWR-NORM

OMR ALL

(b) Put the EQUIP COOLING SUPPLY Switch to NORM (21-27 TASK SUPPORT Figure 301).

(c) Do a check for 28V DC at pin 5 of connector D3470.

(d) If If you do not find 28V DC at pin 5, do these steps:

1) Do a check for 28V DC at pin X1 of connector D3466 of the Normal Supply Fan Control Relay R347.

2) If you find 28V DC at pin X1 of connector D3466, then replace the Normal Supply Fan Control Relay R347.

3) If you do not find 28V DC at pin X1 of connector D3466, then repair the wiring between the pins:

C1116		D3470
term L	. . . . .	pin 5

4) Re-connect connector D3470 to the Normal Supply Fan.

5) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	12	C01116	EQPT COOLING SUPPLY FAN CONT-NORMAL

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
OMR 701-799, 802			
A	8	C00934	EQPT CLG SPLY FAN PWR-NORM
OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
D	10	C00934	EQPT CLG SPLY FAN PWR-NORM

OMR ALL

6) Do the Repair Confirmation at the end of this task.

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(e) If you find 28V DC at pin 5, then continue.

(f) Install a jumper wire between pin 5 and pin 6 on D3470.

(g) Do a check for 3-phase 115V AC at pins 1, 2 and 3 of connector D3470.

(h) If you do not find 115V AC at pins 1, 2 and 3 of D3470, then do these steps:

1) Repair the wiring between these connectors:

C934		D3470
term A	. . . . .	pin 1
term B	. . . . .	pin 2
term C	. . . . .	pin 3

2) Remove the jumper wire from connector D3470.

3) Re-connect connector D3470 to the Normal Supply fan.

4) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	12	C01116	EQPT COOLING SUPPLY FAN CONT-NORMAL

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
OMR 701-799, 802			
A	8	C00934	EQPT CLG SPLY FAN PWR-NORM
OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
D	10	C00934	EQPT CLG SPLY FAN PWR-NORM

OMR ALL

5) Do the Repair Confirmation at the end of this task..

G. Fault Isolation Procedure - Airflow or Indication Circuit Problem

(1) Clean the Equipment Cooling Supply Low Flow Detector. These are the tasks:

- Equipment Cooling Low Flow Sensors Removal, AMM TASK 21-27-03-000-802
- Equipment Cooling Low Flow Sensors Cleaning, AMM TASK 21-27-03-400-803
- Equipment Cooling Low Flow Sensors Installation, AMM TASK 21-27-03-400-802

(2) Replace the Equipment Cooling Air Filter. These are the tasks:

- Equipment Cooling Air Filter Removal, AMM TASK 21-27-01-000-801
- Equipment Cooling Air Filter Installation, AMM TASK 21-27-01-400-801

(a) Do the Repair Confirmation at the end of this task.

(3) Replace the Equipment Cooling Supply Low Flow Detector. These are the tasks:

- Equipment Cooling Low Flow Sensors Removal, AMM TASK 21-27-03-000-802
- Equipment Cooling Low Flow Sensors Installation, AMM TASK 21-27-03-400-802

(a) Dot the Repair confirmation at the end of this task.

(4) Do this check of the wiring (WDM 21-27-31):

(a) Disconnect connector D3464 from the Supply Low Flow Detector, M1915.

(b) Disconnect connector D11452 from the P92 Power Distribution Panel 2.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Lower the P5 Forward Overhead Panel to get access to the electrical connections on the EQUIP COOLING SUPPLY OFF Light, L1141.

(d) Do a check of the wiring between these pins of connector D3464 and connector D11452.

D3464		D11452
pin 9	. . . . .	pin 7

(e) Do a check of the wiring between these pins of connector D3464 and L1141:

D3464		L1141
pin 9	. . . . .	term 2

(f) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.
2) Raise and close the P5 Forward Overhead Panel to the closed position.
3) Re-connect connector D11452 to the P92 Power Distribution Panel 2.
4) Re-connect connector D3464 to the Supply Low Flow connector, M1915.
5) Do the Repair Confirmation at the end of this task.

(g) If you do not find a problem with the wiring, then do these steps:

1) Raise and close the P5 Forward Overhead Panel.
2) Re-connect connector D11452 to the P92 Power Distribution Panel 2 in the EE Compartment.
3) Re-connect connector D3464 to the Supply Low Flow Detector, M1915.

(5) Examine all of the ducts of the Equipment Cooling Supply System for Leakage or Blockage.

(a) Repair the leakage or blockage.

(b) Do the Repair Confirmation at the end of this task.

H. Repair Confirmation

(1) Put the EQUIP COOLING SUPPLY Switch to NORM (21-27 TASK SUPPORT Figure 301).

(2) Make sure the Normal Supply Fan operates .

(a) If the EQUIP COOLING SUPPLY OFF Light goes OFF in 20 seconds, then you corrected the problem.

(b) If the EQUIP COOLING SUPPLY OFF Light does not go OFF in 20 seconds, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

809. Equipment Cooling Overboard Exhaust Valve Does Not Open When Commanded - Fault Isolation

A. This task is for observed fault 222 222 22.

B. Description

OMR 701-799

(1) The flight crew must open the overboard exhaust valve when smoke/fumes are sensed in the aircraft. This causes differential pressure to remove the smoke/fumes from the aircraft. This procedure isolates the fault for conditions that prevent the overboard exhaust valve from opening when commanded. Set the L and R PACK switches to HIGH and the recirculation fan switch to OFF for the smoke removal.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826

(2) The flight crew must open the overboard exhaust valve when smoke/fumes are sensed in the aircraft. This causes differential pressure to remove the smoke/fumes from the aircraft. This procedure isolates the fault for conditions that prevent the overboard exhaust valve from opening when commanded. Set the L and R PACK switches to HIGH and the right recirculation fan switch to OFF for the smoke removal.

OMR 807

(3) The flight crew must open the overboard exhaust valve when smoke/fumes are sensed in the aircraft. This causes differential pressure to remove the smoke/fumes from the aircraft. This procedure isolates the fault for conditions that prevent the overboard exhaust valve from opening when commanded. Set the L and R PACK switches to HIGH for the smoke removal.

OMR ALL

(4) The acronym OEV is used in this procedure as an abbreviation for Overboard Exhaust Valve.

C. Possible Causes

(1) OEV, V157

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826; OMR 701-799, 802 POST SB 737-26A1122

(2) Smoke control relay R648

(3) OEV open enable relay, R650

OMR ALL

(4) Smoke control relay, R648

(5) Ground sensing relay, R592

OMR 701-799

(6) P5-10 air conditioning module (L or R PACK switch, RECIRC FAN switch, wiring)

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(7) P5-10 air conditioning module (L or R PACK switch, R RECIRC FAN switch, wiring)

OMR 807

(8) P5-10 air conditioning module (L or R PACK switch, wiring).

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826; OMR 701-799, 802 POST SB 737-26A1122

(9) Forward fire alarm latch relay, R945

(10) Recirculation fan switch bypass relay, R644

(11) Ground sensing relay, R589

OMR ALL

(12) Wiring

D. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

(a) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	15	C01355	LANDING GEAR AIR/GND SYS 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT
D	2	C01445	A/C OVERBOARD EXH VALVE RECONFIG CONT

(b) Make sure that this circuit breaker is closed:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	2	C01445	A/C OVERBOARD EXH VALVE RECONFIG CONT

E. Related Data

(1) SSM 21-43-21

(2) WDM 21-43-21

F. Preparation for the Initial Evaluation

(1) Do the steps that follow to get access to the equipment cooling OEV:

(a) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

(b) Remove the access panel in the EE bay on the top of the raised platform that is just aft and outboard of the EE compartment access door 117A

(c) If necessary, remove the crew oxygen system cylinder. Do this task: Crew Oxygen Cylinder Replacement, AMM TASK 12-15-21-600-801 or Crew Oxygen Cylinder Replacement, AMM TASK 12-15-21-600-801-003.

OMR 901-999; OMR 701-799, 802 PRE SB 737-26A1122

G. Initial Evaluation

(1) Do an operational test of the equipment cooling OEV smoke clearance mode. To do a test, do this task: Smoke Clearance Mode - Operational Test, AMM TASK 21-27-00-700-807

(a) If the OEV moved to the SMOKE position when commanded, then there was an intermittent fault.

(b) If the OEV did not move to the SMOKE position when commanded, then do the Fault Isolation Procedure - OEV Does Not Move to SMOKE Position When Commanded.

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826; OMR 701-799, 802 POST SB 737-26A1122

H. Initial Evaluation for Condition 1

(1) Condition 1 involves testing the operation of the OEV when the close signal goes through the OEV open control relay R649.

(2) Do the steps that follow to command the OEV to move to the SMOKE position:

(a) Put the airplane in the air mode.

(b) Make sure pneumatic power is removed.

(c) Put the L PACK switch in HIGH with R PACK switch in AUTO or put the R PACK switch in HIGH with L PACK switch in AUTO.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 805, 809, 810, 812, 813, 815, 816, 818-826; OMR 802 POST SB 737-26A1122

(d) Put the R RECIRC FAN switch in the AUTO position.

OMR 701-799 POST SB 737-26A1122

(e) Put the RECIRC FAN switch in the AUTO position.

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826; OMR 701-799, 802 POST SB 737-26A1122

(3) If the OEV moved to the SMOKE position when commanded, then there was an intermittent fault.

(4) If the OEV did not move to the SMOKE position when commanded, then do the Fault Isolation Procedure - OEV Does Not Move to SMOKE Position When Commanded for Condition 1.

I. Initial Evaluation for Condition 2

(1) Condition 2 involves evaluating the operation of the OEV when the close signal goes through the smoke control relay R648.

(2) Do the steps that follow to command the OEV to move to the SMOKE position:

(a) Put the airplane in the air mode.

(b) Make sure pneumatic power is removed.

(c) Put the L PACK switch in HIGH.

(d) Put the R PACK switch to OFF.

OMR 805, 809, 810, 812, 813, 815, 816, 818-826; OMR 802 POST SB 737-26A1122

(e) Put the R RECIRC FAN switch to OFF.

OMR 701-799 POST SB 737-26A1122

(f) Put the RECIRC FAN switch to OFF.

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826; OMR 701-799, 802 POST SB 737-26A1122

(g) Push and hold the TEST switch on the P8 Cargo Fire Control Panel until the FWD and AFT red cargo fire lights come on.

(3) Make sure that the DCPC display shows CARGO HT VLV HIGH which indicates the OEV moved to the SMOKE position.

(a) If the OEV moved to the SMOKE position when commanded, then there was an intermittent fault.

(b) If the OEV did not move to the SMOKE position when commanded, then do the Fault Isolation Procedure - OEV Does Not Move to SMOKE Position When Commanded for Condition 2.

OMR 901-999; OMR 701-799, 802 PRE SB 737-26A1122

J. Fault Isolation Procedure - OEV Does Not Move to SMOKE Position When Commanded.

(1) Make sure the airplane is not pressurized when you do this fault isolation.

(2) Do the steps that follow to get access to the equipment cooling OEV:

(a) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

(b) Remove the access panel in the EE bay on the top of the raised platform that is just aft and outboard of the EE access door, 117A.

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OMR 901-999; OMR 701-799, 802 PRE SB 737-26A1122 (Continued)

(c) If necessary, remove the crew oxygen system cylinder. To remove the oxygen cyclinder, do this task: AMM TASK 12-15-21-600-801.

WARNING

OBEY THE PROCEDURE THAT PUTS THE AIRPLANE IN THE AIR MODE. IF YOU DO THE PROCEDURE INCORRECTLY, INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Put the airplane in the air mode: To put the airplane in the air mode, do this task: AMM TASK 32-09-00-860-801.

(4) Set the L PACK switch on the P5-10 air conditioning panel to HIGH.

OMR 701-799 PRE SB 737-26A1122

(5) Set the RECIRC FAN switch on the P5-10 panel to OFF.

OMR 901-999; OMR 802 PRE SB 737-26A1122

(6) Set the R RECIRC FAN switch on the P5-10 panel to OFF.

OMR 901-999; OMR 701-799, 802 PRE SB 737-26A1122

(7) Make sure that the OEV position indicator shows that the valve is in the SMOKE postion.

(8) Do this check for 28 Vdc at the OEV:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

(b) Disconnect electrical connector D11880 from the OEV.

(c) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

(d) Do a check for 28 Vdc on the aircraft's wiring side of connector D11880 between pins 2 and 3 (ground).

(9) If there is 28 Vdc betweens pins 2 and 3 of connector D11880, then replace the equipment cooling OEV. To replace the valve, do these tasks: AMM TASK 21-27-05-000-801 and AMM TASK 21-27-05-400-801.

(a) Do the Repair Confirmation at the end of this task.

1) If the Repair Confirmation was satisfactory, then you corrected the fault.

2) If the Repair Confirmation was not satisfactory, then continue.

(10) If 28 Vdc is not present between pins 1 and 3 of connector D11880, then do this check of the ground circuit wiring:

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OMR 901-999; OMR 701-799, 802 PRE SB 737-26A1122 (Continued)

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

(b) Do a check of the wiring between pin 3 of connector D11880 and the ground GD1008-DC.

(c) If you find a problem with the wiring, then do these steps:

1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

2) Repair the wire.

3) Connect electrical connector D11880 to the OEV.

4) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

5) Do the Repair Confirmation at the end of this task:

a) If the Repair Confirmation is satisfactory, then you corrected the problem.

b) If the Repair Confirmation is not satisfactory, then continue.

(d) If the ground circuit wiring check is satisfactory, then continue.

(11) Do this check of the OEV open enable relay, R648, in the J24 junction box:

(a) Open this access panel:

Number	Name/Location
114AC	Fwd Nose Wheel Well Upper Access Panel

(b) Make sure that the airplane is still in the air mode.

(c) Make sure that the L PACK switch is set to HIGH.

OMR 701-799 PRE SB 737-26A1122

(d) Make sure that the RECIRC FAN switch is set to OFF.

OMR 901-999; OMR 802 PRE SB 737-26A1122

(e) Make sure that the R RECIRC FAN switch is set to OFF.

OMR 901-999; OMR 701-799, 802 PRE SB 737-26A1122

(f) Remove the cover from the J24 junction box.

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OMR 901-999; OMR 701-799, 802 PRE SB 737-26A1122 (Continued)

(g) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

(h) Remove relay R648 from the A11 position.

(i) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

(j) Do a check for 28 Vdc between pins X1 and X2 of connector D11874.

1) If 28 Vdc is present between pins X1 and X2 of connector D11874, then replace relay R648.

a) Do the Repair Confirmation at the end of this task.

<1> If the Repair Confirmation is satisfactory, then you corrected the fault.

<2> If the Repair Confirmation was not satisfactory, then continue.

2) If 28 Vdc is not present between pins X1 and X2 of connector D11874, then do these checks:

a) Make sure that there is 28 Vdc available on the load side of circuit breaker C1444 in the P6-4 circuit breaker panel.

<1> If 28 Vdc is not present between pins X1 and X2 of connector D11874, fault isolate the 28 Vdc electrical distribution system.

b) Get access to the back of the P5-10 air conditioning panel.

c) Disconnect electrical connector D680 from the back of the P5-10 air conditioning panel.

d) Do a continuity check between these connections (refer to WDM 21-43-21):

P5-10	**J24**
D680	**D11874**
51	X1

P5-10	**P5-10**
D680	**D680**
52	48

e) Repair problems that you find.

(12) Do these checks of the wiring as follows:

(a) Make sure that this circuit breaker is open and has safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 901-999; OMR 701-799, 802 PRE SB 737-26A1122 (Continued)

(b) Do a continuity check between these connections:

J24	**P6-4**
	Circuit Breaker
D11874	**C1444**
C2	Circuit Breaker load side

	Overboard
	Exahaust Valve,
J24	**V157**
D11784	**D11880**
C1	pin 2

(c) Repair problems that you find.

(d) If you made repairs to the wiring, do the Repair Confirmation at the end of this task.

1) If the Repair Confirmation is satisfactory, then you corrected the fault.

2) If the Repair Confirmation is not satisfactory, then continue.

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826; OMR 701-799, 802 POST SB 737-26A1122

K. Fault Isolation Procedure for Condition 1- OEV Does Not Move to Smoke Position When Commanded

(1) This procedure fault isolates the Condition 1 when the OEV does not move to the smoke position when commanded under the conditions that follow:

(a) Airplane in the air mode

(b) Pneumatic power is removed

(c) L PACK switch is in HIGH with R PACK switch in AUTO or R PACK switch is in HIGH with L PACK switch in AUTO.

OMR 805, 809, 810, 812, 813, 815, 816, 818-826; OMR 802 POST SB 737-26A1122

(d) R RECIRC FAN switch is in AUTO

OMR 701-799 POST SB 737-26A1122

(e) RECIRC FAN switch is in AUTO

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826; OMR 701-799, 802 POST SB 737-26A1122

(2) Make sure the airplane is not pressurized when you do this fault isolation.

(3) Get access to the equipment cooling OEV as follows:

(a) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

(b) Remove the access panel in the EE bay on the top of the raised platform that is just aft and outboard of the EE access door, 117A.

(c) If necessary, remove the crew oxygen system cylinder. To remove the oxygen cylinder, do this task: Crew Oxygen Cylinder Replacement, AMM TASK 12-15-21-600-801 or Crew Oxygen Cylinder Replacement, AMM TASK 12-15-21-600-801-003.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING OBEY THE PROCEDURE THAT PUTS THE AIRPLANE IN THE AIR MODE. IF YOU DO THE PROCEDURE INCORRECTLY, INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

(4) Put the airplane in the air mode. Do this task: Put the Airplane in the Air Mode, AMM TASK 32-09-00-860-801.

(5) Set the L PACK switch on the P5-10 air conditioning panel to HIGH.

OMR 701-799 POST SB 737-26A1122

(6) Set the RECIRC FAN switch on P5-10 air conditioning panel to AUTO.

OMR 805, 809, 810, 812, 813, 815, 816, 818-826; OMR 802 POST SB 737-26A1122

(7) Set the R RECIRC FAN switch on P5-10 air conditioning panel to AUTO.

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826; OMR 701-799, 802 POST SB 737-26A1122

(8) Do this check for 28 V dc at the OEV connector D11880:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

(b) Disconnect electrical connector D11880 from the OEV V157.

(c) Close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

(d) Do a check for 28 Vdc on the aircraft's wiring side of connector D11880 between pins 2 and 3.

(9) If there is 28 Vdc between pins 2 and 3 of connector D11880, then replace the equipment cooling OEV, V157. To replace the valve, do these tasks: Equipment Cooling Overboard Exhaust Valve Removal, AMM TASK 21-27-05-000-801 and Equipment Cooling Overboard Exhaust Valve Installation, AMM TASK 21-27-05-400-801.

(a) Do the Repair Confirmation at the end of this task.

1) If the Repair Confirmation was satisfactory, then you corrected the fault.

2) If the Repair Confirmation was not satisfactory, then continue.

(10) If 28 Vdc is not present between pins 2 and 3 of connector D11880, then do this check of the ground circuit wiring:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Disconnect electrical connector D11880 from the OEV V157.

(c) Do a check of the wiring between pin 3 of connector D11880 and ground GD 1008-DC.

(d) If you find a problem with the wiring or the ground, then do these steps:

1) Repair problems that you find.

2) Connect the electrical connector D11880 to the OEV, V157.

3) Close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

4) Do the Repair Confirmation at the end of this task:

a) If the Repair Confirmation is satisfactory, then you corrected the problem.

b) If the Repair Confirmation is not satisfactory, then continue.

(e) If the ground circuit wiring check is satisfactory, then continue.

(11) Do this check for 28 Vdc at the OEV command relay R650 in the J24 junction box:

(a) Open this access panel:

Number	Name/Location
114AC	Fwd Nose Wheel Well Upper Access Panel

(b) Remove the cover from the J24 junction box.

(c) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

(d) Make sure the airplane is still in the air mode.

(e) Make sure the L PACK switch is still in the HIGH position.

OMR 805, 809, 810, 812, 813, 815, 816, 818-826; OMR 802 POST SB 737-26A1122

(f) Make sure the R RECIRC FAN switch is in the AUTO position.

OMR 701-799 POST SB 737-26A1122

(g) Make sure the RECIRC FAN switch is in the AUTO position.

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826; OMR 701-799, 802 POST SB 737-26A1122

(h) Remove relay R650 from the B06B position.

(i) Close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

(j) Do a check for 28 Vdc between pins X1 and X2 of connector D11878.

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1) If 28 Vdc is present between pins X1 and X2 of connector D11878, then replace relay R650.
   a) Do the Repair Confirmation at the end of this task.
      <1> If the Repair Confirmation is satisfactory, then you corrected the fault.
      <2> If the Repair Confirmation is not satisfactory, then continue.
2) If 28 Vdc is not present between pins X1 and X2 of connector D11878, then do these checks:
   a) Open this circuit breaker and install safety tag:

      F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

   b) Do a check of the ground from pin X2 of connector D11878 on the aircraft's wiring to the ground GD544-DC.
      <1> Repair problems you find.
   c) Do a check for 28 Vdc at pin B2 of connector D11878.
      <1> If 28 Vdc is not present at pin B2 of connector D11878, do a check of the wiring between pin B2 of connector D11878.
      <2> Repair problems you find.
3) Install relay R650.
4) Do the Repair Confirmation at the end of this task.
   a) If the Repair Confirmation is satisfactory, then you corrected the fault.

L. Fault Isolation Procedure for Condition 2 - OEV Does Not Move to SMOKE Position When Commanded

NOTE: Condition 2 isolates for faults when the close signal goes through the smoke control relay R648.

(1) Do the steps that follow to get access to the equipment cooling OEV:
   (a) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

   (b) Remove the access panel in the EE bay on the top of the raised platform that is aft and outboard of the EE access door, 117A.
   (c) If necessary, remove the crew oxygen system cylinder. To remove the oxygen cyclinder, do this task: AMM TASK 12-15-21-600-801 or AMM TASK 12-15-21-600-801-003.
(2) Put the airplane in the air mode. Do this task: Put the Airplane in the Air Mode, AMM TASK 32-09-00-860-801.
(3) Set the L PACK switch on the P5-10 air conditioning panel to HIGH.
(4) Set the R PACK switch on the P5-10 air conditioning panel to OFF.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799 POST SB 737-26A1122

(5) Set the RECIRC FAN switch on the P5-10 air conditioning panel to OFF.

OMR 805, 809, 810, 812, 813, 815, 816, 818-826; OMR 802 POST SB 737-26A1122

(6) Set the R RECIRC FAN switch on the P5-10 air conditioning panel to OFF.

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826; OMR 701-799, 802 POST SB 737-26A1122

(7) Do this check for 28 Vdc at the OEV:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

(b) Disconnect electrical connector D11880 from the OEV V157.

(c) Close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

(d) Do a check for 28 Vdc on the aircraft's wiring side of connector D11880 between pins 2 and 3 (ground).

(8) If 28 Vdc is present between pins 2 and 3 of connector D11880, then replace the OEV V157 as follows:

(a) Do these tasks to replace the valve: Equipment Cooling Overboard Exhaust Valve Removal, AMM TASK 21-27-05-000-801 and Equipment Cooling Overboard Exhaust Valve Installation, AMM TASK 21-27-05-400-801.

(b) Do the Repair Confirmation at the end of this task.

1) If the Repair Confirmation was satisfactory, then you corrected the fault.

2) If the Repair Confirmation was not satisfactory, then continue.

(9) If 28 Vdc is not present between pins 2 and 3 of connector D11880, then do this check of the ground circuit wiring:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

(b) Do a check of the wiring between pin 3 of connector D11880 on the aircraft's side and the ground GD1008-DC.

(c) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Connect electrical connector D11880 to the OEV V157.

3) Close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

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4) Do the Repair Confirmation at the end of this task:
   a) If the Repair Confirmation is satisfactory, then you corrected the fault.
   b) If the Repair Confirmation is not satisfactory, then continue.

(10) Do this check of the smoke control relay R648 in the J24 junction box:

(a) Open this access panel:

Number	Name/Location
114AC	Fwd Nose Wheel Well Upper Access Panel

(b) Remove the cover from the J24 junction box.

(c) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

(d) Remove relay R648 from the A11 position.

(e) Close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

(f) Do a check for 28 Vdc between pins X1 and X2 of connector D11874 on the aircraft's side wiring.

(11) If 28 Vdc is present between pins X1 and X2 of connector D11874, then replace relay R648.

(a) Do the Repair Confirmation at the end of this task.
   1) If the Repair Confirmation is satisfactory, then you corrected the fault.
   2) If the Repair Confirmation is not satisfactory, then continue.

(12) If 28 Vdc is not present between pins X1 and X2 of connector D11874, then do this check of the ground circuit wiring:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

(b) Do a check of the wiring between pin X2 of connector D11874 and the ground GD544-DC.

(c) If you find a problem with the wiring, then do these steps:
   1) Repair the wiring.
   2) Connect the electrical connector D11874.

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3) Close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

4) Do the Repair Confirmation at the end of this task:
   a) If the Repair Confirmation is satisfactory, then you corrected the fault.
   b) If the Repair Confirmation is not satisfactory, then continue.

(13) Do this check for 28 Vdc to pin B2 on connector D11874 on smoke control relay R648:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

(b) Open this access panel:

Number	Name/Location
114AC	Fwd Nose Wheel Well Upper Access Panel

(c) Remove the cover from the J24 junction box.

(d) Remove smoke control relay R648.

(e) Close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

(f) Do a check for 28 Vdc at pin B2 of connector D11874.

(14) If 28 Vdc is not present at pin B2 of connector D11874, make sure that 28 Vdc is available on the load side of circuit breaker C01444.

(a) If 28 Vdc is not available on the load side of circuit breaker C01444, fault isolate the 28 Vdc battery bus, section 2.
   1) Repair problems you find.
   2) Do the Repair Confirmation at the end of this task:
      a) If the Repair Confirmation is satisfactory, then you corrected the fault.
      b) f the Repair Confirmation is not satisfactory, then continue.

(b) If 28 Vdc is available on the load side of circuit breaker C01444, do these steps:
   1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

   2) Do this continuity check:

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	J24 Junction
P6-4 C/B Panel	**Box**
C01444	**D11874**
Load side	pin B2

3) Repair problems you find.
4) Reinstall relay R648.
5) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

6) Do the Repair Confirmation at the end of this task:
   a) If the Repair Confirmation is satisfactory, then you corrected the fault.
   b) If the Repair Confirmation is not satisfactory, then continue.

(15) Do this check for 28 Vdc input to the P5-10 air conditioning panel:
   (a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

   (b) Get access to the back of the P5-10 air conditioning panel.
   (c) Disconnect connector D646.
   (d) Close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

   (e) Do a check for 28 Vdc at pin 30 of connector D646 on the aircraft's wiring.

(16) If 28 Vdc is not present at pin 30 of connector D646 on the aircraft's wiring, do these steps:
   (a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

   (b) Do a continuity check between the load side of circuit breaker C01444 and pin 30 of connector D646 on the aircraft's wiring:
      1) If you find a problem, do these steps:
         a) Repair the wiring.
         b) Do the Repair Confirmation at the end of this task:
            <1> If the Repair Confirmation is satisfactory, then you corrected the fault.
            <2> If the Repair Confirmation is not satisfactory, then continue.

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2) If there is continuity, do these steps:
   a) Reconnect connector D646.
   b) Continue.

(17) Do this check for 28 Vdc at ground sensing relay air/gnd system 1 R592 in the J22 junction box:
   (a) Open this access panel:

Number	Name/Location
114AC	Fwd Nose Wheel Well Upper Access Panel

   (b) Remove the cover from the J22 junction box.
   (c) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

   (d) Remove the air/gnd system 1 relay R592 from position A13.
   (e) Close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

   (f) Do a check for 28 Vdc between pin B2 of connector D11018 and a structural ground.

(18) If 28 Vdc is not present between pin B2 of connector D11018 and a structural ground, do these steps:
   (a) Get access to the back of the P5-10 air conditioning panel.
   (b) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

   (c) Disconnect electrical connector D680 from the back of the P5-10 air conditioning panel.
   (d) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

   (e) Do a check for 28 Vdc between pin 51 of connector D680 and a structural ground.

(19) If 28 Vdc is present between pin 51 of connector D680 and a structural ground, do a check of the wiring between pin 51 of connector D680 and pin B2 of connector D11018 on the air/gnd relay R592.
   (a) Repair problems you find.
   (b) If you made repairs to the wiring, do these steps:

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1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

2) Connect the connector D680 to the back of the P5-10 air conditioning panel.

3) Install the air/gnd relay R592 in its position in the J22 junction box.

4) Do the Repair Confirmation at the end of this task.

a) If the Repair Confirmation is satisfactory, then you corrected the fault.

b) If the Repair Confirmation is not satisfactory, then continue.

(20) If 28 Vdc is not present between pin 51 of connector D680 and a structural ground, do these steps:

(a) Disconnect connector D646 from the back of the P5-10 air conditioning panel.

(b) Do a continuity check of the P5-10 panel internal wiring between pin 51 of connector D680 and pin 30 of connector D646.

(c) If there is no continuity, do these tasks to replace the air conditioning module: Air Conditioning Module Removal, AMM TASK 21-51-65-000-801 and Air Conditioning Module Installation, AMM TASK 21-51-65-400-801.

(d) Do the Repair Confirmation at the end of this task:

1) If the Repair Confirmation is satisfactory, then you corrected the fault.

2) If the Repair Confirmation is not satisfactory, then continue.

(21) Do this check for 28 Vdc at recirculation fan switch bypass relay R644 in the J24 junction box as follows:

(a) If not already done, do these steps:

1) Open this access panel:

Number	Name/Location
114AC	Fwd Nose Wheel Well Upper Access Panel

2) Remove the cover from the J24 junction box.

(b) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	2	C01445	A/C OVERBOARD EXH VALVE RECONFIG CONT

(c) Remove recirc fan switch bypass relay R644.

(d) Close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	2	C01445	A/C OVERBOARD EXH VALVE RECONFIG CONT

(e) Do a check for 28 Vdc between pins 6 and 12 of connector D11866 on the aircraft's side wiring.

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(22) If 28 Vdc is present between pins 6 and 12 of connector D11866, then replace relay R644.

(a) Do the Repair Confirmation at the end of this task.

1) If the Repair Confirmation is satisfactory, then you corrected the fault.

2) If the Repair Confirmation is not satisfactory, then continue.

(23) If 28 Vdc is not present between pins 6 and 12 of connector D11866, then do this check of the ground circuit wiring:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	2	C01445	A/C OVERBOARD EXH VALVE RECONFIG CONT

(b) Do a check of the wiring between pin 12 of connector D11866 and the ground GD544-DC.

(c) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Reinstall the relay R644.

3) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	2	C01445	A/C OVERBOARD EXH VALVE RECONFIG CONT

4) Do the Repair Confirmation at the end of this task.

a) If the Repair Confirmation is satisfactory, then you corrected the fault.

b) If the Repair Confirmation is not satisfactory, then continue.

(24) Do this check for 28 Vdc at the forward fire alarm latch relay R945 in the J24 junction box as follows:

(a) If not already done, do these steps:

1) Open this access panel:

Number	Name/Location
114AC	Fwd Nose Wheel Well Upper Access Panel

2) Remove the cover from the J24 junction box.

(b) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

(c) Remove relay R945.

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(d) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

(e) Do a check for 28 Vdc between pin 7 of D12225 and a ground to structure.

(25) If there is 28 Vdc between pin 7 on connector D12225 and a ground, then replace relay R945.

(a) Do the Repair Confirmation at the end of this task:

1) If the Repair Confirmation is satisfactory, then you corrected the fault.

2) If the Repair Confirmation is not satisfactory, then continue.

(26) Do this check for 28 Vdc at the aft fire alarm latch relay R946 in the J24 junction box as follows:

(a) If not already done, do these steps:

1) Open this access panel:

Number	Name/Location
114AC	Fwd Nose Wheel Well Upper Access Panel

2) Remove the cover from the J24 junction box.

(b) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

(c) Remove relay R946.

(d) Close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

(e) Do a check for 28 Vdc between pin 7 of D12227 and a ground to structure.

(27) If there is 28 Vdc between pin 6 connector D12227 and a ground, then replace relay R946.

(a) Do the Repair Confirmation at the end of this task.

1) If the Repair Confirmation is satisfactory, then you corrected the fault.

2) If the Repair Confirmation is not satisfactory, then continue.

(28) Do these steps to make sure the voltage to energize R945 finds the ground in the P8-75 cargo smoke detection & fire suppression module:

(a) Make sure the airplane is in the air mode.

(b) Make sure these switches are in the positions noted:

1) L PACK is in HIGH

2) R PACK is in OFF

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OMR 701-799 POST SB 737-26A1122

3) RECIRC FAN switch is in OFF

OMR 805, 809, 810, 812, 813, 815, 816, 818-826; OMR 802 POST SB 737-26A1122

4) R RECIRC FAN switch is in OFF

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826; OMR 701-799, 802 POST SB 737-26A1122

(c) Push and hold the TEST switch on the P8 Cargo Fire Control Panel until the FWD and AFT red cargo fire lights come on.

(29) If the FWD and AFT red cargo fire lights do not come on, do these steps:

(a) If not already done, do these steps:

1) Open this access panel:

Number	Name/Location
114AC	Fwd Nose Wheel Well Upper Access Panel

2) Remove the covers from the J22 and J24 junction boxes.

3) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

4) Remove ground sensing relay R589 from J22 junction box.

5) Remove forward fire alarm latch relay R945 from J24 junction box.

6) Make sure that there is continuity between pins B2 and B3 and between pins A2 and A3 of relay R589.

a) Secure a new relay if continuity does not exist.

7) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	16	C01523	CARGO FIRE FWD DET B
C	17	C01522	CARGO FIRE FWD DET A
C	18	C01525	CARGO FIRE AFT DET B
C	19	C01524	CARGO FIRE AFT DET A

8) Disconnect connector D12760 from the P8-75 cargo smoke detection & fire suppression module.

9) Do the wiring checks that follow:

J24/J22 Junction Boxes

R945		R589
D12225		D11012
pin 12	. . . . .	pin B3

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J22 Junction Box/P8-75 Module

Relay R589		P8-75 Module
D11012		D12760
pin B2	. . . . .	pin 58

10) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Install the relays R945 and R589.

c) Connect the connector D12760 to the P8-75 cargo smoke detection & fire suppression module.

d) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	16	C01523	CARGO FIRE FWD DET B
C	17	C01522	CARGO FIRE FWD DET A
C	18	C01525	CARGO FIRE AFT DET B
C	19	C01524	CARGO FIRE AFT DET A

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

e) Do the Repair Confirmation at the end of this task:

<1> If the Repair Confirmation is satisfactory, then you corrected the fault.

<2> If the Repair Confirmation is not satisfactory, then continue.

(30) Refer to WDM 21-43-21 before you do wiring checks as follows:

If there is an isolation diode in the wiring run to be checked, put the positive lead of the multimeter on the contact point closest to the anode side of the diode and the negative lead on the contact point closest to the cathode side of the diode.

NOTE: The cathode end of the diode is the end with the stripe on it.

(31) Do the wiring checks that follow :

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

(b) Do a continuity check between these connections:

J24 Junction Box

R648		R644
D11874		D11866
pin B1	. . . . .	pin 5

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J24 Junction Box

R644	**R945**
D11866	**D12225**
pin 4	pin 7

J24 Junction Box

R945	**R946**
D12225	**D12227**
pin 14	pin 7

J24 Junction Box

R946	**R650**
D12227	**D11878**
pin 14	X1

R650	**OEV V157**
D11878	**D11880**
pin B1	pin 2

(c) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

3) Do the Repair Confirmation at the end of this task:

a) If the Repair Confirmation is satisfactory, then you corrected the fault.

OMR ALL

M. Repair Confirmation

(1) Reconnect all electrical connectors that were disconnected.

(2) Reinstall all components that were removed.

(3) Close all circuit breakers that were opened.

(4) Do an operational test of the OEV. Do this task: Smoke Clearance Mode - Operational Test, AMM TASK 21-27-00-700-807.

(5) If the Repair Confirmation is unsatisfactory, return to the step in the Fault Isolation Procedure that you were at prior to performing the Repair Confirmation and continue the Fault Isolation Procedure.

(6) If the Repair Confirmation is satisfactory, make sure that you do all the applicable steps that follow to return the system to a serviceable condition:

(7) Install all access panels that were removed.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826; OMR 701-799, 802 POST SB 737-26A1122

810. Equipment Cooling Supply Fans Do Not Shut Off When Commanded During a Cargo Compartment Smoke Alert - Fault Isolation

A. Description

OMR 805, 809, 810, 812, 813, 815, 816, 818-826; OMR 701-799, 802 POST SB 737-26A1122

(1) The flight crew must open the overboard exhaust valve when smoke/fumes are sensed in the aircraft. This causes differential pressure to remove the smoke/fumes from the aircraft. This procedure also closes the equipment cooling supply fan for five minutes. This procedure isolates the fault for conditions that prevent the equipment cooling supply fan from shutting down when commanded. Set one of the PACK switches to HIGH, the recirculation fan switch to OFF, and the necessary procedure on the P8 Cargo Fire Control Panel for the smoke removal.

OMR 807

(2) The flight crew must open the overboard exhaust valve when smoke/fumes are sensed in the aircraft. This causes differential pressure to remove the smoke/fumes from the aircraft. This procedure also closes the equipment cooling supply fan for five minutes. This procedure isolates the fault for conditions that prevent the equipment cooling supply fan from shutting down when commanded. Set one of the PACK switches to HIGH and the necessary procedure on the P8 Cargo Fire Control Panel for the smoke removal.

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826; OMR 701-799, 802 POST SB 737-26A1122

(3) This fault isolation task examines the circuit from the forward fire alarm latch relay R945 in the J24 junction box to the supply system control interrupt relay R645 in the J20 junction box. Faults in the system circuitry at the R945 relay and upstream of the R945 relay are examined in FIM 21-27, Task 809.

B. Possible Causes

(1) Supply System Control Interrupt Relay R645

(2) Wiring

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

(a) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	12	C01116	EQPT COOLING SUPPLY FAN CONT-NORMAL
C	13	C01117	EQPT COOLING SUPPLY FAN CONTROL-ALTN
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT
D	2	C01445	A/C OVERBOARD EXH VALVE RECONFIG CONT

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
A	8	C00935	EQPT CLG SPLY FAN PWR-ALTN

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Power Distribution Panel Number 2, P92

Row	Col	Number	Name
OMR 701-799, 802 POST SB 737-26A1122			
A	8	C00934	EQPT CLG SPLY FAN PWR-NORM
OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826			
D	10	C00934	EQPT CLG SPLY FAN PWR-NORM

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826; OMR 701-799, 802 POST SB 737-26A1122

D. Related Data

(1) SSM 21-43-21

(2) SSM 21-27-31

(3) WDM 21-43-21

(4) WDM 21-27-31

E. Initial Evaluation

(1) Do an operational test of the equipment cooling overboard exhaust valve smoke clearance mode. Do this task: Smoke Clearance Mode - Operational Test, AMM TASK 21-27-00-700-807.

(a) If the equipment cooling supply fan shuts off when commanded, then there was an intermittent fault.

(2) If the equipment cooling supply fan does not shut off when commanded, do the Fault Isolation Procedure - Equipment Cooling Supply Fans Do Not Shut Off When Commanded During a Cargo Compartment Smoke Alert.

F. Fault Isolation Procedure - Equipment Cooling Supply Fans Do Not Shut Off When Commanded During a Cargo Compartment Smoke Alert.

(1) Do the steps that follow to command the overboard exhaust valve to the SMOKE position which also shuts off the operating supply fan for 5 minutes:

(a) Make sure that pneumatic power is removed:

1) Make sure that the L and R indicators on the duct pressure gauge on the P5-10 air conditioning panel indicate zero psi.

(b) Put the L PACK switch in HIGH.

(c) Put the R PACK switch in OFF.

OMR 701-799 POST SB 737-26A1122

(d) Put the RECIRC FAN switch in OFF.

OMR 805, 809, 810, 812, 813, 815, 816, 818-826; OMR 802 POST SB 737-26A1122

(e) Put the R RECIRC FAN switch in OFF.

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826; OMR 701-799, 802 POST SB 737-26A1122

(f) Do this task to put the airplane in the air mode: Put the Airplane in the Air Mode, AMM TASK 32-09-00-860-801.

(g) Push and hold the TEST switch on the P8 Cargo Fire Control Panel until the FWD and AFT red cargo fire lights come on.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(2) Do this check for 28 Vdc at the supply system control interrupt relay R645 in the J20 junction box:

(a) Open this access panel:

Number	Name/Location
114AC	Fwd Nose Wheel Well Upper Access Panel

(b) Remove the cover from the J20 junction box.

(c) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

(d) Remove supply system control interrupt relay R645.

(e) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

(f) Push and hold the TEST switch on the P8 Cargo Fire Control Panel until the FWD and AFT red cargo fire lights come on.

(g) Do a check for 28 Vdc between pins X1 and X2 of connector D11868.

(3) If 28 Vdc is present between pins X1 and X2 of connector D11868, then replace relay R645.

(a) Do the Repair Confirmation at the end of the task.

1) If the Repair Confirmation is satisfactory, then you corrected the fault.

2) If the Repair Confirmation is not satisfactory, then continue.

(4) If 28 Vdc is not present between pins X1 and X2 of connector D11868, then do this check of the ground circuit wiring:

Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

(a) Do a check of the wiring between pin X2 of connector D11874 and the ground GD528-DC.

(b) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Install the relay R645.

3) Reconnect connector D11868.

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4) Close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

5) Do the Repair Confirmation at the end of this task:
   a) If the Repair Confirmation is satisfactory, then you corrected the fault.
   b) If the Repair Confirmation is not satisfactory, then continue.

(5) Do a check of the wiring (refer to WDMs 21-27-31 and 21-43-21) between the supply system control interrupt relay R645 in the J20 junction box and the forward fire alarm latch relay R945 in the J24 junction box as follows:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

(b) Remove the cover from the J24 junction box.

(c) Remove forward fire alarm latch relay R945.

(d) Remove supply system control interrupt relay R645 from J20 junction box.

(e) Do a continuity check from pin 8 of relay R945 to pin X1 of relay R645.

(f) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Install relay R645 in the J20 junction box and relay R945 in the J24 junction box.

3) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

4) Do the Repair Confirmation at the end of this task:
   a) If the Repair Confirmation is satisfactory, then you corrected the fault.

G. Repair Confirmation

(1) Connect all electrical connectors that were disconnected.

(2) Install all components that were removed.

(3) Close all circuit breakers that were opened.

(4) Do an operational test of the overboard exhaust valve. To do the operational test, do this task: Smoke Clearance Mode - Operational Test, AMM TASK 21-27-00-700-807.

(5) If the Repair Confirmation is unsatisfactory, return to the step in the Fault Isolation Procedure that you were at prior to performing the Repair Confirmation and continue the Fault Isolation Procedure.

(6) If the Repair Confirmation is satisfactory, make sure that you do all the necessary steps to return the system to a serviceable condition:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(7) Do this task to put the airplane in the ground mode: Return the Airplane to the Ground Mode, AMM TASK 32-09-00-860-802.

(8) Install all access panels that were removed.

——— END OF TASK ———

811. Equipment Cooling Exhaust Fans Do Not Shut Off When Commanded During a Cargo Compartment Smoke Alert - Fault Isolation

A. Description

OMR 805, 809, 810, 812, 813, 815, 816, 818-826; OMR 802 POST SB 737-26A1122

(1) The flight crew must open the overboard exhaust valve when smoke/fumes are sensed in the aircraft. This causes differential pressure to remove the smoke/fumes from the aircraft. This procedure also closes the equipment cooling exhaust fan during the flight. This procedure isolates the fault for conditions that prevent the equipment cooling exhaust fan from shutting down when commanded. Set one of the PACK switches to HIGH, the recirculation fan switch to OFF, and the necessary procedure on the P8 cargo fire control panel for the smoke removal.

OMR 807

(2) The flight crew must open the overboard exhaust valve when smoke/fumes are sensed in the aircraft. This causes differential pressure to remove the smoke/fumes from the aircraft. This procedure also closes the equipment cooling exhaust fan during the flight. This procedure isolates the fault for conditions that prevent the equipment cooling exhaust fan from shutting down when commanded. Set one of the PACK switches to HIGH and the necessary procedure on the P8 cargo fire control panel for the smoke removal.

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826; OMR 701-799, 802 POST SB 737-26A1122

(3) This fault isolation task examines the circuit from the forward fire alarm latch relay R945 in the J24 junction box to the exhaust system control interrupt relay R949 in the J24 junction box. Faults in the system circuitry at the R945 relay and upstream of the R945 relay are examined in FIM 21-27, Task 809.

B. Possible Causes

(1) Exhaust system control interrupt relay R949

(2) Wiring

C. Circuit Breakers

(1) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	14	C01435	EQPT COOLING EXHAUST FAN CONT NORMAL
C	15	C01436	EQPT COOLING EXHAUST FAN CONT ALTN
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT
D	2	C01445	A/C OVERBOARD EXH VALVE RECONFIG CONT

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
E	1	C00836	EQPT CLG EXH FAN PWR-NORM

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
OMR 701-799, 802 POST SB 737-26A1122			
A	10	C00837	EQPT CLG EXH FAN PWR-ALTN
OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826			
D	12	C00837	EQPT CLG EXH FAN PWR-ALTN

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826; OMR 701-799, 802 POST SB 737-26A1122

D. Related Data

(1) SSM 21-43-21

(2) SSM 21-27-21

(3) WDM 21-43-21

(4) WDM 21-27-21

E. Initial Evaluation

(1) Do an operational test of the equipment cooling overboard exhaust valve smoke clearance mode. Do this task: Smoke Clearance Mode - Operational Test, AMM TASK 21-27-00-700-807.

(a) If the equipment cooling exhaust fan shuts off when commanded, then there was an intermittent fault.

(2) If the equipment cooling exhaust fan does not shut off when commanded, then do the Fault Isolation Procedure - Equipment Cooling Supply Fans Do Not Shut Off When Commanded During a Cargo Compartment Smoke Alert.

F. Fault Isolation Procedure - Equipment Cooling Exhaust Fans Do Not Shut Off When Commanded During a Cargo Compartment Smoke Alert.

(1) Do the steps that follow to command the overboard exhaust valve to the SMOKE position which also shuts off the operating supply fan for the duration of the flight:

(a) Make sure that pneumatic power is removed:

1) Make sure that the L and R indicators on the duct pressure gauge on the P5-10 air conditioning panel indicate zero psi.

OMR 701-799 POST SB 737-26A1122

(b) Put the RECIRC FAN switch in OFF.

OMR 805, 809, 810, 812, 813, 815, 816, 818-826; OMR 802 POST SB 737-26A1122

(c) Put the R RECIRC FAN switch in OFF.

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826; OMR 701-799, 802 POST SB 737-26A1122

(d) Do this task to put the airplane in the air mode: Put the Airplane in the Air Mode, AMM TASK 32-09-00-860-801

(e) Push and hold the TEST switch on the P8 Cargo Fire Control Panel until the FWD and AFT red cargo fire lights come on.

(2) Do this check for 28 Vdc at the exhaust system control interrupt relay R949 in the J24 junction box:

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(a) Open this access panel:

Number	Name/Location
114AC	Fwd Nose Wheel Well Upper Access Panel

(b) Remove the cover from the J24 junction box.

(c) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

(d) Remove exhaust system control interrupt relay R949.

(e) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

(f) Push and hold the TEST switch on the P8 Cargo Fire Control Panel until the FWD and AFT red cargo fire lights come on.

(g) Do a check for 28 Vdc between pins X1 and X2 of connector D12229.

(h) Push and hold the TEST switch on the P8 Cargo Fire Control Panel until the FWD and AFT red cargo fire lights come on.

(3) If 28 Vdc is present between pins X1 and X2 of connector D12229, then replace relay R949.

(a) Do the Repair Confirmation at the end of the task.

1) If the Repair Confirmation is satisfactory, then you corrected the fault.

2) If the Repair Confirmation is not satisfactory, then continue.

(4) If 28 Vdc is not present between pins X1 and X2 of connector D12229, then do this check of the ground circuit wiring:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

(b) Remove the relay R949.

(c) Do a check of the wiring between pin X2 of connector D12229 and the ground GD544-DC.

(d) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Install the relay R949.

3) Connect the connector D12229.

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4) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

5) Do the Repair Confirmation at the end of this task:
   a) If the Repair Confirmation is satisfactory, then you corrected the fault.
   b) If the Repair Confirmation is not satisfactory, then continue.

(5) Do a check of the wiring between the forward fire alarm latch relay R945 in the J24 junction box and the exhaust system control interrupt relay R949 in the J24 junction box as follows:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

(b) Remove the cover from the J24 junction box.

(c) Remove the forward fire alarm latch relay R945.

(d) Remove the exhaust system control interrupt relay R949.

(e) Do a continuity check from pin 8 of relay R945 to pin X1 of relay R949 (refer to WDMs .21-27-21 and 21-43-21).

(f) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Install relays R945 and R949 in the J24

3) Close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

4) Do the Repair Confirmation at the end of this task
   a) If the Repair Confirmation is satisfactory, then you corrected the fault.

G. Repair Confirmation

(1) Connect all electrical connectors that were disconnected.

(2) Install all components that were removed.

(3) Close all circuit breakers that were opened.

(4) Do an operational test of the overboard exhaust valve. To do the operational test, do this task: Smoke Clearance Mode - Operational Test, AMM TASK 21-27-00-700-807.

(5) If the Repair Confirmation is unsatisfactory, return to the step in the Fault Isolation Procedure that you were at prior to performing the Repair Confirmation and continue the Fault Isolation Procedure.

(6) If the Repair Confirmation is satisfactory, make sure that you do all the necessary steps to return the system to a serviceable condition:

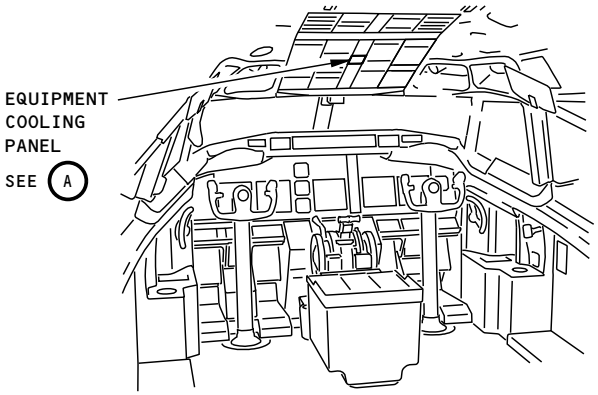
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(7) Do this task to return the airplane to the ground mode: Return the Airplane to the Ground Mode, AMM TASK 32-09-00-860-802.

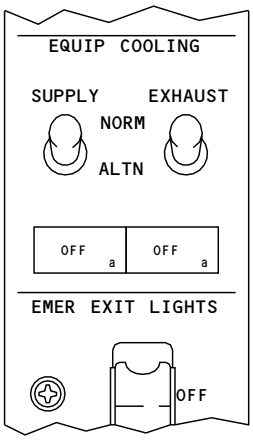
(8) Install all access panels that were removed.

——— END OF TASK ———

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FLIGHT COMPARTMENT



EQUIPMENT COOLING PANEL
A

2865637 S0000674503_V1

Equipment Cooling System - Component Location
Figure 301/21-27-00-990-801

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801. Digital Cabin Pressure Controller (DCPC) BITE Procedure

(Figure 201, Figure 202)

A. General

(1) There are certain conditions that a pilot report might address that do not require fault isolation. For example, if the cabin pressure climbs briefly and then stabilizes because of lavatory usage, it is not necessary to do a BITE procedure for such a condition. Other conditions such as aircraft operation beyond the capability of the pressurization control system also do not require fault operation. For example, aircraft descent at a rate such that the pressurization control system can not adjust quickly enough can result in the observation that the CABIN CLIMB gage indicates an unusually high rate.

(2) There are two identical Digital Cabin Pressure Controller (DCPC).

(a) DCPC No. 1 is on the E2-2 shelf in the electronic equipment bay.

OMR 701-799, 802

(b) DCPC No. 2 is on the E4-1 shelf in the electronic equipment bay.

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(c) DCPC No. 2 is on the E1-1 shelf in the electronic equipment bay.

OMR ALL

(3) If the DCPC shows "FAULT" on the front panel display, there is a fault currently being detected by the DCPC. There may also be faults recorded in the fault history of the DCPC.

(a) The cabin pressure controller should not be replaced unless additional fault isolation indicates that replacement is required.

(4) This BITE procedure must be done on both pressure controllers. Shortly after aircraft touchdown, the controller in charge of control status will switch to the standby controller in the last flight leg.

(5) If the DCPC front panel display is blank, there may be faults currently being detected or faults recorded in fault history or both current faults and recorded faults may exist.

(a) The complete BITE procedure must be completed to determine if maintenance needs to be done on the digital cabin pressurization control system components, the air conditioning pack, the pneumatic system, air leakage problems, or wiring.

(6) The steps that follow provide a summary of the DCPC button functions.

(a) ON/OFF button

1) The ON/OFF button will activate or de-activate the BITE functions in the DCPC.

(b) MENU button

1) The MENU button is used to display the BITE menus in the DCPC. If you push the MENU button, it will take you up one level in the BITE menus.

(c) YES button

1) Used to respond YES to questions on the DCPC.

(d) NO button

1) Used to respond NO to questions on the DCPC.

(e) Up arrow button

1) Used to scroll up through the menu or the results of a BITE test.

(f) Down arrow button

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1) Used to scroll down through the menu or the results of a BITE test.

(7) The steps that follow provide a summary of the main menu selections:

(a) EXISTING FAULTS

1) The EXISTING FAULTS main menu selection provides a list of all faults that are currently detected by the DCPC.

NOTE: Subsequent maintenance action is needed to resolve those maintenance message(s) that show in the EXISTING FAULTS.

(b) FAULT HISTORY

1) The FAULT HISTORY main menu selection provides a list of previous flight legs for which faults have been recorded. It also provides a list of faults for the present flight leg. Each flight leg is numbered. The present flight leg is numbered 00. The previous flight leg is 01. Each previous flight leg is numbered successively higher.

2) If a fault was active during a flight leg and subsequently became inactive, it will be recorded in FAULT HISTORY. It may not be in EXISTING FAULTS.

3) FAULT HISTORY can be cleared when you run a SYSTEM TEST from the GROUND TESTS main menu.

NOTE: Maintenance message(s) that show in the FAULT HISTORY are for reference only and do not require maintenance action.

(c) GROUND TESTS

1) The GROUND TESTS main menu selection provides the test selections that follow:

a) A DISPLAY test selection that tests the LED display on the front panel of the DCPC.

b) A SYSTEM test selection that is used to perform a system test of the DCPC and the associated components.

NOTE: During the SYSTEM test when the DCPC shows "SYS TEST AND CLR?", and you press the YES button, all faults from all flight legs will be cleared from FAULT HISTORY.

(d) SYSTEM STATUS

1) The SYSTEM STATUS main menu selection has the selections that follow:

a) A PRESENT STATUS selection that shows the current input parameters to the DCPC such as air conditioning pack state (for example, L PACK ON or OFF).

b) A SYSTEM CONFIG selection that shows system configuration information such as LRU elapsed time and program pin states.

(e) SYSTEM TEST AND CLEAR

1) The SYSTEM TEST AND CLEAR main menu selection is used to prepare the digital cabin pressurization control system for a system test and to clear the FAULT HISTORY.

NOTE: When the test is complete, all faults from all flight legs will be cleared from FAULT HISTORY.

(8) There are two types of faults identified by the BITE:

(a) Maintenance messages

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1) Maintenance messages are listed in the BITE Message Index at the end of this task and are found when you examine for faults from the EXISTING FAULTS and FAULT HISTORY main menu selections. There are nineteen maintenance messages. An example of the text displayed for a maintenance message is "OFV LRU FAIL". Each of the maintenance messages has a specified FIM task associated with it. From the main menu EXISTING FAULTS selection, the maintenance message is displayed when you press the YES button. From the main menu FAULT HISTORY selection, the maintenance message is displayed after you push the YES button to select the flight leg you want, and then push the YES button again to examine faults for that particular flight leg.

(b) Fault details or fault numbers

1) Fault details or fault numbers are listed and described in the DCPC Fault Number List (Table 201). They are found when you push the YES button when a maintenance message is displayed. Some maintenance messages can have different fault details (or fault numbers), depending on the type of fault that occurred. There are approximately 112 fault numbers. An example of the text displayed for a fault detail or fault number is "FAULT: 088".

NOTE: The DCPC Fault Number List is to be used for reference only. It is not intended to be a trouble shooting guide. Its' purpose is to list information that may provide insight into a solution of cabin pressurization system problems.

2) All fault code numbers except numbers 001, 009, 010 and 052 are RESET recoverable.

3) Fault details or fault numbers should be recorded on your maintenance records. This information could be used later to help identify faults that are difficult to trouble shoot.

4) Fault numbers 001 through 015 relate to pressurization system issues.

5) Fault numbers 016 through 031 relate to issues with systems that interface with the pressurization control system.

6) Fault numbers 032 through 063 relate to issues with the pressurization system controllers, M1653 and M1654.

7) Fault numbers 064 through 095 relate to issues with the aft outflow valve, V48.

8) Fault numbers 096 through 111 relate to issues with the P5-6 cabin pressure selector panel.

B. Initial Evaluation

(1) If FAULT shows on the DCPC front panel display, then do these steps:

NOTE: If FAULT shows on the DCPC front panel display, there are existing faults in the digital cabin pressurization control system.

(a) Push the ON/OFF button on the DCPC front panel.

(b) If the airplane is not in the ground mode, "A/P NOT IN GND" will show for two seconds; then "BITE ABORTED" will show for two seconds.

1) Put the airplane in the ground mode. This is the task: Return the Airplane to the Ground Mode, AMM TASK 32-09-00-860-802.

(c) Reset the cabin pressure controller for any reset recoverable faults as follows:

1) Set the mode selector on the cabin pressure control module on the P5-6 panel to MAN for 10 seconds.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Set the mode selector on the cabin pressure control module to AUTO.

(d) Do the BITE procedure for EXISTING FAULTS (step C.) below.

(2) If the DCPC front panel display is blank, then do these steps:

(a) Push the ON/OFF button on the DCPC front panel.

(b) If the airplane is not in the ground mode, "A/P NOT IN GND" will show for two seconds; then "BITE ABORTED" will show for two seconds.

1) Put the airplane in the ground mode. This is the task: Return the Airplane to the Ground Mode, AMM TASK 32-09-00-860-802.

(c) If the DCPC front panel display shows "SYSTEM OK", then do these steps:

NOTE: If SYSTEM OK is shown, there are no existing faults or faults in FAULT HISTORY present.

1) Push the MENU button.

2) Make sure that the DCPC front panel display shows EXISTING FAULTS.

3) Push the down arrow button until the DCPC front panel display shows GROUND TEST.

4) Do the GROUND TEST BITE procedure (step E.) below.

(d) If the DCPC front panel display shows nn EXIST FAULTS for two seconds, then nn PREV FAULTS for two seconds, and then EXISTING FAULTS, do the BITE procedure for EXISTING FAULTS (step C.) below.

NOTE: The nn is the number of faults.

(e) If the DCPC front panel display shows nn PREV FAULTS for two seconds, then FAULT HISTORY, do the BITE procedure for FAULT HISTORY (step D.) below.

NOTE: The nn is the number of faults.

C. BITE Procedure for EXISTING FAULTS

(1) Make sure that the DCPC front panel display shows EXISTING FAULTS.

(2) Push the YES button on the DCPC front panel.

(3) The DCPC front panel display shows the first fault (maintenance message) in the existing fault list.

NOTE: The text of the fault will be the same as one of the maintenance messages listed in the BITE Message Index at the end of this task. These existing faults are referred to as maintenance messages.

(4) Find and record the existing faults as follows:

(a) Make a record of the existing fault (maintenance message).

(b) Push the YES button on the DCPC front panel to see the fault details (fault numbers) for the maintenance message.

NOTE: The fault details for a maintenance message will be the same as one or more of the fault numbers that are listed in the DCPC Fault Number List (Table 201). The number of fault details (fault numbers) varies with each maintenance message.

(c) Make a record of the fault detail (fault number) for the maintenance message.

(d) Push the down arrow button on the DCPC front panel to look for the next fault detail (fault number).

(e) Make a record of this fault detail (fault number) also.

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(f) Do the above two steps again until the DCPC front panel display shows BOTTOM OF LIST for two seconds, then shows the maintenance message again.

(g) Push the MENU button on the DCPC front panel to go back to see the maintenance message in the existing fault list.

(h) Push the down arrow button on the DCPC front panel to see the next maintenance message in the EXISTING FAULTS list.

NOTE: If there are no more faults, the display will show "BOTTOM OF LIST" for two seconds, then the previous maintenance message will show.

(i) Do the eight steps above until the DCPC front panel display shows BOTTOM OF LIST for two seconds so that all maintenance messages and associated fault numbers have been recorded.

(5) Push the MENU button on the DCPC front panel.

(6) Refer to the BITE Message Index at the end of this task to find the fault isolation task for the applicable maintenance message.

(7) Push the down arrow button on the DCPC front panel until FAULT HISTORY shows and do the BITE procedure for FAULT HISTORY below.

D. BITE Procedure for FAULT HISTORY

(1) Make sure that the DCPC front panel display shows FAULT HISTORY.

(2) Push the YES button on the DCPC front panel.

(3) The DCPC front panel display shows FLIGHT LEG XX.

NOTE: XX is the last flight leg that faults were recorded in history.

(4) If there are no FLIGHT LEGS displayed on the DCPC front panel, go to step D.(11).

NOTE: This indicates that there are no faults recorded in FAULT HISTORY.

(5) Push the YES button on the DCPC front panel.

(6) The DCPC front panel display shows the first fault (maintenance message) in the flight leg fault list.

NOTE: The text of the fault will be the same as one of the maintenance messages listed in the BITE Message Index at the end of this task. These faults are referred to as maintenance messages.

(7) Find and record the faults in the flight leg fault list as follows:

(a) Make a record of the fault (maintenance message).

(b) Push the YES button on the DCPC front panel to see the fault details (fault numbers) for the maintenance message.

NOTE: The fault details for a maintenance message will be the same as one or more of the fault numbers that are listed in the DCPC Fault Number List (Table 201). The number of fault details (fault numbers) varies with each maintenance message.

(c) Make a record of the fault detail (fault number) for the maintenance message.

(d) Push the down arrow button on the DCPC front panel to look for the next fault detail (fault number) for the maintenance message.

(e) Make a record of this fault detail (fault number) also.

(f) Do the above two steps again until the DCPC front panel display shows BOTTOM OF LIST for two seconds, then returns to the maintenance message in the flight leg fault list.

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(g) Push the down arrow button on the DCPC front panel to look for the next fault (maintenance message) for the flight leg.

(h) Do the seven steps above until all fault entries in the flight leg fault list have been examined and recorded for the flight leg.

(8) Push the MENU button on the DCPC front panel until the display shows FLIGHT LEG XX.

(9) Push the down arrow button on the DCPC front panel to look for the next flight leg that recorded faults in fault history.

(10) Do above steps D.(6) through D.(8) until all flight legs with faults in fault history have been examined and recorded.

(11) Refer to the BITE Message Index at the end of this task to find the fault isolation task for the applicable maintenance message to find which FIM task(s) to use to fault isolate the maintenance message(s) that was recorded.

(12) If the DCPC front panel display is not at a main menu selection, push the MENU button until you are in the main menu.

E. GROUND TEST BITE Procedure

(1) Push the up or down arrow as necessary until the DCPC No. 1 front panel display shows GROUND TEST.

(2) Do the GROUND TEST for the DCPC No. 1 as follows:

CAUTION MAKE SURE ONE OF THE EXTERNAL DOORS OR WINDOWS IS OPEN BEFORE YOU CONTINUE WITH THE TEST. UNCONTROLLED PRESSURIZATION OF THE CABIN CAN OCCUR WHEN THE OUTFLOW VALVE IS CLOSED. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(a) Make sure at least one external door or window is open or that the L and R PACK switches on the P5-10 air conditioning panel are OFF.

(b) Push the YES button on the DCPC front panel.

(c) Make sure the DCPC front panel display shows DISPLAY TEST.

(d) Push the down arrow button on the DCPC front panel until the display shows SYSTEM TEST.

(e) Push the YES button on the DCPC front panel to start the system test.

(f) Make sure the DCPC front panel display shows "DOORS OPEN?".

(g) Make sure that at least one external airplane door is open.

(h) Push the YES button on the DCPC front panel.

(i) Make sure the DCPC front panel display shows "DADC ON?".

(j) Make sure that both Air Data Inertial Reference Unit (ADIRU) L and R are on.

(k) Push the YES button on the DCPC front panel.

(l) Make sure the DCPC front panel display shows "SMC ON?".

(m) Make sure that these circuit breakers are closed:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	5	C01204	SMYD-1 CMPTR DC
E	6	C01205	SMYD-1 SNSR EXC AC

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F/O Electrical System Panel, P6-1

Row	Col	Number	Name
B	4	C01207	SMYD-2 SNSR EXC AC
B	5	C01206	SMYD-2 CMPTR DC

(n) Push the YES button on the DCPC front panel.

(o) Make sure the DCPC front panel display shows "OFV CLEAR?".

(p) Do these steps in less than one minute or the test will fail.

NOTE: You will try to do a system test on pressure controller No. 2 while the test for pressure controller No. 1 is in progress. The system test cannot be done on the two controllers at the same time. This step makes sure that if you try to do the system test on pressure controller No. 2 when the test is in progress on pressure controller No. 1, that the test on pressure controller No. 2 will be aborted.

WARNING

DO NOT PUT YOUR HANDS OR TOOLS IN THE OUTLET OF THE OUTFLOW VALVE. THE VALVE IS MOTOR-OPERATED. INJURY TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

1) Push the YES button on the DCPC No. 1 front panel after you make sure the outflow valve is not blocked.

(q) Make sure the DCPC front panel display on pressure controller No. 1 shows TESTING.

NOTE: The lower digits will come on below the word "TESTING" at intervals of approximately 12 seconds. For older controllers, each of the 8 lower digits come on. For newer controllers, only five of the lower digits come on. If faults are detected during this testing, the DCPC front panel display will show nn EXIST FAULTS for two seconds, then the main menu selection EXISTING FAULTS will show.

(r) If this function finds faults, the DCPC front panel display will show nn EXIST FAULTS for two seconds, then show main menu item EXISTING FAULTS.

1) Do the procedure for existing faults in step C.

(s) Do these steps on the DCPC front panel on pressure controller No. 2:

1) Push the ON/OFF button on the DCPC front panel.
2) Push the MENU button on the DCPC front panel.
3) Continue to push the down arrow button until the DCPC front panel display shows GROUND TEST.
4) Push the YES button on the DCPC front panel.
5) Make sure the DCPC front panel display shows DISPLAY TEST.
6) Continue to push the down arrow button until the DCPC front panel display shows SYSTEM TEST.
7) Push the YES button on the DCPC front panel to start the system test.
8) Make sure the DCPC front panel display shows these messages:

NOTE: The messages will show one after the other with a two second delay between the first and the second messages. This shows that the test on pressure controller No. 2 is aborted.

a) BOTH SYS IN IBIT

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b) IBIT ABORTED

9) Push the ON/OFF button on the DCPC front panel on pressure controller No. 2.

(t) Make sure that five or eight digits are on for pressure controller No. 1.

NOTE: The lower digits come on sequentially to show that the test is in progress. For older controllers, each of the 8 lower digits come on at 12 second intervals. This takes approximately 100 seconds. For newer controllers, only five of the lower digits come on.

(u) Make sure the display on the DCPC front panel on pressure controller No. 1 shows these messages:

NOTE: The messages will show one after the other with a two second delay between the first and the second messages.

1) SYSTEM OK

2) SYSTEM TEST & CLEAR

NOTE: This clears all faults in EXISTING FAULTS and all flight leg faults from FAULT HISTORY.

(v) Push the YES button on the DCPC front panel on pressure controller No. 1.

(w) Push the ON/OFF button on the DCPC No. 1 front panel.

(3) Do the DCPC BITE procedure for the other pressure controller.

Table 201/21-31-00-993-803 DCPCS Fault Number List

BCM FAULT NUMBER	FAULT DISPLAYED	AUTO FAIL *[1]	FAULT DESCRIPTION	REFER TO FIM TASK
001	SYSTEM WIRING	X	A communication fail (no bus activity, parity error, short/long word, faulty data) is detected by the CTR and the ARINC wraparound of the CTR and the PNL is passed.	804
001	SYSTEM WIRING	X	A communication fail (no bus activity, parity error, short/long word, faulty data) is detected by the CTR and the ARINC wraparound of the CTR and the PNL is passed.	804
002	SYSTEM WIRING	X	No RS422 activity is detected by the CTR and the fault is isolated to the interconnect (CTR RS422 wrap around passed and CONNECT_ACT_CTR_IN = false).	
003	HI CAB ALT	X	Cabin pressure lower than 8 psi and the OFV position is still above 5 degrees after 1 second.	817
004	CAB DIFF PRESS HI	X	Cabin differential pressure exceeded 8.65 psi .	816
005	CAB RATE HI	X	Cabin pressure rate greater than ±2000 SLFPM and the OFV position is still above (below) 5 degrees after 5 seconds.	818

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Table 201/21-31-00-993-803 DCPCS Fault Number List (Continued)

BCM FAULT NUMBER	FAULT DISPLAYED	AUTO FAIL *[1]	FAULT DESCRIPTION	REFER TO FIM TASK
006	SYSTEM WIRING	X	OFV cannot follow the state of the CTR and the failure is not isolated to the PNLs sever switch or the CTRs HSS switch. Interconnect failure in the DRIVE_ENABLE_CTR_PNL line, the DRIVE_ENABLE_PNL_ACT line or MNL_MODE_PNL_CTR line.	804
007	OFV LRU FAIL	X	OFV cannot follow the CTR position command (OFV_LOOP_CLOSURE_STATUS is set to TRUE) and the OFV position is below 20 degrees.	802
008	AUTO/MAN ERROR	-	More than one or none of the AUTO, ALTN or MANUAL indications are received by the CTR.	821
009	OFV LRU FAIL	X	OFV does not reach the mechanical stop within time or the calibration constant is out of range. (IBIT only.) Mechanical blockage/restriction of the OFV or OFV failure.	802
010	OFV LRU FAIL	X	OFV is not able to drive fully open to fully closed or vice versa within the given time constrains. (IBIT only.)	802
011	OFV LRU FAIL	X	OFV is not able to move from fully open to 105 degrees and back within the given time constrains. (SCBIT only.)	802
012	CONTRLR BBJ PIN [2]	X	BBJ_SEL_IN is not tied to Ground. May be caused by a wire failure or connection failure outside of the CTR (Fault Detail msg CONTRLR BBJ PIN).	804
012	[3]		Unused	
013	SYSTEM WIRING	X	CTR MASTER/SLAVE pin programming is contradictory. May be caused by a wire failure or connection failure outside of the CTR.	804
014	SYSTEM WIRING	X	CTR receives consecutively for 5 minutes an indication that the other CTR performs IBIT while in flight.	804
015	OFV OR WIRING	X	CTR cannot detect OFV RS422 activity and the CTR RS422 wraparound is passed and the CONNECT_ACT_CTR_IN is true.	805
016	SEL PNL LRU FAIL		The CTR detects that the PNL was unable to open the sever switch. The test is performed if a switch from AUTO to MANUAL or from ALTN to MANUAL is detected (CTRs opmode switched to MANUAL). This test cannot be rerun during IBIT.	803
017			Unused	
018			Unused	

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Table 201/21-31-00-993-803 DCPCS Fault Number List (Continued)

BCM FAULT NUMBER	FAULT DISPLAYED	AUTO FAIL *[1]	FAULT DESCRIPTION	REFER TO FIM TASK
019	OFV LRU FAIL	X	The CTR receives faulty data from the OFV for longer than 3 seconds. This failure may be caused by an ELACT hardware or software failure.	802
020	SYSTEM WIRING	-	A mismatch between the discrete OK_PNL_CTR signal and the PNL health information on the PNL fault word is detected.	
021	SYSTEM WIRING	-	A failure of the discrete CONNECT_ACT _CTR line is detected (RS422 activity is present and CONNECT_ACT_CTR = false).	804
022	DADC#1 INVALID	-	Computed Airspeed (CAS) received from DADC #1 is invalid or not updated.	806
023	DADC#2 INVALID	-	CAS received from DADC #2 is invalid or not updated.	807
024	DADC#1 INVALID	X	No bus activity of DADC1 is detected, pressure altitude or baro-corrected pressure altitude are not received or invalid for longer than 2 seconds. Auto channel fail will be set if DADC2 FAIL is set too. Auto channel fail will not be set if true air speed only is missing or faulty for longer than 2 seconds.	806
025	DADC#2 INVALID	X	No bus activity of DADC2 is detected, pressure altitude or baro-corrected pressure altitude are not received or invalid for longer than 2 seconds. Auto channel fail will be set if DADC1 FAIL is set too. Auto channel fail will not be set if true air speed only is missing or faulty for longer than 2 seconds.	807
026	SMC #1 INVALID	-	No bus activity of SMC1 is detected, or N1 value or N2 value is not received or invalid for longer than 2 seconds. Auto channel fail will be set if SMC2 FAIL is set too. Auto channel fail will only be set as long as the mode is GR or TO.	808
027	SMC #2 INVALID	-	No bus activity of SMC2 is detected, or N1 value or N2 value is not received or invalid for longer than 2 seconds. Auto channel fail will be set if SMC1 FAIL is set too. Auto channel fail will only be set as long as the mode is GR or TO.	809
028	LDG/ GEAR DISAGREE	-	A mode switch GR to CI or TO to CI occurs due to ambiguous landing gears using the condition for true air speed. May be caused by a failure of the LEFT_WHEELS_ON_GND or RIGHT_ WHEELS_ON_GND signals outside of the CTR.	822

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Table 201/21-31-00-993-803 DCPCS Fault Number List (Continued)

BCM FAULT NUMBER	FAULT DISPLAYED	AUTO FAIL *[1]	FAULT DESCRIPTION	REFER TO FIM TASK
029	N1/N2 LOW	-	A mode switch GR to CI occurred due to no valid N1 or N2 value available or a valid N1 or N2 value below the value required for the mode switch. May be caused by a lower throttle lever setting during take off run or a SMC fail.	810
030	LO INFL/HI LEAKG	-	The OFV position was below 3 degrees for longer than 5 seconds. This is no failure of the DCPCS. May be caused by a low inflow situation or a high leakage.	823
031	ACFT RATE HI	-	Cabin pressure rate is above the allowed selected hard limit for more than 5 seconds caused by a high a/c rate in conjunction with maximum dp (7.95 psi) or negative dp (-0.05 psi).	815
032	CONTRL LRU FAIL	X	The CPC97 gate array tests detected a failure in the CPC97 gate array.	824
033	CONTRL LRU FAIL	X	The non destructive RAM test detected a failure in the internal RAM of the 8097µP.	824
034	CONTRL LRU FAIL	X	The non destructive RAM test detected a failure in the external CTR RAM.	824
035	CONTRL LRU FAIL	X	The Cyclic Redundancy Check detected a failure of the CTR EPROMs. It has to be assumed that the software contained in the PROMs is not in an operative condition.	824
036	CONTRL LRU FAIL	X [4]	The arithmetic test of the µP detected a failure.	824
036		[3]	Unused	
037	CONTRL LRU FAIL	X	The +15V DC and -15V DC from the power supply are continuously monitored at an analog input port of the A/D converter. The fail flag is set if the difference of both power lines is out of the range required to drive the ARINC transmitters.	824
038	CONTRL LRU FAIL	X	The A/D converter is continuously tested against +5V dc via an analog input. If the measured value differs more than 4% from the nominal value, the fail flag is set.	824
039	CONTRL LRU FAIL	X	No RS422 transmitter interrupt activity is detected.	824
040	CONTRL LRU FAIL	X	The wraparound of the CTR RS422 interface fails.	824
041	CONTRL LRU FAIL	X	No ARINC transmitter interrupt activity is detected.	824

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Table 201/21-31-00-993-803 DCPCS Fault Number List (Continued)

BCM FAULT NUMBER	FAULT DISPLAYED	AUTO FAIL *[1]	FAULT DESCRIPTION	REFER TO FIM TASK
042	CONTRL LRU FAIL	X	The CTR watchdog was not correctly served by the software. May be caused by a software anomaly.	824
043	CONTRL LRU FAIL	X	The ARINC wraparound of ARINC transmitter circuit 1, MUX1 and ARINC receiver 1 fails.	824
044	CONTRL LRU FAIL	X	The ARINC wraparound of ARINC transmitter circuit 2, MUX2 and ARINC receiver 2 fails.	824
045	CONTRL LRU FAIL	X	The ARINC wraparound of ARINC transmitter circuit 3, MUX3 and ARINC receiver 3 fails.	824
046	CONTRL LRU FAIL	X	A write to the NVM cannot be performed correctly.	824
046	[3]	X	Unused	
047	CONTRL LRU FAIL	X	The data read out of the NVM flight buffer or NVM power down buffer after POR or pilot reset not within the valid range.	824
048	CONTRL LRU FAIL	X	The checksum test of the sensor EEPROM data fails.	824
049	CONTRL LRU FAIL	X	The sensed pressure of the sensor is above 26 psi or below 2.471 psi.	824
050	CONTRL LRU FAIL	X	The pressure value is not refreshed by the pressure sensor within the valid time frame.	824
051	CONTRL LRU FAIL	X	The temperature value is not refreshed by the temperature sensor within the valid time frame.	824
052	CONTRL LRU FAIL	X	The PC differs from the PA value more than 0.2 psi absolute. This test is part of the IBIT and the SCBIT only.	824
053			Unused	
054			Unused	
055			Unused	
056	CONTRL LRU FAIL	X	The DRIVE_ENABLE_CTR_PNL_OUT line is powered while the HSS should be open.	824
057	CONTRL LRU FAIL	X	The DRIVE_ENABLE_CTR_PNL_OUT line is unpowered while the HSS should be closed.	824
058 [4]			Unused	
058 [3]			MANUAL_SWITCH_MESSAGE Set if system operates in manual mode.	
059	CONTRL LRU FAIL	X	The wraparound of the OSD_CTR_OUT line fails.	824

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Table 201/21-31-00-993-803 DCPCS Fault Number List (Continued)

BCM FAULT NUMBER	FAULT DISPLAYED	AUTO FAIL *[1]	FAULT DESCRIPTION	REFER TO FIM TASK
060	CONTRL LRU FAIL	X	The wraparound of the AUTO_UNDER_TEST _CTR_CTR_OUT line fails.	824
061	CONTRL LRU FAIL	X	The wraparound of the series discrete output AUTO_FAIL_CTR_OUT failed.	824
062			Unused	
063	-	-	The self test of the BCM fails.	-
064	OFV LRU FAIL	X	The non destructive test of the OFV µP RAM fails during CBIT.	802
065	OFV LRU FAIL	X	The non destructive test of the OFV µP external RAM fails during CBIT.	802
066	OFV LRU FAIL	X	The non destructive test of the OFV µP RAM fails during PUBIT.	802
067	OFV LRU FAIL	X	The non destructive test of the OFV µP external RAM fails during PUBIT.	802
068	OFV LRU FAIL	X	The current limitation for the auto- matic motor fails.	802
069	OFV LRU FAIL	X	The +15V dc of the ELACT feed into an A/D converter input is not in range.	802
070			Unused	
071			Unused	
072	OFV LRU FAIL	X	The cyclic redundancy check of the ELACT EPROMs fails.	802
073	OFV LRU FAIL	X	The wraparound of the ELACT RS422 interface fails.	802
074	OFV OR WIRING	X	The ELACT does not detect RS422 activity of the CTR and the CTR RS422 wraparound is passed.	805
075	OFV LRU FAIL	X	No ELACT RS422 transmitter interrupt activity is detected by the ELACT.	802
076	OFV LRU FAIL	X	The wraparound of the discrete input MUX in the ELACT failed.	802
077	OFV LRU FAIL	X	The ELACT internal discrete wraparound fails.	802
078	OFV LRU FAIL	X	The RVT reference voltage for the position feedback is detected out of range.	802
079	OFV LRU FAIL	X	OFV connector J4 internal pin programming is not set to B737 mode.	802
080	OFV LRU FAIL	X	The A/D conversion is not performed within the defined time frame.	802
081 [4]	-	-	This status fail flag will be set on the OFV fault word by the ELACT and evaluated by the CTR. If the additional conditions are fulfilled the CTR sets as result fail flag 007 on the system fault word.	-

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Table 201/21-31-00-993-803 DCPCS Fault Number List (Continued)

BCM FAULT NUMBER	FAULT DISPLAYED	AUTO FAIL *[1]	FAULT DESCRIPTION	REFER TO FIM TASK
081 [3]			Unused	
082			Unused	
083	OFV LRU FAIL	X	The arithmetic operation tests of the ALU of the ELACT fails.	802
084			Unused	
085			Unused	
086			Unused	
087	OFV LRU FAIL	X	The -15V dc feed into the A/D con- verter is out of range. May be caused by a power supply or A/D converter fail.	802
088	OFV LRU FAIL	X	The hall sensor signals are detected in incorrect combinations.	802
089			Unused	
090	CAB PRES SW ACTIV	-	This status flag will be set as long as the cabin pressure switch of the ELACT is active.	-
091 [4]	CONTRL LRU FAIL	X	The ELACT receives invalid P messages from the CTR for longer than 3 seconds.	824
091 [3]			Unused	
092	OFV LRU FAIL	X	ELACT internal pin programming shows a forbidden combination.	802
093			Unused	
094			Unused	
095			Unused	
096	SEL PNL LRU FAIL	-	The cyclic redundancy check of the PNL PROMs fails.	803
097			Unused	
098	SEL PNL LRU FAIL	-	The test of the PNL RAM fails.	803
099	SEL PNL LRU FAIL	-	The PNL µP arithmetic operation test fails.	803
100	SEL PNL LRU FAIL	-	The PNL detects a wraparound fail of the ARINC interface to the master CTR.	803
101	SEL PNL LRU FAIL	-	The PNL detects a wraparound fail of the ARINC interface to the slave CTR.	803
102	SEL PNL LRU FAIL	-	The PNL A/D converter test fails.	803

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Table 201/21-31-00-993-803 DCPCS Fault Number List (Continued)

BCM FAULT NUMBER	FAULT DISPLAYED	AUTO FAIL *[1]	FAULT DESCRIPTION	REFER TO FIM TASK
103	SEL PNL LRU FAIL	-	Wraparound of the discrete output AUTO_FAIL_PNL_OUT fails.	803
104	SEL PNL LRU FAIL	-	The PNL resets 3 times without detected reason within 6 minutes.	803
105	SEL PNL LRU FAIL	-	The PNL watchdog timer expires.	803
106	SEL PNL LRU FAIL	-	One discrete input value is out of range.	803
107	SEL PNL LRU FAIL	X	The CTR state is SB and the OFV state is OP while the CTRs opmode is not MANUAL and the PNL indicates a high voltage drop from either sever switch relay.	803
108	SYSTEM WIRING	X	The PNL detects no ARINC activity or receives faulty data from the CTR1 and no wraparound failure of the PNL and the CTR ARINC interface is detected.	804
109	DRV ENBL ERROR	X	The PNL detects both CTRs are operational at the same time.	819
110	SYSTEM WIRING	X	The PNL detects no ARINC activity or receives faulty data from the CTR2 and no wraparound failure of the PNL and the CTR ARINC interface is detected.	804
111	-	-	A PNL internal fail (fail flags 96 to 106) is detected.	-

*[1] X in column indicates "AUTO FAIL" comes on.
[2] Pressure controller P/N 7121-19971-01BA
[3] Pressure controllers P/N 21933-01AA, 21933-01AB, 21933-01AC, 21933-01AD, 25436-01AD
[4] Pressure controllers P/N 7121-19971-01AA, 7121-19971-01AB, 7121-19971-01AC, 7121-19971-01BA

F. SYSTEM STATUS Bite Procedure

(1) If the DCPC front panel display is not at a main menu selection, push the MENU button until you are in the main menu.

(2) Push the ↑ or ↓ button as necessary until the DCPC front panel display shows SYSTEM STATUS.

(3) Close these circuit breakers:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
E	5	C01009	ADIRU LEFT DC
E	7	C01007	ADIRU LEFT AC

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	5	C01204	SMYD-1 CMPTR DC
E	6	C01205	SMYD-1 SNSR EXC AC

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F/O Electrical System Panel, P6-1

Row	Col	Number	Name
B	4	C01207	SMYD-2 SNSR EXC AC
B	5	C01206	SMYD-2 CMPTR DC
C	14	C01008	ADIRU RIGHT AC
C	17	C01010	ADIRU RIGHT DC

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	16	C01356	LANDING GEAR AIR/GND SYS 1
D	1	C01399	PSEU PRI
D	2	C01400	PSEU ALTN
D	15	C01401	LANDING GEAR AIR/GND RELAY

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT
D	2	C01445	A/C OVERBOARD EXH VALVE RECONFIG CONT
F	1	C01273	PRESSURIZATION CONTROL LCD LTG
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2

(4) Set the pressurization mode selector switch on the P5-6 panel to AUTO.

NOTE: The AUTO FAIL light may momentarily flash when AUTO is selected.

(5) If you do SYSTEM STATUS Bite Procedure for the pressure controller No.1 (M1653), do these steps:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	5	C01271	PRESSURIZATION CONTROL AUTO 2

(b) Continue after a minimum of one second.

(c) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	5	C01271	PRESSURIZATION CONTROL AUTO 2

(6) If you do SYSTEM STATUS Bite Procedure for the pressure controller No.2 (M1654), do these steps:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	3	C01270	PRESSURIZATION CONTROL AUTO 1

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(b) Continue after a minimum of one second.

(c) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	3	C01270	PRESSURIZATION CONTROL AUTO 1

(7) Set 33,000 feet in the FLT ALT window of the P5-6 pressurization control panel.

(8) Set 500 feet in the LAND ALT window of the P5-6 pressurization control panel.

OMR 701-799

(9) Set the RECIRC FAN switch on the P5-10 air conditioning panel to OFF.

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(10) Set the R RECIRC FAN switch on the P5-10 air conditioning panel to OFF.

OMR ALL

(11) Set the L PACK and R PACK switches on the P5-10 air conditioning panel to OFF.

(12) If you do SYSTEM STATUS Bite Procedure for the pressure controller No.1 (M1653), do this step:

(a) Push the ON/OFF button on the front of the pressure controller No.1 (M1653) on the E2-2 shelf of the E/E compartment.

(13) If you do SYSTEM STATUS Bite Procedure for the pressure controller No.2 (M1654), do this step:

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(a) Push the ON/OFF button on the front of the pressure controller No.2 (M1654) on the E1-1 shelf of the E/E compartment.

OMR 701-799, 802

(b) Push the ON/OFF button on the front of the pressure controller No.2 (M1654) on the E4-1 shelf of the E/E compartment.

OMR ALL

(14) Push the MENU button.

(15) Continue to push the ↓ button until the display shows SYSTEM STATUS.

(16) Push the YES button.

(17) Make sure the display shows PRESENT STATUS after two seconds.

(18) Push the YES button.

(19) Make sure the display shows FLT ALT +33000FT.

(20) Push the ↓ button.

(21) Make sure the display shows LAND ALT +500FT.

(22) Push the ↓ button.

(23) Make sure the display shows AUTO IN CTRL.

(24) Push the ↓ button.

(25) Make sure the display is between OFV + 100 and OFV + 122 DEG OPEN DEG OPEN.

(26) Push the ↓ button.

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(27) Make sure the display shows NO AUTO FAIL.

(28) Push the ↓ button.

(29) Make sure the display is between CAB PRES and CAB PRES 13.70 PSI 15.70 PSI.

(30) Push the ↓ button.

(31) Make sure the display shows CARGO HT VALVE CL.

(32) Push the ↓ button.

(33) Make sure the display shows LFT PACK FLOW OFF.

(34) Push the ↓ button.

(35) Make sure the display shows RT PACK FLOW OFF.

(36) Push the ↓ button.

(37) Make sure the display is BOTTOM OF LIST.

(38) Push the menu button.

(39) Push the ↓ button continuously until the display shows SYSTEM CONFIG.

(40) Push the YES button.

(41) Push the ↓ button continuously until the display shows this message:

NOTE: While you push the button, the display will show CONTRLR XXX HRS and PNL TIME XXX HRS. XXX is the number of hours.

(a) ROC SEL HIGH

(42) Push the ↓ button.

(43) Make sure the display shows this message:

(a) ROD SEL2 HIGH

(44) Push the ↓ button.

(45) If you do SYSTEM STATUS Bite Procedure for the pressure controller No.1 (M1653), make sure that the display shows MASTERID LOW.

(46) If you do SYSTEM STATUS Bite Procedure for the pressure controller No.2 (M1654), make sure that the display shows MASTERID HIGH.

(47) Push the ↓ button.

(48) Make sure the display shows this message:

(a) ROD SEL1 LOW

(49) Push the ↓ button.

(50) Make sure the display shows QFE SEL HIGH

(51) Push the ↓ button.

(52) If you do SYSTEM STATUS Bite Procedure for the pressure controller No.1 (M1653), make sure that the display shows SLAVE ID HIGH.

(53) If you do SYSTEM STATUS Bite Procedure for the pressure controller No.2 (M1654), make sure that the display shows SLAVE ID LOW.

(54) Push the ↓ button.

(55) Make sure the display shows OFV CONN LOW.

(56) Push the ↓ button.

(57) Make sure the display shows IBIT IN HIGH.

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(58) Push the ↓ button.

(59) Make sure the display shows CARGO HT VLV LOW.

(60) Push the ↓ button.

(61) Make sure the display shows MANUAL IN HIGH.

(62) Push the ↓ button.

(63) Make sure the display shows LEFT A/G LOW.

(64) Push the ↓ button.

(65) Make sure the display shows RT A/G LOW.

(66) Push the ↓ button.

(67) Make sure the display shows SHOP MODE HIGH.

(68) Push the ↓ button.

(69) Make sure the display shows L PACK VLV LOW.

(70) Push the ↓ button.

(71) Make sure the display shows R PACK VLV LOW.

(72) Push the ↓ button.

(73) Make sure the display shows 41K SEL LOW.

(74) Push the ↓ button.

(75) Make sure the display shows AUTO CTL IN LOW.

(76) Push the ↓ button.

(77) Make sure the display shows PNL OK HIGH.

(78) Push the ↓ button.

(79) Make sure the display shows BBJ SEL HIGH.

NOTE: Except aircraft with pressure controllers P/N 7121-19971-01AA, 7121-19971-01AB, and 7121-19971-01AC.

(80) Push the ↓ button.

(81) Make sure the display shows HIGH LFE HIGH.

NOTE: For aircraft with pressure controllers P/N 21933-series.

(82) Push the ↓ button.

(83) Make sure the display shows BOTTOM OF LIST.

(84) Push the ON/OFF button.

(85) Do these steps if the passenger cabin pressure controller passes the BITE test and the passenger cabin temperature can not be made colder or warmer with the temperature selector in the AUTO or MANUAL mode:

(a) For the applicable (forward or aft) cabin zone temperature sensors:

1) Do this task: Cabin Temperature Sensor Removal, AMM TASK 21-61-10-000-802

2) Do this task: Cabin Temperature Sensor Installation, AMM TASK 21-61-10-400-802.

(b) For the applicable (forward or aft) cabin duct temperature sensors:

1) Do this task: Duct Temperature Limit Sensor Removal, AMM TASK 21-61-04-000-802-001 or Duct Temperature Limit Sensor Removal, AMM TASK 21-61-04-000-803-002

ECCN 9E991 BOEING PROPRIETARY - See title page for details

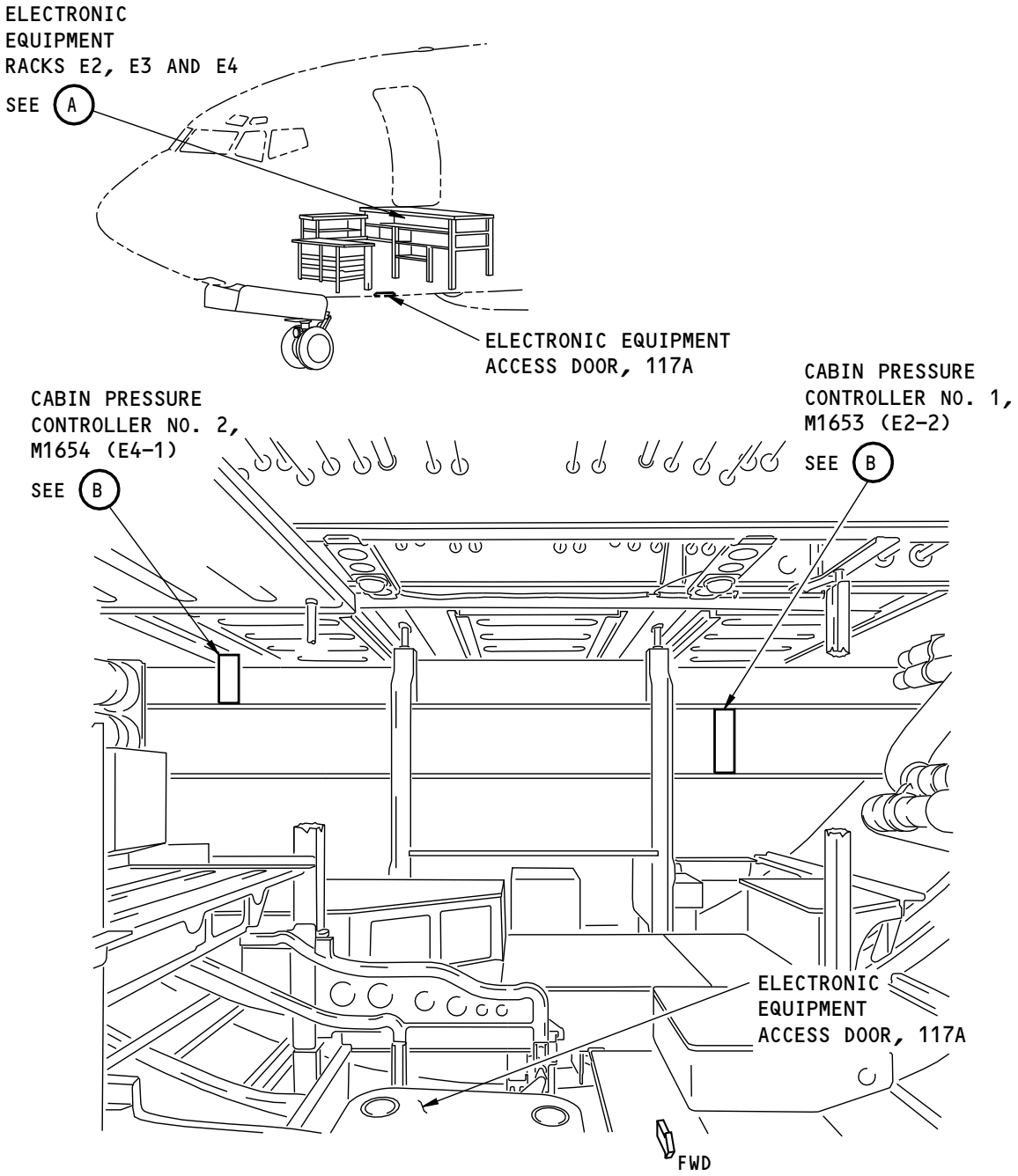
2) Do this task: Duct Temperature Limit Sensor Installation, AMM TASK 21-61-04-400-802-001 or Duct Temperature Limit Sensor Installation, AMM TASK 21-61-04-400-803-002.

(86) Do the SYSTEM STATUS Bite procedure for the other pressure controller.

LRU/SYSTEM	MAINTENANCE MESSAGE	GO TO FIM TASK
CAB PRESS CON	ACFT RATE HI	21-31 TASK 815
CAB PRESS CON	AUTO/MAN ERROR	21-31 TASK 821
CAB PRESS CON	CAB DIFF PRESS HI	21-31 TASK 816
CAB PRESS CON	CAB PRESS SW ACTIV	21-31 TASK 820
CAB PRESS CON	CAB RATE HI	21-31 TASK 818
CAB PRESS CON	CONTRL LRU FAIL	21-31 TASK 801
CAB PRESS CON	DADC #1 INVALID	21-31 TASK 806
CAB PRESS CON	DADC #2 INVALID	21-31 TASK 807
CAB PRESS CON	DRV ENBL ERROR	21-31 TASK 819
CAB PRESS CON	HI CAB ALT	21-31 TASK 817
CAB PRESS CON	LDG GEAR DISAGREE	21-31 TASK 822
CAB PRESS CON	LO INFL/HI LEAKG	21-31 TASK 823
CAB PRESS CON	N1/N2 LOWRU FAIL	21-31 TASK 810
CAB PRESS CON	OFV LRU FAIL	21-31 TASK 802
CAB PRESS CON	OFV OR WIRING	21-31 TASK 805
CAB PRESS CON	SEL PNL LRU FAIL	21-31 TASK 803
CAB PRESS CON	SMC #1 INVALID	21-31 TASK 808
CAB PRESS CON	SMC #2 INVALID	21-31 TASK 809
CAB PRESS CON	SYSTEM WIRING	21-31 TASK 804

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



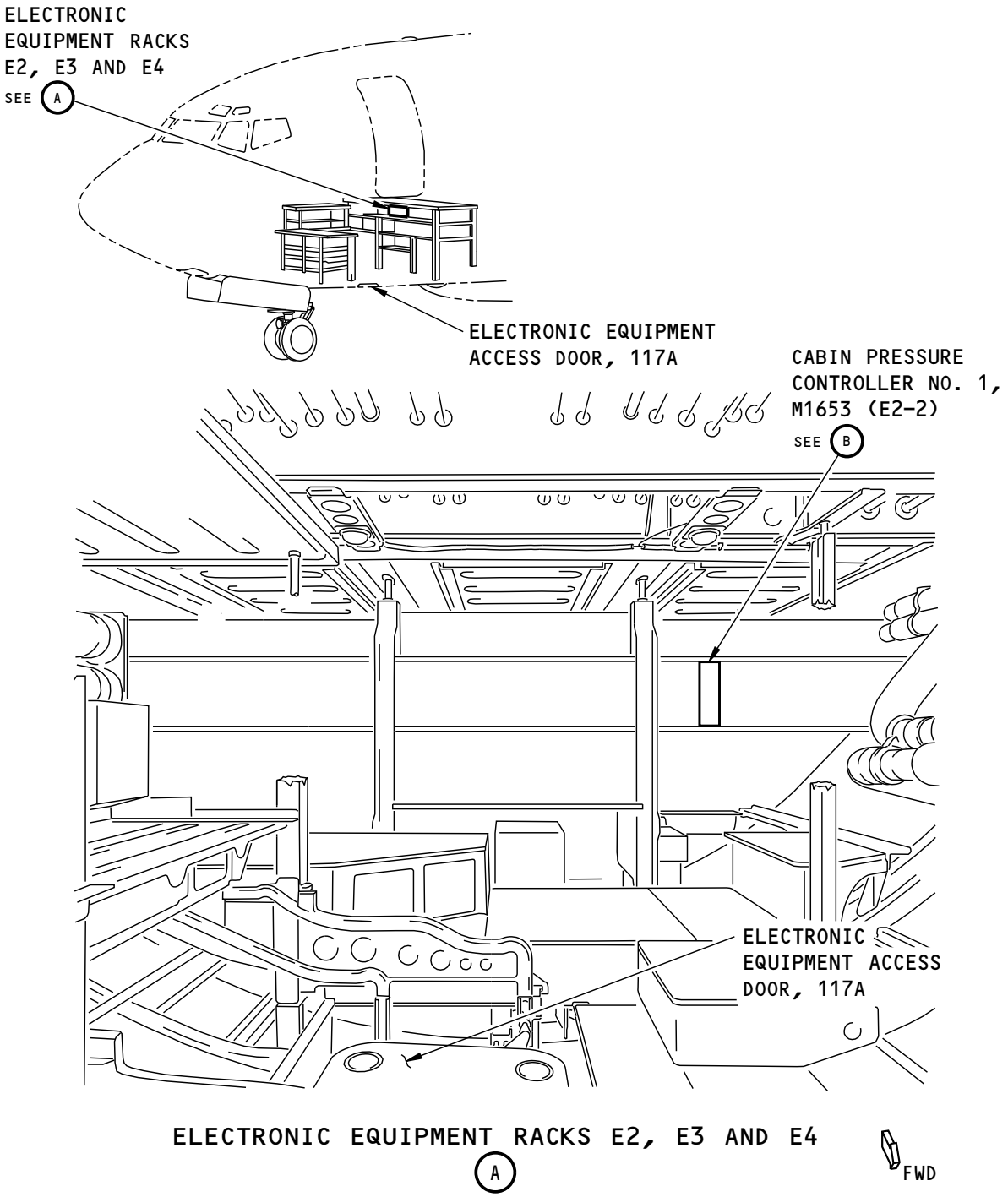
ELECTRONIC EQUIPMENT RACKS E2, E3 AND E4

A

G57907 S0006740546_V1

Cabin Pressure Control System BITE Test
Figure 201/21-31-00-990-807 (Sheet 1 of 4)

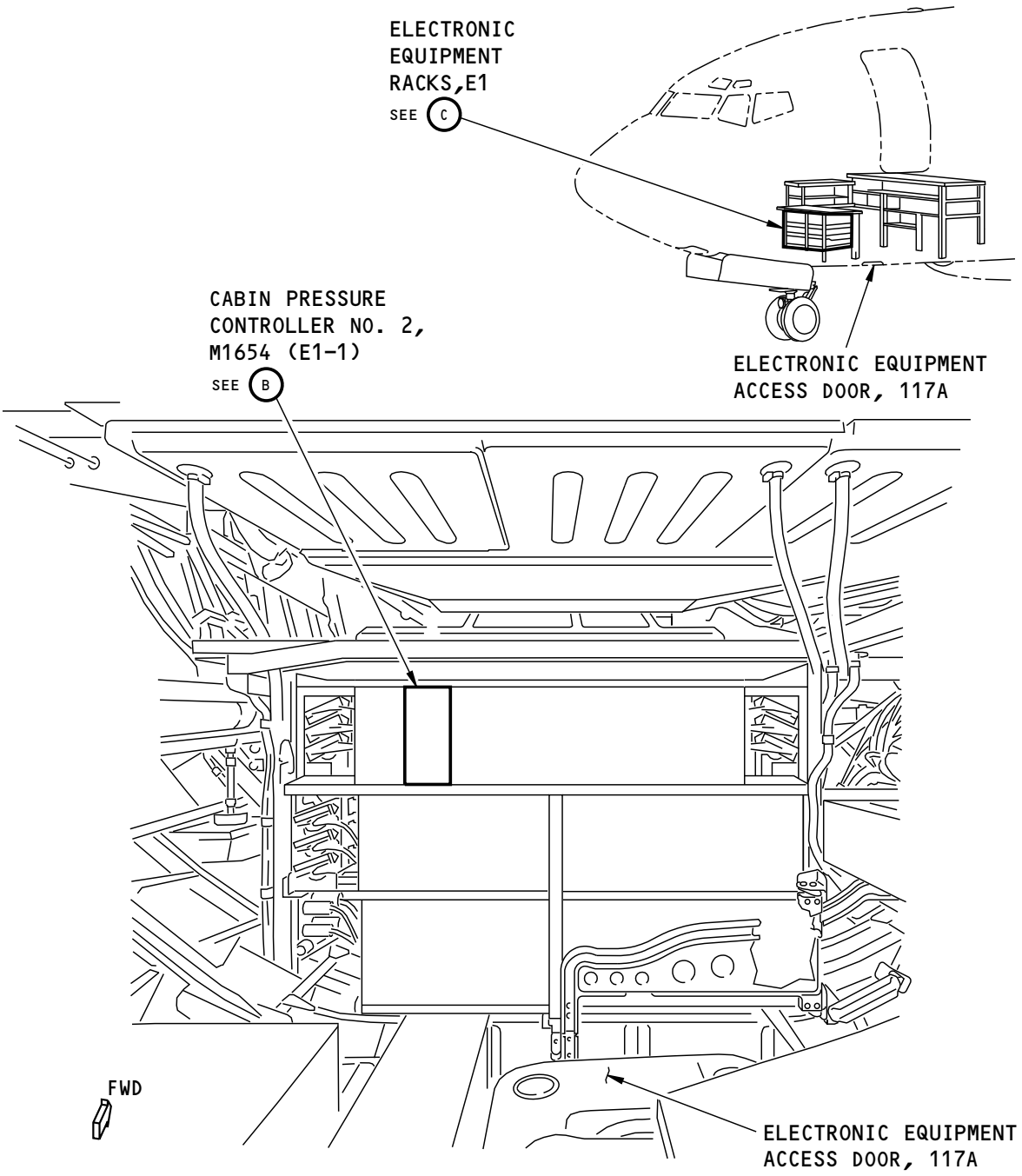
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2809630 S0000645673_V1

Cabin Pressure Control System BITE Test
Figure 201/21-31-00-990-807 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

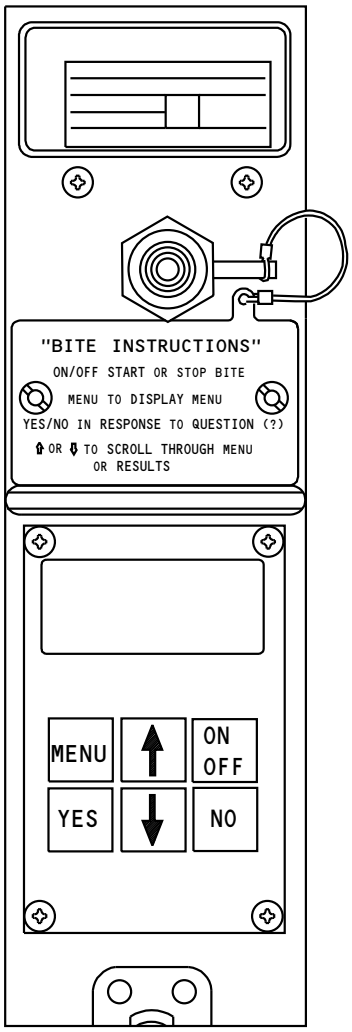


ELECTRONIC EQUIPMENT RACK, E1
C

2809643 S0000645674_V1

Cabin Pressure Control System BITE Test
Figure 201/21-31-00-990-807 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



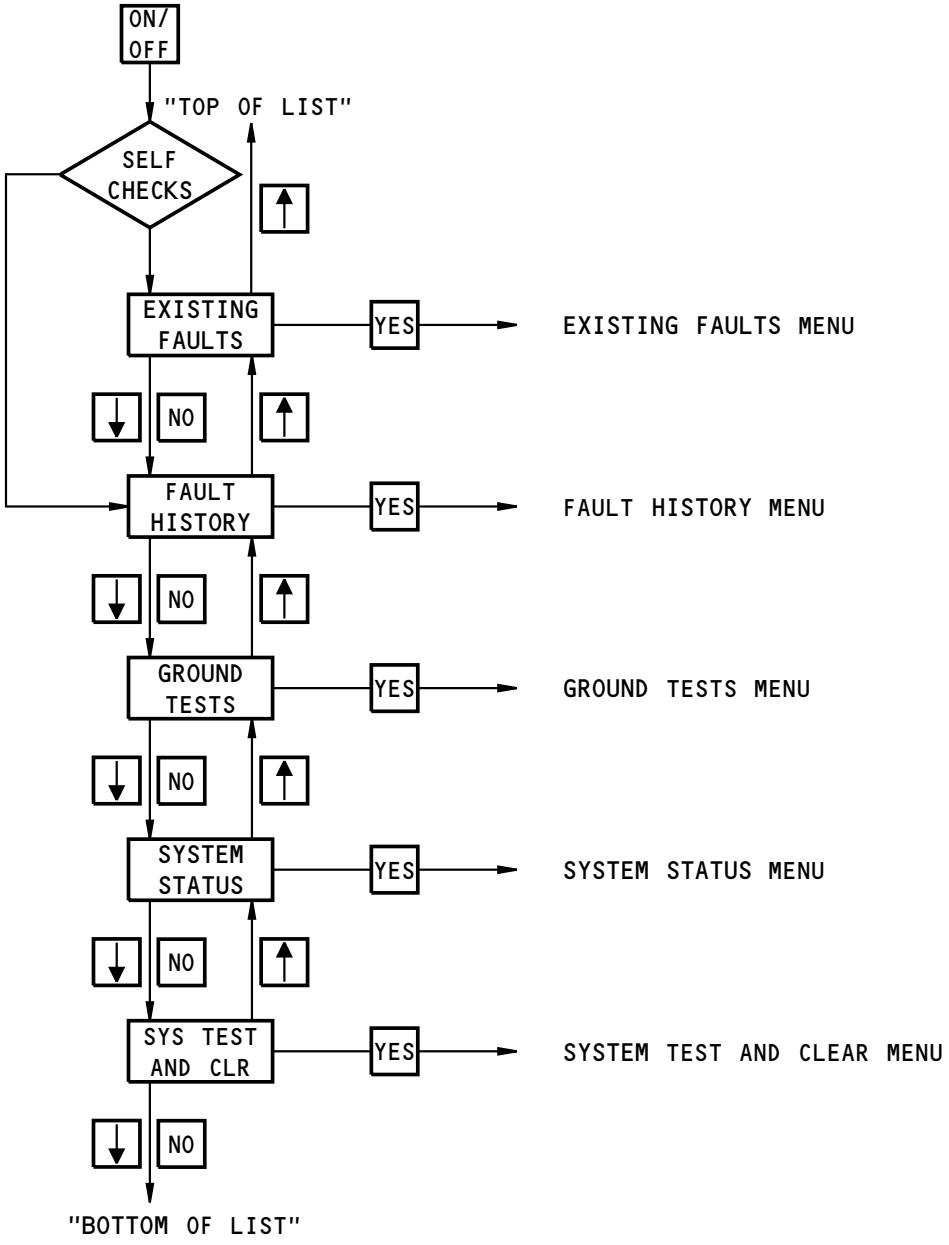
CABIN PRESSURE CONTROLLER

B

G57910 S0006740547_V1

Cabin Pressure Control System BITE Test
Figure 201/21-31-00-990-807 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

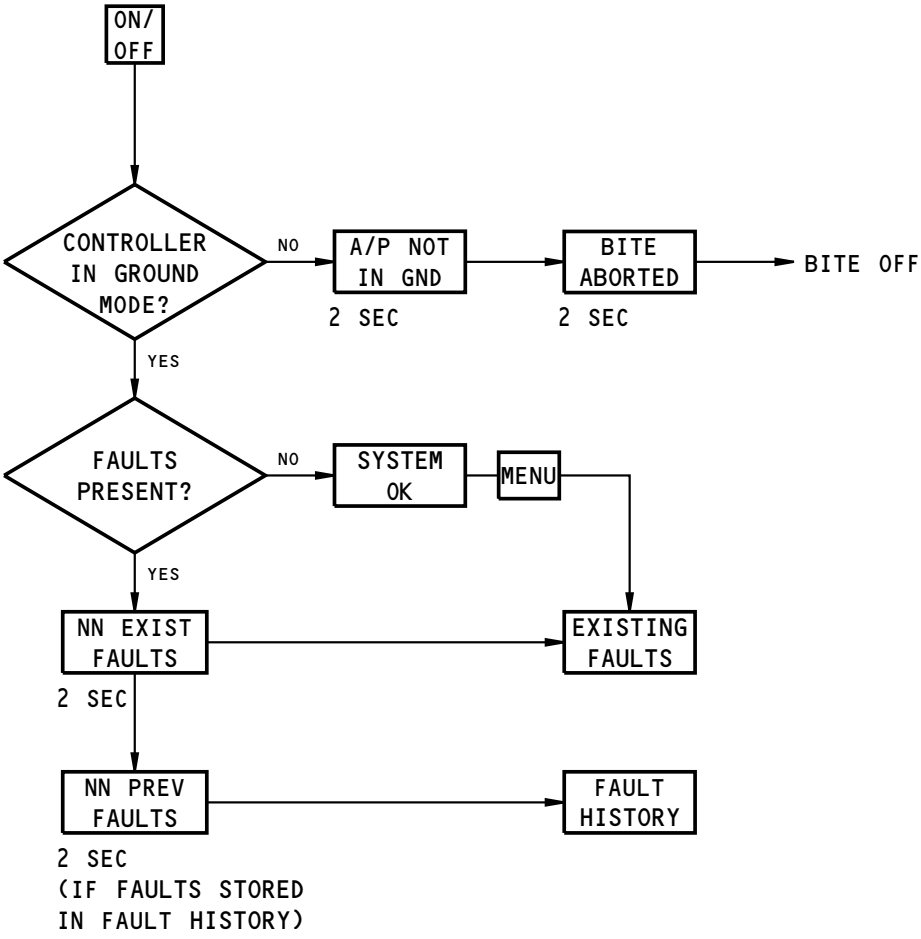


DCPC BITE MAIN MENUS

K43609 S0006740548_V1

Cabin Pressure Control System BITE Procedure
Figure 202/21-31-00-990-808 (Sheet 1 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

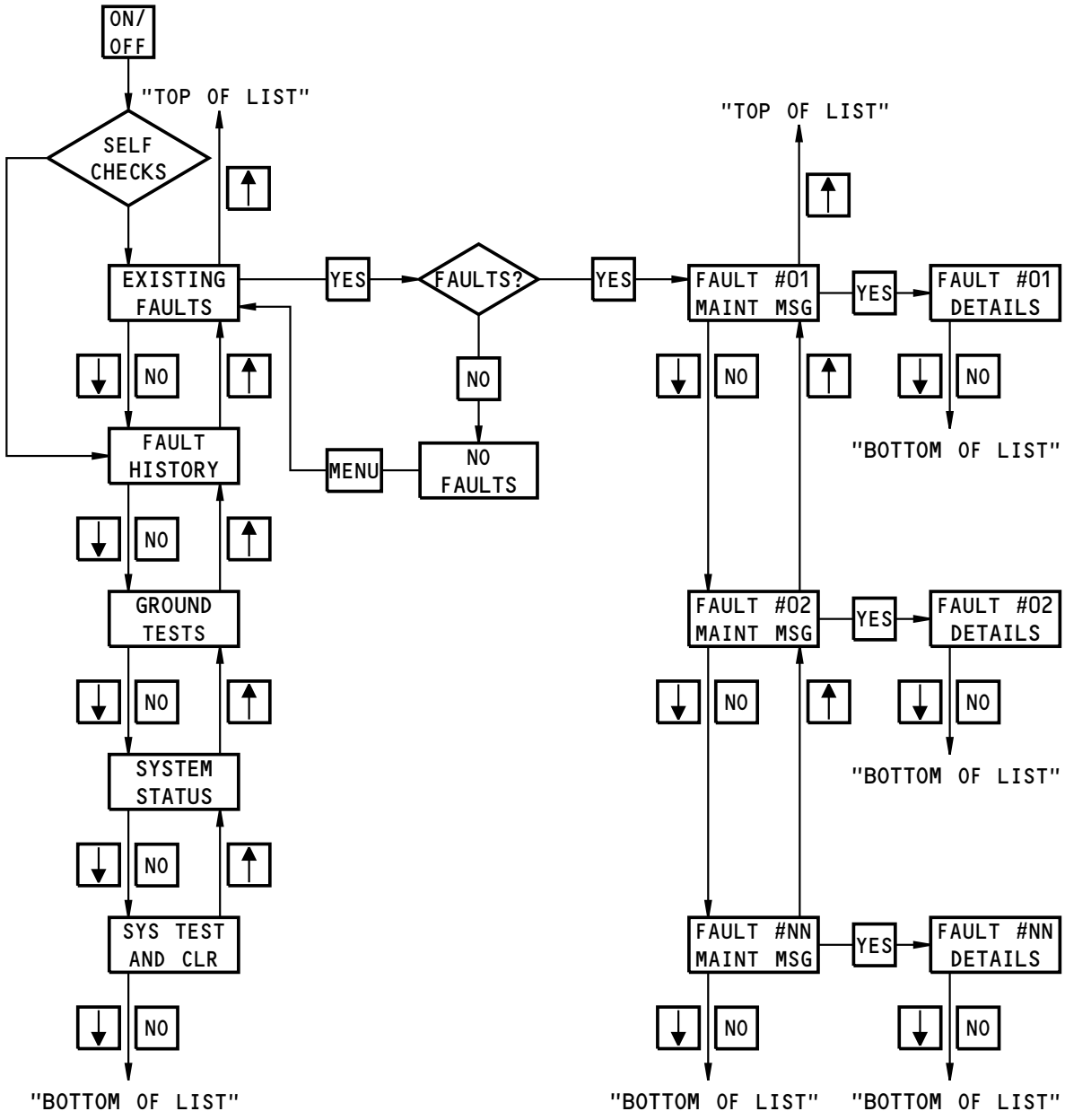


BITE ACCESS

K47402 S0006740549_V1

Cabin Pressure Control System BITE Procedure
Figure 202/21-31-00-990-808 (Sheet 2 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

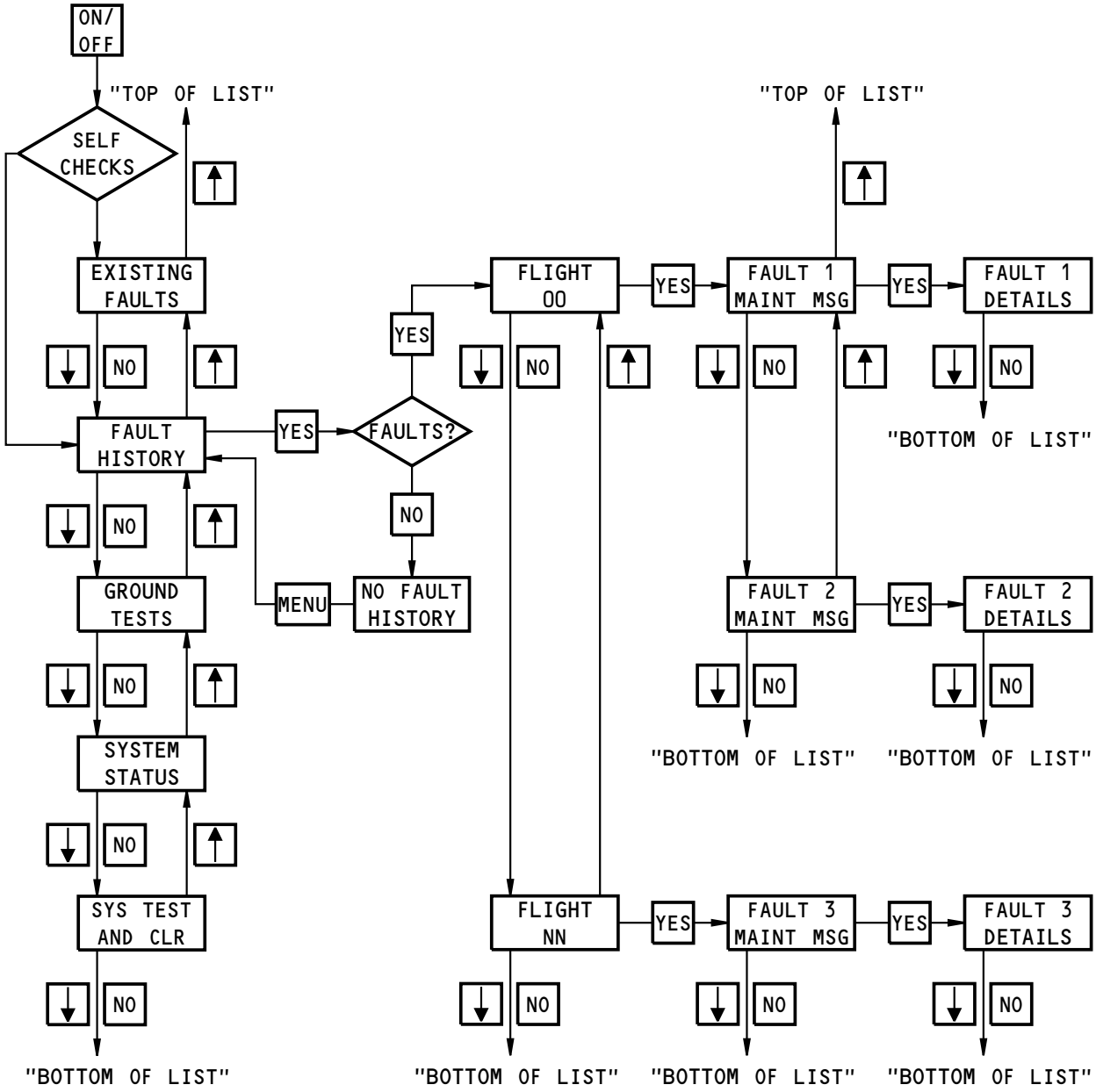


EXISTING FAULTS MENU

K43633 S0006740550_V1

Cabin Pressure Control System BITE Procedure
Figure 202/21-31-00-990-808 (Sheet 3 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

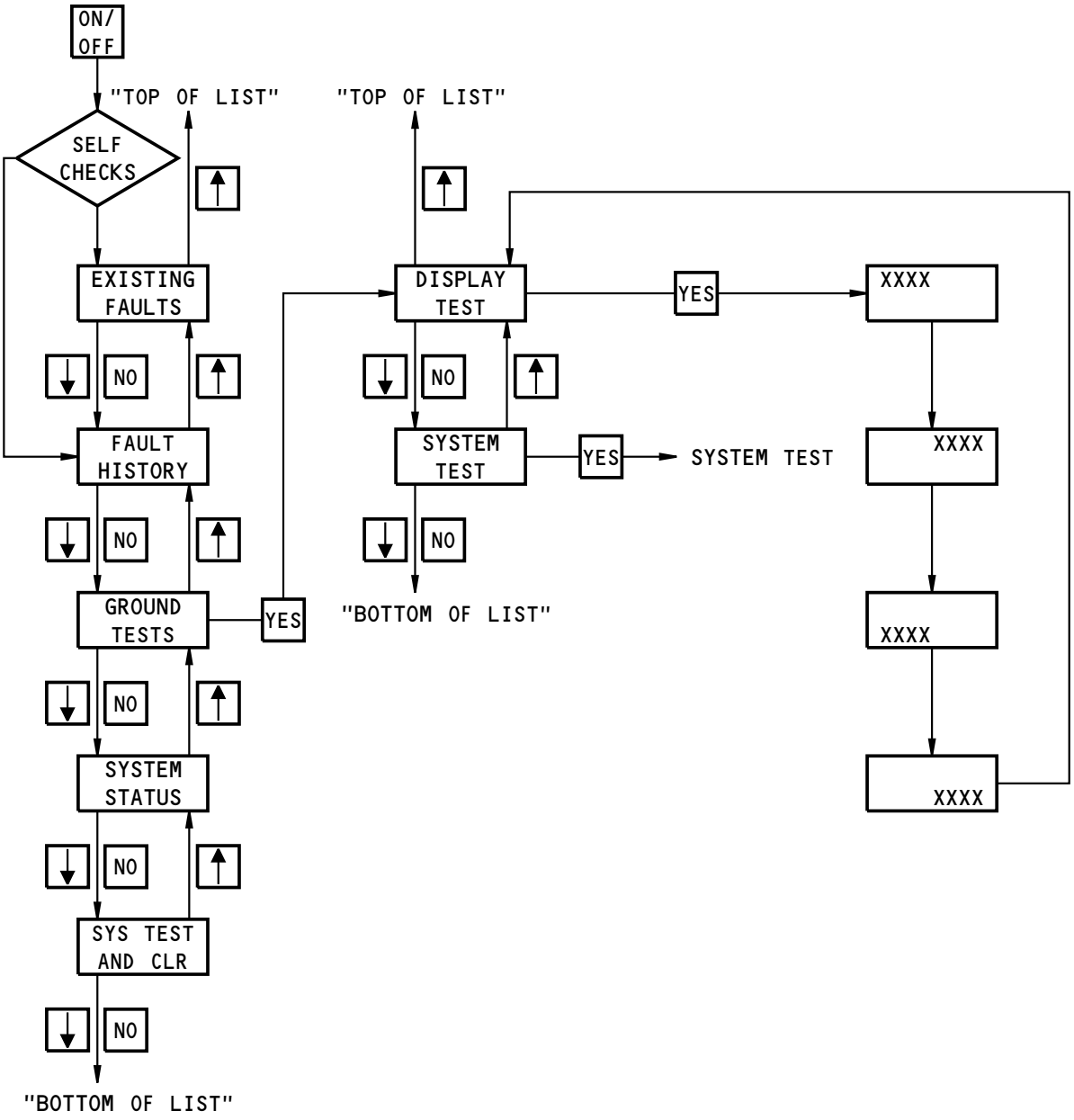


FAULT HISTORY MENU

K43825 S0006740551_V1

Cabin Pressure Control System BITE Procedure
Figure 202/21-31-00-990-808 (Sheet 4 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

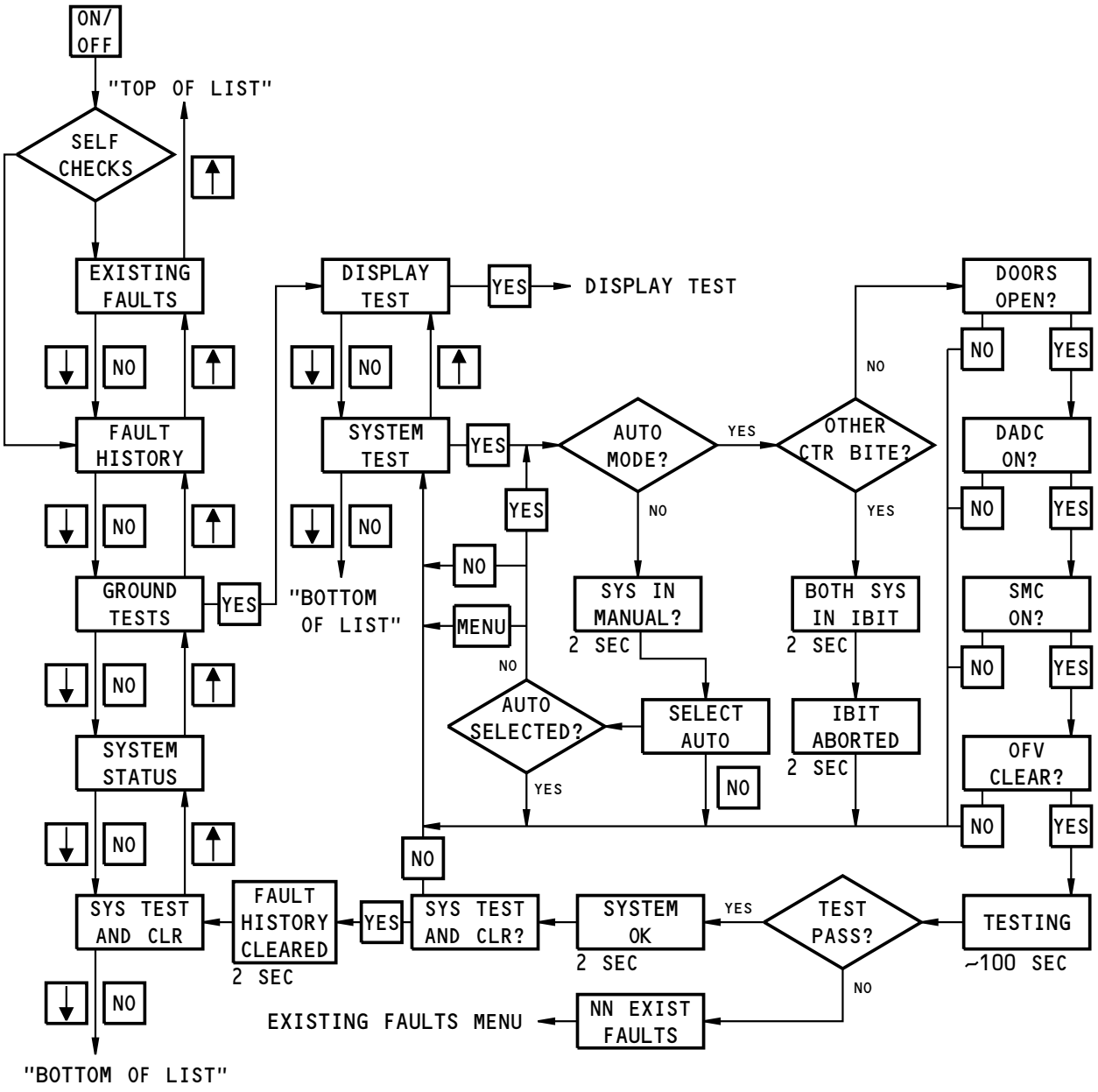


GROUND TESTS MENUS - DISPLAYS

K47449 S0006740552_V1

Cabin Pressure Control System BITE Procedure
Figure 202/21-31-00-990-808 (Sheet 5 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

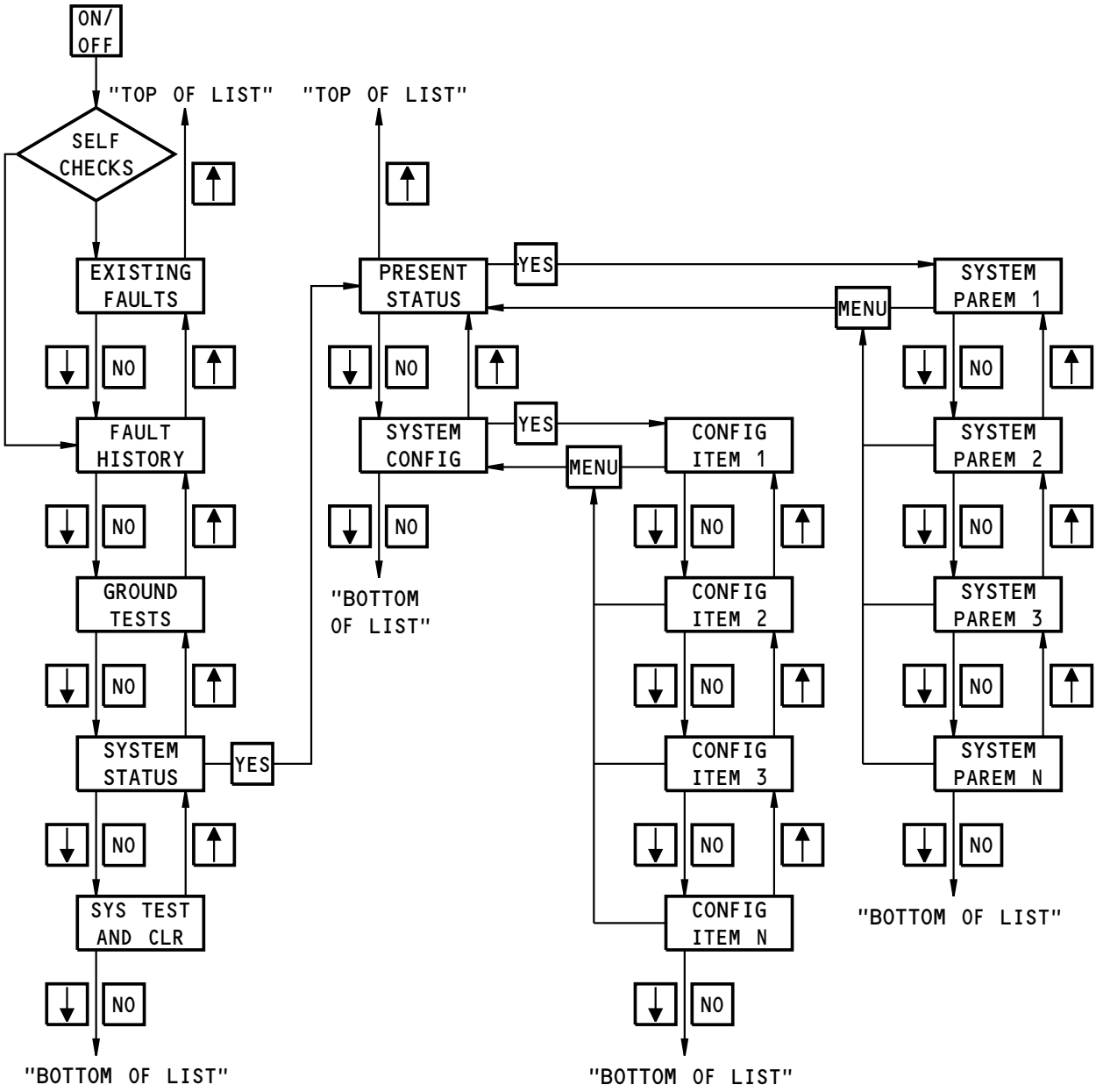


GROUND TESTS MENUS - SYSTEMS TEST

K47461 S0006740553_V1

Cabin Pressure Control System BITE Procedure
Figure 202/21-31-00-990-808 (Sheet 6 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

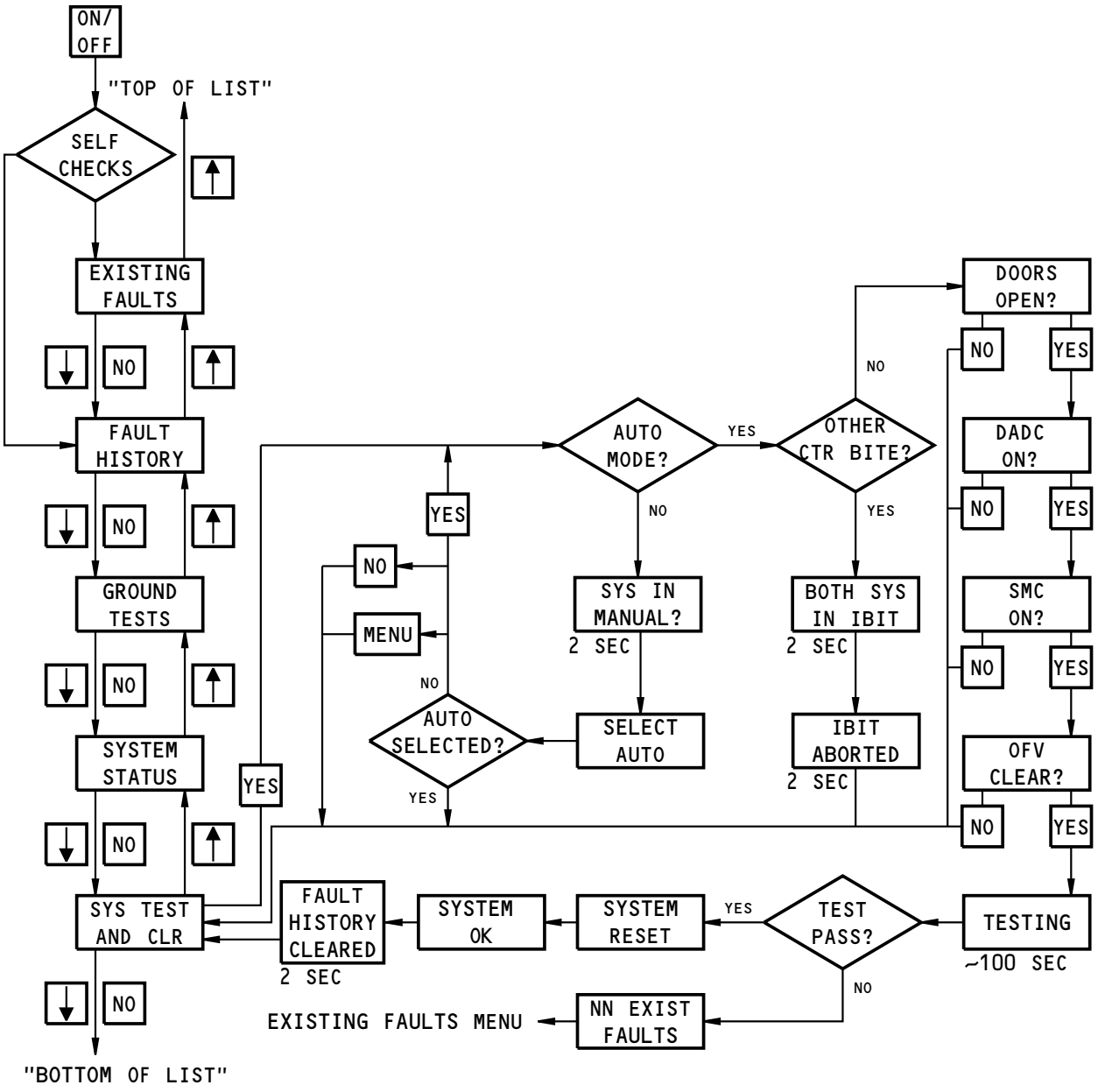


SYSTEM STATUS MENUS

K51277 S0006740554_V1

Cabin Pressure Control System BITE Procedure
Figure 202/21-31-00-990-808 (Sheet 7 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



SYSTEM TEST & CLEAR MENUS

K51289 S0006740555_V1

Cabin Pressure Control System BITE Procedure
Figure 202/21-31-00-990-808 (Sheet 8 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

802. Cabin Pressure Outflow Valve Fail - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) OFV LRU FAIL

B. Possible Causes

(1) Cabin pressure outflow valve, V48

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	1	C01273	PRESSURIZATION CONTROL LCD LTG
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL
F	7	C01272	PRESSURIZATION CONTROL IND

D. Related Data

(1) SSM 21-31-22

(2) SSM 21-31-23

(3) WDM 21-31-22

(4) WDM 21-31-23

E. Initial Evaluation

(1) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

(2) If the maintenance message does not show, then there was an intermittent fault.

(3) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Replace the cabin pressure outflow valve.

These are the tasks:

Aft Outflow Valve Assembly Removal, AMM TASK 21-31-03-000-801,

Aft Outflow Valve Assembly Installation, AMM TASK 21-31-03-400-801.

(a) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

1) If the maintenance message does not show, then you corrected the fault.

——— END OF TASK ———

803. Cabin Pressure Control Module Display Faults - Fault Isolation

A. Description

(1) Scope: The display faults covered in this task include improper displays of the FLT ALT and LAND ALT settings, improper illumination of the panel light plate or improper movement of the valve position indicator.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(2) Fault Indication: Depending on the type of failure, display failures of the pressure control module may or may not generate a BITE control module fault code during a cabin pressure control system BITE test.

(3) LCD Displays were replaced by LED Displays on cabin pressure control module (pressure selector panel) part number 1019439-1-001 as part of an obsolescence redesign. Fault Isolation procedures are the same, except where noted.

B. Possible Causes

(1) Pressure control module failure (P5-6 panel)

(2) Defective connectors (D10724, D10726)

(3) Dual AUTO FAIL light indication

(4) Loss of power or delayed power transfer to the cabin pressure controllers, M1653 and M1654

(5) Loss of power to the cabin pressure control module (P5-6 panel)

(6) System wiring fault or communication failure between the cabin pressure controllers and the cabin pressure control module

(7) System wiring fault between the cabin pressure control module (P5-6 panel) and the outflow valve, V48

(8) Outflow valve fault

C. Related Data

(1) SDSs 21-30-00

(2) SSMs 21-31-11 thru 21-31-25

(3) WDMs 21-31-11 thru 21-31-25

D. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	1	C01273	PRESSURIZATION CONTROL LCD LTG
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL
F	7	C01272	PRESSURIZATION CONTROL IND

E. Initial Evaluation

(1) Normal Conditions: If no cabin pressure control module faults are currently present, the FLT ALT and LAND ALT displays should be showing the last selected values. These values should be clearly shown with backlighting that responds to the panel lighting controls. The faceplate backlighting should be present and responsive to the panel lighting controls. The valve position indicator should be showing that the aft outflow valve is fully open with the aircraft on the ground.

(a) If the above conditions do not reflect the existing pressure control module displays, then do the Fault Isolation Procedure - Abnormal Conditions.

(b) If the above conditions reflect the existing pressure control module displays, the reported cabin pressure control module fault may no longer be present or may have been a normal condition reported as a fault. Continue to the next subtask to do functional checks of the module to make sure that it functions satisfactorily.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(2) Do these checks to make sure that the cabin pressure control module functions properly:

(a) Do these checks of the LCD (LED) segment displays on the cabin pressure control module:

1) Do this task: Master Dim and Test - Operational Test, AMM TASK 33-18-00-710-802.

2) Make sure that all of the LCD (LED) segments (all eights) on the LAND ALT and FLT ALT displays come on for two seconds and then go off for one second again and again.

3) If any LCD (LED) segments do not come on or go off properly:

a) For LCD displays, do this task: Fault Isolation Procedure - LCD Display Is Scrambled or One or More Segments Are Missing or Display Is Improperly Illuminated.

b) For LED displays, do this task: Fault Isolation Procedure - LCD (LED) Display is Blank.

4) If all LCD (LED) segments come on and go off properly, continue.

(b) Do a check of the cabin pressure control module LCD (LED) lighting:

1) Rotate the PANEL light control knob on the upper center section of the P5 forward overhead panel to the maximum BRIGHT position (fully clockwise).

2) Make sure that both the FLT ALT and LAND ALT displays increase in intensity.

3) Rotate the PANEL light control knob to the minimum BRIGHT position (fully counterclockwise).

4) Make sure that both the FLT ALT and LAND ALT displays decrease in intensity.

5) If one or more LCD (LED) displays fail to properly increase or decrease in intensity:

a) For LCD displays, do this task: Fault Isolation Procedure - LCD Display Backlighting Is Off or Dim.

b) For LED displays, do this task: Fault Isolation Procedure - LCD (LED) Display is Blank.

6) If the light intensity of both LCD (LED) displays increase and decrease properly, continue.

(c) Do this check of the displays of the altitude range settings in the FLT ALT and LAND ALT windows:

1) Set altitudes in the FLT ALT window from -1000 to 42000.

2) Make sure that all the digits are complete and clearly visible at each selection.

3) Set altitudes in the LAND ALT window from -1000 to 14000.

4) Make sure that all the digits are complete and clearly visible at each selection.

5) If any digit is incomplete:

a) For LCD displays, do this task: Fault Isolation Procedure - LCD Display Is Scrambled, One or More Segments Are Missing or Display Is Improperly Illuminated.

b) For LED displays, do this task: Fault Isolation Procedure - LCD (LED) Display is Blank.

6) If either display fails to adjust across the specified range, do this task: Fault Isolation Procedure - LCD (LED) Display Cannot Be Adjusted Above or Below Certain Values.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

7) If either display fails to adjust at all (display is frozen or full bright), do this task: Fault Isolation Procedure - LCD (LED) Display Is Frozen or Full Bright (LCD (LED) does not respond to rotation of the corresponding knob).

8) If all digits are complete and both displays adjust across the specified range, then continue.

(d) Do this check of the valve position indicator on the cabin pressure control module:

1) Turn the mode selector switch to MAN.

2) Make sure that these switches on the P5-10 forward overhead panel are set to OFF:

a) L PACK

b) R PACK

c) BLEED 1

d) BLEED 2

e) APU BLEED.

3) Make sure that an outside cabin door is open.

4) Push and hold the CLOSE/OPEN toggle switch on the pressure control module to the CLOSE position until the outflow valve is fully closed.

a) Make sure that the valve position indicator needle moves smoothly and freely.

b) Visually observe that the valve has fully closed.

5) Push and hold the CLOSE/OPEN toggle switch on the pressure control module to the OPEN position until the outflow valve is fully open.

a) Make sure that the valve position indicator needle moves smoothly and freely.

b) Visually observe that the valve has fully opened.

6) Turn the mode selector switch to AUTO.

7) If the valve position indicator needle did not move, do this task: Fault Isolation Procedure - Valve Position Indicator Needle Does Not Move When Outflow Valve Movement Is Commanded.

8) If the valve position indicator needle did not move freely (sticks), do this task: Fault Isolation Procedure - Valve Position Indicator Needle Is Sticking.

9) If the valve position indicator needle moves satisfactorily, then continue.

(3) Do the DCPC BITE procedure on both cabin pressure controllers, M1653 and M1654. To do the BITE, do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

(a) If the BITE test shows the maintenance message SEL PNL LRU FAIL in EXISTING FAULTS or FAULT HISTORY, then do this task: Cabin Pressure Control Module Display Faults - Fault Isolation, 21-31 TASK 803.

(b) If the BITE test shows a different maintenance message, then do the appropriate task as indicated in Table 201 of this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

(c) If no fault codes are displayed in EXISTING FAULTS or FAULT HISTORY, then the system and the pressure control module are performing correctly. No additional maintenance is required.

(d) If fault isolation of the displayed faults does not resolve the fault in the cabin pressure control module, then proceed to Fault Isolation Procedure - Abnormal Conditions.

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(4) If the BITE test shows occurrences of Fault Code 06 or a combination of Fault Codes 06 and 101, do these checks:

(a) Do a check that the power to both cabin pressure controllers M1653 and M1654 is satisfactory.

(b) Perform a SYSTEM TEST or SYSTEM TEST and CLEAR to do a check of the system health status.

(c) If these checks are satisfactory, there is no requirement to replace any digital cabin pressurization control components.

(d) If the Fault Codes still show in the EXISTING FAULTS menu and cannot be reset, replace pressure controller M1654 as follows:

These are the tasks:

Cabin Pressure Controller (CPC) Removal, AMM TASK 21-31-01-000-801,

Cabin Pressure Controller (CPC) Installation, AMM TASK 21-31-01-400-801.

F. Fault Isolation Procedure - Abnormal Display Conditions

(1) Cabin pressure control module display faults may take these forms:

Cabin Pressure Control Module Display Faults - Conditions

NO.	FAULT CONDITION	FAULT ISOLATION PROCEDURE
1	LCD (LED) Display is blank or too dim to read	LCD (LED) Display Is Blank or Too Dim To Read
2	LCD (LED) display is “88888”	LCD (LED) Display Shows “88888”
3	LCD (LED) display is showing “-----” (dashes)	LCD (LED) Display Shows “-----” (all dashes)
4	LCD display is scrambled or has one or more segments missing or improperly illuminated	LCD Display Is Scrambled or One or More Segments Are Missing or Display Is Improperly Illuminated
5	LCD (LED) display is frozen or full bright	LCD (LED) Display is Frozen or Full Bright
6	LCD (LED) display cannot be adjusted above or below certain values	LCD (LED) Display Cannot Be Adjusted Above or Below Certain Values
7	LCD display backlighting is off or dim	LCD Display Backlighting Is Off or Dim
8	Light panel backlighting is off or dim	Light Panel Backlighting Is Off or Dim
9	Valve position indicator needle is sticking	Valve Position Indicator Needle Is Sticking
10	Valve position indicator needle does not move when outflow valve movement is commanded	Valve Position Indicator Needle Does Not Move When Outflow Valve Movement Is Commanded

(2) Do the fault isolation procedure listed in the table for the specific fault condition.

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G. Fault Isolation Procedure - LCD (LED) Display Is Blank or Too Dim To Read

(1) Possible causes of a blank or very dim LCD (LED) display are a fault in the backlighting, lack of power to the panel in the AUTO mode or internal control module failure.

NOTE: For LED display panels, the display will be very dim and may be appear to be blank during daylight hours with the Master Dim & Test switch in the Dim position. Also, the battery switch needs to be on for the valve position indicator to be displayed. These are normal operating characteristics for the LED displays.

(2) For cabin pressure control modules with LED display panels, if the FLT ALT, LAND ALT or Outflow Valve (OFV) position indicator displays are blank, do this check to attempt to restore the displays to operation.

(a) If displays are blank, open these circuit breakers and then close them again after three seconds as follows:

1) Open and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	1	C01273	PRESSURIZATION CONTROL LCD LTG
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL
F	7	C01272	PRESSURIZATION CONTROL IND

(b) If the displays now show valid values for FLT ALT, LAND ALT and OFV position indicator, then there was an intermittent fault and the system is serviceable.

(3) If both the FLT ALT and LAND ALT displays are blank, do this check to make sure that power is being supplied to the panel:

(a) Make sure that all circuit breakers are closed as follows:

1) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	1	C01273	PRESSURIZATION CONTROL LCD LTG
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL
F	7	C01272	PRESSURIZATION CONTROL IND

(b) Do a check that power is available to the pressure control module backlighting and to the cabin pressure controllers as follows:

1) Do Fault Isolation Procedure - Panel Power Input Check.

(c) If the power check is satisfactory, the pressure control module has an internal failure. Remove and replace the pressure control module with a serviceable unit as follows: Cabin Pressure Control Module - Removal, AMM TASK 21-31-02-000-801 and Cabin Pressure Control Module - Installation, AMM TASK 21-31-02-400-801.

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(4) If only one display is blank and the other display operates properly, an internal failure in the pressure control module has occurred. Do these tasks to remove and install a serviceable pressure control module: Cabin Pressure Control Module - Removal, AMM TASK 21-31-02-000-801 and Cabin Pressure Control Module - Installation, AMM TASK 21-31-02-400-801.

(5) Do the Repair Confirmation at the end of this task.

H. Fault Isolation Procedure - LCD (LED) Displays Show “88888”

(1) There are some conditions that occur with normal power transfers that cause disturbances in the cabin pressure control module displays but are normal conditions that do not require maintenance action. After a power interruption the panel will perform an internal self-test. During the self-test, the FLT ALT and LAND ATL windows will display “88888”, the FLT ALT and LAND ALT adjustment knobs will not be able to adjust the display and the AUTO FAIL indication will be extinguished. The self-test should not last more than one second unless failures are detected. When the self-test is successfully completed, the FLT ALT and LAND ALT displays should show the last selected values. If either of the FLT ALT or LAND ALT displays currently shows “88888”, proceed with the steps that follow:

(a) If both displays show “88888” (all eights), open these circuit breakers and then close them again after three seconds as follows:

1) Open and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	1	C01273	PRESSURIZATION CONTROL LCD LTG
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL
F	7	C01272	PRESSURIZATION CONTROL IND

(b) If both displays continue to show “88888” (all eights), do these steps to replace the pressure control module:

1) Do this task to remove the pressure control module: Cabin Pressure Control Module - Removal, AMM TASK 21-31-02-000-801.

2) Do this task to install a serviceable pressure control module: Cabin Pressure Control Module - Installation, AMM TASK 21-31-02-400-801.

(c) If only one display shows “88888” (all eights) and the other display operates correctly, do these steps to replace the pressure control module:

1) Do this task to remove the pressure control module: Cabin Pressure Control Module - Removal, AMM TASK 21-31-02-000-801.

2) Do this task to install a serviceable pressure control module: Cabin Pressure Control Module - Installation, AMM TASK 21-31-02-400-801

(2) Do the Repair Confirmation at the end of this task.

I. Fault Isolation Procedure - LCD (LED) Display Shows “-----” (all dashes)

(1) If both displays show “-----” (all dashes) and there is no AUTO FAIL indication present or reported, do these steps:

(a) Make sure the mode selector switch is set to the AUTO position.

(b) Perform a power reset to the system as follows:

1) Open the circuit breakers that follow and close them again after three seconds:

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a) Open and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	1	C01273	PRESSURIZATION CONTROL LCD LTG
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL
F	7	C01272	PRESSURIZATION CONTROL IND

2) Wait five seconds for the system to reset:

a) Make sure that engine, ground or APU power is present for the power reset.

(2) If the display now shows valid values for FLT ALT and LAND ALT, then there was an intermittent fault and the system is serviceable.

(3) If both displays continue to show "-----" (all dashes) 5 seconds after the reset and no AUTO FAIL indication is present, an internal failure in the pressure control module has occurred. Do these tasks to remove and install a serviceable pressure control module: Cabin Pressure Control Module - Removal, AMM TASK 21-31-02-000-801 and Cabin Pressure Control Module - Installation, AMM TASK 21-31-02-400-801.

(4) If both displays show "-----" with AUTO FAIL indication present, do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

(a) If the BITE test shows that faults are present in either EXISTING FAULTS or FAULT HISTORY, then do the appropriate task as indicated in Table 201 of Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

(5) If only one display shows "-----" (all dashes) and the other display operates normally, an internal failure in the pressure control module has occurred. Do these tasks to remove and install a serviceable pressure control module: Cabin Pressure Control Module - Removal, AMM TASK 21-31-02-000-801 and Cabin Pressure Control Module - Installation, AMM TASK 21-31-02-400-801.

(6) Do the Repair Confirmation at the end of this task.

J. Fault Isolation Procedure - LCD Display Is Scrambled or One or More Segments Are Missing or Improperly Illuminated

(1) If one or both displays appear scrambled or show one or more segments improperly illuminated or extinguished, do these steps to find if the pressure control module is serviceable:

(a) There are some pressure control module display failures caused by internal failures of the pressure control module that have no effect on the operation or performance of the pressurization control system.

1) Operators must determine if the FLT ALT and LAND ALT values can be accurately selected and then make a decision if the pressure control module is serviceable.

2) Operators may find that pressure control modules with the display failures listed below can be considered serviceable for dispatch:

a) Loss or degradation of LCD backlighting: the LCD displays are of the transflective type so that ambient light or an external light source may make a display readable despite degraded or failed backlighting.

b) Degradation of the light plate backlighting: ambient light or an external light source may compensate for failed or degraded light plate illumination.

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c) Illumination of non-digit portions of the displays may degrade display contrast but the display may still be readable.

d) Improper illumination of some display digit segments does not necessarily make the pressure control module unserviceable. For example, the two digits on the right end of the FLT ALT display are always zero “0” under normal conditions as is the most right digit of the LAND ALT display. A missing or improperly lit segment in these digits should not significantly affect the ability to read the display. Similarly, the most left digit of the LAND ALT display is always either a “1” or blank during normal operation so that improper operation of one of the unused segments should not significantly affect the ability to read the display.

(2) If the pressure control module is found not serviceable, do these tasks to replace the pressure control module with a serviceable unit: Cabin Pressure Control Module - Removal, AMM TASK 21-31-02-000-801 and Cabin Pressure Control Module - Installation, AMM TASK 21-31-02-400-801.

(3) Do the Repair Confirmation at the end of this task.

K. Fault Isolation Procedure - LCD (LED) Display Is Frozen or Full Bright

(1) If both displays appear frozen or full bright, open all circuit breakers and close them again after 3 seconds to perform a power reset to the system:

(a) Open and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	1	C01273	PRESSURIZATION CONTROL LCD LTG
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL
F	7	C01272	PRESSURIZATION CONTROL IND

(2) If both displays still appear frozen or full bright or if a single display appears frozen or full bright, an internal failure in the cabin pressure control panel has occurred. Do these tasks to remove and install a serviceable pressure control module: Cabin Pressure Control Module - Removal, AMM TASK 21-31-02-000-801 and Cabin Pressure Control Module - Installation, AMM TASK 21-31-02-400-801.

(3) Do the Repair Confirmation at the end of this task.

L. Fault Isolation Procedure - LCD (LED) Display Cannot Be Adjusted Above or Below Certain Values

(1) If one or both displays do not adjust across the full range (FLT ALT: -1000 to 42000; LAND ALT: -1000 to 14000), an internal failure in the pressure control module has occurred. Do these tasks to replace the cabin pressure control module with a serviceable unit: Cabin Pressure Control Module - Removal, AMM TASK 21-31-02-000-801 and Cabin Pressure Control Module - Installation, AMM TASK 21-31-02-400-801.

(2) Do the Repair Confirmation at the end of this task.

M. Fault Isolation Procedure - LCD Display Backlighting is Off or Dim

(1) If the backlighting for both the FLT ALT and LAND ALT displays is off or dim, make sure that the PANEL light control knob on the upper center section of the P5 forward overhead panel is set to maximum brightness.

(2) If the backlighting is still off or dim, do this check for the availability of power as follows:

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(a) Do Fault Isolation Procedure - Panel Power Input Check.

(3) If the power check is satisfactory, the pressure control module has an internal failure. Remove and replace the pressure control module with a serviceable module as follows:Cabin Pressure Control Module - Removal, AMM TASK 21-31-02-000-801 and Cabin Pressure Control Module - Installation, AMM TASK 21-31-02-400-801.

(4) If only one display backlighting is off or dim, do these steps to find if the pressure control module is serviceable:

(a) There are some pressure control module display failures caused by internal failures of the pressure control module that have no effect on the operation or performance of the pressurization control system.

1) Operators must determine if the FLT ALT and LAND ALT values can be accurately selected and then make a decision if the pressure control module is serviceable.

2) Operators may find that pressure control modules with the display failures listed below can be considered serviceable for dispatch:

a) Loss or degradation of LCD backlighting: the LCD displays are of the transflective type so that ambient light or an external light source may make a display readable despite degraded or failed backlighting.

b) Degradation of the light plate backlighting: ambient light or an external light source may compensate for failed or degraded light plate illumination.

c) Illumination of non-digit portions of the displays may degrade display contrast but the display may still be readable.

d) Improper illumination of some display digit segments does not necessarily make the pressure control module unserviceable. For example, the two digits on the right end of the FLT ALT display are always zero “0” under normal conditions as is the most right digit of the LAND ALT display. A missing or improperly lit segment in these digits should not significantly affect the ability to read the display. Similarly, the most left digit of the LAND ALT display is always either a “1” or blank during normal operation so that improper operation of one of the unused segments should not significantly affect the ability to read the display.

(5) If the pressure control module is found not serviceable, do these tasks to replace the pressure control module with a serviceable unit: Cabin Pressure Control Module - Removal, AMM TASK 21-31-02-000-801 and Cabin Pressure Control Module - Installation, AMM TASK 21-31-02-400-801.

(6) Do the Repair Confirmation at the end of this task.

N. Fault Isolation Procedure - Light Panel Backlighting Is Off or Dim

(1) If the pressure control module light plate indications are dim or appear to be off, make sure the PANEL light control knob on the upper center section of the P5 forward overhead panel is set to maximum brightness.

(2) If the backlighting is still off or dim, do this check for the availability of power to the light panel as follows:

(a) Make sure that all circuit breakers are closed as follows:

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1) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	1	C01273	PRESSURIZATION CONTROL LCD LTG
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL
F	7	C01272	PRESSURIZATION CONTROL IND

(b) Do Fault Isolation Procedure - Panel Power Input Check.

(3) If the power check is satisfactory, do these steps to find if the pressure control module is serviceable:

(a) There are some pressure control module display failures caused by internal failures of the pressure control module that have no effect on the operation or performance of the pressurization control system.

1) Operators must determine if the FLT ALT and LAND ALT values can be accurately selected and then make a decision if the pressure control module is serviceable.

2) Operators may find that pressure control modules with the display failures listed below can be considered serviceable for dispatch:

a) Loss or degradation of LCD backlighting: the LCD displays are of the transflective type so that ambient light or an external light source may make a display readable despite degraded or failed backlighting.

b) Degradation of the light plate backlighting: ambient light or an external light source may compensate for failed or degraded light plate illumination.

c) Illumination of non-digit portions of the displays may degrade display contrast but the display may still be readable.

d) Improper illumination of some display digit segments does not necessarily make the pressure control module unserviceable. For example, the two digits on the right end of the FLT ALT display are always zero “0” under normal conditions as is the most right digit of the LAND ALT display. A missing or improperly lit segment in these digits should not significantly affect the ability to read the display. Similarly, the most left digit of the LAND ALT display is always either a “1” or blank during normal operation so that improper operation of one of the unused segments should not significantly affect the ability to read the display.

(4) If the pressure control module is found not serviceable, do these tasks to replace the pressure control module with a serviceable unit: Cabin Pressure Control Module - Removal, AMM TASK 21-31-02-000-801 and Cabin Pressure Control Module - Installation, AMM TASK 21-31-02-400-801.

(5) Do the Repair Confirmation at the end of this task.

O. Fault Isolation Procedure - Valve Position Indicator Is Sticking

(1) If not already done in the Initial Evaluation, do these steps:

(a) Turn the mode selector switch on the pressure control module to MAN.

(b) Make sure that these switches are set to OFF:

1) L PACK

2) R PACK

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3) BLEED 1

4) BLEED 2

5) APU BLEED

(c) Make sure that an external cabin door is open.

(d) Push and hold the CLOSE/OPEN toggle switch on the pressure control panel to the CLOSE position until the outflow valve is fully closed.

1) Make sure that the valve position indicator needle movement is smooth and the needle does not stick.

2) Visually observe that the valve has fully closed.

(e) Push and hold the CLOSE/OPEN toggle switch on the pressure control panel to the OPEN position until the outflow valve is fully open.

1) Make sure that the valve position indicator needle movement is smooth and the needle does not stick.

2) Visually observe that the valve has fully opened.

(f) Set the mode selector switch to AUTO.

(2) If the valve position indicator needle moves satisfactorily, there was an intermittent fault and the system is serviceable.

(3) If the needle hesitates after passing the fifth hash mark from the left, this is a known unusual operation that is considered acceptable; the pressure control module is considered satisfactory for continued service.

(4) If the needle sticks, the pressure control module has an internal failure. Do these tasks to replace the pressure control module with a serviceable unit: Cabin Pressure Control Module - Removal, AMM TASK 21-31-02-000-801 and Cabin Pressure Control Module - Installation, AMM TASK 21-31-02-400-801.

(5) Do the Repair Confirmation at the end of this task.

P. Fault Isolation Procedure - Valve Position Indicator Needle Does Not Move When Outflow Valve Movement Is Commanded

(1) Do a test of the pressurization system manual mode. To do the test, do this task: Pressurization System Manual Mode Test, AMM TASK 21-31-00-710-801 or Pressurization System Manual Mode Test, AMM TASK 21-31-00-710-803.

(a) Make sure that an observer near the aft outflow valve at the rear of the aircraft sees that the outflow valve goes fully closed and then fully open.

(2) If the test was satisfactory, there was an intermittent fault and the cabin pressure control system is serviceable.

(3) If the outflow valve did not move when commanded, do this task: Manual Mode Control of Aft Outflow Valve Does Not Operate - Fault Isolation, 21-31 TASK 811.

(4) If the outflow valve moved but the valve position indicator needle did not move, do this check of the wiring between the cabin pressure control module and the aft outflow valve:

(a) Lower the forward overhead P5 panel.

1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	1	C01273	PRESSURIZATION CONTROL LCD LTG

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(Continued)

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL
F	7	C01272	PRESSURIZATION CONTROL IND

(b) Disconnect connector D10724 from the back of the cabin pressure selector panel.

(c) Do this task to remove the aft outflow valve to get access to electrical connectors D10728 and D10730: Aft Outflow Valve Assembly Removal, AMM TASK 21-31-03-000-801.

(d) Do a wiring check between these pins of connector D10724 of the P5-6 cabin pressure selector panel and connectors D10728 and D10730 of the cabin pressure outflow valve.

P5-6 Panel D10724	**OFV D10728**
pin 18	pin F
pin 30	pin H
pin 31	pin G

P5-6 D10724	**OFV D10730**
pin 25	pin A
pin 24	pin B

(f) Inspect all pins on all connectors to make sure they are not bent or pushed back.

(g) Repair any problems you find with the wiring or connectors.

(5) If you repaired the wiring or connectors, then do these steps:

(a) Reconnect the electrical connector D10724 to the cabin pressure control module.

(b) Raise the cabin pressure control module into position and secure it to the P5 forward overhead panel.

(c) Install the aft outflow valve as follows: Aft Outflow Valve Assembly Installation, AMM TASK 21-31-03-400-801.

(6) If no wiring faults are found and the valve position indicator needle does not move when outflow valve movement is observed, do this step:

(a) Remove the cabin pressure control module and install a serviceable unit as follows: Cabin Pressure Control Module - Removal, AMM TASK 21-31-02-000-801 and Cabin Pressure Control Module - Installation, AMM TASK 21-31-02-400-801.

(7) If there were no wiring faults and the replacement of the cabin pressure control module did not correct the fault, the outflow valve has an internal fault and must be replaced as follows: Aft Outflow Valve Assembly Removal, AMM TASK 21-31-03-000-801 and Aft Outflow Valve Assembly Installation, AMM TASK 21-31-03-400-801.

(8) Do the Repair Confirmation at the end of this task.

Q. Fault Isolation Procedure - Panel Power Input Fault

(1) Do a check for 28V DC power to the P5-6 cabin pressure selector panel as follows:

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(a) Open these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	1	C01273	PRESSURIZATION CONTROL LCD LTG
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL
F	7	C01272	PRESSURIZATION CONTROL IND

(b) Get access to the electrical connectors on the back of the P5-6 cabin pressure control module as follows:

1) Loosen the 1/4-turn fasteners that hold the P5 forward overhead panel in position and hold the panel as it moves downward.

(c) Disconnect electrical connectors D10724 and D10726.

(d) Close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	1	C01273	PRESSURIZATION CONTROL LCD LTG
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL
F	7	C01272	PRESSURIZATION CONTROL IND

(e) Do these checks for power to the P5-6 pressure selector panel:

1) Do a check for a nominal 28V DC from pin 20 of connector D10726 to ground at pin 21 of connector D10726.

2) Do a check for a nominal 28V DC from pin 15 of connector D10724 to ground at pin 28 of connector D10724.

3) Do a check for a nominal 28V DC from pin 16 of connector D10724 to ground at pin 29 of connector D10724.

4) Make sure that the PANEL light control knob on the upper center section of the P5 forward overhead panel is set to maximum brightness.

5) Make sure that the master dim and test switch (TEST/BRT/DIM) on the P1-3 main instrument panel is set to BRT.

6) Do a check for a nominal 5V AC from pin 12 of connector D10726 to ground at pin 13 of connector D10726.

7) Do a check for a nominal 5V AC from pin 24 of connector D10726 to ground at pin 13 of connector D10726.

8) If power is not available, repair wiring or fault isolate the power distribution system to the 28V battery buses.

(f) Inspect the connectors as follows:

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1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL
F	7	C01272	PRESSURIZATION CONTROL IND

2) Examine the plug and receptacle of each connector to make sure that they are not damaged.

3) Inspect the pins to make sure they are not bent or pushed back.

4) Repair any problems that you find.

(g) If you made a repair, do the Repair Confirmation at the end of this procedure.

(h) If you did not find any problems, then return to the step that directed you to this Fault Isolation Procedure - Panel Power Input Fault and continue.

R. Repair Confirmation

(1) If the pressure control module or outflow valve has been removed and replaced or if any wiring or connectors have been repaired, do these steps to make sure that the repair is satisfactory:

(a) Do the Pressurization System Ground Test if not already done. To do the test, do this task: Pressurization System Ground Test, AMM TASK 21-31-00-700-802.

(b) Do a Pressurization System Manual Mode test if not already done. To do the test, do this task: Pressurization System Manual Mode Test, AMM TASK 21-31-00-710-801 or Pressurization System Manual Mode Test, AMM TASK 21-31-00-710-803.

(c) Do steps E.(1) and E.(2) of the Initial Evaluation to make sure that the displays and functions of the cabin pressure control module operate properly.

(d) If any abnormal condition still continues, do the fault isolation procedure as directed in the specific checks listed in the Initial Evaluation.

——— END OF TASK ———

804. System Wiring Failure - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) SYSTEM WIRING

B. Possible Causes

(1) Wiring

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	1	C01273	PRESSURIZATION CONTROL LCD LTG
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL

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(Continued)

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	7	C01272	PRESSURIZATION CONTROL IND

D. Related Data

(1) SSM 21-31-11

(2) SSM 21-31-22

(3) SSM 21-31-23

(4) SSM 21-31-24

(5) WDM 21-31-11

(6) WDM 21-31-22

(7) WDM 21-31-23

(8) WDM 21-31-24

E. Initial Evaluation

(1) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

(a) If the maintenance message does not show, then there was an intermittent fault.

(b) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Do these steps for the fault detail CPC-PNL WIRING on Cabin Pressure Controller 1:

(a) Do this check of the wiring between the Cabin Pressure Controller (CPC) 1, M1653 in the Electronic Equipment (EE) Bay and the Cabin Pressure Control Module, P5-6 in the Flight Deck:

1) Remove the CPC 1. This is the task: Cabin Pressure Controller (CPC) Removal, AMM TASK 21-31-01-000-801.

2) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	6	C01269	PRESSURIZATION CONTROL MANUAL

3) Disconnect connector D10732 from the Cabin Pressure Outflow Valve.

NOTE: Reach through the open valve gates to get access to the connector.

4) Lower the Forward Overhead Panel, P5.

5) Disconnect connector D10724 from the Cabin Pressure Control Module.

6) Do a wiring check between these pins of connector D10724 of the Cabin Pressure Control Module and connectors D10738B and D10738A of the CPC 1 (WDM 21-31-11, WDM 21-31-22):

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CABIN PRESSURE CONTROL MODULE, P5-6 D10724	CPC 1, M1653 D10738B
pin 20	pin A7
pin 21	pin A8
pin 1	pin B3
pin 4	pin B7

D10724	D10738A
pin 9	pin A1
pin 22	pin A2
pin 26	pin D15

D10724	Ground
pin 11	GND
pin 12	GND

7) Do a wiring check between these pins of connector D10732 and Structure Ground (WDM 21-31-22):

CABIN PRESSURE OUTFLOW VALVE, V48 D10732	Ground
pin B-	GD3738-DC
pin J	GD3738-DC

8) If you find a problem with the wiring, then do these steps:
   a) Repair the wiring.
   b) Re-connect connector D10724 to the Cabin Pressure Control Module.
   c) Close the Forward Overhead Panel, P5.
   d) Re-connect connector D10732 to the Cabin Pressure Outflow Valve.
   e) Re-install the CPC 1. Do this task: Cabin Pressure Controller (CPC) Installation, AMM TASK 21-31-01-400-801.
   f) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	6	C01269	PRESSURIZATION CONTROL MANUAL

   g) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.
   h) If the maintenance message does not show, then you corrected the problem.

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i) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

(2) Do these steps for the fault detail CPC-PNL WIRING on Cabin Pressure Controller 2:

(a) Do this check of the wiring between the CPC 2, M1654 in the EE Bay and the Cabin Pressure Control Module, P5-6 in the Flight Deck:

1) Remove the CPC 2. This is the task: Cabin Pressure Controller (CPC) Removal, AMM TASK 21-31-01-000-801.

2) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	6	C01269	PRESSURIZATION CONTROL MANUAL

3) Disconnect connector D10734 from the Cabin Pressure Outflow Valve.

NOTE: Reach through the open valve gates to get access to the connector.

4) Lower the Forward Overhead Panel, P5.

5) Disconnect connector D10726 from the Cabin Pressure Control Module.

6) Do a wiring check between these pins of connector D10726 of the Cabin Pressure Control Module and connectors D10740B and D10740A of the CPC 2 (WDM 21-31-11, WDM 21-31-23):

CABIN PRESSURE CONTROL MODULE, P5-6 D10726	**CPC 2, M1654 D10740B**
pin 28	pin A7
pin 29	pin A8
pin 8	pin B3
pin 9	pin B7

D10726	**D10740A**
pin 16	pin A1
pin 17	pin A2
pin 19	pin D15

D10726	**Ground**
pin 18	GD506-ST
pin 30	GD506-ST

7) Do a wiring check between these pins of connector D10734 and Structure Ground (WDM 21-31-23):

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CABIN PRESSURE OUTFLOW VALVE, V48

D10734	**Ground**
pin B-	GD3740-DC
pin J	GD3740-DC

8) If you find a problem with the wiring, then do these steps:
   a) Repair the wiring.
   b) Re-connect connector D10726 to the Cabin Pressure Control Module.
   c) Close the Forward Overhead Panel, P5.
   d) Re-connect connector D10734 to the Cabin Pressure Outflow Valve.
   e) Re-install the CPC 2. Do this task: Cabin Pressure Controller (CPC) Installation, AMM TASK 21-31-01-400-801.
   f) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	6	C01269	PRESSURIZATION CONTROL MANUAL

   g) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.
   h) If the maintenance message does not show, then you corrected the problem.
   i) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

(3) Do these steps for the fault detail CPC/OFV WIRING on Cabin Pressure Controller 1:
   (a) Do this check of the wiring between the CPC 1, M1653 in the EE Bay and the Cabin Pressure Outflow Valve, V48:
      1) Remove the CPC 1. This is the task: Cabin Pressure Controller (CPC) Removal, AMM TASK 21-31-01-000-801.
      2) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	6	C01269	PRESSURIZATION CONTROL MANUAL

      3) Disconnect connector D10732 from the Cabin Pressure Outflow Valve.

         NOTE: Reach through the open valve gates to get access to the connector.

      4) Do a wiring check between these pins of connector D10738A of the CPC 1 and connector D10732 of the Cabin Pressure Outflow Valve (WDM 21-31-22):

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CPC 1, M1653 D10738A		CABIN PRESSURE OUTFLOW VALVE, V48 D10732
pin A7	. . . . .	pin R
pin A8	. . . . .	pin P
pin A9	. . . . .	pin N
pin A10	. . . . .	pin M

5) If you find a problem with the wiring, then do these steps:
   a) Repair the wiring.
   b) Re-connect connector D10732 to the Cabin Pressure Outflow Valve.
   c) Re-install the CPC 1. Do this task: Cabin Pressure Controller (CPC) Installation, AMM TASK 21-31-01-400-801.
   d) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	6	C01269	PRESSURIZATION CONTROL MANUAL

   e) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.
   f) If the maintenance message does not show, then you corrected the problem.
   g) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

(4) Do these steps for the fault detail CPC/OFV WIRING on Cabin Pressure Controller 2:
   (a) Do this check of the wiring between the CPC 2, M1654 in the EE Bay and the Cabin Pressure Outflow Valve, V48:
      1) Remove the CPC 2. This is the task: Cabin Pressure Controller (CPC) Removal, AMM TASK 21-31-01-000-801.
      2) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	6	C01269	PRESSURIZATION CONTROL MANUAL

      3) Disconnect connector D10734 from the Cabin Pressure Outflow Valve.

NOTE: Reach through the open valve gates to get access to the connector.

      4) Do a wiring check between these pins of connector D10740A of the CPC 2 and connector D10734 of the Cabin Pressure Outflow Valve (WDM 21-31-23):

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CPC 2, M1654 D10740A		CABIN PRESSURE OUTFLOW VALVE, V48 D10734
pin A7	. . . . .	pin R
pin A8	. . . . .	pin P
pin A9	. . . . .	pin N
pin A10	. . . . .	pin M

5) If you find a problem with the wiring, then do these steps:
   a) Repair the wiring.
   b) Re-connect connector D10734 to the Cabin Pressure Outflow Valve.
   c) Re-install the CPC 2. Do this task: Cabin Pressure Controller (CPC) Installation, AMM TASK 21-31-01-400-801.
   d) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	6	C01269	PRESSURIZATION CONTROL MANUAL

   e) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.
   f) If the maintenance message does not show, then you corrected the problem.
   g) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

(5) Do these steps for the fault detail CPC-OFV on Cabin Pressure Controller 1:
   (a) Do this check of the wiring between the CPC 1, M1653 in the EE Bay and the Cabin Pressure Outflow Valve, V48:
      1) Remove the CPC 1. This is the task: Cabin Pressure Controller (CPC) Removal, AMM TASK 21-31-01-000-801.
      2) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	6	C01269	PRESSURIZATION CONTROL MANUAL

      3) Disconnect connector D10732 from the Cabin Pressure Outflow Valve.

NOTE: Reach through the open valve gates to get access to the connector.

      4) Do a wiring check between these pins of connector D10738B of the CPC 1 and connector D10732 of the Cabin Pressure Outflow Valve (WDM 21-31-22):

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CPC 1, M1653 D10738B		CABIN PRESSURE OUTFLOW VALVE, V48 D10732
pin B5	. . . . .	pin C-

5) If you find a problem with the wiring, then do these steps:
   a) Repair the wiring.
   b) Re-connect connector D10732 to the Cabin Pressure Outflow Valve.
   c) Re-install the CPC 1. Do this task: Cabin Pressure Controller (CPC) Installation, AMM TASK 21-31-01-400-801.
   d) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	6	C01269	PRESSURIZATION CONTROL MANUAL

   e) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.
   f) If the maintenance message does not show, then you corrected the problem.
   g) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

(6) Do these steps for the fault detail CPC-OFV on Cabin Pressure Controller 2:
   (a) Do this check of the wiring between the CPC 2, M1654 in the EE Bay and the Cabin Pressure Outflow Valve, V48:
      1) Remove the CPC 2. This is the task: Cabin Pressure Controller (CPC) Removal, AMM TASK 21-31-01-000-801.
      2) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	6	C01269	PRESSURIZATION CONTROL MANUAL

      3) Disconnect connector D10734 from the Cabin Pressure Outflow Valve.

         NOTE: Reach through the open valve gates to get access to the connector.

      4) Do a wiring check between these pins of connector D10740B of the CPC 2 and connector D10734 of the Cabin Pressure Outflow Valve (WDM 21-31-23):

CPC 2, M1654 D10740B		CABIN PRESSURE OUTFLOW VALVE, V48 D10734
pin B5	. . . . .	pin C-

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5) If you find a problem with the wiring, then do these steps:
   a) Repair the wiring.
   b) Re-connect connector D10734 to the Cabin Pressure Outflow Valve.
   c) Re-install the CPC 2. Do this task: Cabin Pressure Controller (CPC) Installation, AMM TASK 21-31-01-400-801.
   d) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	6	C01269	PRESSURIZATION CONTROL MANUAL

   e) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.
   f) If the maintenance message does not show, then you corrected the problem.
   g) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

(7) Do these steps for the fault detail CONTRLR PROG PIN on Cabin Pressure Controller 1 or 2:
   (a) Do this check of the wiring between the CPC 1, M1653 and the CPC 2, M1654 in the EE Bay:
      1) Remove the CPC 1 and 2. This is the task: Cabin Pressure Controller (CPC) Removal, AMM TASK 21-31-01-000-801.
      2) Do a wiring check between these pins of connectors D10738B and D10738A of the CPC 1 and connectors D10740B and D10740A of the CPC 2 (WDM 21-31-24):

CPC 1, M1653		CPC 2, M1654
D10738B		**D10740A**
pin B6	. . . . .	pin A15
D10738A		**D10740B**
pin A15	. . . . .	pin B6

      3) If you find a problem with the wiring, then do these steps:
         a) Repair the wiring.
         b) Re-install the CPC 1 and 2. This is the task: Cabin Pressure Controller (CPC) Installation, AMM TASK 21-31-01-400-801.
         c) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.
         d) If the maintenance message does not show, then you corrected the problem.
         e) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

(8) Do these steps for the fault detail CPC-CPC WIRING on Cabin Pressure Controller 1 or 2:

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(a) Do this check of the wiring between the CPC 1, M1653 and the CPC 2, M1654 in the EE Bay:

1) Remove the CPC 1 and 2. This is the task: Cabin Pressure Controller (CPC) Removal, AMM TASK 21-31-01-000-801.

2) Do a wiring check between these pins of connectors D10738B and D10738A of the CPC 1 and connectors D10740B and D10740A of the CPC 2 (WDM 21-31-24):

CPC 1, M1653	CPC 2, M1654
D10738B	**D10740A**
pin B2	pin D3
D10738A	**D10740B**
pin D3	pin B2

3) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-install the CPC 1 and 2. This is the task: Cabin Pressure Controller (CPC) Installation, AMM TASK 21-31-01-400-801.

c) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

d) If the maintenance message does not show, then you corrected the problem.

e) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

(9) Do these steps for the fault detail CPC-OFV WIRING on Cabin Pressure Controller 1:

(a) Do this check of the wiring between the CPC 1, M1653 in the EE Bay and the Cabin Pressure Outflow Valve, V48:

1) Remove the CPC 1. This is the task: Cabin Pressure Controller (CPC) Removal, AMM TASK 21-31-01-000-801.

2) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	6	C01269	PRESSURIZATION CONTROL MANUAL

3) Disconnect connector D10732 from the Cabin Pressure Outflow Valve.

NOTE: Reach through the open valve gates to get access to the connector.

4) Do a wiring check between these pins of connector D10738B of the CPC 1 and connector D10732 of the Cabin Pressure Outflow Valve (WDM 21-31-22):

CPC 1, M1653	CABIN PRESSURE OUTFLOW VALVE, V48
D10738B	**D10732**
pin B5	pin C-

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5) If you find a problem with the wiring, then do these steps:
   a) Repair the wiring.
   b) Re-connect connector D10732 to the Cabin Pressure Outflow Valve.
   c) Re-install the CPC 1. Do this task: Cabin Pressure Controller (CPC) Installation, AMM TASK 21-31-01-400-801.
   d) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	6	C01269	PRESSURIZATION CONTROL MANUAL

   e) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.
   f) If the maintenance message does not show, then you corrected the problem.
   g) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

(10) Do these steps for the fault detail CPC-OFV WIRING on Cabin Pressure Controller 2:
   (a) Do this check of the wiring between the CPC 2, M1654 in the EE Bay and the Cabin Pressure Outflow Valve, V48:
      1) Remove the CPC 2. This is the task: Cabin Pressure Controller (CPC) Removal, AMM TASK 21-31-01-000-801.
      2) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	6	C01269	PRESSURIZATION CONTROL MANUAL

      3) Disconnect connector D10734 from the Cabin Pressure Outflow Valve.

         NOTE: Reach through the open valve gates to get access to the connector.

      4) Do a wiring check between these pins of connector D10740B of the CPC 2 and connector D10734 of the Cabin Pressure Outflow Valve (WDM 21-31-23):

CPC 2, M1654 D10740B		CABIN PRESSURE OUTFLOW VALVE, V48 D10734
pin B5	. . . . .	pin C-

      5) If you find a problem with the wiring, then do these steps:
         a) Repair the wiring.
         b) Re-connect connector D10734 to the Cabin Pressure Outflow Valve.
         c) Re-install the CPC 2. Do this task: Cabin Pressure Controller (CPC) Installation, AMM TASK 21-31-01-400-801.

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d) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	6	C01269	PRESSURIZATION CONTROL MANUAL

e) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

f) If the maintenance message does not show, then you corrected the problem.

g) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

(11) Do these steps for the fault detail PNL-OFV on Cabin Pressure Controller 1:

(a) Do this check of the wiring between the Cabin Pressure Control Module, P5-6 and the Cabin Pressure Outflow Valve, V48:

1) Lower the Forward Overhead Panel, P5.

2) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL

3) Disconnect connector D10724 from the Cabin Pressure Control Module.

4) Disconnect connector D10732 from the Electronic Actuator 1.

5) Do a wiring check between pin 27 of connector D10724 of the Cabin Pressure Control Module and pin L of connector D10732 of Electronic Actuator 1 (WDM 21-31-22).

6) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-connect connector D10724 to the Cabin Pressure Control Module.

c) Close the Forward Overhead Panel, P5.

d) Re-connect connector D10732 to the Electronic Actuator 1.

e) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL

f) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

g) If the maintenance message does not show, then you corrected the problem.

(12) Do these steps for the fault detail PNL-OFV on Cabin Pressure Controller 2:

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(a) Do this check of the wiring between the Cabin Pressure Control Module, P5-6 and the Cabin Pressure Outflow Valve, V48:

1) Lower the Forward Overhead Panel, P5.

2) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL

3) Disconnect connector D10726 from the Cabin Pressure Control Module.

4) Disconnect connector D10734 from the Electronic Actuator 2.

5) Do a wiring check between pin 31 of connector D10726 of the Cabin Pressure Control Module and pin L of connector D10734 of Electronic Actuator 2 (WDM 21-31-23).

6) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-connect connector D10726 to the Cabin Pressure Control Module.

c) Close the Forward Overhead Panel, P5.

d) Re-connect connector D10734 to the Electronic Actuator 2.

e) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL

f) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

g) If the maintenance message does not show, then you corrected the problem.

(13) Do these steps for the fault detail MNL MODE PNL CTR:

(a) Do this check of the wiring between the Cabin Pressure Control Module, P5-6 in the Flight Deck and the CPC 1, M1653 in the EE Bay:

1) Remove the CPC 1. This is the task: Cabin Pressure Controller (CPC) Removal, AMM TASK 21-31-01-000-801.

2) Lower the Forward Overhead Panel, P5.

3) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	6	C01269	PRESSURIZATION CONTROL MANUAL

4) Disconnect connector D10724 from the Cabin Pressure Control Module.

5) Do a wiring check between pin 1 of connector D10724 of the Cabin Pressure Control Module and pin B3 of connector D10738B (WDM 21-31-11).

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6) If you find a problem with the wiring, then do these steps:
   a) Repair the wiring.
   b) Re-connect connector D10724 to the Cabin Pressure Control Module.
   c) Close the Forward Overhead Panel, P5.
   d) Re-install the CPC 1. This is the task: Cabin Pressure Controller (CPC) Installation, AMM TASK 21-31-01-400-801.
   e) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	6	C01269	PRESSURIZATION CONTROL MANUAL

   f) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.
   g) If the maintenance message does not show, then you corrected the problem.

(b) Do this check of the wiring between the Cabin Pressure Control Module, P5-6 in the Flight Deck and the CPC 2, M1654 in the EE Bay:
   1) Remove the CPC 2. This is the task: Cabin Pressure Controller (CPC) Removal, AMM TASK 21-31-01-000-801.
   2) Lower the Forward Overhead Panel, P5.
   3) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	6	C01269	PRESSURIZATION CONTROL MANUAL

   4) Disconnect connector D10726 from the Cabin Pressure Control Module.
   5) Do a wiring check between pin 8 of connector D10726 of the Cabin Pressure Control Module and pin B3 of connector D10740B (WDM 21-31-11).
   6) If you find a problem with the wiring, then do these steps:
      a) Repair the wiring.
      b) Re-connect connector D10726 to the Cabin Pressure Control Module.
      c) Close the Forward Overhead Panel, P5.
      d) Re-install the CPC 2. This is the task: Cabin Pressure Controller (CPC) Installation, AMM TASK 21-31-01-400-801.
      e) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	6	C01269	PRESSURIZATION CONTROL MANUAL

      f) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.
      g) If the maintenance message does not show, then you corrected the problem.

END OF TASK

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805. Cabin Pressure Outflow Valve or Wiring - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) OFV OR WIRING

B. Possible Causes

(1) Cabin pressure outflow valve, V48

(2) Wiring

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	1	C01273	PRESSURIZATION CONTROL LCD LTG
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL
F	7	C01272	PRESSURIZATION CONTROL IND

D. Related Data

(1) SSM 21-31-22

(2) SSM 21-31-23

(3) WDM 21-31-22

(4) WDM 21-31-23

E. Initial Evaluation

(1) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

(2) If the maintenance message does not show, then there was an intermittent fault.

(3) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Replace the cabin pressure outflow valve.

These are the tasks:

Aft Outflow Valve Assembly Removal, AMM TASK 21-31-03-000-801,

Aft Outflow Valve Assembly Installation, AMM TASK 21-31-03-400-801.

(a) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

1) If the maintenance message does not show, then you corrected the fault.

2) If the maintenance message shows, then continue.

(2) Do these steps if the maintenance message shows on Cabin Pressure Controller 1:

(a) Do this check of the wiring between cabin pressure controller 1, M1653 in the electronic equipment bay and the cabin pressure outflow valve, V48:

1) Remove cabin pressure controller 1. To remove the controller, do this task: Cabin Pressure Controller (CPC) Removal, AMM TASK 21-31-01-000-801.

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2) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL

3) Disconnect connector D10732 from the cabin pressure outflow valve.

NOTE: Reach through the open valve gates to get access to the connector.

4) Do a wiring check between these pins of connector D10738A of cabin pressure controller 1 and connector D10732 of the cabin pressure outflow valve:

D10738A		D10732
pin A7	. . . . .	pin R
pin A8	. . . . .	pin P
pin A9	. . . . .	pin N
pin A10	. . . . .	pin M

5) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	3	C01270	PRESSURIZATION CONTROL AUTO 1

6) Do a check for 28 VDC between pin K and pin J of connector D10732 at the Cabin Pressure Outflow Valve, V48.

7) If you find a problem with the wiring, then do these steps:

a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	3	C01270	PRESSURIZATION CONTROL AUTO 1

b) Repair the wiring.

c) Re-connect connector D10732 to the cabin pressure outflow valve.

d) Close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL

e) Re-install the cabin pressure controller. To install the controller, do this task: Cabin Pressure Controller (CPC) Installation, AMM TASK 21-31-01-400-801.

f) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

g) If the maintenance message does not show, then you corrected the fault.

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h) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

(3) Do these steps if the maintenance message shows on Cabin Pressure Controller 2:

(a) Do this check of the wiring between cabin pressure controller 2, M1654 in the electronic equipment bay and the cabin pressure outflow valve, V48:

1) Remove cabin pressure controller 2. To remove the controller, do this task: Cabin Pressure Controller (CPC) Removal, AMM TASK 21-31-01-000-801.

2) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL

3) Disconnect connector D10734 from the cabin pressure outflow valve.

NOTE: Reach through the open valve gates to get access to the connector.

4) Do a wiring check between these pins of connector D10740A of cabin pressure controller 2 and connector D10734 of the cabin pressure outflow valve:

D10740A		D10734
pin A7	. . . . .	pin R
pin A8	. . . . .	pin P
pin A9	. . . . .	pin N
pin A10	. . . . .	pin M

5) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	5	C01271	PRESSURIZATION CONTROL AUTO 2

6) Do a check for 28 VDC between pin K and pin J of connector D10734 at the Cabin Pressure Outflow Valve, V48.

7) If you find a problem with the wiring, then do these steps:

a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	5	C01271	PRESSURIZATION CONTROL AUTO 2

b) Repair the wiring.

c) Re-connect connector D10734 to the cabin pressure outflow valve.

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d) Close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL

e) Re-install the cabin pressure controller. To install the controller, do this task: Cabin Pressure Controller (CPC) Installation, AMM TASK 21-31-01-400-801.

f) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

g) If the maintenance message does not show, then you corrected the fault.

h) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

806. Air Data Inertial Reference Unit 1 Input Problem - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) DADC #1 INVALID

B. Possible Causes

(1) Air data inertial reference unit (ADIRU) 1, M1749

(2) Wiring

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	1	C01273	PRESSURIZATION CONTROL LCD LTG
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL
F	7	C01272	PRESSURIZATION CONTROL IND

D. Related Data

(1) SSM 21-31-24

(2) WDM 21-31-24

E. Initial Evaluation

(1) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

(2) If the maintenance message does not show, then there was an intermittent fault.

(3) If the maintenance message shows, then do the Fault Isolation Procedure below.

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F. Fault Isolation Procedure

(1) Do the BITE test on air data inertial reference unit (ADIRU) 1, M1749. To do the BITE test, do this task: ADIRS BITE Procedure, 34-21 TASK 801.

(a) If the BITE test shows any maintenance messages, then do the corrective action for the message.

1) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

2) If the maintenance message does not show, then you corrected the fault.

(b) If the BITE test does not show any maintenance messages, then continue.

(2) Do this check of the wiring between the cabin pressure controllers and ADIRU 1:

(a) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

(b) Remove cabin pressure controllers 1 and 2. To remove the controllers, do this task: Cabin Pressure Controller (CPC) Removal, AMM TASK 21-31-01-000-801.

(c) Remove ADIRU 1. To remove ADIRU 1, do this task: Air Data Inertial Reference Unit Removal, AMM TASK 34-21-01-000-801.

(d) Do a wiring check between these pins of connectors D10738B and D10740B of cabin pressure controllers 1 and 2 and connector D3687A of ADIRU 1:

D10738B		D3687A
pin A3	. . . . .	pin A9
pin A4	. . . . .	pin B9

D10740B		D3687A
pin A3	. . . . .	pin A9
pin A4	. . . . .	pin B9

(e) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-install ADIRU 1. To install it, do this task: Air Data Inertial Reference Unit - Installation, AMM TASK 34-21-01-400-801.

3) Re-install the cabin pressure controllers. To install the controllers, do this task: Cabin Pressure Controller (CPC) Installation, AMM TASK 21-31-01-400-801.

4) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

5) If the maintenance message does not show, then you corrected the fault.

6) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

END OF TASK

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807. Air Data Inertial Reference Unit 2 Input Problem - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) DADC #2 INVALID

B. Possible Causes

(1) Air data inertial reference unit (ADIRU) 2, M1752

(2) Wiring

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	1	C01273	PRESSURIZATION CONTROL LCD LTG
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL
F	7	C01272	PRESSURIZATION CONTROL IND

D. Related Data

(1) SSM 21-31-24

(2) WDM 21-31-24

E. Initial Evaluation

(1) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

(2) If the maintenance message does not show, then there was an intermittent fault.

(3) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Do the BITE test on air data inertial reference unit (ADIRU) 2, M1752. To do the BITE test, do this task: ADIRS BITE Procedure, 34-21 TASK 801.

(a) If the BITE test shows any maintenance messages, then do the corrective action for the message.

1) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

2) If the maintenance message does not show, then you corrected the fault.

(b) If the BITE test does not show any maintenance messages, then continue.

(2) Do this check of the wiring between the cabin pressure controllers and ADIRU 2:

(a) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

(b) Remove cabin pressure controllers 1 and 2. To remove the controllers, do this task: Cabin Pressure Controller (CPC) Removal, AMM TASK 21-31-01-000-801.

(c) Remove ADIRU 2. To remove ADIRU 2, do this task: Air Data Inertial Reference Unit Removal, AMM TASK 34-21-01-000-801.

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(d) Do a wiring check between these pins of connectors D10738B and D10740B of cabin pressure controllers 1 and 2 and connector D3693A of ADIRU 2:

D10738B	D3693A
pin A9	pin A9
pin A10	pin B9

D10740B	D3693A
pin A9	pin A9
pin A10	pin B9

(e) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.
2) Re-install ADIRU 2. To install it, do this task: Air Data Inertial Reference Unit - Installation, AMM TASK 34-21-01-400-801.
3) Re-install the cabin pressure controllers. To install the controllers, do this task: Cabin Pressure Controller (CPC) Installation, AMM TASK 21-31-01-400-801.
4) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.
5) If the maintenance message does not show, then you corrected the fault.
6) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

808. Stall Management Yaw Damper Computer 1 Input Problem - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) SMC #1 INVALID

B. Possible Causes

(1) Stall management yaw damper computer (SMYD) 1, M1747

(2) Wiring

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	1	C01273	PRESSURIZATION CONTROL LCD LTG
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL
F	7	C01272	PRESSURIZATION CONTROL IND

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D. Related Data

(1) SSM 21-31-24

(2) WDM 21-31-24

E. Initial Evaluation

(1) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

(2) If the maintenance message does not show, then there was an intermittent fault.

(3) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Do the BITE test for stall management yaw damper computer (SMYD) 1, M1747. To do the BITE test, do this task: Stall Management Yaw Damper BITE Procedure, 27-32 TASK 801.

(a) If the BITE test shows any maintenance messages, then do the corrective action for the message.

1) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

2) If the maintenance message does not show, then you corrected the fault.

(b) If the BITE test does not show any maintenance messages, then continue.

(2) Do this check of the wiring between the cabin pressure controllers and SMYD 1:

(a) Remove cabin pressure controllers 1 and 2. To remove the controllers, do this task: Cabin Pressure Controller (CPC) Removal, AMM TASK 21-31-01-000-801.

(b) Remove SMYD 1. To remove SMYD 1, do this task: Stall Management Yaw Damper (SMYD) Removal, AMM TASK 27-32-42-000-801.

(c) Do a wiring check between these pins of connectors D10738B and D10740B of cabin pressure controllers 1 and 2 and connector D3683A of SMYD 1:

D10738B	**D3683A**
pin A5	pin 95
pin A6	pin 94

D10740B	**D3683A**
pin A5	pin 95
pin A6	pin 94

(d) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-install SMYD 1. To install it, do this task: Stall Management Yaw Damper (SMYD) Installation, AMM TASK 27-32-42-400-801.

3) Re-install the cabin pressure controllers. To install the controllers, do this task: Cabin Pressure Controller (CPC) Installation, AMM TASK 21-31-01-400-801.

4) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

5) If the maintenance message does not show, then you corrected the fault.

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6) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

809. Stall Management Yaw Damper Computer 2 Input Problem - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) SMC #2 INVALID

B. Possible Causes

(1) Stall management yaw damper computer (SMYD) 2, M1748

(2) Wiring

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	1	C01273	PRESSURIZATION CONTROL LCD LTG
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL
F	7	C01272	PRESSURIZATION CONTROL IND

D. Related Data

(1) SSM 21-31-24

(2) WDM 21-31-24

E. Initial Evaluation

(1) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

(2) If the maintenance message does not show, then there was an intermittent fault.

(3) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Do the BITE test for stall management yaw damper computer (SMYD) 2, M1748. To do the BITE test, do this task: Stall Management Yaw Damper BITE Procedure, 27-32 TASK 801.

(a) If the BITE test shows any maintenance messages, then do the corrective action for the message.

1) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

2) If the maintenance message does not show, then you corrected the fault.

(b) If the BITE test does not show any maintenance messages, then continue.

(2) Do this check of the wiring between the cabin pressure controllers and SMYD 2:

(a) Remove cabin pressure controllers 1 and 2. To remove the controllers, do this task: Cabin Pressure Controller (CPC) Removal, AMM TASK 21-31-01-000-801.

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(b) Remove SMYD 2. To remove SMYD 2, do this task: Stall Management Yaw Damper (SMYD) Removal, AMM TASK 27-32-42-000-801.

(c) Do a wiring check between these pins of connectors D10738B and D10740B of cabin pressure controllers 1 and 2 and connector D3685A of SMYD 2:

D10738B	D3685A
pin A13	pin 95
pin A14	pin 94

D10740B	D3685A
pin A13	pin 95
pin A14	pin 94

(d) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.
2) Re-install SMYD 2. To install it, do this task: Stall Management Yaw Damper (SMYD) Installation, AMM TASK 27-32-42-400-801.
3) Re-install the cabin pressure controllers. To install the controllers, do this task: Cabin Pressure Controller (CPC) Installation, AMM TASK 21-31-01-400-801.
4) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.
5) If the maintenance message does not show, then you corrected the fault.
6) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

810. Engine Speed Input Problem - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) N1/N2 LOWRU FAIL

B. Possible Causes

(1) Stall management yaw damper computer (SMYD), M1747 (SMYD-1) or M1748 (SMYD-2)

(2) Wiring

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	1	C01273	PRESSURIZATION CONTROL LCD LTG
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL
F	7	C01272	PRESSURIZATION CONTROL IND

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D. Related Data

(1) SSM 21-31-24

(2) WDM 21-31-24

E. Initial Evaluation

(1) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

(2) If the maintenance message does not show, then there was an intermittent fault.

(3) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Do the BITE test on stall management yaw damper computers (SMYD) 1 and 2, M1747 and M1748. To do the BITE test, do this task: Stall Management Yaw Damper BITE Procedure, 27-32 TASK 801.

(a) If the BITE test shows any maintenance messages, then do the corrective action for the message.

1) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

2) If the maintenance message does not show, then you corrected the fault.

(b) If the BITE test does not show any maintenance messages, then continue.

(2) Do this check of the wiring between the cabin pressure controllers and SMYD 1 and 2:

(a) Remove cabin pressure controllers 1 and 2. To remove the controllers, do this task: Cabin Pressure Controller (CPC) Removal, AMM TASK 21-31-01-000-801.

(b) Remove SMYD 1 and 2. To remove SMYD 1 and 2, do this task: Stall Management Yaw Damper (SMYD) Removal, AMM TASK 27-32-42-000-801.

(c) Do a wiring check between these pins of connectors D10738B and D10740B of cabin pressure controllers 1 and 2 and connector D3683A of SMYD 1:

D10738B	**D3683A**
pin A5	pin 95
pin A6	pin 94

D10740B	**D3683A**
pin A5	pin 95
pin A6	pin 94

(d) Do a wiring check between these pins of connectors D10738B and D10740B of cabin pressure controllers 1 and 2 and connector D3685A of SMYD 2:

D10738B	**D3685A**
pin A13	pin 95
pin A14	pin 94

D10740B	**D3685A**
pin A13	pin 95
pin A14	pin 94

(e) If you find a problem with the wiring, then do these steps:

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1) Repair the wiring.
2) Re-install SMYD 1 and 2. To install them, do this task: Stall Management Yaw Damper (SMYD) Installation, AMM TASK 27-32-42-400-801.
3) Re-install the cabin pressure controllers. To install the controllers, do this task: Cabin Pressure Controller (CPC) Installation, AMM TASK 21-31-01-400-801.
4) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.
5) If the maintenance message does not show, then you corrected the fault.
6) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

811. Manual Mode Control of Aft Outflow Valve Does Not Operate - Fault Isolation

A. Description

(1) The aft outflow valve cannot be moved towards open or closed using the manual toggle switch and with the cabin pressure mode select switch in MAN.

B. Possible Causes

(1) Cabin pressure outflow valve, V48
(2) Cabin pressure control module, P5-6
(3) Wiring problem

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	1	C01273	PRESSURIZATION CONTROL LCD LTG
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL
F	7	C01272	PRESSURIZATION CONTROL IND

D. Related Data

(1) (SSM 21-31-11)
(2) (WDM 21-31-11)

E. Initial Evaluation

(1) Put the cabin pressure mode select switch on the pressure control module to MAN.
(2) Put a person near the outflow valve located at the lower surface of the airplane near the aft entry door.

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WARNING MAKE SURE PERSONNEL ARE CLEAR OF THE AFT OUTFLOW VALVE WHEN YOU MOVE IT. IF PERSONNEL ARE NOT CLEAR OF THE OUTFLOW VALVE WHEN IT MOVES INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Push and hold the manual control toggle switch to CLOSE.

(4) Speak to the person near the outflow valve to make sure that the outflow valve moved to the closed position.

(5) If the outflow valve moved to the closed position, push and hold the manual control toggle switch to OPEN.

(6) Speak to the person near the outflow valve to make sure that the outflow valve moved to the open position.

(7) If the outflow valve moved towards the closed position and the open position, then there was an intermittent fault.

(a) Make sure the mode selector switch on the cabin pressurization control panel, P5-6, is returned to the AUTO position.

(8) If the outflow valve does not move towards the closed position or the open position, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL
F	7	C01272	PRESSURIZATION CONTROL IND

(2) Do this check for 48 VDC power to the DC manual motor as follows:

(a) Disconnect electrical connector D10730P from the manual DC motor.

NOTE: The P in D10730P refers to the connector plug (male connector) on the ship's wiring.

(b) Close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	6	C01269	PRESSURIZATION CONTROL MANUAL

(c) Do a check for 48 V DC at connector D10730P as follows:

1) Connect the red lead of a multimeter to pin A of D10730P and the black lead of the multimeter to pin B of D10730P.

2) Push and hold the manual control toggle switch to OPEN.

3) Make sure the multimeter indicates +48 VDC.

4) Push and hold the manual control toggle switch to CLOSE.

5) Make sure the multimeter indicates -48 VDC.

6) If the voltages are satisfactory, then continue.

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(3) If the voltage checks were satisfactory, then do these checks;

(a) Examine connector D10730P/D10730J for bent or pushed back pins/sockets.

(b) Repair any problems that you find.

(c) If you did any wiring repair, then do the Repair Confirmation at the end of this task.

1) If the repair confirmation was not satisfactory, then continue.

(4) If the checks of the voltages and connector D10730 were satisfactory, then remove the defective valve and install a serviceable outflow valve as follows:

(a) Do this task to remove the outflow valve: Aft Outflow Valve Assembly Removal, AMM TASK 21-31-03-000-801.

(b) Do this task to install a serviceable outflow valve: Aft Outflow Valve Assembly Installation, AMM TASK 21-31-03-400-801.

(5) If 48V DC is not available at pin A or B of connector D10730P, do these checks of the wiring between the cabin pressure control module and the aft outflow valve:

(a) Lower the P5-6 cabin pressure control module to get access to the connectors on the back of the module.

(b) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	6	C01269	PRESSURIZATION CONTROL MANUAL

(c) Disconnect the D10724P electrical connector from the P5-6 cabin pressure control module.

(d) Do these continuity checks of the wiring between connectors D10724P and D10730P:

P5-6 Panel	**Outflow Valve**
D10724	**V48**
	D10730
Pin 25	Pin A
Pin 24	Pin B

(e) Do a continuity check from pin 28 of connector D10724P to a ground.

(f) If the continuity checks are not satisfactory, then do these steps:

1) Repair any problems that you find.

2) Do the Repair Confirmation at the end of this task.

3) If the Repair Confirmation is not satisfactory, then continue.

(g) If the continuity checks are satisfactory, then continue.

(6) Do this check for 28 V DC power to the cabin pressure selector panel:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	6	C01269	PRESSURIZATION CONTROL MANUAL

(b) Lower the P5-6 cabin pressure control module to get access to the connectors on the back of the module.

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(c) Disconnect the D10724P electrical connector from the P5-6 cabin pressure control module.

(d) Close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	6	C01269	PRESSURIZATION CONTROL MANUAL

(e) Do a check for 28V DC at pin 15 of connector D10724P on the ship's wiring.

(f) If 28V DC is not available at pin 15 of connector D10724P, repair wiring or fault isolate the power distribution system to the 28V DC battery bus section 2.

(g) If there was 28V DC at pin 15 of connector D10724P, replace the cabin pressure control module as follows:

1) Do this task to remove the cabin pressure control module: Cabin Pressure Control Module - Removal, AMM TASK 21-31-02-000-801.

2) Do this task to install a serviceable cabin pressure control module: Cabin Pressure Control Module - Installation, AMM TASK 21-31-02-400-801.

G. Repair Confirmation

(1) If applicable, reconnect all disconnected electrical connectors and close all circuit breakers that were opened.

(2) Do a test of the manual mode. To test the manual mode, do this task

: Pressurization System Manual Mode Test, AMM TASK 21-31-00-710-801 or Pressurization System Manual Mode Test, AMM TASK 21-31-00-710-803.

(3) Do a cabin pressure control system ground test. To do the ground test, do this task: Pressurization System Ground Test, AMM TASK 21-31-00-700-802.

——— END OF TASK ———

815. Aircraft Rate Too High - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) ACFT RATE HI

(2) This maintenance message does not indicate a fault with the digital cabin pressurization control system.

B. Possible Causes

(1) The aircraft rate of climb or descent was too high.

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	1	C01273	PRESSURIZATION CONTROL LCD LTG
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL
F	7	C01272	PRESSURIZATION CONTROL IND

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D. Related Data

(1) (SSM 21-31-22)

(2) (WDM 21-31-22)

(3) Airplane Pack Operation - Confidence Check, AMM TASK 21-00-05-780-801

(4) Engine Bleed Air System Health Check, AMM TASK 36-11-00-700-801

E. Initial Evaluation

(1) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

(a) If there are no maintenance messages then there was an intermittent fault.

(b) If there are any maintenance messages, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) The problem was due to the aircraft making a climb or descent at too high a rate and is not a fault of the cabin pressure control system.

(a) Do these steps to clear the fault:

1) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

NOTE: Do the System Test and Clear in the BITE test.

(2) If the Digital Cabin Pressure Controller shows a Fault Code 30 in the history when performing the BITE check or if pilots report cabin altitude is above pressurization schedule during cruise, cabin altitude climbs during cruise, or cabin altitude climbs during initial descent, do the following.

(a) Verify the performance of the engine bleed system (Engine Bleed Air System Health Check, AMM TASK 36-11-00-700-801).

(b) Check the air conditioning pack air flow and the fuselage leakage rate (Airplane Pack Operation - Confidence Check, AMM TASK 21-00-05-780-801).

(c) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

——— END OF TASK ———

816. Hi Cabin Differential Pressure - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) CAB DIFF PRESS HI

B. Possible Causes

(1) Cabin pressure outflow valve, V48

(2) Cabin pressure controller, M1653 (No. 1) or M1654 (No. 2)

(3) Cabin altitude and differential pressure indicator (located on P5-16 panel)

(4) Flight altitude exceeded FLT ALT window setting

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C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	1	C01273	PRESSURIZATION CONTROL LCD LTG
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL
F	7	C01272	PRESSURIZATION CONTROL IND

D. Related Data

(1) (SSM 21-31-22)

(2) (SSM 21-31-23)

(3) (WDM 21-31-22)

(4) (WDM 21-31-23)

(5) Airplane Pack Operation - Confidence Check, AMM TASK 21-00-05-780-801

(6) Engine Bleed Air System Health Check, AMM TASK 36-11-00-700-801

E. Initial Evaluation

OMR ALL; AIRPLANES WITH CABIN PRESSURE CONTROLLERS P/N 7121-19971-01AA/01AB/01AC/01BA

(1) Visually inspect the copper tape on the outside edges of the cabin pressure controller as follows:

(a) Make sure that the copper tape extends onto the front outer edges of the pressure controller 1.5 to 2.2 mm.

(b) Make sure that there are no cracks, holes or dents in the tape.

(c) Make sure that the tape is smoothly attached to the face of the controller.

(d) Make sure that there are no openings into the controller where the controller front cover connects with the controller sides.

(e) If any of the above conditions exist, replace all of the cooper tape or controller as follows:

These are the tasks:

Cabin Pressure Controller (CPC) Removal, AMM TASK 21-31-01-000-801,

Cabin Pressure Controller (CPC) Installation, AMM TASK 21-31-01-400-801.

OMR ALL

(2) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

(3) If there are no maintenance messages, then there was an intermittent fault.

(4) If the maintenance message shows, and there is an additional maintenance message, then do the Fault Isolation Procedure for the other maintenance message.

(5) If the maintenance message shows, and it is the only maintenance message, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Replace the cabin pressure outflow valve.

These are the tasks:

Aft Outflow Valve Assembly Removal, AMM TASK 21-31-03-000-801,

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Aft Outflow Valve Assembly Installation, AMM TASK 21-31-03-400-801.

(a) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

1) If the maintenance message does not show, then you corrected the fault.

2) If the maintenance message shows, then continue.

(2) Replace the cabin pressure controller that has the maintenance message.

These are the tasks:

Cabin Pressure Controller (CPC) Removal, AMM TASK 21-31-01-000-801,

Cabin Pressure Controller (CPC) Installation, AMM TASK 21-31-01-400-801.

(a) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

1) If the maintenance message does not show, then you corrected the fault.

2) If the maintenance message shows, then continue.

(3) Replace the cabin altitude and differential pressure indicator as follows:

(a) Do these tasks: Cabin Altitude and Differential Pressure Indicator Removal, AMM TASK 21-33-02-000-801Cabin Altitude and Differential Pressure Indicator Installation, AMM TASK 21-33-02-400-801.

1) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

a) If the maintenance message does not show, then you corrected the fault.

b) If the maintenance message shows, do this task: Alternate Static System Low-range Leak Test, AMM TASK 34-11-00-790-808.

<1> Correct any problems that you find.

(4) Do this check to make sure the airplane did not exceed the altitude set into the FLT ALT window on the pressurization control panel:

(a) Find the flight leg when this maintenance message occurred.

(b) Do this task: Reference Not Currently Available.

(c) If the controller download procedure shows that the airplane altitude for the flight leg was more than the FLT ALT setting, no further action is required.

——— END OF TASK ———

817. Hi Cabin Altitude - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) HI CAB ALT

B. Possible Causes

(1) Cabin pressure outflow valve, V48

(2) Cabin pressure controller, M1653 (No. 1) or M1654 (No. 2)

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C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	1	C01273	PRESSURIZATION CONTROL LCD LTG
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL
F	7	C01272	PRESSURIZATION CONTROL IND

D. Related Data

(1) (SSM 21-31-22)

(2) (SSM 21-31-23)

(3) (WDM 21-31-22)

(4) (WDM 21-31-23)

(5) Airplane Pack Operation - Confidence Check, AMM TASK 21-00-05-780-801

(6) Engine Bleed Air System Health Check, AMM TASK 36-11-00-700-801

E. Initial Evaluation

OMR ALL; AIRPLANES WITH CABIN PRESSURE CONTROLLERS P/N 7121-19971-01AA/01AB/01AC/01BA

(1) Visually inspect the copper tape on the outside edges of the cabin pressure controller as follows:

(a) Make sure that the copper tape extends onto the front outer edges of the pressure controller 1.5 to 2.2 mm.

(b) Make sure that there are no cracks, holes or dents in the tape.

(c) Make sure that the tape is smoothly attached to the face of the controller.

(d) Make sure that there are no openings into the controller where the controller front cover connects with the controller sides.

(e) If any of the above conditions exist, replace all of the cooper tape or controller as follows:

These are the tasks:

Cabin Pressure Controller (CPC) Removal, AMM TASK 21-31-01-000-801,

Cabin Pressure Controller (CPC) Installation, AMM TASK 21-31-01-400-801.

OMR ALL

(2) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

(3) If there are no maintenance messages, then there was an intermittent fault.

(4) If the Digital Cabin Pressure Controller shows a Fault Code 30 in the history when performing the BITE check or if pilots report cabin altitude is above pressurization schedule during cruise, cabin altitude climbs during cruise, or cabin altitude climbs during initial descent, do the following.

(a) Verify the performance of the engine bleed system (Engine Bleed Air System Health Check, AMM TASK 36-11-00-700-801).

(b) Check the air conditioning pack air flow and the fuselage leakage rate (Airplane Pack Operation - Confidence Check, AMM TASK 21-00-05-780-801).

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(c) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

(5) If the maintenance message shows, and there is an additional maintenance message, then do the Fault Isolation Procedure for the other maintenance message.

(6) If the maintenance message shows, and it is the only maintenance message, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Replace the cabin pressure outflow valve.

These are the tasks:

Aft Outflow Valve Assembly Removal, AMM TASK 21-31-03-000-801,

Aft Outflow Valve Assembly Installation, AMM TASK 21-31-03-400-801.

(a) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

1) If the maintenance message does not show, then you corrected the fault.

2) If the maintenance message shows, then continue.

(2) Replace the cabin pressure controller that has the maintenance message.

These are the tasks:

Cabin Pressure Controller (CPC) Removal, AMM TASK 21-31-01-000-801,

Cabin Pressure Controller (CPC) Installation, AMM TASK 21-31-01-400-801.

(a) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

1) If the maintenance message does not show, then you corrected the fault.

——— END OF TASK ———

818. Hi Cabin Altitude Rate Change - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) CAB RATE HI

B. Possible Causes

(1) Cabin pressure outflow valve, V48

(2) Cabin pressure controller, M1653 (No. 1) or M1654 (No. 2)

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	1	C01273	PRESSURIZATION CONTROL LCD LTG
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL
F	7	C01272	PRESSURIZATION CONTROL IND

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D. Related Data

(1) (SSM 21-31-22)

(2) (SSM 21-31-23)

(3) (WDM 21-31-22)

(4) (WDM 21-31-23)

(5) Airplane Pack Operation - Confidence Check, AMM TASK 21-00-05-780-801

(6) Engine Bleed Air System Health Check, AMM TASK 36-11-00-700-801

E. Initial Evaluation

OMR ALL; AIRPLANES WITH CABIN PRESSURE CONTROLLERS P/N 7121-19971-01AA/01AB/01AC/01BA

(1) Visually inspect the copper tape on the outside edges of the cabin pressure controller as follows:

(a) Make sure that the copper tape extends onto the front outer edges of the pressure controller 1.5 to 2.2 mm.

(b) Make sure that there are no cracks, holes or dents in the tape.

(c) Make sure that the tape is smoothly attached to the face of the controller.

(d) Make sure that there are no openings into the controller where the controller front cover connects with the controller sides.

(e) If any of the above conditions exist, replace all of the cooper tape or controller as follows:

These are the tasks:

Cabin Pressure Controller (CPC) Removal, AMM TASK 21-31-01-000-801,

Cabin Pressure Controller (CPC) Installation, AMM TASK 21-31-01-400-801.

OMR ALL

(2) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

(3) If there are no maintenance messages, then there was an intermittent fault.

(4) If the Digital Cabin Pressure Controller shows a Fault Code 30 in the history when performing the BITE check or if pilots report cabin altitude is above pressurization schedule during cruise, cabin altitude climbs during cruise, or cabin altitude climbs during initial descent, do the following.

(a) Verify the performance of the engine bleed system (Engine Bleed Air System Health Check, AMM TASK 36-11-00-700-801).

(b) Check the air conditioning pack air flow and the fuselage leakage rate (Airplane Pack Operation - Confidence Check, AMM TASK 21-00-05-780-801).

(c) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

(5) If the maintenance message shows, and there is an additional maintenance message, then do the Fault Isolation Procedure for the other maintenance message.

(6) If the maintenance message shows, and it is the only maintenance message, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Replace the cabin pressure outflow valve.

These are the tasks:

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Aft Outflow Valve Assembly Removal, AMM TASK 21-31-03-000-801,

Aft Outflow Valve Assembly Installation, AMM TASK 21-31-03-400-801.

(a) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

1) If the maintenance message does not show, then you corrected the fault.

2) If the maintenance message shows, then continue.

(2) Replace the cabin pressure controller that has the maintenance message.

These are the tasks:

Cabin Pressure Controller (CPC) Removal, AMM TASK 21-31-01-000-801,

Cabin Pressure Controller (CPC) Installation, AMM TASK 21-31-01-400-801.

(a) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

1) If the maintenance message does not show, then you corrected the fault.

——— END OF TASK ———

819. Drive Enable Signal Problem - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) DRV ENBL ERROR

B. Possible Causes

(1) Cabin pressure controller, M1653 (No. 1) or M1654 (No. 2)

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	1	C01273	PRESSURIZATION CONTROL LCD LTG
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL
F	7	C01272	PRESSURIZATION CONTROL IND

D. Related Data

(1) (SSM 21-31-22)

(2) (WDM 21-31-22)

E. Initial Evaluation

OMR ALL; AIRPLANES WITH CABIN PRESSURE CONTROLLERS P/N 7121-19971-01AA/01AB/01AC/01BA

(1) Visually inspect the copper tape on the outside edges of the cabin pressure controller as follows:

(a) Make sure that the copper tape extends onto the front outer edges of the pressure controller 1.5 to 2.2 mm.

(b) Make sure that there are no cracks, holes or dents in the tape.

(c) Make sure that the tape is smoothly attached to the face of the controller.

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OMR ALL; AIRPLANES WITH CABIN PRESSURE CONTROLLERS P/N 7121-19971-01AA/01AB/01AC/01BA (Continued)

(d) Make sure that there are no openings into the controller where the controller front cover connects with the controller sides.

(e) If any of the above conditions exist, replace all of the cooper tape or controller as follows:

These are the tasks:

Cabin Pressure Controller (CPC) Removal, AMM TASK 21-31-01-000-801,

Cabin Pressure Controller (CPC) Installation, AMM TASK 21-31-01-400-801.

OMR ALL

(2) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

NOTE: Do the BITE test on both cabin pressure controllers.

(3) If the maintenance message does not show, then there was an intermittent fault.

(4) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Replace the cabin pressure controller in the aft EE bay the shows a controller fault.

These are the tasks:

Cabin Pressure Controller (CPC) Removal, AMM TASK 21-31-01-000-801,

Cabin Pressure Controller (CPC) Installation, AMM TASK 21-31-01-400-801.

(a) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

NOTE: Do the BITE test on both controllers.

1) If the maintenance message does not show, then you corrected the fault.

——— END OF TASK ———

820. Cabin Pressure Outflow Valve Altitude Switch Activity - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) CAB PRESS SW ACTIV

B. Possible Causes

(1) Cabin pressure outflow valve, V48

(2) Cabin pressure controller, M1653 (No. 1) or M1654 (No. 2)

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	1	C01273	PRESSURIZATION CONTROL LCD LTG
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL
F	7	C01272	PRESSURIZATION CONTROL IND

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D. Related Data

(1) (SSM 21-31-22)

(2) (SSM 21-31-23)

(3) (WDM 21-31-22)

(4) (WDM 21-31-23)

(5) Airplane Pack Operation - Confidence Check, AMM TASK 21-00-05-780-801

(6) Engine Bleed Air System Health Check, AMM TASK 36-11-00-700-801

E. Initial Evaluation

OMR ALL; AIRPLANES WITH CABIN PRESSURE CONTROLLERS P/N 7121-19971-01AA/01AB/01AC/01BA

(1) Visually inspect the copper tape on the outside edges of the cabin pressure controller as follows:

(a) Make sure that the copper tape extends onto the front outer edges of the pressure controller 1.5 to 2.2 mm.

(b) Make sure that there are no cracks, holes or dents in the tape.

(c) Make sure that the tape is smoothly attached to the face of the controller.

(d) Make sure that there are no openings into the controller where the controller front cover connects with the controller sides.

(e) If any of the above conditions exist, replace all of the cooper tape or controller as follows:

These are the tasks:

Cabin Pressure Controller (CPC) Removal, AMM TASK 21-31-01-000-801,

Cabin Pressure Controller (CPC) Installation, AMM TASK 21-31-01-400-801.

OMR ALL

(2) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

(3) If there are no maintenance messages, then there was an intermittent fault.

(4) If the Digital Cabin Pressure Controller shows a Fault Code 30 in the history when performing the BITE check or if pilots report cabin altitude is above pressurization schedule during cruise, cabin altitude climbs during cruise, or cabin altitude climbs during initial descent, do the following.

(a) Verify the performance of the engine bleed system (Engine Bleed Air System Health Check, AMM TASK 36-11-00-700-801).

(b) Check the air conditioning pack air flow and the fuselage leakage rate (Airplane Pack Operation - Confidence Check, AMM TASK 21-00-05-780-801).

(c) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

(5) If the maintenance message shows, and there is an additional maintenance message, then do the Fault Isolation Procedure for the other maintenance message.

(6) If the maintenance message shows, and it is the only maintenance message, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Replace the cabin pressure outflow valve.

These are the tasks:

Aft Outflow Valve Assembly Removal, AMM TASK 21-31-03-000-801,

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Aft Outflow Valve Assembly Installation, AMM TASK 21-31-03-400-801.

(a) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

1) If the maintenance message does not show, then you corrected the fault.

2) If the maintenance message shows, then continue.

(2) Replace the cabin pressure controller that has the maintenance message.

These are the tasks:

Cabin Pressure Controller (CPC) Removal, AMM TASK 21-31-01-000-801,

Cabin Pressure Controller (CPC) Installation, AMM TASK 21-31-01-400-801.

(a) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

1) If the maintenance message does not show, then you corrected the fault.

——— END OF TASK ———

821. Auto Manual Input to Controller Problem - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) AUTO/MAN ERROR

B. Possible Causes

(1) Cabin pressure control module, P5-6

(2) Wiring problem

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	1	C01273	PRESSURIZATION CONTROL LCD LTG
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL
F	7	C01272	PRESSURIZATION CONTROL IND

D. Related Data

(1) (SSM 21-31-11)

(2) (WDM 21-31-11)

E. Initial Evaluation

(1) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

(2) If the maintenance message does not show, then there was an intermittent fault.

(3) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Replace the cabin pressure control module.

These are the tasks:

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Cabin Pressure Control Module - Removal, AMM TASK 21-31-02-000-801,

Cabin Pressure Control Module - Installation, AMM TASK 21-31-02-400-801.

(a) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

1) If the maintenance message does not show, then you corrected the fault.

2) If the maintenance message shows, then continue.

(2) Do this check of the wiring between cabin pressure controllers 1 and 2, M1653 and M1654, in the electronic equipment bay and the cabin pressure control module, P5-6 in the control compartment:

(a) Remove cabin pressure controller 1 and 2. To remove the controllers, do this task: Cabin Pressure Controller (CPC) Removal, AMM TASK 21-31-01-000-801.

(b) Lower the overhead P5 panel.

(c) Disconnect connector D10724 and D10726 from the cabin pressure control module.

(d) Do a wiring check between these pins of connector D10724 of the cabin pressure control module and connectors D10738B and D10740B of cabin pressure controllers 1 and 2:

D10724		D10738B
pin 1	. . . . .	pin B3

D10726		D10740B
pin 8	. . . . .	pin B3

(e) Repair the problem that you find.

(f) Re-connect connector D10724 and D10726 to the cabin pressure control module.

(g) Lift (close) the overhead P5 panel.

(h) Re-install cabin pressure controllers 1 and 2. To install the controllers, do this task: Cabin Pressure Controller (CPC) Installation, AMM TASK 21-31-01-400-801.

(i) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

1) If the maintenance message does not show, then you corrected the fault.

2) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

822. Wheels On Ground Disagreement Problem - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) LDG GEAR DISAGREE

B. Possible Causes

(1) Proximity switch electronics unit (PSEU), M2061

(2) Wiring

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C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	1	C01273	PRESSURIZATION CONTROL LCD LTG
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL
F	7	C01272	PRESSURIZATION CONTROL IND

D. Related Data

(1) (SSM 21-31-24)

(2) (WDM 21-31-24)

E. Initial Evaluation

(1) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

(2) If the maintenance message does not show, then there was an intermittent fault.

(3) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Do the BITE test for the proximity switch electronics unit (PSEU), M2061. To do the BITE test, do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

(a) If the PSEU BITE test shows any maintenance messages, then do the corrective action for the message.

1) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

2) If the maintenance message does not show, then you corrected the fault.

(b) If the PSEU BITE test does not show any maintenance messages, then continue.

(2) Do this check of the wiring between cabin pressure controllers 1 and 2 and the PSEU:

(a) Remove cabin pressure controllers 1 and 2. To remove the controllers, do this task: Cabin Pressure Controller (CPC) Removal, AMM TASK 21-31-01-000-801.

(b) Open this access panel:

Number	Name/Location
112A	Forward Access Door

(c) Do a wiring check between these pins of connectors D10738B and D10740B of cabin pressure controllers 1 and 2 and connectors D11138 and D10988 of the PSEU:

D10738B		D11138
pin B1	. . . . .	pin 12

D10738B		D10988
pin B4	. . . . .	pin 12

D10740B		D11138
pin B1	. . . . .	pin 12

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D10740B		D10988
pin B4	. . . . .	pin 12

(d) Repair the problem with the wiring that you find.

(e) Close this access panel:

Number	Name/Location
112A	Forward Access Door

(f) Re-install the cabin pressure controllers. To install the controllers, do this task: Cabin Pressure Controller (CPC) Installation, AMM TASK 21-31-01-400-801.

(g) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

(h) If the maintenance message does not show, then you corrected the fault.

Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

823. Low Pack Air Inflow Or Airplane Cabin Leak Problem - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) LO INFL/HI LEAKG

(2) This maintenance message does not indicate a fault with the Digital Cabin Pressurization Control System.

B. Possible Causes

(1) Low pneumatic pressure

OMR 701-799

(2) Dirty Water Separator Coalescer Bag

OMR ALL

(3) L (R) Pack Flow Control And Shutoff Valve, V18 (V19)

(4) Low Limit Temperature (35°F) Sensor

(5) Air Cycle Machine (ACM)

(6) Airplane cabin leak

(a) Cargo door seal failure at high differential pressure

NOTE: This may not be detected during a leak test procedure as the differential pressure is not high enough.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(7) Flexible duct

OMR ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	1	C01273	PRESSURIZATION CONTROL LCD LTG
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL
F	7	C01272	PRESSURIZATION CONTROL IND

D. Related Data

(1) SSM 21-51-11

(2) WDM 21-51-11

(3) Airplane Pack Operation - Confidence Check, AMM TASK 21-00-05-780-801

(4) Engine Bleed Air System Health Check, AMM TASK 36-11-00-700-801

E. Initial Evaluation

(1) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

(2) If the maintenance message does not show, then there was an intermittent fault.

(3) If the DCPC shows a Fault Code 30 in the history when performing the Built-In-Test Equipment (BITE) check or if pilots report cabin altitude is above pressurization schedule during cruise, cabin altitude climbs during cruise, or cabin altitude climbs during initial descent, do the following:

(a) Do a check of the airplane's ability to maintain cabin pressure in flight. This is the task: Airplane Pack Operation - Confidence Check, AMM TASK 21-00-05-780-801.

(b) If the check was not satisfactory, do this Fault Isolation Procedure: Airplane Cannot Maintain Cabin Pressure On a Single Pack Within Specified Limits - Fault Isolation, 21-00 TASK 807.

(c) Verify the performance of the Engine Bleed System. This is the task: Engine Bleed Air System Health Check, AMM TASK 36-11-00-700-801.

(d) Check the Air Conditioning Pack air flow and the fuselage leakage rate. This is the task: Airplane Pack Operation - Confidence Check, AMM TASK 21-00-05-780-801.

1) If a pressurization problem occurs during flight (high cabin altitude), inspect the cargo door seals before the doors are opened after landing.

a) The seal may have deformed in flight. This may not be detected if the seal is inspected after the door is opened as the seal may return to its installed position when the door is opened.

2) This may not show during the check of AMM TASK 21-00-05-780-801 as the differential pressure during the check may not be high enough.

(e) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

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OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(f) Inspect the condition of the flexible duct between the High Pressure Water Separator (HPWS) Mix Muff and the HPWS Condenser:

1) Do the applicable step of this task to inspect the flexible duct: Air Cycle Machine Removal, AMM TASK 21-51-04-000-802-002.

2) Replace the flexible duct if the duct fails the inspection.

OMR ALL

(4) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Check to see if there was a report of low pneumatic duct pressure.

(a) If there was a report of low duct pressure, do this task: Duct Pressure Low or Duct Pressure Very Low / Zero, the Engine is the Bleed Source - Fault Isolation, 36-10 TASK 810.

1) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

NOTE: Do this step to clear the fault.

(b) If there was not a report of low duct pressure, then continue.

OMR 701-799

(2) Do an inspection of the Left Pack And Right Pack Coalescer Bags. This is the task: Coalescer Bag Pressure Indicator Inspection, AMM TASK 21-51-05-400-803.

(a) If the indicator disk on the pressure indicator assembly is in the red area, clean the Coalescer Bag. This is the task: Coalescer Bag Cleaning, AMM TASK 21-51-05-100-801.

1) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

NOTE: Do this step to clear the fault.

(b) If the indicator disk on the pressure indicator assembly is in the black area, then continue.

(3) If a air pack - air flow and temperature diagnostic, SPL-6191, is available, do the check of the Pack Flow Control and Pack Cooling System:

(a) To do the check, do this task: Pack Flow Control and Pack Cooling System - Check, AMM TASK 21-51-00-943-801.

(b) If the temperature of the pack outlet air is 34F or lower, then do this task: Pack Outlet Temperature Too Hot or Too Cold - Fault Isolation, 21-52 TASK 856.

(c) If the pack air flow was too low, then do the applicable procedure of this task: Pack Air Flow is Too Low or Too High - Fault Isolation, 21-52 TASK 857.

OMR ALL

(4) Do a check of the Air Conditioning Packs for proper flow. This is the task: Pack Flow Control Valve - Operational Test, AMM TASK 21-51-00-700-802 or Pack Flow Control Valve - Operational Test, AMM TASK 21-51-00-710-801.

(a) If a Pack Flow Control And Shutoff Valve does not operate correctly, replace the valve. These are the tasks:

- Flow Control and Shutoff Valve Removal, AMM TASK 21-51-01-000-801-001 or Right Flow Control and Shutoff Valve - Removal, AMM TASK 21-51-01-000-805-002 or Right Flow Control and Shutoff Valve Removal, AMM TASK 21-51-01-000-807-003

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- Flow Control and Shutoff Valve Installation, AMM TASK 21-51-01-400-801-001 or Left Flow Control and Shutoff Valve - Installation, AMM TASK 21-51-01-400-805-002 or Right Flow Control and Shutoff Valve - Installation, AMM TASK 21-51-01-400-802-002 or Right Flow Control and Shutoff Valve Installation, AMM TASK 21-51-01-400-806-003 or Left Flow Control and Shutoff Valve Installation, AMM TASK 21-51-01-400-807-003

(b) Make sure that air flows from the ram air exhaust outlets when the packs are in operation.

1) If air does not flow from the ram air exhaust outlet for a pack, then replace the ACM. These are the tasks:

- ACM Removal, AMM TASK 21-51-04-000-801-001 or Air Cycle Machine Removal, AMM TASK 21-51-04-000-802-002
- ACM Installation, AMM TASK 21-51-04-400-801-001 or Air Cycle Machine Installation, AMM TASK 21-51-04-400-802-002

2) If air flows from the ram air exhaust outlet, then continue.

3) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

NOTE: Do this step to clear the fault.

(c) If the pack flow control and shutoff valves operate correctly, then continue.

(5) Do this task: Cargo Door Adjustment, AMM TASK 52-31-00-820-801.

(6) Do a check of the airplane cabin for leaks. This is the task: Cabin Pressure Leak Test, AMM TASK 05-51-91-790-801.

(a) Repair the source of the leak that you find.

1) Do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

NOTE: Do this step to clear the fault.

END OF TASK

824. Cabin Pressure Controller Fail - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) CONTRL LRU FAIL

B. Possible Causes

(1) Cabin pressure controller, M1653 (No. 1) or M1654 (No. 2)

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	1	C01273	PRESSURIZATION CONTROL LCD LTG
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL
F	7	C01272	PRESSURIZATION CONTROL IND

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Related Data

(1) SSM 21-31-24

(2) WDM 21-31-24

E. Initial Evaluation

OMR ALL; AIRPLANES WITH CABIN PRESSURE CONTROLLERS P/N 7121-19971-01AA/01AB/01AC/01BA

(1) Visually inspect the copper tape on the outside edges of the cabin pressure controller as follows:

(a) Make sure that the copper tape extends onto the front outer edges of the pressure controller 1.5 to 2.2 mm.

(b) Make sure that there are no cracks, holes or dents in the tape.

(c) Make sure that the tape is smoothly attached to the face of the controller.

(d) Make sure that there are no openings into the controller where the controller front cover connects with the controller sides.

(e) If any of the above conditions exist, replace all of the cooper tape or controller as follows:

These are the tasks:

- Cabin Pressure Controller (CPC) Removal, AMM TASK 21-31-01-000-801
- Cabin Pressure Controller (CPC) Installation, AMM TASK 21-31-01-400-801

OMR ALL

(2) Do the Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

(3) If the maintenance message does not show, then there was an intermittent fault.

(4) If the BITE has shown occasional occurrences of fault code 42, perform the SYSTEM TEST on the ground:

(a) If the SYSTEM TEST status is satisfactory, the pressure controller does not need to be replaced.

(5) If the BITE has shown frequent occurrences of Fault Code 42 within a short time interval such as once per flight leg, replace the controller as follows:

These are the tasks:

- Cabin Pressure Controller (CPC) Removal, AMM TASK 21-31-01-000-801
- Cabin Pressure Controller (CPC) Installation, AMM TASK 21-31-01-400-801

(6) For occurrences of Fault Code 42 on part number controllers 7121-19971-AA or 7121-19971-AB, it is recommended that the software be upgraded to “AC” as directed by Nord-Micro Service Bulletin SB 19971-21-006.

(7) If the BITE has shown occurrences of Fault Code 43 on the ground or during testing, perform the SYSTEM TEST or SYSTEM TEST and CLEAR to do a check of the system health status:

(a) If the SYSTEM TEST or SYSTEM TEST and CLEAR is satisfactory, the pressure controller does not need to be replaced.

(8) If the BITE has shown frequent occurrences of Fault Code 43, replace the pressure controller as follows:

These are the tasks:

- Cabin Pressure Controller (CPC) Removal, AMM TASK 21-31-01-000-801
- Cabin Pressure Controller (CPC) Installation, AMM TASK 21-31-01-400-801

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(9) If the BITE has shown occasional occurrences of Fault Code 57 on the ground or during testing, perform the SYSTEM TEST or SYSTEM TEST and CLEAR to do a check of the system health status:

(a) If the SYSTEM TEST or SYSTEM TEST and CLEAR is satisfactory, the pressure controller does not need to be replaced.

(10) If the BITE has shown frequent occurrences of Fault Code 57 within a short time interval such as once per flight leg, replace the pressure controller as follows:

These are the tasks:

- Cabin Pressure Controller (CPC) Removal, AMM TASK 21-31-01-000-801
- Cabin Pressure Controller (CPC) Installation, AMM TASK 21-31-01-400-801

(11) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Replace the cabin pressure controller in the aft EE bay.

These are the tasks:

- Cabin Pressure Controller (CPC) Removal, AMM TASK 21-31-01-000-801
- Cabin Pressure Controller (CPC) Installation, AMM TASK 21-31-01-400-801

(a) Do the Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

1) If the maintenance message does not show, then you corrected the fault.

——— END OF TASK ———

827. AUTO FAIL Light On - Fault Isolation

A. Description

(1) These are the indications that one channel is not serviceable when the system is in the automatic mode:

(a) The amber AUTO FAIL light comes on

(b) The MASTER CAUTION and AIR COND lights come on

(c) The green ALTN light comes on.

B. Possible Causes

(1) Pressure Controller Fault, M1653 and M1654

(2) Outflow Valve Fault, V48

(3) Cabin Pressure Selector Panel Fault, P5-6

(4) Wiring

(5) Aircraft Operation Beyond DCPC Limits Parameters

C. Circuit Breakers

(1) Circuit Breakers

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	1	C01273	PRESSURIZATION CONTROL LCD LTG
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	7	C01272	PRESSURIZATION CONTROL IND

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D. Related Data

(1) WDM 21-31-22

(2) WDM 21-31-23

E. Initial Evaluation

(1) Momentarily position the Pressurization Mode Selector to MAN and then back to AUTO.

NOTE: Some faults indicated by an AUTO FAIL light are recoverable.

(a) If the AUTO FAIL light remains extinguished with the Pressurization Mode Selector in AUTO, the fault has been corrected and the controller is operating normally.

(b) Do the Repair Confirmation at the end of this procedure.

(2) Examine the circuit breakers shown in the circuit breaker topic to make sure that no circuit breaker is open.

(3) If you find an open circuit breaker, reset the circuit breaker. To reset the circuit breaker, do this task: Circuit Breaker Reset, AMM TASK 20-60-01-860-801.

(a) The reset procedure does checks to find the cause of the tripped circuit breaker. You must not reset a circuit breaker until you know the cause it tripped.

(4) Examine the display window on each of the cabin pressure controllers.

(5) If FAULT shows in the display window, do the DCPC BITE. To do the BITE, do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

(6) Make a record of BCM fault numbers shown in the cabin pressure controller BITE:

(a) All the fault numbers shown in the steps that follow in this subtask will cause an AUTO FAIL indication.

(b) AIRPLANES WITH CABIN PRESSURE CONTROLLERS P/N 7121-19971-01XX:

Fault numbers 19, 24 and 25 together in IBIT, 032 through 052, 56, 57, 59 through 61, 91 and 108 through 110 show problems with the pressure controllers, M1653 and M1654.

1) Do the Fault Isolation Procedure - Maintenance Message CONTRL LRU FAIL for these fault numbers.

(c) AIRPLANES WITH CABIN PRESSURE CONTROLLERS P/N 21933-01XX:

Fault numbers 19, 24 and 25 together in IBIT, 33 through 35, 37 through 45, 47 through 50, 52, 56, 57, 59 through 61, 108, 109 and 110 show problems with the pressure controllers, M1653 and M1654.

1) Do the Fault Isolation Procedure - Maintenance Message CONTRL LRU FAIL for these fault numbers.

(d) AIRPLANES WITH CABIN PRESSURE CONTROLLERS P/N 7121-19971-01XX:

Fault numbers 7, 9, 10, 11, 019, 064 through 069, 072 through 080, 083, 087, 088, 091 and 092 show problems with the aft outflow valve

1) Do the Fault Isolation Procedure - OFV LRU FAIL Maintenance Message for these fault numbers.

(e) AIRPLANES WITH CABIN PRESSURE CONTROLLERS P/N 21933-01XX:

Fault numbers 7, 9, 10, 11, 15, 19, 64, 65, 68, 69, 72-80, 80, 83, 87, 88 and 92 show problems with the aft outflow valve.

1) Do the Fault Isolation Procedure - OFV LRU FAIL Maintenance Message for these fault numbers.

(f) Fault number 107 shows problems with the pressure control panel.

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1) Do the Fault Isolation Procedure - SEL PNL LRU FAIL Maintenance Message for these fault numbers.

(g) Fault numbers 15 and 74 show problems with the outflow valve or wires between the valve and the controller.

1) Do the Fault Isolation Procedure - OFV OR WIRING Maintenance Message for these fault numbers.

(h) AIRPLANES WITH CABIN PRESSURE CONTROLLERS P/N 7121-19971-01XX:

Fault numbers 1, 2, 6, 13, 14, 15 and 110 show problems with system wire runs.

1) Do the Fault Isolation Procedure - SYSTEM WIRING Maintenance Message for these fault numbers.

(i) AIRPLANES WITH CABIN PRESSURE CONTROLLERS P/N 21933-01XX:

Fault numbers 1, 2, 6, 13, 14, 15, 108 and 110 show problems with system wire runs.

1) Do the Fault Isolation Procedure - SYSTEM WIRING Maintenance Message for these fault numbers.

(j) Fault number 3 shows a problem with the cabin altitude.

1) Do the Fault Isolation Procedure - HI CAB ALT Maintenance Message.

(k) Fault number 4 shows a problem with the cabin differential pressure.

1) Do the Fault Isolation Procedure - CAB DIFF PRESS HI Maintenance Message.

(l) Fault number 5 shows a problem with the controllers or the outflow valve.

1) Do the Fault Isolation Procedure - CAB RATE HI Maintenance Message.

(m) Fault number 109 shows a problem with the controllers.

1) Do the Fault Isolation Procedure - DRV ENBL ERROR Maintenance Message.

(n) Fault numbers 024 and 025 will cause an AUTO FAIL indication only if triggered together during IBIT. This fault number combination shows problems with the DADC #1 and DADC #2.

1) Do the Fault Isolation Procedure - DADC #1 INVALID Maintenance Message.

2) Do the Fault Isolation Procedure - DADC #2 INVALID Maintenance Message.

F. Fault Isolation Procedure - CONTRL LRU FAIL Maintenance Message

(1) Do this task: Cabin Pressure Controller Fail - Fault Isolation, 21-31 TASK 824.

(2) Do the Repair Confirmation at the end of this procedure.

G. Fault Isolation Procedure - OFV LRU FAIL Maintenance Message

(1) Do this task: Cabin Pressure Outflow Valve Fail - Fault Isolation, 21-31 TASK 802.

(2) Do the Repair Confirmation at the end of this procedure.

H. Fault Isolation Procedure - SEL PNL LRU FAIL Maintenance Message

(1) Do this task: Cabin Pressure Control Module Display Faults - Fault Isolation, 21-31 TASK 803.

(2) Do the Repair Confirmation at the end of this procedure.

I. Fault Isolation Procedure - SYSTEM WIRING Maintenance Message

(1) Do this task: System Wiring Failure - Fault Isolation, 21-31 TASK 804.

(2) Do the Repair Confirmation at the end of this procedure.

J. Fault Isolation Procedure- HI CAB ALT Maintenance Message

(1) Do this task: Hi Cabin Altitude - Fault Isolation, 21-31 TASK 817.

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(2) Do the Repair Confirmation at the end of this procedure.

K. Fault Isolation Procedure - CAB RATE HI Maintenance Message

(1) Do this task: Hi Cabin Altitude - Fault Isolation, 21-31 TASK 817.

(2) Do the Repair Confirmation at the end of this procedure.

L. Fault Isolation Procedure - CAB DIFF PRESS HI Maintenance Message

(1) Do this task: Hi Cabin Differential Pressure - Fault Isolation, 21-31 TASK 816.

(2) Do the Repair Confirmation at the end of this procedure.

M. Fault Isolation Procedure - DRV ENBL ERROR Maintenance Message

(1) Do this task: Drive Enable Signal Problem - Fault Isolation, 21-31 TASK 819.

(2) Do the Repair Confirmation at the end of this procedure.

N. Fault Isolation Procedure - CAB RATE HI Maintenance Message

(1) Do this task: Hi Cabin Altitude Rate Change - Fault Isolation, 21-31 TASK 818.

(2) Do the Repair Confirmation at the end of this procedure.

O. Fault Isolation Procedure - OFV OR WIRING Maintenance Message

(1) Do this task: Cabin Pressure Outflow Valve or Wiring - Fault Isolation, 21-31 TASK 805.

(2) Do the Repair Confirmation at the end of this procedure.

P. Fault Isolation Procedure - DADC #1 INVALID Maintenance Message

(1) Do this task: Air Data Inertial Reference Unit 1 Input Problem - Fault Isolation, 21-31 TASK 806.

(2) Do the Repair Confirmation at the end of this procedure.

Q. Fault Isolation Procedure - DADC #2 INVALID Maintenance Message

(1) Do this task: Air Data Inertial Reference Unit 2 Input Problem - Fault Isolation, 21-31 TASK 807.

(2) Do the Repair Confirmation at the end of this procedure.

R. Repair Confirmation

(1) If not already done, do the DCPC BITE. To do the BITE, do this task: Digital Cabin Pressure Controller (DCPC) BITE Procedure, 21-31 TASK 801.

(a) If these conditions are present, then you corrected the fault:

1) No maintenance message or fault codes are present

2) AUTO FAIL light is off.

(b) If the problem still exists, continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

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801. Cabin Altitude Warning System Does Not Operate - Fault Isolation

A. Description

OMR 805, 807 PRE SB 737-31A1332

(1) The CABIN ALTITUDE Indicator Lights on the Captain's Instrument Panel, P1-3 and the First Officer's Instrument Panel, P3-1 have been placarded inoperative. However, if the LIGHTS Switch on the Captain's Instrument Panel, P1-3 is set to the TEST position, the CABIN ALTITUDE Indicator Lights will come ON.

OMR ALL

(2) The aural warning does not operate as expected when the cabin altitude is more than 10,000 feet (3048 meters).

OMR 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 802, 805, 807 POST SB 737-31A1332

(3) The CABIN ALTITUDE indicators on the captains's instrument panel P1-3 and the first officers's instrument panel P3-1 do not come on when the cabin altitude is more than 10,000 feet (3048 meters).

OMR ALL

(4) For additional information on Cabin Altitude Warning System, refer to SDS 21-30-00.

B. Possible Causes

(1) Cabin Altitude Warning Switch, S128

OMR 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 802, 805, 807 POST SB 737-21-1165

(2) Cabin Altitude Warning Switch, S1153

OMR ALL

(3) Warning Cutout Relay, K1 (P5-16 Panel)

(4) Altitude Horn Cutout Switch, S1 (P5-16 Panel)

(5) Wiring.

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	17	C00129	LANDING GEAR LATCH & PRESS WARN

D. Related Data

(1) WDM 21-33-11.

(2) SSM 21-33-11

E. Initial Evaluation

(1) Do an operational test of the cabin altitude warning system. To do this, do this task: Cabin Altitude Warning Switch Functional Test, AMM TASK 21-33-00-000-801.

(a) If the test is satisfactory, then there was an intermittent fault.

(b) If the test is not satisfactory, then do the Fault Isolation Procedure below.

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F. Aural Warning Fault Isolation Procedure

(1) Replace the Cabin Altitude Warning Switch, S128. These are the tasks:

- Cabin Altitude Warning Switch Removal, AMM TASK 21-33-04-000-801
- Cabin Altitude Warning Switch Installation, AMM TASK 21-33-04-400-801

(a) Do the Repair Confirmation at the end of this task.

OMR 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 802, 805, 807 POST SB 737-21-1165

(2) Replace the Cabin Altitude Warning Switch, S1153. These are the tasks:

- Cabin Altitude Warning Switch Removal, AMM TASK 21-33-04-000-801
- Cabin Altitude Warning Switch Installation, AMM TASK 21-33-04-400-801

(a) Do the Repair Confirmation at the end of this task.

OMR ALL

(3) Replace the Warning Cutout Relay, K1.

(a) Do the Repair Confirmation at the end of this task.

(4) Replace the Altitude Horn Cutout Switch, S1.

(a) Do the Repair Confirmation at the end of this task.

(5) Do these checks of the wiring for a fault with the aural warning:

(a) Lower the P5 overhead panel to get access to the electrical connector on the back of the P5-16 panel.

(b) Disconnect the electrical connector, D654, from the P5-16, cabin altitude and rate of climb module.

(c) Do a check for 28 VDC at connector D654 from pin 8 to structure ground.

(d) If there is not 28 VDC from pin 8 to structure ground on connector D654, then do these steps:

1) Repair the wiring from pin 8 on connector D654 to circuit breaker C129, LANDING GEAR LATCH & PRESS WARN.
2) Re-connect electrical connector D654 on the P5-16 panel.
3) Close the P5 overhead panel.
4) Do the Repair Confirmation at the end of this task.

(e) If there is 28 VDC from pin 8 on connector D654 to structure ground, then continue.

(f) Do a check for 28 VDC at connector D654 from pin 6 to structure ground.

(g) If there is not 28 VDC from pin 6 to structure ground on connector D654, then do these steps:

1) Repair the wiring from pin 6 on connector D654 to the aural warning module, M315.
2) Re-connect electrical connector D654 on the P5-16 panel.
3) Close the P5 overhead panel.
4) Do the Repair Confirmation at the end of this task.

(h) If there is 28 VDC from pin 6 on connector D654 to structure ground, then continue.

(i) Disconnect connector D776 from the cabin altitude warning switch, S128.

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OMR 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 802, 805, 807 POST SB 737-21-1165

(j) Disconnect connector D14284 from the cabin altitude warning switch, S1153.

OMR ALL

(k) Do a continuity check between this pin of connector D654 at the P5-16 panel, and this pin of connector D776 at the cabin altitude warning switch, S128:

D654		D776
pin 7	. . . . .	pin A

OMR 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 802, 805, 807 POST SB 737-21-1165

(l) Do a continuity check between this pin of connector D654 at the P5-16 panel, and this pin of connector D14284 at the cabin altitude warning switch, S1153:

D654		D14284
7	. . . . .	A

OMR ALL

(m) If there is not continuity from pin 7 on connector D654 to pin A on connector D776, then do these steps:

1) Repair the wiring.
2) Re-connect electrical connector D776 on the cabin altitude warning switch, S128.
3) Re-connect electrical connector D654 on the P5-16 panel.
4) Close the P5 Overhead Panel.
5) Do the Repair Confirmation at the end of this task.

OMR 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 802, 805, 807 POST SB 737-21-1165

(n) If there is not continuity from pin 7 on connector D654 to pin A on connector D14284, then do these steps:

1) Repair the wiring.
2) Re-connect electrical connector D14284 on the cabin altitude warning switch, S1153.
3) Re-connect electrical connector D654 on the P5-16 panel.
4) Close the P5 Overhead Panel.
5) Do the Repair Confirmation at the end of this task.

OMR ALL

(o) If there is continuity from pin 7 on connector D654 to pin A on connector D776, then continue.

OMR 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 802, 805, 807 POST SB 737-21-1165

(p) If there is continuity from pin 7 on connector D654 to pin A on connector D14284, then continue.

OMR ALL

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(q) Do a continuity check between pin B of connector D776 at the cabin altitude warning switch, S128 and structure ground.

OMR 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 802, 805, 807 POST SB 737-21-1165

(r) Do a continuity check between pin B of connector D14284 at the cabin altitude warning switch S1153 and structure ground.

OMR ALL

(s) If there is not continuity from pin B on connector D776 to structure ground on connector D776, then do these steps:

1) Repair the wiring.
2) Re-connect electrical connector D776 on the cabin altitude warning switch, S128.
3) Re-connect electrical connector D654 on the P5-16 panel.
4) Close the P5 Overhead Panel.
5) Do the Repair Confirmation at the end of this task.

OMR 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 802, 805, 807 POST SB 737-21-1165

(t) If there is not continuity from pin B on connector D14284 to structure ground, then do these steps:

1) Repair the wiring.
2) Re-connect electrical connector D14284 on the cabin altitude warning switch, S1153.
3) Re-connect electrical connector D654 on the P5-16 panel.
4) Close the P5 Overhead Panel.
5) Do the operational test of the cabin altitude warning system. To do this test, do this task: Cabin Altitude Warning Switch Functional Test, AMM TASK 21-33-00-000-801.
6) If the operational test is satisfactory, then you corrected the fault.

OMR ALL

(u) If there is continuity from pin B on connector D776 to structure ground on connector D776, then continue:

1) Replace the cabin altimeter and rate of climb module, P5-16.
2) Re-connect electrical connector D776 on the cabin altitude warning switch, S128.
3) Re-connect electrical connector D654 on the P5-16 panel.
4) Close the P5 Overhead Panel.
5) Do the Repair Confirmation at the end of this task.
6) If the operational test is satisfactory, then you corrected the fault.

OMR 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 802, 805, 807 POST SB 737-21-1165

(v) If there is continuity from pin B on connector D14284 to structure ground, then continue:

1) Replace the cabin altimeter and rate of climb module, P5-16.
2) Re-connect electrical connector D14284 on the cabin altitude switch, S1153.
3) Re-connect electrical connector D654 on the P5-16 Panel.

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OMR 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 802, 805, 807 POST SB 737-21-1165 (Continued)

4) Close the P5 Overhead Panel.

5) Do the Repair Confirmation at the end of this task.

(w) Re-connect electrical connector D14284 on the Cabin Altitude Warning Switch, S1153.

OMR 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 802, 805, 807 POST SB 737-31A1332

(6) Do these checks of the wiring for a fault with the CABIN ALTITUDE indicators:

(a) Make sure these circuit breakers are closed:

Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	11	C00133	INDICATOR MASTER DIM DIM/TST CONT
E	12	C00314	INDICATOR MASTER DIM SECT 2
F	12	C00318	INDICATOR MASTER DIM SECT 6

(b) Set the LIGHTS switch on the captain's instrument panel P1-3 to the TEST position.

(c) Make sure that the red CABIN ALTITUDE indicators on the captain's instrument panel P1-3 and the first officer's instrument panel P3-1 come on.

(d) If a CABIN ALTITUDE indicator lamp does not come on, do the applicable portion of this fault isolation task:

1) Flight Compartment Lighting Problem - Fault Isolation, 33-10 TASK 801.

NOTE: This fault isolation task does a check of the MASTER DIM & TEST system and makes sure that 28 VDC is at the indicator.

2) Do the Repair Confirmation at the end of this task.

(e) If the CABIN ALTITUDE indicators came on, then do these steps:

1) Get access to the back of the captain's instrument panel P1-3 or the back of the first officer's instrument panel P3-1, as applicable.

2) Disconnect electrical connector D3999 from the P3-1 panel or electrical connector D895 from the P1-3 panel, as applicable.

(f) Get access to the cabin pressure switch, S128, on the ceiling in the lower compartment forward of the nose landing gear.

(g) Disconnect electrical connector D776 from the cabin pressure switch.

(h) If the faulty indicator is on the P1-3 panel, do a check of the wiring between electrical connector D895, pin 13 and connector D776, pin A.

1) Repair any wiring problems that you find.

(i) If the faulty indicator is on the P3-1 panel, do a check of the wiring between connector D3999, pin 17 and connector D776, pin A .

1) Repair any wiring problems that you find.

(j) Connect all connectors that were disconnected.

(k) Do the Repair Confirmation at the end of this task.

OMR ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Repair Confirmation

(1) Do this task: Cabin Altitude Warning Switch Functional Test, AMM TASK 21-33-00-000-801.

(a) If the test is satisfactory, then you corrected the problem.

(b) If the test is not satisfactory, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

802. CABIN CLIMB Indicator Does not Operate or Fluctuates - Fault Isolation

A. Description

(1) This task is for these Observed Faults:

(a) The CABIN CLIMB Indicator does not operate.

(b) The CABIN CLIMB Pointer does not move.

OMR 820-826

(c) The CABIN CLIMB Pointer fluctuates.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(2) The CABIN CLIMB Indicator reacts to Cabin Pressure Altitude Rate Changes during ascent or descent in feet per minute.

NOTE: The gauge does not indicate Zero on the ground with one of the Entry Doors open to make sure that the cabin is not pressurized.

(3) Flush operations of the Vacuum Toilet System can cause the CABIN CLIMB INDICATOR to momentarily show a rate of climb indication of 300-500 feet per minute. This is normal.

B. Possible Causes

OMR 820-826; AIRPLANES WITH PFTC P/N 51090248-002/-003

(1) Pack Flow Temperature Controller (PFTC)

(2) Cabin Rate of Climb Indicator, N55

(3) Leakage in the Pack Bay Area

(4) L (R) Temperature Control Valve (TCV), V139 (V141)

(5) L (R) Pack Flow Sensor, M2953 (M2954)

(6) L (R) Flow Control and Shutoff Valve, V184 (V185)

(7) Wiring

OMR 820-826; AIRPLANES WITH PFTC P/N 51090248-005

(8) PFTC BITE

(9) Leakage in the Pack Bay Area

(10) Pack Water Drain Leakage or Blockage

(11) Dirty Heat Exchangers

(12) Pack Temperature Sensor

(13) L (R) Pack Flow Sensor, M2953 (M2954)

(14) L (R) Flow Control and Shutoff Valve, V184 (V185)

(15) Bleed Air System

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 820-826; AIRPLANES WITH PFTC P/N 51090248-005 (Continued)

(16) Wiring

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999; AIRPLANES WITH PZTC P/N 51090160-1 OR 622814-5.

(17) Pack/Zone Temperature Controller (PZTC) BITE

(18) Cabin Rate of Climb Indicator, N55

(19) Bleed Air System

(20) Leakage in the Pack Bay Area

(21) L (R) Flow Control and Shutoff Valve, V18 (V19)

(22) L (R) TCV, V139 (V141)

(23) Zone Trim Air Valve (TAV)s (V143, V144, V145)

OMR 820-826; AIRPLANES WITH PFTC P/N 51090248-002/-003

C. Fault Isolation Procedure

(1) Do the following steps on the PFTC:

NOTE: If one side of the bleed duct pressure gauge fluctuates during cabin rate of climb indicator oscillation, do the fault isolation on that side.

NOTE: If you do not know which side has the oscillations, do the fault isolation on both sides.

(a) Do the FAULT HISTORY review on the affected PFTC (M2951 or M2952) for the last 10 or more flight legs, this is the task: Pack Flow Temperature Controller BITE Test, AMM TASK 21-61-00-740-801-002.

1) If a fault is found, do the corrective action for the fault.

2) If the FAULT HISTORY is not active, continue.

(b) Do the GROUND TEST BITE procedure on the affected PFTC (M2951 or M2952), this is the task: Pack Flow Temperature Controller BITE Test, AMM TASK 21-61-00-740-801-002.

1) Do not bypass the flow sensor drift check.

2) If a fault is found, do the corrective action for the fault.

3) If a fault is not found, continue.

(2) Do a check of the Cabin Rate of Climb Indicator.

(a) If the Cabin Rate of Climb Indicator Needle shows a value other than Zero (0) when the airplane is on the ground, then do these steps:

1) Make sure that one of the Entry Doors has been open for more than 5 minutes so that the cabin is not pressurized.

2) On the lower left corner of the gauge, turn the Zero Adjustment Screw and make sure that the pointer can be adjusted to a minimum of 400 ft/min on both sides of the Zero graduation.

3) Reset the pointer to Zero.

4) If the problem continues, replace the Cabin Rate of Climb Indicator, N55. These are the tasks:

- Cabin Rate of Climb Indicator Removal, AMM TASK 21-33-01-000-801
- Cabin Rate of Climb Indicator Installation, AMM TASK 21-33-01-400-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 820-826; AIRPLANES WITH PFTC P/N 51090248-002/-003 (Continued)

(b) If the Cabin Rate of Climb Indicator Needle shows zero (0), then you have corrected the problem.

(c) If the fault shows on subsequent flights, then continue.

(3) Do the following leak checks in the Pack Bay:

(a) Do a soap bubble test of the applicable duct joints at the condition air check valves, TCV, Standby Temperature Control Valve (STCV), TAVs, joints at the condenser, reheater, water extractor and applicable sense lines.

(b) If a leak is found, repair the leak.

1) Do the Repair Confirmation at the end of this task.

(c) If a leak is not found, then continue

(4) Replace the L (R) TCV, V139 (V141). These are the tasks:

- Temperature Control Valve Removal, AMM TASK 21-51-10-000-801
- Temperature Control Valve Installation, AMM TASK 21-51-10-400-801

(a) Do the Repair Confirmation at the end of this task.

(5) Replace the applicable L (R) Pack Flow Sensor, M2953 (M2954). These are the tasks:

- Pack Flow Sensor Removal, AMM TASK 21-51-43-000-801
- Pack Flow Sensor Installation, AMM TASK 21-51-43-400-801

(a) Do the Repair Confirmation at the end of this task.

(6) Replace the applicable L (R) Flow Control and Shutoff Valve, V184 (V185). These are the tasks:

- Left Flow Control and Shutoff Valve Removal, AMM TASK 21-51-01-000-806-003
- Left Flow Control and Shutoff Valve Installation, AMM TASK 21-51-01-400-807-003
- Right Flow Control and Shutoff Valve Removal, AMM TASK 21-51-01-000-807-003
- Right Flow Control and Shutoff Valve Installation, AMM TASK 21-51-01-400-806-003

(a) Do the Repair Confirmation at the end of this task.

(7) Do this check of the wiring (WDM 21-51-12, WDM 21-51-22):

(a) Remove the L (R) PFTC, M2951 (M2952). This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

(b) Disconnect connector D15822 (D15824) from the L (R) Pack Flow Sensor, M2953 (M2954).

(c) Do a wiring check as follows:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 820-826; AIRPLANES WITH PFTC P/N 51090248-002/-003 (Continued)

LEFT AIR CONDITIONING

LEFT PACK FLOW TEMP CONTROLLER, M2951 D15814B	**LEFT PACK FLOW SENSOR, M2953 D15822**
pin A11	pin 7
pin B10	pin 5
pin A12	pin 9
pin B11	pin 11
pin B12	pin 8
pin D1	pin 2
pin D2	pin 4

LEFT PACK FLOW SENSOR, M2953 D15822	
pin 12	GND

RIGHT AIR CONDITIONING

RIGHT PACK FLOW TEMP CONTROLLER, M2952 D15816B	**RIGHT PACK FLOW SENSOR, M2954 D15824**
pin A11	pin 7
pin B10	pin 5
pin A12	pin 9
pin B11	pin 11
pin B12	pin 8
pin D1	pin 2
pin D2	pin 4

RIGHT PACK FLOW SENSOR, M2954 D15824	
pin 12	GND

1) If you find a problem with the wiring, then do these steps:
   a) Repair the wiring.
   b) Re-connect connector D15822 (D15824) to the L (R) Pack Flow Sensor, M2953 (M2954).
   c) Re-install the L (R) PFTC, M2951 (M2952). This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 820-826; AIRPLANES WITH PFTC P/N 51090248-002/-003 (Continued)

d) Do the Repair Confirmation at the end of this task.

OMR 820-826; AIRPLANES WITH PFTC P/N 51090248-005

D. Fault Isolation Procedure

(1) Do the following steps on the PFTC:

NOTE: If one side of the bleed duct pressure gauge fluctuates during cabin rate of climb indicator oscillation, do the fault isolation on that side.

NOTE: If you do not know which side has the oscillations, do the fault isolation on both sides.

(a) Do the FAULT HISTORY review on the affected PFTC (M2951 or M2952) for the last 10 or more flight legs, this is the task: Pack Flow Temperature Controller BITE Test, AMM TASK 21-61-00-740-801-002.

1) If a fault is found, do the corrective action for the fault.

2) If the FAULT HISTORY is not active, continue.

(b) Do the GROUND TEST BITE procedure on the affected PFTC (M2951 or M2952), this is the task: Pack Flow Temperature Controller BITE Test, AMM TASK 21-61-00-740-801-002.

1) Do not bypass the flow sensor drift check.

2) If a fault is found, do the corrective action for the fault.

3) If a fault is not found, continue.

(2) Do the following leak checks in the Pack Bay:

(a) Do a soap bubble test of the applicable duct joints at the condition air check valves, TCV, STCV, TAVs, joints at the condenser, reheater, water extractor and applicable sense lines.

(b) If a leak is found, repair the leak.

1) Do the Repair Confirmation at the end of this task.

(c) If a leak is not found, then continue.

(3) Do a water spray nozzle check. This is the task: Water Spray Nozzle Check, AMM TASK 21-51-16-200-801.

(a) Do the Repair Confirmation at the end of this task.

(4) Clean the Heat Exchangers. This is the task: Heat Exchanger - Blowing Out Debris/Blockages with Nitrogen Gas/Compressed Air, AMM TASK 21-51-03-160-811.

(a) Do the Repair Confirmation at the end of this task.

(5) Replace Pack Temperature Sensor. These are the tasks:

- Pack Temperature Sensor Removal, AMM TASK 21-51-52-000-801
- Pack Temperature Sensor Installation, AMM TASK 21-51-52-400-801

(a) Do the Repair Confirmation at the end of this task.

(6) If the problem continues, and if the L (R) Pack Flow Sensor, M2953 (M2954) was not replaced at a previous step, replace the L (R) Pack Flow Sensor, M2953 (M2954). These are the tasks:

- Pack Flow Sensor Removal, AMM TASK 21-51-43-000-801
- Pack Flow Sensor Removal, AMM TASK 21-51-43-000-801

(a) Do the Repair Confirmation at the end of this task.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 820-826; AIRPLANES WITH PFTC P/N 51090248-005 (Continued)

(7) Replace the L (R) Flow Control Shutoff Valve, V184 (V185). These are the tasks:

- Left Flow Control and Shutoff Valve Removal, AMM TASK 21-51-01-000-806-003
- Left Flow Control and Shutoff Valve Installation, AMM TASK 21-51-01-400-807-003
- Right Flow Control and Shutoff Valve Removal, AMM TASK 21-51-01-000-807-003
- Right Flow Control and Shutoff Valve Installation, AMM TASK 21-51-01-400-806-003

(a) Do the Repair Confirmation at the end of this task.

(8) Do the bleed air system health check. This is the task: Engine Bleed Air System Health Check, AMM TASK 36-11-00-700-801.

(a) If you find a problem or defective component, repair the problem or replace the defective component.

(b) Do the Repair Confirmation at the end of this task.

(9) Do this check of the wiring (WDM 21-51-12, WDM 21-51-22):

(a) Remove the L (R) PFTC, M2951 (M2952). This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

(b) Disconnect connector D15822 (D15824) from the L (R) Pack Flow Sensor, M2953 (M2954).

(c) Do a wiring check as follows:

LEFT AIR CONDITIONING

LEFT PACK FLOW TEMP CONTROLLER, M2951 D15814B	**LEFT PACK FLOW SENSOR, M2953 D15822**
pin A11	pin 7
pin B10	pin 5
pin A12	pin 9
pin B11	pin 11
pin B12	pin 8
pin D1	pin 2
pin D2	pin 4

LEFT PACK FLOW SENSOR, M2953 D15822	
pin 12	GND

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 820-826; AIRPLANES WITH PFTC P/N 51090248-005 (Continued)

RIGHT AIR CONDITIONING

RIGHT PACK FLOW TEMP CONTROLLER, M2952 D15816B	**RIGHT PACK FLOW SENSOR, M2954 D15824**
pin A11	pin 7
pin B10	pin 5
pin A12	pin 9
pin B11	pin 11
pin B12	pin 8
pin D1	pin 2
pin D2	pin 4

RIGHT PACK FLOW SENSOR, M2954 D15824	
pin 12	GND

1) If you find a problem with the wiring, then do these steps:
   a) Repair the wiring.
   b) Re-connect connector D15822 (D15824) to the L (R) Pack Flow Sensor, M2953 (M2954).
   c) Re-install the L (R) PFTC, M2951 (M2952). This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.
   d) Do the Repair Confirmation at the end of this task.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999; AIRPLANES WITH PZTC P/N 51090160-1 OR 622814-5.

E. Fault Isolation Procedure

(1) Perform the PZTC Built-In-Test Equipment (BITE) test, this is the task: Pack/Zone Temperature Controller BITE Test, AMM TASK 21-61-00-700-806-002.
   (a) If a fault is found, do the corrective action for the fault.
   (b) If a fault is not found, continue.

(2) Do a check of the Cabin Rate of Climb Indicator.
   (a) If the Cabin Rate of Climb Indicator Needle shows a value other than Zero (0) when the airplane is on the ground, then do these steps:
      1) Make sure that one of the Entry Doors has been open for more than 5 minutes so that the cabin is not pressurized.
      2) On the lower left corner of the gauge, turn the Zero Adjustment Screw and make sure that the pointer can be adjusted to a minimum of 400 ft/min on both sides of the Zero graduation.
      3) Reset the pointer to Zero.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999; AIRPLANES WITH PZTC P/N 51090160-1 OR 622814-5. (Continued)

4) If the problem continues, replace the Cabin Rate of Climb Indicator, N55. These are the tasks:
   - Cabin Rate of Climb Indicator Removal, AMM TASK 21-33-01-000-801
   - Cabin Rate of Climb Indicator Installation, AMM TASK 21-33-01-400-801

(b) If the Cabin Rate of Climb Indicator Needle shows zero (0), then you have corrected the problem.

(c) If the fault shows on subsequent flights, then continue.

(3) Do the bleed air system health check. This is the task: Engine Bleed Air System Health Check, AMM TASK 36-11-00-700-801.

(a) If you find a problem or defective component, repair the problem or replace the defective component.

(b) Do the Repair Confirmation at the end of this task.

(4) Do the following leak checks in the Pack Bay:

(a) Do a soap bubble test of the applicable duct joints at the condition air check valves, TCV, STCV, TAVs, joints at the condenser, reheater, water extractor and applicable sense lines.

(b) If a leak is found, repair the leak.

1) Do the Repair Confirmation at the end of this task.

(c) If a leak is not found, then continue.

(5) Replace the L (R) Flow Control Shutoff Valve, V18 (V19). These are the tasks:
- Left Flow Control and Shutoff Valve - Removal, AMM TASK 21-51-01-000-802-002
- Left Flow Control and Shutoff Valve - Installation, AMM TASK 21-51-01-400-805-002
- Right Flow Control and Shutoff Valve - Removal, AMM TASK 21-51-01-000-805-002
- Right Flow Control and Shutoff Valve - Installation, AMM TASK 21-51-01-400-802-002

(a) Do the Repair Confirmation at the end of this task.

(6) Replace the L (R) TCV, V139 (V141). These are the tasks:
- Temperature Control Valve Removal, AMM TASK 21-51-10-000-801
- Temperature Control Valve Installation, AMM TASK 21-51-10-400-801

(a) Do the Repair Confirmation at the end of this task.

(7) Replace the zone TAVs (V143, V144, and V145). These are the tasks:
- Zone Trim Air Modulating Valve Removal, AMM TASK 21-61-21-000-801
- Zone Trim Air Modulating Valve Installation, AMM TASK 21-61-21-400-801

(a) Do the Repair Confirmation at the end of this task.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999 (Continued)

F. Repair Confirmation

(1) If the fault does not show on subsequent flights, then you have corrected the problem.

(2) If the fault shows on subsequent flights, then continue the Fault Isolation Procedure at the subsequent step.

OMR ALL

——— END OF TASK ———

803. Cabin Altitude and Differential Pressure Indicator Indicates High Cabin Pressure Differential and Low Cabin Altitude - Fault Isolation

A. Description

(1) The gauges do not indicate as normally expected. For example, see below:

(a) The cabin altitude gauge indicates 6000 feet while at cruise altitude of FL 370.

NOTE: Expected cabin altitude at cruise would be about 8000 feet.

(b) The cabin differential pressure gauge indicates in the amber arc and does not decrease as the cabin altitude decreases.

(c) Both gauges do not indicate as expected for the altitude.

B. Possible Causes

(1) Cabin Altitude and Differential Pressure Indicator

C. Initial Evaluation

(1) Open one the main entry doors to make sure the airplane is not pressurized.

(2) Do this check of the cabin altimeter gauge on the P5-16 cabin pressure control panel:

(a) Set the barometric pressure for the pilot's altimeter to 29.92 inches of Hg.

(b) Observe the altitude on the pilot's altimeter.

(c) Compare the altitude on the pilot's altimeter to the altitude on the cabin altimeter.

(d) If the cabin altitude is within 1000 feet of the pilot's altimeter, the cabin altimeter is satisfactory.

(e) If the cabin altitude is not within 1000 feet of the pilot's altimeter, go to the Fault Isolation procedure.

(3) Do this check of the cabin differential pressure gauge on the P5-16 cabin pressure control panel:

(a) Observe the differential pressure on the DIFF PRESS gauge.

(b) If the differential pressure is 0 (+/- 0.2) psid, the indicator is satisfactory.

(c) If the differential pressure is greater than 0 (+/- 0.2) psid, go to the Fault Isolation procedure..

D. Fault Isolation Procedure

(1) If the cabin altitude is not within 1000 feet of the pilot's altimeter, replace the cabin altitude and differential pressure indicator. To replace the indicator, do these tasks: Cabin Altitude and Differential Pressure Indicator Removal, AMM TASK 21-33-02-000-801 and Cabin Altitude and Differential Pressure Indicator Installation, AMM TASK 21-33-02-400-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(2) If the differential pressure is greater than 0 (+/- 0.2) psid, replace the cabin altitude and differential pressure indicator. To replace the indicator, do these tasks: Cabin Altitude and Differential Pressure Indicator Removal, AMM TASK 21-33-02-000-801 and Cabin Altitude and Differential Pressure Indicator Installation, AMM TASK 21-33-02-400-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

801. Door Area Heater Does Not Operate - Fault Isolation

A. Description

(1) One of the door area heaters does not operate as expected.

(2) (SDS SUBJECT 21-45-00)

B. Possible Causes

(1) Door area heater, M1708 (forward) or M1681 (aft)

OMR 901-999

(2) Door area heater, M2637 (mid)

(3) Mid door heater blankets:

(a) L mid door surround, M2633 or M2634

(b) L mid door lining, M2629 or M2630

(c) R mid door surround, M2635 or M2636

(d) R mid door lining, M2629 or M2630

OMR ALL

(4) Door area heater power relay, R560

OMR 901-999

(5) Door area heater power relay, R982

OMR ALL

(6) Wiring

(7) Sys 1 air/gnd relay, R593

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
E	11	C01274	AIR CONDITIONING DOOR AREA HEAT CONT

F/O Electrical System Panel, P6-12

Row	Col	Number	Name
OMR 901-999			
D	5	C01675	MID EXIT DOOR AREA HEATER

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
OMR 701-799, 802			
A	14	C01279	DOOR AREA HTR-AFT
OMR 807			
A	14	C01279	DOOR AREA HTR-AFT (INOP)
OMR 701-799, 802			
A	16	C01280	DOOR AREA HTR-FWD
OMR 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
D	7	C01279	DOOR AREA HTR-AFT

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OMR 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999 (Continued)

(Continued)

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
D	9	C01280	DOOR AREA HTR-FWD

OMR ALL

D. Related Data

(1) (SSM 21-42-11)

(2) (WDM 21-42-11)

OMR 701-799, 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826

E. Initial Evaluation

(1) AIRPLANES WITH CAB/UTIL SWITCH ON THE P5-13 PANEL;

Make sure that the CAB/UTIL switch on the P5-13 panel is set to the ON position.

(2) Do an operational test of the two door area heaters. To do this, do this task: Door Area Heater Operational Test, AMM TASK 21-45-00-700-802.

(a) If the operational test is not satisfactory for one of the door area heaters, then do the Fault Isolation Procedure - One Door Area Heater Does Not Operate.

(b) If the operational test is not satisfactory for the two door area heaters, then do the Fault Isolation Procedure - Both Door Area Heaters Do Not Operate.

(c) If the operational test is satisfactory, then there was an intermittent fault.

OMR 901-999

F. Initial Evaluation

(1) AIRPLANES WITH CAB/UTIL SWITCH ON THE P5-13 PANEL;

Make sure that the CAB/UTIL switch on the P5-13 panel is set to the ON position.

(2) Do an operational test of the door area heaters. To do this, do this task: Door Area Heater Operational Test, AMM TASK 21-45-00-700-802.

(a) If the operational test is not satisfactory for one of the Fwd or Aft door area heaters, then do the Fault Isolation Procedure - One Door Area Heater Does Not Operate.

(b) If the operational test is not satisfactory for the both of the Fwd and Aft door area heaters, then do the Fault Isolation Procedure - Both Door Area Heaters Do Not Operate.

(c) If the operational test is not satisfactory for the Mid door area heater or Mid door heater blankets, then do the Fault Isolation Procedure - Mid Door Area Heaters Do Not Operate.

(d) If the operational test is satisfactory, then there was an intermittent fault.

OMR ALL

G. Fault Isolation Procedure - One Door Area Heater Does Not Operate

NOTE: You must do the steps in the Initial Evaluation before you do these steps.

NOTE: This procedure isolates the fault to one of these possible causes:

- forward or aft door area heater
- door area heater power relay

ECCN 9E991 BOEING PROPRIETARY - See title page for details

- wiring

(1) Replace the applicable door area heater, M1708 (forward) or M1681 (aft).
   (a) Do the Repair Confirmation at the end of this task.
   (b) If the Repair Confirmation is not satisfactory, then continue.

(2) Replace the door area heater power relay, R560:
   (a) Do the Repair Confirmation at the end of this task.
   (b) If the Repair Confirmation is not satisfactory, then continue.

(3) Do these checks of the wiring:
   (a) Make sure conditioned air is supplied through one of the packs. To do this, do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.
   (b) Make sure the airplane is in air mode. To do this, do this task: Put the Airplane in the Air Mode, AMM TASK 32-09-00-860-801.
   (c) FOR A FAULT WITH THE FORWARD DOOR AREA HEATER, M1708, do these steps:
      1) Remove the door area heater power relay, R560.
      2) Do a check for 200 VAC (phase-to-phase) on connector D10882 from pin A2 to pin B2.
      3) Do a continuity check between these pins of connector D10882 in the P91 panel and these pins of connector D10884 on the forward door area heater:

D10882		D10884
pin A1	. . . . .	pin 1
pin B1	. . . . .	pin 3

   (d) FOR A FAULT WITH THE AFT DOOR AREA HEATER, M1681, do these steps:
      1) Remove the door area heater power relay, R560.
      2) Do a check for 200 VAC (phase-to-phase) on connector D10882 from pin C2 to pin D2.
      3) Do a continuity check between these pins of connector D10882 in the P91 panel and these pins of connector D10770 on the aft door area heater:

D10882		D10770
pin C1	. . . . .	pin 2
pin C1	. . . . .	pin 3
pin D1	. . . . .	pin 1

   (e) Repair the wiring.
   (f) Re-install the door area heater power relay, R560.
   (g) Re-connect connector D10884 or D10770.
   (h) Do the Repair Confirmation at the end of this task.

H. Fault Isolation Procedure - Both Door Area Heaters Do Not Operate

NOTE: You must do the steps in the Initial Evaluation before you do these steps.

NOTE: This procedure isolates the fault to one of these possible causes:

- door area heater power relay
- wiring

ECCN 9E991 BOEING PROPRIETARY - See title page for details

- air/gnd relay

(1) Replace the door area heater power relay, R560.

(a) Do the Repair Confirmation at the end of this task.

(b) If the Repair Confirmation is not satisfactory, then continue.

(2) Do these checks of the wiring and the air/gnd relay, R593:

(a) Make sure conditioned air is supplied through one of the packs. To do this, do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(b) Make sure the airplane is in air mode. To do this, do this task: Put the Airplane in the Air Mode, AMM TASK 32-09-00-860-801.

(c) Remove the door area heater power relay, R560.

(d) Do a check for 28 VDC on connector D10882 from pin X1 to structure ground.

(e) If there is not 28 VDC from pin X1 on connector D10882 to structure ground, then do these steps:

1) Repair the wiring from pin X1 on connector D10882 to circuit breaker C1274, AIR CONDITIONING DOOR AREA HEAT CONTROL.

2) Re-install the door area heater power relay, R560.

3) Do the Repair Confirmation at the end of this task.

(f) If there is 28 VDC from pin X1 on connector D10882 to structure ground, then continue.

(g) Do a continuity check from pin X2 on connector D10882 to structure ground.

(h) If there is not continuity from pin X2 on connector D10882 to structure ground, then do these steps:

1) Replace SYS 1 AIR/GND RELAY, R593.

2) Re-install the door area heater power relay, R560.

3) Do the Repair Confirmation at the end of this task.

4) If the Repair Confirmation is not satisfactory, then repair the wiring from pin X2 on connector D10882 to terminal block, TB22Y.

(i) Re-install the door area heater power relay, R560.

(j) Do the Repair Confirmation at the end of this task.

OMR 901-999

I. Fault Isolation Procedure - Mid Door Area Heater Does Not Operate

NOTE: You must do the steps in the Initial Evaluation before you do these steps.

NOTE: This procedure isolates the fault to one of these possible causes:

- mid door area heater
- mid door heater blankets
- door area heater power relay
- wiring

(1) If the mid door area heater does not operate correctly, but the mid door heater blankets operate normally. Replace the door area heater, M2637.

(a) Do the Repair Confirmation at the end of this task.

(b) If the Repair Confirmation is not satisfactory, then continue.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 901-999 (Continued)

(2) If the mid door area heater and mid door heater blankets do not operate correctly. Replace the door area heater power relay, R982:

(a) Do the Repair Confirmation at the end of this task.

(b) If the Repair Confirmation is not satisfactory, then continue.

(3) If one of the mid door heater blankets does not operate correctly. Replace the applicable door heater blanket:

(a) Do the Repair Confirmation at the end of this task.

(b) If the Repair Confirmation is not satisfactory, then continue.

(4) Do these checks of the wiring:

(a) Make sure conditioned air is supplied through one of the packs. To do this, do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(b) Make sure the airplane is in air mode. To do this, do this task: Put the Airplane in the Air Mode, AMM TASK 32-09-00-860-801.

(c) FOR A FAULT WITH THE MID DOOR AREA HEATER, M2637, do these steps:

1) Remove the door area heater power relay, R982.

2) Do a check for 200 VAC (phase-to-phase) on connector D14064 from pin A2 to pin B2.

3) Do a continuity check between these pins of connector D14064 in the J39 panel and these pins of connector D14098 on the mid door area heater:

D14064		**14098**
pin A1	. . . . .	pin 2
pin A1	. . . . .	pin 3
pin B1	. . . . .	pin 1

(d) FOR A FAULT WITH THE MID DOOR HEATER BLANKET, M2629, do these steps:

1) Remove the door area heater power relay, R982.

2) Do a check for 200 VAC (phase-to-phase) on connector D14064 from pin A2 to pin B2.

3) Do a continuity check between these pins of connector D14064 in the J39 panel and these pins of connector D14066 on the L or R mid door heater blanket:

D14064		**D14066**
pin A1	. . . . .	pin 1

(e) FOR A FAULT WITH THE MID DOOR HEATER BLANKET, M2630, do these steps:

1) Remove the door area heater power relay, R982.

2) Do a check for 200 VAC (phase-to-phase) on connector D14064 from pin A2 to pin B2.

3) Do a continuity check between these pins of connector D14064 in the J39 panel and these pins of connector D14070 on the L or R mid door heater blanket:

D14064		**D14070**
pin A1	. . . . .	pin 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 901-999 (Continued)

(f) FOR A FAULT WITH THE MID DOOR HEATER BLANKET, M2633, do these steps:

1) Remove the door area heater power relay, R982.
2) Do a check for 200 VAC (phase-to-phase) on connector D14064 from pin A2 to pin B2.
3) Do a continuity check between these pins of connector D14064 in the J39 panel and these pins of connector D14082 on the mid door heater blanket:

D14064		D14082
pin A1	. . . . .	pin 1

(g) FOR A FAULT WITH THE MID DOOR HEATER BLANKET, M2634, do these steps:

1) Remove the door area heater power relay, R982.
2) Do a check for 200 VAC (phase-to-phase) on connector D14064 from pin A2 to pin B2.
3) Do a continuity check between these pins of connector D14064 in the J39 panel and these pins of connector D14086 on the mid door heater blanket:

D14064		D14086
pin A1	. . . . .	pin 1

(h) FOR A FAULT WITH THE MID DOOR HEATER BLANKET, M2635, do these steps:

1) Remove the door area heater power relay, R982.
2) Do a check for 200 VAC (phase-to-phase) on connector D14064 from pin A2 to pin B2.
3) Do a continuity check between these pins of connector D14064 in the J39 panel and these pins of connector D14090 on the mid door heater blanket:

D14064		D14090
pin B1	. . . . .	pin 1

(i) FOR A FAULT WITH THE MID DOOR HEATER BLANKET, M2636, do these steps:

1) Remove the door area heater power relay, R982.
2) Do a check for 200 VAC (phase-to-phase) on connector D14064 from pin A2 to pin B2.
3) Do a continuity check between these pins of connector D14064 in the J39 panel and these pins of connector D14094 on the mid door heater blanket:

D14064		D14094
pin B1	. . . . .	pin 1

(j) Repair the wiring.

(k) Re-install the door area heater power relay, R982.

(l) Re-connect the applicable connectors.

(m) Do the Repair Confirmation at the end of this task.

OMR ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

J. Repair Confirmation

(1) Do an operational test of the two door area heaters. To do this, do this task: Door Area Heater Operational Test, AMM TASK 21-45-00-700-802.

(a) If the operational test is satisfactory, then you corrected the fault.

——— END OF TASK ———

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

802. Emergency Exit Door Area Too Cold - Fault Isolation

A. Description

(1) The area around the emergency exit door (EED) is too cold when the airplane is in flight.

NOTE: The emergency exit door will be referred to as the EED during this procedure.

(2) (SDS SUBJECT 21-45-00)

B. Possible Causes

OMR 701-799

(1) Left Overwing Door Lining Heater, M2176 (FWD) or M2204 (AFT)

(2) Right Overwing Door Lining Heater, M2174 (FWD) or M2201 (AFT)

(3) Left Overwing Door Trim Heater, M2205 (FWD) or M2206 (AFT)

(4) Right Overwing Door Trim Heater, M2202 (FWD) or M2203 (AFT)

(5) Left Closeout Panel Heater, M2177

(6) Right Closeout Panel Heater, M2175

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(7) Left AFT Overwing Door Lining Heater, M2176 (FWD) or M2204 (AFT)

(8) Right AFT Overwing Door Lining Heater, M2174 (FWD) or M2201 (AFT)

(9) Left AFT Overwing Door Trim Heater, M2205 (FWD) or M2206 (AFT)

(10) Right AFT Overwing Door Trim Heater, M2202 (FWD) or M2203 (AFT)

(11) Left FWD Overwing Door Lining Heater, M2182 (FWD) or M2213 (AFT)

(12) Right FWD Overwing Door Lining Heater, M2178 (FWD) or M2207 (AFT)

(13) Left FWD Overwing Door Trim Heater, M2214 (FWD) or M2215 (AFT)

(14) Right FWD Overwing Door Trim Heater, M2208 (FWD) or M2209 (AFT)

(15) Left FWD Closeout Panel Heater, M2183

(16) Right FWD Closeout Panel Heater, M2179

(17) Left AFT Closeout Panel Heater, M2177

(18) Right AFT Closeout Panel Heater, M2175

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(19) Door Area Heater Power Relay, R560

(20) SYS 1 AIR/GND Relay, R593

(21) Wiring

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
E	11	C01274	AIR CONDITIONING DOOR AREA HEAT CONT

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
OMR 701-799, 802			
A	14	C01279	DOOR AREA HTR-AFT
A	16	C01280	DOOR AREA HTR-FWD
OMR 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
D	7	C01279	DOOR AREA HTR-AFT
D	9	C01280	DOOR AREA HTR-FWD

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

D. Related Data

(1) (SSM 21-42-11)

(2) (WDM 21-42-11)

E. Initial Evaluation

(1) Do this task: Door Area Heater Operational Test, AMM TASK 21-45-00-700-802.

(a) If the forward door area heater does not operate, then it must be repaired before you do this task, do this task: Door Area Heater Does Not Operate - Fault Isolation, 21-45 TASK 801.

NOTE: The emergency exit door heaters use the same source of power as the forward door area heater. If there is a problem with the wire bundle for the forward door area heater it can be the source of problems with the EED heaters.

(b) If the two door area heaters operate, then you must do a test of each EED heater and its wire bundle. There are five heaters installed for each door. To test the EED heaters, do this task: Emergency Exit Hatch Heaters Operational Test, AMM TASK 21-45-00-700-801. If one of the heaters do not operate, then do the Fault Isolation Procedure - Emergency Exit Door Heater Does Not Operate below. This procedure isolates the fault to one of these possible causes:

1) Individual Heaters of the Emergency Exit Doors
2) Wiring

(c) If the two door area heaters do not operate, then do the Fault Isolation Procedure - All Emergency Exit Door Area Heaters Do Not Operate below. This procedure isolates the fault to one of these possible causes:

1) Door Area Heater Power Relay, R560
2) SYS 1 AIR/GND Relay, R593
3) Wiring

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799

F. Fault Isolation Procedure - Emergency Exit Door Heater Does Not Operate

NOTE: You must do the steps in the Initial Evaluation before you do these steps.

(1) If one of the EED Trim Heaters (M2202, M2203, M2205 and M2206) did not operate, replace the applicable heater. These are the tasks:

- Emergency Exit Door Surround Panel Heater Blanket Removal, AMM TASK 21-45-02-000-802
- Emergency Exit Door Surround Panel Heater Blanket Installation, AMM TASK 21-45-02-400-802

(a) Do the Repair Confirmation at the end of this task.

(2) If one of the EED Lining Heaters (M2176, M2204, M2174 and M2201) did not operate, replace the applicable heater. These are the tasks:

- Emergency Exit Door Heater Blanket Removal, AMM TASK 21-45-02-000-801
- Emergency Exit Door Heater Blanket Installation, AMM TASK 21-45-02-400-801

(a) Do the Repair Confirmation at the end of this task.

(3) If one of the EED Closeout Panel Heaters (M2175 and M2177) did not operate, replace the applicable heater. These are the tasks:

- Emergency Exit Door Closeout Panel Heater Blanket Removal, AMM TASK 21-45-02-000-803
- Emergency Exit Door Closeout Panel Heater Blanket Installation, AMM TASK 21-45-02-400-803

(a) Do the Repair Confirmation at the end of this task.

(4) Do this check of the wiring for the applicable EED heater:

(a) Do the steps in these tasks to get access to the applicable heater. These are the tasks:

- Emergency Exit Door Heater Blanket Removal, AMM TASK 21-45-02-000-801
- Emergency Exit Door Surround Panel Heater Blanket Removal, AMM TASK 21-45-02-000-802
- Emergency Exit Door Closeout Panel Heater Blanket Removal, AMM TASK 21-45-02-000-803

(b) Do this check of the wiring for the applicable EED heater:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799 (Continued)

	HEATER	
	DOOR AREA HEATER POWER RELAY, R560	HEATER CONNECTOR
L OVERWING DOOR TRIM AFT HEATER, M2206	D10882	D12672
	pin A1	pin 1
L OVERWING DOOR TRIM FWD HEATER, M2205	D10882	D12668
	pin A1	pin 1
L OVERWING DOOR LINING AFT HEATER, M2204	D10882	D12664
	pin A1	pin 1
L OVERWING DOOR LINING FWD HEATER, M2176	D10882	D12552
	pin A1	pin 1
L CLOSEOUT PANEL HEATER, M2177	D10882	WHT/BLU
	pin A1	pin 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799 (Continued)

	HEATER	
	DOOR AREA HEATER POWER RELAY, R560	HEATER CONNECTOR
R OVERWING DOOR TRIM AFT HEATER, M2203	D10882	D12660
	pin B1	pin 1
R OVERWING DOOR TRIM FWD HEATER, M2202	D10882	D12656
	pin B1	pin 1
R OVERWING DOOR LINING AFT HEATER, M2201	D10882	D12652
	pin B1	pin 1
R OVERWING DOOR LINING FWD HEATER, M2174	D10882	D12544
	pin B1	pin 1
R CLOSEOUT PANEL HEATER, M2175	D10882	WHT/BLU
	pin B1	pin 1

1) Make sure that the other heater connector pin has continuity to ground.
2) If you find a problem, then repair the wiring.
3) Do the applicable steps in these tasks to complete installation of the heater. These are the tasks:
   - Emergency Exit Door Heater Blanket Installation, AMM TASK 21-45-02-400-801
   - Emergency Exit Door Surround Panel Heater Blanket Installation, AMM TASK 21-45-02-400-802
   - Emergency Exit Door Closeout Panel Heater Blanket Installation, AMM TASK 21-45-02-400-803
4) Do the Repair Confirmation at the end of this task.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

G. Fault Isolation Procedure - Emergency Exit Door Heater Does Not Operate

NOTE: You must do the steps in the Initial Evaluation before you do these steps.

(1) If one of the EED Trim Heaters (M2202, M2203, M2205, M2206, M2208, M2209, M2214, M2215) did not operate, replace the applicable heater. These are the tasks:

- Emergency Exit Door Surround Panel Heater Blanket Removal, AMM TASK 21-45-02-000-802
- Emergency Exit Door Surround Panel Heater Blanket Installation, AMM TASK 21-45-02-400-802

(a) Do the Repair Confirmation at the end of this task.

(2) If one of the EED Lining Heaters (M2201, M2204, M2207, M2213, M2174, M2176, M2178, M2182) did not operate, replace the applicable heater. These are the tasks:

- Emergency Exit Doorway Lining Removal, AMM TASK 25-21-20-000-801
- Emergency Exit Doorway Lining Installation, AMM TASK 25-21-20-400-801

(a) Do the Repair Confirmation at the end of this task.

(3) If one of the EED Closeout Panel Heaters (M2177, M2179, M2175, M2183) did not operate, replace the applicable heater. These are the tasks:

- Emergency Exit Door Closeout Panel Heater Blanket Removal, AMM TASK 21-45-02-000-803
- Emergency Exit Door Closeout Panel Heater Blanket Installation, AMM TASK 21-45-02-400-803

(a) Do the Repair Confirmation at the end of this task.

(4) Do this check of the wiring for the applicable EED heater:

(a) Do the steps in these tasks to get access to the applicable heater. These are the tasks:

- Emergency Exit Door Heater Blanket Removal, AMM TASK 21-45-02-000-801
- Emergency Exit Door Surround Panel Heater Blanket Removal, AMM TASK 21-45-02-000-802
- Emergency Exit Door Closeout Panel Heater Blanket Removal, AMM TASK 21-45-02-000-803

(b) Do this check of the wiring for the applicable EED heater:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

	HEATER	
	DOOR AREA HEATER POWER RELAY, R560	HEATER CONNECTOR
L FWD OVERWING DOOR TRIM AFT HEATER, M2215	D10882	D12708
	pin A1	pin 1
L FWD OVERWING DOOR TRIM FWD HEATER, M2214	D10882	D12704
	pin A1	pin 1
L OVERWING DOOR LINING AFT HEATER, M2213	D10882	D12700
	pin A1	pin 1
L OVERWING DOOR LINING FWD, M2182	D10882	D12576
	pin A1	pin 1
L FWD CLOSEOUT PANEL HEATER, M2183	D10882	WHT/BLU
	pin A1	pin 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

	HEATER	
	DOOR AREA HEATER POWER RELAY, R560	HEATER CONNECTOR
L AFT OVERWING DOOR TRIM AFT HEATER, M2206	D10882 pin A1	D12672 pin 1
L AFT OVERWING DOOR TRIM FWD HEATER, M2205	D10882 pin A1	D12668 pin 1
L AFT OVERWING DOOR LINING AFT HEATER, M2204	D10882 pin A1	D12664 pin 1
L AFT OVERWING DOOR LINING FWD HEATER, M2176	D10882 pin A1	D12552 pin 1
L AFT CLOSEOUT PANEL HEATER, M2177	D10882 pin A1	WHT/BLU pin 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

	HEATER	
	DOOR AREA HEATER POWER RELAY, R560	HEATER CONNECTOR
R FWD OVERWING DOOR TRIM AFT HEATER, M2209	D10882 pin B1	D12684 pin 1
R FWD OVERWING DOOR TRIM FWD HEATER, M2208	D10882 pin B1	D12680 pin 1
R FWD OVERWING DOOR LINING AFT HEATER, M2207	D10882 pin B1	D12676 pin 1
R FWD OVERWING DOOR LINING FWD HEATER, M2178	D10882 pin B1	D12560 pin 1
R FWD CLOSEOUT PANEL HEATER, M2179	D10882 pin B1	WHT/BLU pin 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

	HEATER	
	DOOR AREA HEATER POWER RELAY, R560	HEATER CONNECTOR
R AFT OVERWING DOOR TRIM AFT HEATER, M2203	D10882 pin B1	D12660 pin 1
R AFT OVERWING DOOR TRIM FWD HEATER, M2202	D10882 pin B1	D12656 pin 1
R AFT OVERWING DOOR LINING AFT HEATER, M2201	D10882 pin B1	D12652 pin 1
R AFT OVERWING DOOR LINING FWD HEATER, M2174	D10882 pin B1	D12544 pin 1
R AFT CLOSEOUT PANEL HEATER, M2175	D10882 pin B1	WHT/BLU pin 1

1) Make sure that the other heater connector pin has continuity to ground.
2) If you find a problem, then repair the wiring.
3) Do the applicable steps in these tasks to complete installation of the heater. These are the tasks:
   - Emergency Exit Door Heater Blanket Installation, AMM TASK 21-45-02-400-801
   - Emergency Exit Door Surround Panel Heater Blanket Installation, AMM TASK 21-45-02-400-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

- Emergency Exit Door Closeout Panel Heater Blanket Installation, AMM TASK 21-45-02-400-803

4) Do the Repair Confirmation at the end of this task.

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

H. Fault Isolation Procedure - All Emergency Exit Door Heaters Do Not Operate

NOTE: You must do the steps in the Initial Evaluation before you do these steps.

NOTE: This procedure isolates the fault to one of these possible causes:

- door area heater power relay
- wiring
- air/gnd relay

(1) Replace the Door Area Heater Power Relay, R560. The door area heater power relay is located on the P91 panel in the electrical and electronics compartment.

(a) Do the Repair Confirmation at the end of this task.

(2) Replace the SYS 1 AIR/GND RELAY, R593.

(a) Do the Repair Confirmation at the end of this task.

(3) Do this check of the wiring (WDM 21-42-11):

(a) Disconnect connector D10882 from the Door Area Heater Power Relay, R560.

(b) Do a continuity check between the load terminal of circuit breaker C1274 and SYS 1 AIR/GND Relay, R593 as follows:

AIR CONDITIONING DOOR AREA HEAT CONTROL, P18-3 C1274		**DOOR AREA HEATER POWER RELAY, R560 D10882**
Load terminal	. . . . .	pin X1

1) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-connect connector D10882 to the Door Area Heater Power Relay, R560.

c) Do the Repair Confirmation at the end of this task.

(4) Do this check of the wiring (WDM 21-42-11):

(a) Disconnect connector D11020 from the SYS 1 AIR/GND Relay, R593.

(b) Do a continuity check between the Door Area Heater Power Relay, R560 and SYS 1 AIR/GND Relay, R593 as follows:

DOOR AREA HEATER POWER RELAY, R560 D10882		**SYS 1 AIR/GND RELAY, R593 D11020**
pin X2	. . . . .	pin A2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) If you find a problem with the wiring, then do these steps:
   a) Repair the wiring.
   b) Re-connect connector D11020 to the SYS 1 AIR/GND Relay, R593.
   c) Do the Repair Confirmation at the end of this task.

(5) Do this check of the wiring (WDM 21-42-11):

(a) If there is an isolation diode in the wiring run to be checked, put the positive lead of the multimeter on the contact point closest to the anode side of the diode and the negative lead on the contact point closest to the cathode side of the diode.

NOTE: The cathode end of the diode is the end with the stripe on it.

(b) Disconnect connector D11020 from the SYS 1 AIR/GND Relay, R593.

(c) Do a continuity check between the SYS 1 AIR/GND Relay, R593 and Flow Control Valve, V18 (V19) as follows:

LEFT PACK

SYS 1 AIR/GND RELAY, R593 D11020		**FLOW CONTROL VALVE, V18 D486**
pin A1	. . . . .	pin 2

RIGHT PACK

SYS 1 AIR/GND RELAY, R593 D11020		**FLOW CONTROL VALVE, V19 D490**
pin A1	. . . . .	pin 2

1) If you find a problem with the wiring, then do these steps:
   a) Repair the wiring.
   b) Re-connect connector D11020 to the SYS 1 AIR/GND Relay, R593.
   c) Do the Repair Confirmation at the end of this task.

I. Repair Confirmation

(1) Do these tasks: .

- Door Area Heater Operational Test, AMM TASK 21-45-00-700-802
- Emergency Exit Hatch Heaters Operational Test, AMM TASK 21-45-00-700-801

(a) If the heaters operate, then you have corrected the fault.

(b) If the heaters do not operate, then continue at the next fault isolation step.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

801. Fog, High Humidity or Ice in Cabin - Fault Isolation

A. Description

(1) There is fog, unusually high humidity, or ice in the passenger cabin or the flight compartment. Conditioned air for the flight compartment is usually supplied by the left pack. Conditioned air for the passenger cabin is usually supplied by the right pack.

B. Possible Causes

OMR 701-799

(1) Water separator

(2) 35 degree F control valve, V27 (left) or V28 (right)

(3) 35 degree F sensor, T430 (left) or T431 (right)

(4) 35 degree F control module, M223 (left) or M224 (right).

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(5) Pack temperature control system

(6) Water separator

(7) Water spray nozzle.

OMR ALL

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

OMR 701-799

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	1	C00344	AIR CONDITIONING TEMP CONT 35 DEG F L
B	1	C00345	AIR CONDITIONING TEMP CONT 35 DEG F R

These are the primary circuit breakers related to the fault:

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	1	C01157	A/C TEMP CONT VALVE CLOSE LEFT
B	1	C01160	A/C TEMP CONT VALVE CLOSE RIGHT

OMR ALL

D. Related Data

OMR 701-799

(1) (SSM 21-51-16)

(2) (SSM 21-51-26)

(3) (WDM 21-51-16)

(4) (WDM 21-51-26).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(5) (SSM 21-51-12)

(6) (SSM 21-51-22)

(7) (WDM 21-51-12)

(8) (WDM 21-51-22).

OMR ALL

E. Fault Isolation Procedure

OMR 701-799

(1) Do this check of the 35 degree F sensor, control valve, and control module for the applicable pack:

NOTE: Conditioned air for the flight compartment is supplied by the left pack. Conditioned air for the passenger cabin is supplied by the right pack.

(a) Do this task: Low Limit (35 °F) Controller BITE Procedure, 21-52 TASK 801.

(b) If the BITE test shows a fault, then do the Fault Isolation Procedure for the applicable maintenance message.

1) If the problem does not occur on the subsequent flight, then you corrected the fault.

(c) If the BITE test does not show a fault, then continue.

(2) Do this check of the applicable water separator:

(a) For the coalescer bag on the applicable water separator, do this task: Coalescer Bag Pressure Indicator Inspection, AMM TASK 21-51-05-400-803.

(b) If the indicator disk on the pressure indicator assembly for the coalescer bag is in the red area, then do these steps:

1) Replace the coalescer bag.

These are the tasks:

Coalescer Bag Removal, AMM TASK 21-51-05-020-801,

Coalescer Bag Installation, AMM TASK 21-51-05-400-801.

2) If the problem does not occur on the subsequent flight, then you corrected the fault.

3) If the problem occurs on the subsequent flight, then replace the applicable water separator.

These are the tasks:

Water Separator Removal, AMM TASK 21-51-05-000-801,

Water Separator Installation, AMM TASK 21-51-05-400-802.

(c) If the indicator disc for the coalescer bag is in the white area, then do these steps:

1) Replace the applicable water separator.

These are the tasks:

Water Separator Removal, AMM TASK 21-51-05-000-801,

Water Separator Installation, AMM TASK 21-51-05-400-802.

2) If the problem does not occur on the subsequent flight, then you corrected the fault.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(3) Do the BITE test on the pack/zone temperature controller (M1442 left or M1443 right) for the applicable pack:

NOTE: Conditioned air for the flight compartment is supplied by the left pack. Conditioned air for the passenger cabin is supplied by the right pack.

(a) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(b) If the BITE test shows a fault, then do the Fault Isolation Procedure for the applicable maintenance message.

1) If the problem does not occur on the subsequent flight, then you corrected the fault.

(c) If the BITE test does not show a fault, then continue.

(4) Do this check of the water spray nozzle:

(a) Remove the applicable

Number	Name/Location
192BL	ECS Ram Air Inlet Mixing Duct Panel - Forward

or open this access panel:

Number	Name/Location
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward

to get access to the water spray nozzle.

(b) Remove the access panel on the ram air inlet duct (AMM TASK 21-51-03-160-807).

(c) Operate the applicable air conditioning pack. To operate the pack, do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(d) Make sure a spray of water flows from the water spray nozzle.

(e) If water does not flow from the water spray nozzle, do these steps:

1) Stop the air conditioning pack. To stop the pack, do this task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

2) Replace the water spray nozzle.

These are the tasks:

Water Spray Nozzle Removal, AMM TASK 21-51-16-000-801,

Water Spray Nozzle Installation, AMM TASK 21-51-16-400-801.

3) Operate the applicable air conditioning pack. To operate the pack, do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

4) Make sure a spray of water flows from the water spray nozzle.

5) If the problem does not occur on the subsequent flight, then you corrected the fault.

(f) If water does flow from the water spray nozzle, then continue:

1) Stop the air conditioning pack. To stop the pack, do this task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

2) Install the access panel on the ram air inlet duct.

3) Install the applicable access panel:

Number	Name/Location
192BL	ECS Ram Air Inlet Mixing Duct Panel - Forward

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999 (Continued)

or close this access panel:

Number	Name/Location
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward

(5) Replace the water extractors on the applicable pack.

These are the tasks:

Water Extractor Removal, AMM TASK 21-51-14-000-801,

Water Extractor Installation, AMM TASK 21-51-14-400-801.

(a) If the problem does not occur on the subsequent flight, then you corrected the fault.

(b) If the problem occurs on the subsequent flight, then replace the applicable water separator.

These are the tasks:

High Pressure Water Separator Removal, AMM TASK 21-51-08-000-802,

High Pressure Water Separator Installation, AMM TASK 21-51-08-400-802.

OMR ALL

——— END OF TASK ———

802. Noise or Pounding Heard Through the Air Distribution System - Fault Isolation

A. Description

OMR 701-799

(1) An unusual noise is heard in the air distribution system. This can be caused by ice formation in the duct system. Ice formation is usually caused by a problem with the low limit (35F) control system. Noises in the air distribution system can also be caused by leakage in the air ducts, air ducts which are loose or completely disconnected, or by blockage in the air ducts.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(2) An unusual noise is heard in the air distribution system. This can be caused by ice formation in the duct system. Ice formation is usually caused by a problem with the pack temperature control system. Noises in the air distribution system can also be caused by leakage in the air ducts, air ducts which are loose or completely disconnected, or by blockage in the air ducts.

OMR 701-799

B. Possible Causes

(1) 35 degree F control valve, V27 (left) or V28 (right)

(2) 35 degree F sensor, T430 (left) or T431 (right)

(3) 35 degree F control module, M223 (left) or M224 (right)

(4) Air distribution ducts

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

C. Possible Causes

(1) Pack temperature control system

(2) Air distribution ducts

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799

D. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	1	C00344	AIR CONDITIONING TEMP CONT 35 DEG F L
B	1	C00345	AIR CONDITIONING TEMP CONT 35 DEG F R

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

E. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	1	C01157	A/C TEMP CONT VALVE CLOSE LEFT
B	1	C01160	A/C TEMP CONT VALVE CLOSE RIGHT

OMR 701-799

F. Related Data

(1) (SSM 21-51-16)

(2) (SSM 21-51-26)

(3) (WDM 21-51-16)

(4) (WDM 21-51-26)

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

G. Related Data

(1) (SSM 21-51-12)

(2) (SSM 21-51-22)

(3) (WDM 21-51-12)

(4) (WDM 21-51-22)

OMR 701-799

H. Fault Isolation Procedure

(1) Do this check of the 35 degree F sensor, control valve, and control module for each pack:

NOTE: Conditioned air for the flight compartment is supplied by the left pack. Conditioned air for the passenger cabin is supplied by the right pack.

(a) Do this task: Low Limit (35 °F) Controller BITE Procedure, 21-52 TASK 801.

(b) If the BITE test shows a fault, then do the do the Fault Isolation Procedure for the applicable maintenance message.

1) If the problem does not occur on the subsequent flight, then you corrected the fault.

(c) If the BITE test does not show a fault, then continue.

(2) Do this check for leakage or blockage in the air distribution ducts:

(a) Supply conditioned air from the left and the right pack. To do this, do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799 (Continued)

(b) Find the source of the noise in the problem report.

(c) Get access to the air distribution duct in that location.

(d) Look for indications of leakage or a loose connection in that location.

(e) If you find indications of leakage or a loose connection, then repair the air distribution duct.

(f) If you do not find indications of leakage or a loose connection, then disassemble the ducts in that location and look for indications of blockage or unwanted material inside the ducts.

(g) If the problem does not occur on the subsequent flight, then you corrected the fault.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

I. Fault Isolation Procedure

(1) Do the BITE test on the pack/zone temperature controller (M1442 left or M1443 right) for the applicable pack:

NOTE: Conditioned air for the flight compartment is supplied by the left pack. Conditioned air for the passenger cabin is supplied by the right pack.

(a) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(b) If the BITE test shows a fault, then do the Fault Isolation Procedure for the applicable maintenance message.

1) If the problem does not occur on the subsequent flight, then you corrected the fault.

(c) If the BITE test does not show a fault, then continue.

(2) Do this check for leakage or blockage in the air distribution ducts:

(a) Supply conditioned air from the left and the right pack. To do this, do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(b) Find the source of the noise in the problem report.

(c) Get access to the air distribution duct in that location.

(d) Look for indications of leakage or a loose connection in that location.

(e) If you find indications of leakage or a loose connection, then repair the air distribution duct.

(f) If you do not find indications of leakage or a loose connection, then disassemble the ducts in that location and look for indications of blockage or unwanted material inside the ducts.

(g) If the problem does not occur on the subsequent flight, then you corrected the fault.

OMR ALL

——— END OF TASK ———

803. RAM DOOR FULL OPEN Light Stays ON During Flight - Fault Isolation

A. Description

(1) The RAM DOOR FULL OPEN Light on the P5-10 Air Conditioning Panel should go OFF when the airplane is in the air with the Trailing Edge Flaps fully retracted.

(a) This is an Advisory Light.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(2) There are two RAM DOOR FULL OPEN Lights, one for the Left Ram Air Inlet Modulation Doors and one for the Right Ram Air Inlet Modulation Doors.

OMR 701-799; 737-600 OR 737-700

(3) Dirty Heat Exchangers or unwanted material like paper that blocks the flow of air through the Heat Exchangers could prevent the Modulation Doors from closing and allow the RAM DOOR FULL OPEN Light to go OFF.

(a) Dirty or blocked Heat Exchangers could cause a subsequent decrease in heat transfer from the Pack Air side to the Ram Air side of the Heat Exchangers and higher temperatures at the Ram Air Sensor and an open signal from the Ram Air Controller to the Ram Air Actuator.

(b) The blue RAM DOOR FULL OPEN Advisory Light indicates that the temperature at the Ram Air Control Temperature Sensor is higher than the desired 230 °F and the Ram Air Controller has signaled the actuator to open the Modulation Panels to provide more Ram Air cooling.

(c) If the blue RAM DOOR FULL OPEN Advisory Light is ON continuously, it also indicates that the Ram Air System does not operate as efficiently as it was designed to.

(d) Do the Fault Isolation as soon as possible to prevent a PACK TRIP OFF Light indication that could possibly affect dispatch.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999; 737-800 OR 737-900

(4) Dirty Heat Exchangers or unwanted material such as paper that blocks the flow of air through the Heat Exchangers could prevent the Modulation Doors from closing and allowing the RAM DOOR FULL OPEN Light to go OFF.

(a) Dirty or blocked Heat Exchangers could result in a subsequent decrease in heat transfer from the Pack Air side to the Ram Air side of the Heat Exchangers and higher temperatures at the Ram Air Sensor and an open signal from the Pack/Zone Temperature Controller to the Ram Air Actuator.

(b) The blue RAM DOOR FULL OPEN Advisory Light indicates that the temperature at the Ram Air Control Temperature Sensor is higher than the desired 230 °F and the Pack/Zone Temperature Controller has signaled the actuator to open the Modulation Panels to provide more Ram Air cooling. It also indicates that the Ram Air System is not operating as efficiently as it was designed to operate if the light is ON continuously. Perform fault isolation at the earliest maintenance opportunity to avoid a PACK Light indication occuring that could possibly affect dispatch.

OMR 815, 816, 818-826, 901-999

(5) Ram Air Exit Louvers that do not open and restrict the flow of air through the Heat Exchangers could prevent the Modulation Doors from closing and allowing the RAM DOOR FULL OPEN Light to go OFF.

(a) There are two Smart Ram Air Door Actuators (Inlet and Exit SRADAs) on the Left and Right Ram Air Systems.

(b) The Inlet and Exit SRADAs include four Indication Lights to give a visual display of Ram Air System LRU faults.

(c) The continuous BIT detects and shows the fault status of the Ram Air Temperature Sensor, the Inlet SRADA and the Exit SRADA.

(d) The fault indication shows when there is power to the SRADAs and a valid Weight-on-Wheels (WOW) signal is present.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 815, 816, 818-826, 901-999 (Continued)

1) If there are no faults, a Green NO LRU FAULT Light comes ON, and the three LRU Lights are OFF.
2) If there is a fault, the NO LRU FAULT Light is OFF and the light related to the defective LRU comes ON Red.

OMR ALL

B. Possible Causes

OMR 701-799

(1) 737-600 or 737-700 Airplanes

(a) Left (Right) Ram Air Door Actuator (RADA), M326 (M327)
(b) Dirty Heat Exchangers or Blocked Airflow to Heat Exchangers
(c) Air Conditioning Accessory Unit (ACAU), M324
(d) Left Trailing Edge Flaps UP Switch, S1051
(e) Flap/Slat Electronics Unit (FSEU), M1746
(f) Left (Right) Ram Air Sensor, T438 (T439)
(g) Left (Right) Ram Air Controller, M328 (M329)
(h) Wiring

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(2) 737-800 or 737-900 Airplanes

OMR 815, 816, 818-826, 901-999

(a) Ram Air Actuators:
- Left Forward Ram Air Actuator, M2795
- Right Forward Ram Air Actuator, M2797
- Left Aft Ram Air Actuator, M2796
- Right Aft Ram Air Actuator, M2798

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(b) Dirty Heat Exchangers or Blocked Airflow to Heat Exchangers
(c) Left (Right) ACAU, M324 (M1455)
(d) Left Trailing Edge Flaps UP Switch, S1051
(e) FSEU, M1746
(f) Left (Right) Ram Air Sensor, T481 (T482)
(g) Left (Right) Pack Zone Temperature Controller (PZTC), M1442 (M1443)
(h) Fan Bypass Check Valve
(i) Air Cycle Machine (ACM)
(j) Wiring

OMR ALL

C. Circuit Breakers

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799

(1) 737-600 or 737-700 Airplanes

(a) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C00399	AIR CONDITIONING RAM AIR MOD LEFT
A	8	C00265	AIR CONDITIONING RAM AIR MOD CONT L
B	4	C00400	AIR CONDITIONING RAM AIR MOD RIGHT
B	8	C00266	AIR CONDITIONING RAM AIR MOD CONT R

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(2) 737-800 or 737-900 Airplanes

(a) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	11	C00313	INDICATOR MASTER DIM SECT 1

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
OMR 802, 805, 807, 809, 810, 812, 813			
A	4	C00399	AIR CONDITIONING RAM AIR MOD LEFT
OMR 815, 816, 818-826, 901-999			
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
A	8	C00555	AIR CONDITIONING RAM AIR MOD CONT L
OMR 802, 805, 807, 809, 810, 812, 813			
B	4	C00400	AIR CONDITIONING RAM AIR MOD RIGHT
OMR 815, 816, 818-826, 901-999			
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
B	8	C01178	AIR CONDITIONING RAM AIR MOD CONT R

OMR ALL

D. Related Data

(1) WDM 21-51-14

(2) WDM 21-51-24

(3) SSM 21-51-24

(4) SSM 21-51-14

E. Initial Evaluation

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(1) Do the Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) Make sure that the Left and Right Ram Air Systems do not show failures.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 815, 816, 818-826, 901-999

(2) Open the applicable Left (Right) Pack Ram Air Inlet Actuator Panels:

Number	Name/Location
191FL	Forward Wing To Body Fairing Panel - Mid Fairing, Above Ram Air Inlet
191FR	Forward Wing To Body Fairing Panel - Mid Fairing, Above Ram Air Inlet
191GL	Ram Air Actuator Panel - Forward
191GR	Ram Air Actuator Panel - Forward

(3) Make sure that no SRADA Red Fault Indicator Lights are ON.

(a) If a Red Fault Lights is ON, replace the associated SRADA or Ram Air Sensor.

OMR ALL

(4) Make sure that the Left and Right Blue RAM DOOR FULL OPEN Lights are ON.

(5) Make sure that the Left and Right Ram Air Inlet Deflector Doors are extended.

(6) Extend the Trailing Edge Flaps to FLAPS 1 position. This is the task: Extend the Trailing Edge Flaps, AMM TASK 27-51-00-860-803.

(7) Put the airplane in the AIR Mode. This is the task: Put the Airplane in the Air Mode, AMM TASK 32-09-00-860-801.

(8) Make sure that these results occur:

(a) Make sure that the Ram Air Inlet Deflector Doors fully retract.

(b) Make sure that the Ram Air Inlet Modulation Panels move toward the CLOSED position.

NOTE: The movement of the Ram Air Inlet Modulation Panels can not be easy to notice.

(c) Make sure that the Left and Right Blue RAM DOOR FULL OPEN Lights stay ON.

OMR 815, 816, 818-826, 901-999

(d) Make sure that the Ram Air Exit Louvers stay in the OPEN position.

OMR ALL

(e) If the above results do not occur, then do the “Fault Isolation Procedure - Deflector Door Does Not Retract” below.

(f) If the above results occur, continue with the Initial Evaluation.

(9) Retract the Trailing Edge Flaps to the FULLY UP position. This is the task: Retract the Trailing Edge Flaps, AMM TASK 27-51-00-860-804.

(10) Make sure that these results occur:

- Make sure that the Ram Air Inlet Modulation Panels move toward the CLOSED position.
- Make sure that the RAM DOOR FULL OPEN Lights go OFF.

OMR 815, 816, 818-826, 901-999

- Make sure that the Ram Air Exit Louvers move towards the CLOSED position.

OMR ALL

(a) If the above results do not occur, then do the “Fault Isolation Procedure - Modulation Panels Do Not Move to Cruise Position Problem”.

(b) If the RAM DOOR FULL OPEN Light goes OFF, then there was an intermittent fault.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(11) If the RAM DOOR FULL OPEN Light does not go OFF but the Modulation Panels moved toward the CLOSED position, then do the “Fault Isolation Procedure - Indication Circuit Problem” below.

(12) Do this check of the Fan Bypass Check Valve:

(a) To get access to the fan bypass check valve in the left pack primary plenum/diffuser, do this step:

1) Open this access panel:

Number	Name/Location
192CL	ECS Access Door

2) Remove the access panel on the bottom of the primary plenum/diffuser.

(b) To get access to the fan bypass check valve in the right pack primary plenum/diffuser, do this step:

1) Open these access panels:

Number	Name/Location
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

2) Remove the access panel on the bottom of the Primary Plenum/Diffuser.

(c) Do these checks:

1) Make sure that the Fan Bypass Check Valve Door is installed and it operates correctly. This is the task: Fan Bypass Check Valve Door - Operational Check, AMM TASK 21-51-03-710-801.

a) If the Fan Bypass Check Valve Door is not installed or does not operate correctly, then replace the Diffuser Assembly with a serviceable unit or replace the Check Valve Door off aircraft per the vendor instructions.

NOTE: Allied Signal Service Bulletin 2215240-21-2721 or 2215240-21-2722 gives the instructions to install a replacement Left (Right) Diffuser Assembly Check Valve Door.

2) Make sure that the Fan Bypass Check Valve Door is installed on the bottom of the Primary Diffuser.

a) If the Fan Bypass Check Valve Door is not installed on the bottom of the Primary Diffuser, replace the Bypass Door on the bottom of the diffuser. These are the tasks:

- Heat Exchanger and Plenum/Diffuser Assembly - Removal, AMM TASK 21-51-03-000-802-001 or Heat Exchanger and Plenum/Diffuser Assembly - Removal, AMM TASK 21-51-03-000-805-002
- Heat Exchanger and Plenum/Diffuser Assembly - Installation, AMM TASK 21-51-03-400-801-001 or Heat Exchanger and Plenum/Diffuser Assembly - Installation, AMM TASK 21-51-03-400-804-002

(d) Close or install the applicable access panels or doors that you opened or removed.

(13) If you do not find a problem, then do these steps:

(a) Open the access panel to the Heat Exchangers.

(b) Remove the Heat Exchanger Inlet Duct Inspection Panel.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Examine the Ram Air Duct for unwanted material that can decrease air flow to the Heat Exchangers.

1) If you find no unwanted material, then do one of these tasks to clean the Heat Exchangers:

- Heat Exchanger and Plenum/Diffuser Assembly Cleaning Using Backflush Equipment, AMM TASK 21-51-03-160-807
- Heat Exchanger - Blowing Out Debris/Blockages with Nitrogen Gas/Compressed Air, AMM TASK 21-51-03-160-811
- Heat Exchanger - Cleaning with Soap-Water-Air, AMM TASK 21-51-03-160-809

OMR 701-799; 737-600 OR 737-700

F. Fault Isolation Procedure - Deflector Door Does Not Retract Problem

OMR 701-799

(1) Do this check for 115V AC at the Ram Air Actuator:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C00399	AIR CONDITIONING RAM AIR MOD LEFT
B	4	C00400	AIR CONDITIONING RAM AIR MOD RIGHT

(b) For the left ram air inlet system, open this panel to get access to the actuator:

Number	Name/Location
191GL	Ram Air Actuator Panel - Forward

(c) For the right ram air inlet system, open this panel to get access to the actuator:

Number	Name/Location
191GR	Ram Air Actuator Panel - Forward

(d) Disconnect connector D966 (D968) from the L (R) Ram Air Actuator.

(e) Close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C00399	AIR CONDITIONING RAM AIR MOD LEFT
B	4	C00400	AIR CONDITIONING RAM AIR MOD RIGHT

(f) Do a check for 115V AC from between pins 5 and 11 (Ground) of the applicable connector D966 (D968).

1) If you find 115V AC, then replace the Left (Right) Ram Air Actuator, M326 (M327). These are the tasks:

- Ram Air Inlet Actuator Removal, AMM TASK 21-51-21-000-801
- Ram Air Inlet Actuator Installation, AMM TASK 21-51-21-400-801

a) Do the “Repair Confirmation - Deflector Door Does Not Retract Problem” at the end of this Fault Isolation Procedure.

2) If you do not find 115V AC between pins 5 and 11, then continue.

(2) Do this check of the Ram Air Actuator Ground Circuit Wiring:

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OMR 701-799 (Continued)

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C00399	AIR CONDITIONING RAM AIR MOD LEFT
B	4	C00400	AIR CONDITIONING RAM AIR MOD RIGHT

(b) Do a check of the wiring between connector D966 (Left) or D968 (Right), pin 11 and the Ground GD68-AC (Left) or GD72-AC (Right).

(c) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-connect connector D966 (left) or D968 (right) to the ram air actuator.

3) Close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C00399	AIR CONDITIONING RAM AIR MOD LEFT
B	4	C00400	AIR CONDITIONING RAM AIR MOD RIGHT

a) Do the “Repair Confirmation - Deflector Door Does Not Retract Problem” at the end of this Fault Isolation Procedure.

(d) If you do not find a problem with the wiring, then do these steps and continue:

1) Re-connect connector D966 (Left) or D968 (Right) to the Ram Air Actuator.

2) Close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C00399	AIR CONDITIONING RAM AIR MOD LEFT
B	4	C00400	AIR CONDITIONING RAM AIR MOD RIGHT

(3) Do this check of the Air/Ground Logic Circuit for the Air Conditioning Accessory Unit (ACAU), M324:

(a) Remove the ACAU. This is the task: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801.

(b) For the Left System, do a continuity check from pin 44 of connector D458B on the ships' wiring to a structural ground.

(c) For the Right System, do a continuity check from pin 7 of connector D458A on the ships' wiring to a structural ground.

(d) If there is continuity, then do these steps:

1) Do this task: Air/Ground Relay Fault - Fault Isolation, 32-09 TASK 804.

2) Do the corrective action for the problems that you find.

3) Re-install the ACAU. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

4) Do the “Repair Confirmation - Deflector Door Does Not Retract Problem” at the end of this Fault Isolation Procedure.

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OMR 701-799 (Continued)

(e) If there is no continuity, then continue.

(4) Do this check of the wiring from the ACAU to the Left (Right) Ram Air Actuator, M326 (M327):

(a) Disconnect connector D966 (D968) from the Left (Right) Ram Air Actuator.

(b) Do a check for an open circuit between these pins:

LEFT RAM AIR SYSTEM

ACAU CONNECTOR D458B	RAM AIR ACTUATOR CONNECTOR D966
pin 34	pin 5
pin 34	pin 6

RIGHT RAM AIR SYSTEM

ACAU CONNECTOR D458A	RAM AIR ACTUATOR CONNECTOR D968
pin 34	pin 5
pin 34	pin 6

(c) If there is an open circuit, then do these steps:

1) Repair the wiring.

2) Re-install the ACAU. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

3) Re-connect connector D966 (D968) from the Left (Right) Ram Air Actuator.

4) Do the “Repair Confirmation - Deflector Door Does Not Retract Problem” at the end of this Fault Isolation Procedure.

(d) If there is continuity, then do this step and continue.

1) Re-connect connector D966 (D968) from the Left (Right) Ram Air Actuator.

OMR 701-799 PRE SB 737-27-1290 AND PRE SB 737-27-1304

(5) For a problem in the Left Ram Air Inlet System, do this check of the Ground Circuit from the ACAU to the Trailing Edge Flaps UP Switch, S1051:

(a) Do a continuity check between pin 28 of connector D458B and structure ground.

NOTE: There should be continuity through a grounded Trailing Edge Flaps UP Switch, S1051.

(b) If there is no continuity, then do these steps:

1) Disconnect connector D46040J in the Main Wheel Well.

2) Do a check for an open circuit between these pins of connector D458B and connector D46040J:

D458B	D46040J
pin 28	pin 14

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799 PRE SB 737-27-1290 AND PRE SB 737-27-1304 (Continued)

3) If there is an open circuit, then do these steps:
   a) Repair the wiring.
   b) Re-connect connector D46040J.
   c) Re-install the ACAU. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.
   d) Do the “Repair Confirmation - Deflector Door Does Not Retract Problem” at the end of this Fault Isolation Procedure.
4) If there is continuity, then do these steps:
   a) Replace the Trailing Edge Flaps UP Switch, S1051. These are the tasks:
      • Flap Limit Switch Removal, AMM TASK 27-51-63-000-801
      • Flap Limit Switch Installation, AMM TASK 27-51-63-400-801
   b) Re-connect connector D46040J.
   c) Re-install the ACAU. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.
   d) Do the “Repair Confirmation - Deflector Door Does Not Retract Problem” at the end of this Fault Isolation Procedure.

(c) If there is continuity, then continue.

OMR 701-799 POST SB 737-27-1290 OR POST SB 737-27-1304

(6) For a problem in the Left Ram Air Inlet system, do this check of the Ground Circuit from the ACAU to the Trailing Edge Flaps UP Switch, S1051:

(a) Do a continuity check between pin 28 of connector D458B and structure ground.

NOTE: There should be continuity through a grounded Trailing Edge Flaps UP Switch, S1051.

(b) If there is no continuity, then do these steps:

1) Disconnect connector D14360 from the S1051 Switch in the Main Wheel Well.

2) Do a check for an open circuit between these pins of connector D458B and connector D14360:

D458B	D14360
pin 28	pin F

3) If there is an open circuit, then do these steps:
   a) Repair the wiring.
   b) Re-connect connector D14360.
   c) Re-install the ACAU. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.
   d) Do the “Repair Confirmation - Deflector Door Does Not Retract Problem” at the end of this Fault Isolation Procedure.
4) If there is continuity, then do these steps:
   a) Replace the Trailing Edge Flaps UP Switch, S1051. These are the tasks:
      • Flap Limit Switch Removal, AMM TASK 27-51-63-000-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799 POST SB 737-27-1290 OR POST SB 737-27-1304 (Continued)

- Flap Limit Switch Installation, AMM TASK 27-51-63-400-801

b) Re-connect connector D14360.

c) Re-install the ACAU. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

d) Do the “Repair Confirmation - Deflector Door Does Not Retract Problem” at the end of this Fault Isolation Procedure.

(c) If there is continuity, then continue.

OMR 701-799

(7) For a problem in the Right Ram Air Inlet System, do this check of the Ground Circuit from the ACAU to the FSEU, M1746:

(a) Do a continuity test from pin 28 of connector D458A to structure ground.

NOTE: There should be continuity through a grounded FSEU, M1746.

(b) If there is no continuity, then do these steps:

1) Remove the FSEU, M1746. This is the task: Flap/Slat Electronics Unit Removal, AMM TASK 27-51-01-000-801.

2) Do a check for an open circuit between these pins of connector D458A and connector D728A:

D458A	D728A
pin 28	pin C11

3) If there is an open circuit, then do these steps:

a) Repair the wiring.

b) Re-install the FSEU, M1746. This is the task: Flap/Slat Electronics Unit Installation, AMM TASK 27-51-01-400-801.

c) Re-install the ACAU. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

d) Do the “Repair Confirmation - Deflector Door Does Not Retract Problem” at the end of this Fault Isolation Procedure.

4) If there is continuity, then do these steps:

a) Replace the FSEU, M1746. These are the tasks:

- Flap/Slat Electronics Unit Removal, AMM TASK 27-51-01-000-801
- Flap/Slat Electronics Unit Installation, AMM TASK 27-51-01-400-801

b) Re-install the ACAU. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

c) Do the Repair Confirmation at the end of this task.

(c) If there is continuity, then continue.

(8) Do this check for 115V AC to the ACAU:

(a) Remove the ACAU. This is the task: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801.

(b) Do a check for 115V AC between pin 15 of connector D458B (Left) or pin 11 of connector D458A (Right) and structure ground.

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OMR 701-799 (Continued)

(c) If there is 115V AC, then do these steps:

1) Install a new ACAU, M324. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

2) Do the “Repair Confirmation - Deflector Door Does Not Retract Problem” at the end of this Fault Isolation Procedure.

(d) If there is not 115V AC, then continue:

(9) Do this check for 115V AC at the circuit breaker:

(a) Make sure there is 115V AC at the load side of circuit breaker C399 (Left) or C400 (Right) on the P6-4 Panel.

1) If there is 115V AC, then do these steps:

a) Repair the wiring between the applicable circuit breaker and the ACAU connector:

LEFT RAM AIR SYSTEM

CIRCUIT BREAKER	ACAU CONNECTOR
C00399	D458B
term 2	pin 15

RIGHT RAM AIR SYSTEM

CIRCUIT BREAKER	ACAU CONNECTOR
C00400	D458A
term 2	pin 11

b) Re-install the ACAU. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

c) Do the Repair Confirmation at the end of this task.

2) If there is not 115V AC, then do these steps:

a) Do this task: DIST/BUS FAULT For GCU 1 - Fault Isolation, 24-21 TASK 815.

b) Re-install the ACAU. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

c) Do the “Repair Confirmation - Deflector Door Does Not Retract Problem” at the end of this Fault Isolation Procedure.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999; 737-800 OR 737-900

G. Fault Isolation Procedure - Deflector Door Does Not Retract Problem

OMR 802, 805, 807, 809, 810, 812, 813

(1) Do this check for 115V AC at the Ram Air Actuator:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C00399	AIR CONDITIONING RAM AIR MOD LEFT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 802, 805, 807, 809, 810, 812, 813 (Continued)

(Continued)

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	4	C00400	AIR CONDITIONING RAM AIR MOD RIGHT

(b) For the left ram air inlet system, open this panel to get access to the actuator:

Number	Name/Location
191GL	Ram Air Actuator Panel - Forward

(c) For the right ram air inlet system, open this panel to get access to the actuator:

Number	Name/Location
191GR	Ram Air Actuator Panel - Forward

(d) Disconnect connector D966 (left) or D968 (right) from the ram air actuator.

(e) Close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C00399	AIR CONDITIONING RAM AIR MOD LEFT
B	4	C00400	AIR CONDITIONING RAM AIR MOD RIGHT

(f) Do a check for 115 VAC from pin 5 to pin 11 (ground) of D966 (left) or D968 (right).

(g) If there is 115V AC from pin 5 to pin 11, then replace the Left (Right) Ram Air Actuator, M326 (M327). These are the tasks:

- Ram Air Inlet Actuator Removal, AMM TASK 21-51-21-000-801
- Ram Air Inlet Actuator Installation, AMM TASK 21-51-21-400-801

1) Do the “Repair Confirmation - Deflector Door Does Not Retract Problem” at the end of this Fault Isolation Procedure.

(h) If there is not 115V AC from pin 5 to pin 11, then continue.

(2) Do this check of the Ram Air Actuator Ground Wiring:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C00399	AIR CONDITIONING RAM AIR MOD LEFT
B	4	C00400	AIR CONDITIONING RAM AIR MOD RIGHT

(b) Do a check of the wiring between connector D966 (left) or D968 (right), pin 11 and the ground GD68-AC (left) or GD72-AC (right).

(c) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-connect connector D966 (left) or D968 (right) to the ram air actuator.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 802, 805, 807, 809, 810, 812, 813 (Continued)

3) Close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C00399	AIR CONDITIONING RAM AIR MOD LEFT
B	4	C00400	AIR CONDITIONING RAM AIR MOD RIGHT

4) Do the “Repair Confirmation - Deflector Door Does Not Retract Problem” at the end of this Fault Isolation Procedure.

(d) If you do not find a problem with the wiring, then do these steps and continue:

1) Re-connect connector D966 (left) or D968 (right) to the ram air actuator.

2) Close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C00399	AIR CONDITIONING RAM AIR MOD LEFT
B	4	C00400	AIR CONDITIONING RAM AIR MOD RIGHT

(3) Do this check of the air/ground logic circuit for the air conditioning accessory unit (ACAU), M324 and M1455:

(a) Remove ACAU No.1 (left) or ACAU No.2 (right). This is the task: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801.

(b) For the left system, connect an ohmmeter between pin 35 of connector D458A and a structural ground.

(c) For the right system, connect an ohmmeter between pin 35 of connector D10002A to a structural ground.

(d) If there is continuity, then do these steps:

1) Do this task: Air/Ground Relay Fault - Fault Isolation, 32-09 TASK 804.

2) Do the corrective action for any fault found.

3) Re-install the ACAU. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

4) Do the “Repair Confirmation - Deflector Door Does Not Retract Problem” at the end of this Fault Isolation Procedure.

(e) If there is no continuity, then continue.

(4) Do this check of the wiring from ACAU No. 1 (left) or ACAU No. 2 (right) to the ram air actuator, M326 (left) or M327 (right):

(a) Disconnect connector D966 (left) or D968 (right) from the ram air actuator.

(b) Do a check for an open circuit between these pins:

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OMR 802, 805, 807, 809, 810, 812, 813 (Continued)

	RAM AIR SYSTEM	
	ACAU CONNECTOR	RAM AIR ACTUATOR CONNECTOR
LEFT	D458A	D966
	pin 24	pin 5
	pin 24	pin 6

	RAM AIR SYSTEM	
	ACAU CONNECTOR	RAM AIR ACTUATOR CONNECTOR
RIGHT	D10002A	D968
	pin 24	pin 5
	pin 24	pin 6

(c) If there is an open circuit, then do these steps:

1) Repair the wiring.

2) Re-install the ACAU. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

3) Re-connect connector D966 (left) or D968 (right) to the ram air actuator.

4) Do the “Repair Confirmation - Deflector Door Does Not Retract Problem” at the end of this Fault Isolation Procedure.

(d) If there is continuity, then do this step and continue.

1) Re-connect connector D966 (left) or D968 (right) to the ram air actuator.

OMR 802, 805, 807, 809, 810 PRE SB 737-27-1290 AND PRE SB 737-27-1304

(5) For a problem in the left ram air inlet system, do this check of the ground circuit from ACAU No. 1 to the trailing edge flaps up switch, S1051:

(a) Do a continuity test between pin 33 of connector D458B and structure ground.

NOTE: There should be continuity through a grounded trailing edge flaps up switch, S1051.

(b) If there is no continuity, then do these steps:

1) Disconnect connector D46040J in the main wheel well.

2) Do a check for an open circuit between these pins of connector D458B and connector D46040J:

D458B	D46040J
33	14

3) If there is an open circuit, then do these steps:

a) Repair the wiring.

b) Re-connect connector D46040J.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

c) Re-install ACAU No. 1. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

d) Do the “Repair Confirmation - Deflector Door Does Not Retract Problem” at the end of this Fault Isolation Procedure.

4) If there is continuity, then do these steps:

a) Replace the trailing edge flaps up switch, S1051.

These are the tasks:

Flap Limit Switch Removal, AMM TASK 27-51-63-000-801,

Flap Limit Switch Installation, AMM TASK 27-51-63-400-801.

b) Re-connect connector D46040J.

c) Re-install ACAU No 1. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

d) Do the “Repair Confirmation - Deflector Door Does Not Retract Problem” at the end of this Fault Isolation Procedure.

(c) If there is continuity, then continue.

OMR 812, 813; OMR 802, 805, 807, 809, 810 POST SB 737-27-1290 OR POST SB 737-27-1304

(6) For a problem in the left ram air inlet system, do this check of the ground circuit from ACAU No. 1 to the trailing edge flaps up switch, S1051:

(a) Make sure that the trailing edge flaps are not fully retracted.

(b) Do a continuity test between pin 33 of connector D458B and structure ground.

NOTE: There should be continuity through a grounded trailing edge flaps up switch, S1051.

(c) If there is no continuity, then do these steps:

1) Disconnect connector D14360 from the trailing edge flaps up switch S1051 in the main wheel well.

2) Do a check for an open circuit between these pins of connector D458B and connector D14360:

D458B	D14360
pin 33	pin F

3) If there is an open circuit, then do these steps:

a) Repair the wiring.

b) Re-connect connector D14360.

c) Re-install ACAU No. 1. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

d) Do the “Repair Confirmation - Deflector Door Does Not Retract Problem” at the end of this Fault Isolation Procedure.

4) If there is continuity, then do these steps:

a) Replace the trailing edge flaps up switch, S1051. These are the tasks:

- Flap Limit Switch Removal, AMM TASK 27-51-63-000-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

- Flap Limit Switch Installation, AMM TASK 27-51-63-400-801

b) Re-connect connector D14360.

c) Re-install ACAU No 1. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

d) Do the “Repair Confirmation - Deflector Door Does Not Retract Problem” at the end of this Fault Isolation Procedure.

(d) If there is continuity, then continue.

OMR 802, 805, 807, 809, 810, 812, 813

(7) For a problem in the right ram air inlet system, do this check of the ground circuit from ACAU No. 2 to the flap/slat electronics unit, M1746:

(a) Do a continuity test from pin 33 of connector D10002B to structure ground.

NOTE: There should be continuity through a grounded flap/slat electronics unit, M1746.

(b) If there is no continuity, then do these steps:

1) Remove the flap slat electronic module, M1746. To remove the module, do this task: Flap/Slat Electronics Unit Removal, AMM TASK 27-51-01-000-801.

2) Do a check for an open circuit between these pins of connector D10002B and connector D728A:

M1455	**M1746**
D10002B	**D728A**
pin 33	pin C11

3) If there is an open circuit, then do these steps:

a) Repair the wiring.

b) Re-install the flap slat electronic unit, M1746. This is the task: Flap/Slat Electronics Unit Installation, AMM TASK 27-51-01-400-801.

c) Re-install the ACAU. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

d) Do the “Repair Confirmation - Deflector Door Does Not Retract Problem” at the end of this Fault Isolation Procedure.

4) If there is continuity, then do these steps:

a) Replace the flap/ slat electronic unit, M1746. These are the tasks:

- Flap/Slat Electronics Unit Removal, AMM TASK 27-51-01-000-801
- Flap/Slat Electronics Unit Installation, AMM TASK 27-51-01-400-801

b) Re-install the ACAU. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

c) Do the “Repair Confirmation - Deflector Door Does Not Retract Problem” at the end of this Fault Isolation Procedure.

(c) If there is continuity, then continue.

(8) Do this check for 115 VAC to the ACAU No. 1 (left) or ACAU No. 2 (right):

(a) Remove ACAU No. 1 or ACAU No. 2. This is the task: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 802, 805, 807, 809, 810, 812, 813 (Continued)

(b) Do a check for 115 VAC between pin 49 of connector D458A (left) or pin 49 of connector D10002A (right) and structure ground.

(c) If there is 115 VAC, then do these steps:

1) Install a new air conditioning accessory unit, M324 (No. 1) or M1455 (No. 2). This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

2) Do the “Repair Confirmation - Deflector Door Does Not Retract Problem” at the end of this Fault Isolation Procedure.

(d) If there is not 115 VAC, then continue:

(9) Do this check for 115 VAC at the circuit breaker:

(a) Make sure that there is 115V AC at the load side of these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C00399	AIR CONDITIONING RAM AIR MOD LEFT
B	4	C00400	AIR CONDITIONING RAM AIR MOD RIGHT

1) If there is 115 VAC, then do these steps:

a) Repair the wiring between the applicable circuit breaker and the connector for the ACAU:

	RAM AIR SYSTEM	
	CIRCUIT BREAKER	**ACAU CONNECTOR**
LEFT	**C00399**	**D458A**
	term 2	pin 49

	RAM AIR SYSTEM	
	CIRCUIT BREAKER	**ACAU CONNECTOR**
RIGHT	**C00400**	**D10002A**
	term 2	pin 49

b) Re-install the ACAU. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

c) Do the “Repair Confirmation - Deflector Door Does Not Retract Problem” at the end of this Fault Isolation Procedure.

2) If there is not 115 VAC, then do these steps:

a) Do this task: DIST/BUS FAULT For GCU 1 - Fault Isolation, 24-21 TASK 815.

b) Re-install the ACAU. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

c) Do the “Repair Confirmation - Deflector Door Does Not Retract Problem” at the end of this Fault Isolation Procedure.

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OMR 815, 816, 818-826, 901-999

(10) Do this check for 28 VDC at the ram air inlet actuator:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT

(b) For the left ram air inlet system, open this panel to get access to the actuator:

Open this access panel:

Number	Name/Location
191GL	Ram Air Actuator Panel - Forward

(c) For the right ram air inlet system, open this panel to get access to the actuator:

Open this access panel:

Number	Name/Location
191GR	Ram Air Actuator Panel - Forward

(d) Disconnect electrical connector D14386 from the left ram air inlet actuator M2795 or electrical connector D14394 from the right ram air inlet actuator.

(e) Close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT

(f) Do a check for 28 VDC from pin 4 to pin 16 of connector D14386 (left) or D14394 (right).

(g) If there is 28 VDC between pins 4 and 16, then do these steps:

1) Replace the ram air inlet actuator M2795 (left) or M2797 (right). These are the tasks:

- Ram Air Inlet Actuator Removal, AMM TASK 21-51-21-000-801
- Ram Air Inlet Actuator Installation, AMM TASK 21-51-21-400-801.

2) Do the “Repair Confirmation - Deflector Door Does Not Retract Problem” at the end of this Fault Isolation Procedure.

(h) If 28V DC is not present between pins 4 and 16 of connector D14386 (left) or D14394 (right), then continue.

(11) Do this check of the ground circuit wiring for the ram air inlet actuator:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT

(b) Do a check of the left actuator ground circuit wiring as follows:

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1) Connect an ohmmeter between pins 16 and 24 of connector D14386.
2) Make sure the resistance is between 0 and 40 ohms.

NOTE: This makes sure that the GD712-DC ground is satisfactory.

(c) Do a check of the right actuator ground circuit wiring as follows:
1) Connect an ohmmeter between pins 16 and 24 of connector D14394.
2) Make sure the resistance is between 0 and 40 ohms.

NOTE: This makes sure that the GD716-DC ground is satisfactory.

(d) If you find a problem with the wiring, then do these steps:
1) Repair the wiring.
2) Reconnect electrical connectors D14386 (left) or D14394 (right) to the ram air inlet actuator.
3) Close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT

4) Do the “Repair Confirmation - Deflector Door Does Not Retract Problem” at the end of this Fault Isolation Procedure.

(e) If you do not find a problem with the wiring, then do these steps and continue:
1) Reconnect electrical connector D14386 (left) or D14394 (right) to the ram air inlet actuator.
2) Close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT

(12) Do this check of the air/ground logic circuit for the air conditioning accessory unit (ACAU), M324 and M1455:
(a) Remove ACAU No. 1 (left) or ACAU No. 2 (right). This is the task: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801.
(b) For the left system, do a continuity check from pin 35 of connector D458 to a structural ground.
(c) For the right system, do a continuity check from pin 35 of connector D10002A to a structural ground.
(d) If there is continuity, then do these steps:
1) Do this task: Air/Ground Relay Fault - Fault Isolation, 32-09 TASK 804.
2) Do the corrective action for any fault found.
3) Re-install the ACAU. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

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4) Do the “Repair Confirmation - Deflector Door Does Not Retract Problem” at the end of this Fault Isolation Procedure.

(e) If there is no continuity, continue.

(f) For the left system, do these steps:

1) Install a jumper wire between pin 35 of connector D458A and an aircraft ground.

2) Connect an ohmmeter between pin 1 of connector D14386 and an aircraft ground.

3) Make sure the measured resistance is between 0 and 40 ohms.

4) Repair any wiring problems that you find.

(g) For the right system, do these steps:

1) Install a jumper wire between pin 35 of connector D10002A and an aircraft ground.

2) Connect an ohmmeter between pin 1 of connector D14394 and an aircraft ground.

3) Make sure the measured resistance is between 0 and 40 ohms.

(h) If there is no continuity, then continue.

(13) Do this check of the wiring from ACAU No. 1 (left) or ACAU No. 2 (right) to the ram air inlet actuator, M2795 (left) or M2797 (right):

(a) Disconnect connector D14386 (left) or D14394 (right) from the ram air inlet actuator M2795.

(b) Do these checks for open circuits between ACAU No.1 M324 and left ram air inlet actuator M2795:

1) Install a jumper between pin 47 of connector D458A and a structural ground.

2) Connect an ohmmeter between pin 19 of connector D14386 and a structural ground.

3) Make sure the ohmmeter indicates between 0 and 40 ohms.

4) Install a jumper between pin 33 of connector D458B and a structural ground.

5) Connect an ohmmeter between pin 9 of connector D14386 and a structural ground.

6) Make sure the ohmmeter indicates between 0 and 40 ohms.

7) Put the airplane in the ground mode.

8) Connect an ohmmeter between pin 1 of connector D14386 and a structural ground.

9) Make sure the ohmmeter indicates between 0 and 40 ohms.

(c) If there is an open circuit, then do these steps:

1) Repair the wiring.

2) Re-install the ACAU. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

3) Re-connect connector D14386 to the ram air actuator.

4) Do the “Repair Confirmation - Deflector Door Does Not Retract Problem” at the end of this Fault Isolation Procedure.

(d) Do these checks for open circuits between ACAU No.2 (M1455) and right ram air inlet actuator M2797:

1) Install a jumper between pin 47 of connector D10002A and a structural ground.

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2) Connect an ohmmeter between pin 19 of connector D14394 and a structural ground.
3) Make sure the ohmmeter indicates between 0 and 40 ohms.
4) Install a jumper between pin 33 of connector D10002B and a structural ground.
5) Connect an ohmmeter between pin 9 of connector D14394 and a structural ground.
6) Make sure the ohmmeter indicates between 0 and 40 ohms.
7) Put the airplane in the ground mode.
8) Connect an ohmmeter between pin 1 of connector D14394 and a structural ground.
9) Make sure the ohmmeter indicates between 0 and 40 ohms.

(e) If there is an open circuit, then do these steps:
1) Repair the wiring.
2) Re-install the ACAU. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.
3) Re-connect connector D14386 to the ram air actuator.
4) Do the “Repair Confirmation - Deflector Door Does Not Retract Problem” at the end of this Fault Isolation Procedure.

(14) For a problem in the left ram air inlet system, do this check of the ground circuit from ACAU No. 1 to the trailing edge flaps up switch, S1051:
(a) Make sure that the trailing edge flaps are not fully retracted.
(b) Do a continuity test between pin 33 of connector D458B and structure ground.

NOTE: There should be continuity through a grounded trailing edge flaps up switch, S1051.

(c) If there is no continuity, then do these steps:
1) Disconnect connector D14360 from the trailing edge flaps up switch S1051 in the main wheel well.
2) Do a check for an open circuit between these pins of connector D458B and connector D14360:

D458B	D14360
pin 33	pin F

3) If there is an open circuit, then do these steps:
a) Repair the wiring.
b) Re-connect connector D14360.
c) Re-install ACAU No. 1. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.
d) Do the “Repair Confirmation - Deflector Door Does Not Retract Problem” at the end of this Fault Isolation Procedure.
4) If there is continuity, then do these steps:
a) Replace the trailing edge flaps up switch, S1051. These are the tasks:
• Flap Limit Switch Removal, AMM TASK 27-51-63-000-801

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OMR 815, 816, 818-826, 901-999 (Continued)

- Flap Limit Switch Installation, AMM TASK 27-51-63-400-801

b) Re-connect connector D14360.

c) Re-install ACAU No 1. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

d) Do the “Repair Confirmation - Deflector Door Does Not Retract Problem” at the end of this Fault Isolation Procedure.

(d) If there is continuity, then continue.

(15) For a problem in the right ram air inlet system, do this check of the ground circuit from ACAU No. 2 to the flap/slat electronics unit, M1746:

(a) Do a continuity test from pin 33 of connector D10002B to structure ground.

NOTE: There should be continuity through a grounded flap/slat electronics unit, M1746.

(b) If there is no continuity, then do these steps:

1) Remove the flap slat electronic module, M1746. This is the task: Flap/Slat Electronics Unit Removal, AMM TASK 27-51-01-000-801.

2) Do a check for an open circuit between these pins of connector D10002B and connector D728A:

M1455	**M1746**
D10002B	**D728A**
pin 33	pin C11

3) If there is an open circuit, then do these steps:

a) Repair the wiring.

b) Re-install the flap slat electronic unit, M1746. This is the task: Flap/Slat Electronics Unit Installation, AMM TASK 27-51-01-400-801.

c) Re-install the ACAU. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

d) Do the “Repair Confirmation - Deflector Door Does Not Retract Problem” at the end of this Fault Isolation Procedure.

4) If there is continuity, then do these steps:

a) Replace the flap/ slat electronic unit, M1746. These are the tasks:

- Flap/Slat Electronics Unit Removal, AMM TASK 27-51-01-000-801
- Flap/Slat Electronics Unit Installation, AMM TASK 27-51-01-400-801

b) Re-install the ACAU. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

c) Do the “Repair Confirmation - Deflector Door Does Not Retract Problem” at the end of this Fault Isolation Procedure.

(c) If there is continuity, then continue.

(16) Do this check for 28 VDC to the ACAU No. 1 (left) or ACAU No. 2 (right):

(a) Remove ACAU No. 1 or ACAU No. 2. This is the task: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801.

(b) Do a check for 28 VDC between pin 59 of connector D458A (left) or pin 59 of connector D10002A (right) and structure ground.

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OMR 815, 816, 818-826, 901-999 (Continued)

(c) If there is 28 VDC, then do these steps:

1) Install a new air conditioning accessory unit, M324 (No. 1) or M1455 (No. 2). This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

2) Do the “Repair Confirmation - Deflector Door Does Not Retract Problem” at the end of this Fault Isolation Procedure.

(d) If there is not 28 VDC, then continue:

(17) Do this check for 28 VDC at the circuit breaker:

(a) Make sure that there is 28 VDC at the load side of the following circuit breakers;

1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT

2) If there is 28 VDC, then do these steps:

a) Repair the wiring between the applicable circuit breaker and electrical connectors D14386 (left) and D14394 (right) ram air inlet actuators:

RAM AIR SYSTEM

CIRCUIT BREAKER C01950 Terminal 2		**M2795 Ram Air Actuator D14386**
Terminal 2	. . . . .	Pin 4

RAM AIR SYSTEM

CIRCUIT BREAKER C01951 Terminal 2		**M2797 D14394**
Terminal 2	. . . . .	Pin 4

b) If the ram air inlet deflector door moves to the fully retracted positon, then you corrected the fault.

3) If there is not 28 VDC, then do these steps:

a) Fault isolate the power distribution system as follows:

<1> Do a check of the wiring from the main 28V DC BUS 1 to 28V DC BUS 1 SECT 2.

<2> Fix any problems that you find.

b) Re-connect electrical connector D14386 (left) or D14394 (right).

c) Do the “Repair Confirmation - Deflector Door Does Not Retract Problem” at the end of this Fault Isolation Procedure.

OMR ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Repair Confirmation - Deflector Door Does Not Retract Problem

(1) If the Ram Air Inlet Deflector Door moves to the fully retracted position, then you corrected the problem. Do this step to complete the task:

(a) Put the airplane in the Ground Mode. This is the task: Return the Airplane to the Ground Mode, AMM TASK 32-09-00-860-802.

(2) If the Ram Air Inlet Deflector Door does not move to the fully retracted position, then continue the Fault Isolation Procedure at the subsequent step.

OMR 701-799; 737-600 OR 737-700

I. Fault Isolation Procedure - Modulation Panels Do Not Move to Cruise Position

OMR 701-799

(1) Do this check of the ram air temperature sensor:

(a) Open these access panels:

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

(b) Disconnect connector D1288 (D1290) to the Left (Right) Ram Air Temperature Sensor, T438 (T439).

(c) If the RAM DOOR FULL OPEN Light goes OFF and the Modulation Panels close, then do these steps:

1) Replace the Left (Right) Ram Air Temperature Sensor, T438 (T439). These are the tasks:

- Ram Air Control Temperature Sensor - Removal, AMM TASK 21-51-50-000-801
- Ram Air Control Temperature Sensor - Installation, AMM TASK 21-51-50-400-801

2) Do the “Repair Confirmation - Modulation Panels Do Not Move to Cruise Position” at the end of this Fault Isolation Procedure.

(d) If the RAM DOOR FULL OPEN Light stays ON and the Inlet Modulation Doors close, then do these steps:

1) Re-connect connector D1288 (D1290) to the Left (Right) Ram Air Temperature Sensor, T438 (T439).

2) Replace the Left (Right) Ram Air Actuator, M326 (M327). These are the tasks:

- Ram Air Inlet Actuator Removal, AMM TASK 21-51-21-000-801
- Ram Air Inlet Actuator Installation, AMM TASK 21-51-21-400-801

3) Do the “Repair Confirmation - Modulation Panels Do Not Move to Cruise Position” at the end of this Fault Isolation Procedure.

4) If the RAM DOOR FULL OPEN light does not go off and the modulation panels do not close, then continue.

(e) If the RAM DOOR FULL OPEN light stays on and the modulation panels do not close, then do this step and continue:

1) Re-connect connector D1288 (D1290) to the Left (Right) Ram Air Temperature Sensor, T438 (T439).

(2) Do this check for 115 VAC at the ram air actuator:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799 (Continued)

(a) Disconnect connector D966 (left) or D968 (right) from the ram air actuator, M326 (left) or M327 (right).

(b) Do a check for 115 VAC between pins 10 and 11 (ground) of connector D966 (left) or D968 (right).

(c) If there is 115 VAC at pin 10, then do these steps:

1) Replace the ram air actuator, M326 (left) or M327 (right). These are the tasks:

- Ram Air Inlet Actuator Removal, AMM TASK 21-51-21-000-801
- Ram Air Inlet Actuator Installation, AMM TASK 21-51-21-400-801

2) Do the “Repair Confirmation - Modulation Panels Do Not Move to Cruise Position” at the end of this Fault Isolation Procedure.

(d) If there is not 115 VAC, then do this step and continue:

1) Re-connect connector D966 (left) or D968 (right) to the ram air actuator.

(3) Do this check for 115 VAC at the ram air controller:

(a) Disconnect connector D970 (left) or D972 (right) from the ram air controller M328 (left) or M329 (right).

(b) Do a check for 115 VAC between pins 4 and 5 (ground) of connector D970 (left) or D972 (right).

(c) If there is 115 VAC at pin 4, then do these steps:

1) Replace the ram air controller, M328 (left) or M329 (right). These are the tasks:

- Ram Air Inlet Controller Removal, AMM TASK 21-51-20-000-801
- Ram Air Inlet Controller Installation, AMM TASK 21-51-20-400-801

2) Do the “Repair Confirmation - Modulation Panels Do Not Move to Cruise Position” at the end of this Fault Isolation Procedure.

(d) If there is not 115 VAC at pin 4, then continue.

(4) Do this check of the ground circuit wiring for the ram air controller:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C00399	AIR CONDITIONING RAM AIR MOD LEFT
B	4	C00400	AIR CONDITIONING RAM AIR MOD RIGHT

(b) Do a check of the wiring between connector D970 (left) or D972 (right), pin 5 and the ground GD12-AC (left) or GD16-AC (right).

(c) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-connect connector D970 (left) or D972 (right) to the ram air controller M328 (left) or M329 (right).

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OMR 701-799 (Continued)

3) Close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C00399	AIR CONDITIONING RAM AIR MOD LEFT
B	4	C00400	AIR CONDITIONING RAM AIR MOD RIGHT

4) Do the “Repair Confirmation - Modulation Panels Do Not Move to Cruise Position” at the end of this Fault Isolation Procedure.

(d) If you do not find a problem with the ground wiring, then continue:

(5) Do this check of the wiring between the ACAU and the ram air controller:

(a) Remove the ACAU. This is the task: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801.

(b) Do a check for an open circuit between these pins:

	RAM AIR SYSTEM	
	ACAU CONNECTOR	RAM AIR CONTROLLER CONNECTOR
LEFT	D458B	D970
	pin 6	pin 4

	RAM AIR SYSTEM	
	ACAU CONNECTOR	RAM AIR CONTROLLER CONNECTOR
RIGHT	D458A	D972
	pin 14	pin 4

(c) If there is an open circuit, then do these steps:

1) Repair the wiring.

2) Re-install the ACAU. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

3) Re-connect connector D970 (left) or D972 (right) to the ram air controller.

4) Do the “Repair Confirmation - Modulation Panels Do Not Move to Cruise Position” at the end of this Fault Isolation Procedure.

(d) If there is continuity, then do this step and continue:

1) Re-connect connector D970 (left) or D972 (right) to the ram air controller.

OMR 701-799 PRE SB 737-27-1290 AND PRE SB 737-27-1304

(6) For a problem in the left ram air inlet system, do this check of the circuit from the ACAU to the trailing edge flaps up switch, S1051:

(a) Do a continuity check of the wiring between pin 28 of connector D458B and a structural ground.

(b) If there is continuity, then do these steps:

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OMR 701-799 PRE SB 737-27-1290 AND PRE SB 737-27-1304 (Continued)

1) Disconnect connector D46040J in the main wheel well.
2) Do a continuity check between pin 14 of connector D46040P and a structural ground.
3) If there is continuity, then do these steps:
   a) Replace the trailing edge flaps up switch, S1051. These are the tasks:
      • Flap Limit Switch Removal, AMM TASK 27-51-63-000-801
      • Flap Limit Switch Installation, AMM TASK 27-51-63-400-801
   b) Re-connect connector D46040.
   c) Re-install the ACAU. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.
   d) Do the “Repair Confirmation - Modulation Panels Do Not Move to Cruise Position” at the end of this Fault Isolation Procedure.
4) If there is no continuity, then do these steps:
   a) Repair the wiring between pin 28 of connector D458B and connector D46040J.
   b) Re-connect connector D46040.
   c) Re-install the ACAU. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.
   d) Do the “Repair Confirmation - Modulation Panels Do Not Move to Cruise Position” at the end of this Fault Isolation Procedure.

(c) If there is no continuity, then continue.

OMR 701-799 POST SB 737-27-1290 OR POST SB 737-27-1304

(7) For a problem in the left ram air inlet system, do this check of the circuit from the ACAU to the trailing edge flaps up switch, S1051:
   (a) Do a continuity check of the wiring between pin 28 of connector D458B and a structural ground.
   (b) If there is continuity, then do these steps:
      1) Disconnect connector D14360 in the main wheel well.
      2) Do a continuity check between pin F of connector D14360 and a structural ground.
      3) If there is continuity, then do these steps:
         a) Replace the trailing edge flaps up switch, S1051. These are the tasks:
            • Flap Limit Switch Removal, AMM TASK 27-51-63-000-801
            • Flap Limit Switch Installation, AMM TASK 27-51-63-400-801
         b) Re-connect connector D14360.
         c) Re-install the ACAU. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.
         d) Do the “Repair Confirmation - Modulation Panels Do Not Move to Cruise Position” at the end of this Fault Isolation Procedure.
      4) If there is no continuity, then do these steps:
         a) Repair the wiring between pin 28 of connector D458B and connector D14360.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799 POST SB 737-27-1290 OR POST SB 737-27-1304 (Continued)

b) Re-connect connector D14360.

c) Re-install the ACAU. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

d) Do the “Repair Confirmation - Modulation Panels Do Not Move to Cruise Position” at the end of this Fault Isolation Procedure.

(c) If there is no continuity, then continue.

OMR 701-799

(8) For a problem in the right ram air inlet system, do this check of the circuit from the ACAU to the flap/slat electronics unit, M1746:

(a) Do a continuity check of the wiring between pin 28 of connector D458A and a structural ground.

(b) If there is continuity, then do these steps:

1) Remove the flap/slat electronics unit, M1746. This is the task: Flap/Slat Electronics Unit Removal, AMM TASK 27-51-01-000-801.

2) Do a check of the wiring between pin 28 of connector D458A and pin C11 of connector D728A.

3) If the wiring check indicates a short to ground, then do these steps:

a) Repair the wiring.

b) Re-install the ACAU. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801

c) Re-install the flap/slat electronics unit, M1746. This is the task: Flap/Slat Electronics Unit Installation, AMM TASK 27-51-01-400-801.

d) Do the “Repair Confirmation - Modulation Panels Do Not Move to Cruise Position” at the end of this Fault Isolation Procedure.

4) If the wiring check does not indicate a problem, then do these steps:

a) Replace the flat/slat electronics unit, M1746. These are the tasks:

- Flap/Slat Electronics Unit Removal, AMM TASK 27-51-01-000-801
- Flap/Slat Electronics Unit Installation, AMM TASK 27-51-01-400-801

b) Re-install the ACAU. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

c) Do the “Repair Confirmation - Modulation Panels Do Not Move to Cruise Position” at the end of this Fault Isolation Procedure.

(c) If there is no continuity, then continue.

(9) Replace the ACAU, M324. These are the tasks:

- Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801
- Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801

(a) Do the “Repair Confirmation - Modulation Panels Do Not Move to Cruise Position” at the end of this Fault Isolation Procedure.

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OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999; 737-800 OR 737-900

J. Fault Isolation Procedure - Modulation Panels Do Not Move to Cruise Position

OMR 815, 816, 818-826, 901-999

(1) Do this check of the ram air temperature sensor:

(a) Open these access panels:

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

(b) Disconnect the connector D3916 (left) or D3918 (right) from the ram air temperature sensor, T481 (left) or T482 (right).

(c) If the RAM DOOR FULL OPEN light goes off and the modulation panels close, then do these steps:

1) Replace the ram air temperature sensor, T481 (left) or T482 (right). These are the tasks:

- Ram Air Control Temperature Sensor - Removal, AMM TASK 21-51-50-000-801
- Ram Air Control Temperature Sensor - Installation, AMM TASK 21-51-50-400-801

(d) Do the BITE test on the left pack/zone temperature controller M1442 if you replaced the sensor for the aft passenger compartment zone.

1) Do the “Repair Confirmation - Modulation Panels Do Not Move to Cruise Position” at the end of this Fault Isolation Procedure.

(e) If the RAM DOOR FULL OPEN light stays on and the inlet modulation doors close, then do these steps:

1) Re-connect the connector D3916 (left) or D3918 (right) to the ram air temperature sensor, T481 (left) or T482 (right).

2) Replace the ram air inlet actuator, M2795 (left) or M2797 (right). These are the tasks:

- Ram Air Inlet Actuator Removal, AMM TASK 21-51-21-000-801,
- Ram Air Inlet Actuator Installation, AMM TASK 21-51-21-400-801

3) Do the “Repair Confirmation - Modulation Panels Do Not Move to Cruise Position” at the end of this Fault Isolation Procedure.

(f) If the RAM DOOR FULL OPEN light stays on and the modulation panels do not close, then do this step and continue:

1) Re-connect the connector D3916 (left) or D3918 (right) to the ram air temperature sensor.

(2) Do this check for 28 VDC at the forward ram air actuator:

(a) Disconnect electrical connector D14386 (left) or D14394 (right) from the ram air actuator M2795 (left) or M2797 (right).

(b) Do a check for 28 VDC between pins 4 and 16 of connector D14386 (left) or connector D14394.

(c) If there is 28 VDC at pin 4, then replace the ram air inlet actuator, M2795 (left) or M2797 (right). These are the tasks:

- Ram Air Inlet Actuator Removal, AMM TASK 21-51-21-000-801,

ECCN 9E991 BOEING PROPRIETARY - See title page for details

- Ram Air Inlet Actuator Installation, AMM TASK 21-51-21-400-801.

1) Do the “Repair Confirmation - Modulation Panels Do Not Move to Cruise Position” at the end of this Fault Isolation Procedure.

(d) If 28 VDC is not present, then do this step and continue:

1) Re-connect electrical connector D14386 (left) or D14394 (right).

(3) For a problem in the left ram air inlet system, do this check of the circuit from the ACAU No. 1 to the trailing edge flaps up switch, S1051:

(a) Do a continuity check of the wiring between pin 33 of connector D458B and a structural ground.

(b) If there is continuity, then do these steps:

1) Disconnect connector D14360 in the main wheel well.

2) Do a continuity check between pin F of connector D14360 and a structural ground.

3) If there is continuity, then do these steps:

a) Replace the trailing edge flaps up switch, S1051. These are the tasks:

- Flap Limit Switch Removal, AMM TASK 27-51-63-000-801
- Flap Limit Switch Installation, AMM TASK 27-51-63-400-801

b) Re-connect connector D14360.

c) Re-install the ACAU. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

d) Do the “Repair Confirmation - Modulation Panels Do Not Move to Cruise Position” at the end of this Fault Isolation Procedure.

4) If there is no continuity, then do these steps:

a) Repair the wiring between pin 33 of connector D458B and connector D14360.

b) Re-connect connector D14360.

c) Re-install the ACAU. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

d) Do the “Repair Confirmation - Modulation Panels Do Not Move to Cruise Position” at the end of this Fault Isolation Procedure.

(c) If there is no continuity, then continue.

(4) For a problem in the right ram air inlet system, do this check of the circuit from the ACAU No. 2 to the flap/slat electronics unit, M1746:

(a) Do a continuity check of the wiring between pin 33 of connector D10002B and a structural ground.

(b) If there is continuity, then do these steps:

1) Remove the flap/slat electronics unit, M1746. This is the task: Flap/Slat Electronics Unit Removal, AMM TASK 27-51-01-000-801.

2) Do a check of the wiring between pin 33 of connector D10002B and pin C11 of connector D728A.

3) If the wiring check indicates a short to ground, then do these steps:

a) Repair the wiring.

b) Re-install the ACAU. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801

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OMR 815, 816, 818-826, 901-999 (Continued)

c) Re-install the flap/slat electronics unit, M1746. This is the task: Flap/Slat Electronics Unit Installation, AMM TASK 27-51-01-400-801.

d) Do the “Repair Confirmation - Modulation Panels Do Not Move to Cruise Position” at the end of this Fault Isolation Procedure.

4) If the wiring check does not indicate a problem, then do these steps:

a) Replace the flat/slat electronics unit, M1746. These are the tasks:

- Flap/Slat Electronics Unit Removal, AMM TASK 27-51-01-000-801
- Flap/Slat Electronics Unit Installation, AMM TASK 27-51-01-400-801

b) Re-install the ACAU. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

c) Do the “Repair Confirmation - Modulation Panels Do Not Move to Cruise Position” at the end of this Fault Isolation Procedure.

(c) If there is no continuity, then continue.

(5) Replace ACAU No. 1, M324, or ACAU No. 2, M1455. These are the tasks:

- Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801
- Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801

(a) Do the “Repair Confirmation - Modulation Panels Do Not Move to Cruise Position” at the end of this Fault Isolation Procedure.

OMR 802, 805, 807, 809, 810, 812, 813

(6) Do this check of the ram air temperature sensor:

(a) Open these access panels:

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

(b) Disconnect the connector D3916 (left) or D3918 (right) from the ram air temperature sensor, T481 (left) or T482 (right).

(c) If the RAM DOOR FULL OPEN light goes off and the modulation panels close, then do these steps:

1) Replace the ram air temperature sensor, T481 (left) or T482 (right). These are the tasks:

- Ram Air Control Temperature Sensor - Removal, AMM TASK 21-51-50-000-801
- Ram Air Control Temperature Sensor - Installation, AMM TASK 21-51-50-400-801

(d) Do the BITE test on the left pack/zone temperature controller M1442 if you replaced the sensor for the aft passenger compartment zone.

1) Do the “Repair Confirmation - Modulation Panels Do Not Move to Cruise Position” at the end of this Fault Isolation Procedure.

2) If the RAM DOOR FULL OPEN light does not stay off, then continue.

(e) If the RAM DOOR FULL OPEN light stays on and the inlet modulation doors close, then do these steps:

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OMR 802, 805, 807, 809, 810, 812, 813 (Continued)

1) Re-connect the connector D3916 (left) or D3918 (right) to the ram air temperature sensor, T481 (left) or T482 (right).
2) Replace the ram air actuator, M326 (left) or M327 (right). These are the tasks:
   - Ram Air Inlet Actuator Removal, AMM TASK 21-51-21-000-801
   - Ram Air Inlet Actuator Installation, AMM TASK 21-51-21-400-801
3) Do the “Repair Confirmation - Modulation Panels Do Not Move to Cruise Position” at the end of this Fault Isolation Procedure.

(f) If the RAM DOOR FULL OPEN light stays on and the modulation panels do not close, then do this step and continue:
1) Re-connect the connector D3916 (left) or D3918 (right) to the ram air temperature sensor.

(7) Do this check for 115 VAC at the ram air actuator:
(a) Disconnect connector D966 (left) or D968 (right) from the ram air actuator, M326 (left) or M327 (right).
(b) Do a check for 115 VAC between pins 10 and 11 (ground) of connector D966 (left) or D968 (right).
(c) If there is 115 VAC at pin 10, then do these steps:
1) Replace the ram air actuator, M326 (left) or M327 (right). These are the tasks
   - Ram Air Inlet Actuator Removal, AMM TASK 21-51-21-000-801
   - Ram Air Inlet Actuator Installation, AMM TASK 21-51-21-400-801
2) Do the “Repair Confirmation - Modulation Panels Do Not Move to Cruise Position” at the end of this Fault Isolation Procedure.
(d) If there is not 115 VAC, then do this step and continue:
1) Re-connect connector D966 (left) or D968 (right) to the ram air actuator.

(8) Do this check for 115 VAC at the pack/zone temperature controller:
(a) Remove the Pack/zone temperature controller No. 1, M1442 or No. 2, M1443. This is the task: Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801.
(b) Do a check for 115 VAC between pins B15 and A15 (ground) of connector D3858B (No. 1) or D3860B (No. 2).
(c) If there is 115 VAC at pin B15, then do these steps:
1) Install a new pack/zone temperature controller. This is the task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.
2) Do the “Repair Confirmation - Modulation Panels Do Not Move to Cruise Position” at the end of this Fault Isolation Procedure.
(d) If there is not 115 VAC at pin B15, then continue.

(9) Do this check of the ground circuit wiring for the pack/zone temperature controller:
(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C00399	AIR CONDITIONING RAM AIR MOD LEFT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 802, 805, 807, 809, 810, 812, 813 (Continued)

(Continued)

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	4	C00400	AIR CONDITIONING RAM AIR MOD RIGHT

(b) Do a check of the wiring between connector D3858B (No. 1) or D3860B (No. 2), pin A15 and the ground GD3303-AC (No. 1) or GD3304-AC (No. 2).

(c) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-install the pack/zone temperature controller. This is the task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.

3) Close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C00399	AIR CONDITIONING RAM AIR MOD LEFT
B	4	C00400	AIR CONDITIONING RAM AIR MOD RIGHT

4) Do the “Repair Confirmation - Modulation Panels Do Not Move to Cruise Position” at the end of this Fault Isolation Procedure.

(d) If you do not find a problem with the ground wiring, then continue:

(10) Do this check of the wiring between ACAU No. 1 (left) or ACAU No. 2 (right) and the pack/zone temperature controller 1, M1442 or the pack/zone temperature controller 2, M1443:

(a) Remove ACAU No 1 or No 2, M324 or M1445. This is the task: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801.

(b) Remove the Pack/zone temperature controller No. 1, M1442 or No. 2, M1443. This is the task: Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801.

(c) Do a check for an open circuit between these pins:

	RAM AIR SYSTEM	
	ACAU CONNECTOR	PACK/ZONE CONTROLLER CONNECTOR
LEFT	D458A	D3858B
	pin 12	pin A13
	pin 13	pin A12

	RAM AIR SYSTEM	
	ACAU CONNECTOR	PACK/ZONE CONTROLLER CONNECTOR
RIGHT	D10002A	D3860B
	pin 12	pin A13
	pin 13	pin A12

(d) If there is an open circuit, then do these steps:

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OMR 802, 805, 807, 809, 810, 812, 813 (Continued)

1) Repair the wiring.
2) Re-install the ACAU. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.
3) Re-install the pack/zone temperature controller. This is the task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.
4) Do the “Repair Confirmation - Modulation Panels Do Not Move to Cruise Position” at the end of this Fault Isolation Procedure.

(e) If there is continuity, then do this step and continue:

1) Re-install the pack/zone temperature controller. This is the task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.

OMR 802, 805, 807, 809, 810 PRE SB 737-27-1290 AND PRE SB 737-27-1304

(11) For a problem in the left ram air inlet system, do this check of the circuit from the ACAU No. 1 to the trailing edge flaps up switch, S1051:

(a) Do a continuity check of the wiring between pin 33 of connector D458B and a structural ground.

(b) If there is continuity, then do these steps:

1) Disconnect connector D46040J in the main wheel well.
2) Do a continuity check between pin 14 of connector D46040P and a structural ground.
3) If there is continuity, then do these steps:
   a) Replace the trailing edge flaps up switch, S1051. These are the tasks:
      - Flap Limit Switch Removal, AMM TASK 27-51-63-000-801
      - Flap Limit Switch Installation, AMM TASK 27-51-63-400-801
   b) Re-connect connector D46040.
   c) Re-install the ACAU. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.
   d) Do the “Repair Confirmation - Modulation Panels Do Not Move to Cruise Position” at the end of this Fault Isolation Procedure.
4) If there is no continuity, then do these steps:
   a) Repair the wiring between pin 33 of connector D458B and connector D46040J.
   b) Re-connect connector D46040.
   c) Re-install the ACAU. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.
   d) Do the “Repair Confirmation - Modulation Panels Do Not Move to Cruise Position” at the end of this Fault Isolation Procedure.

(c) If there is no continuity, then continue.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 812, 813; OMR 802, 805, 807, 809, 810 POST SB 737-27-1290 OR POST SB 737-27-1304

(12) For a problem in the left ram air inlet system, do this check of the circuit from the ACAU No. 1 to the trailing edge flaps up switch, S1051:

(a) Do a continuity check of the wiring between pin 33 of connector D458B and a structural ground.

(b) If there is continuity, then do these steps:

1) Disconnect connector D14360 in the main wheel well.

2) Do a continuity check between pin F of connector D14360 and a structural ground.

3) If there is continuity, then do these steps:

a) Replace the trailing edge flaps up switch, S1051. These are the tasks:

- Flap Limit Switch Removal, AMM TASK 27-51-63-000-801
- Flap Limit Switch Installation, AMM TASK 27-51-63-400-801

b) Re-connect connector D14360.

c) Re-install the ACAU. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

d) Do the “Repair Confirmation - Modulation Panels Do Not Move to Cruise Position” at the end of this Fault Isolation Procedure.

4) If there is no continuity, then do these steps:

a) Repair the wiring between pin 33 of connector D458B and connector D14360.

b) Re-connect connector D14360.

c) Re-install the ACAU. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

d) Do the “Repair Confirmation - Modulation Panels Do Not Move to Cruise Position” at the end of this Fault Isolation Procedure.

(c) If there is no continuity, then continue.

OMR 802, 805, 807, 809, 810, 812, 813

(13) For a problem in the right ram air inlet system, do this check of the circuit from the ACAU No. 2 to the flap/slat electronics unit, M1746:

(a) Do a continuity check of the wiring between pin 33 of connector D10002B and a structural ground.

(b) If there is continuity, then do these steps:

1) Remove the flap/slat electronics unit, M1746. This is the task: Flap/Slat Electronics Unit Removal, AMM TASK 27-51-01-000-801.

2) Do a check of the wiring between pin 33 of connector D10002B and pin C11 of connector D728A.

3) If the wiring check indicates a short to ground, then do these steps:

a) Repair the wiring.

b) Re-install the ACAU. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801

c) Re-install the flap/slat electronics unit, M1746. This is the task: Flap/Slat Electronics Unit Installation, AMM TASK 27-51-01-400-801.

d) Do the “Repair Confirmation - Modulation Panels Do Not Move to Cruise Position” at the end of this Fault Isolation Procedure.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 802, 805, 807, 809, 810, 812, 813 (Continued)

4) If the wiring check does not indicate a problem, then do these steps:
   a) Replace the flat/slat electronics unit, M1746. These are the tasks:
      • Flap/Slat Electronics Unit Removal, AMM TASK 27-51-01-000-801
      • Flap/Slat Electronics Unit Installation, AMM TASK 27-51-01-400-801
   b) Re-install the ACAU. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.
   c) Do the “Repair Confirmation - Modulation Panels Do Not Move to Cruise Position” at the end of this Fault Isolation Procedure.

(c) If there is no continuity, then continue.

(14) Replace ACAU No. 1, M324, or ACAU No. 2, M1455. These are the tasks:
   • Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801
   • Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801
   (a) Do the “Repair Confirmation - Modulation Panels Do Not Move to Cruise Position” at the end of this Fault Isolation Procedure.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999; 737-800 OR 737-900

(15) Replace the ACM. These are the tasks:
   • Air Cycle Machine Removal, AMM TASK 21-51-04-000-802-002
   • Air Cycle Machine Installation, AMM TASK 21-51-04-400-802-002
   (a) Do the “Repair Confirmation - Modulation Panels Do Not Move to Cruise Position” at the end of this Fault Isolation Procedure.

OMR ALL

K. Repair Confirmation - Modulation Panels Do Not Move to Cruise Position

(1) If the RAM DOOR FULL OPEN Light goes OFF and the Modulation Panels close, then you corrected the problem. Do these steps to complete the task:
   (a) Close the applicable access panel:

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

   (b) Put the airplane in the Ground Mode. This is the task: Return the Airplane to the Ground Mode, AMM TASK 32-09-00-860-802.

(2) If the RAM DOOR FULL OPEN Light does not go OFF and/or the Modulation Panels do not close, then continue the Fault Isolation Procedure at the subsequent step.

OMR 701-799; 737-600 OR 737-700

L. Fault Isolation Procedure - Indication Circuit Problem

OMR 701-799

(1) Do a check of the wiring between the RAM DOOR FULL OPEN light and the ram air actuator:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799 (Continued)

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C00399	AIR CONDITIONING RAM AIR MOD LEFT
A	8	C00265	AIR CONDITIONING RAM AIR MOD CONT L
B	4	C00400	AIR CONDITIONING RAM AIR MOD RIGHT
B	8	C00266	AIR CONDITIONING RAM AIR MOD CONT R

(b) Open this panel to get access to the actuator for the left ram air inlet system:

Number	Name/Location
191GL	Ram Air Actuator Panel - Forward

(c) Open this panel to get access to the actuator for the right ram air inlet system:

Number	Name/Location
191GR	Ram Air Actuator Panel - Forward

(d) Disconnect connector D966 (left) or D968 (right) from the ram air actuator.

(e) Lower the P5 forward overhead panel to get access to the electrical connector on the RAM DOOR FULL OPEN light, L12 (left) or L13 (right).

(f) Disconnect connector D646 (left) or D680 (right) from the RAM DOOR FULL OPEN light, L12 (left) or L13 (right).

(g) Do a check of the wiring between these pins:

	RAM AIR SYSTEM	
	RAM DOOR FULL OPEN LIGHT CONN	**RAM AIR ACTUATOR CONNECTOR**
LEFT	**D646**	**D966**
	pin 28	pin 7

	RAM AIR SYSTEM	
	RAM DOOR FULL OPEN LIGHT CONN	**RAM AIR ACTUATOR CONNECTOR**
RIGHT	**D680**	**D968**
	pin 3	pin 7

1) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-connect connector D966 (left) or D968 (right) to the ram air actuator, M326 (left) or M327 (right).

c) Re-connect connector D646 (left) or D680 (right) to the RAM DOOR FULL OPEN light, L12 (left) or the RAM DOOR FULL OPEN light, L13 on the back of the P5-10 Air Conditioning panel.

d) Raise the P5 forward overhead panel into position and secure the fasteners.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799 (Continued)

e) Close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C00399	AIR CONDITIONING RAM AIR MOD LEFT
A	8	C00265	AIR CONDITIONING RAM AIR MOD CONT L
B	4	C00400	AIR CONDITIONING RAM AIR MOD RIGHT
B	8	C00266	AIR CONDITIONING RAM AIR MOD CONT R

f) Do the “Repair Confirmation - Indication Circuit Problem” at the end of this Fault Isolation Procedure.

2) If you do not find a problem with the wiring, then replace the ram air actuator. These are the tasks:

- Ram Air Inlet Actuator Removal, AMM TASK 21-51-21-000-801
- Ram Air Inlet Actuator Installation, AMM TASK 21-51-21-400-801

a) Re-connect connector D646 (left) or D680 (right) to the RAM DOOR FULL OPEN light, L12 (left) or the RAM DOOR FULL OPEN light, L13 on the back of the P5-10 Air Conditioning panel.

b) Raise the P5 forward overhead panel into position and secure the fasteners.

c) Close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C00399	AIR CONDITIONING RAM AIR MOD LEFT
B	4	C00400	AIR CONDITIONING RAM AIR MOD RIGHT

d) Do the “Repair Confirmation - Indication Circuit Problem” at the end of this Fault Isolation Procedure.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999; 737-800 OR 737-900

M. Fault Isolation Procedure - Indication Circuit Problem

OMR 815, 816, 818-826, 901-999

(1) Do a check of the wiring between the RAM DOOR FULL OPEN light and the ram air actuator:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	8	C00555	AIR CONDITIONING RAM AIR MOD CONT L
B	8	C01178	AIR CONDITIONING RAM AIR MOD CONT R
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L

(b) Open this panel to get access to the actuator for the left ram air inlet system:

Number	Name/Location
191GL	Ram Air Actuator Panel - Forward

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 815, 816, 818-826, 901-999 (Continued)

(c) Open this panel to get access to the actuator for the right ram air inlet system:

Number	Name/Location
191GR	Ram Air Actuator Panel - Forward

(d) Lower the P5 forward overhead panel to get access to the electrical connector on the RAM DOOR FULL OPEN light, L12 (left) or L13 (right).

(e) Disconnect connector D646 (left) or D680 (right) from the RAM DOOR FULL OPEN light, L12 (left) or L13 (right).

(f) Do a check of the wiring between these pins:

RAM AIR SYSTEM

	RAM DOOR FULL OPEN LIGHT CONN		RAM AIR ACTUATOR CONNECTOR
LEFT	**D646**		**D966**
	pin28	. . . . .	pin 7

RAM AIR SYSTEM

	RAM DOOR FULL OPEN LIGHT CONN		RAM AIR ACTUATOR CONNECTOR
RIGHT	**D680**		**D968**
	pin 3	. . . . .	pin 7

1) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-connect connector D966 (left) or D968 (right) to the ram air actuator, M326 (left) or M327 (right).

c) Re-connect connector D646 (left) or D680 (right) to the RAM DOOR FULL OPEN light, L12 (left) or the RAM DOOR FULL OPEN light, L13 on the back of the P5-10 Air Conditioning panel.

d) Raise the P5 forward overhead panel into position and secure the fasteners.

e) Close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	8	C00555	AIR CONDITIONING RAM AIR MOD CONT L
B	8	C01178	AIR CONDITIONING RAM AIR MOD CONT R

f) If the RAM DOOR FULL OPEN light goes out, then you corrected the fault.

g) Close these access panels:

Number	Name/Location
191GL	Ram Air Actuator Panel - Forward
191GR	Ram Air Actuator Panel - Forward

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 815, 816, 818-826, 901-999 (Continued)

2) If you do not find a problem with the wiring, then replace the ram air actuator. These are the tasks:
   - Ram Air Inlet Actuator Removal, AMM TASK 21-51-21-000-801
   - Ram Air Inlet Actuator Installation, AMM TASK 21-51-21-400-801

   a) Re-connect connector D646 (left) or D680 (right) to the RAM DOOR FULL OPEN light, L12 (left) or the RAM DOOR FULL OPEN light, L13 on the back of the P5-10 Air Conditioning panel.

   b) Raise the P5 forward overhead panel into position and secure the fasteners.

   c) Close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	8	C00555	AIR CONDITIONING RAM AIR MOD CONT L
B	8	C01178	AIR CONDITIONING RAM AIR MOD CONT R

   d) Do the “Repair Confirmation - Indication Circuit Problem” at the end of this Fault Isolation Procedure.

(g) Do a check of the wiring between these pins:

	RAM AIR SYSTEM		
	P5-10 Panel	Left Ram Air Inlet Actuator	
L12	D646	D14386	
	pin 28	pin 11	Continuity

	RAM AIR SYSTEM		
	P5-10 Panel	Right Ram Air Inlet Actuator	
RIGHT	D680	D14394	
	Pin 3	Pin 11	Continuity

1) If you find a problem with the wiring, then do these steps:

   a) Repair the wiring.

   b) Re-connect connector D14386 (left) or D14394 (right) to the ram air actuator, M2795 (left) or M2797 (right).

   c) Re-connect connector D646 (left) or D680 (right) to the RAM DOOR FULL OPEN light, L12 (left) or the RAM DOOR FULL OPEN light, L13 (right) on the back of the P5-10 Air Conditioning panel.

   d) Raise the P5 forward overhead panel into position and secure the fasteners.

   e) Close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	8	C00555	AIR CONDITIONING RAM AIR MOD CONT L
B	8	C01178	AIR CONDITIONING RAM AIR MOD CONT R
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 815, 816, 818-826, 901-999 (Continued)

(Continued)

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L

f) Do the “Repair Confirmation - Indication Circuit Problem” at the end of this Fault Isolation Procedure.

2) If you do not find a problem with the wiring, then replace the ram air inlet actuator. These are the tasks:

- Ram Air Inlet Actuator Removal, AMM TASK 21-51-21-000-801
- Ram Air Inlet Actuator Installation, AMM TASK 21-51-21-400-801

a) Re-connect connector D646 (left) or D680 (right) to the RAM DOOR FULL OPEN light, L12 (left) or the RAM DOOR FULL OPEN light, L13 (right) on the back of the P5-10 Air Conditioning panel.

b) Raise the P5 forward overhead panel into position and secure the fasteners.

c) Close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	8	C00555	AIR CONDITIONING RAM AIR MOD CONT L
B	8	C01178	AIR CONDITIONING RAM AIR MOD CONT R
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L

d) Do the “Repair Confirmation - Indication Circuit Problem” at the end of this Fault Isolation Procedure.

OMR 802, 805, 807, 809, 810, 812, 813

(2) Do a check of the wiring between the RAM DOOR FULL OPEN light and the ram air actuator:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C00399	AIR CONDITIONING RAM AIR MOD LEFT
A	8	C00555	AIR CONDITIONING RAM AIR MOD CONT L
B	4	C00400	AIR CONDITIONING RAM AIR MOD RIGHT
B	8	C01178	AIR CONDITIONING RAM AIR MOD CONT R
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L

(b) Open this panel to get access to the actuator for the left ram air inlet system:

Number	Name/Location
191GL	Ram Air Actuator Panel - Forward

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 802, 805, 807, 809, 810, 812, 813 (Continued)

(c) Open this panel to get access to the actuator for the right ram air inlet system:

Number	Name/Location
191GR	Ram Air Actuator Panel - Forward

(d) Disconnect connector D966 (left) or D968 (right) from the ram air actuator.

(e) Lower the P5 forward overhead panel to get access to the electrical connector on the RAM DOOR FULL OPEN light, L12 (left) or L13 (right).

(f) Disconnect connector D646 (left) or D680 (right) from the RAM DOOR FULL OPEN light, L12 (left) or L13 (right).

(g) Do a check of the wiring between these pins:

RAM AIR SYSTEM

	RAM DOOR FULL OPEN LIGHT CONN		RAM AIR ACTUATOR CONNECTOR
LEFT	**D646**		**D966**
	pin28	. . . . .	pin 7

RAM AIR SYSTEM

	RAM DOOR FULL OPEN LIGHT CONN		RAM AIR ACTUATOR CONNECTOR
RIGHT	**D680**		**D968**
	pin 3	. . . . .	pin 7

1) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-connect connector D966 (left) or D968 (right) to the ram air actuator, M326 (left) or M327 (right).

c) Re-connect connector D646 (left) or D680 (right) to the RAM DOOR FULL OPEN light, L12 (left) or the RAM DOOR FULL OPEN light, L13 on the back of the P5-10 Air Conditioning panel.

d) Raise the P5 forward overhead panel into position and secure the fasteners.

e) Close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C00399	AIR CONDITIONING RAM AIR MOD LEFT
A	8	C00555	AIR CONDITIONING RAM AIR MOD CONT L
B	4	C00400	AIR CONDITIONING RAM AIR MOD RIGHT
B	8	C01178	AIR CONDITIONING RAM AIR MOD CONT R

f) Do the “Repair Confirmation - Indication Circuit Problem” at the end of this Fault Isolation Procedure.

2) If you do not find a problem with the wiring, then replace the ram air actuator. These are the tasks:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 802, 805, 807, 809, 810, 812, 813 (Continued)

- Ram Air Inlet Actuator Removal, AMM TASK 21-51-21-000-801
- Ram Air Inlet Actuator Installation, AMM TASK 21-51-21-400-801

a) Re-connect connector D646 (left) or D680 (right) to the RAM DOOR FULL OPEN light, L12 (left) or the RAM DOOR FULL OPEN light, L13 on the back of the P5-10 Air Conditioning panel.

b) Raise the P5 forward overhead panel into position and secure the fasteners.

c) Close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C00399	AIR CONDITIONING RAM AIR MOD LEFT
A	8	C00555	AIR CONDITIONING RAM AIR MOD CONT L
B	4	C00400	AIR CONDITIONING RAM AIR MOD RIGHT
B	8	C01178	AIR CONDITIONING RAM AIR MOD CONT R

d) Do the “Repair Confirmation - Indication Circuit Problem” at the end of this Fault Isolation Procedure.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999; 737-800 OR 737-900

(3) Replace the ACM. These are the tasks:

- Air Cycle Machine Removal, AMM TASK 21-51-04-000-802-002
- Air Cycle Machine Installation, AMM TASK 21-51-04-400-802-002

(a) Do the “Repair Confirmation - Indication Circuit Problem” at the end of this Fault Isolation Procedure.

OMR ALL

N. Repair Confirmation - Indication Circuit Problem

(1) If the RAM DOOR FULL OPEN Light goes OFF, then you corrected the problem. Do these steps to complete the task:

(a) Close these access panels:

Number	Name/Location
191GL	Ram Air Actuator Panel - Forward
191GR	Ram Air Actuator Panel - Forward

(b) Put the airplane in the Ground Mode. This is the task: Return the Airplane to the Ground Mode, AMM TASK 32-09-00-860-802.

(2) If the RAM DOOR FULL OPEN Light does not go OFF, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

804. RAM DOOR FULL OPEN Light is Not On, Airplane is On the Ground - Fault Isolation

A. Description

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799, 802, 805, 807, 809, 810, 812, 813

(1) The RAM DOOR FULL OPEN light on the P5-10 air conditioning panel should be on when the airplane is on the ground. In addition, the ram air inlet modulations panels should be fully open and the deflector door should be fully extended. This is the correct configuration for these components when the airplane is on the ground.

OMR 815, 816, 818-826, 901-999

(2) The RAM DOOR FULL OPEN light on the P5-10 air conditioning panel should be on when the airplane is on the ground. In addition, the ram air inlet modulations panels should be fully open, the deflector doors should be fully extended and the ram air exit louvers should be fully open. This is the correct configuration for these components when the airplane is on the ground.

OMR ALL

(3) The light normally will come on when the air/ground logic retracts the ram air actuator which closes a switch in the actuator to provide a ground for the light.

(4) If the master dim and test system does not provide 28 VDC to the light or if the light is burned out, the light will also fail to come on.

(5) There are two RAM DOOR FULL OPEN lights, one for the left ram air inlet modulation doors and one for the right ram air inlet modulation doors.

OMR 815, 816, 818-826, 901-999

(6) There are two smart ram air door actuators (inlet and exit SRADAs) on both the left and right ram air systems. Both the inlet and exit SRADAs incorporate four indication lights to provide a visual display of LRU faults of the ram air system. The continuous BIT detects and displays the fault status of the ram air temperature sensor, the inlet SRADA and the exit SRADA. The fault indication is provided when the SRADAs are powered and a valid weight-on-wheels (WOW) signal is present. If there is no fault condition present, a NO LRU FAULT light is illuminated green and the three LRU lights are off. If there is a fault present, the NO LRU FAULT light is off and the light associated with the failed LRU is illuminated red.

OMR 701-799

B. Possible Causes

(1) RAM DOOR FULL OPEN light, L12 (left) or L13 (right)

(2) Ram air actuator, M326 (left) or M327 (right)

(3) Air conditioning accessory unit, M324

(4) Proximity switch electronics unit (PSEU), E11

(5) Wiring problem

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

C. Possible Causes

(1) RAM DOOR FULL OPEN light, L12 (left) or L13 (right)

OMR 802, 805, 807, 809, 810, 812, 813

(2) Ram air actuator, M326 (left) or M327 (right)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 815, 816, 818-826, 901-999

(3) Ram Air Actuators

(a) Left Forward Ram Air Actuator, M2795

(b) Right Forward Ram Air Actuator, M2797

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(4) Air conditioning accessory unit, M324 (left) or M1455 (right)

(5) Proximity switch electronics unit (PSEU), M2061

(6) Wiring problem

OMR 701-799

D. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	11	C00133	INDICATOR MASTER DIM DIM/TST CONT
E	11	C00313	INDICATOR MASTER DIM SECT 1

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C00399	AIR CONDITIONING RAM AIR MOD LEFT
A	8	C00265	AIR CONDITIONING RAM AIR MOD CONT L
B	4	C00400	AIR CONDITIONING RAM AIR MOD RIGHT
B	8	C00266	AIR CONDITIONING RAM AIR MOD CONT R

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

E. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	11	C00313	INDICATOR MASTER DIM SECT 1

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
OMR 802, 805, 807, 809, 810, 812, 813			
A	4	C00399	AIR CONDITIONING RAM AIR MOD LEFT
OMR 815, 816, 818-826, 901-999			
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
A	8	C00555	AIR CONDITIONING RAM AIR MOD CONT L
OMR 802, 805, 807, 809, 810, 812, 813			
B	4	C00400	AIR CONDITIONING RAM AIR MOD RIGHT
OMR 815, 816, 818-826, 901-999			
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 815, 816, 818-826, 901-999 (Continued)

(Continued)

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
B	8	C01178	AIR CONDITIONING RAM AIR MOD CONT R

OMR 701-799

F. Related Data

(1) (SSM 21-51-14)

(2) (SSM 21-51-24)

(3) (SSM 33-18-11)

(4) (SSM 33-18-36)

(5) (WDM 21-51-14)

(6) (WDM 21-51-24)

(7) (WDM 33-18-36)

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

G. Related Data

(1) (SSM 21-51-14)

(2) (WDM 21-51-14)

(3) (SSM 21-51-24)

(4) (WDM 21-51-14)

(5) (SSM 33-18-36)

(6) (WDM 33-18-36)

OMR ALL

H. Initial Evaluation

(1) If the RAM DOOR FULL OPEN light is on, then there was an intermittent fault.

(2) If the RAM DOOR FULL OPEN light is not on, put the LIGHTS TEST/BRT/DIM switch on the P1-3 captain's instrument panel to TEST.

(a) If the RAM DOOR FULL OPEN light comes on, then do the Fault Isolation Procedure - Ram Air Actuator Circuit Problem below.

(b) If the RAM DOOR FULL OPEN light does not come on, then do the Fault Isolation Procedure - Indication Circuit Problem below.

OMR 701-799

I. Fault Isolation Procedure - Ram Air Actuator Circuit Problem

(1) Do this check for 115 VAC to the ram air actuator:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C00399	AIR CONDITIONING RAM AIR MOD LEFT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799 (Continued)

(Continued)

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	8	C00265	AIR CONDITIONING RAM AIR MOD CONT L
B	4	C00400	AIR CONDITIONING RAM AIR MOD RIGHT
B	8	C00266	AIR CONDITIONING RAM AIR MOD CONT R

(b) Get access to the left ram air actuator:

1) Open this access panel:

Number	Name/Location
191GL	Ram Air Actuator Panel - Forward

(c) Get access to the right ram air actuator:

1) Open this access panel:

Number	Name/Location
191GR	Ram Air Actuator Panel - Forward

(d) Disconnect the connector D966 (left) or D968 (right) from the ram air actuator.

(e) Close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C00399	AIR CONDITIONING RAM AIR MOD LEFT
A	8	C00265	AIR CONDITIONING RAM AIR MOD CONT L
B	4	C00400	AIR CONDITIONING RAM AIR MOD RIGHT
B	8	C00266	AIR CONDITIONING RAM AIR MOD CONT R

(f) Do a check for 115 VAC between pin 1 of connector D966 (left) or D968 (right) and structure ground.

(g) If there is 115 VAC between pin 1 and structure ground, then do to the Fault Isolation Procedure - Indication Circuit Problem.

(h) If there is not 115 VAC between pin 1 and structure ground, then continue.

(2) Do this check of the ram air actuator ground circuit wiring:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C00399	AIR CONDITIONING RAM AIR MOD LEFT
B	4	C00400	AIR CONDITIONING RAM AIR MOD RIGHT

(b) Do a check of the wiring between pin 11 of connector D966 (left) or D968 (right) and the ground GD68-AC (left) or GD72-AC (right).

(c) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-connect connector D966 (left) or D968 (right) to the ram air actuator.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799 (Continued)

3) Close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C00399	AIR CONDITIONING RAM AIR MOD LEFT
B	4	C00400	AIR CONDITIONING RAM AIR MOD RIGHT

4) If the RAM DOOR FULL OPEN light comes on, then you corrected the problem.

5) Close this access panel:

Number	Name/Location
191GL	Ram Air Actuator Panel - Forward

or open this access panel:

Number	Name/Location
191GR	Ram Air Actuator Panel - Forward

as applicable.

(d) If you do not find a problem with the wiring, then continue.

(3) Do this check of the wiring from the ACAU to the ram air actuator, M326 (left) or M327 (right):

(a) Remove the ACAU. To remove the ACAU, do this task: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801.

(b) Do a check for an open circuit between these pins:

	RAM AIR SYSTEM	
	ACAU CONNECTOR	**RAM AIR ACTUATOR CONNECTOR**
LEFT	**D458B**	**D966**
	pin 32	pin 1

	RAM AIR SYSTEM	
	ACAU CONNECTOR	**RA AIR ACTUATOR CONNECTOR**
RIGHT	**D458A**	**D968**
	pin 32	pin 1

(c) If there is an open circuit, then do these steps:

1) Repair the wiring.

2) Re-install the ACAU. To install it, do this task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

3) Re-connect connector D966 (left) or D968 (right) to the ram air actuator.

4) If the RAM DOOR FULL OPEN light comes on, then you corrected the fault. Close

Number	Name/Location
191GL	Ram Air Actuator Panel - Forward

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799 (Continued)

or open this access panel:

Number	Name/Location
191GR	Ram Air Actuator Panel - Forward

as applicable.

(d) If there is continuity, then do these steps and continue:

1) Re-connect connector D966 (left) or D968 (right) to the ram air actuator.

2) Close this access panel:

Number	Name/Location
191GL	Ram Air Actuator Panel - Forward

or open this access panel:

Number	Name/Location
191GR	Ram Air Actuator Panel - Forward

as applicable.

(4) Do this check of the air/ground logic circuit for the air conditioning accessory unit (ACAU), M324:

(a) For the left system, do a continuity check from pin 44 of connector D458B to a structural ground.

(b) For the right system, do a continuity check from pin 7 of connector D458A to a structural ground.

(c) If there is no continuity, then do these steps:

1) Do this task: Air/Ground Relay Fault - Fault Isolation, 32-09 TASK 804.

2) Do the corrective action for any fault found.

3) Re-install the ACAU. To install the ACAU, do this task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

4) If the RAM DOOR FULL OPEN light comes on, then you corrected the fault.

(d) If there is continuity, then continue.

(5) Do this check for voltage to the air conditioning accessory unit (ACAU), M324:

(a) Do a check for 115 VAC between pin 15 of connector D458B (left) or pin 11 of connector D458A (right) and structure ground.

(b) Do a check for 28 VDC between pin 41 of connector D458B (left) or D458A (right) and structure ground.

(c) If there is voltage, then do these steps:

1) Install a new ACAU. To install the ACAU, do this task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

2) If the RAM DOOR FULL OPEN light comes on, then you corrected the fault.

(d) If there is no voltage, then continue.

(6) Do this check for 115 VAC at the circuit breaker:

(a) Make sure there is 115 VAC at the load side of circuit breaker C399 (left) or C400 (right) on the P6-4 panel.

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OMR 701-799 (Continued)

1) If there is 115 VAC, then do these steps:

a) Repair the wiring between the applicable circuit breaker and the connector for the ACAU:

	RAM AIR SYSTEM	
	CIRCUIT BREAKER	**ACAU CONNECTOR**
LEFT	**C00399**	**D458B**
	term 2	pin 15

	RAM AIR SYSTEM	
	CIRCUIT BREAKER	**ACAU CONNECTOR**
RIGHT	**C00400**	**D458A**
	term 2	pin 11

b) Re-install the ACAU. To install it, do this task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

c) If the RAM DOOR FULL OPEN light comes on, then you corrected the fault.

2) If there is not 115 VAC, then do these steps:

a) Do this task: DIST/BUS FAULT For GCU 1 - Fault Isolation, 24-21 TASK 815.

b) Re-install the ACAU. To install it, do this task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

c) If the RAM DOOR FULL OPEN light comes on, then you corrected the fault.

(7) Do this check for 28 VDC at the circuit breaker:

(a) Make sure there is 28 VDC at the load side of circuit breaker C265 (left) or C266 (right) on the P6-4 panel.

1) If there is 28 VDC, then do these steps:

a) Repair the wiring between the applicable circuit breaker and the connector for the ACAU:

	RAM AIR SYSTEM	
	CIRCUIT BREAKER	**ACAU CONNECTOR**
LEFT	**C00265**	**D458B**
	term L	pin 15

	RAM AIR SYSTEM	
	CIRCUIT BREAKER	**ACAU CONNECTOR**
RIGHT	**C00266**	**D458A**
	term L	pin 11

b) Re-install the ACAU. To install it, do this task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799 (Continued)

c) If the RAM DOOR FULL OPEN light comes on, then you corrected the fault.

2) If there is not 28 VDC, then do these steps:

a) Do this task: DIST/BUS FAULT For GCU 1 - Fault Isolation, 24-21 TASK 815.

b) Re-install the ACAU. To install it, do this task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

c) If the RAM DOOR FULL OPEN light comes on, then you corrected the fault.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

J. Fault Isolation Procedure - Ram Air Actuator Circuit Problem

OMR 802, 805, 807, 809, 810, 812, 813

(1) Do this check for 115 VAC to the ram air actuator:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C00399	AIR CONDITIONING RAM AIR MOD LEFT
A	8	C00555	AIR CONDITIONING RAM AIR MOD CONT L
B	4	C00400	AIR CONDITIONING RAM AIR MOD RIGHT
B	8	C01178	AIR CONDITIONING RAM AIR MOD CONT R

(b) Open this access panel:

Number	Name/Location
191GL	Ram Air Actuator Panel - Forward

to get access to the left ram air actuator.

(c) Open this access panel:

Number	Name/Location
191GR	Ram Air Actuator Panel - Forward

to get access to the right ram air actuator.

(d) Disconnect the connector D966 (left) or D968 (right) from the ram air actuator.

(e) Close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C00399	AIR CONDITIONING RAM AIR MOD LEFT
A	8	C00555	AIR CONDITIONING RAM AIR MOD CONT L
B	4	C00400	AIR CONDITIONING RAM AIR MOD RIGHT
B	8	C01178	AIR CONDITIONING RAM AIR MOD CONT R

(f) Do a check for 115 VAC between pin 1 of connector D966 (left) or D968 (right) and structure ground.

(g) If there is 115 VAC between pin 1 and structure ground, then do to the Fault Isolation Procedure - Indication Circuit Problem.

(h) If there is not 115 VAC between pin 1 and structure ground, then continue.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 815, 816, 818-826, 901-999

(2) Do this check for 28 VDC to the ram air inlet actuator:

(a) Open these circuit breakers and install safety tags:

1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	8	C01178	AIR CONDITIONING RAM AIR MOD CONT R
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L

(b) Open this access panel:

Number	Name/Location
191GL	Ram Air Actuator Panel - Forward

(c) Open this access panel:

Number	Name/Location
191GR	Ram Air Actuator Panel - Forward

(d) Disconnect the electrical connector D14386 (left) or D14394 (right) from the ram air inlet actuator.

(e) Close these circuit breakers:

Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	8	C00555	AIR CONDITIONING RAM AIR MOD CONT L
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L

(f) Do a check for 28 VDC between pin 4 of connector D14386 (left) or D14394 (right) and a structural ground.

(g) If there is 28 VDC between pin 4 and a structural ground, then do the Fault Isolation Procedure - Indication Circuit Problem.

(h) If there is not 28 VDC between pin 4 and a structural ground, then continue.

OMR 802, 805, 807, 809, 810, 812, 813

(3) Do this check of the ram air actuator ground circuit wiring:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C00399	AIR CONDITIONING RAM AIR MOD LEFT
B	4	C00400	AIR CONDITIONING RAM AIR MOD RIGHT

(b) Do a check of the wiring between pin 11 of connector D966 (left) or D968 (right) and the ground GD68-AC (left) or GD72-AC (right).

(c) If you find a problem with the wiring, then do these steps:

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OMR 802, 805, 807, 809, 810, 812, 813 (Continued)

1) Repair the wiring.
2) Re-connect connector D966 (left) or D968 (right) to the ram air actuator.
3) Close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C00399	AIR CONDITIONING RAM AIR MOD LEFT
B	4	C00400	AIR CONDITIONING RAM AIR MOD RIGHT

4) If the RAM DOOR FULL OPEN light comes on, then you corrected the fault.
5) Close this access panel:

Number	Name/Location
191GL	Ram Air Actuator Panel - Forward

or open this access panel:

Number	Name/Location
191GR	Ram Air Actuator Panel - Forward

as applicable.

(d) If you do not find a problem with the wiring, then continue.

OMR 815, 816, 818-826, 901-999

(4) Do this check of the ground circuit wiring of the ram air inlet actuator:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	8	C00555	AIR CONDITIONING RAM AIR MOD CONT L
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	8	C01178	AIR CONDITIONING RAM AIR MOD CONT R
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L

(b) Do a check of the wiring between pin 16 of connector D14386 (left) or D14394 (right) and the ground GD712-DC (left) or GD716-DC (right).

(c) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.
2) Re-connect connector D14386 (left) or D14394 (right) to the ram air inlet actuator.
3) Close these circuit breakers:

Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	8	C00555	AIR CONDITIONING RAM AIR MOD CONT L

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OMR 815, 816, 818-826, 901-999 (Continued)

(Continued)

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	8	C01178	AIR CONDITIONING RAM AIR MOD CONT R
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L

4) If the RAM DOOR FULL OPEN light comes on, then you corrected the fault.

5) Close these access panels:

Number	Name/Location
191GL	Ram Air Actuator Panel - Forward
191GR	Ram Air Actuator Panel - Forward

(d) If you do not find a problem with the wiring, then continue.

OMR 802, 805, 807, 809, 810, 812, 813

(5) Do this check of the wiring from the left or right ACAU, M324 (left) or M1455 (right) to the ram air actuator, M326 (left) or M327 (right):

(a) Remove the left or right ACAU. To remove the ACAU, do this task: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801.

(b) Do a check for an open circuit between these pins:

	RAM AIR SYSTEM	
	ACAU CONNECTOR	RAM AIR ACTUATOR CONNECTOR
LEFT	D458A	D966
	pin 36	pin 1

	RAM AIR SYSTEM	
	ACAU CONNECTOR	RAM AIR ACTUATOR CONNECTOR
RIGHT	D10002A	D968
	pin 36	pin 1

(c) If there is an open circuit, then do these steps:

1) Repair the wiring.

2) Re-install the ACAU. To install it, do this task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

3) Re-connect connector D966 (left) or D968 (right) to the ram air actuator.

4) If the RAM DOOR FULL OPEN light comes on, then you corrected the fault. Close

Number	Name/Location
191GL	Ram Air Actuator Panel - Forward

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OMR 802, 805, 807, 809, 810, 812, 813 (Continued)

or open this access panel:

Number	Name/Location
191GR	Ram Air Actuator Panel - Forward

as applicable.

(d) If there is continuity, then do these steps and continue:

1) Re-connect connector D966 (left) or D968 (right) to the ram air actuator.

2) Close this access panel:

Number	Name/Location
191GL	Ram Air Actuator Panel - Forward

or open this access panel:

Number	Name/Location
191GR	Ram Air Actuator Panel - Forward

as applicable.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(6) Do this check of the air/ground logic circuit for the air conditioning accessory unit (ACAU), M324 (left), or M1455 (right):

(a) For the left system, do a continuity check from pin 35 of connector D458A to a structural ground.

(b) For the right system, do a continuity check from pin 35 of connector D10002A to a structural ground.

(c) If there is no continuity, then do these steps:

1) Do this task: Air/Ground Relay Fault - Fault Isolation, 32-09 TASK 804.

2) Do the corrective action for any fault found.

3) Re-install the ACAU. To install the ACAU, do this task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

4) If the RAM DOOR FULL OPEN light comes on, then you corrected the fault.

(d) If there is continuity, then continue.

OMR 815, 816, 818-826, 901-999

(7) Do a check of the air/ground logic circuit input signal to the ram air inlet actuator:

(a) Remove ACAU No.1 (M324) or No.2 (M1455), as applicable. Do this task: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801.

(b) Do a continuity check between pin 1 of connector D14386 (left) or D14394 (right) to a structural ground.

(c) If there is no continuity, then do these steps:

1) Do this task: Air/Ground Relay Fault - Fault Isolation, 32-09 TASK 804.

2) Do the corrective action for any fault found.

3) Re-install the ACAU. Do this task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

4) If the RAM DOOR FULL OPEN light comes on, then you corrected the fault.

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OMR 815, 816, 818-826, 901-999 (Continued)

(d) If there is continuity, then continue.

OMR 802, 805, 807, 809, 810, 812, 813

(8) Do this check for 115 VAC to the air conditioning accessory unit (ACAU), M324 (left), or M1455 (right):

(a) Do a check for 115 VAC between pin 49 of connector D458A (left) or D10002A (right) and structure ground.

(b) If there is not 115 VAC at pin 49, then do these steps:

1) Do a check for 115 VAC at the load side of circuit breaker C399 (left) or C400 (right) on the P6-4 panel.

2) If there is 115 VAC, then do these steps:

a) Repair the wiring between the applicable circuit breaker and the connector for the ACAU, M324 (left) or M1455 (right):

	RAM AIR SYSTEM	
	CIRCUIT BREAKER	**ACAU CONNECTOR**
LEFT	**C00399**	**D458A**
	term L	pin 49

	RAM AIR SYSTEM	
	CIRCUIT BREAKER	**ACAU CONNECTOR**
RIGHT	**C00400**	**D10002A**
	term L	pin 49

b) Re-install the ACAU. To install it, do this task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

c) If the RAM DOOR FULL OPEN light comes on, then you corrected the fault.

3) If there is not 115 VAC, then do these steps:

a) Do this task: DIST/BUS FAULT For GCU 1 - Fault Isolation, 24-21 TASK 815.

b) Re-install the ACAU. To install it, do this task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

c) If the RAM DOOR FULL OPEN light comes on, then you corrected the fault.

(c) If there is 115 VAC at pin 49, then continue.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(9) Do this check for 28 VDC to the air conditioning accessory unit (ACAU), M324 (left), or M1455 (right):

(a) Do a check for 28 VDC between pin 59 of connector D458A (left) or D10002A (right) and structure ground.

(b) If there is not 28 VDC at pin 59, then do these steps:

1) Make sure there is 28 VDC at the load side of circuit breaker C555 (left) or C1178 (right) on the P6-4 panel.

2) If there is 28 VDC, then do these steps:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999 (Continued)

a) Repair the wiring between the applicable circuit breaker and the connector for the ACAU, M324 (left) or M1455 (right):

	RAM AIR SYSTEM	
	CIRCUIT BREAKER	**ACAU CONNECTOR**
LEFT	**C00555**	**D458A**
	term L	pin 59

	RAM AIR SYSTEM	
	CIRCUIT BREAKER	**ACAU CONNECTOR**
RIGHT	**C01178**	**D10002A**
	term L	pin 59

b) Re-install the ACAU. To install it, do this task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

c) If the RAM DOOR FULL OPEN light comes on, then you corrected the fault.

3) If there is not 28 VDC, then do these steps:

a) Do this task: DIST/BUS FAULT For GCU 1 - Fault Isolation, 24-21 TASK 815.

b) Re-install the ACAU. To install it, do this task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

c) If the RAM DOOR FULL OPEN light comes on, then you corrected the fault.

(c) If there is 28 VDC at pin 59, then continue.

(10) Install a new ACAU. To install the ACAU, do this task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

(a) If the RAM DOOR FULL OPEN light comes on, then you corrected the fault.

OMR 701-799

K. Fault Isolation Procedure - Indication Circuit Problem

(1) If both the left and right RAM DOOR FULL OPEN lights do not come on, then do this step:

(a) Replace the P5-10 air conditioning panel.

(b) If the RAM DOOR FULL OPEN lights come on, then you corrected the fault.

(c) If the RAM DOOR FULL OPEN lights do not come on, then continue.

(2) Do this check for 28 VDC to the RAM DOOR FULL OPEN light:

(a) Release the fasteners that hold the P5-10 forward overhead panel in position and let the panel swing down.

(b) Disconnect connector D646 from the P5-10 air conditioning panel.

(c) Do a check for 28 VDC between pin 5 of connector D646 and structure ground.

(d) If there is 28 VDC between pin 5 of connector D646 and ground, then do these steps:

1) Replace the P5-10 air conditioning panel.

2) If the RAM DOOR FULL OPEN light comes on, then you corrected the fault.

3) If the RAM DOOR FULL OPEN light does not come on, then continue.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799 (Continued)

(e) If there is not 28 VDC between pin 5 of connector D646 and ground, then continue.

(3) Do this check for 28 VDC at the circuit breaker that supplies power to the RAM DOOR FULL OPEN light:

(a) Make sure there is 28 VDC at the load side of circuit breaker C313 on the P6-3 panel.

(b) If there is 28 VDC, then do these steps:

1) Repair the wiring between the load side of circuit breaker C313 on the P6-3 panel and pin 5 of connector D646 on the P5-10 air conditioning panel.

2) Re-connect connector D646 to the P5-10 air conditioning panel.

3) If the RAM DOOR FULL OPEN light comes on, then you corrected the fault.

4) If the RAM DOOR FULL OPEN light does not come on, then continue.

(c) If there is not 28 VDC, then do these steps:

1) Do this task: DIST/BUS FAULT For GCU 1 - Fault Isolation, 24-21 TASK 815.

2) Re-connect connector D646 to the P5-10 air conditioning panel.

3) If the RAM DOOR FULL OPEN light comes on, then you corrected the fault.

(4) Do this check of the wiring between the RAM DOOR FULL OPEN light and the ram air actuator:

(a) Disconnect connector D680 (right) from the P5-10 air conditioning panel.

(b) Open this access panel:

Number	Name/Location
191GL	Ram Air Actuator Panel - Forward

to get access to the left ram air actuator.

(c) Open this access panel:

Number	Name/Location
191GR	Ram Air Actuator Panel - Forward

to get access to the right ram air actuator.

(d) Remove connector D966 (left) or D968 (right) from the ram air actuator.

(e) Do a check of the wiring between these pins:

	RAM AIR SYSTEM	
	P5-10 PANEL CONNECTOR	RAM AIR ACTUATOR CONNECTOR
LEFT	D646	D966
	pin 28	pin 7

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799 (Continued)

	RAM AIR SYSTEM	
	P5-10 PANEL CONNECTOR D680	RAM AIR ACTUATOR CONNECTOR D968
RIGHT	pin 3	pin 7

(f) If there is a problem with the wiring, then do these steps:

1) Repair the wiring.
2) Re-connect connector D966 (left) or D968 (right) to the ram air actuator.
3) Re-connect connector D646 (left) or D680 (right) to the P5-10 air conditioning panel
4) If the RAM DOOR FULL OPEN light comes on, then you corrected the fault.
5) Do these steps to complete the task:
   a) Close this access panel:

Number	Name/Location
191GR	Ram Air Actuator Panel - Forward

   or open this access panel:

Number	Name/Location
191GR	Ram Air Actuator Panel - Forward

   as applicable.
   b) Raise the P5 forward overhead panel into position and tighten the fasteners that hold it there.
6) Raise the P5 forward overhead panel into its normal position and secure the fasteners that hold it in place.
7) If the RAM DOOR FULL OPEN light comes on, then you corrected the fault.

(g) If there is continuity, then do these steps and continue:

1) Re-connect connector D646 (left) or D680 (right).
2) Raise the P5 forward overhead panel into its normal position and secure the fasteners that hold it in place.

(5) Do this check of the actuator ground wiring for the RAM DOOR FULL OPEN light:

(a) Do a check of the wiring between pin 8 of connector D966 (left) or D968 (right) and the ground GD68-DC (left) or GD72-DC (right).

(b) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.
2) Re-connect connector D966 (left) or D968 (right) to the actuator.
3) If the RAM DOOR FULL OPEN light comes on, then you corrected the fault.
4) Close this access panel:

Number	Name/Location
191GL	Ram Air Actuator Panel - Forward

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799 (Continued)

or open this access panel:

Number	Name/Location
191GR	Ram Air Actuator Panel - Forward

(c) If you do not find a problem with the wiring, then continue.

(6) Replace the ram air actuator M326 or M327.

These are the tasks:

Ram Air Inlet Actuator Removal, AMM TASK 21-51-21-000-801,

Ram Air Inlet Actuator Installation, AMM TASK 21-51-21-400-801.

(a) If the RAM DOOR FULL OPEN light comes on, then you corrected the fault.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

L. Fault Isolation Procedure - Indication Circuit Problem

(1) If both the left and right RAM DOOR FULL OPEN lights do not come on, then do this step:

(a) Replace the P5-10 air conditioning panel. To replace the P5-10 panel, do these tasks:

Air Conditioning Module Removal, AMM TASK 21-51-65-000-801,

Air Conditioning Module Installation, AMM TASK 21-51-65-400-801.

(b) If the RAM DOOR FULL OPEN lights come on, then you corrected the fault.

(c) If the RAM DOOR FULL OPEN lights do not come on, then continue.

(2) Do this check for 28 VDC to the RAM DOOR FULL OPEN light:

(a) Release the fasteners that hold the P5-10 forward overhead panel in position and let the panel swing down.

(b) Disconnect connector D646 from the P5-10 air conditioning panel.

(c) Do a check for 28 VDC between pin 5 of connector D646 on the ship's wiring and structure ground.

(d) If there is 28 VDC between pin 5 of connector D646 and ground, then do these steps:

1) Replace the P5-10 air conditioning panel. To replace the P5-10 panel, do these tasks:

Air Conditioning Module Removal, AMM TASK 21-51-65-000-801,

Air Conditioning Module Installation, AMM TASK 21-51-65-400-801.

2) If the RAM DOOR FULL OPEN light comes on, then you corrected the fault.

3) If the RAM DOOR FULL OPEN light does not come on, then continue.

(e) If there is not 28 VDC between pin 5 of connector D646 and ground, then continue.

(3) Do this check for 28 VDC at the circuit breaker that supplies power to the RAM DOOR FULL OPEN light:

(a) Make sure there is 28 VDC at the load side of circuit breaker C313 on the P6-3 panel.

(b) If there is 28 VDC, then do these steps:

1) Repair the wiring between the load side of circuit breaker C313 on the P6-3 panel and pin 5 of connector D646 on the P5-10 air conditioning panel.

2) Re-connect connector D646 to the P5-10 air conditioning panel.

3) If the RAM DOOR FULL OPEN light comes on, then you corrected the fault.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999 (Continued)

4) If the RAM DOOR FULL OPEN light does not come on, then continue.

(c) If there is not 28 VDC, then do these steps:

1) Do this task: DIST/BUS FAULT For GCU 1 - Fault Isolation, 24-21 TASK 815.

2) Re-connect connector D646 to the P5-10 air conditioning panel.

3) If the RAM DOOR FULL OPEN light comes on, then you corrected the fault.

(4) Do this check of the wiring between the RAM DOOR FULL OPEN light and the ram air actuator:

(a) Disconnect connector D646 (left) and connector D680 (right) from the P5-10 air conditioning panel.

(b) To get access to the left ram air actuator, do this step:

1) Open this access panel:

Number	Name/Location
191GL	Ram Air Actuator Panel - Forward

(c) To get access to the right ram air actuator, do this step:

1) Open this access panel:

Number	Name/Location
191GR	Ram Air Actuator Panel - Forward

OMR 802, 805, 807, 809, 810, 812, 813

(d) Remove connector D966 (left) or D968 (right) from the ram air actuator.

OMR 815, 816, 818-826, 901-999

(e) Remove connector D14386 (left) or D14394 (right) from the ram air inlet actuator.

OMR 802, 805, 807, 809, 810, 812, 813

(f) Do a check of the wiring between these pins:

	RAM AIR SYSTEM	
	P5-10 PANEL CONNECTOR	**RAM AIR ACTUATOR CONNECTOR**
LEFT	**D646**	**D966**
	pin 28	pin 7

	RAM AIR SYSTEM	
	P5-10 PANEL CONNECTOR	**RAM AIR ACTUATOR CONNECTOR**
RIGHT	**D680**	**D968**
	pin 3	pin 7

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OMR 815, 816, 818-826, 901-999

(g) Do a check of the wiring between these pins:

RAM AIR SYSTEM		
P5 PANEL CONNECTOR D646		**RAM AIR ACTUATOR CONNECTOR D14386**
Pin 28	. . . . .	Pin 11

RAM AIR SYSTEM		
P5-10 PANEL D680		**RAM AIR ACTUATOR D14394**
Pin 3	. . . . .	Pin 11

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(h) If there is a problem with the wiring, then do these steps:

1) Repair the wiring.

OMR 802, 805, 807, 809, 810, 812, 813

2) Re-connect connector D966 (left) or D968 (right) to the ram air actuator.

OMR 815, 816, 818-826, 901-999

3) Re-connect connector D14386 (left) or D14394 (right) to the ram air inlet actuator.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

4) Re-connect connector D646 (left) or D680 (right) to the P5-10 air conditioning panel

5) If the RAM DOOR FULL OPEN light comes on, then you corrected the fault.

6) Do these steps to complete the task:

a) Close this access panel:

Number	Name/Location
191GR	Ram Air Actuator Panel - Forward

or open this access panel:

Number	Name/Location
191GR	Ram Air Actuator Panel - Forward

as applicable.

b) Raise the P5 forward overhead panel into position and tighten the fasteners that hold it there.

7) Raise the P5 forward overhead panel into its normal position and secure the fasteners that hold it in place.

8) If the RAM DOOR FULL OPEN light comes on, then you corrected the fault.

(i) If there is continuity, then do these steps and continue:

1) Re-connect connector D646 (left) or D680 (right).

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OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999 (Continued)

2) Raise the P5 forward overhead panel into its normal position and secure the fasteners that hold it in place.

(5) Do this check of the actuator ground wiring for the RAM DOOR FULL OPEN light:

OMR 802, 805, 807, 809, 810, 812, 813

(a) Do a check of the wiring between pin 8 of connector D966 (left) or D968 (right) and the ground GD68-DC (left) or GD72-DC (right).

OMR 815, 816, 818-826, 901-999

(b) Do a check of the wiring between pin 24 of connector D14386 (left) or D14394 (right) and the ground GD710-ST (left) or GD714-ST (right).

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(c) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

OMR 802, 805, 807, 809, 810, 812, 813

2) Re-connect connector D966 (left) or D968 (right) to the actuator.

OMR 815, 816, 818-826, 901-999

3) Re-connect connector D14386 (left) or D14394 (right).

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

4) If the RAM DOOR FULL OPEN light comes on, then you corrected the fault.

5) Close this access panel:

Number	Name/Location
191GL	Ram Air Actuator Panel - Forward

or open this access panel:

Number	Name/Location
191GR	Ram Air Actuator Panel - Forward

(d) If you do not find a problem with the wiring, then continue.

OMR 802, 805, 807, 809, 810, 812, 813

(6) Replace the ram air actuator M326 or M327.

These are the tasks:

Ram Air Inlet Actuator Removal, AMM TASK 21-51-21-000-801,

Ram Air Inlet Actuator Installation, AMM TASK 21-51-21-400-801.

(a) If the RAM DOOR FULL OPEN light comes on, then you corrected the fault.

OMR 815, 816, 818-826, 901-999

(7) Replace the ram air inlet actuator M2795 (left) or M2797 (right) as follows:

(a) Do these tasks: Ram Air Inlet Actuator Removal, AMM TASK 21-51-21-000-801, Ram Air Inlet Actuator Installation, AMM TASK 21-51-21-400-801.

(b) If the RAM DOOR FULL OPEN light comes on, then you corrected the fault.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

801. Low Limit (35 °F) Controller BITE Procedure

A. General

(1) There are two identical 35° F Controllers, one for each pack. The controllers are located in the respective pack bay just outboard of the Water Separator.

(2) You use the rotary switch on the controller to do the 35° F Controller BITE.

(3) The BITE Test does an internal check of the 35° F Controller and the Line Replaceable Unit (LRU) Interface checks for the 35° F Control Valve and the 35° F Temperature Sensor. A Red NO GO light identifies a specific failure which is detected by the 35° F Controller.

B. BITE Procedure

(1) Do the BITE Procedure for the 35° F Controller as follows (Figure 201):

(a) Do these steps to do the BITE on the Left 35° F Controller:

1) Put the L PACK to the OFF position.

2) Open this access panel:

Number	Name/Location
192CL	ECS Access Door

(b) Do these steps to do the BITE on the Right 35° F Controller:

1) Put the R PACK to the OFF position.

2) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

(c) Push the GO and the NO GO lights on the controller.

1) Make sure that the lights come ON when you push them.

(d) Rotate the BITE Switch to position 1.

1) Make sure that the GO light comes ON.

(e) Rotate the BITE Switch to position 2.

1) Make sure that the GO light comes ON.

2) Make sure that the 35° F Control Valve goes to the fully open position.

(f) Rotate the BITE Switch to position 3.

1) Make sure that the GO light comes ON.

(g) Rotate the BITE Switch to position 4.

1) Make sure that the GO light comes ON.

2) Make sure that the 35° F Control Valve goes to the fully closed position.

(h) Rotate the BITE Switch to position 5.

1) Make sure that the GO light comes ON.

(i) Rotate the BITE Switch to the FLIGHT position.

(2) Refer to the table at the end of this task to find the applicable Fault Isolation Manual (FIM) Tasks for the maintenance messages below, which are indicated by the BITE Switch position:

(a) Red NO GO Light Indicator comes ON.

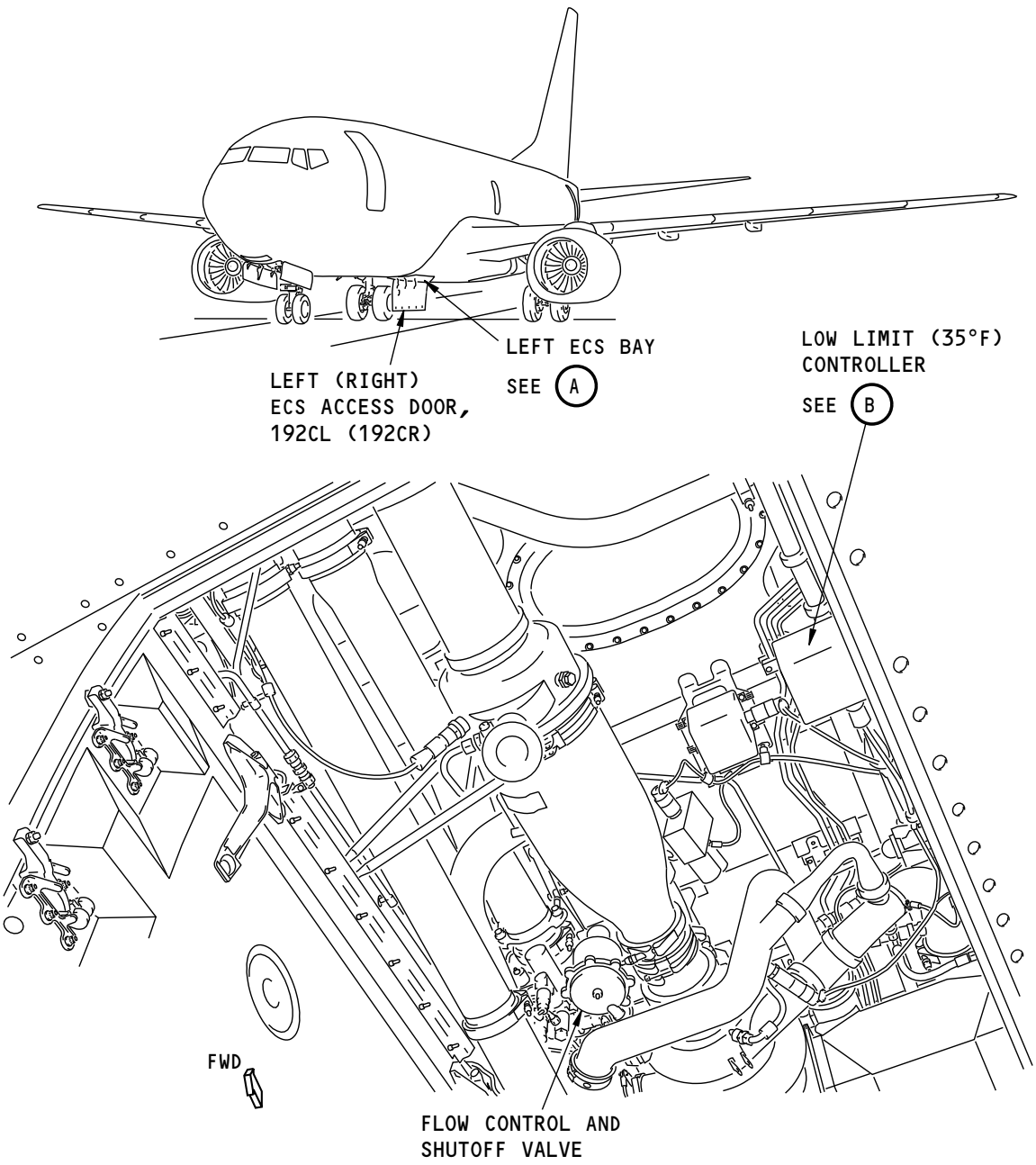
(b) Green GO, 35° F CONTROL VALVE NOT OPEN Light Indicator comes ON.

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(c) Green GO, 35° F CONTROL VALVE NOT CLOSED Light Indicator comes ON.

——— END OF TASK ———

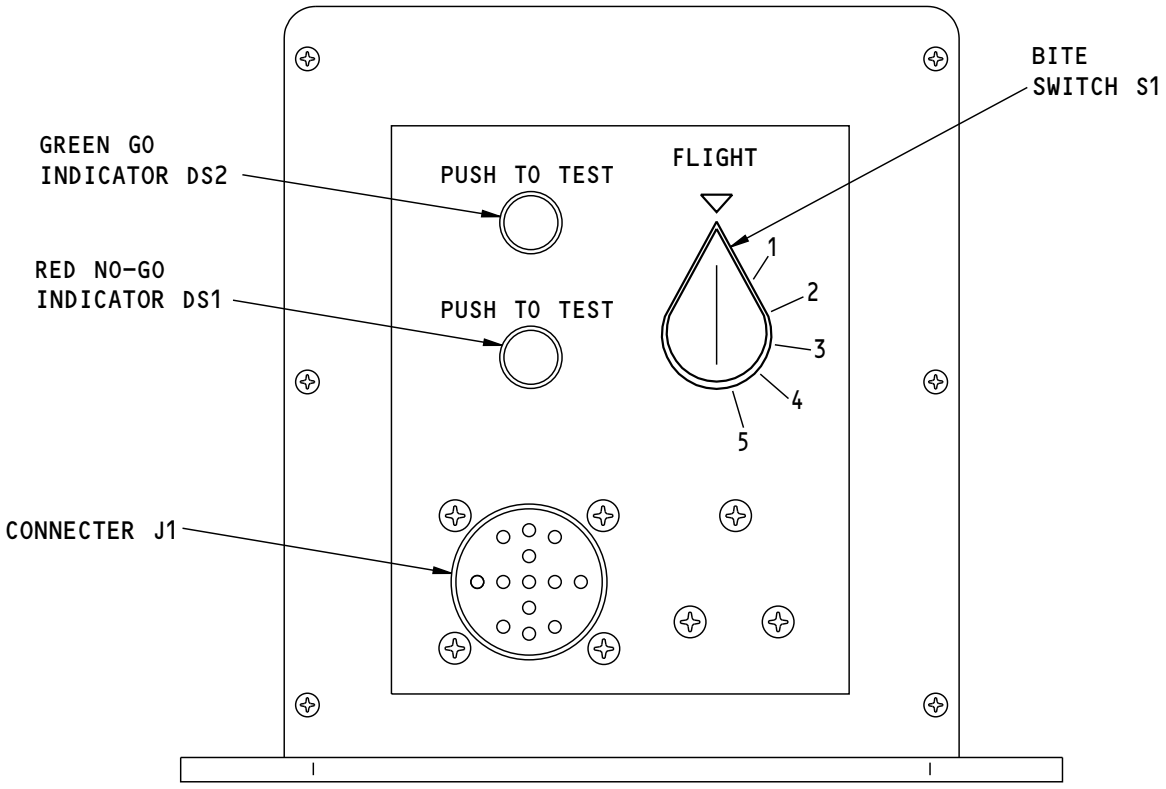
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L19743 S0006740653_V1

Low Limit 35° F Controller
Figure 201/21-52-00-990-803 (Sheet 1 of 2)

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L19744 S0006740654_V1

Low Limit 35° F Controller
Figure 201/21-52-00-990-803 (Sheet 2 of 2)

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LRU/SYSTEM	MAINTENANCE MESSAGE	GO TO FIM TASK
35 DEG CONT L	POSN 1 - NO GO	21-52 TASK 802
35 DEG CONT L	POSN 2 - GO, 35° F CONTROL VALVE NOT OPEN	21-52 TASK 812
35 DEG CONT L	POSN 2 - NO GO	21-52 TASK 802
35 DEG CONT L	POSN 3 - NO GO	21-52 TASK 802
35 DEG CONT L	POSN 4 - GO, 35° F CONTROL VALVE NOT CLOSED	21-52 TASK 813
35 DEG CONT L	POSN 4 - NO GO	21-52 TASK 802
35 DEG CONT L	POSN 5 - NO GO	21-52 TASK 810
35 DEG CONT R	POSN 1 - NO GO	21-52 TASK 802
35 DEG CONT R	POSN 2 - GO, 35° F CONTROL VALVE NOT OPEN	21-52 TASK 812
35 DEG CONT R	POSN 2 - NO GO	21-52 TASK 802
35 DEG CONT R	POSN 3 - NO GO	21-52 TASK 802
35 DEG CONT R	POSN 4 - GO, 35° F CONTROL VALVE NOT CLOSED	21-52 TASK 813
35 DEG CONT R	POSN 4 - NO GO	21-52 TASK 802
35 DEG CONT R	POSN 5 - NO GO	21-52 TASK 810

802. Left (Right) 35 °F Controller Failed - Fault Isolation

A. Description

(1) This task is for these Left (Right) 35 °F Controller maintenance messages:

(a) POSN 1 - NO GO

(b) POSN 2 - NO GO

(c) POSN 3 - NO GO

(d) POSN 4 - NO GO

B. Possible Causes

(1) Left (Right) 35 °F Controller, M223 (M224)

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	1	C00344	AIR CONDITIONING TEMP CONT 35 DEG F L
B	1	C00345	AIR CONDITIONING TEMP CONT 35 DEG F R

D. Related Data

(1) WDM 21-51-16

(2) WDM 21-51-26

(3) SSM 21-51-16

(4) SSM 21-51-26

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E. Initial Evaluation

(1) Do this task on the applicable Left (Right) 35 °F Controller: Low Limit (35 °F) Controller BITE Procedure, 21-52 TASK 801.

(a) If the maintenance message does not show, then there was an intermittent fault.

(b) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Replace Left (Right) 35 °F Controller, M223 (M224). These are the tasks:

- Low Limit (35 Degree F) Controller Removal, AMM TASK 21-51-31-000-801
- Low Limit (35 Degree F) Controller Installation, AMM TASK 21-51-31-400-801

(a) Do the Low Limit (35 °F) Controller BITE Procedure, 21-52 TASK 801 again.

1) If the maintenance message does not show, then you corrected the problem.

——— END OF TASK ———

810. Left (Right) 35 °F Temp Sensor Failed - Fault Isolation

A. Description

(1) This task is for this Left (Right) 35 °F Controller maintenance message:

(a) POSN 5 - NO GO

B. Possible Causes

(1) Left (Right) 35 °F Temperature Sensor, T430 (T431)

(2) Wiring

(3) Left (Right) 35 °F Controller, M223 (M224)

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	1	C00344	AIR CONDITIONING TEMP CONT 35 DEG F L
B	1	C00345	AIR CONDITIONING TEMP CONT 35 DEG F R

D. Related Data

(1) WDM 21-51-16

(2) WDM 21-51-26

(3) SSM 21-51-16

(4) SSM 21-51-26

E. Initial Evaluation

(1) Do this task on the applicable Left (Right) 35 °F Controller: Low Limit (35 °F) Controller BITE Procedure, 21-52 TASK 801.

(a) If the maintenance message does not show, then there was an intermittent fault.

(b) If the maintenance message shows, then do the Fault Isolation Procedure below.

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F. Fault Isolation Procedure

(1) Replace the applicable Left (Right) 35 °F Temperature Sensor, T430 (T431) in the Left (Right) Air Conditioning Equipment Bay. These are the tasks:

- Low Limit (35 Degree Fahrenheit) Temperature Sensor Removal, AMM TASK 21-51-32-000-801
- Low Limit (35 Degree Fahrenheit) Temperature Sensor Installation, AMM TASK 21-51-32-400-801

(a) Do the Repair Confirmation at the end of this task.

(2) Do this check of the wiring:

(a) Disconnect connector D706 (708) from the Left (Right) 35 °F Controller, M223 (M224).

(b) Disconnect connector D710 (712) from the Left (Right) 35 °F Temperature Sensor, T430 (T431).

(c) Measure the Resistance of the wires between the Left (Right) 35 °F Controller and the Left (Right) 35 °F Temperature Sensor as follows (WDM 21-51-16, WDM 21-51-26):

LEFT 35 °F TEMP SENSOR

L CNTRLR D706		L SENSOR D710	RESISTANCE
pin 6	. . . . .	pin 1	9.5 - 10.5 KOhms
pin 7	. . . . .	pin 1	1.7 - 1.9 KOhms
pin 6	. . . . .	pin 2	0 Ohms

RIGHT 35 °F TEMP SENSOR

R CNTRLR D708		R SENSOR D712	RESISTANCE
pin 6	. . . . .	pin 1	9.5 - 10.5 KOhms
pin 7	. . . . .	pin 1	1.7 - 1.9 KOhms
pin 6	. . . . .	pin 2	0 Ohms

(d) If you find a problem in the wiring, do these steps:

- If the Resistance between pins 6 and 1 is not as specified above, replace the Left (Right) 35 °F Control Parallel Resistor, R665 (R667) in the Left (Right) ECS Bay.
- If the Resistance between pins 7 and 1 is not as specified above, replace the Left (Right) 35 °F Control Series Resistor, R666 (R668) in the Left (Right) ECS Bay.
- If the Resistance between pins 6 and 2 is not as specified above, repair the applicable wiring.

1) Re-connect connector D706 (708) to the Left (Right) 35 °F Controller, M223 (M224).
2) Re-connect connector D710 (712) to the Left (Right) 35 °F Temperature Sensor, T430 (T431).
3) Do the Repair Confirmation at the end of this task.

(e) If you do not find a problem in the wiring, do these steps and then continue.

1) Re-connect connector D706 (708) to the Left (Right) 35 °F Controller, M223 (M224).
2) Re-connect connector D710 (712) to the Left (Right) 35 °F Temperature Sensor, T430 (T431).

(3) Install a new Left (Right) 35 °F Controller, M223 (M224). These are the tasks:

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- Low Limit (35 Degree F) Controller Removal, AMM TASK 21-51-31-000-801
- Low Limit (35 Degree F) Controller Installation, AMM TASK 21-51-31-400-801

(a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do this task on the applicable Left (Right) 35 °F Controller: Low Limit (35 °F) Controller BITE Procedure, 21-52 TASK 801.

(a) If the maintenance message does not show, then you corrected the problem.

(b) If the maintenance message still shows, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

812. Left (Right) 35 °F Control Valve Failed (Not Open) - Fault Isolation

A. Description

(1) This task is for this Left (Right) 35 °F Controller maintenance message:

(a) POSN 2 - GO, 35 °F CONTROL VALVE NOT OPEN

B. Possible Causes

(1) Left (Right) 35 °F Control Valve, V27 (V28)

(2) Wiring

(3) Left (Right) 35 °F Controller, M223 (M224)

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	1	C00344	AIR CONDITIONING TEMP CONT 35 DEG F L
B	1	C00345	AIR CONDITIONING TEMP CONT 35 DEG F R

D. Related Data

(1) WDM 21-51-16

(2) WDM 21-51-26

(3) SSM 21-51-16

(4) SSM 21-51-26

E. Initial Evaluation

(1) Do this task on the applicable Left (Right) 35 °F Controller: Low Limit (35 °F) Controller BITE Procedure, 21-52 TASK 801.

(a) If the maintenance message does not show, then there was an intermittent fault.

(b) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Replace the applicable Left (Right) 35 °F Control Valve, V27 (V28) in the Left (Right) Air Conditioning Equipment Bay. These are the tasks:

- Low Limit (35 Degree F) Valve Removal, AMM TASK 21-51-30-000-801
- Low Limit (35 Degree F) Valve Installation, AMM TASK 21-51-30-400-801

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(a) Do the Repair Confirmation at the end of this task.

(2) Do this check of the wiring:

(a) Disconnect connector D706 (708) from the Left (Right) 35 °F Controller, M223 (M224).

(b) Disconnect connector D702 (704) from the Left (Right) 35 °F Control Valve, V27 (V28).

(c) Examine the wiring between the Left (Right) 35 °F Controller and the Left (Right) 35 °F Control Valve as follows (WDM 21-51-16, WDM 21-51-26):

LEFT 35 °F CONTROL VALVE

L CNTRLR		L CTRL VLV
D706		D702
pin 2	. . . . .	pin 1
pin 3	. . . . .	pin 2

RIGHT 35 °F CONTROL VALVE

R CNTRLR		R CTRL VLV
D706		D704
pin 2	. . . . .	pin 1
pin 3	. . . . .	pin 2

(d) If you find a problem in the wiring, do these steps:

1) Repair the wiring.

2) Re-connect connector D706 (708) to the Left (Right) 35 °F Controller, M223 (M224).

3) Re-connect connector D702 (704) to the Left (Right) 35 °F Control Valve, V27 (V28).

4) Do the Repair Confirmation at the end of this task.

(e) If you do not find a problem in the wiring, do these steps and then continue.

1) Re-connect connector D706 (708) to the Left (Right) 35 °F Controller, M223 (M224).

2) Re-connect connector D702 (704) to the Left (Right) 35 °F Control Valve, V27 (V28).

(3) Install a new Left (Right) 35 °F Controller, M223 (M224). These are the tasks:

- Low Limit (35 Degree F) Controller Removal, AMM TASK 21-51-31-000-801
- Low Limit (35 Degree F) Controller Installation, AMM TASK 21-51-31-400-801

(a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do this task on the applicable Left (Right) 35 °F Controller: Low Limit (35 °F) Controller BITE Procedure, 21-52 TASK 801.

(a) If the maintenance message does not show, then you corrected the problem.

(b) If the maintenance message still shows, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

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813. Left (Right) 35 °F Control Valve Failed (Not Closed) - Fault Isolation

A. Description

(1) This task is for this Left (Right) 35 °F Controller maintenance message:

(a) POSN 4 - GO, 35 °F CONTROL VALVE NOT CLOSED.

B. Possible Causes

(1) Left (Right) 35 °F Control Valve, V27 (V28)

(2) Wiring

(3) Left (Right) 35 °F Controller, M223 (M224)

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	1	C00344	AIR CONDITIONING TEMP CONT 35 DEG F L
B	1	C00345	AIR CONDITIONING TEMP CONT 35 DEG F R

D. Related Data

(1) WDM 21-51-16

(2) WDM 21-51-26

(3) SSM 21-51-16

(4) SSM 21-51-26

E. Initial Evaluation

(1) Do this task on the applicable Left (Right) 35 °F Controller: Low Limit (35 °F) Controller BITE Procedure, 21-52 TASK 801.

(a) If the maintenance message does not show, then there was an intermittent fault.

(b) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Replace the applicable Left (Right) 35 °F Control Valve, V27 (V28) in the Left (Right) Air Conditioning Equipment Bay. These are the tasks:

- Low Limit (35 Degree F) Controller Removal, AMM TASK 21-51-31-000-801
- Low Limit (35 Degree F) Controller Installation, AMM TASK 21-51-31-400-801

(a) Do the Repair Confirmation at the end of this task.

(2) Do the electrical checks as follows (WDM 21-51-16, WDM 21-51-26):

(a) Disconnect connector D706 (708) from the Left (Right) 35 °F Controller, M223 (M224).

(b) Disconnect connector D702 (704) from the Left (Right) 35 °F Control Valve, V27 (V28).

(c) Examine the wiring between the Left (Right) 35 °F Controller and the Left (Right) 35 °F Control Valve:

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LEFT 35 °F CONTROL VALVE

L CONTROLLER D706		L CTRL VLV D702
pin 2	. . . . .	pin 1
pin 3	. . . . .	pin 2

RIGHT 35 °F CONTROL VALVE

R CONTROLLER D706		R CTRL VLV D704
pin 2	. . . . .	pin 1
pin 3	. . . . .	pin 2

(d) If you find a problem in the wiring, do these steps:

1) Repair the wiring.
2) Re-connect connector D706 (708) to the Left (Right) 35 °F Controller, M223 (M224).
3) Re-connect connector D702 (704) to the Left (Right) 35 °F Control Valve, V27 (V28).
4) Do the Repair Confirmation at the end of this task.

(e) If you do not find a problem in the wiring, do these steps and then continue.

1) Re-connect connector D706 (708) to the Left (Right) 35 °F Controller, M223 (M224).
2) Re-connect connector D702 (704) to the Left (Right) 35 °F Control Valve, V27 (V28).

(3) Install a new Left (Right) 35 °F Controller, M223 (M224). These are the tasks:

- Low Limit (35 Degree F) Controller Removal, AMM TASK 21-51-31-000-801
- Low Limit (35 Degree F) Controller Installation, AMM TASK 21-51-31-400-801

(a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do this task on the applicable Left (Right) 35 °F Controller: Low Limit (35 °F) Controller BITE Procedure, 21-52 TASK 801.

(a) If the maintenance message does not show, then you corrected the problem.

(b) If the maintenance message still shows, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

816. Left PACK TRIP OFF Light Illuminated and Will Not Reset - Fault Isolation

A. Description

(1) The left PACK TRIP OFF light on the P5-10 air conditioning module is on and will not reset.

B. Possible Causes

(1) Left compressor discharge overheat switch, S2

(2) Left turbine inlet overheat switch, S1

(3) Control cabin duct overheat switch, S3

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(4) Wiring problem

(5) Air conditioning accessory unit (left pack overheat relay, K8), M324

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	4	C00257	AIR CONDITIONING OVERHEAT

D. Related Data

(1) (SSM 21-51-15)

(2) (WDM 21-51-15)

E. Initial Evaluation

(1) Push the TRIP RESET switch on the air conditioning module on the P5 panel.

(a) If the PACK TRIP OFF light goes off, then there was an intermittent fault.

(b) If the PACK TRIP OFF light stays on, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Do these steps to do a check of the left compressor discharge overheat switch, S2:

(a) Open this access panel:

Number	Name/Location
192CL	ECS Access Door

(b) Disconnect connector D532 from the left compressor discharge overheat switch.

(c) Push the TRIP RESET switch.

(d) If the PACK TRIP OFF light goes out, then do these steps:

1) Replace the compressor discharge overheat switch, S2. These are the tasks:

- Compressor Discharge Overheat Switch Removal, AMM TASK 21-51-40-000-801,
- Compressor Discharge Overheat Switch Installation, AMM TASK 21-51-40-400-801.

2) Re-connect connector D532 to the left compressor discharge overheat switch, S2.

3) If the PACK TRIP OFF light is still off, then you corrected the fault. Do this step to complete the task:

a) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

(e) If the PACK TRIP OFF light is still on, then continue.

1) Re-connect connector D532 to the left compressor discharge overheat switch, S2.

(2) Do these steps to do a check of the left turbine inlet overheat switch, S1:

(a) Disconnect connector D530 from the left turbine inlet overheat switch, S1.

(b) Push the TRIP RESET switch.

(c) If the PACK TRIP OFF light goes out, then do these steps:

1) Replace the turbine inlet overheat switch, S1. These are the tasks:

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- Turbine Inlet Overheat Switch Removal, AMM TASK 21-51-41-000-801,
- Turbine Inlet Overheat Switch Installation, AMM TASK 21-51-41-400-801.

2) Re-connect connector D530 to the left turbine inlet overheat switch, S1.

3) If the PACK TRIP OFF light is still off, then you corrected the fault. Do this step to complete the task:

a) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

(d) If the PACK TRIP OFF light is still on, then continue.

1) Re-connect connector D530 to the left turbine inlet overheat switch, S1.

(3) Do these steps to do a check of the control cabin duct overheat switch, S3:

(a) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

(b) Disconnect connector D534 from the control cabin duct overheat switch, S3.

(c) Push the TRIP RESET switch.

(d) If the PACK TRIP OFF light goes out, then do these steps:

1) Replace the control cabin duct overheat switch, S3. These are the tasks:

- Pack Discharge Duct Overheat Switch Removal, AMM TASK 21-51-42-000-804-001,
- Pack Discharge Duct Overheat Switch Installation, AMM TASK 21-51-42-400-804-001.

2) Re-connect connector D534 to the control cabin duct overheat switch, S3.

3) If the PACK TRIP OFF light is still off, then you corrected the fault. Do these steps to complete the task:

a) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

b) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

(e) If the PACK TRIP OFF light stays on, then continue.

1) Re-connect connector D534 to the control cabin duct overheat switch, S3.

(4) Do this check of the wiring:

(a) Do this task: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801.

(b) Do a check for a short from pin 33 at connector D458B on the E4-1 shelf to structure ground (WDM 21-51-15).

(c) If there is a short to ground, then do these steps:

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1) Examine and repair the wiring between the air conditioning accessory unit, M324, and the left pack and control cabin duct overheat switches, S1, S2, and S3 (WDM 21-51-15).

2) Install the Air Conditioning Accessory Unit (ACAU). To install the ACAU, do this task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801

3) Push the PACK RESET switch.

4) If the PACK TRIP OFF light is off, then you corrected the fault. Do these steps to complete the task:

a) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

b) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

(d) If there is not a short to ground, then continue.

(5) Install a new ACAU. These are the tasks:

- Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801,
- Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

(a) Push the TRIP RESET switch.

(b) If the PACK TRIP OFF light is off, then you corrected the fault. Do these steps to complete the task:

1) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

2) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

——— END OF TASK ———

817. Right PACK TRIP OFF Light Illuminated and Will Not Reset - Fault Isolation

A. Description

(1) The right PACK TRIP OFF light on the P5-10 air conditioning module is on and will not reset.

B. Possible Causes

(1) Right compressor discharge overheat switch, S5

(2) Right turbine inlet overheat switch, S4

(3) Passenger cabin duct overheat switch, S6

(4) Wiring problem

(5) Air conditioning accessory unit (right pack overheat relay, K3), M324

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C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	4	C00257	AIR CONDITIONING OVERHEAT

D. Related Data

(1) (SSM 21-51-25)

(2) (WDM 21-51-25)

E. Initial Evaluation

(1) Push the TRIP RESET switch on the air conditioning module on the P5 panel.

(a) If the PACK TRIP OFF light goes off, then there was an intermittent fault.

(b) If the PACK TRIP OFF light stays on, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Do these steps to do a check of the right compressor discharge overheat switch, S5:

(a) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

(b) Disconnect connector D538 from the right compressor discharge overheat switch.

(c) Push the TRIP RESET switch.

(d) If the PACK TRIP OFF light goes out, then do these steps:

1) Replace the compressor discharge overheat switch, S5. These are the tasks:

- Compressor Discharge Overheat Switch Removal, AMM TASK 21-51-40-000-801,
- Compressor Discharge Overheat Switch Installation, AMM TASK 21-51-40-400-801.

2) Re-connect connector D538 to the right compressor discharge overheat switch, S5.

3) If the PACK TRIP OFF light is still off, then you corrected the fault. Do this step to complete the task:

Close this access panel:

Number	Name/Location
192CR	ECS Access Door

(e) If the PACK TRIP OFF light is still on, then continue.

1) Re-connect connector D538 to the right compressor discharge overheat switch, S5.

(2) Do these steps to do a check of the right turbine inlet overheat switch, S4:

(a) Disconnect connector D536 from the right turbine inlet overheat switch, S4.

(b) Push the TRIP RESET switch.

(c) If the PACK TRIP OFF light goes out, then do these steps:

1) Replace the turbine inlet overheat switch, S4. These are the tasks:

- Turbine Inlet Overheat Switch Removal, AMM TASK 21-51-41-000-801,
- Turbine Inlet Overheat Switch Installation, AMM TASK 21-51-41-400-801.

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2) Re-connect connector D536 to the right turbine inlet overheat switch, S4.

3) If the PACK TRIP OFF light is still off, then you corrected the fault. Do this step to complete the task:

a) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

(d) If the PACK TRIP OFF light is still on, then continue.

1) Re-connect connector D536 to the right turbine inlet overheat switch, S4.

(3) Do these steps to do a check of the passenger cabin duct overheat switch, S6:

(a) Remove the ceiling liner that is five windows forward of the overwing escape hatches.

1) To remove the liner, do this task: Ceiling Panel - Removal, AMM TASK 25-21-45-000-801.

(b) Disconnect connector D540 from the passenger cabin duct overheat switch, S6.

(c) Push the TRIP RESET switch.

(d) If the PACK TRIP OFF light goes out, then do these steps:

1) Replace the passenger cabin duct overheat switch, S6. These are the tasks:

- Pack Discharge Duct Overheat Switch Removal, AMM TASK 21-51-42-000-804-001,
- Pack Discharge Duct Overheat Switch Installation, AMM TASK 21-51-42-400-804-001.

2) Re-connect connector D540 to the passenger cabin duct overheat switch, S6.

3) If the PACK TRIP OFF light is still off, then you corrected the fault. Do these steps to complete the task:

a) Install the ceiling liner that is five windows forward of the overwing escape hatches.

To install the liner, do this task: Ceiling Panel - Installation, AMM TASK 25-21-45-400-801.

b) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

(e) If the PACK TRIP OFF light stays on, then continue.

1) Re-connect connector D540 to the passenger cabin duct overheat switch, S6.

(4) Do this check of the wiring:

(a) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

(b) Do this task: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801.

(c) Do a check for a short from pin 33 at connector D458A on the E4-1 shelf to structure ground (WDM 21-51-25).

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(d) If there is a short to ground, then do these steps:

1) Examine and repair the wiring between the air conditioning accessory unit, M324, and the right pack and passenger cabin duct overheat switches, S4, S5, and S6 (WDM 21-51-25).

2) Re-install the ACAU. To install the ACAU, do this task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801

3) Push the PACK RESET switch.

4) If the PACK TRIP OFF light is off, then you corrected the fault. Do these steps to complete the task:

a) Install the ceiling liner that is five windows forward of the overwing escape hatches.

To install the liner, do this task: Ceiling Panel - Installation, AMM TASK 25-21-45-400-801.

b) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

c) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

(e) If there is not a short to ground, then continue.

1) Install the ceiling liner that is five windows forward of the overwing escape hatches.

To install the liner, do this task: Ceiling Panel - Installation, AMM TASK 25-21-45-400-801.

2) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

(5) Install a new ACAU. These are the tasks:

- Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801,
- Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

(a) Push the TRIP RESET switch.

(b) If the PACK TRIP OFF light is off, then you corrected the fault. Do these steps to complete the task:

Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

818. Left PACK TRIP OFF Light On During Ground Operation and Stays OFF After Reset - Fault Isolation

A. Description

(1) The Left PACK TRIP OFF Light on the P5-10 air conditioning module comes ON during ground operation and stays off after reset.

B. Possible Causes

(1) Fan Bypass Check Valve Door in the Primary Plenum/Diffuser Assembly - damaged or missing

(a) An indication of this fault would be little or no airflow from the ram air exhaust duct outlet.

(2) Dirty heat exchangers

(3) Left Ram Air Door not fully open

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	4	C00257	AIR CONDITIONING OVERHEAT

D. Related Data

(1) SSM 21-51-15

(2) WDM 21-51-15

E. Initial Evaluation

(1) Operate the Left Air Conditioning Pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) If the PACK TRIP OFF Light comes ON, then do the Fault Isolation Procedure below.

(a) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(3) If the PACK TRIP OFF Light does not come ON, then there was an intermittent fault.

(a) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

F. Fault Isolation Procedure

(1) Do this check of the Fan Bypass Check Valve in the Primary Plenum/Diffuser Assembly:

(a) Remove conditioned air from the airplane. This is the task: Remove Conditioned Air from the Airplane, AMM TASK 21-00-00-800-802.

(b) Get access to the Fan Bypass Check Valve in the Primary Plenum/Diffuser Assembly as follows:

1) Open this access panel:

Number	Name/Location
192CL	ECS Access Door

(c) Open the access panel on the bottom of the Primary Plenum/Diffuser Assembly.

(d) Make sure that the Fan Bypass Check Valve Door is in place and that the door hinge is not damaged.

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(e) If the Fan Bypass Check Valve Door is gone or damaged, replace the Primary Plenum/Diffuser Assembly. These are the tasks:

- Heat Exchanger and Plenum/Diffuser Assembly - Removal, AMM TASK 21-51-03-000-802-001
- Heat Exchanger and Plenum/Diffuser Assembly - Installation, AMM TASK 21-51-03-400-801-001

(f) Make sure that the access panel is installed on the bottom of the Primary Plenum/Diffuser Assembly.

(g) Operate the left air conditioning pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(h) If the PACK TRIP OFF Light does not come ON, then you corrected the problem.

1) Remove conditioned air from the airplane. This is the task: Remove Conditioned Air from the Airplane, AMM TASK 21-00-00-800-802.

(i) If the PACK TRIP OFF Light comes ON, then continue.

(2) Look at the RAM DOOR FULL OPEN Light on the P5-10 Air Conditioning Panel.

(a) If the RAM DOOR FULL OPEN Light is not ON, do this task: RAM DOOR FULL OPEN Light is Not On, Airplane is On the Ground - Fault Isolation, 21-51 TASK 804.

1) If the task shows a problem with the Ram Air System, do the applicable corrective action.

2) Do the Repair Confirmation at the end of this task.

(b) If the RAM DOOR FULL OPEN Light is ON, then continue.

(3) Clean the Left Pack Heat Exchangers. This is the task: Heat Exchanger - Blowing Out Debris/Blockages with Nitrogen Gas/Compressed Air, AMM TASK 21-51-03-160-811.

(a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Operate the Left Air Conditioning Pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(a) If the PACK TRIP OFF Light does not come ON, then you corrected the problem.

1) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(b) If the PACK TRIP OFF Light comes ON, then continue the Fault Isolation Procedure at the subsequent step.

1) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

END OF TASK

819. Right PACK TRIP OFF Light On During Ground Operation and Stays OFF After Reset - Fault Isolation

A. Description

(1) The Right PACK TRIP OFF Light on the P5-10 air conditioning module comes ON during ground operation and stays off after reset.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

B. Possible Causes

(1) Fan Bypass Check Valve Door in the Primary Plenum/Diffuser Assembly - damaged or missing

(a) An indication of this condition would be little or no airflow from the Ram Air Exhaust Outlet.

(2) Dirty Heat Exchangers

(3) Right Ram Air Door not fully open

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	4	C00257	AIR CONDITIONING OVERHEAT

D. Related Data

(1) SSM 21-51-25

(2) WDM 21-51-25

E. Initial Evaluation

(1) Operate the Right Air Conditioning Pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) If the PACK TRIP OFF Light comes ON, then do the Fault Isolation Procedure below.

(a) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(3) If the PACK TRIP OFF Light does not come ON, then there was an intermittent fault.

(a) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

F. Fault Isolation Procedure

(1) Do this check of the Fan Bypass Check Valve in the Primary Plenum/Diffuser Assembly:

(a) Remove conditioned air from the airplane. This is the task: Remove Conditioned Air from the Airplane, AMM TASK 21-00-00-800-802.

(b) Get access to the Fan Bypass Check Valve in the Primary Plenum/Diffuser Assembly as follows:

1) Open this access panel:

Number	Name/Location
192CL	ECS Access Door

(c) Open the access panel on the bottom of the Primary Plenum/Diffuser Assembly.

(d) Make sure that the Fan Bypass Check Valve Door is in place and that the door hinge is not damaged.

(e) If the Fan Bypass Check Valve Door is gone or damaged, do these tasks to replace the Primary Plenum/Diffuser Assembly:

- Heat Exchanger and Plenum/Diffuser Assembly - Removal, AMM TASK 21-51-03-000-802-001
- Heat Exchanger and Plenum/Diffuser Assembly - Installation, AMM TASK 21-51-03-400-801-001

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(f) Make sure that the access panel is installed on the bottom of the Primary Plenum/Diffuser Assembly.

(g) Operate the Right Air Conditioning Pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(h) If the PACK TRIP OFF Light does not come ON, then you corrected the fault.

1) Remove conditioned air from the airplane. This is the task: Remove Conditioned Air from the Airplane, AMM TASK 21-00-00-800-802.

(i) If the PACK TRIP OFF Light comes ON, then continue.

(2) Look at the RAM DOOR FULL OPEN Light on the P5-10 Air Conditioning Panel.

(a) If the RAM DOOR FULL OPEN Light is not ON, do this task: RAM DOOR FULL OPEN Light is Not On, Airplane is On the Ground - Fault Isolation, 21-51 TASK 804.

1) If the task shows a problem with the Ram Air System, then do the applicable corrective action.

2) Do the Repair Confirmation at the end of this task.

(b) If the RAM DOOR FULL OPEN Light is ON, then continue.

(3) Clean the Right Pack Heat Exchangers. This is the task: Heat Exchanger - Blowing Out Debris/Blockages with Nitrogen Gas/Compressed Air, AMM TASK 21-51-03-160-811.

(a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Operate the Right Air Conditioning Pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(a) If the PACK TRIP OFF Light does not come ON, then you corrected the problem.

1) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(b) If the PACK TRIP OFF Light comes ON, then continue the Fault Isolation Procedure at the subsequent step.

1) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

——— END OF TASK ———

820. Left PACK TRIP OFF Light On During Climb and Stays OFF After Reset - Fault Isolation

A. Description

(1) The Left PACK TRIP OFF Light on the P5-10 Air Conditioning Module comes ON during climb and stays off after reset.

B. Possible Causes

(1) Dirty Heat Exchangers

(2) Left Ram Air Door not fully open

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	4	C00257	AIR CONDITIONING OVERHEAT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Related Data

(1) SSM 21-51-15

(2) WDM 21-51-15

E. Initial Evaluation

(1) Operate the Left Air Conditioning Pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) If the PACK TRIP OFF Light comes ON, then do the Fault Isolation Procedure below.

(a) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(3) If the PACK TRIP OFF Light does not come ON, then there was an intermittent fault.

(a) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

F. Fault Isolation Procedure

(1) Look at the RAM DOOR FULL OPEN Light on the P5-10 Air Conditioning Panel.

(a) If the RAM DOOR FULL OPEN Light is not ON, do this task: RAM DOOR FULL OPEN Light is Not On, Airplane is On the Ground - Fault Isolation, 21-51 TASK 804.

1) If the task shows a problem with the ram air system, then do the applicable corrective action.

2) Do the Repair Confirmation at the end of this task.

(b) If the RAM DOOR FULL OPEN Light is ON, then continue.

(2) Clean the Left Pack Heat Exchangers. This is the task: Heat Exchanger - Blowing Out Debris/Blockages with Nitrogen Gas/Compressed Air, AMM TASK 21-51-03-160-811.

(a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Operate the Left Air Conditioning Pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(a) If the PACK TRIP OFF Light does not come ON, then you corrected the problem.

1) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(b) If the PACK TRIP OFF Light comes ON, then continue the Fault Isolation Procedure at the subsequent step.

1) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

——— END OF TASK ———

821. Right PACK TRIP OFF Light On During Climb and Stays OFF After Reset - Fault Isolation

A. Description

(1) The Right PACK TRIP OFF Light on the P5-10 Air Conditioning Module comes ON during climb and stays off after reset.

B. Possible Causes

(1) Dirty Heat Exchangers

(2) Left Ram Air Door not fully open

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	4	C00257	AIR CONDITIONING OVERHEAT

D. Related Data

(1) SSM 21-51-25

(2) WDM 21-51-25

E. Initial Evaluation

(1) Operate the Right Air Conditioning Pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) If the PACK TRIP OFF Light comes ON, then do the Fault Isolation Procedure below.

(a) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(3) If the PACK TRIP OFF Light does not come ON, then there was an intermittent fault.

(a) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

F. Fault Isolation Procedure

(1) Look at the RAM DOOR FULL OPEN Light on the P5-10 Air Conditioning Panel.

(a) If the RAM DOOR FULL OPEN Light is not ON, do this task: RAM DOOR FULL OPEN Light is Not On, Airplane is On the Ground - Fault Isolation, 21-51 TASK 804.

1) If the task shows a problem with the Ram Air System, then do the applicable corrective action.

2) Do the Repair Confirmation at the end of this task.

(b) If the RAM DOOR FULL OPEN Light is ON, then continue.

(2) Clean the Right Pack Heat Exchangers. This is the task: Heat Exchanger - Blowing Out Debris/Blockages with Nitrogen Gas/Compressed Air, AMM TASK 21-51-03-160-811.

(a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Operate the Right Air Conditioning Pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(a) If the PACK TRIP OFF Light does not come ON, then you corrected the fault.

1) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(b) If the PACK TRIP OFF Light comes ON, then continue the Fault Isolation Procedure at the subsequent step.

1) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

822. Left PACK TRIP OFF Light On During Approach and Stays OFF After Reset - Fault Isolation

A. Description

(1) The Left PACK TRIP OFF Light on the P5-10 Air Conditioning Module comes ON during approach and stays off after reset.

B. Possible Causes

(1) Dirty Heat Exchangers

(2) Left Ram Air Door not fully open

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	4	C00257	AIR CONDITIONING OVERHEAT

D. Related Data

(1) SSM 21-51-15

(2) WDM 21-51-15

E. Initial Evaluation

(1) Operate the Left Air Conditioning Pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) If the PACK TRIP OFF Light comes ON, then do the Fault Isolation Procedure below.

(a) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(3) If the PACK TRIP OFF Light does not come ON, then there was an intermittent fault.

(a) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

F. Fault Isolation Procedure

(1) Look at the RAM DOOR FULL OPEN Light on the P5-10 Air Conditioning Panel.

(a) If the RAM DOOR FULL OPEN Light is not ON, do this task: RAM DOOR FULL OPEN Light is Not On, Airplane is On the Ground - Fault Isolation, 21-51 TASK 804.

1) If the task shows a problem with the Ram Air System, then do the applicable corrective action.

2) Do the Repair Confirmation at the end of this task.

(b) If the RAM DOOR FULL OPEN Light is ON, then continue.

(2) Clean the Left Pack Heat Exchangers. This is the task: Heat Exchanger - Blowing Out Debris/Blockages with Nitrogen Gas/Compressed Air, AMM TASK 21-51-03-160-811.

(a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Operate the Left Air Conditioning Pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(a) If the PACK TRIP OFF Light does not come ON, then you corrected the problem.

1) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) If the PACK TRIP OFF Light comes ON, then continue the Fault Isolation Procedure at the subsequent step.

1) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

——— END OF TASK ———

823. Right PACK TRIP OFF Light On During Approach and Stays OFF After Reset - Fault Isolation

A. Description

(1) The Right PACK TRIP OFF Light on the P5-10 Air Conditioning Module comes ON during approach and stays off after reset.

B. Possible Causes

(1) Dirty Heat Exchangers

(2) Left Ram Air Door not fully open

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	4	C00257	AIR CONDITIONING OVERHEAT

D. Related Data

(1) SSM 21-51-25

(2) WDM 21-51-25

E. Initial Evaluation

(1) Operate the Right Air Conditioning Pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) If the PACK TRIP OFF Light comes ON, then do the Fault Isolation Procedure below.

(a) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(3) If the PACK TRIP OFF Light does not come ON, then there was an intermittent fault.

(a) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

F. Fault Isolation Procedure

(1) Look at the RAM DOOR FULL OPEN Light on the P5-10 Air Conditioning Panel.

(a) If the RAM DOOR FULL OPEN Light is not ON, do this task: RAM DOOR FULL OPEN Light is Not On, Airplane is On the Ground - Fault Isolation, 21-51 TASK 804.

1) If the task shows a problem with the Ram Air System, then do the applicable corrective action.

2) Do the Repair Confirmation at the end of this task.

(b) If the RAM DOOR FULL OPEN Light is ON, then continue.

(2) Clean the Right Pack Heat Exchangers. This is the task: Heat Exchanger - Blowing Out Debris/Blockages with Nitrogen Gas/Compressed Air, AMM TASK 21-51-03-160-811.

(a) Do the Repair Confirmation at the end of this task.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Repair Confirmation

(1) Operate the Right Air Conditioning Pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(a) If the PACK TRIP OFF Light does not come ON, then you corrected the problem.

1) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(b) If the PACK TRIP OFF Light comes ON, then continue the Fault Isolation Procedure at the subsequent step.

1) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

——— END OF TASK ———

824. Left PACK TRIP OFF Light On During Cruise and Stays OFF After Reset - Fault Isolation

A. Description

(1) The Left PACK TRIP OFF Light on the P5-10 Air Conditioning Module comes ON during cruise and stays off after reset.

B. Possible Causes

(1) Dirty Heat Exchangers

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	4	C00257	AIR CONDITIONING OVERHEAT

D. Related Data

(1) SSM 21-51-15

(2) WDM 21-51-15

E. Initial Evaluation

(1) Operate the Left Air Conditioning Pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) If the PACK TRIP OFF Light comes ON, then do the Fault Isolation Procedure below.

(a) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(3) If the PACK TRIP OFF Light does not come ON, then there was an intermittent fault.

(a) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

F. Fault Isolation Procedure

(1) Clean the Left Pack Heat Exchangers. This is the task: Heat Exchanger - Blowing Out Debris/Blockages with Nitrogen Gas/Compressed Air, AMM TASK 21-51-03-160-811.

(a) Operate the Left Air Conditioning Pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(b) If the PACK TRIP OFF Light does not come ON, then you corrected the problem.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

——— END OF TASK ———

825. Right PACK TRIP OFF Light On During Cruise and Stays OFF After Reset - Fault Isolation

A. Description

(1) The Right PACK TRIP OFF Light on the P5-10 Air Conditioning Module comes ON during cruise and stays off after reset.

B. Possible Causes

(1) Dirty Heat Exchangers

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	4	C00257	AIR CONDITIONING OVERHEAT

D. Related Data

(1) SSM 21-51-25

(2) WDM 21-51-25

E. Initial Evaluation

(1) Operate the Right Air Conditioning Pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(a) If the PACK TRIP OFF Light comes ON, then do the Fault Isolation Procedure below.

1) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(2) If the PACK TRIP OFF Light does not come ON, then there was an intermittent fault.

(a) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

F. Fault Isolation Procedure

(1) Clean the Right Pack Heat Exchangers. This is the task: Heat Exchanger - Blowing Out Debris/Blockages with Nitrogen Gas/Compressed Air, AMM TASK 21-51-03-160-811.

(a) Operate the Right Air Conditioning Pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(b) If the PACK TRIP OFF Light does not come ON, then you corrected the problem.

1) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

——— END OF TASK ———

828. Left PACK TRIP OFF Light On On the Ground and Comes on Again After Reset - Fault Isolation

A. Description

(1) The Left PACK TRIP OFF Light on the P5-10 Air Conditioning Module comes ON during ground operation and comes ON again after reset.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

B. Possible Causes

(1) Fan Bypass Check Valve Door in the Primary Plenum/Diffuser Assembly - damaged or missing

(2) Air Cycle Machine (ACM), high drag during start or seized.

(3) Dirty Heat Exchangers

(4) Blockage or unwanted material in the ram air ducting

(5) Ram Air Control System failure

(6) Pack overheat switches:

(a) Compressor Discharge Overheat Switch, S2

(b) Turbine Inlet Overheat Switch, S1

(c) Pack Discharge Duct Overheat Switch, S3

(7) Air Mix Valve, V23 stuck open

(8) 35°F Control System failure

(9) Bleed air too hot

(10) Cabin Temperature Control System failure

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	1	C00344	AIR CONDITIONING TEMP CONT 35 DEG F L
A	2	C00268	AIR CONDITIONING TEMP CONTROL AUTO L
A	3	C00267	AIR CONDITIONING TEMP CONTROL MANUAL
A	4	C00399	AIR CONDITIONING RAM AIR MOD LEFT
A	8	C00265	AIR CONDITIONING RAM AIR MOD CONT L
C	4	C00257	AIR CONDITIONING OVERHEAT
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L

D. Related Data

(1) SSM 21-51-14

(2) WDM 21-51-14

(3) SSM 21-51-15

(4) WDM 21-51-15

(5) SSM 21-51-16

(6) WDM 21-51-16

(7) SSM 21-61-21

(8) WDM 21-61-21

E. Initial Evaluation

(1) Do these steps to make sure the operation of the ACM is satisfactory:

NOTE: With the APU as the pneumatic source this check will make sure the ACM starts with the highest duct pressure available.

(a) Set the BLEED 1 and BLEED 2 switches on the P5-10 Air Conditioning Panel to OFF.

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(b) Do this task: Supply Pressure to the Pneumatic System with the APU, AMM TASK 36-00-00-860-803.

(c) Operate the Left Air Conditioning Pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(d) Do a check of the airflow at the exhaust duct outlet for the Left Ram Air System.

1) If there is little or no airflow at the exhaust duct outlet, then do these steps:

a) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

b) Go to the Fault Isolation Procedure.

2) If the airflow at the exhaust duct outlet is satisfactory, then continue.

(2) If the PACK TRIP OFF Light comes ON, then do the Fault Isolation Procedure below.

(a) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(3) If the PACK TRIP OFF Light does not come ON, then there was an intermittent fault.

(a) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

F. Fault Isolation Procedure

(1) Do this check of the Fan Bypass Check Valve in the Primary Plenum/Diffuser Assembly:

(a) Remove conditioned air from the airplane as follows:

1) Do this task: Remove Conditioned Air from the Airplane, AMM TASK 21-00-00-800-802.

(b) Get access to the Fan Bypass Check Valve in the Primary Plenum/Diffuser Assembly as follows:

1) Open this access panel:

Number	Name/Location
192CL	ECS Access Door

(c) Open the access panel on the bottom of the Primary Plenum/Diffuser Assembly.

(d) Make sure the Fan Bypass Check Valve Door is in place and that the door hinge is not damaged.

(e) If the Fan Bypass Check Valve Door is gone or damaged, do these tasks to replace the Primary Plenum/Diffuser Assembly:

- Heat Exchanger and Plenum/Diffuser Assembly - Removal, AMM TASK 21-51-03-000-802-001
- Heat Exchanger and Plenum/Diffuser Assembly - Installation, AMM TASK 21-51-03-400-801-001

(f) Make sure that the access panel is installed on the bottom of Primary Plenum/Diffuser Assembly.

(g) Operate the Left Air Conditioning Pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(h) If the PACK TRIP OFF Light does not come ON, then you corrected the problem.

1) Remove conditioned air from the airplane as follows:

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a) Do this task: Remove Conditioned Air from the Airplane, AMM TASK 21-00-00-800-802.

(i) If the PACK TRIP OFF Light comes ON, then continue.

(2) If there was little or no airflow at the exhaust duct outlet for the Left Ram Air System when you performed the Initial Evaluation, then do this torque check of the Left Pack ACM or replace the ACM.

(a) Remove the ACM. This is the task: ACM Removal, AMM TASK 21-51-04-000-801-001.

(b) Put the ACM in a horizontal position, similar to its installed position.

(c) Make sure that you can turn the rotating components of the ACM with 20 inch-pounds of torque applied on the axial nut just aft of the fan impeller.

CAUTION

TURN THE ROTATING COMPONENTS OF THE ACM ONLY IN THE DIRECTION THAT THEY TURN WHEN IN OPERATION. IF YOU TURN THE ACM ROTATING COMPONENTS IN THE INCORRECT DIRECTION, YOU CAN CAUSE DAMAGE TO THE FOIL AIR BEARINGS.

(d) If more than 20 inch-pounds of torque is required to turn the ACM, then install a new ACM and the pneumatic system should be checked for sources of contamination. This is the task: ACM Installation, AMM TASK 21-51-04-400-801-001.

1) Do the Repair Confirmation at the end of this task.

(e) If the ACM does rotate with 20 inch-pounds of torque, then do the steps that follow:

1) Do this task to install the ACM: ACM Installation, AMM TASK 21-51-04-400-801-001.

2) Do the Repair Confirmation at the end of this task.

(3) Examine the left pack ram air ducts for blockage or unwanted materials.

(a) If the ducts have blockage or unwanted materials, do these steps:

1) Remove the blockage or unwanted materials.

2) Do the Repair Confirmation at the end of this task.

(b) If the ducts do not have blockage or unwanted materials, then continue.

(4) Do the BITE test of the left pack 35°F Control System. This is the task: Low Limit (35 °F) Controller BITE Procedure, 21-52 TASK 801.

(a) If the BITE test shows a problem, do these steps:

1) Do the corrective action for the maintenance message shown.

2) Do the Repair Confirmation at the end of this task.

(b) If the BITE test does not show a problem, then continue.

(5) Do the BITE test of the Cabin Temperature Control System. This is the task: Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.

(a) If the BITE test shows a problem, do these steps:

1) Do the corrective action for the maintenance message shown.

2) Do the Repair Confirmation at the end of this task.

(b) If the BITE test does not show a problem, then continue.

(6) Clean the Left Pack Heat Exchangers. This is the task: Heat Exchanger - Blowing Out Debris/Blockages with Nitrogen Gas/Compressed Air, AMM TASK 21-51-03-160-811.

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(a) Do the Repair Confirmation at the end of this task.

(7) Do these steps to do a check of the Ram Air Modulation System:

(a) Look at the Left Pack Ram Air Doors.

(b) If the Ram Air Doors are not in the full open position, do this task: RAM DOOR FULL OPEN Light is Not On, Airplane is On the Ground - Fault Isolation, 21-51 TASK 804.

1) Do the Repair Confirmation at the end of this task.

(c) If the Ram Air Doors are in the full open position, then continue.

(8) Check to see if there was a report for bleed air that is too hot (a bleed trip off).

(a) If there was a bleed trip off, then do these steps:

1) Do the Fault Isolation Procedure for a bleed trip off. This is the task: BLEED TRIP OFF Light ON - Fault Isolation, 36-10 TASK 809.

2) Do the Repair Confirmation at the end of this task.

(b) If there was not a bleed trip off, then continue.

(9) Replace the Left Pack Compressor Discharge Overheat Switch, S2. These are the tasks:

- Compressor Discharge Overheat Switch Removal, AMM TASK 21-51-40-000-801
- Compressor Discharge Overheat Switch Installation, AMM TASK 21-51-40-400-801

(a) Do the Repair Confirmation at the end of this task.

(10) Replace the Left Pack Turbine Inlet Overheat Switch, S1. These are the tasks:

- Turbine Inlet Overheat Switch Removal, AMM TASK 21-51-41-000-801
- Turbine Inlet Overheat Switch Installation, AMM TASK 21-51-41-400-801

(a) Do the Repair Confirmation at the end of this task.

(11) Replace the Left Pack Outlet Duct Overheat Switch, S3. These are the tasks:

- Pack Discharge Duct Overheat Switch Removal, AMM TASK 21-51-42-000-804-001
- Pack Discharge Duct Overheat Switch Installation, AMM TASK 21-51-42-400-804-001

(a) Do the Repair Confirmation at the end of this task.

(12) Do these steps to do a check of the Temperature Control (air mix) Valve:

(a) Look at the CONT CABIN AIR MIX VALVE position indicator on the temperature control module in the control cabin.

(b) If the position indicator is not in the full COLD position, do these steps:

1) Replace the Left Pack Air Mix Valve, V23. These are the tasks:

- Air Mix Valve Removal, AMM TASK 21-61-02-000-801
- Air Mix Valve Installation, AMM TASK 21-61-02-400-801

2) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Operate the Left Air Conditioning Pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(a) If the PACK TRIP OFF Light does not come ON, then you corrected the problem.

1) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(b) If the PACK TRIP OFF Light comes ON, then continue the Fault Isolation Procedure at the subsequent step.

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1) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

——— END OF TASK ———

829. Right PACK TRIP OFF Light On On the Ground and Comes on Again After Reset - Fault Isolation

A. Description

(1) The Right PACK TRIP OFF Light on the P5-10 Air Conditioning Module comes ON during ground operation and comes ON again after reset.

B. Possible Causes

(1) Fan Bypass Check Valve Door in the Primary Plenum/Diffuser Assembly - damaged or missing
(2) ACM
(3) Dirty Heat Exchangers
(4) Blockage or unwanted material in the ram air ducting
(5) Ram Air Control System failure
(6) Pack overheat switches:
  (a) Compressor Discharge Overheat Switch, S5
  (b) Turbine Inlet Overheat Switch, S4
  (c) Pack Discharge Duct Overheat Switch, S6
(7) Air Mix Valve, V22
(8) 35°F Control System failure
(9) Bleed air too hot
(10) Cabin Temperature Control System failure

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	3	C00267	AIR CONDITIONING TEMP CONTROL MANUAL
B	1	C00345	AIR CONDITIONING TEMP CONT 35 DEG F R
B	2	C00258	AIR CONDITIONING TEMP CONTROL AUTO R
B	4	C00400	AIR CONDITIONING RAM AIR MOD RIGHT
B	8	C00266	AIR CONDITIONING RAM AIR MOD CONT R
C	4	C00257	AIR CONDITIONING OVERHEAT
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R

D. Related Data

(1) SSM 21-51-24
(2) WDM 21-51-24
(3) SSM 21-51-25
(4) WDM 21-51-25
(5) SSM 21-51-26
(6) WDM 21-51-26
(7) SSM 21-61-22

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(8) WDM 21-61-22

E. Initial Evaluation

(1) Do these steps to make sure the operation of the Right ACM is satisfactory:

(a) Set the BLEED 1 and BLEED 2 switches on the P5-10 Air Conditioning Panel to OFF.

(b) Do this task: Supply Pressure to the Pneumatic System with the APU, AMM TASK 36-00-00-860-803.

(c) Operate the Right Air Conditioning Pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(d) Do a check of the airflow at the exhaust duct outlet for the Right Ram Air System:

1) If there is little or no airflow at the exhaust duct outlet, then do these steps:

a) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

b) Go to the Fault Isolation Procedure.

2) If the airflow at the exhaust duct outlet is satisfactory, then continue.

(2) If the PACK TRIP OFF Light comes ON, then do the Fault Isolation Procedure below.

(a) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(3) If the PACK TRIP OFF Light does not come ON, then there was an intermittent fault.

(a) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

F. Fault Isolation Procedure

(1) Do this check of the Fan Bypass Check Valve in the Primary Plenum/Diffuser Assembly as follows:

(a) Remove conditioned air from the airplane. This is the task: Remove Conditioned Air from the Airplane, AMM TASK 21-00-00-800-802.

(b) Get access to the Fan Bypass Check Valve in the Primary Plenum/Diffuser Assembly as follows:

1) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

(c) Open the access panel on the bottom of the Primary Plenum/Diffuser Assembly.

(d) Make sure that the Fan Bypass Check Valve Door is in place and that the door hinge is not damaged.

(e) If the Fan Bypass Check Valve Door is gone or damaged, do these tasks to replace the Primary Plenum/Diffuser Assembly:

- Heat Exchanger and Plenum/Diffuser Assembly - Removal, AMM TASK 21-51-03-000-802-001
- Heat Exchanger and Plenum/Diffuser Assembly - Installation, AMM TASK 21-51-03-400-801-001

(f) Make sure that the access panel is installed on the bottom of the Primary Plenum/Diffuser Assembly.

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(g) Operate the Right Air Conditioning Pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(h) If the PACK TRIP OFF Light does not come ON, then you corrected the problem.

1) Remove conditioned air from the airplane. This is the task: Remove Conditioned Air from the Airplane, AMM TASK 21-00-00-800-802.

(i) If the PACK TRIP OFF Light comes ON, then continue.

(2) If there was little or no airflow at the exhaust duct outlet for the Right Ram Air System when you performed the Initial Evaluation, then do this torque check of the Right Pack ACM or replace the ACM.

(a) Remove the ACM. This is the task: ACM Removal, AMM TASK 21-51-04-000-801-001.

(b) Put the ACM in a horizontal position, similar to its installed position.

(c) Make sure that you can turn the rotating components of the ACM with 20 inch-pounds of torque applied on the axial nut just aft of the fan impeller.

(d) If more than 20 inch-pounds of torque is required to turn the ACM, then install a new ACM and the pneumatic system should be checked for sources of contamination. This is the task: ACM Installation, AMM TASK 21-51-04-400-801-001.

1) Do the Repair Confirmation at the end of this task.

(e) If the ACM does rotate with 20 inch-pounds of torque, then do the steps that follow:

1) Do this task to install the ACM: ACM Installation, AMM TASK 21-51-04-400-801-001.

2) Do the Repair Confirmation at the end of this task.

(3) Examine the right pack ram air ducts for blockage or unwanted materials.

(a) If the ducts have blockage or unwanted materials, do these steps:

1) Remove the blockage or unwanted materials.

2) Do the Repair Confirmation at the end of this task.

(b) If the ducts do not have blockage or unwanted materials, then continue.

(4) Do the BITE test of the Right Pack 35°F Control System. This is the task: Low Limit (35 °F) Controller BITE Procedure, 21-52 TASK 801.

(a) If the BITE test shows a problem, do these steps:

1) Do the corrective action for the maintenance message shown.

2) Do the Repair Confirmation at the end of this task.

(b) If the BITE test does not show a problem, then continue.

(5) Do the BITE test of the Cabin Temperature Control System. This is the task: Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.

(a) If the BITE test shows a problem, do these steps:

1) Do the corrective action for the maintenance message shown.

2) Do the Repair Confirmation at the end of this task.

(b) If the BITE test does not show a problem, then continue.

(6) Clean the Right Pack Heat Exchangers. This is the task: Heat Exchanger - Blowing Out Debris/Blockages with Nitrogen Gas/Compressed Air, AMM TASK 21-51-03-160-811.

(a) Do the Repair Confirmation at the end of this task.

(7) Do these steps to do a check of the Ram Air Modulation System:

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(a) Look at the Right Pack Ram Air Doors.

(b) If the Ram Air Doors are not in the full open position, do this task: RAM DOOR FULL OPEN Light is Not On, Airplane is On the Ground - Fault Isolation, 21-51 TASK 804.

1) Do the Repair Confirmation at the end of this task.

(c) If the Ram Air Doors are in the full open position, then continue.

(8) Check to see if there was a report for bleed air that is too hot (a bleed trip off).

(a) If there was a bleed trip off, then do these steps:

1) Do the fault isolation procedure for a bleed trip off. This is the task: BLEED TRIP OFF Light ON - Fault Isolation, 36-10 TASK 809.

2) Do the Repair Confirmation at the end of this task.

(b) If there was not a bleed trip off, then continue.

(9) Replace the Right Pack Compressor Discharge Overheat Switch, S5. These are the tasks:

- Compressor Discharge Overheat Switch Removal, AMM TASK 21-51-40-000-801
- Compressor Discharge Overheat Switch Installation, AMM TASK 21-51-40-400-801

(a) Do the Repair Confirmation at the end of this task.

(10) Replace the Right Pack Turbine Inlet Overheat Switch, S4. These are the tasks:

- Turbine Inlet Overheat Switch Removal, AMM TASK 21-51-41-000-801
- Turbine Inlet Overheat Switch Installation, AMM TASK 21-51-41-400-801

(a) Do the Repair Confirmation at the end of this task.

(11) Replace the Right Pack Outlet Duct Overheat Switch, S6. These are the tasks:

- Pack Discharge Duct Overheat Switch Removal, AMM TASK 21-51-42-000-804-001
- Pack Discharge Duct Overheat Switch Installation, AMM TASK 21-51-42-400-804-001

(a) Do the Repair Confirmation at the end of this task.

(12) Do these steps to do a check of the Temperature Control (air mix) Valve:

(a) Look at the PASS CABIN AIR MIX VALVE position indicator on the temperature control module in the control cabin.

(b) If the position indicator is not in the full COLD position, do these steps:

1) Replace the Right Pack Air Mix Valve, V22. These are the tasks:

- Air Mix Valve Removal, AMM TASK 21-61-02-000-801
- Air Mix Valve Installation, AMM TASK 21-61-02-400-801

2) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Operate the Right Air Conditioning Pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(a) If the PACK TRIP OFF Light does not come ON, then you corrected the problem.

1) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(b) If the PACK TRIP OFF Light comes ON, then continue the Fault Isolation Procedure at the subsequent step.

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1) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

——— END OF TASK ———

830. Left PACK TRIP OFF Light On at Cruise and Comes on Again After Reset - Fault Isolation

A. Description

(1) The Left PACK TRIP OFF Light on the P5-10 Air Conditioning Module comes ON during cruise and comes ON again after reset.

B. Possible Causes

(1) Dirty Heat Exchangers

(2) Blockage or unwanted material in the ram air ducting

(3) Pack overheat switches:

(a) Compressor Discharge Overheat Switch, S2

(b) Turbine Inlet Overheat Switch, S1

(c) Pack Discharge Duct Overheat Switch, S3

(4) Air Mix Valve, V23

(5) 35°F Control System failure

(6) Bleed air too hot

(7) Cabin Temperature Control System failure

(8) ACM

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	1	C00344	AIR CONDITIONING TEMP CONT 35 DEG F L
A	2	C00268	AIR CONDITIONING TEMP CONTROL AUTO L
A	3	C00267	AIR CONDITIONING TEMP CONTROL MANUAL
C	4	C00257	AIR CONDITIONING OVERHEAT
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L

D. Related Data

(1) SSM 21-51-15

(2) WDM 21-51-15

(3) SSM 21-51-16

(4) WDM 21-51-16

(5) SSM 21-61-21

(6) WDM 21-61-21

E. Initial Evaluation

(1) Operate the Left Air Conditioning Pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) If the PACK TRIP OFF Light comes ON, then do the Fault Isolation Procedure below.

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(a) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(3) If the PACK TRIP OFF Light does not come ON, then there was an intermittent fault.

(a) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

F. Fault Isolation Procedure

(1) Examine the left pack ram air ducts for blockage or unwanted materials.

(a) If the ducts have blockage or unwanted materials, do these steps:

1) Remove the blockage or unwanted materials.

2) Do the Repair Confirmation at the end of this task.

(b) If the ducts do not have blockage or unwanted materials, then continue.

(2) Do the BITE test of the Left Pack 35°F Control System. This is the task: Low Limit (35 °F) Controller BITE Procedure, 21-52 TASK 801.

(a) If the BITE test shows a problem, do these steps:

1) Do the corrective action for the maintenance message shown.

2) Do the Repair Confirmation at the end of this task.

(b) If the BITE test does not show a problem, then continue.

(3) Do the BITE test of the Cabin Temperature Control System. This is the task: Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.

(a) If the BITE test shows a problem, do these steps:

1) Do the corrective action for the maintenance message shown.

2) Do the Repair Confirmation at the end of this task.

(b) If the BITE test does not show a problem, then continue.

(4) Clean the Left Pack Heat Exchangers. This is the task: Heat Exchanger - Blowing Out Debris/Blockages with Nitrogen Gas/Compressed Air, AMM TASK 21-51-03-160-811.

(a) Do the Repair Confirmation at the end of this task.

(5) Check to see if there was a report for bleed air that is too hot (a bleed trip off).

(a) If there was a bleed trip off, then do these steps:

1) Do the fault isolation procedure for a bleed trip off. This is the task: BLEED TRIP OFF Light ON - Fault Isolation, 36-10 TASK 809.

2) Do the Repair Confirmation at the end of this task.

(b) If there was not a bleed trip off, then continue.

(6) Replace the Left Pack Compressor Discharge Overheat Switch, S2. These are the tasks:

- Compressor Discharge Overheat Switch Removal, AMM TASK 21-51-40-000-801
- Compressor Discharge Overheat Switch Installation, AMM TASK 21-51-40-400-801

(a) Do the Repair Confirmation at the end of this task.

(7) Replace the Left Pack Turbine Inlet Overheat Switch, S1. These are the tasks:

- Turbine Inlet Overheat Switch Removal, AMM TASK 21-51-41-000-801
- Turbine Inlet Overheat Switch Installation, AMM TASK 21-51-41-400-801

(a) Do the Repair Confirmation at the end of this task.

(8) Replace the left pack outlet duct overheat switch, S3. These are the tasks:

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- Pack Discharge Duct Overheat Switch Removal, AMM TASK 21-51-42-000-804-001
- Pack Discharge Duct Overheat Switch Installation, AMM TASK 21-51-42-400-804-001

(a) Do the Repair Confirmation at the end of this task.

(9) Do these steps to do a check of the Temperature Control (air mix) Valve:

(a) Look at the CONT CABIN AIR MIX VALVE position indicator on the Temperature Control Module in the control cabin.

(b) If the position indicator is not in the full COLD position, do these steps:

1) Replace the Left Pack Air Mix Valve, V23. These are the tasks:

- Air Mix Valve Removal, AMM TASK 21-61-02-000-801,
- Air Mix Valve Installation, AMM TASK 21-61-02-400-801.

2) Do the Repair Confirmation at the end of this task.

(c) If the position indicator is in the full COLD position, then continue.

(10) Replace the Left Pack ACM. These are the tasks:

- ACM Removal, AMM TASK 21-51-04-000-801-001
- ACM Installation, AMM TASK 21-51-04-400-801-001

(a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Operate the Left Air Conditioning Pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(a) If the PACK TRIP OFF Light does not come ON, then you corrected the problem.

1) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(b) If the PACK TRIP OFF Light comes ON, then continue the Fault Isolation Procedure at the subsequent step.

1) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

——— END OF TASK ———

831. Right PACK TRIP OFF Light On at Cruise and Comes on Again After Reset - Fault Isolation

A. Description

(1) The Right PACK TRIP OFF Light on the P5-10 Air Conditioning Module comes ON during cruise and comes ON again after reset.

B. Possible Causes

(1) Dirty Heat Exchangers

(2) Blockage or unwanted material in the ram air ducting

(a) Pack overheat switches:

1) Compressor Discharge Overheat Switch, S5

2) Turbine Inlet Overheat Switch, S4

3) Pack Discharge Duct Overheat Switch, S6

(3) Air Mix Valve, V22

(4) 35°F Control System failure

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(5) Bleed air too hot

(6) Cabin Temperature Control System failure

(7) ACM

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	3	C00267	AIR CONDITIONING TEMP CONTROL MANUAL
B	1	C00345	AIR CONDITIONING TEMP CONT 35 DEG F R
B	2	C00258	AIR CONDITIONING TEMP CONTROL AUTO R
C	4	C00257	AIR CONDITIONING OVERHEAT
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R

D. Related Data

(1) SSM 21-51-25

(2) WDM 21-51-25

(3) SSM 21-51-26

(4) WDM 21-51-26

(5) SSM 21-61-22

(6) WDM 21-61-22

E. Initial Evaluation

(1) Operate the Right Air Conditioning Pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) If the PACK TRIP OFF Light comes ON, then do the Fault Isolation Procedure below.

(a) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(3) If the PACK TRIP OFF Light does not come ON, then there was an intermittent fault.

(a) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

F. Fault Isolation Procedure

(1) Examine the right pack ram air ducts for blockage or unwanted materials.

(a) If the ducts have blockage or unwanted materials, do these steps:

1) Remove the blockage or unwanted materials.

2) Do the Repair Confirmation at the end of this task.

(b) If the ducts do not have blockage or unwanted materials, then continue.

(2) Do the BITE test of the Right Pack 35°F Control System. This is the task: Low Limit (35 °F) Controller BITE Procedure, 21-52 TASK 801.

(a) If the BITE test shows a problem, do these steps:

1) Do the corrective action for the maintenance message shown.

2) Do the Repair Confirmation at the end of this task.

(b) If the BITE test does not show a problem, then continue.

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(3) Do the BITE test of the Cabin Temperature Control System. This is the task: Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.

(a) If the BITE test shows a problem, do these steps:

1) Do the corrective action for the maintenance message shown.

2) Do the Repair Confirmation at the end of this task.

(b) If the BITE test does not show a problem, then continue.

(4) Clean the Right Pack Heat Exchangers. This is the task: Heat Exchanger - Blowing Out Debris/Blockages with Nitrogen Gas/Compressed Air, AMM TASK 21-51-03-160-811.

(a) Do the Repair Confirmation at the end of this task.

(5) Check to see if there was a report for bleed air that is too hot (a bleed trip off).

(a) If there was a bleed trip off, then do these steps:

1) Do the Fault Isolation Procedure for a bleed trip off. This is the task: BLEED TRIP OFF Light ON - Fault Isolation, 36-10 TASK 809.

2) Do the Repair Confirmation at the end of this task.

(b) If there was not a bleed trip off, then continue.

(6) Replace the Right Pack Compressor Discharge Overheat Switch, S5. These are the tasks:

- Compressor Discharge Overheat Switch Removal, AMM TASK 21-51-40-000-801
- Compressor Discharge Overheat Switch Installation, AMM TASK 21-51-40-400-801

(a) Do the Repair Confirmation at the end of this task.

(7) Replace the Right Pack Turbine Inlet Overheat Switch, S4. These are the tasks:

- Turbine Inlet Overheat Switch Removal, AMM TASK 21-51-41-000-801
- Turbine Inlet Overheat Switch Installation, AMM TASK 21-51-41-400-801

(a) Do the Repair Confirmation at the end of this task.

(8) Replace the Right Pack Outlet Duct Overheat Switch, S6. These are the tasks:

- Pack Discharge Duct Overheat Switch Removal, AMM TASK 21-51-42-000-804-001
- Pack Discharge Duct Overheat Switch Installation, AMM TASK 21-51-42-400-804-001

(a) Do the Repair Confirmation at the end of this task.

(b) If the PACK TRIP OFF Light comes ON, then continue.

(9) Do these steps to do a check of the Temperature Control (air mix) Valve:

(a) Look at the PASS CABIN AIR MIX VALVE position indicator on the Temperature Control Module in the control cabin.

(b) If the position indicator is not in the full COLD position, do these steps:

1) Replace the Right Pack Air Mix Valve, V22. These are the tasks:

- Air Mix Valve Removal, AMM TASK 21-61-02-000-801
- Air Mix Valve Installation, AMM TASK 21-61-02-400-801

2) Do the Repair Confirmation at the end of this task.

(c) If the position indicator is in the full COLD position, then continue.

(10) Replace the Right Pack ACM. These are the tasks:

- ACM Removal, AMM TASK 21-51-04-000-801-001
- ACM Installation, AMM TASK 21-51-04-400-801-001

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(a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Operate the Right Air Conditioning Pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(a) If the PACK TRIP OFF Light does not come ON, then you corrected the problem.

1) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(b) if the PACK TRIP OFF Light comes ON, then continue the Fault Isolation Procedure at the subsequent step.

1) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

——— END OF TASK ———

832. Left PACK TRIP OFF Light On During Approach and Comes on Again After Reset - Fault Isolation

A. Description

(1) The Left PACK TRIP OFF Light on the P5-10 Air Conditioning Module comes ON during approach and comes ON again after reset.

B. Possible Causes

(1) Dirty Heat Exchangers

(2) Blockage or unwanted material in the ram air ducting

(3) Ram Air Control System failure

(4) Pack overheat switches:

(a) Compressor Discharge Overheat Switch, S2

(b) Turbine Inlet Overheat Switch, S1

(c) Pack Discharge Duct Overheat Switch, S3

(5) Air Mix Valve, V23

(6) 35°F Control System failure

(7) Bleed air too hot

(8) Cabin Temperature Control System failure

(9) ACM

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	1	C00344	AIR CONDITIONING TEMP CONT 35 DEG F L
A	2	C00268	AIR CONDITIONING TEMP CONTROL AUTO L
A	3	C00267	AIR CONDITIONING TEMP CONTROL MANUAL
A	4	C00399	AIR CONDITIONING RAM AIR MOD LEFT
A	8	C00265	AIR CONDITIONING RAM AIR MOD CONT L
C	4	C00257	AIR CONDITIONING OVERHEAT
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L

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D. Related Data

(1) SSM 21-51-14

(2) WDM 21-51-14

(3) SSM 21-51-15

(4) WDM 21-51-15

(5) SSM 21-51-16

(6) WDM 21-51-16

(7) SSM 21-61-21

(8) WDM 21-61-21

E. Initial Evaluation

(1) Operate the Left Air Conditioning Pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) If the PACK TRIP OFF Light comes ON, then do the Fault Isolation Procedure below.

(a) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(3) If the PACK TRIP OFF Light does not come ON, then there was an intermittent fault.

(a) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

F. Fault Isolation Procedure

(1) Examine the left pack ram air ducts for blockage or unwanted materials.

(a) If the ducts have blockage or unwanted materials, do these steps:

1) Remove the blockage or unwanted materials.

2) Do the Repair Confirmation at the end of this task.

(b) If the ducts do not have blockage or unwanted materials, then continue.

(2) Do the BITE test of the Left Pack 35°F Control System. This is the task: Low Limit (35 °F) Controller BITE Procedure, 21-52 TASK 801.

(a) If the BITE test shows a problem, do these steps:

1) Do the corrective action for the maintenance message shown.

2) Do the Repair Confirmation at the end of this task.

(b) If the BITE test does not show a problem, then continue.

(3) Do the BITE test of the Cabin Temperature Control System. This is the task: Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.

(a) If the BITE test shows a problem, do these steps:

1) Do the corrective action for the maintenance message shown.

2) Do the Repair Confirmation at the end of this task.

(b) If the BITE test does not show a problem, then continue.

(4) Clean the Left Pack Heat Exchangers. This is the task: Heat Exchanger - Blowing Out Debris/Blockages with Nitrogen Gas/Compressed Air, AMM TASK 21-51-03-160-811.

(a) Do the Repair Confirmation at the end of this task.

(5) Do these steps to do a check of the Ram Air Modulation System:

(a) Look at the Left Pack Ram Air Doors.

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(b) If the Ram Air Doors are not in the full open position, do this task: RAM DOOR FULL OPEN Light is Not On, Airplane is On the Ground - Fault Isolation, 21-51 TASK 804.

1) Do the Repair Confirmation at the end of this task.

(c) If the Ram Air Doors are in the full open position, then continue.

(6) Check to see if there was a report for bleed air that is too hot (a bleed trip off).

(a) If there was a bleed trip off, then do these steps:

1) Do the Fault Isolation Procedure for a bleed trip off. This is the task: BLEED TRIP OFF Light ON - Fault Isolation, 36-10 TASK 809.

2) Do the Repair Confirmation at the end of this task.

(b) If there was not a bleed trip off, then continue.

(7) Replace the Left Pack Compressor Discharge Overheat Switch, S2. These are the tasks:

- Compressor Discharge Overheat Switch Removal, AMM TASK 21-51-40-000-801
- Compressor Discharge Overheat Switch Installation, AMM TASK 21-51-40-400-801

(a) Do the Repair Confirmation at the end of this task.

(8) Replace the Left Pack Turbine Inlet Overheat Switch, S1. These are the tasks:

- Turbine Inlet Overheat Switch Removal, AMM TASK 21-51-41-000-801
- Turbine Inlet Overheat Switch Installation, AMM TASK 21-51-41-400-801

(a) Do the Repair Confirmation at the end of this task.

(9) Replace the Left Pack Outlet Duct Overheat switch, S3. These are the tasks:

- Pack Discharge Duct Overheat Switch Removal, AMM TASK 21-51-42-000-804-001
- Pack Discharge Duct Overheat Switch Installation, AMM TASK 21-51-42-400-804-001

(a) Do the Repair Confirmation at the end of this task.

(10) Do these steps to do a check of the Temperature Control (air mix) Valve:

(a) Look at the CONT CABIN AIR MIX VALVE position indicator on the Temperature Control Module in the control cabin.

(b) If the position indicator is not in the full COLD position, do these steps:

1) Replace the Left Pack Air Mix Valve, V23. These are the tasks:

- Air Mix Valve Removal, AMM TASK 21-61-02-000-801
- Air Mix Valve Installation, AMM TASK 21-61-02-400-801

2) Do the Repair Confirmation at the end of this task.

(c) If the position indicator is in the full COLD position, then continue.

(11) Replace the Left Pack ACM. These are the tasks:

- ACM Removal, AMM TASK 21-51-04-000-801-001
- ACM Installation, AMM TASK 21-51-04-400-801-001

(a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Operate the Left Air Conditioning Pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(a) If the PACK TRIP OFF Light does not come ON, then you corrected the problem.

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1) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(b) If the PACK TRIP OFF Light comes ON, then continue the Fault Isolation Procedure at the subsequent step.

1) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

——— END OF TASK ———

833. Right PACK TRIP OFF Light On During Approach and Comes on Again After Reset - Fault Isolation

A. Description

(1) The Right PACK TRIP OFF Light on the P5-10 Air Conditioning Module comes ON during approach and comes ON again after reset.

B. Possible Causes

(1) Dirty Heat Exchangers

(2) Blockage or unwanted material in the ram air ducting

(3) Ram Air Control System failure

(4) Pack overheat switches:

(a) Compressor Discharge Overheat Switch, S5

(b) Turbine Inlet Overheat Switch, S4

(c) Pack Discharge Duct Overheat Switch, S6

(5) Air Mix Valve, V22

(6) 35°F Control System failure

(7) Bleed air too hot

(8) Cabin Temperature Control System failure

(9) ACM

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	3	C00267	AIR CONDITIONING TEMP CONTROL MANUAL
B	1	C00345	AIR CONDITIONING TEMP CONT 35 DEG F R
B	2	C00258	AIR CONDITIONING TEMP CONTROL AUTO R
B	4	C00400	AIR CONDITIONING RAM AIR MOD RIGHT
B	8	C00266	AIR CONDITIONING RAM AIR MOD CONT R
C	4	C00257	AIR CONDITIONING OVERHEAT
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R

D. Related Data

(1) SSM 21-51-24

(2) WDM 21-51-24

(3) SSM 21-51-25

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(4) WDM 21-51-25

(5) SSM 21-51-26

(6) WDM 21-51-26

(7) SSM 21-61-22

(8) WDM 21-61-22

E. Initial Evaluation

(1) Operate the Right Air Conditioning Pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) If the PACK TRIP OFF Light comes ON, then do the Fault Isolation Procedure below.

(a) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(3) If the PACK TRIP OFF Light does not come ON, then there was an intermittent fault.

(a) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

F. Fault Isolation Procedure

(1) Examine the right pack ram air ducts for blockage or unwanted materials.

(a) If the ducts have blockage or unwanted materials, do these steps:

1) Remove the blockage or unwanted materials.

2) Do the Repair Confirmation at the end of this task.

(b) If the ducts do not have blockage or unwanted materials, then continue.

(2) Do the BITE test of the right pack 35°F Control System. This is the task: Low Limit (35 °F) Controller BITE Procedure, 21-52 TASK 801.

(a) If the BITE test shows a problem, do these steps:

1) Do the corrective action for the maintenance message shown.

2) Do the Repair Confirmation at the end of this task.

(b) If the BITE test does not show a problem, then continue.

(3) Do the BITE test of the Cabin Temperature Control System. This is the task: Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.

(a) If the BITE test shows a problem, do these steps:

1) Do the corrective action for the maintenance message shown.

2) Do the Repair Confirmation at the end of this task.

(b) If the BITE test does not show a problem, then continue.

(4) Clean the Right Pack Heat Exchangers. This is the task: Heat Exchanger - Blowing Out Debris/Blockages with Nitrogen Gas/Compressed Air, AMM TASK 21-51-03-160-811.

(a) Do the Repair Confirmation at the end of this task.

(5) Do these steps to do a check of the Ram Air Modulation System:

(a) Look at the Right Pack Ram Air Doors.

(b) If the Ram Air Doors are not in the full open position, do this task: RAM DOOR FULL OPEN Light is Not On, Airplane is On the Ground - Fault Isolation, 21-51 TASK 804.

1) Do the Repair Confirmation at the end of this task.

(c) If the ram air doors are in the full open position, then continue.

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(6) Check to see if there was a report for bleed air that is too hot (a bleed trip off).

(a) If there was a bleed trip off, then do these steps:

1) Do the fault isolation procedure for a bleed trip off. This is the task: BLEED TRIP OFF Light ON - Fault Isolation, 36-10 TASK 809.

2) Do the Repair Confirmation at the end of this task.

(b) If there was not a bleed trip off, then continue.

(7) Replace the Right Pack Compressor Discharge Overheat Switch, S5. These are the tasks:

- Compressor Discharge Overheat Switch Removal, AMM TASK 21-51-40-000-801
- Compressor Discharge Overheat Switch Installation, AMM TASK 21-51-40-400-801

(a) Do the Repair Confirmation at the end of this task.

(8) Replace the Right Pack Turbine Inlet Overheat Switch, S4. These are the tasks:

- Turbine Inlet Overheat Switch Removal, AMM TASK 21-51-41-000-801
- Turbine Inlet Overheat Switch Installation, AMM TASK 21-51-41-400-801

(a) Do the Repair Confirmation at the end of this task.

(9) Replace the Right Pack Outlet Duct Overheat Switch, S6. These are the tasks:

- Pack Discharge Duct Overheat Switch Removal, AMM TASK 21-51-42-000-804-001
- Pack Discharge Duct Overheat Switch Installation, AMM TASK 21-51-42-400-804-001

(a) Do the Repair Confirmation at the end of this task.

(10) Do these steps to do a check of the Temperature Control (air mix) Valve:

(a) Look at the PASS CABIN AIR MIX VALVE position indicator on the Temperature Control Module in the control cabin.

(b) If the position indicator is not in the full COLD position, do these steps:

1) Replace the Right Pack Air Mix Valve, V22. These are the tasks:

- Air Mix Valve Removal, AMM TASK 21-61-02-000-801
- Air Mix Valve Installation, AMM TASK 21-61-02-400-801

2) Do the Repair Confirmation at the end of this task.

(c) If the position indicator is in the full COLD position, then continue.

(11) Replace the Right Pack ACM. These are the tasks:

- ACM Removal, AMM TASK 21-51-04-000-801-001
- ACM Installation, AMM TASK 21-51-04-400-801-001

(a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Operate the Right Air Conditioning Pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(a) If the PACK TRIP OFF Light does not come ON, then you corrected the problem.

1) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(b) If the PACK TRIP OFF Light comes ON, then continue the Fault Isolation Procedure at the subsequent step.

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1) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

——— END OF TASK ———

834. Left PACK TRIP OFF Light On During Climb and Comes on Again After Reset - Fault Isolation

A. Description

(1) The left PACK TRIP OFF light on the P5-10 air conditioning module comes on during climb and comes on again after reset.

B. Possible Causes

(1) Blockage or unwanted material in the ram air ducting

(2) Ram air control system failure

(3) Pack overheat switches:

(a) Compressor discharge overheat switch, S2

(b) Turbine inlet overheat switch, S1

(c) Pack discharge duct overheat switch, S3

(4) Air mix valve, V23

(5) 35°F control system failure

(6) Bleed air too hot

(7) Cabin temperature control system failure

(8) Air cycle machine

C.

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	1	C00344	AIR CONDITIONING TEMP CONT 35 DEG F L
A	2	C00268	AIR CONDITIONING TEMP CONTROL AUTO L
A	3	C00267	AIR CONDITIONING TEMP CONTROL MANUAL
A	4	C00399	AIR CONDITIONING RAM AIR MOD LEFT
A	8	C00265	AIR CONDITIONING RAM AIR MOD CONT L
C	4	C00257	AIR CONDITIONING OVERHEAT
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L

D. Related Data

(1) (SSM 21-51-14)

(2) (WDM 21-51-14)

(3) (SSM 21-51-15)

(4) (WDM 21-51-15)

(5) (SSM 21-51-16)

(6) (WDM 21-51-16)

(7) (SSM 21-61-21)

(8) (WDM 21-61-21)

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E. Initial Evaluation

(1) Operate the left air conditioning pack. To operate the pack, do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) If the PACK TRIP OFF light comes on, then do the Fault Isolation Procedure below.

(a) Stop the pack. To stop the pack, do this task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(3) If the PACK TRIP OFF light does not come on, then there was an intermittent fault.

(a) Stop the pack. To stop the pack, do this task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

F. Fault Isolation Procedure

(1) Examine the left pack ram air ducts for blockage or unwanted materials.

(a) If the ducts have blockage or unwanted materials, do these steps:

1) Remove the blockage or unwanted materials.

2) Operate the left air conditioning pack. To operate the pack, do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

3) If the PACK TRIP OFF light does not come on, then you corrected the fault.

4) Stop the pack. To stop the pack, do this task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(b) If the ducts do not have blockage or unwanted materials, then continue.

(2) Do the BITE test of the left pack 35°F control system. To do the BITE test, do this task: Low Limit (35 °F) Controller BITE Procedure, 21-52 TASK 801.

(a) If the BITE test shows a problem, do these steps:

1) Do the corrective action for the maintenance message shown.

2) Operate the left air conditioning pack. To operate the pack, do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

3) If the PACK TRIP OFF light does not come on, then you corrected the fault.

4) Stop the pack. To stop the pack, do this task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(b) If the BITE test does not show a problem, then continue.

(3) Do the BITE test of the cabin temperature control system. To do the BITE test, do this task: Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.

(a) If the BITE test shows a problem, do these steps:

1) Do the corrective action for the maintenance message shown.

2) Operate the left air conditioning pack. To operate the pack, do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

3) If the PACK TRIP OFF light does not come on, then you corrected the fault.

4) Stop the pack. To stop the pack, do this task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(b) If the BITE test does not show a problem, then continue.

(4) Do these steps to do a check of the ram air modulation system:

(a) Look at the left pack ram air doors.

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(b) If the ram air doors are not in the full open position, do this task: RAM DOOR FULL OPEN Light is Not On, Airplane is On the Ground - Fault Isolation, 21-51 TASK 804.

1) Operate the left air conditioning pack. To operate the pack, do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

2) If the PACK TRIP OFF light does not come on, then you corrected the fault.

3) Stop the pack. To stop the pack, do this task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(c) If the ram air doors are in the full open position, then continue.

(5) Check to see if there was a report for bleed air that is too hot (a bleed trip off).

(a) If there was a bleed trip off, then do these steps:

1) Do the fault isolation procedure for a bleed trip off. To do the procedure, do this task: BLEED TRIP OFF Light ON - Fault Isolation, 36-10 TASK 809.

2) Operate the left air conditioning pack. To operate the pack, do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

3) If the PACK TRIP OFF light does not come on, then you corrected the fault.

4) Stop the pack. To stop the pack, do this task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(b) If there was not a bleed trip off, then continue.

(6) Replace the left pack compressor discharge overheat switch, S2.

These are the tasks:

Compressor Discharge Overheat Switch Removal, AMM TASK 21-51-40-000-801,

Compressor Discharge Overheat Switch Installation, AMM TASK 21-51-40-400-801.

(a) Operate the left air conditioning pack. To operate the pack, do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(b) If the PACK TRIP OFF light does not come on, then you corrected the fault.

1) Stop the pack. To stop the pack, do this task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(c) If the PACK TRIP OFF light comes on, then continue.

(7) Replace the left pack turbine inlet overheat switch, S1.

These are the tasks:

Turbine Inlet Overheat Switch Removal, AMM TASK 21-51-41-000-801,

Turbine Inlet Overheat Switch Installation, AMM TASK 21-51-41-400-801.

(a) Operate the left air conditioning pack. To operate the pack, do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(b) If the PACK TRIP OFF light does not come on, then you corrected the fault.

1) Stop the pack. To stop the pack, do this task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(c) If the PACK TRIP OFF light comes on, then continue.

(8) Replace the left pack outlet duct overheat switch, S3.

These are the tasks:

Pack Discharge Duct Overheat Switch Removal, AMM TASK 21-51-42-000-804-001,

Pack Discharge Duct Overheat Switch Installation, AMM TASK 21-51-42-400-804-001.

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(a) Operate the left air conditioning pack. To operate the pack, do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(b) If the PACK TRIP OFF light does not come on, then you corrected the fault.

1) Stop the pack. To stop the pack, do this task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(c) If the PACK TRIP OFF light comes on, then continue.

(9) Do these steps to do a check of the temperature control (air mix) valve:

(a) Look at the CONT CABIN AIR MIX VALVE position indicator on the temperature control module in the control cabin.

(b) If the position indicator is not in the full COLD position, do these steps:

1) Replace the left pack air mix valve, V23.

These are the tasks:

Air Mix Valve Removal, AMM TASK 21-61-02-000-801,

Air Mix Valve Installation, AMM TASK 21-61-02-400-801.

2) Operate the left air conditioning pack. To operate the pack, do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

3) If the PACK TRIP OFF light does not come on, then you corrected the fault.

4) Stop the pack. To stop the pack, do this task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(c) If the position indicator is in the full COLD position, then continue.

(10) Replace the left pack air cycle machine.

These are the tasks:

ACM Removal, AMM TASK 21-51-04-000-801-001,

ACM Installation, AMM TASK 21-51-04-400-801-001.

(a) Operate the left air conditioning pack. To operate the pack, do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(b) If the PACK TRIP OFF light does not come on, then you corrected the fault.

(c) Stop the pack. To stop the pack, do this task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

——— END OF TASK ———

835. Right PACK TRIP OFF Light On During Climb and Comes on Again After Reset - Fault Isolation

A. Description

(1) The right PACK TRIP OFF light on the P5-10 air conditioning module comes on during climb and comes on again after reset.

B. Possible Causes

(1) Blockage or unwanted material in the ram air ducting

(2) Ram air control system failure

(3) Pack overheat switches:

(a) Compressor discharge overheat switch, S5

(b) Turbine inlet overheat switch, S4

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(c) Pack discharge duct overheat switch, S6

(4) Air mix valve, V22

(5) 35°F control system failure

(6) Bleed air too hot

(7) Cabin temperature control system failure

(8) Air cycle machine

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	3	C00267	AIR CONDITIONING TEMP CONTROL MANUAL
B	1	C00345	AIR CONDITIONING TEMP CONT 35 DEG F R
B	2	C00258	AIR CONDITIONING TEMP CONTROL AUTO R
B	4	C00400	AIR CONDITIONING RAM AIR MOD RIGHT
B	8	C00266	AIR CONDITIONING RAM AIR MOD CONT R
C	4	C00257	AIR CONDITIONING OVERHEAT
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R

D. Related Data

(1) (SSM 21-51-24)

(2) (WDM 21-51-24)

(3) (SSM 21-51-25)

(4) (WDM 21-51-25)

(5) (SSM 21-51-26)

(6) (WDM 21-51-26)

(7) (SSM 21-61-22)

(8) (WDM 21-61-22)

E. Initial Evaluation

(1) Operate the right air conditioning pack. To operate the pack, do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) If the PACK TRIP OFF light comes on, then do the Fault Isolation Procedure below.

(a) Stop the pack. To stop the pack, do this task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(3) If the PACK TRIP OFF light does not come on, then there was an intermittent fault.

(a) Stop the pack. To stop the pack, do this task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

F. Fault Isolation Procedure

(1) Examine the right pack ram air ducts for blockage or unwanted materials.

(a) If the ducts have blockage or unwanted materials, do these steps:

1) Remove the blockage or unwanted materials.

2) Operate the right air conditioning pack. To operate the pack, do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

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3) If the PACK TRIP OFF light does not come on, then you corrected the fault.

4) Stop the pack. To stop the pack, do this task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(b) If the ducts do not have blockage or unwanted materials, then continue.

(2) Do the BITE test of the right pack 35°F control system. To do the BITE test, do this task: Low Limit (35 °F) Controller BITE Procedure, 21-52 TASK 801.

(a) If the BITE test shows a problem, do these steps:

1) Do the corrective action for the maintenance message shown.

2) Operate the right air conditioning pack. To operate the pack, do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

3) If the PACK TRIP OFF light does not come on, then you corrected the fault.

4) Stop the pack. To stop the pack, do this task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(b) If the BITE test does not show a problem, then continue.

(3) Do the BITE test of the cabin temperature control system. To do the BITE test, do this task: Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.

(a) If the BITE test shows a problem, do these steps:

1) Do the corrective action for the maintenance message shown.

2) Operate the right air conditioning pack. To operate the pack, do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

3) If the PACK TRIP OFF light does not come on, then you corrected the fault.

4) Stop the pack. To stop the pack, do this task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(b) If the BITE test does not show a problem, then continue.

(4) Do these steps to do a check of the ram air modulation system:

(a) Look at the right pack ram air doors.

(b) If the ram air doors are not in the full open position, do this task: RAM DOOR FULL OPEN Light is Not On, Airplane is On the Ground - Fault Isolation, 21-51 TASK 804.

1) Operate the right air conditioning pack. To operate the pack, do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

2) If the PACK TRIP OFF light does not come on, then you corrected the fault.

3) Stop the pack. To stop the pack, do this task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(c) If the ram air doors are in the full open position, then continue.

(5) Check to see if there was a report for bleed air that is too hot (a bleed trip off).

(a) If there was a bleed trip off, then do these steps:

1) Do the fault isolation procedure for a bleed trip off. To do the procedure, do this task: BLEED TRIP OFF Light ON - Fault Isolation, 36-10 TASK 809.

2) Operate the right air conditioning pack. To operate the pack, do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

3) If the PACK TRIP OFF light does not come on, then you corrected the fault.

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4) Stop the pack. To stop the pack, do this task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(b) If there was not a bleed trip off, then continue.

(6) Replace the right pack compressor discharge overheat switch, S5.

These are the tasks:

Compressor Discharge Overheat Switch Removal, AMM TASK 21-51-40-000-801,

Compressor Discharge Overheat Switch Installation, AMM TASK 21-51-40-400-801.

(a) Operate the right air conditioning pack. To operate the pack, do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(b) If the PACK TRIP OFF light does not come on, then you corrected the fault.

1) Stop the pack. To stop the pack, do this task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(c) If the PACK TRIP OFF light comes on, then continue.

(7) Replace the right pack turbine inlet overheat switch, S4.

These are the tasks:

Turbine Inlet Overheat Switch Removal, AMM TASK 21-51-41-000-801,

Turbine Inlet Overheat Switch Installation, AMM TASK 21-51-41-400-801.

(a) Operate the left air conditioning pack. To operate the pack, do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(b) If the PACK TRIP OFF light does not come on, then you corrected the fault.

1) Stop the pack. To stop the pack, do this task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(c) If the PACK TRIP OFF light comes on, then continue.

(8) Replace the right pack outlet duct overheat switch, S6.

These are the tasks:

Pack Discharge Duct Overheat Switch Removal, AMM TASK 21-51-42-000-804-001,

Pack Discharge Duct Overheat Switch Installation, AMM TASK 21-51-42-400-804-001.

(a) Operate the right air conditioning pack. To operate the pack, do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(b) If the PACK TRIP OFF light does not come on, then you corrected the fault.

1) Stop the pack. To stop the pack, do this task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(c) If the PACK TRIP OFF light comes on, then continue.

(9) Do these steps to do a check of the temperature control (air mix) valve:

(a) Look at the PASS CABIN AIR MIX VALVE position indicator on the temperature control module in the control cabin.

(b) If the position indicator is not in the full COLD position, do these steps:

1) Replace the right pack air mix valve, V22.

These are the tasks:

Air Mix Valve Removal, AMM TASK 21-61-02-000-801,

Air Mix Valve Installation, AMM TASK 21-61-02-400-801.

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2) Operate the right air conditioning pack. To operate the pack, do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

3) If the PACK TRIP OFF light does not come on, then you corrected the fault.

4) Stop the pack. To stop the pack, do this task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(c) If the position indicator is in the full COLD position, then continue.

(10) Replace the right pack air cycle machine.

These are the tasks:

ACM Removal, AMM TASK 21-51-04-000-801-001,

ACM Installation, AMM TASK 21-51-04-400-801-001.

(a) Operate the right air conditioning pack. To operate the pack, do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(b) If the PACK TRIP OFF light does not come on, then you corrected the fault.

(c) Stop the pack. To stop the pack, do this task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

——— END OF TASK ———

856. Pack Outlet Temperature Too Hot or Too Cold - Fault Isolation

A. Description

(1) This task is for fault code 215 350 00.

(2) This fault isolation procedure is used if the data obtained from the Pack Flow Control and Pack Cooling System Check (AMM TASK 21-51-00-943-801) indicates a problem with the temperature of the pack outlet air before it is conditioned for distribution to the airplane flight deck and passenger compartment.

B. Possible Causes

(1) 35°F control valve

(2) 35°F control module

(3) 35°F sensor

(4) Air mix valve V23 (V22).

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	1	C00344	AIR CONDITIONING TEMP CONT 35 DEG F L
B	1	C00345	AIR CONDITIONING TEMP CONT 35 DEG F R

D. Related Data

(1) SSM 21-51-16

(2) WDM 21-51-16

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E. Initial Evaluation

(1) Evaluate the pack outlet temperature and flow for the AUTO, HIGH and APU HIGH pack flow modes. Do this task:

AMM TASK 21-51-00-943-801.

(2) If the temperature of the pack outlet air is within the range of 35°F to 39°F, then there was an intermittent fault.

(3) If the temperature of the pack outlet air is too low (34°F or lower), then do the Pack Outlet Temperature Is 34°F or Lower - Fault Isolation procedure.

(4) If the temperature of the pack outlet air is too high (40°F or higher), then do the Pack Outlet Temperature is 40°F or Higher - Fault Isolation procedure.

F. Pack Outlet Temperature Is 34°F or Lower - Fault Isolation Procedure

(1) Supply electrical power. Do this task::

AMM TASK 24-22-00-860-811.

(2) Open the applicable air conditioning bay door:

(a) Open these access panels:

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door

(3) Do the low limit (35°F) controller BITE procedure. Do this task:

Low Limit (35 °F) Controller BITE Procedure, 21-52 TASK 801.

G. Pack Outlet Temperature Is 40°F or Higher - Fault Isolation Procedure

(1) If the ambient temperature is greater than 110°F and/or the humidity is greater than 50%, the outlet pack temperature is satisfactory.

(2) If the ambient air temperature is less than 110°F, the ambient humidity is less than 50% and the pack airflow output is lower than expected, then do this task:

Pack Air Flow is Too Low or Too High - Fault Isolation, 21-52 TASK 857.

(3) If the ambient air temperature is less than 110°F, the ambient humidity is less than 50% and the pack airflow output is satisfactory, then do the steps for the Air Mix Valve Installation Test in this task:

AMM TASK 21-61-02-400-801.

H. Repair Confirmation

(1) Evaluate the pack outlet temperature and flow for the AUTO, HIGH and APU HIGH pack flow modes. Do this task::

AMM TASK 21-51-00-943-801.

(2) If the temperature of the air through the pack airflow/temperature diagnostics tool is within the range of 35°F to 39°F, then you corrected the fault.

——— END OF TASK ———

857. Pack Air Flow is Too Low or Too High - Fault Isolation

A. Description

(1) This task is for these Observed faults:

(a) Pack Air Flow is too Low

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(b) Pack Air Flow is too High

(2) Low Air Conditioning Pack flow can contribute to pressurization problems.

(3) High Air Conditioning Pack flow is an indication of water droplets in the Pack Outlet air.

(a) This condition is an indication that the performance of the Water Separator Coalescer Bag is not satisfactory.

B. Possible Causes

(1) Coalescer Bag

(2) Pack Flow Control and Shutoff Valve

(3) Blocked or Contaminated Heat Exchangers

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L

D. Related Data

(1) 737-SL-21-075

E. Initial Evaluation

(1) Do the Pack Flow Control and Pack Cooling System Check (AMM TASK 21-51-00-943-801).

(a) If the Pack Air Flow was satisfactory for the three modes of pack operation, then there was an intermittent problem.

(b) If the Pack Air Flow was too high or too low, then do the applicable Fault Isolation Procedure below.

F. Pack Air Flow is Too High - Fault Isolation Procedure

(1) If the Pack Flow is too high, then do this task to clean the Coalescer Bag: Coalescer Bag Cleaning, AMM TASK 21-51-05-100-801.

(a) Do the Repair Confirmation at the end of this task.

G. Pack Air Flow is Too Low - Fault Isolation Procedure.

(1) If the Pack Control and Shutoff Valve was open in the APU High Flow Mode, then do these steps:

(a) Inspect the Coalescer Bag.

1) If necessary, replace the bag if it is dirty.

(b) Clean the Heat Exchangers. This is the task: Heat Exchanger - Blowing Out Debris/Blockages with Nitrogen Gas/Compressed Air, AMM TASK 21-51-03-160-811.

(c) Do the Repair Confirmation at the end of this task.

(2) If the Pack Flow Control and Shutoff Valve was not open in the APU High Flow Mode, then do the Operational Test of the Pack Flow Control Valve (AMM TASK 21-51-00-700-802).

(a) If the Operational Test was not satisfactory, then replace the Pack Flow Control and Shutoff Valve: These are the tasks:

- Flow Control and Shutoff Valve Removal, AMM TASK 21-51-01-000-801-001
- Flow Control and Shutoff Valve Installation, AMM TASK 21-51-01-400-801-001

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1) Do the Repair Confirmation at the end of this task.

(b) If the Operational Test of the Pack Flow Control Valve was satisfactory, continue.

(3) Visually inspect the Coalescer Bag.

(a) If the Coalescer Bag appears to be damaged or contaminated, then replace the bag. These are the tasks:

- Coalescer Bag Removal, AMM TASK 21-51-05-020-801
- Coalescer Bag Installation, AMM TASK 21-51-05-400-801

1) Do the Repair Confirmation at the end of this task.

(4) Do these steps to clean the Heat Exchangers off-wing:

(a) Remove the Heat Exchangers/Plenum Diffuser for the applicable pack: This is the task: Heat Exchanger and Plenum/Diffuser Assembly - Removal, AMM TASK 21-51-03-000-802-001.

(b) Clean the Heat Exchangers in accordance with the manufacturer's instructions.

(c) Do a Pressure Flow Test in accordance with the manufacturer's instructions to make sure that there is sufficient air flow through the Cooling Air side and the Bleed Air side of the Heat Exchanger.

(d) Install the Heat Exchangers/Plenum Diffuser for the applicable pack: This is the task: Heat Exchanger and Plenum/Diffuser Assembly - Installation, AMM TASK 21-51-03-400-801-001.

(e) If time constraints do not allow for cleaning and testing the Heat Exchanger, then replace the Heat Exchangers with either new or serviceable units. These are the tasks:

- Heat Exchanger and Plenum/Diffuser Assembly - Removal, AMM TASK 21-51-03-000-802-001
- Heat Exchanger and Plenum/Diffuser Assembly - Installation, AMM TASK 21-51-03-400-801-001

(f) Do the Repair Confirmation at the end of this task.

H. Repair Confirmation

(1) Do the Confidence Check of the Airplane Ability to Maintain Cabin Pressure in Flight on Single Pack Operation. This is the task: Airplane Pack Operation - Confidence Check, AMM TASK 21-00-05-780-801.

NOTE: If the Confidence Check is satisfactory, then the Pack Flow Rate is acceptable.

(a) If the Confidence Check is satisfactory, then you corrected the problem.

(b) If the Confidence Check is not satisfactory, then continue the Fault Isolation Procedure at the applicable subsequent step.

END OF TASK

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819. Divergent Pack Outlet Temperatures Noted Between Left and Right Packs

A. Description

(1) The flight deck or forward/aft passenger cabin is consistently 5°F or more warmer than the other temperature zones with the temperature selector switches all set to the same temperature.

(2) You can observe the divergent temperatures by selecting the AIR TEMP selector to SUPPLY DUCT CONT CAB, SUPPLY DUCT FWD or SUPPLY DUCT AFT.

B. Possible Causes

(1) Mix manifold temperature sensor, T472 or T475 (drifted)

(2) Wiring

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

D. Related Data

(1) SSM 21-51-12

(2) SSM 21-51-22

(3) WDM 21-51-12

(4) WDM 21-51-22

E. Initial Evaluation

(1) Operate the left and right air conditioning packs. To operate the packs, do this task:Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) Make sure that the ISOLATION VALVE switch on the P5-10 air conditioning panel is set to OPEN.

(3) Set the TRIM AIR switch on the P5-17 temperature control panel to ON.

(4) Set the CONT CAB, FWD CAB and AFT CAB temperature selectors on the P5-17 panel to the full C position.

(5) Allow the temperatures to stabilize.

(6) Set the AIR TEMP selector on the P5-17 temperature control panel to SUPPLY DUCT CONT CAB and record the temperature indicated on the gage.

(7) Set the AIR TEMP selector on the P5-17 temperature control panel to SUPPLY DUCT FWD and record the temperature indicated on the gage.

(8) Set the AIR TEMP selector on the P5-17 temperature control panel to SUPPLY DUCT AFT and record the temperature indicated on the gage.

(9) If there is not a variation (divergence) in the supply duct temperature indications, then there was an intermittent fault.

(a) Stop the operation of the packs. To stop the packs operation, do this task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

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(10) If there is a wide variation (divergence) between any of the supply duct temperatures, then do the Fault Isolation procedure below.

F. Fault Isolation Procedure

(1) Remove conditioned air from the airplane. To remove conditioned air, do this task: Remove Conditioned Air from the Airplane, AMM TASK 21-00-00-800-802.

(2) If the temperature indication of the SUPPLY DUCT CONT CAB was much greater than the temperature of the SUPPLY DUCT FWD or SUPPLY DUCT AFT, then replace the left mix manifold temperature sensor, T472. To replace sensor, T472, do these tasks: Mix Manifold Temperature Sensor Removal, AMM TASK 21-51-51-000-803 and Mix Manifold Temperature Sensor Installation, AMM TASK 21-51-51-400-802.

(3) If the temperature indication of the SUPPLY DUCT FWD OR SUPPLY DUCT AFT was much greater than the temperature indication of the SUPPLY DUCT CONT CAB, then replace the right mix manifold temperature sensor, T475. To replace sensor, T475, do these tasks: Mix Manifold Temperature Sensor Removal, AMM TASK 21-51-51-000-803 and Mix Manifold Temperature Sensor Installation, AMM TASK 21-51-51-400-802.

(4) Do the Repair Confirmation at the end of this procedure.

(a) If the Repair Confirmation was satisfactory, then you corrected the fault.

(b) If the Repair Confirmation was not satisfactory, then continue.

(5) Do these wiring checks:

(a) Make sure that these circuit breakers are open and have safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

(b) Open the forward cargo door.

(c) Remove the center bulkhead liner on the aft bulkhead of the forward cargo compartment. Do this task to remove the center bulkhead liner, Forward Cargo Compartment Aft Bulkhead Liner - Removal, AMM TASK 25-52-17-000-801.

(d) Disconnect electrical connector D3868 from the left mix manifold temperature sensor, T472, and electrical connector D6878 from the right mix manifold temperature sensor, T475.

(e) Get access to the electronic equipment racks in the E/E bay.

(f) Remove the left and right pack/zone temperature controllers. To remove the controllers, do this task:Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801 or Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

(g) Do a continuity check between these connections on wire bundle W5104:

D3858A		D3868
Pin B7	. . . . .	Pin 1
Pin A7	. . . . .	Pin 2

1) Repair any problems that you find.

(h) Do a continuity check between these connections on wire bundle W6230:

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D3860A		D3878
Pin B7	. . . . .	Pin 1
Pin A7	. . . . .	Pin 2

1) Repair any problems that you find.

(i) Reconnect connectors D3868 and D3878 to temperature sensors T472 and T475, respectively.

(j) Install the left and right zone/temperature controllers M1442 and M1443, respectively. To install the controllers, do this task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801 or Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

(k) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

(l) Do the Repair Confirmation at the end of this procedure.

1) If the Repair Confirmation is satisfactory, then you corrected the fault.

(6) Do a Pack/Zone Temperature Controller BITE test as follows: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(7) Close all access panels and doors opened for access.

G. Repair Confirmation

(1) Operate the left and right air conditioning packs. To operate the packs, do this task:Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) Make sure that the ISOLATION VALVE switch on the P5-10 air conditioning panel is set to OPEN.

(3) Set the CONT CAB, FWD CAB and AFT CAB temperature selectors on the P5-17 panel to the full C position.

(4) Allow the temperatures to stabilize.

(5) Set the AIR TEMP selector on the P5-17 temperature control panel to SUPPLY DUCT CONT CAB and record the temperature indicated on the gage.

(6) Set the AIR TEMP selector on the P5-17 temperature control panel to SUPPLY DUCT FWD and record the temperature indicated on the gage.

(7) Set the AIR TEMP selector on the P5-17 temperature control panel to SUPPLY DUCT AFT and record the temperature indicated on the gage.

(8) If the supply duct temperatures are almost the same, then you have corrected the fault.

(9) Stop the operation of the packs. To stop the packs operation, do this task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

——— END OF TASK ———

820. L (R) PACK Light is ON (Without MASTER CAUTION Recall) - Fault Isolation

NOTE: You can refer to 21-53 TASK SUPPORT Figure 301 for an illustration of this procedure in a Block Diagram.

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A. Description

(1) This task is for these Observed Faults:

(a) The L PACK Light on the Air Conditioning Module, P5-10, comes ON without MASTER CAUTION Recall.

(b) The R PACK Light on the Air Conditioning Module, P5-10, comes ON without MASTER CAUTION Recall.

B. Possible Causes

(1) Damaged or missing Left (Right) Fan Bypass Check Valve Door in the Primary Plenum/Diffuser Assembly

OMR 815, 816, 818-826, 901-999; AIRPLANES WITH SRADA

(2) Left (Right) Aft (Exhaust, Exit) Smart Ram Air Door Actuator (SRADA), M2796 (M2798)

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(3) Flap Switch, S1051

OMR 802, 805, 807, 809, 810, 812, 813; AIRPLANES WITH RADA

(4) Left (Right) Ram Air Door Actuator (RADA), M326 (M327)

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(5) Blocked Left (Right) Heat Exchangers

(6) Left (Right) Air Cycle Machine (ACM)

(7) Standby Temperature Control Valve (STCV) Sense Lines

(8) Left (Right) STCV, V140 (V142)

(9) Loss of 28V DC Power to the Air Conditioning Panel, P5-10

(10) Wiring

(11) Left (Right) Compressor Discharge Overheat Switch, S2 (S5)

(12) Left (Right) Pack Discharge Duct Overheat Switch, S934 (S935)

(13) Left (Right) Turbine Inlet Overheat Switch, S938 (S939)

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	7	C01177	A/C PACK/ENGINE BLEED AIR OVHT RIGHT
C	8	C01176	A/C PACK/ENGINE BLEED AIR OVHT LEFT

D. Related Data

(1) WDM 21-51-15

(2) WDM 21-51-25

E. Initial Evaluation

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 815, 816, 818-826, 901-999; AIRPLANES WITH SRADA

(1) Do a visual check and make sure that the gap at the Inlet and Exit Doors is correct.

- The Deflector Door should be deployed, and the Inlet Door angle should be less than 180 degrees.
- Exit Door: The correct gap between two louvers is 3.8 ±0.5 in. (9.7 ±1.3 cm).

(a) If the gap is not correct, replace the applicable SRADA. These are the tasks:

- Ram Air Exhaust Actuator - Removal, AMM TASK 21-51-27-000-801
- Ram Air Exhaust Actuator - Installation, AMM TASK 21-51-27-400-801

(b) If the gap at the Inlet and Exit Doors is correct, then continue.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(2) Push the TRIP RESET Switch on the Air Conditioning Module, P5-10 on the Forward Overhead Panel, P5.

NOTE: If the TRIP RESET does not extinguish the PACK light, there may be a nuisance Temperature Control Valve (TCV) fault after the BITE procedure is performed.

(3) Do the Pack Temperature Control - BITE Procedure, 21-62 TASK 801 on the Left and the Right Temperature Controllers.

(a) If maintenance messages show, refer to the table at the end of the BITE Procedure to find the applicable Fault Isolation Manual (FIM) Tasks for the messages that show, do the indicated corrective actions, then continue at the Fault Isolation Procedure below.

1) If you find a Dual Channel Fault, then do the applicable FIM Task Procedures to correct the problem. There is no need to continue at the Fault Isolation Procedure below.

NOTE: A Dual Channel Fault is a combination of two different maintenance messages (one per each Temperature Controller).

(b) If maintenance messages do not show, then continue.

F. Fault Isolation Procedure

NOTE: You can refer to 21-53 TASK SUPPORT Figure 301 for an illustration of this procedure in a Block Diagram.

(1) Do this check of the Left (Right) Fan Bypass Check Valve in the Primary Plenum/Diffuser Assembly:

(a) Get access to the applicable Left (Right) Fan Bypass Check Valve:

Open the applicable Access Panel:

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door

(b) Open the access panel at the bottom of the Primary Plenum/Diffuser Assembly.

(c) Make sure that the Fan Bypass Check Valve Door is installed on the bottom of the Primary Diffuser and that the Door Hinge is not damaged.

1) If the Fan Bypass Check Valve Door is damaged or not installed at the bottom of the Primary Diffuser, replace the Primary Plenum/Diffuser Assembly. These are the tasks:

- Heat Exchanger and Plenum/Diffuser Assembly - Removal, AMM TASK 21-51-03-000-805-002

ECCN 9E991 BOEING PROPRIETARY - See title page for details

- Heat Exchanger and Plenum/Diffuser Assembly - Installation, AMM TASK 21-51-03-400-804-002

a) Make sure that the access panel is installed on the bottom of the Primary Plenum/Diffuser Assembly.

2) If the Fan Bypass Check Valve Door is satisfactory, continue.

OMR 815, 816, 818-826, 901-999; AIRPLANES WITH SRADA

(2) Examine the Left (Right) Aft (Exhaust, Exit) SRADA:

(a) Do this task: Ram Air System (Smart Ram Air Door Actuator SRADA) - Operational Test, AMM TASK 21-51-00-730-801.

1) If the Operational Test is not satisfactory, replace defective components like SRADA Linkage, Bearings, Flap Switch, S1051.

2) If you find no defective components, and you did not already replace the Exit SRADA in the Initial Evaluation, then replace the Exit SRADA. These are the applicable tasks:

- Ram Air Exhaust Actuator - Removal, AMM TASK 21-51-27-000-801
- Ram Air Exhaust Actuator - Installation, AMM TASK 21-51-27-400-801

(b) If you did not replace defective components, then continue.

OMR 802, 805, 807, 809, 810, 812, 813; AIRPLANES WITH RADA

(3) Examine the Left (Right) Ram Air Door Actuator (RADA). This is the task: Ram Air System (Ram Air Door Actuator RADA) - Operational Test, AMM TASK 21-51-00-730-802.

(a) If the RADA Operational Test is not satisfactory, replace defective components like RADA Linkage, Bearings, Flap Switch, S1051.

(b) If the RADA Operational Test is satisfactory, then continue.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(4) Clean the Left (Right) Heat Exchangers. This is the task: Heat Exchanger - Blowing Out Debris/Blockages with Nitrogen Gas/Compressed Air, AMM TASK 21-51-03-160-811.

(5) Replace the Left (Right) Pack ACM. These are the tasks:

- Air Cycle Machine Removal, AMM TASK 21-51-04-000-802-002
- Air Cycle Machine Installation, AMM TASK 21-51-04-400-802-002

NOTE: If this is the first occurrence of a pack overheat condition/PACK Light without Master Caution recall, and the ACM passed the inspection criteria at the Air Cycle Machine Removal, AMM TASK 21-51-04-000-802-002, the removed ACM can be re-installed.

(6) Examine all Left (Right) STCV Sense Lines for cracks and blockage as follows:

(a) Remove the four STCV Sense Lines from each end.

(b) If you find blockage or unwanted materials, remove the blockage and/or unwanted materials.

1) Re-install the Sense Lines.

NOTE: Apply a light coat of Pure Nickel Special compound, D00006 to the threads of the Sense Lines.

Use torque wrench, STD-1015 and tighten to 160 ±8 in-lb (18 ±1 N·m).

2) Do the STCV Installation Test in this task: Standby Temperature Control Valve Installation, AMM TASK 21-51-11-400-801.

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(c) If you find damage, replace the defective Sense Line(s).

NOTE: Apply a light coat of Pure Nickel Special compound, D00006 to the threads of the Sense Lines.

Use torque wrench, STD-1015 and tighten to 160 ±8 in-lb (18 ±1 N·m).

1) Do the STCV Installation Test in this task: Standby Temperature Control Valve Installation, AMM TASK 21-51-11-400-801.

(d) If you do not find blockage or damage in the Sense Lines, then replace the Left (Right) STCV. These are the tasks:

NOTE: This step can be skipped if there are no reported temperature control issues in either the flight deck or passenger cabin. This is because if the STCV were to result in pack overheat, it would also result in inability to cool the flight deck or passenger cabin.

- Standby Temperature Control Valve Removal, AMM TASK 21-51-11-000-801
- Standby Temperature Control Valve Installation, AMM TASK 21-51-11-400-801

(7) Do a check for 28V DC Power for the L PACK (R PACK) Light at the AIR CONDITIONING Panel, P5-10 as follows (WDM 21-51-15, WDM 21-51-25):

- L PACK: pin 29 of connector D646
- R PACK: pin 25 of connector D680

(a) If you do not find 28V DC, examine the wiring between the Air Conditioning Accessory Unit (ACAU) 1, M324 and the Air Conditioning Panel, P5-10, as follows:

L PACK LIGHT

ACAU 1	**P5-10 PANEL**
D458B	**D646**
pin 36	pin 29

R PACK LIGHT

ACAU 2	**P5-10 PANEL**
D10002B	**D680**
pin 36	pin 25

1) Repair the problems that you find.

(b) If you find 28V DC, then continue.

(8) Replace the Overheat Switches as shown below:

(a) Left (Right) Compressor Discharge Overheat Switch, S2 (S5)

- Compressor Discharge Overheat Switch Removal, AMM TASK 21-51-40-000-801
- Compressor Discharge Overheat Switch Installation, AMM TASK 21-51-40-400-801

(b) Left (Right) Pack Discharge Duct Overheat Switch, S934 (S935)

- Pack Discharge Overheat Switch Removal, AMM TASK 21-51-42-000-803-002
- Pack Discharge Overheat Switch Installation, AMM TASK 21-51-42-400-803-002

(c) Left (Right) Turbine Inlet Overheat Switch, S938 (S939)

- Turbine Inlet Overheat Switch Removal, AMM TASK 21-51-41-000-801
- Turbine Inlet Overheat Switch Installation, AMM TASK 21-51-41-400-801

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

821. L (R) PACK PACK Light is ON During MASTER CAUTION Recall - Fault Isolation

A. Description

(1) This task is for these Observed Faults:

(a) The L PACK Light on the Air Conditioning Module, P5-10, comes ON during MASTER CAUTION Recall.

(b) The R PACK Light on the Air Conditioning Module, P5-10, comes ON during MASTER CAUTION Recall.

(2) The MASTER CAUTION Recall is required during Airplane Departure and Landing Approach.

B. Possible Causes

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

(1) Fault(s) in L (R) Pack/Zone Temperature Controller (PZTC), M1442 (M1443)

OMR 820-826

(2) Fault(s) in L (R) Pack Flow Temperature Controller (PFTC), M2951 (M2952)

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

C. Related Data

(1) SSM 31-52-51

D. Initial Evaluation

OMR 820-826; AIRPLANES WITH PFTC P/N 51090248-002/-003

(1) Do these steps to clear the fault:

NOTE: PACK and/or or ZONE TEMP Lights can come ON, if one or all of the Temperature Selectors (CONT CAB, FWD CAB, AFT CAB) are set to OFF.

(a) On the Cabin Temperature Panel, P5-17, make sure that all three Temperature Selectors (CONT CAB, FWD CAB, AFT CAB) are not set to OFF.

1) If a selector is set to OFF, move it from the OFF position.

(b) On the Front Panel Display of the applicable Pack Flow Temperature Controller, M2951 (M2952), clear the EXISTING FAULTS.

(c) On the Glareshield Panel, P7, push the MASTER CAUTION switch to reset it.

(d) Wait Thirty (30) seconds.

(e) Make sure that the PACK Lights stay OFF.

(f) Look at the EXISTING FAULTS again.

1) If the maintenance message does not show, then there was an intermittent fault.

2) If a maintenance message shows or if a PACK Light stays ON, then continue.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(2) On the P7 Glareshield Panel, push and release the Left (Right) MASTER CAUTION switch.

(3) If one or more PACK Lights on the Temperature Control Panel intermittently comes ON, make a record and then do the Fault Isolation Procedure below.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Fault Isolation Procedure

(1) Do the Pack Temperature Control - BITE Procedure, 21-62 TASK 801 on the Left and the Right Temperature Controllers.

(a) If maintenance messages show, refer to the table at the end of the BITE Task to find the applicable FIM Tasks for the messages that show.

(2) Do the Pack Temperature Control - BITE Procedure, 21-62 TASK 801 again to make sure that you corrected the problem.

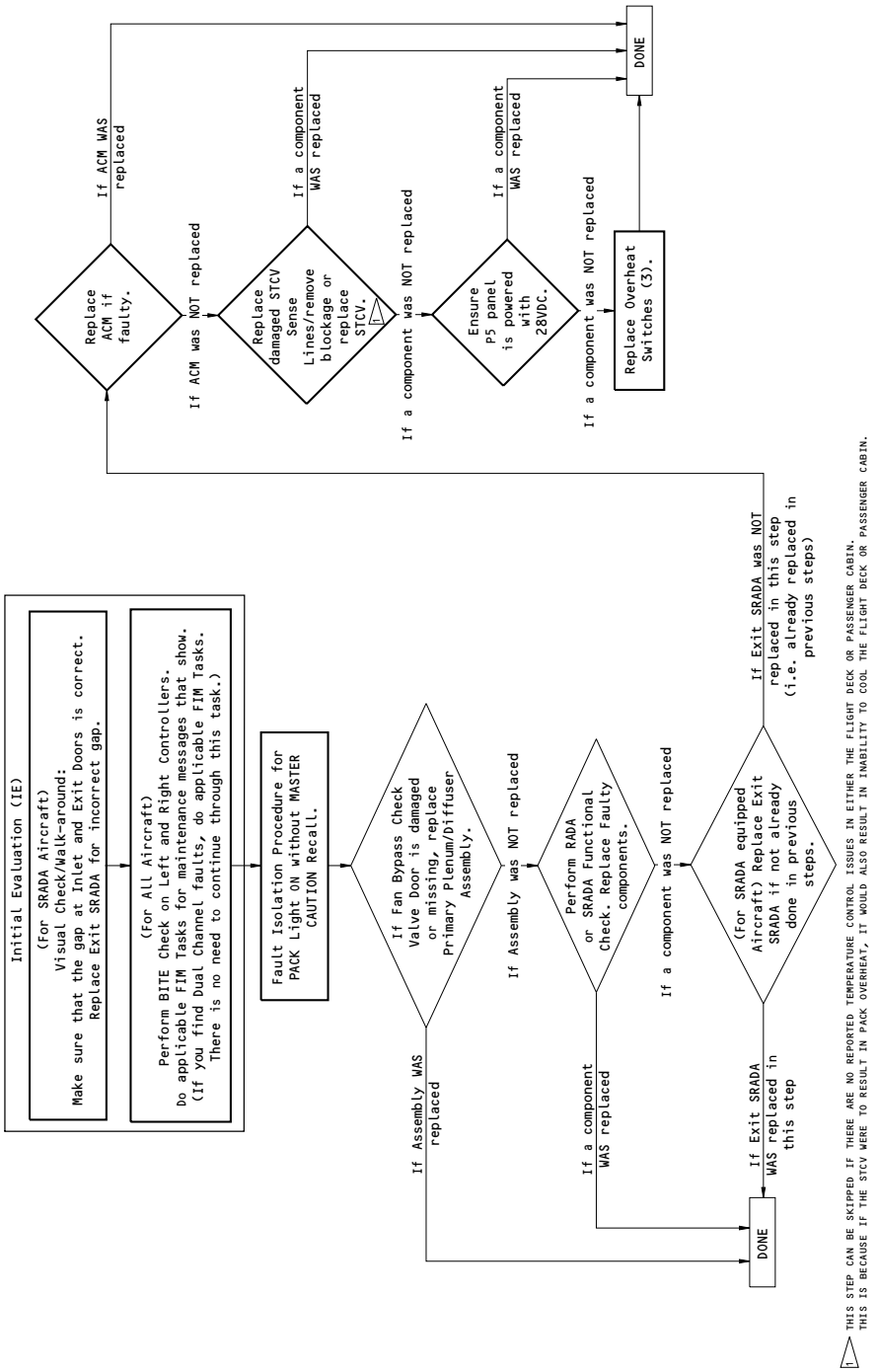
(3) On the Glareshield Panel, P7, push and release the Left (Right) MASTER CAUTION Switch.

(4) Make sure that the L (R) PACK Light on the Air Conditioning Module, P5-10, does not come ON.

——— END OF TASK ———

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PACK Light ON - Fault Isolation Flowchart
Figure 301/21-53-00-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

801. Cabin Temperature Controller BITE Procedure

A. General

OMR 701-799

(1) The Cabin Temperature Controller (CTC) is located on the top shelf of the E4 Rack in the Electronics Equipment Compartment.

(2) You do the CTC Built-In-Test Equipment (BITE) Procedure on the front panel of the of the CTC. To start the test, you push and rotate the TEST Switch clockwise (Figure 201).

(3) The BITE Test does these checks:

(a) An internal check of the CTC and

(b) BITE Interface checks for the:

- Control Cabin Temperature Sensor
- Control Cabin Duct Limit Sensor
- Control Cabin Duct Anticipator
- Passenger Cabin Temperature Sensor
- Passenger Cabin Duct Limit Sensor
- Passenger Cabin Duct Anticipator
- Control and Passenger Cabin Temperature Selectors.

(c) A Red NO GO Light indicates a specific failure which is identified by the CTC.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(4) There is no CTC BITE Procedure for the 737-800 or 737-900 airplanes.

B. BITE Procedure

(1) There is no CTC BITE Procedure for the 737-800 or 737-900 airplanes.

OMR 701-799

C. BITE Procedure

(1) Do the BITE Procedure for the CTC as follows (Figure 201):

(a) Put the BLEED 1 and BLEED 2 Switches on the Air Conditioning Module of the P5 Overhead Panel to the OFF position.

(b) Put the APU BLEED Switch on the Air Conditioning Module to the OFF position.

(c) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	4	C00257	AIR CONDITIONING OVERHEAT

(d) Put the CONT CABIN and PASS CABIN Temperature Selectors on the Temperature Control Module on the P5 Overhead Panel to the AUTO (12 o'clock) position.

(e) Put the L PACK and R PACK Switches on the Temperature Control Module to AUTO.

(f) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799 (Continued)

(g) Get access to the CTC located on the top shelf of the E4 Rack in the Electronics Equipment Compartment.

(h) Put the BITE Switch on the controller to the START position.

(i) Push the GO and NO GO Lights on the controller.

1) Make sure that the lights come ON when you push them.

(j) Push and rotate the BITE Switch to each of the TEMP CONTROL BOX positions.

NOTE: Make sure that you push the BITE Switch.

1) Make sure that the Green GO Light comes ON for each pack.

NOTE: The Left Pack Light is for the Control Cabin Sensors and the Right Pack Light is for the Passenger Cabin Sensors.

(k) Push and rotate the BITE Switch to the fourth BITE Switch position.

1) Make sure that the Green GO Light comes ON for each pack.

(l) Push and rotate the BITE Switch to the CABIN SENSOR position.

1) Make sure that the Green GO Light comes ON for each pack.

(m) Push and rotate the BITE Switch to the ANTICIPATOR SENSOR position.

1) Make sure that the Green GO Light comes ON for each pack.

(n) Push and rotate the BITE Switch to the DUCT LIMIT SENSOR position.

1) Make sure that the Green GO Light comes ON for each pack.

(o) Push and rotate the BITE Switch to the TEMP SELECTOR position.

1) Make sure that the green GO Light comes ON for each pack.

(p) Put the BITE Switch to the START position.

1) If a NO GO Light comes ON, then refer to the table at the end of this task to find the applicable Fault Isolation Manual (FIM) Task for the maintenance messages that show.

2) If the CTC passes the BITE Test and the Passenger Cabin Temperature can not be made colder or warmer with the Temperature Selector in the AUTO or MANUAL Mode, then do these steps:

a) For the applicable FWD (AFT) Cabin Zone Temperature Sensors, do this task: Cabin Temperature Sensor Resistance Check, AMM TASK 21-61-10-200-801.

<1> If the Cabin Zone Temperature Sensor Resistance values are unsatisfactory, replace the applicable sensor(s). These are the tasks:

- Cabin Temperature Sensor Removal, AMM TASK 21-61-10-000-802
- Cabin Temperature Sensor Installation, AMM TASK 21-61-10-400-802.

b) For the applicable FWD (AFT) Cabin Duct Temperature Sensors, do this task: Duct Temperature Limit Sensor Test, AMM TASK 21-61-04-000-805-001.

<1> If the Cabin Duct Temperature Sensor Resistance values are unsatisfactory, replace the applicable sensor(s). These are the tasks:

- Duct Temperature Limit Sensor Removal, AMM TASK 21-61-04-000-802-001

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799 (Continued)

- Duct Temperature Limit Sensor Installation, AMM TASK 21-61-04-400-802-001.

c) For the applicable Flight Deck (Passenger Cabin) Duct Temperature Anticipator Sensors, do this task: Duct Temperature Anticipator Sensor Resistance Check, AMM TASK 21-61-08-200-801.

<1> If the Flight Deck (Passenger Cabin) Duct Temperature Anticipator Sensor Resistance values are unsatisfactory, replace the applicable sensor(s). These are the tasks:

- Duct Temperature Anticipator Sensor Removal, AMM TASK 21-61-08-000-801
- Duct Temperature Anticipator Sensor Installation, AMM TASK 21-61-08-400-801.

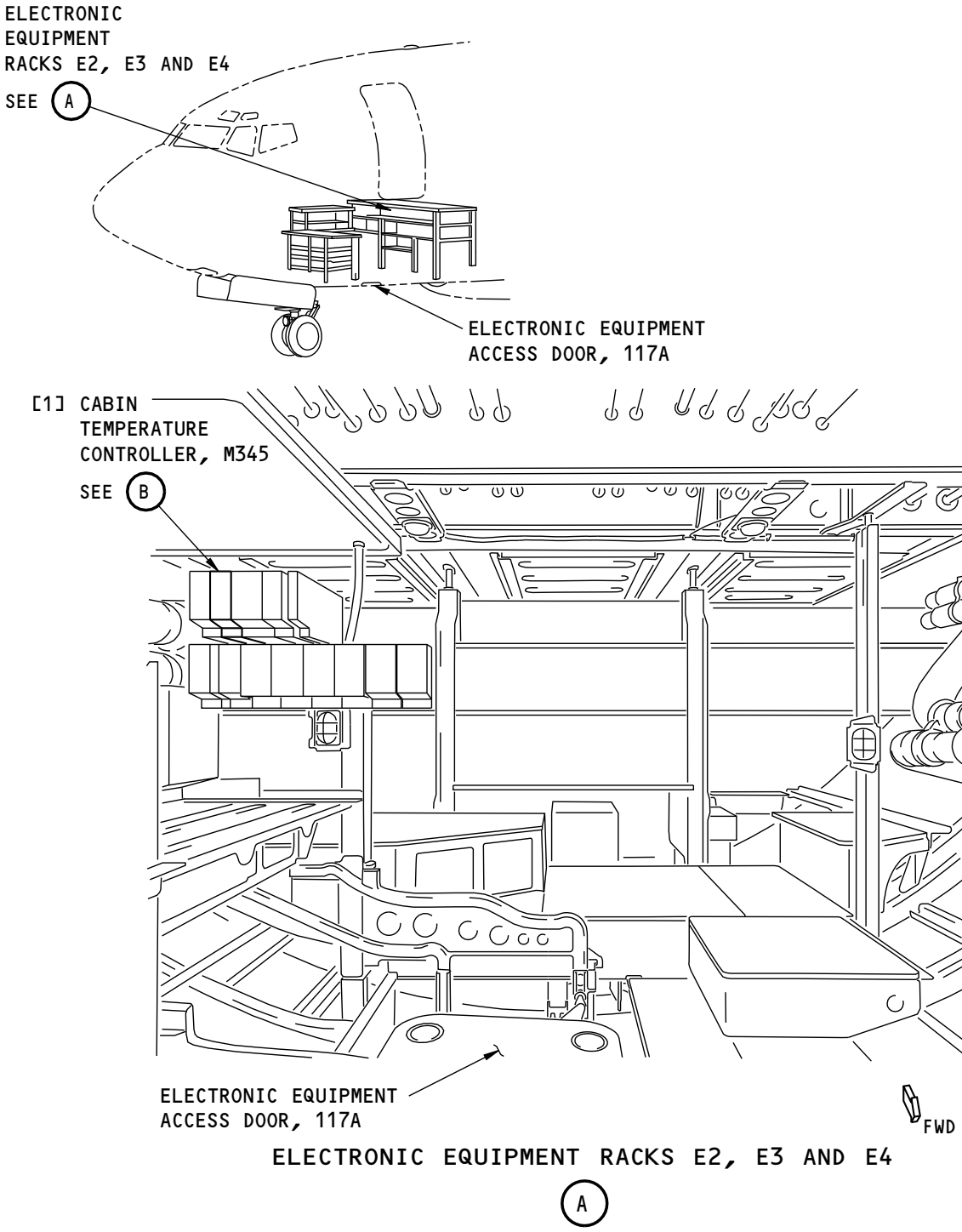
(q) Make sure that this circuit breaker is closed:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	4	C00257	AIR CONDITIONING OVERHEAT

——— END OF TASK ———

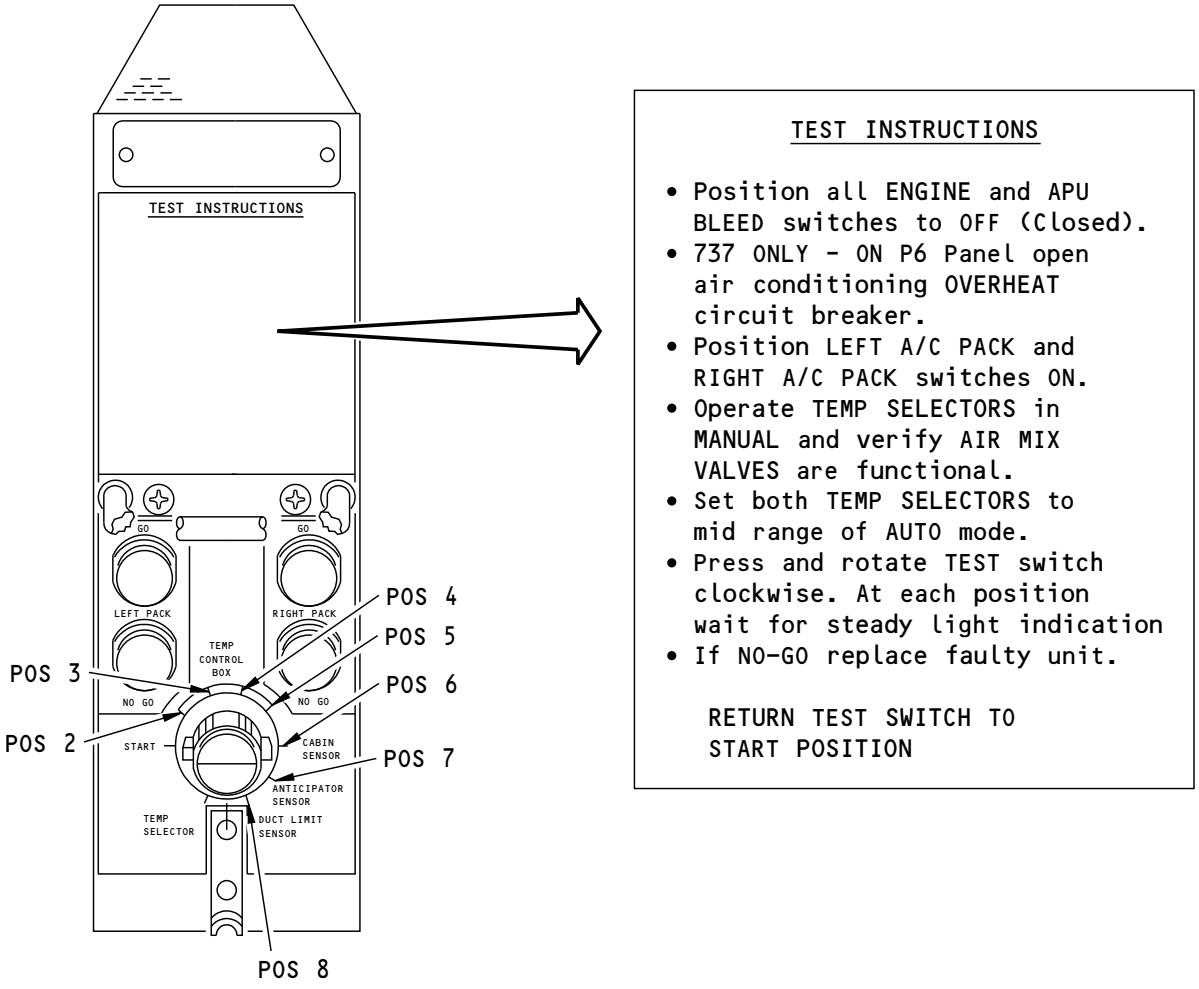
ECCN 9E991 BOEING PROPRIETARY - See title page for details



H03967 S0006740722_V1

Cabin Temperature Controller BITE Test
Figure 201/21-61-00-990-807 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



CABIN TEMPERATURE CONTROLLER

B

H03969 S0006740723_V2

Cabin Temperature Controller BITE Test
Figure 201/21-61-00-990-807 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR ALL

LRU/SYSTEM	MAINTENANCE MESSAGE	GO TO FIM TASK
CAB TEMP CONT	ANTICIPATOR SENSOR - LEFT PACK - NO GO (Position 7)	21-61 TASK 831
CAB TEMP CONT	ANTICIPATOR SENSOR - RIGHT PACK - NO GO (Position 7)	21-61 TASK 832
CAB TEMP CONT	CABIN SENSOR - LEFT PACK - NO GO (Position 6)	21-61 TASK 833
CAB TEMP CONT	CABIN SENSOR - RIGHT PACK - NO GO (Position 6)	21-61 TASK 834
CAB TEMP CONT	CLOSE VALVE DRIVE CHECK - LEFT PACK - NO GO (Position 5)	21-61 TASK 865
CAB TEMP CONT	CLOSE VALVE DRIVE CHECK - RIGHT PACK - NO GO (Position 5)	21-61 TASK 865
CAB TEMP CONT	DEADBAND CHECK - LEFT PACK - NO GO (Position 4)	21-61 TASK 865
CAB TEMP CONT	DEADBAND CHECK - RIGHT PACK - NO GO (Position 4)	21-61 TASK 865
CAB TEMP CONT	DUCT LIMIT SENSOR - LEFT PACK - NO GO (Position 8)	21-61 TASK 829
CAB TEMP CONT	DUCT LIMIT SENSOR - RIGHT PACK - NO GO (Position 8)	21-61 TASK 830
CAB TEMP CONT	OPEN VALVE DRIVE CHECK - LEFT PACK - NO GO (Position 3)	21-61 TASK 865
CAB TEMP CONT	OPEN VALVE DRIVE CHECK - RIGHT PACK - NO GO (Position 3)	21-61 TASK 865
CAB TEMP CONT	POWER CHECK - LEFT PACK - NO GO (Position 2)	21-61 TASK 865
CAB TEMP CONT	POWER CHECK - RIGHT PACK - NO GO (Position 2)	21-61 TASK 865
CAB TEMP CONT	TEMP SELECTOR - LEFT PACK - NO GO	21-61 TASK 835
CAB TEMP CONT	TEMP SELECTOR - RIGHT PACK - NO GO	21-61 TASK 836

OMR 701-799

829. Flight Compartment Duct Limit Sensor Failed - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) DUCT LIMIT SENSOR - LEFT PACK - NO GO

B. Possible Causes

(1) Control cabin duct limit sensor, T419

(2) Cabin temperature controller, M345

(3) Wiring Problem

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	2	C00268	AIR CONDITIONING TEMP CONTROL AUTO L
B	2	C00258	AIR CONDITIONING TEMP CONTROL AUTO R

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OMR 701-799 (Continued)

D. Related Data

(1) SSM 21-61-21

(2) WDM 21-61-21

E. Initial Evaluation

(1) Do this task: Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.

(2) If the maintenance message does not show, then there was an intermittent fault.

(3) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Replace the control cabin duct limit sensor T419 in the aft EE bay.

These are the tasks:

Duct Temperature Limit Sensor Removal, AMM TASK 21-61-04-000-802-001,

Duct Temperature Limit Sensor Installation, AMM TASK 21-61-04-400-802-001.

(a) Do this task: Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.

1) If the maintenance message does not show, then you corrected the fault.

2) If the maintenance message shows, then continue.

(2) Do this check of the wiring between the control cabin duct limit sensor and the cabin temperature controller, M345 on the E4-1 shelf:

(a) Disconnect connector D512 from the control cabin duct limit sensor.

(b) Remove the cabin temperature controller. To remove the controller, do this task: Cabin Temperature Controller (CTC) Removal, AMM TASK 21-61-01-000-801.

(c) Do a wiring check between these pins of connector D512 of the control cabin duct limit sensor and the connector D454 of the cabin temperature controller:

D512		D454
pin 1	. . . . .	pin 22
pin 3	. . . . .	pin 21

(d) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-connect connector D512 to the control cabin duct limit sensor.

3) Re-install the cabin temperature controller. To install the controller, do this task: Cabin Temperature Controller (CTC) Installation, AMM TASK 21-61-01-400-801.

4) Do this task: Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.

5) If the maintenance message does not show, then you corrected the fault.

(e) If you do not find a problem with the wiring, then continue.

1) Reconnect connector D512 to the control cabin duct limit sensor.

(3) Install a new cabin temperature controller. To install the controller, do this task: Cabin Temperature Controller (CTC) Installation, AMM TASK 21-61-01-400-801.

(a) Do this task: Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799 (Continued)

(b) If the maintenance message does not show, then you corrected the fault.

——— END OF TASK ———

830. Passenger Cabin Duct Limit Sensor Failed - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) DUCT LIMIT SENSOR - RIGHT PACK - NO GO

B. Possible Causes

(1) Passenger cabin duct limit sensor, T416

(2) Cabin temperature controller, M345

(3) Wiring Problem

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	2	C00268	AIR CONDITIONING TEMP CONTROL AUTO L
B	2	C00258	AIR CONDITIONING TEMP CONTROL AUTO R

D. Related Data

(1) SSM 21-61-22

(2) WDM 21-61-22

E. Initial Evaluation

(1) Do this task: Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.

(2) If the maintenance message does not show, then there was an intermittent fault.

(3) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Replace the passenger cabin duct limit sensor T416 in the overhead distribution duct.

These are the tasks:

Duct Temperature Limit Sensor Removal, AMM TASK 21-61-04-000-802-001,

Duct Temperature Limit Sensor Installation, AMM TASK 21-61-04-400-802-001.

(a) Do this task: Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.

1) If the maintenance message does not show, then you corrected the fault.

2) If the maintenance message shows, then continue.

(2) Do this check of the wiring between the passenger cabin duct limit sensor and the cabin temperature controller, M345 on the E4-1 shelf:

(a) Disconnect connector D518 from the passenger cabin duct limit sensor.

(b) Remove the cabin temperature controller. To remove the controller, do this task: Cabin Temperature Controller (CTC) Removal, AMM TASK 21-61-01-000-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799 (Continued)

(c) Do a wiring check between these pins of connector D518 of the control cabin duct limit sensor and the connector D456 of the cabin temperature controller:

D518		D456
pin 1	. . . . .	pin 22
pin 3	. . . . .	pin 21

(d) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.
2) Re-connect connector D518 to the passenger cabin duct limit sensor.
3) Re-install the cabin temperature controller. To install the controller, do this task: Cabin Temperature Controller (CTC) Installation, AMM TASK 21-61-01-400-801.
4) Do this task: Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.
5) If the maintenance message does not show, then you corrected the fault.

(e) If you do not find a problem with the wiring, then continue.

1) Re-connect connector D518 to the passenger cabin duct limit sensor.

(3) Install a new cabin temperature controller. To install the controller, do this task: Cabin Temperature Controller (CTC) Installation, AMM TASK 21-61-01-400-801.

(a) Do this task: Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.

(b) If the maintenance message does not show, then you corrected the fault.

——— END OF TASK ———

831. Flight Compartment Duct Anticipator Sensor Failed - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) ANTICIPATOR SENSOR - LEFT PACK - NO GO

B. Possible Causes

(1) Control cabin duct anticipator sensor, T420

(2) Cabin temperature controller, M345

(3) Wiring Problem

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	2	C00268	AIR CONDITIONING TEMP CONTROL AUTO L
B	2	C00258	AIR CONDITIONING TEMP CONTROL AUTO R

D. Related Data

(1) SSM 21-61-21

(2) WDM 21-61-21

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OMR 701-799 (Continued)

E. Initial Evaluation

(1) Do this task: Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.

(2) If the maintenance message does not show, then there was an intermittent fault.

(3) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Replace the control cabin duct anticipator sensor T420 in the aft EE bay.

These are the tasks:

Duct Temperature Anticipator Sensor Removal, AMM TASK 21-61-08-000-801,

Duct Temperature Anticipator Sensor Installation, AMM TASK 21-61-08-400-801.

(a) Do this task: Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.

1) If the maintenance message does not show, then you corrected the fault.

2) If the maintenance message shows, then continue.

(2) Do this check of the wiring between the control cabin duct anticipator sensor and the cabin temperature controller, M345 on the E4-1 shelf:

(a) Disconnect connector D514 from the control cabin duct anticipator sensor.

(b) Remove the cabin temperature controller. To remove the controller, do this task: Cabin Temperature Controller (CTC) Removal, AMM TASK 21-61-01-000-801.

(c) Do a wiring check between these pins of connector D514 of the control cabin duct limit sensor and the connector D454 of the cabin temperature controller:

D514		D454
pin 1	. . . . .	pin 14
pin 2	. . . . .	pin 15
pin 3	. . . . .	pin 16

(d) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-connect connector D514 to the control cabin duct anticipator sensor.

3) Re-install the cabin temperature controller. To install the controller, do this task: Cabin Temperature Controller (CTC) Installation, AMM TASK 21-61-01-400-801.

4) Do this task: Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.

5) If the maintenance message does not show, then you corrected the fault.

(e) If you do not find a problem with the wiring, then continue.

1) Re-connect connector D514 to the control cabin duct anticipator sensor.

(3) Install a new cabin temperature controller. To install the controller, do this task: Cabin Temperature Controller (CTC) Installation, AMM TASK 21-61-01-400-801.

(a) Do this task: Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.

(b) If the maintenance message does not show, then you corrected the fault.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

832. Passenger Cabin Duct Anticipator Sensor Failed - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) ANTICIPATOR SENSOR - RIGHT PACK - NO GO

B. Possible Causes

(1) Passenger cabin duct anticipator sensor, T417

(2) Cabin temperature controller, M345

(3) Wiring Problem

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	2	C00268	AIR CONDITIONING TEMP CONTROL AUTO L
B	2	C00258	AIR CONDITIONING TEMP CONTROL AUTO R

D. Related Data

(1) SSM 21-61-22

(2) WDM 21-61-22

E. Initial Evaluation

(1) Do this task: Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.

(2) If the maintenance message does not show, then there was an intermittent fault.

(3) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Replace the passenger cabin duct anticipator sensor T417 in the overhead distribution duct.

These are the tasks:

Duct Temperature Anticipator Sensor Removal, AMM TASK 21-61-08-000-801,

Duct Temperature Anticipator Sensor Installation, AMM TASK 21-61-08-400-801.

(a) Do this task: Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.

1) If the maintenance message does not show, then you corrected the fault.

2) If the maintenance message shows, then continue.

(2) Do this check of the wiring between the passenger cabin duct anticipator sensor and the cabin temperature controller, M345 on the E4-1 shelf:

(a) Disconnect connector D520 from the passenger cabin duct anticipator sensor.

(b) Remove the cabin temperature controller. To remove the controller, do this task: Cabin Temperature Controller (CTC) Removal, AMM TASK 21-61-01-000-801.

(c) Do a wiring check between these pins of connector D520 of the control cabin duct limit sensor and the connector D456 of the cabin temperature controller:

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OMR 701-799 (Continued)

D520		D456
pin 1	. . . . .	pin 14
pin 2	. . . . .	pin 15
pin 3	. . . . .	pin 16

(d) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-connect connector D520 to the passenger cabin duct anticipator sensor.

3) Re-install the cabin temperature controller. To install the controller, do this task: Cabin Temperature Controller (CTC) Installation, AMM TASK 21-61-01-400-801.

4) Do this task: Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.

5) If the maintenance message does not show, then you corrected the fault.

(e) If you do not find a problem with the wiring, then continue.

1) Reconnect connector D518 to the passenger cabin duct limit sensor.

(3) Install a new cabin temperature controller. To install the controller, do this task: Cabin Temperature Controller (CTC) Installation, AMM TASK 21-61-01-400-801.

(a) Do this task: Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.

(b) If the maintenance message does not show, then you corrected the fault.

——— END OF TASK ———

833. Flight Compartment Temperature Sensor Failed - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) CABIN SENSOR - LEFT PACK - NO GO

B. Possible Causes

(1) Control cabin temperature sensor, M1718

(2) Cabin temperature controller, M345

(3) Wiring Problem

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	2	C00268	AIR CONDITIONING TEMP CONTROL AUTO L
B	2	C00258	AIR CONDITIONING TEMP CONTROL AUTO R

D. Related Data

(1) SSM 21-61-21

(2) WDM 21-61-21

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OMR 701-799 (Continued)

E. Initial Evaluation

(1) Do this task: Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.

(2) If the maintenance message does not show, then there was an intermittent fault.

(3) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Replace the control cabin temperature sensor in the M1718 module in the control cabin.

These are the tasks:

Cabin Temperature Sensor Removal, AMM TASK 21-61-10-000-802,

Cabin Temperature Sensor Installation, AMM TASK 21-61-10-400-802.

(a) Do this task: Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.

1) If the maintenance message does not show, then you corrected the fault.

2) If the maintenance message shows, then continue.

(2) Do this check of the wiring between the control cabin temperature sensor and the cabin temperature controller, M345 on the E4-1 shelf:

(a) Disconnect connector D10958 from the control cabin temperature sensor.

(b) Remove the cabin temperature controller. To remove the controller, do this task: Cabin Temperature Controller (CTC) Removal, AMM TASK 21-61-01-000-801.

(c) Do a wiring check between these pins of connector D10958 of the control cabin temperature sensor and the connector D454 of the cabin temperature controller:

D10958		D454
pin 1	. . . . .	pin 23
pin 3	. . . . .	pin 24

(d) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-connect connector D10958 to the control cabin temperature sensor.

3) Re-install the cabin temperature controller. To install the controller, do this task: Cabin Temperature Controller (CTC) Installation, AMM TASK 21-61-01-400-801.

4) Do this task: Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.

5) If the maintenance message does not show, then you corrected the fault.

(e) If you do not find a problem with the wiring, then continue:

1) Reconnect connector D10958 to the control cabin temperature sensor.

(3) Install a new cabin temperature controller. To install the controller, do this task: Cabin Temperature Controller (CTC) Installation, AMM TASK 21-61-01-400-801.

(a) Do this task: Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.

(b) If the maintenance message does not show, then you corrected the fault.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799 (Continued)

834. Passenger Cabin Temperature Sensor Failed - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) CABIN SENSOR - RIGHT PACK - NO GO

B. Possible Causes

(1) Passenger cabin temperature sensor, M1719

(2) Cabin temperature controller, M345

(3) Wiring Problem

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	2	C00268	AIR CONDITIONING TEMP CONTROL AUTO L
B	2	C00258	AIR CONDITIONING TEMP CONTROL AUTO R

D. Related Data

(1) SSM 21-61-22

(2) WDM 21-61-22

E. Initial Evaluation

(1) Do this task: Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.

(2) If the maintenance message does not show, then there was an intermittent fault.

(3) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Replace the passenger cabin temperature sensor in the M1719 module in the main passenger cabin.

These are the tasks:

Cabin Temperature Sensor Removal, AMM TASK 21-61-10-000-802,

Cabin Temperature Sensor Installation, AMM TASK 21-61-10-400-802.

(a) Do this task: Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.

1) If the maintenance message does not show, then you corrected the fault.

2) If the maintenance message shows, then continue.

(2) Do this check of the wiring between the passenger cabin temperature sensor and the cabin temperature controller, M345 on the E4-1 shelf:

(a) Disconnect connector D10962 from the passenger cabin temperature sensor.

(b) Remove the cabin temperature controller. To remove the controller, do this task: Cabin Temperature Controller (CTC) Removal, AMM TASK 21-61-01-000-801.

(c) Do a wiring check between these pins of connector D10962 of the control cabin temperature sensor and the connector D456 of the cabin temperature controller:

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OMR 701-799 (Continued)

D10962		D456
pin 1	. . . . .	pin 23
pin 3	. . . . .	pin 24

(d) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-connect connector D10962 to the passenger cabin temperature sensor.

3) Re-install the cabin temperature controller. To install the controller, do this task: Cabin Temperature Controller (CTC) Installation, AMM TASK 21-61-01-400-801.

4) Do this task: Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.

5) If the maintenance message does not show, then you corrected the fault.

(e) If you do not find a problem with the wiring, then continue.

1) Re-connect connector D10962 to the passenger cabin temperature sensor.

(3) Install a new cabin temperature controller. To install the controller, do this task: Cabin Temperature Controller (CTC) Installation, AMM TASK 21-61-01-400-801.

(a) Do this task: Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.

(b) If the maintenance message does not show, then you corrected the fault.

——— END OF TASK ———

835. Flight Compartment Temperature Selector Failed - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) TEMP SELECTOR - LEFT PACK - NO GO

B. Possible Causes

(1) Control cabin temperature selector, M392

(2) Cabin temperature controller, M345

(3) Wiring Problem

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	2	C00268	AIR CONDITIONING TEMP CONTROL AUTO L
A	3	C00267	AIR CONDITIONING TEMP CONTROL MANUAL
B	2	C00258	AIR CONDITIONING TEMP CONTROL AUTO R

D. Related Data

(1) SSM 21-61-21

(2) WDM 21-61-21

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799 (Continued)

E. Initial Evaluation

(1) Do this task: Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.

(2) If the maintenance message does not show, then there was an intermittent fault.

(3) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Replace the control cabin temperature selector in the P5-17 panel.

These are the tasks:

Cabin Temperature Selector Removal, AMM TASK 21-61-07-000-804-001,

Cabin Temperature Selector Installation, AMM TASK 21-61-07-400-804-001.

(a) Do this task: Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.

1) If the maintenance message does not show, then you corrected the fault.

2) If the maintenance message shows, then continue.

(2) Do this check of the wiring between the control cabin temperature selector and the cabin temperature controller, M345 on the E4-1 shelf:

(a) Disconnect connector D618 from the control cabin temperature selector.

(b) Remove the cabin temperature controller. To remove the controller, do this task: Cabin Temperature Controller (CTC) Removal, AMM TASK 21-61-01-000-801.

(c) Do a wiring check between these pins of connector D618 of the control cabin temperature selector and the connector D454 of the cabin temperature controller:

D618		D454
pin 1	. . . . .	pin 19
pin 2	. . . . .	pin 20
pin 3	. . . . .	pin 17
pin 4	. . . . .	pin 10
pin 5	. . . . .	pin 25

(d) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-connect connector D618 to the control cabin temperature selector.

3) Re-install the cabin temperature controller. To install the controller, do this task: Cabin Temperature Controller (CTC) Installation, AMM TASK 21-61-01-400-801.

4) Do this task: Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.

5) If the maintenance message does not show, then you corrected the fault.

(e) If you do not find a problem with the wiring, then continue.

1) Re-connect connector D618 to the control cabin temperature sensor.

(3) Install a new cabin temperature controller. To install the controller, do this task: Cabin Temperature Controller (CTC) Installation, AMM TASK 21-61-01-400-801.

(a) Do this task: Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.

(b) If the maintenance message does not show, then you corrected the fault.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799 (Continued)

836. Passenger Cabin Temperature Selector Failed - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) TEMP SELECTOR - RIGHT PACK - NO GO

B. Possible Causes

(1) passenger cabin temperature selector, M393

(2) Cabin temperature controller, M345

(3) Wiring Problem

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	2	C00268	AIR CONDITIONING TEMP CONTROL AUTO L
A	3	C00267	AIR CONDITIONING TEMP CONTROL MANUAL
B	2	C00258	AIR CONDITIONING TEMP CONTROL AUTO R

D. Related Data

(1) SSM 21-61-22

(2) WDM 21-61-22

E. Initial Evaluation

(1) Do this task: Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.

(2) If the maintenance message does not show, then there was an intermittent fault.

(3) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Replace the passenger cabin temperature selector in the P5-17 panel.

These are the tasks:

Cabin Temperature Selector Removal, AMM TASK 21-61-07-000-804-001,

Cabin Temperature Selector Installation, AMM TASK 21-61-07-400-804-001.

(a) Do this task: Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.

1) If the maintenance message does not show, then you corrected the fault.

2) If the maintenance message shows, then continue.

(2) Do this check of the wiring between the passenger cabin temperature selector and the cabin temperature controller, M345 on the E4-1 shelf:

(a) Disconnect connector D620 from the passenger cabin temperature selector.

(b) Remove the cabin temperature controller. To remove the controller, do this task: Cabin Temperature Controller (CTC) Removal, AMM TASK 21-61-01-000-801.

(c) Do a wiring check between these pins of connector D620 of the control cabin temperature selector and the connector D454 of the cabin temperature controller:

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OMR 701-799 (Continued)

D620		D454
pin 1	. . . . .	pin 19
pin 2	. . . . .	pin 20
pin 3	. . . . .	pin 17
pin 4	. . . . .	pin 10
pin 5	. . . . .	pin 25

(d) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-connect connector D620 to the passenger cabin temperature selector.

3) Re-install the cabin temperature controller. To install the controller, do this task: Cabin Temperature Controller (CTC) Installation, AMM TASK 21-61-01-400-801.

4) Do this task: Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.

5) If the maintenance message does not show, then you corrected the fault.

(e) If you do not find a problem with the wiring, then continue.

1) Re-connect connector D620 to the passenger cabin temperature sensor.

(3) Install a new cabin temperature controller. To install the controller, do this task: Cabin Temperature Controller (CTC) Installation, AMM TASK 21-61-01-400-801.

(a) Do this task: Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.

(b) If the maintenance message does not show, then you corrected the fault.

——— END OF TASK ———

837. DUCT OVERHEAT Light On, Cannot Reset, CONT CABIN - Fault Isolation

A. Description

(1) The DUCT OVERHEAT light for the control cabin on the P5-17 cabin temperature control module is on and will not reset.

B. Possible Causes

(1) Control cabin duct overheat switch, S11

(2) Air conditioning accessory unit, M324

(3) Wiring problem

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	7	C00124	AIR CONDITIONING MIX VALVE POS IND

D. Related Data

(1) (SSM 21-61-21)

(2) (WDM 21-61-21)

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OMR 701-799 (Continued)

E. Initial Evaluation

(1) Push the TRIP RESET switch on the air conditioning module on the P5 panel.

(2) If the DUCT OVERHEAT light goes off, then there was an intermittent fault.

(3) If the DUCT OVERHEAT light stays on, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Do this check of the control cabin duct overheat switch, S11:

(a) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

(b) Disconnect connector D11360 from the passenger cabin duct overheat switch, S11.

(c) Push the TRIP RESET switch.

(d) If the DUCT OVERHEAT light goes out, then do these steps:

1) Replace the control cabin duct overheat switch, S11.

These are the tasks:

Duct Overheat Switch Removal, AMM TASK 21-61-05-000-801,

Duct Overheat Switch Installation, AMM TASK 21-61-05-400-801.

2) Re-connect connector D11360 to the passenger cabin duct overheat switch, S11.

3) If the DUCT OVERHEAT light is still off, then you corrected the fault.

4) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

(e) If the DUCT OVERHEAT light comes on, then continue.

(2) Replace the air conditioning accessory unit, M324. To replace it,

These are the tasks:

Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801,

Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

(a) Push the TRIP RESET switch.

(b) If the DUCT OVERHEAT light goes out, then you corrected the fault. Do this step to complete the task:

Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

(c) If the DUCT OVERHEAT light is still on, then continue.

(3) Do this check of the wiring between the ACAU and the control cabin duct temperature sensor:

(a) Do this task: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801.

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OMR 701-799 (Continued)

(b) Do a check for a short from pin 56 at connector D458B on the E4-1 shelf to structure ground, (WDM 21-61-21).

(c) If there is a short to ground, then do these steps:

1) Repair the wiring between the air conditioning accessory unit, M324, and control cabin duct overheat switch, S11, (WDM 21-61-21).

2) Re-install the ACAU, M324. To install it, do this task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

3) Push the TRIP RESET switch.

4) If the DUCT OVERHEAT light is off, then you corrected the fault. Do this step to complete the task:

Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

(d) If there is not a short to ground, then continue.

(4) Do this check of the wiring between the ACAU and the pilot's overhead panel, P5:

(a) Do a check for a short from pin 40 at connector D458B on the E4-1 shelf to structure ground, (WDM 21-61-22).

(b) If there is a short to ground, then do these steps:

1) Repair the wiring between the air conditioning accessory unit, M324, and the pilot's overhead panel, P5 (WDM 21-61-21).

2) Re-install the ACAU, M324. To install it, do this task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

3) Push the TRIP RESET switch.

4) If the DUCT OVERHEAT light is off, then you corrected the fault. Do this step to complete the task:

Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

838. DUCT OVERHEAT Light On, Cannot Reset, PASS CABIN - Fault Isolation

A. Description

(1) The DUCT OVERHEAT light for the passenger cabin on the P5-10 air conditioning module is on and will not reset.

B. Possible Causes

(1) Passenger cabin duct overheat switch, S14

(2) Air conditioning accessory unit, M324

(3) Wiring problem

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799 (Continued)

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	7	C00124	AIR CONDITIONING MIX VALVE POS IND

D. Related Data

(1) (SSM 21-61-22)

(2) (WDM 21-61-22)

E. Initial Evaluation

(1) Push the TRIP RESET switch on the air conditioning module on the P5 panel.

(2) If the DUCT OVERHEAT light goes off, then there was an intermittent fault.

(3) If the DUCT OVERHEAT light stays on, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Do these steps to do a check of the passenger cabin duct overheat switch, S14:

(a) Remove the ceiling panel that is five windows forward of the overwing escape hatches.

To remove the panel, do this task: Ceiling Panel - Removal, AMM TASK 25-21-45-000-801.

(b) Disconnect connector D11362 from the passenger cabin duct overheat switch, S14.

(c) Push the TRIP RESET switch.

(d) If the DUCT OVERHEAT light goes out, then do these steps:

1) Replace the passenger cabin duct overheat switch, S14.

These are the tasks:

Duct Overheat Switch Removal, AMM TASK 21-61-05-000-801,

Duct Overheat Switch Installation, AMM TASK 21-61-05-400-801.

2) Re-connect connector D11362 to the passenger cabin duct overheat switch, S14.

3) If the DUCT OVERHEAT light is still off, then you corrected the fault.

4) Install the ceiling liner.

To install the liner, do this task: Ceiling Panel - Installation, AMM TASK 25-21-45-400-801.

(e) If the DUCT OVERHEAT light comes on, then continue.

(2) Replace the air conditioning accessory unit, M324. To replace it,

These are the tasks:

Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801,

Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

(a) Push the TRIP RESET switch.

(b) If the DUCT OVERHEAT light goes out, then you corrected the fault.

(c) If the DUCT OVERHEAT light is still on, then continue.

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OMR 701-799 (Continued)

(3) Do this check of the wiring between the ACAU and the passenger cabin duct temperature sensor:

(a) Do this task: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801.

(b) Do a check for a short from pin 56 at connector D458A on the E4-1 shelf to structure ground, (WDM 21-61-22).

(c) If there is a short to ground, then do these steps:

1) Repair the wiring between the air conditioning accessory unit, M324, and passenger cabin duct overheat switch, S14, (WDM 21-61-22).

2) Re-install the ACAU, M324. To install it, do this task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

3) Push the TRIP RESET switch.

4) If the DUCT OVERHEAT light is off, then you corrected the fault. Do these steps to complete the task:

Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

(d) If there is not a short to ground, then continue.

(4) Do this check of the wiring between the ACAU and the pilot's overhead panel, P5:

(a) Do a check for a short from pin 40 at connector D458A on the E4-1 shelf to structure ground, (WDM 21-61-22).

(b) If there is a short to ground, then do these steps:

1) Repair the wiring between the air conditioning accessory unit, M324, and the pilot's overhead panel, P5 (WDM 21-61-22).

2) Re-install the ACAU, M324. To install it, do this task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

3) Push the TRIP RESET switch.

4) If the DUCT OVERHEAT light is off, then you corrected the fault. Do these steps to complete the task:

a) Install the ceiling liner.

To install the liner, do this task: Ceiling Panel - Installation, AMM TASK 25-21-45-400-801.

b) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

839. Control Cabin DUCT OVERHEAT Light On and Stays OFF After Reset - Fault Isolation

A. Description

(1) The control cabin DUCT OVERHEAT light on the P5-10 air conditioning module comes on and stays off after reset.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799 (Continued)

B. Possible Causes

(1) Heat Exchangers

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	4	C00257	AIR CONDITIONING OVERHEAT

D. Related Data

(1) SSM 21-61-21

(2) WDM 21-61-21

E. Initial Evaluation

(1) Operate the Left Air Conditioning Pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) If the DUCT OVERHEAT light comes ON, then do the Fault Isolation Procedure below.

(a) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(3) If the DUCT OVERHEAT light does not come ON, then there was an intermittent fault.

(a) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

F. Fault Isolation Procedure

(1) Clean the Left Pack Heat Exchangers. This is the task: Heat Exchanger - Blowing Out Debris/Blockages with Nitrogen Gas/Compressed Air, AMM TASK 21-51-03-160-811.

NOTE: Make sure that the heat exchangers are not blocked before you clean them.

(a) Operate the Left Air Conditioning Pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(b) If the DUCT OVERHEAT light does not come ON, then you corrected the problem.

1) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

——— END OF TASK ———

840. Passenger Cabin DUCT OVERHEAT Light On and Stays OFF After Reset - Fault Isolation

A. Description

(1) The passenger cabin DUCT OVERHEAT light on the P5-10 air conditioning module comes on and stays off after reset.

B. Possible Causes

(1) Heat Exchangers

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799 (Continued)

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	4	C00257	AIR CONDITIONING OVERHEAT

D. Related Data

(1) SSM 21-61-22

(2) WDM 21-61-22

E. Initial Evaluation

(1) Operate the Right Air Conditioning Pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) If the DUCT OVERHEAT light comes ON, then do the Fault Isolation Procedure below.

(a) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(3) If the DUCT OVERHEAT light does not come ON, then there was an intermittent fault.

(a) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

F. Fault Isolation Procedure

(1) Clean the Right Pack Heat Exchangers. This is the task: Heat Exchanger - Blowing Out Debris/Blockages with Nitrogen Gas/Compressed Air, AMM TASK 21-51-03-160-811.

NOTE: Make sure that the heat exchangers are not blocked before you clean them.

(a) Operate the Right Air Conditioning Pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(b) If the DUCT OVERHEAT light does not come ON, then you corrected the problem.

1) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

——— END OF TASK ———

841. Air Mix Valve Indicator Does Not Operate, Pointer Does Not Move, Temperature Is OK - Fault Isolation

A. Description

(1) There are two air mix valves. The air mix valve for the flight compartment (control cabin) is found in the left ECS bay. The other air mix valve for the passenger cabin is found in the right ECS bay. One of the two air mix valve position indicators on the temperature control panel (P5-17) does not move when you change the setting of the applicable temperature selector on the temperature control panel. But the temperature in the flight compartment and the passenger cabin is still controlled. This indicates that the applicable air mix valve operates correctly and the problem is with the air mix valve position indication, not with the air mix valve.

(2) SDS 21–61–00

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OMR 701-799 (Continued)

B. Possible Causes

(1) Air mix valve position indicator, N1 (flight compartment) or N2 (passenger cabin)

(2) Air mix valve position transmitter, T406 (flight compartment) or T407 (passenger cabin)

(3) Wiring

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	3	C00267	AIR CONDITIONING TEMP CONTROL MANUAL
B	2	C00258	AIR CONDITIONING TEMP CONTROL AUTO R
D	7	C00124	AIR CONDITIONING MIX VALVE POS IND

D. Related Data

(1) (SSM 21-61-21) (flight compartment)

(2) (SSM 21-61-22) (passenger cabin)

(3) (WDM 21-61-21) (flight compartment)

(4) (WDM 21-61-22) (passenger cabin)

E. Initial Evaluation

(1) Do this task: Supply Conditioned Air to the Airplane, AMM TASK 21-00-00-800-801.

(2) Put the L PACK and R PACK switches on the P5-10 panel in the AUTO position.

(3) For a problem with the flight compartment (control cabin) mix valve position indicator, N1, do these steps:

(a) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	2	C00268	AIR CONDITIONING TEMP CONTROL AUTO L
A	3	C00267	AIR CONDITIONING TEMP CONTROL MANUAL
D	7	C00124	AIR CONDITIONING MIX VALVE POS IND

(b) Turn the control cabin temperature selector on the P5-17 panel to the AUTO COOL position.

(c) Make sure that the control cabin AIR MIX VALVE indicator on the P5-17 panel shows COLD.

(d) Open this access panel:

Number	Name/Location
192CL	ECS Access Door

(e) Make sure that the position indicator on the left air mix valve is in the COLD position.

NOTE: Use an inspection mirror to see the mix valve position indicator.

1) If the position indicator on the left air mix valve is not in the COLD position, then do the Fault Isolation Procedure below.

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OMR 701-799 (Continued)

(f) Turn the control cabin temperature selector on the P5-17 panel to the AUTO WARM position.

(g) Make sure that the control cabin AIR MIX VALVE indicator on the P5-17 panel shows HOT.

(h) Make sure that the position indicator on the left air mix valve is in the HOT position.

NOTE: Use an inspection mirror to see the mix valve position indicator.

1) If the position indicator on the left air mix valve is not in the HOT position, then do the Fault Isolation Procedure below.

(i) If these items are true, then there was an intermittent fault:

1) The AIR MIX VALVE indicator on the P5-17 panel agrees with the position indicator on the air mix valve with the applicable temperature selector set to AUTO COOL.

2) The AIR MIX VALVE indicator on the P5-17 panel agrees with the position indicator on the air mix valve with the applicable temperature selector set to AUTO WARM.

(4) For a problem with the passenger cabin mix valve position indicator, N2, do these steps:

(a) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	3	C00267	AIR CONDITIONING TEMP CONTROL MANUAL
B	2	C00258	AIR CONDITIONING TEMP CONTROL AUTO R
D	7	C00124	AIR CONDITIONING MIX VALVE POS IND

(b) Turn the passenger cabin temperature selector on the P5-17 panel to the AUTO COOL position.

(c) Make sure that the passenger cabin AIR MIX VALVE indicator on the P5-17 panel shows COLD.

(d) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

(e) Make sure that the position indicator on the right air mix valve is in the COLD position.

NOTE: Use an inspection mirror to see the mix valve position indicator.

1) If the position indicator on the right air mix valve is not in the COLD position, then do the Fault Isolation Procedure below.

(f) Turn the passenger cabin temperature selector on the P5-17 panel to the AUTO WARM position.

(g) Make sure that the passenger cabin AIR MIX VALVE indicator on the P5-17 panel shows HOT.

(h) Make sure that the position indicator on the right air mix valve is in the HOT position.

NOTE: Use an inspection mirror to see the mix valve position indicator.

1) If the position indicator on the right air mix valve is not in the HOT position, then do the Fault Isolation Procedure below.

(i) If these items are true, then there was an intermittent fault:

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OMR 701-799 (Continued)

1) The AIR MIX VALVE indicator on the P5-17 panel agrees with the position indicator on the air mix valve with the applicable temperature selector set to AUTO COOL.

2) The AIR MIX VALVE indicator on the P5-17 panel agrees with the position indicator on the air mix valve with the applicable temperature selector set to AUTO WARM.

F. Fault Isolation Procedure

(1) Do these steps to do a check of the air mix valve position indicator, N1 or N2:

(a) These are the tasks:

Air Mix Valve Position Indicator Removal, AMM TASK 21-61-33-000-801,

Air Mix Valve Position Indicator Installation, AMM TASK 21-61-33-400-801.

(b) Do the Repair Confirmation at the end of this task.

(c) If the Repair Confirmation is not satisfactory, then continue.

(2) Do these steps to do a check of the air mix valve position transmitter:

(a) Do this task: Air Mix Valve Position Transmitter Adjustment, AMM TASK 21-61-34-820-801.

(b) Do this task: Cabin Temperature Module Printed Circuit Assembly Adjustment, AMM TASK 21-61-34-820-802.

(c) If the adjustment does not repair the fault, then replace the air mix valve position transmitter.

(d) Do the Repair Confirmation at the end of this task.

(e) If the Repair Confirmation is not satisfactory, then continue.

(3) Do this check of the wiring:

(a) For a problem with the control cabin mix valve indication, do these steps:

1) Do a check for an open circuit from these pins on connector D172 on the mix valve position transmitter to these pins on connector D656 on the P5-17 cabin temperature panel:

D172		D656
pin 1	. . . . .	pin 3
pin 2	. . . . .	pin 4
pin 3	. . . . .	pin 2

2) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Do the Repair Confirmation at the end of this task.

(b) For a problem with the passenger cabin mix valve indication, do these steps:

1) Do a check for an open circuit from these pins on connector D170 on the mix valve position transmitter to these pins on connector D656 on the P5-17 cabin temperature panel:

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OMR 701-799 (Continued)

D170		D656
pin 1	. . . . .	pin 3
pin 2	. . . . .	pin 4
pin 3	. . . . .	pin 16

2) If you find a problem with the wiring, then do these steps:
   a) Repair the wiring.
   b) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do this task: Supply Conditioned Air to the Airplane, AMM TASK 21-00-00-800-801.

(2) Put the L PACK and R PACK switches on the P5-10 panel in the AUTO position.

(3) For a problem with the flight compartment (control cabin) mix valve position indicator, do these steps:
   (a) Turn the control cabin temperature selector on the P5-17 panel to the AUTO COOL position.
   (b) Make sure that the control cabin AIR MIX VALVE indicator on the P5-17 panel shows COLD.
   (c) Make sure that the position indicator on the left air mix valve is in the COLD position.

   NOTE: Use an inspection mirror to see the mix valve position indicator.

   (d) Turn the control cabin temperature selector on the P5-17 panel to the AUTO WARM position.
   (e) Make sure that the control cabin AIR MIX VALVE indicator on the P5-17 panel shows HOT.
   (f) Make sure that the position indicator on the left air mix valve is in the HOT position.

   NOTE: Use an inspection mirror to see the mix valve position indicator.

   (g) If these items are true, then you corrected the fault:
      1) The AIR MIX VALVE indicator on the P5-17 panel agrees with the position indicator on the air mix valve with the applicable temperature selector set to AUTO COOL.
      2) The AIR MIX VALVE indicator on the P5-17 panel agrees with the position indicator on the air mix valve with the applicable temperature selector set to AUTO WARM.

(4) For a problem with the passenger cabin mix valve position indicator, N2, do these steps:
   (a) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	3	C00267	AIR CONDITIONING TEMP CONTROL MANUAL
B	2	C00258	AIR CONDITIONING TEMP CONTROL AUTO R
D	7	C00124	AIR CONDITIONING MIX VALVE POS IND

   (b) Turn the passenger cabin temperature selector on the P5-17 panel to the AUTO COOL position.
   (c) Make sure that the passenger cabin AIR MIX VALVE indicator on the P5-17 panel shows COLD.

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OMR 701-799 (Continued)

(d) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

(e) Make sure that the position indicator on the right air mix valve is in the COLD position.

NOTE: Use an inspection mirror to see the mix valve position indicator.

(f) Turn the passenger cabin temperature selector on the P5-17 panel to the AUTO WARM position.

(g) Make sure that the passenger cabin AIR MIX VALVE indicator on the P5-17 panel shows HOT.

(h) Make sure that the position indicator on the right air mix valve is in the HOT position.

NOTE: Use an inspection mirror to see the mix valve position indicator.

(i) If these items are true, then there was an intermittent fault:

1) The AIR MIX VALVE indicator on the P5-17 panel agrees with the position indicator on the air mix valve with the applicable temperature selector set to AUTO COOL.

2) The AIR MIX VALVE indicator on the P5-17 panel agrees with the position indicator on the air mix valve with the applicable temperature selector set to AUTO WARM.

(5) If the control cabin mix valve position indicator was replaced, then do these steps:

(a) Put the control cabin temperature selector in the AUTO NORMAL position.

(b) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

(6) If the passenger cabin mix valve position indicator was replaced, then do these steps:

(a) Put the passenger cabin temperature selector in the AUTO NORMAL position.

(b) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

(7) Do this task: Remove Conditioned Air from the Airplane, AMM TASK 21-00-00-800-802.

——— END OF TASK ———

842. Supply Duct Temperature Indication Problem - Fault Isolation

A. Description

(1) There is a problem with the SUPPLY DUCT air temperature indication shown on the P5-17 Cabin Temperature Panel.

NOTE: The temperature bulb is for indication only. A defective temperature bulb or wiring will not affect the temperature control system.

NOTE: There is no on-wing check of the temperature bulb.

B. Possible Causes

(1) Wiring

(2) Passenger cabin supply duct temperature bulb, T402

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OMR 701-799 (Continued)

(3) Air temperature indicator, N11

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	8	C00076	AIR CONDITIONING TEMP IND

D. Related Data

(1) SSM 21-61-51

(2) WDM 21-61-51

E. Initial Evaluation

(1) Do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) On the P5-17 cabin temperature panel, put the CONT CABIN and PASS CABIN temperature selectors to the AUTO NORMAL position.

(3) Let the pack operate for two minutes to let the output temperature become stable.

(4) Put the air temperature selector on the P5-17 cabin temperature panel to the SUPPLY DUCT position.

(5) Move the air temperature selector on the P5-17 Cabin Temperature Panel to the PASS CABIN position.

(6) If the temperature indication problem exists for the two positions, do the steps below to replace the indicator.

(7) If the temperature indication problem no longer exists for the PASS CABIN position, but it still exists for the SUPPLY DUCT position, do the task below.

NOTE: When the temperature indicating system operates normally, the SUPPLY DUCT temperature will be ±41° F (5° C) of the PASS CABIN temperature.

F. Fault Isolation Procedure

(1) Do this check of the wiring for P5-17 cabin temperature panel and the temperature bulb for the passenger cabin supply duct:

(a) Lower the P5 overhead panel to get access to connectors D90 and D656.

(b) Disconnect connector D90 from the panel.

(c) Do a check for 28 VDC at pin 2 on connector D90.

(d) If there is not 28 VDC at pin 2, then do these steps:

1) Repair the wiring between this circuit breaker and connector D90:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	8	C00076	AIR CONDITIONING TEMP IND

2) Re-connect connector D90 to the panel.

3) Close the P5 overhead panel.

4) Do the Repair Confirmation at the end of this task.

(e) If there is 28 VDC at pin 2, then do these steps:

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OMR 701-799 (Continued)

1) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

2) Disconnect connector D94 from the temperature bulb.
3) Disconnect connectors D656 from the panel.
4) Do a wiring check between these pins:

D94		**D90**
pin A	. . . . .	pin 4

D94		**D656**
pin B	. . . . .	pin 9

5) Repair the wiring.
6) Re-connect connector D94 to the temperature bulb.
7) Re-connect connectors D90 and D656 to the panel.
8) Close the P5 overhead panel.
9) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

10) Do the Repair Confirmation at the end of this task.

(2) Replace the temperature bulb for the passenger cabin supply duct. These are the tasks:
- Duct Temperature Bulb Removal, AMM TASK 21-61-30-000-804-001
- Duct Temperature Bulb Installation, AMM TASK 21-61-30-400-804-001

(a) Do the Repair Confirmation at the end of this task.

(3) Replace the cabin temperature indicator. These are the tasks:
- Cabin Temperature Indicator Removal, AMM TASK 21-61-32-000-801
- Cabin Temperature Indicator Installation, AMM TASK 21-61-32-400-801

(a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.
(2) On the P5-17 cabin temperature panel, put the CONT CABIN and PASS CABIN temperature selectors to the AUTO NORMAL position.
(3) Operate the pack for two minutes to let output temperature become stable.
(4) Put the air temperature selector on the P5-17 cabin temperature panel to the SUPPLY DUCT position.
(5) Make sure that the temperature indicator problem no longer exists for the SUPPLY DUCT position.

(a) If Supply Duct Temperature Indication problem no longer exists for the SUPPLY DUCT position, then you corrected the problem.

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OMR 701-799 (Continued)

(b) If Supply Duct Temperature Indication problem still exists for the SUPPLY DUCT, then continue the Fault Isolation Procedure at the subsequent step

END OF TASK

843. Passenger Cabin Temperature Indication Problem - Fault Isolation

A. Description

(1) There is a problem with the PASS CABIN air temperature indication shown on the P5-17 Cabin Temperature Panel.

NOTE: The temperature bulb is for indication only. A defective temperature bulb or wiring will not affect the temperature control system.

NOTE: There is no on-wing check of the temperature bulb.

B. Possible Causes

(1) Passenger cabin temperature sensor fan

(2) Passenger cabin temperature bulb, T401

(3) Air temperature indicator, N11

(4) Wiring

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	8	C00076	AIR CONDITIONING TEMP IND

D. Related Data

(1) SSM 21-61-51

(2) WDM 21-61-51

E. Initial Evaluation

(1) Do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) On the P5-17 Cabin Temperature Panel, put the CONT CABIN and PASS CABIN temperature selectors to the AUTO NORMAL position.

(3) Operate the pack for two minutes to let output temperature become stable.

(4) Use a thermometer, STD-1122 to determine the ambient air temperature in the control cabin and the passenger cabin.

(5) Put the air temperature selector on the P5-17 Cabin Temperature Panel to the SUPPLY DUCT position.

(6) Move the air temperature selector on the P5-17 Cabin Temperature Panel to the PASS CABIN position.

(7) If the temperature indication problem exists for the two positions, do the steps below to replace the indicator.

(8) If the temperature indication problem no longer exists for the SUPPLY DUCT position, but it still exists for the PASS CABIN position, do the task below.

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OMR 701-799 (Continued)

F. Fault Isolation Procedure

(1) Do this test of the fan for the temperature sensor assembly:

(a) Hold a 3 inch by 4 inch sheet of lightweight paper over the air inlet grille of the temperature sensor assembly.

(b) Make sure that the fan holds the paper in its position.

(c) If the vacuum from the fan will not hold the paper in its position, then do these steps:

1) Inspect the temperature sensor assembly filter for blockage.

a) If blockage is found, then clean the filter and do the Repair Confirmation.

b) If no blockage is found, then continue.

2) Inspect the duct from the filter to the temperature sensor for blockage:

a) If blockage is found, then clean the duct.

b) If no blockage is found then replace the fan. These are the tasks:

- Cabin Temperature Sensor Fan - Removal, AMM TASK 21-61-11-000-801
- Cabin Temperature Sensor Fan - Installation, AMM TASK 21-61-11-400-801

c) Do the Repair Confirmation at the end of this task.

(d) If the vacuum from the fan will hold the paper, remove the paper then continue.

(2) Do this check of the wiring for P5-17 cabin temperature panel and the temperature bulb for the passenger cabin:

(a) Lower the P5 overhead panel to get access to connectors D90 and D656.

(b) Disconnect connector D90 from the panel.

(c) Do a check for 28 VDC at pin 2 on connector D90.

(d) If there is not 28 VDC at pin 2, then do these steps:

1) Repair the wiring between the following circuit breaker and connector D90:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	8	C00076	AIR CONDITIONING TEMP IND

2) Re-connect connector D90 to the panel.

3) Close the P5 overhead panel.

4) Do the Repair Confirmation at the end of this task.

(e) If 28 VDC is found at pin 2, then do these steps:

1) Re-connect connector D90 to the panel.

2) Remove the bullnose cover to get access to the cabin temperature bulb. To remove the bullnose cover, do the applicable steps of this task: (Cabin Temperature Bulb Removal, AMM TASK 21-61-31-000-801).

3) Disconnect connector D10964 from the temperature bulb.

4) Disconnect connector D656 from the P5-17 cabin temperature panel.

5) Do a check for ground to structure of pin A for connector D10964.

6) Do a wiring check between these pins:

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OMR 701-799 (Continued)

D10964		D656
pin B	. . . . .	pin 10

7) Repair the wiring.
8) Re-connect connector D10964 to the temperature bulb.
9) Re-connect connector D656 to the panel.
10) Close the P5 overhead panel.
11) Install the bullnose cover that was removed to get access to the cabin temperature bulb. To install the bullnose cover, do the applicable steps of this task: (Cabin Temperature Bulb Installation, AMM TASK 21-61-31-400-801).
12) Do the Repair Confirmation at the end of this task.

(3) Replace the temperature bulb for the passenger cabin. These are the tasks:
- Cabin Temperature Bulb Removal, AMM TASK 21-61-31-000-801
- Cabin Temperature Bulb Installation, AMM TASK 21-61-31-400-801

(a) Do the Repair Confirmation at the end of this task.

(4) Replace the cabin temperature indicator. These are the tasks:
- Cabin Temperature Indicator Removal, AMM TASK 21-61-32-000-801
- Cabin Temperature Indicator Installation, AMM TASK 21-61-32-400-801

(a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.
(2) On the P5-17 cabin temperature panel, put the CONT CABIN and PASS CABIN temperature selectors to the AUTO NORMAL position.
(3) Operate the pack for two minutes to let output temperature become stable.
(4) Put the air temperature selector on the P5-17 cabin temperature panel to the PASS CABIN position.
(5) Make sure the temperature indication problem no longer exists for the PASS CABIN position.
(a) If temperature indication problem no longer exists, then you corrected the problem.
(b) If temperature indication problem still exists, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

844. Control Cabin Too Hot, No Temperature Control in AUTO or MANUAL - Fault Isolation

A. Description

(1) The temperature in the control cabin is too hot. Control cabin temperature can not be made colder with the temperature selector in the AUTO or the MANUAL mode. The air mix valve does not move to the COLD position.

B. Possible Causes

(1) Air mix valve, V23
(2) Control cabin temperature selector, M392

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OMR 701-799 (Continued)

(3) Wiring

(4) Air conditioning accessory unit (ACAU), M324

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	2	C00268	AIR CONDITIONING TEMP CONTROL AUTO L
A	3	C00267	AIR CONDITIONING TEMP CONTROL MANUAL
D	7	C00124	AIR CONDITIONING MIX VALVE POS IND

D. Related Data

(1) SSM 21-61-21

(2) WDM 21-61-21

E. Initial Evaluation

(1) Do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) On the P5-17 cabin temperature panel, put the CONT CABIN temperature selector to the AUTO NORMAL position.

(3) Let the pack operate for a minute to let output temperature become stable.

(4) Put the CONT CABIN temperature selector to the AUTO COOL position.

(5) Make sure the position indicator for the CONT CABIN AIR MIX VALVE moves to the COLD position.

(6) Put the CONT CABIN temperature selector to the MANUAL COOL position.

(7) Make sure the position indicator for the CONT CABIN AIR MIX VALVE stays in the COLD position.

(8) If the position indicator for the CONT CABIN AIR MIX VALVE did move to the COLD position, then there was an intermittent fault. Do this task, do this task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(9) If the position indicator for the CONT CABIN AIR MIX VALVE did not move to the COLD position, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Open this access panel:

Number	Name/Location
192CL	ECS Access Door

(2) Do this test of the air mix valve V23:

(a) Remove connector D524 from the air mix valve.

(b) Do a check for 115 VAC at pins 1 and 2 of D524.

(c) If there is 115 VAC at pins 1 and 2 of D524, then do these steps:

1) Replace the air mix valve.

These are the tasks:

Air Mix Valve Removal, AMM TASK 21-61-02-000-801,

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Air Mix Valve Installation, AMM TASK 21-61-02-400-801.

2) Do the Repair Confirmation at the end of this task.

(d) If there is not 115 VAC at pins 1 and 2 of D524, then continue. Re-connect connector D524 to the air mix valve.

(3) Do this test of the control cabin temperature selector:

(a) Do this task: Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.

(b) If the BITE test shows a failure of the control cabin temperature selector, then do these steps:

1) Replace the control cabin temperature selector, M392.

These are the tasks:

Cabin Temperature Selector Removal, AMM TASK 21-61-07-000-804-001,

Cabin Temperature Selector Installation, AMM TASK 21-61-07-400-804-001.

2) Do the Repair Confirmation at the end of this task.

(c) If the control cabin temperature selector passes the BITE test, then continue.

(4) Do this check of the wiring between air conditioning accessory unit and the air mix valve:

(a) Disconnect connector D524 from the air mix valve.

(b) Disconnect connectors D458B from the air conditioning accessory unit.

(c) Make sure there is continuity from pin 3 of D524 to the structure ground.

(d) Do a wiring check between these pins:

D458B	D524
pin 22	pin 2
pin 39	pin 1

(e) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-connect connector D458B to the air conditioning accessory unit.

3) Re-connect connectors D524 to the air mix valve.

4) Do the Repair Confirmation at the end of this task.

(f) If you do not find a problem with the wiring, then continue.

(5) Replace the air conditioning accessory unit.

These are the tasks:

Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801,

Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

(a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) On the P5-17 cabin temperature panel, put the CONT CABIN temperature selector to the AUTO NORMAL position.

(3) Operate the pack for a minute to let output temperature become stable.

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(4) Move the CONT CABIN temperature selector to the COOl NORMAL position.

(5) Make sure the AIR MIX VALVE position indicator for the CONT CABIN moves to the COLD position and the control cabin temperature gets colder.

(6) Move the CONT CABIN temperature selector to the COOL MANUAL position.

(7) Make sure the AIR MIX VALVE position indicator for the CONT CABIN stays at the COLD position.

(8) If the cabin air temperature can be changed in the manual and the automatic modes, then you corrected the fault.

(9) Do this task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(10) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

(11) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

END OF TASK

845. Control Cabin Too Cold, No Temperature Control in AUTO or MANUAL - Fault Isolation

A. Description

(1) The temperature in the control cabin is too cold. Control cabin temperature can not be made warmer with the temperature selector in the AUTO or MANUAL mode. The air mix valve does not move to the HOT position.

B. Possible Causes

(1) Air mix valve, V23

(2) Control cabin temperature selector, M392

(3) Wiring

(4) Air conditioning accessory unit (ACAU), M324

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	2	C00268	AIR CONDITIONING TEMP CONTROL AUTO L
A	3	C00267	AIR CONDITIONING TEMP CONTROL MANUAL
D	7	C00124	AIR CONDITIONING MIX VALVE POS IND

D. Related Data

(1) SSM 21-61-21

(2) WDM 21-61-21

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E. Initial Evaluation

(1) Do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) On the P5-17 cabin temperature panel, put the CONT CABIN temperature selector to the AUTO NORMAL position.

(3) Let the pack operate for a minute to let output temperature become stable.

(4) Put the CONT CABIN temperature selector to the AUTO WARM position.

(5) Make sure the position indicator for the CONT CABIN AIR MIX VALVE moves to the WARM position.

(6) Put the CONT CABIN temperature selector to the MANUAL WARM position.

(7) Make sure the position indicator for the CONT CABIN AIR MIX VALVE stays in the WARM position.

(8) If the position indicator for the CONT CABIN AIR MIX VALVE did move to the WARM position, then there was an intermittent fault. Do this task, do this task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(9) If the position indicator for the CONT CABIN AIR MIX VALVE did not move to the WARM position, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Open this access panel:

Number	Name/Location
192CL	ECS Access Door

(2) Do this test of the air mix valve V23:

(a) Remove connector D524 from the air mix valve.

(b) Do a check for 115 VAC at pins 1 and 2 of D524.

(c) If there is 115 VAC at pins 1 and 2 of D524, then do these steps:

1) Replace the air mix valve.

These are the tasks:

Air Mix Valve Removal, AMM TASK 21-61-02-000-801,

Air Mix Valve Installation, AMM TASK 21-61-02-400-801.

2) Do the Repair Confirmation at the end of this task.

(d) If there is not 115 VAC at pins 1 and 2 of D524, then continue. Re-connect connector D524 to the air mix valve.

(3) Do this test of the control cabin temperature selector:

(a) Do this task: Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.

(b) If the BITE test shows a failure of the control cabin temperature selector, then do these steps:

1) Replace the control cabin temperature selector, M392.

These are the tasks:

Cabin Temperature Selector Removal, AMM TASK 21-61-07-000-804-001,

Cabin Temperature Selector Installation, AMM TASK 21-61-07-400-804-001.

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2) Do the Repair Confirmation at the end of this task.

(c) If the control cabin temperature selector passes the BITE test, then continue.

(4) Do this check of the wiring between air conditioning accessory unit and the air mix valve:

(a) Disconnect connector D524 from the air mix valve.

(b) Disconnect connectors D458B from the air conditioning accessory unit.

(c) Make sure there is continuity from pin 3 of D524 to the structure ground point.

(d) Do a wiring check between these pins:

D458B	D524
pin 22	pin 2
pin 39	pin 1

(e) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-connect connector D458B to the air conditioning accessory unit.

3) Re-connect connectors D524 to the air mix valve.

4) Do the Repair Confirmation at the end of this task.

(f) If you do not find a problem with the wiring, then continue.

(5) Replace the air conditioning accessory unit.

These are the tasks:

Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801,

Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

(a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) On the P5-17 cabin temperature panel, put the CONT CABIN temperature selector to the AUTO NORMAL position.

(3) Let the pack operate for a minute to let output temperature become stable.

(4) Move the CONT CABIN temperature selector to the WARM NORMAL position.

(5) Make sure the AIR MIX VALVE position indicator for the CONT CABIN moves to the HOT position and the control cabin temperature gets warmer.

(6) Move the CONT CABIN temperature selector to the WARM MANUAL position.

(7) Make sure the AIR MIX VALVE position indicator for the CONT CABIN stays at the HOT position.

(8) If the cabin air temperature can be changed in the manual and the automatic modes, then you corrected the fault.

(9) Do this task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

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(10) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

(11) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

846. Passenger Cabin Too Hot, No Temperature Control in AUTO or MANUAL - Fault Isolation

A. Description

(1) The temperature in the passenger cabin is too hot. Passenger cabin temperature can not be made colder with the temperature selector in the AUTO or MANUAL mode. The air mix valve does not move to the COLD position.

B. Possible Causes

(1) Air mix valve, V22

(2) Control cabin temperature selector, M393

(3) Wiring

(4) Air conditioning accessory unit (ACAU), M324

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	2	C00268	AIR CONDITIONING TEMP CONTROL AUTO L
A	3	C00267	AIR CONDITIONING TEMP CONTROL MANUAL
D	7	C00124	AIR CONDITIONING MIX VALVE POS IND

D. Related Data

(1) SSM 21-61-22

(2) WDM 21-61-22

E. Initial Evaluation

(1) Do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) On the P5-17 cabin temperature panel, put the PASS CABIN temperature selector to the AUTO NORMAL position.

(3) Operate the pack for a minute to let output temperature become stable.

(4) Put the PASS CABIN temperature selector to the AUTO COOL position.

(5) Make sure the position indicator for the PASS CABIN AIR MIX VALVE moves to the COLD position.

(6) Put the PASS CABIN temperature selector to the MANUAL COOL position.

(7) Make sure the position indicator for the PASS CABIN AIR MIX VALVE stays in the COLD position.

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OMR 701-799 (Continued)

(8) If the position indicator for the PASS CABIN AIR MIX VALVE did move to the COLD position, then there was an intermittent fault. Do this task, do this task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(9) If the position indicator for the PASS CABIN AIR MIX VALVE did not move to the COLD position, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Open these panels in the specified sequence:

Number	Name/Location
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

(2) Do this test of the air mix valve, V22:

(a) Remove connector D522 from the air mix valve.

(b) Do a check for 115 VAC at pins 1 and 2 of D522.

(c) If there is 115 VAC at pins 1 and 2 of D522, then do these steps:

1) Replace the air mix valve.

These are the tasks:

Air Mix Valve Removal, AMM TASK 21-61-02-000-801,

Air Mix Valve Installation, AMM TASK 21-61-02-400-801.

2) Do the Repair Confirmation at the end of this task.

(d) If there is not 115 VAC at pins 1 and 2 of D522, then continue. Re-connect connector D522 to the air mix valve.

(3) Do this test of the passenger cabin temperature selector:

(a) Do this task: Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.

(b) If the BITE test shows a failure of the passenger cabin temperature selector, then do these steps:

1) Replace the control cabin temperature selector, M393.

These are the tasks:

Cabin Temperature Selector Removal, AMM TASK 21-61-07-000-804-001,

Cabin Temperature Selector Installation, AMM TASK 21-61-07-400-804-001.

2) Do the Repair Confirmation at the end of this task.

(c) If the passenger cabin temperature selector passes the BITE test, then continue.

(4) Do this check of the wiring between air conditioning accessory unit and the air mix valve:

(a) Disconnect connector D522 from the air mix valve.

(b) Disconnect connectors D458A from the air conditioning accessory unit.

(c) Make sure there is continuity from pin 3 of D522 to the structure ground point.

(d) Do a wiring check between these pins:

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D458A	D522
pin 13	pin 2
pin 39	pin 1

(e) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.
2) Re-connect connector D458A to the air conditioning accessory unit.
3) Re-connect connectors D522 to the air mix valve.
4) Do the Repair Confirmation at the end of this task.

(f) If you do not find a problem with the wiring, then continue.

(5) Replace the air conditioning accessory unit.

These are the tasks:

Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801,

Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

(a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) On the P5-17 cabin temperature panel, put the PASS CABIN temperature selector to the AUTO NORMAL position.

(3) Operate the pack for a minute to let output temperature become stable.

(4) Move the PASS CABIN temperature selector to the COOl NORMAL position.

(5) Make sure the AIR MIX VALVE position indicator for the PASS CABIN moves to the COLD position and the passenger cabin temperature gets colder.

(6) Move the PASS CABIN temperature selector to the COOL MANUAL position.

(7) Make sure the AIR MIX VALVE position indicator for the PASS CABIN stays at the COLD position.

(8) If the passenger cabin air temperature can be changed in the manual and the automatic modes, then you corrected the fault.

(9) Do this task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(10) Close these panels in the specified sequence:

Number	Name/Location
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

(11) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

END OF TASK

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OMR 701-799 (Continued)

847. Passenger Cabin Too Cold, No Temperature Control in AUTO or MANUAL - Fault Isolation

A. Description

(1) The temperature in the passenger cabin is too cold. Passenger cabin temperature can not be made warmer with the temperature selector in the AUTO or MANUAL mode. The air mix valve does not move to the HOT position.

B. Possible Causes

(1) Air mix valve, V22

(2) Control cabin temperature selector, M393

(3) Wiring

(4) Air conditioning accessory unit (ACAU), M324

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	2	C00268	AIR CONDITIONING TEMP CONTROL AUTO L
A	3	C00267	AIR CONDITIONING TEMP CONTROL MANUAL
D	7	C00124	AIR CONDITIONING MIX VALVE POS IND

D. Related Data

(1) SSM 21-61-22

(2) WDM 21-61-22

E. Initial Evaluation

(1) Do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) On the P5-17 cabin temperature panel, put the PASS CABIN temperature selector to the AUTO NORMAL position.

(3) Operate the pack for a minute to let output temperature become stable.

(4) Put the PASS CABIN temperature selector to the AUTO WARM position.

(5) Make sure the position indicator for the PASS CABIN AIR MIX VALVE moves to the WARM position.

(6) Put the PASS CABIN temperature selector to the MANUAL WARM position.

(7) Make sure the position indicator for the PASS CABIN AIR MIX VALVE stays in the WARM position.

(8) If the position indicator for the PASS CABIN AIR MIX VALVE did move to the WARM position, then there was an intermittent fault. Do this task, do this task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(9) If the position indicator for the PASS CABIN AIR MIX VALVE did not move to the WARM position, then do the Fault Isolation Procedure below.

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OMR 701-799 (Continued)

F. Fault Isolation Procedure

(1) Open these access panels:

Number	Name/Location
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

(2) Do this test of the air mix valve V22:

(a) Remove connector D522 from the air mix valve.

(b) Do a check for 115 VAC at pins 1 and 2 of D522.

(c) If there is 115 VAC at pins 1 and 2 of D522, then do these steps:

1) Replace the air mix valve.

These are the tasks:

Air Mix Valve Removal, AMM TASK 21-61-02-000-801,

Air Mix Valve Installation, AMM TASK 21-61-02-400-801.

2) Do the Repair Confirmation at the end of this task.

(d) If there is not 115 VAC at pins 1 and 2 of D522, then continue. Re-connect connector D522 to the air mix valve.

(3) Do this test of the passenger cabin temperature selector:

(a) Do this task: Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.

(b) If the BITE test shows a failure of the control cabin temperature selector, then do these steps:

1) Replace the control cabin temperature selector, M393.

These are the tasks:

Cabin Temperature Selector Removal, AMM TASK 21-61-07-000-804-001,

Cabin Temperature Selector Installation, AMM TASK 21-61-07-400-804-001.

2) Do the Repair Confirmation at the end of this task.

(c) If the passenger cabin temperature selector passes the BITE test, then continue.

(4) Do this check of the wiring between air conditioning accessory unit and the air mix valve:

(a) Disconnect connector D522 from the air mix valve.

(b) Disconnect connectors D458A from the air conditioning accessory unit.

(c) Make sure there is continuity from pin 3 of D522 to the structure ground point.

(d) Do a wiring check between these pins:

D458A	D522
pin 13	pin 2
pin 39	pin 1

(e) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-connect connector D458A to the air conditioning accessory unit.

3) Re-connect connectors D522 to the air mix valve.

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4) Do the Repair Confirmation at the end of this task.

(f) If you do not find a problem with the wiring, then continue.

(5) Replace the air conditioning accessory unit.

These are the tasks:

Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801,

Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

(a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) On the P5-17 cabin temperature panel, put the PASS CABIN temperature selector to the AUTO NORMAL position.

(3) Operate the pack for a minute to let output temperature become stable.

(4) Move the CONT CABIN temperature selector to the WARM NORMAL position.

(5) Make sure the AIR MIX VALVE position indicator for the PASS CABIN moves to the HOT position and the passenger cabin temperature gets warmer.

(6) Move the PASS CABIN temperature selector to the WARM MANUAL position.

(7) Make sure the AIR MIX VALVE position indicator for the PASS CABIN stays at the HOT position.

(8) Do this task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(9) If the cabin air temperature can be changed in the manual and the automatic modes, then you corrected the fault.

(10) Close these panels in the specified sequence:

Number	Name/Location
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

(11) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

848. CONT CABIN DUCT OVERHEAT light Is On and Comes On Again After Reset - Fault Isolation

A. Description

(1) The CONT CABIN DUCT OVERHEAT light on the P5-10 air conditioning module is on and comes on again after reset.

B. Possible Causes

(1) Air Mix Valve, V23

(2) Control Cabin Duct Overheat Switch, S11

(3) Cabin temperature control system failure

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(4) Heat Exchangers

(5) Blockage or unwanted material in the ram air ducting

(6) Ram air control system failure

(7) 35°F control system failure

(8) Bleed air too hot

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	1	C00344	AIR CONDITIONING TEMP CONT 35 DEG F L
A	2	C00268	AIR CONDITIONING TEMP CONTROL AUTO L
A	3	C00267	AIR CONDITIONING TEMP CONTROL MANUAL
A	4	C00399	AIR CONDITIONING RAM AIR MOD LEFT
A	8	C00265	AIR CONDITIONING RAM AIR MOD CONT L
C	4	C00257	AIR CONDITIONING OVERHEAT
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L

D. Related Data

(1) SSM 21-51-14

(2) WDM 21-51-14

(3) SSM 21-51-15

(4) WDM 21-51-15

(5) SSM 21-51-16

(6) WDM 21-51-16

(7) SSM 21-61-21

(8) WDM 21-61-21

E. Initial Evaluation

(1) Operate the left air conditioning pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) If the CONT CABIN DUCT OVERHEAT light comes ON, then do the Fault Isolation Procedure below.

(a) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(3) If the CONT CABIN DUCT OVERHEAT light does not come ON, then there was an intermittent fault.

(a) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

F. Fault Isolation Procedure

(1) Do the BITE test of the cabin temperature control system. This is the task: Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.

(a) If the BITE test shows a problem, then do these steps:

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OMR 701-799 (Continued)

1) Do the corrective action for the maintenance message shown.
2) Do the Repair Confirmation at the end of this task.

(b) If the BITE test does not show a problem, then continue.

(2) Do these steps to do a check of the temperature control (air mix) valve:

(a) Look at the CONT CABIN AIR MIX VALVE position indicator on the temperature control module in the control cabin.

(b) If the position indicator is not in the full COLD position, then do these steps:

1) Replace the Left Pack Air Mix Valve, V23. These are the tasks:
- Air Mix Valve Removal, AMM TASK 21-61-02-000-801
- Air Mix Valve Installation, AMM TASK 21-61-02-400-801

2) Do the Repair Confirmation at the end of this task.

(c) If the position indicator is in the full COLD position, then continue.

(3) Do the BITE test of the left pack 35°F control system. This is the task: Low Limit (35 °F) Controller BITE Procedure, 21-52 TASK 801.

(a) If the BITE test shows a problem, then do these steps:

1) Do the corrective action for the maintenance message shown.
2) Do the Repair Confirmation at the end of this task.

(b) If the BITE test does not show a problem, then continue.

(4) Replace the Control Cabin Duct Overheat Switch, S11. These are the tasks:
- Duct Overheat Switch Removal, AMM TASK 21-61-05-000-801
- Duct Overheat Switch Installation, AMM TASK 21-61-05-400-801

(a) Do the Repair Confirmation at the end of this task.

(b) If the CONT CABIN DUCT OVERHEAT light comes ON, then continue.

(5) Examine the left pack ram air ducts for blockage or unwanted materials.

(a) If the ducts have blockage or unwanted materials, then do these steps:

1) Remove the blockage or unwanted materials.
2) Do the Repair Confirmation at the end of this task.

(b) If the ducts do not have blockage or unwanted materials, then continue.

(6) Clean the left pack heat exchangers. This is the task: Heat Exchanger - Blowing Out Debris/Blockages with Nitrogen Gas/Compressed Air, AMM TASK 21-51-03-160-811.

(a) Do the Repair Confirmation at the end of this task.

(b) If the CONT CABIN DUCT OVERHEAT light comes ON, then continue.

(7) Check to see if there was a report for bleed air that is too hot (a bleed trip off).

(a) If there was a bleed trip off, then do these steps:

1) Do the fault isolation procedure for a bleed trip off. This is the task: BLEED TRIP OFF Light ON - Fault Isolation, 36-10 TASK 809.
2) Do the Repair Confirmation at the end of this task.

(b) If there was not a bleed trip off, then continue.

(8) Do these steps to do a check of the ram air modulation system:

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(a) Look at the left pack ram air doors.

(b) If the ram air doors are not in the full open position, do this task: RAM DOOR FULL OPEN Light is Not On, Airplane is On the Ground - Fault Isolation, 21-51 TASK 804.

1) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Operate the left air conditioning pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) If the CONT CABIN DUCT OVERHEAT light does not come ON, then you corrected the problem.

(3) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

——— END OF TASK ———

849. PASS CABIN DUCT OVERHEAT light Is On and Comes On Again After Reset - Fault Isolation

A. Description

(1) The PASS CABIN DUCT OVERHEAT light on the P5-10 air conditioning module is on and comes on again after reset.

B. Possible Causes

(1) Air Mix Valve, V22

(2) Passenger Cabin Duct Overheat Switch, S14

(3) Cabin temperature control system failure

(4) Heat Exchangers

(5) Blockage or unwanted material in the ram air ducting

(6) Ram air control system failure

(7) 35°F control system failure

(8) Bleed air too hot

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	3	C00267	AIR CONDITIONING TEMP CONTROL MANUAL
B	1	C00345	AIR CONDITIONING TEMP CONT 35 DEG F R
B	2	C00258	AIR CONDITIONING TEMP CONTROL AUTO R
B	4	C00400	AIR CONDITIONING RAM AIR MOD RIGHT
B	8	C00266	AIR CONDITIONING RAM AIR MOD CONT R
C	4	C00257	AIR CONDITIONING OVERHEAT
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R

D. Related Data

(1) SSM 21-51-24

(2) WDM 21-51-24

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OMR 701-799 (Continued)

(3) SSM 21-51-25

(4) WDM 21-51-25

(5) SSM 21-51-26

(6) WDM 21-51-26

(7) SSM 21-61-22

(8) WDM 21-61-22

E. Initial Evaluation

(1) Operate the right air conditioning pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) If the PASS CABIN DUCT OVERHEAT light comes ON, then do the Fault Isolation Procedure below.

(a) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(3) If the PASS CABIN DUCT OVERHEAT light does not come ON, then there was an intermittent fault.

(a) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

F. Fault Isolation Procedure

(1) Do the BITE test of the cabin temperature control system. This is the task: Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.

(a) If the BITE test shows a problem, then do these steps:

1) Do the corrective action for the maintenance message shown.

2) Do the Repair Confirmation at the end of this task.

(b) If the BITE test does not show a problem, then continue.

(2) Do these steps to do a check of the temperature control (air mix) valve:

(a) Look at the PASS CABIN AIR MIX VALVE position indicator on the temperature control module in the control cabin.

(b) If the position indicator is not in the full COLD position, then do these steps:

1) Replace the Right Pack Air Mix Valve, V22. These are the tasks:

- Air Mix Valve Removal, AMM TASK 21-61-02-000-801
- Air Mix Valve Installation, AMM TASK 21-61-02-400-801

2) Do the Repair Confirmation at the end of this task.

(c) If the position indicator is in the full COLD position, then continue.

(3) Do the BITE test of the left pack 35°F control system. This is the task: Low Limit (35 °F) Controller BITE Procedure, 21-52 TASK 801.

(a) If the BITE test shows a problem, then do these steps:

1) Do the corrective action for the maintenance message shown.

2) Do the Repair Confirmation at the end of this task.

(b) If the BITE test does not show a problem, then continue.

(4) Replace the Passenger Cabin Duct Overheat Switch, S14. These are the tasks:

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OMR 701-799 (Continued)

- Duct Overheat Switch Removal, AMM TASK 21-61-05-000-801
- Duct Overheat Switch Installation, AMM TASK 21-61-05-400-801

(a) Do the Repair Confirmation at the end of this task.

(b) If the PASS CABIN DUCT OVERHEAT light comes ON, then continue.

(5) Examine the right pack ram air ducts for blockage or unwanted materials.

(a) If the ducts have blockage or unwanted materials, then do these steps:

1) Remove the blockage or unwanted materials.

2) Do the Repair Confirmation at the end of this task..

(b) If the ducts do not have blockage or unwanted materials, then continue.

(6) Clean the Right Pack Heat Exchangers. This is the task: Heat Exchanger - Blowing Out Debris/Blockages with Nitrogen Gas/Compressed Air, AMM TASK 21-51-03-160-811.

(a) Operate the Right Air Conditioning Pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(b) If the PASS CABIN DUCT OVERHEAT light does not come ON, then you corrected the fault.

1) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(c) If the PASS CABIN DUCT OVERHEAT light comes ON, then continue.

(7) Check to see if there was a report for bleed air that is too hot (a bleed trip off).

(a) If there was a bleed trip off, then do these steps:

1) Do the fault isolation procedure for a bleed trip off. This is the task: BLEED TRIP OFF Light ON - Fault Isolation, 36-10 TASK 809.

2) Do the Repair Confirmation at the end of this task.

(b) If there was not a bleed trip off, then continue.

(8) Do these steps to do a check of the ram air modulation system:

(a) Look at the right pack ram air doors.

(b) If the ram air doors are not in the full open position, do this task: RAM DOOR FULL OPEN Light is Not On, Airplane is On the Ground - Fault Isolation, 21-51 TASK 804.

1) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Operate the right air conditioning pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) If the PASS CABIN DUCT OVERHEAT light does not come ON, then you corrected the fault.

(3) Stop the pack. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

——— END OF TASK ———

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OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

854. Left Pack Temperature Indication Problem - Fault Isolation

A. Description

(1) There is a problem with the L PACK Air Temperature Indication shown on the P5-17 Cabin Temperature Panel.

NOTE: The temperature bulb is for indication only. A defective temperature bulb or wiring will not affect the temperature control system.

NOTE: There is no on-wing check of the temperature bulb.

B. Possible Causes

(1) Air Temperature Indicator, N11

(2) Left Pack Temperature Bulb, T488

(3) Air Temperature Selector, S1

(4) Wiring

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	8	C00076	AIR CONDITIONING TEMP IND

D. Related Data

(1) WDM 21-61-51

(2) SSM 21-61-51

E. Initial Evaluation

(1) Supply conditioned air with the Left Cooling Pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) On the P5-17 Cabin Temperature Panel, put the CONT CAB, FWD CAB and AFT CAB Temperature Selectors to the AUTO (12:00) position.

(3) Operate the pack for two minutes to let the Output Temperature become stable.

(4) Use a thermometer, STD-1122 to determine the Ambient Air Temperature in the control cabin.

(5) On the P5-17 Cabin Temperature Panel, put the Air Temperature Selector to the L PACK position.

NOTE: The temperature shown on the indicator should be approximately the same as the Ambient Air Temperature.

(6) On the P5-17 Cabin Temperature Panel, move the Air Temperature Selector to the CONT CAB SUPPLY DUCT position.

NOTE: The temperature shown on the indicator should be approximately the same as the Ambient Air Temperature.

(7) If the temperature indication problem exists for the two positions, then do the Fault Isolation Procedure - Temperature Indication Problem for More Than One Position.

(8) If the temperature indication problem no longer exists for the CONT CAB SUPPLY DUCT position, but it still exists for the L PACK position, then do the Fault Isolation Procedure - Temperature Indication Problem for One Position.

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(9) If the temperature indication problem no longer exists for the CONT CAB SUPPLY DUCT and the L PACK positions, then there was an intermittent problem.

F. Fault Isolation Procedure - Temperature Indication Problem for More Than One Position

NOTE: This procedure isolates the fault to one of these possible causes:

- Cabin Temperature Indicator, N11
- Air Temperature Selector, S1
- Wiring

(1) Do this check for 28 VDC at the Cabin Temperature Indicator:

(a) Lower the P5 Overhead Panel to get access to electrical connector D90 for the Cabin Temperature Indicator N11.

(b) Do a check for 28 VDC between pin 2 of connector D90 and structure ground.

1) If you do not find 28 VDC at pin 2 of connector D90, then do these steps:

a) Repair the wiring between pin 2 of D90 and the this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	8	C00076	AIR CONDITIONING TEMP IND

b) Close the P5 Overhead Panel.

c) Do the Repair Confirmation at the end of this task.

2) If you find 28 VDC at pin 2 of electrical connector D90, then continue.

(2) Do a Resistance check between the bonding jumper at the upper left corner of the P5 FWD Overhead Panel and the airplane structure:

(a) Make sure that the Resistance is not more than the value specified in Standard Wiring Practices Manual 20-20-10, 5.D.

1) If the Resistance is greater than allowed, repair the installation of the bonding jumper with the SWPM 20-20-10 instructions.

(b) Close the P5 FWD Overhead Panel.

(c) Do the Repair Confirmation at the end of this task.

(3) Replace the Cabin Temperature Indicator N11. These are the tasks:

- Cabin Temperature Indicator Removal, AMM TASK 21-61-32-000-801
- Cabin Temperature Indicator Installation, AMM TASK 21-61-32-400-801

(a) Do the Repair Confirmation at the end of this task.

(4) Replace the Air Temperature Selector S1.

(a) Do the Repair Confirmation at the end of this task.

G. Fault Isolation Procedure - Temperature Indication Problem for One Position

NOTE: This procedure isolates the fault to one of these possible causes:

- left pack temperature bulb, T488
- air temperature selector, S1
- wiring

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(1) Do this check of the wiring for the P5-17 Cabin Temperature Panel and the Temperature Bulb for the Left Pack:

(a) Lower the P5 Overhead Panel to get access to electrical connector D3998 for the Air Temperature Selector Switch S1.

(b) Disconnect electrical connector D3930 from the Left Pack Temperature Bulb T488 (AMM TASK 21-51-60-000-801).

(c) Do a check for an open circuit between these pins:

D3998	D3930
pin 24	pin B

(d) Make sure pin A of connector D3930 goes to ground.

(e) If there is an open circuit, then do these steps:

1) Repair the wiring.

2) Close the P5 Overhead Panel.

3) Do the Repair Confirmation at the end of this task.

(f) If there is continuity, then continue.

(2) Replace the Left Pack Temperature Bulb T488. These are the tasks:

- Pack Temperature Bulb Removal, AMM TASK 21-51-60-000-801
- Pack Temperature Bulb Installation, AMM TASK 21-51-60-400-801

(a) Do the Repair Confirmation at the end of this task.

(3) Replace the Air Temperature Selector S1.

(a) Close the P5 Overhead Panel.

(b) Do the Repair Confirmation at the end of this task.

H. Repair Confirmation

(1) Supply conditioned air with the Left Cooling Pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) On the P5-17 Cabin Temperature Panel, put the CONT CAB, FWD CAB and AFT CAB Temperature Selectors to the AUTO (12:00) position.

(3) Operate the pack for two minutes to let the Output Temperature become stable.

(4) On the P5-17 Cabin Temperature Panel, put the Air Temperature Selector to the L PACK position.

(5) Make sure that temperature indication problem no longer exists for the L PACK position.

(a) If Temperature Indication problem no longer exists for the L PACK, then you corrected the problem.

(b) If Temperature Indication problem still exists for the L PACK, then continue the Fault Isolation Procedure at the subsequent step.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

855. Right Pack Temperature Indication Problem - Fault Isolation

A. Description

(1) There is a problem with The R PACK air temperature indication shown on the P5-17 cabin temperature panel.

NOTE: The temperature bulb is for indication only. A defective temperature bulb or wiring will not affect the temperature control system.

NOTE: There is no on-wing check of the temperature bulb.

B. Possible Causes

(1) Air temperature indicator, N11

(2) Right pack temperature bulb, T489

(3) Air temperature selector, S1

(4) Wiring

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	8	C00076	AIR CONDITIONING TEMP IND

D. Related Data

(1) SSM 21-61-51

(2) WDM 21-61-51

E. Initial Evaluation

(1) Supply conditioned air with the right cooling pack. This is the task:: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) On the P5-17 cabin temperature panel, put the CONT CAB, FWD CAB and AFT CAB temperature selectors to the AUTO (12:00) position.

(3) Operate the pack for two minutes to let output temperature become stable.

(4) Use a thermometer, STD-1122 to determine the ambient air temperature in the control cabin.

(5) Put the air temperature selector on the P5-17 cabin temperature panel to the R PACK position.

NOTE: The temperature shown on the indicator should be approximately the same as the ambient air temperature.

(6) Move the air temperature selector on the P5-17 cabin temperature panel to the AFT PASS CAB position.

NOTE: The temperature shown on the indicator should be approximately the same as the ambient air temperature.

(7) If the temperature indication problem exists for the two positions, then do the Fault Isolation Procedure - Temperature Indication Problem for More Than One Position.

(8) If the temperature indication problem no longer exists for the AFT PASS CAB position, but it still exists for the R PACK position, then do the Fault Isolation Procedure - Temperature Indication Problem for One Position.

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(9) If the temperature indication problem exists for the AFT PASS CAB and the R PACK positions, then there was an intermittent fault.

F. Fault Isolation Procedure - Temperature Indication Problem for More Than One Position

NOTE: This procedure isolates the fault to one of these possible causes:

- cabin temperature indicator, N11
- air temperature selector, S1
- wiring

(1) Do this check for 28 VDC at the cabin temperature indicator:

(a) Lower the P5 overhead panel to get access to electrical connector D90 for cabin temperature indicator N11.

(b) Do a check for 28 VDC between pin 2 of connector D90 and structure ground.

(c) If there is not 28 VDC at pin 2 of connector D90, then do these steps:

1) Repair the wiring between pin 2 of D90 and circuit breaker C00076:

This is the circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	8	C00076	AIR CONDITIONING TEMP IND

2) Close the P5 overhead panel.

3) Do the Repair Confirmation at the end of this task.

(d) If there is 28 VDC at pin 2 of electrical connector D90, then continue.

(2) Do a resistance check between the bonding jumper at the upper left corner of the P5 forward overhead panel and the airplane structure:

(a) Make sure that the resistance is not more than the value specified in Standard Wiring Practices Manual 20-20-10, 5.D.

1) If the resistance is greater than allowed, repair the installation of the bonding jumper with the SWPM 20-20-10 instructions.

(b) Close the P5 forward overhead panel.

(c) Do the Repair Confirmation at the end of this task.

1) If the Repair Confirmation is not satisfactory, then continue.

(3) Replace the cabin temperature indicator, N11. These are the tasks:

- Cabin Temperature Indicator Removal, AMM TASK 21-61-32-000-801
- Cabin Temperature Indicator Installation, AMM TASK 21-61-32-400-801

(a) Do the Repair Confirmation at the end of this task.

(4) Replace the air temperature selector, S1.

(a) Do the Repair Confirmation at the end of this task.

G. Fault Isolation Procedure - Temperature Indication Problem for One Position

NOTE: This procedure isolates the fault to one of these possible causes:

- right pack temperature bulb, T489
- air temperature selector, S1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

- wiring

(1) Do this check of the wiring for P5-17 cabin temperature panel and the temperature bulb for the right pack:

(a) Lower the P5 overhead panel to get access to electrical connector D3996 for the air temperature selector switch S1.

(b) Disconnect electrical connector D3982 from the right pack temperature bulb T489. (AMM TASK 21-51-60-000-801)

(c) Do check for an open circuit between these pins:

D3996	D3982
pin 24	pin B

(d) Make sure pin A of connector D3982 goes to ground.

(e) If there is an open circuit, then do these steps:

1) Repair the wiring.

2) Close the P5 overhead panel.

3) Do the Repair Confirmation at the end of this task.

(f) If there is continuity, then continue.

(2) Replace the right pack temperature bulb T489. These are the tasks:

- Pack Temperature Bulb Removal, AMM TASK 21-51-60-000-801
- Pack Temperature Bulb Installation, AMM TASK 21-51-60-400-801

(a) Do the Repair Confirmation at the end of this task.

(3) Replace the air temperature selector S1.

(a) Close the P5 overhead panel.

(b) Do the Repair Confirmation at the end of this task.

H. Repair Confirmation

(1) Supply conditioned air with the right cooling pack. To supply conditioned air, do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) On the P5-17 cabin temperature panel, put the CONT CAB, FWD CAB and AFT CAB temperature selectors to the AUTO (12:00) position.

(3) Operate the pack for two minutes to let output temperature become stable.

(4) Put the air temperature selector on the P5-17 cabin temperature panel to the R PACK position.

(5) Make sure the temperature indication problem no longer exists for the R PACK position.

(a) If the temperature indication problem no longer exists for the R PACK, then you corrected the fault.

(b) If the temperature indication problem still exists for the R PACK, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

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856. Forward Passenger Cabin Temperature Indication Problem - Fault Isolation

A. Description

(1) There is a problem with the FWD PASS CAB air temperature indication shown on P5-17 cabin temperature panel.

NOTE: The temperature bulb is for indication only. A defective temperature bulb or wiring will not affect the temperature control system.

NOTE: There is no on-wing check of the temperature bulb.

B. Possible Causes

(1) Forward Passenger Cabin Temperature Sensor Fan

(2) Forward Passenger Cabin Temperature Bulb

(3) Air Temperature Indicator, N11

(4) Air Temperature Selector, S1

(5) Wiring

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	8	C00076	AIR CONDITIONING TEMP IND

D. Related Data

(1) SSM 21-61-51

(2) WDM 21-61-51

E. Initial Evaluation

(1) Supply conditioned air with the right cooling pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) On the P5-17 cabin temperature panel, put the CONT CAB, FWD CAB and AFT CAB temperature selectors to the AUTO (12:00) position.

(3) Operate the pack for two minutes to let output temperature become stable.

(4) Use a thermometer, STD-1122 to determine the ambient air temperature in the forward and aft zones of the passenger cabin.

(5) Put the air temperature selector on the P5-17 cabin temperature panel to the FWD PASS CAB position.

NOTE: The temperature shown on the indicator should be approximately the same as the ambient air temperature.

(6) Move the air temperature selector on the P5-17 cabin temperature panel to the AFT PASS CAB position.

NOTE: The temperature shown on the indicator should be approximately the same as the ambient air temperature.

(7) If the temperature indication problem exists for the two positions, then do the Fault Isolation Procedure - Temperature Indication Problem for More Than One Position.

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(8) If the temperature indication problem no longer exists for the AFT PASS CAB position, but it still exists for the FWD PASS CAB position, then do the Fault Isolation Procedure - Temperature Indication Problem for One Position.

(9) If the temperature indication problem is no longer exist for the AFT PASS CAB and the FWD PASS CAB positions, then there was an intermittent fault.

F. Fault Isolation Procedure - Temperature Indication Problem for More Than One Position

NOTE: This procedure isolates the fault to one of these possible causes:

- Cabin Temperature Indicator, N11
- Air Temperature Selector, S1
- Wiring

(1) Do this check for 28 VDC at the Cabin Temperature Indicator:

(a) Lower the P5 overhead panel to get access to electrical connector D90 for Cabin Temperature Indicator, N11.

(b) Do a check for 28 VDC at between pin 2 of connector D90 and structure ground.

(c) If there is not 28 VDC at pin 2 of connector D90, then do these steps:

1) Repair the wiring between pin 2 of D90 and circuit breaker C00076.

2) Close the P5 overhead panel.

3) Do the Repair Confirmation at the end of this task.

(d) If there is 28 VDC at pin 2 of connector D90, then continue.

(2) Do a resistance check between the bonding jumper at the upper left corner of the P5 forward overhead panel and the airplane structure:

(a) Make sure that the resistance is not more than the value specified in Standard Wiring Practices Manual 20-20-10, 5.D.

1) If the resistance is greater than allowed, then repair the installation of the bonding jumper with the SWPM 20-20-10 instructions.

(b) Close the P5 forward overhead panel.

(c) Do the Repair Confirmation at the end of this task.

(3) Replace the Cabin Temperature Indicator, N11. These are the tasks:

- Cabin Temperature Indicator Removal, AMM TASK 21-61-32-000-801
- Cabin Temperature Indicator Installation, AMM TASK 21-61-32-400-801

(a) Do the Repair Confirmation at the end of this task.

(4) Replace the Air Temperature Selector, S1.

(a) Do the Repair Confirmation at the end of this task.

G. Fault Isolation Procedure - Temperature Indication Problem for One Position

NOTE: This procedure isolates the fault to one of these possible causes:

- Forward Passenger Cabin Temperature Sensor Fan
- Forward Passenger Cabin Temperature Bulb
- Air Temperature Selector, S1
- Wiring

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999 (Continued)

(1) Do this test of the fan in the temperature sensor assembly for the forward passenger cabin:

(a) Hold a 3 inch by 4 inch sheet of lightweight paper over the air inlet grille of the temperature sensor assembly.

(b) Make sure that the fan holds the paper in its position.

(c) If the vacuum from the fan will not hold the paper in its position, then do these steps:

1) Do this task: Cabin Temperature Sensor Assembly Filter Removal, AMM TASK 21-61-06-000-801.

2) Examine the temperature sensor assembly filter for blockage.

3) If blockage is found, then do these steps:

a) Install a new filter. This is the task: Cabin Temperature Sensor Assembly Filter Installation, AMM TASK 21-61-06-400-801.

b) Do the Repair Confirmation at the end of this task.

4) If no blockage is found, then continue.

5) Inspect the duct from the filter to the temperature sensor for blockage.

6) If blockage is found, then do these steps:

a) Clean the duct

b) Do this task: Cabin Temperature Sensor Assembly Filter Installation, AMM TASK 21-61-06-400-801.

c) Do the Repair Confirmation at the end of this task.

7) If no blockage is found, then do these steps:

a) Do this task: Cabin Temperature Sensor Assembly Filter Installation, AMM TASK 21-61-06-400-801.

b) Replace the Cabin Temperature Sensor Fan. These are the tasks:

- Cabin Temperature Sensor Fan - Removal, AMM TASK 21-61-11-000-801
- Cabin Temperature Sensor Fan - Installation, AMM TASK 21-61-11-400-801

c) Do the Repair Confirmation at the end of this task.

(d) If the vacuum from the fan will hold the paper, then remove the paper and continue.

(2) Do this check of the wiring for P5-17 cabin temperature panel and the temperature bulb for the forward passenger cabin zone:

(a) Lower the P5 overhead panel to get access to electrical connector D3998 for the air temperature selector switch S1.

(b) Do the steps to remove the bullnose cover in this task to get access to the temperature bulb. This is the task: Cabin Temperature Bulb Removal, AMM TASK 21-61-31-000-801 or Cabin Temperature Bulb Removal, AMM TASK 21-61-31-000-802

(c) Disconnect electrical connector D3920 from the forward passenger cabin temperature bulb.

(d) Do a check for an open circuit between these pins:

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OMR 802, 805, 809, 810, 812, 813

P5-17	M1445
D3998	D3920
pin 3	pin B
Ground	pin A

OMR 901-999

P5-17	M2847
D3998	D15018
pin 3	pin B
Ground	pin A

OMR 807

P5-17	M3283
D3998	D16848
pin 3	pin B
Ground	pin A

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(e) If there is an open circuit, then do these steps:

1) Repair the wiring.

2) Close the P5 overhead panel.

3) Do the Repair Confirmation at the end of this task.

(f) If there is continuity, then continue.

(3) Replace the Forward Passenger Cabin Temperature Bulb. These are the tasks:

- Cabin Temperature Bulb Removal, AMM TASK 21-61-31-000-801 or Cabin Temperature Bulb Removal, AMM TASK 21-61-31-000-802
- Cabin Temperature Bulb Installation, AMM TASK 21-61-31-400-801 or Cabin Temperature Bulb Installation, AMM TASK 21-61-31-400-802

(a) Do the Repair Confirmation at the end of this task.

(4) Replace the Air Temperature Selector, S1.

(a) Close the P5 overhead panel.

(b) Do the Repair Confirmation at the end of this task.

H. Repair Confirmation

(1) Supply conditioned air with the right cooling pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) On the P5-17 cabin temperature panel, put the CONT CAB, FWD CAB and AFT CABIN temperature selectors to the AUTO (12:00) position.

(3) Operate the pack for two minutes to let output temperature become stable.

(4) Put the air temperature selector on the P5-17 cabin temperature panel to the FWD PASS CAB position.

(5) Make sure the Cabin Temperature Indication problem no longer exists for the FWD PASS CAB position.

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(a) If Cabin Temperature Indication problem no longer exists for the FWD PASS CAB, then you corrected the fault.

(b) If Cabin Temperature Indication problem still exists for the FWD PASS CAB, then continue the Fault Isolation procedure at the subsequent step.

——— END OF TASK ———

857. Aft Passenger Cabin Temperature Indication Problem - Fault Isolation

A. Description

(1) There is a problem with The AFT PASS CAB air temperature indication shown on the P5-17 cabin temperature panel.

NOTE: The temperature bulb is for indication only. A defective temperature bulb or wiring will not affect the temperature control system.

NOTE: There is no on-wing check of the temperature bulb.

B. Possible Causes

(1) Aft Passenger Cabin Temperature Sensor Fan

(2) Aft Passenger Cabin Temperature Bulb

(3) Air Temperature Indicator, N11

(4) Air Temperature Selector, S1

(5) Wiring

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	8	C00076	AIR CONDITIONING TEMP IND

D. Related Data

(1) SSM 21-61-51

(2) WDM 21-61-51

E. Initial Evaluation

(1) Supply conditioned air with the right cooling pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) On the P5-17 cabin temperature panel, put the CONT CAB, FWD CAB and AFT CAB temperature selectors to the AUTO (12:00) position.

(3) Operate the pack for two minutes to let output temperature become stable.

(4) Use a thermometer, STD-1122 to determine the ambient air temperature in the forward and aft zones of the passenger cabin.

(5) Put the air temperature selector on the P5-17 cabin temperature panel to the AFT PASS CAB position.

NOTE: The temperature shown on the indicator should be approximately the same as the ambient air temperature.

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(6) Move the air temperature selector on the P5-17 cabin temperature panel to the FWD PASS CAB position.

NOTE: The temperature shown on the indicator should be approximately the same as the ambient air temperature.

(7) If the temperature indication problem exists for the two positions, then do the Fault Isolation Procedure - Temperature Indication Problem for More Than One Position.

(8) If the temperature indication problem no longer exists for the FWD PASS CAB position, but it still exists for the AFT PASS CAB position, then do the Fault Isolation Procedure - Temperature Indication Problem for One Position.

(9) If the temperature indication problem no longer exists for the AFT PASS CAB and the FWD PASS CAB positions, then there was an intermittent fault.

F. Fault Isolation Procedure - Temperature Indication problem for More Than One Position

NOTE: This procedure isolates the fault to one of these possible causes:

- Cabin Temperature Indicator, N11
- Air Temperature Selector, S1
- Wiring

(1) Do this check for 28 VDC at the cabin temperature indicator:

(a) Lower the P5 overhead panel to get access to electrical connector D90 for Cabin Temperature Indicator, N11.

(b) Do a check for 28 VDC between pin 2 of connector D90 and structure ground.

(c) If there is not 28 VDC at pin 2 of D90, then do these steps:

1) Repair the wiring between pin 2 of D90 and circuit breaker C00076:

2) Close the P5 overhead panel.

3) Do the Repair Confirmation at the end of this task.

(d) If there is 28 VDC at pin 2 of D90, then continue.

(2) Do a resistance check between the bonding jumper at the upper left corner of the P5 forward overhead panel and the airplane structure:

(a) Make sure that the resistance is not more than the value specified in Standard Wiring Practices Manual 20-20-10, 5.D.

1) If the resistance is greater than allowed, repair the installation of the bonding jumper with the SWPM 20-20-10 instructions.

2) Do the Repair Confirmation at the end of this task.

(b) Close the P5 forward overhead panel.

(3) Replace the Cabin Temperature Indicator N11. These are the tasks:

- Cabin Temperature Indicator Removal, AMM TASK 21-61-32-000-801
- Cabin Temperature Indicator Installation, AMM TASK 21-61-32-400-801

(a) Do the Repair Confirmation at the end of this task.

(4) Replace the Air Temperature Selector, S1.

(a) Do the Repair Confirmation at the end of this task.

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G. Fault Isolation Procedure - Temperature Indication Problem for One Position

NOTE: This procedure isolates the fault to one of these possible causes:

- Aft Passenger Cabin Temperature Sensor Fan
- Aft Passenger Cabin Temperature Bulb
- Air Temperature Selector, S1
- Wiring

(1) Do this test of the fan in the temperature sensor assembly for the aft passenger cabin:

(a) Hold a 3 inch by 4 inch sheet of lightweight paper over the air inlet grille of the temperature sensor assembly.

(b) Make sure that the fan holds the paper in its position.

(c) If the vacuum from the fan will not hold the paper in its position, then do these steps:

1) Do this task: Cabin Temperature Sensor Assembly Filter Removal, AMM TASK 21-61-06-000-801.

2) Examine the temperature sensor assembly filter for blockage.

3) If blockage is found, then do these steps:

a) Install a new filter. This is the task: Cabin Temperature Sensor Assembly Filter Installation, AMM TASK 21-61-06-400-801.

b) Do the Repair Confirmation at the end of this task.

4) If no blockage is found, then continue.

5) Inspect the duct from the filter to the temperature sensor for blockage.

6) If blockage is found, then do these steps:

a) Clean the duct

b) Do this task: Cabin Temperature Sensor Assembly Filter Installation, AMM TASK 21-61-06-400-801.

c) Do the Repair Confirmation at the end of this task.

7) If no blockage is found, then do these steps:

a) Do this task: Cabin Temperature Sensor Assembly Filter Installation, AMM TASK 21-61-06-400-801.

b) Replace the Cabin Temperature Sensor Fan. These are the tasks:

- Cabin Temperature Sensor Fan - Removal, AMM TASK 21-61-11-000-801
- Cabin Temperature Sensor Fan - Installation, AMM TASK 21-61-11-400-801

c) Do the Repair Confirmation at the end of this task.

(d) If the vacuum from the fan will hold the paper, then remove the paper and continue.

(2) Do this check of the wiring for P5-17 cabin temperature panel and the temperature bulb for the aft passenger cabin zone:

(a) Lower the P5 overhead panel to get access to electrical connector D3996 for the air temperature selector switch S1.

(b) Do the steps to remove the bullnose cover in this task to get access to the temperature bulb. This is the task: Cabin Temperature Bulb Removal, AMM TASK 21-61-31-000-801 or Cabin Temperature Bulb Removal, AMM TASK 21-61-31-000-802.

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OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999 (Continued)

(c) Disconnect the electrical connector from the aft passenger cabin temperature bulb.

(d) Do a check for an open circuit between these pins:

OMR 802, 805, 809, 810, 812, 813

P5-17	**M1446**
D3996	**D3922**
pin 13	pin B
Ground	pin A

OMR 901-999

P5-17	**M2848**
D3996	**D15024**
pin 13	pin B
Ground	pin A

OMR 807

P5-17	**M3282**
D3996	**D16850**
pin 13	pin B
Ground	pin A

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(e) If there is an open circuit, then do these steps:

1) Repair the wiring.

2) Close the P5 overhead panel.

3) Do the Repair Confirmation at the end of this task.

(f) If there is continuity, then continue.

(3) Replace the aft passenger cabin temperature bulb. These are the tasks:

- Cabin Temperature Bulb Removal, AMM TASK 21-61-31-000-801 or Cabin Temperature Bulb Removal, AMM TASK 21-61-31-000-802
- Cabin Temperature Bulb Installation, AMM TASK 21-61-31-400-801 or Cabin Temperature Bulb Installation, AMM TASK 21-61-31-400-802

(a) Do the Repair Confirmation at the end of this task.

(4) Replace the Air Temperature Selector, S1.

(a) Close the P5 overhead panel.

(b) Do the Repair Confirmation at the end of this task.

H. Repair Confirmation

(1) Supply conditioned air with the right cooling pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) On the P5-17 cabin temperature panel, put the CONT CAB, FWD CAB and AFT CABIN temperature selectors to the AUTO (12:00) position.

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(3) Operate the pack for two minutes to let output temperature become stable.

(4) Put the air temperature selector on the P5-17 cabin temperature panel to the AFT PASS CAB position.

(5) Make sure the temperature indication problem no longer exists for the AFT PASS CAB position.

(a) If the temperature indication problem no longer exists for the AFT PASS CAB, then you corrected the fault.

(b) If temperature indication problem still exists for the AFT PASS CAB, then continue the Fault Isolation procedure at the subsequent step.

——— END OF TASK ———

858. Control Cabin Supply Duct Temperature Indication Problem - Fault Isolation

A. Description

(1) There is a problem with the CONT CAB SUPPLY DUCT air temperature indication shown on the P5-17 cabin temperature panel.

NOTE: The temperature bulb is for indication only. A defective temperature bulb or wiring will not affect the temperature control system.

NOTE: There is no on-wing check of the temperature bulb.

B. Possible Causes

(1) Wiring

(2) Air temperature indicator, N11

(3) Control cabin supply duct temperature bulb, T485

(4) Air temperature selector, S1

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	8	C00076	AIR CONDITIONING TEMP IND

D. Related Data

(1) SSM 21-61-51

(2) WDM 21-61-51

E. Initial Evaluation

(1) Supply conditioned air with the left cooling pack. To supply conditioned air, do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) On the P5-17 cabin temperature panel, put the CONT CAB, FWD CAB and AFT CAB temperature selectors to the AUTO (12:00) position.

(3) Operate the pack for two minutes to let output temperature become stable.

(4) Use a thermometer, STD-1122 to determine the ambient air temperature in the control cabin.

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(5) Move the air temperature selector on the P5-17 cabin temperature panel to the CONT CAB SUPPLY DUCT position.

NOTE: The temperature shown on the indicator should be approximately the same as the ambient air temperature.

(6) Put the air temperature selector on the P5-17 cabin temperature panel to the L PACK position.

NOTE: The temperature shown on the indicator should be approximately the same as the ambient air temperature.

(7) If the supply duct temperature indication problem exists for the two positions, then do the Fault Isolation Procedure - Temperature Indication Problem for More Than One Position.

(8) If the supply duct temperature indication problem no longer exists for the L PACK position, but it still exists for the CONT CAB SUPPLY DUCT position, the do the Fault Isolation Procedure - Temperature Indication Problem for One Position.

(9) If the supply duct temperature indication problem no longer exists for the CONT CAB SUPPLY DUCT and the L PACK positions, then there was an intermittent fault.

F. Fault Isolation Procedure - Temperature Indication Problem for More Than One Position

NOTE: This procedure isolates the fault to one of these possible causes:

- cabin temperature indicator, N11
- air temperature selector, S1
- wiring

(1) Do this check for 28 VDC at the cabin temperature indicator:

(a) Lower the P5 overhead panel to get access to electrical connector D90 for cabin temperature indicator N11.

(b) Do a check for 28 VDC between pin 2 of D90 and structure ground.

(c) If there is not 28 VDC at pin 2 of D90, then do these steps:

1) Repair the wiring between pin 2 of D90 and circuit breaker C00076:

This is the circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	8	C00076	AIR CONDITIONING TEMP IND

2) Close the P5 overhead panel.

3) Do the Repair Confirmation at the end of this task.

(d) If there is 28 VDC at pin 2 of D90, then continue.

(2) Do a resistance check between the bonding jumper at the upper left corner of the P5 forward overhead panel and the airplane structure:

(a) Make sure that the resistance is not more than the value specified in Standard Wiring Practices Manual 20-20-10, 5.D.

1) If the resistance is greater than allowed, repair the installation of the bonding jumper with the SWPM 20-20-10 instructions.

(b) Close the P5 forward overhead panel.

(c) Do the Repair Confirmation at the end of this task.

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1) If the Repair Confirmation is not satisfactory, then continue.

(3) Replace the cabin temperature indicator N11. These are the tasks:

- Cabin Temperature Indicator Removal, AMM TASK 21-61-32-000-801
- Cabin Temperature Indicator Installation, AMM TASK 21-61-32-400-801

(a) Do the Repair Confirmation at the end of this task.

1) If the Repair Confirmation is not satisfactory, then continue.

(4) Replace the air temperature selector S1.

(a) Close the P5 overhead panel.

(b) Do the Repair Confirmation at the end of this task.

G. Fault Isolation Procedure - Temperature Indication Problem for One Position

NOTE: This procedure isolates the fault to one of these possible causes:

- control cabin supply duct temperature bulb, T485
- air temperature selector, S1
- wiring

(1) Do this check of the wiring for P5-17 cabin temperature panel and the temperature bulb for the control cabin supply duct:

(a) Lower the P5 overhead panel to get access to electrical connector D3998 for the air temperature selector switch S1.

(b) Disconnect electrical connector D3924 from the control cabin supply duct temperature bulb T485. (AMM TASK 21-61-30-000-803-002)

(c) Do check for an open circuit between these pins on connector D3998 and connector D3924:

D3998	D3924
pin 19	pin B

(d) Make sure pin A of connector D3924 goes to ground.

(e) If there is an open circuit, then do these steps:

1) Repair the wiring.

2) Raise the P5 overhead panel to the closed position.

3) Do the Repair Confirmation at the end of this task.

(f) If there is continuity, then continue.

(2) Replace the control cabin supply duct temperature bulb T485. These are the tasks:

- Duct Temperature Bulb Removal, AMM TASK 21-61-30-000-803-002
- Duct Temperature Bulb Installation, AMM TASK 21-61-30-400-803-002

(a) Do the Repair Confirmation at the end of this task.

(3) Replace the air temperature selector S1.

(a) Raise the P5 overhead panel to the closed position.

(b) Do the Repair Confirmation at the end of this task.

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OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999 (Continued)

H. Repair Confirmation

(1) Supply conditioned air with the left cooling pack. To supply conditioned air, do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) On the P5-17 cabin temperature panel, put the CONT CAB, FWD CAB and AFT CAB temperature selectors to the AUTO (12:00) position.

(3) Operate the pack for two minutes to let output temperature become stable.

(4) Put the air temperature selector on the P5-17 cabin temperature panel to the CONT CAB SUPPLY DUCT position.

(5) Make sure the supply duct temperature indication problem no longer exists for the CONT CAB SUPPLY DUCT position.

(a) If the Supply Duct temperature indication problem no longer exists for the CONT CAB SUPPLY DUCT, then you corrected the fault.

(b) If the Supply Duct Temperature Indication problem still exists for the CONT CAB SUPPLY DUCT, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

859. Forward Passenger Cabin Supply Duct Temperature Indication Problem - Fault Isolation

A. Description

(1) There is a problem with the FWD SUPPLY DUCT air temperature indication shown on the P5-17 cabin temperature panel.

NOTE: The temperature bulb is for indication only. A defective temperature bulb or wiring will not affect the temperature control system.

NOTE: There is no on-wing check of the temperature bulb.

B. Possible Causes

(1) Air temperature indicator, N11

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(2) Forward cabin supply duct temperature bulb, T534

OMR 807

(3) Supernumerary supply duct temperature bulb, T552

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(4) Air temperature selector, S1

(5) Wiring

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	8	C00076	AIR CONDITIONING TEMP IND

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Related Data

(1) SSM 21-61-51

(2) WDM 21-61-51

E. Initial Evaluation

(1) Supply conditioned air with the right cooling pack. To supply conditioned air, do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) On the P5-17 cabin temperature panel, put the CONT CAB, FWD CAB and AFT CAB temperature selectors to the AUTO (12:00) position.

(3) Operate the pack for two minutes to let output temperature become stable.

(4) Use a thermometer, STD-1122 to determine the ambient air temperature in the forward passenger cabin zone.

(5) Move the air temperature selector on the P5-17 cabin temperature panel to the FWD SUPPLY DUCT position.

NOTE: The temperature shown on the indicator should be approximately the same as the ambient air temperature.

(6) Put the air temperature selector on the P5-17 cabin temperature panel to the AFT SUPPLY DUCT position.

NOTE: The temperature shown on the indicator should be approximately the same as the ambient air temperature.

(7) If the temperature indication problem exists for two positions, then do the Fault Isolation Procedure - Temperature Indication Problem for More Than One Position.

(8) If the temperature indication problem no longer exists for the AFT SUPPLY DUCT position, but it still a problem for the FWD SUPPLY DUCT position, then do the Fault Isolation Procedure - Temperature Indication Problem for One Position.

(9) If the temperature indication no longer exits for the FWD SUPPLY DUCT and the AFT SUPPLY DUCT positions, then there was an intermittent fault.

F. Fault Isolation Procedure - Temperature Indication Problem for More Than One Position

NOTE: This procedure isolates the fault to one of these possible causes:

- cabin temperature indicator, N11
- air temperature selector, S1
- wiring

(1) Do this check for 28 VDC at the cabin temperature indicator:

(a) Lower the P5 overhead panel to get access to electrical connector D90 for cabin temperature indicator N11.

(b) Do a check for 28 VDC between pin 2 of D90 and structure ground.

(c) If there is not 28 VDC at pin 2 of D90, then do these steps:

1) Repair the wiring between pin 2 of D90 and circuit breaker C00076:

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OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999 (Continued)

This is the circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	8	C00076	AIR CONDITIONING TEMP IND

2) Close the P5 overhead panel.

3) Do the Repair Confirmation at the end of this task.

(d) If there is 28 VDC at pin 2 of D90, then continue.

(2) Do a resistance check between the bonding jumper at the upper left corner of the P5 forward overhead panel and the airplane structure:

(a) Make sure that the resistance is not more than the value specified in Standard Wiring Practices Manual 20-20-10, 5.D.

1) If the resistance is greater than allowed, repair the installation of the bonding jumper with the SWPM 20-20-10 instructions.

(b) Close the P5 forward overhead panel.

(c) Do the Repair Confirmation at the end of this task.

(3) Replace the cabin temperature indicator N11. These are the tasks:

- Cabin Temperature Indicator Removal, AMM TASK 21-61-32-000-801
- Cabin Temperature Indicator Installation, AMM TASK 21-61-32-400-801

(a) Do the Repair Confirmation at the end of this task.

(4) Replace the air temperature selector S1.

(a) Close the P5 overhead panel.

(b) Do the Repair Confirmation at the end of this task.

G. Fault Isolation Procedure - Temperature Indication Problem for One Position

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

NOTE: This procedure isolates the fault to one of these possible causes:

- forward cabin supply duct temperature bulb, T534
- air temperature selector, S1
- wiring

OMR 807

NOTE: This procedure isolates the fault to one of these possible causes:

- Supernumerary supply duct temperature bulb, T552
- air temperature selector, S1
- wiring

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(1) Do this check of the wiring for P5-17 cabin temperature panel and the temperature bulb for the forward passenger cabin supply duct:

(a) Lower the P5 overhead panel to get access to electrical connector D3998 for the air temperature selector switch S1.

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OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(b) Disconnect electrical connector D11832 from the temperature bulb T534 for the forward passenger cabin supply duct. (AMM TASK 21-61-30-000-803-002)

OMR 807

(c) Disconnect electrical connector D16846 from the temperature bulb T552 for the forward passenger cabin supply duct. (Duct Temperature Bulb Removal, AMM TASK 21-61-30-000-803-002)

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(d) Do a check for an open circuit between these pins on connector D3998 and connector D11832:

D3998	D11832
pin 18	pin B

OMR 807

(e) Do a check for an open circuit between these pins on connector D3998 and connector D16846:

D3998	D16846
pin 18	pin B

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(f) Make sure pin A of connector D11832 goes to ground.

OMR 807

(g) Make sure pin A of connector D16846 goes to ground.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(h) If there is an open circuit, then do these steps:

1) Repair the wiring.
2) Close the P5 overhead panel.
3) Do the Repair Confirmation at the end of this task.

(i) If there is continuity, then continue.

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(2) Replace the temperature bulb T534 for forward passenger cabin supply duct. These are the tasks:

- Duct Temperature Bulb Removal, AMM TASK 21-61-30-000-803-002
- Duct Temperature Bulb Installation, AMM TASK 21-61-30-400-803-002

(a) Do the Repair Confirmation at the end of this task.

OMR 807

(3) Replace the temperature bulb T552 for forward passenger cabin supply duct. These are the tasks:

- Duct Temperature Bulb Removal, AMM TASK 21-61-30-000-803-002
- Duct Temperature Bulb Installation, AMM TASK 21-61-30-400-803-002

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OMR 807 (Continued)

(a) Do the Repair Confirmation at the end of this task.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(4) Replace the air temperature selector S1.

(a) Close the P5 overhead panel.

(b) Do the Repair Confirmation at the end of this task.

H. Repair Confirmation

(1) Supply conditioned air with the right cooling pack. To supply conditioned air, do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) On the P5-17 cabin temperature panel, put the CONT CAB, FWD CAB and AFT CAB temperature selectors to the AUTO (12:00) position.

(3) Operate the pack for two minutes to let output temperature become stable.

(4) Put the air temperature selector on the P5-17 cabin temperature panel to the FWD SUPPLY DUCT position.

(5) Make sure temperature indication problem no longer exists for the FWD SUPPLY DUCT position.

(a) If the temperature indication problem no longer exists for the FWD SUPPLY DUCT, then you corrected the fault.

(b) If the temperature indication problem still exists for the FWD SUPPLY DUCT, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

860. Aft Passenger Cabin Supply Duct Temperature Indication Problem - Fault Isolation

A. Description

(1) There is a problem with the AFT SUPPLY DUCT air temperature indication shown on the P5-17 cabin temperature panel.

NOTE: The temperature bulb is for indication only. A defective temperature bulb or wiring will not affect the temperature control system.

NOTE: There is no on-wing check of the temperature bulb.

B. Possible Causes

(1) Air temperature indicator, N11

OMR 807

(2) Main cargo supply duct temperature bulb, T535

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(3) Aft cabin supply duct temperature bulb, T535

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(4) Air temperature selector, S1

(5) Wiring

C. Related Data

(1) SSM 21-61-51

(2) WDM 21-61-51

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D. Initial Evaluation

(1) Supply conditioned air with the right cooling pack. To supply conditioned air, do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) On the P5-17 cabin temperature panel, put the CONT CAB, FWD CAB and AFT CAB temperature selectors to the AUTO (12:00) position.

(3) Operate the pack for two minutes to let output temperature become stable.

(4) Use a thermometer, STD-1122 to determine the ambient air temperature in the forward passenger cabin zone.

(5) Move the air temperature selector on the P5-17 cabin temperature panel to the AFT SUPPLY DUCT position.

NOTE: The temperature shown on the indicator should be approximately the same as the ambient air temperature.

(6) Put the air temperature selector on the P5-17 cabin temperature panel to the FWD SUPPLY DUCT position.

NOTE: The temperature shown on the indicator should be approximately the same as the ambient air temperature.

(7) If the temperature indication problem still exists for the two positions, then do the Fault Isolation Procedure - Temperature Indication Problem for More Than One Position.

(8) If the temperature indication problem no longer exists for the FWD SUPPLY DUCT position, but it still a problem for the AFT SUPPLY DUCT position, then do the Fault Isolation Procedure - Temperature Indication Problem for One Position.

(9) If the temperature indication problem no longer exists for the FWD SUPPLY DUCT and the AFT SUPPLY DUCT positions, then there was an intermittent fault.

E. Fault Isolation Procedure - Temperature Indication Problem for More Than One Position

NOTE: This procedure isolates the fault to one of these possible causes:

- cabin temperature indicator, N11
- air temperature selector, S1
- wiring

(1) Do this check for 28 VDC at the cabin temperature indicator:

(a) Lower the P5 overhead panel to get access to electrical connector D90 for cabin temperature indicator N11.

(b) Do a check for 28 VDC between pin 2 of connector D90 and structure ground.

(c) If there is not 28 VDC at pin 2 of D90, then do these steps:

1) Repair the wiring between pin 2 of D90 and circuit breaker C00076:

This is the circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	8	C00076	AIR CONDITIONING TEMP IND

2) Close the P5 overhead panel.

3) Do the Repair Confirmation at the end of this task.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) If there is 28 VDC at pin 2 of D90, then continue.

(2) Do a resistance check between the bonding jumper at the upper left corner of the P5 forward overhead panel and the airplane structure:

(a) Make sure that the resistance is not more than the value specified in Standard Wiring Practices Manual 20-20-10, 5.D.

1) If the resistance is greater than allowed, repair the installation of the bonding jumper with the SWPM 20-20-10 instructions.

(b) Close the P5 forward overhead panel.

(c) Do the Repair Confirmation at the end of this task.

(3) Replace the cabin temperature indicator N11. These are the tasks:

- Cabin Temperature Indicator Removal, AMM TASK 21-61-32-000-801
- Cabin Temperature Indicator Installation, AMM TASK 21-61-32-400-801

(a) Do the Repair Confirmation at the end of this task.

(4) Replace the air temperature selector S1.

(a) Close the P5 overhead panel.

(b) Do the Repair Confirmation at the end of this task.

F. Fault Isolation Procedure - Temperature Indication Problem for One Position

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

NOTE: This procedure isolates the fault to one of these possible causes:

- aft cabin supply duct temperature bulb, T535
- air temperature selector, S1
- wiring

OMR 807

NOTE: This procedure isolates the fault to one of these possible causes:

- main cargo supply duct temperature bulb, T535
- air temperature selector, S1
- wiring

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(1) Do this check of the wiring for P5-17 cabin temperature panel and the temperature bulb for the aft passenger cabin supply duct:

(a) Lower the P5 overhead panel to get access to electrical connector D3998 for the air temperature selector switch S1.

(b) Disconnect electrical connector D11838 from the left pack temperature bulb T535. (AMM TASK 21-61-30-000-803-002)

(c) Do a check for an open circuit between these pins on connector D3998 and connector D11838:

D3998	D11838
pin 17	pin B

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OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999 (Continued)

(d) Make sure pin A of connector D11838 goes to ground.

(e) If there is an open circuit, then do these steps:

1) Repair the wiring.

2) Close the P5 overhead panel.

3) Do the Repair Confirmation at the end of this task.

(f) If there is continuity, then continue.

(2) Replace the temperature bulb T535 for the aft passenger cabin supply duct. These are the tasks:

- Duct Temperature Bulb Removal, AMM TASK 21-61-30-000-803-002
- Duct Temperature Bulb Installation, AMM TASK 21-61-30-400-803-002

(a) Do the Repair Confirmation at the end of this task.

(3) Replace the air temperature selector S1.

(a) Close the P5 overhead panel.

(b) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Supply conditioned air with the right cooling pack. To supply conditioned air, do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) On the P5-17 cabin temperature panel, put the CONT CAB, FWD CAB and AFT CAB temperature selectors to the AUTO (12:00) position.

(3) Operate the pack for two minutes to let output temperature become stable.

(4) Put the air temperature selector on the P5-17 cabin temperature panel to the AFT SUPPLY DUCT position.

(5) Make sure cabin temperature indication problem no longer exists for the AFT SUPPLY DUCT position.

(a) If the cabin temperature indication problem no longer exist for the AFT SUPPLY DUCT, then you corrected the fault.

(b) If the cabin temperature indication problem still exists for the AFT SUPPLY DUCT, then continue the Fault Isolation Procedure at the subsequent step.

OMR ALL

——— END OF TASK ———

861. Control Cabin AUTO Temperature Control Does Not Operate - Fault Isolation

A. Description

(1) The control cabin automatic temperature control does not operate.

OMR 701-799

B. Possible Causes

(1) Temperature Control System failure

(2) Control Cabin Zone Temperature Sensor Fan, M1718

(3) Control Cabin Zone Temperature Sensor Filter

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

C. Possible Causes

(1) Trim Air PRSOV, V146

(2) Control Cabin Zone Temperature Sensor Filter

(3) Control Cabin Zone Temperature Sensor Fan, M1718

(4) Control Cabin Duct Temperature Sensor (CCDTS), T478

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

(5) Zone Temperature Controller (ZTC) No. 2, M1443

OMR 820-826

(6) R Pack Temperature Controller (PTC), M2952

OMR 701-799

D. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	2	C00268	AIR CONDITIONING TEMP CONTROL AUTO L

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

E. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	3	C01163	A/C ZONE TEMP VALVE/FAN CONT FLT DECK
OMR 802, 805, 807, 809, 810, 812, 813			
B	4	C00400	AIR CONDITIONING RAM AIR MOD RIGHT
OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC
D	9	C01172	AIR CONDITIONING TRIM AIR PRESS

OMR 701-799

F. Related Data

(1) SSM 21-61-21

(2) WDM 21-61-21

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

G. Related Data

(1) WDM 21-61-11

(2) WDM 21-61-12

(3) SSM 21-61-11

(4) SSM 21-61-12

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OMR 701-799

H. Initial Evaluation

(1) Supply conditioned air to the control cabin with both packs. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) Make sure that the CONT CABIN temperature selector on the P5-17 panel is in the AUTO NORMAL position.

(3) Let the air conditioning packs operate for a few minutes to allow the temperature of the air from the packs to stabilize.

(4) Put the control CONT CABIN temperature selector on the P5-17 panel to the AUTO WARM position.

(5) If the temperature in the control cabin does not increase, then do the Fault Isolation Procedure below.

(6) If the temperature in the control cabin increases, then there was an intermittent fault.

(a) Remove conditioned air from the airplane. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

I. Initial Evaluation

(1) Supply conditioned air to the control cabin with both packs. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) Make sure that the CONT CABIN temperature selector on the P5-17 panel is in the AUTO position.

(3) Let the air conditioning packs operate for a few minutes to allow the temperature of the air from the packs to stabilize.

(4) Put the CONT CABIN temperature selector on the P5-17 panel to the W position.

(5) If the temperature in the control cabin does not increase, then do the Fault Isolation Procedure below.

(6) If the temperature in the control cabin increases, then there was an intermittent fault.

(a) Remove conditioned air from the airplane. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

OMR 701-799

J. Fault Isolation Procedure

(1) Do this check of the control cabin zone temperature sensor filter and fan:

(a) Put the L PACK switch on the P5-10 air conditioning panel to the AUTO position.

(b) Hold a 3 in. (8 cm) by 4 in. (10 cm) sheet of lightweight paper over the air inlet grille of the temperature sensor assembly for the control cabin.

(c) If the fan does not hold the paper in its position, then do these steps:

1) Put the L PACK switch on the P5-10 air conditioning panel to the OFF position.

2) Remove the control cabin zone temperature sensor filter. This is the task: Cabin Temperature Sensor Assembly Filter Removal, AMM TASK 21-61-06-000-801.

3) Examine the filter for dirt or other blockage.

4) If the filter is dirty, then do these steps:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799 (Continued)

a) Install a new filter. This is the task: Cabin Temperature Sensor Assembly Filter Installation, AMM TASK 21-61-06-400-801.

b) Supply conditioned air to the control cabin with both packs. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

c) Make sure that the CONT CABIN temperature selector on the P5-17 panel is in the AUTO NORMAL position.

d) Let the air conditioning packs operate for a few minutes to allow the temperature of the air from the packs to stabilize.

e) Put the CONT CABIN temperature selector on the P5-17 panel to the AUTO WARM position.

f) If the temperature in the control cabin increases, then you corrected the fault.

g) Remove conditioned air from the airplane. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

5) If the filter is not dirty, then do these steps:

a) Replace the zone temperature sensor fan for the control cabin zone. These are the tasks:

- Cabin Temperature Sensor Fan - Removal, AMM TASK 21-61-11-000-801
- Cabin Temperature Sensor Fan - Installation, AMM TASK 21-61-11-400-801

b) Supply conditioned air to the control cabin with both packs. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

c) Make sure that the CONT CABIN temperature selector on the P5-17 panel is in the AUTO NORMAL position.

d) Let the air conditioning packs operate for a few minutes to allow the temperature of the air from the packs to stabilize.

e) Put the CONT CABIN temperature selector on the P5-17 panel to the AUTO WARM position.

f) If the temperature in the control cabin increases, then you corrected the fault.

g) Remove conditioned air from the airplane. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(d) If the fan holds the paper in its position, then continue.

1) Remove conditioned air from the airplane. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(2) Do the BITE test of the cabin temperature controller, M345. This is the task: Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.

(a) If the BITE test shows a fault, then do these steps:

1) Do the corrective action for the maintenance message that shows.

2) Supply conditioned air to the control cabin with both packs. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

3) Make sure that the CONT CABIN temperature selector on the P5-17 panel is in the AUTO NORMAL position.

4) Let the air conditioning packs operate for a few minutes to allow the temperature of the air from the packs to stabilize.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799 (Continued)

5) Put the CONT CABIN temperature selector on the P5-17 panel to the AUTO WARM position.

6) If the temperature in the control cabin increases, then you corrected the fault.

7) Remove conditioned air from the airplane. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

K. Fault Isolation Procedure

(1) Do the BITE test of the right pack/zone temperature controller, M1443. This is the task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the BITE test shows a fault, then do these steps:

1) Do the corrective action for the maintenance message that shows.

2) Supply conditioned air to the control cabin with both packs. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

3) Make sure that the CONT CABIN temperature selector on the P5-17 panel is in the AUTO position.

4) Let the air conditioning packs operate for a few minutes to allow the temperature of the air from the packs to stabilize.

5) Put the CONT CABIN temperature selector on the P5-17 panel to the W position.

6) If the temperature in the control cabin increases, then you corrected the fault.

7) Remove conditioned air from the airplane. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(b) If the BITE test did not find a fault, then continue.

(2) Replace the trim air pressure regulating and shutoff valve (PRSOV) as follows:

(a) Do these tasks: Trim Air Pressure Regulating and Shutoff Valve Removal, AMM TASK 21-62-01-000-801 and Trim Air Pressure Regulating and Shutoff Valve Installation, AMM TASK 21-62-01-400-801.

(3) Do this test of the pressure regulating and shutoff valve (PRSOV):

(a) Close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L
D	9	C01172	AIR CONDITIONING TRIM AIR PRESS

(b) Do this task: Supply Pressure to the Pneumatic System with the APU, AMM TASK 36-00-00-860-803.

(c) Do these steps on the P5-10 air conditioning panel:

1) Set the L PACK and R PACK switches to AUTO.

2) Set the BLEED 1 and BLEED 2 switches to ON.

3) Set the APU BLEED switch to ON.

(d) Do these steps on the P5-17 temperature control panel:

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1) Set the FWD CABIN temperature selector to midway between C and AUTO.
2) Set the AFT CABIN temperature selector to midway between C and AUTO.
3) Set the CONT CABIN temperature selector to midway between C and AUTO.
4) Set the TRIM AIR switch to ON.
5) Set the AIR TEMP selector switch on the P5-17 temperature control panel to the CONT CABIN position for SUPPLY DUCT temperature indications.

(e) Operate the packs until the temperature indication for the control cabin supply duct does not change.

(f) Set the AIR TEMP selector switch on the P5-17 temperature control panel to each of the three SUPPLY DUCT positions (CONT CABIN, FWD and AFT).

(g) Make a record of the temperatures shown on the indicator for each of the three positions.

(h) Set the CONT CABIN temperature selector switch to W (full warm).

(i) Make sure that the supply duct temperature for the control cabin zone is higher than the supply duct temperatures for the forward and aft zones.

(j) If the supply duct temperature for the control cabin zone is higher than the supply duct temperatures for the forward and aft zone, then you corrected the fault.

NOTE: The supply duct temperature for the control cabin zone should be at least 10°F (-12°C) warmer than the supply duct temperatures for the forward and aft zones.

(k) If the supply duct temperature for the control cabin zone is not higher than the supply duct temperatures for the forward and aft zones, then continue.

(4) Do this check of the control cabin zone temperature sensor filter and fan:

(a) Hold a 3 in. (8 cm) by 4 in. (10 cm) sheet of lightweight paper over the air inlet grille of the temperature sensor assembly for the control cabin.

(b) If the fan does not hold the paper in its position, then do these steps:

1) Remove the control cabin zone temperature sensor filter. This is the task: Cabin Temperature Sensor Assembly Filter Removal, AMM TASK 21-61-06-000-801.
2) Examine the filter for dirt or other blockage.
3) If the filter is dirty, then do these steps:

a) Install a new filter. This is the task: Cabin Temperature Sensor Assembly Filter Installation, AMM TASK 21-61-06-400-801.

b) Supply conditioned air to the control cabin with both packs. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

c) Make sure that the CONT CABIN temperature selector on the P5-17 panel is in the AUTO position.

d) Let the air conditioning packs operate for a few minutes to allow the temperature of the air from the packs to stabilize.

e) Put the CONT CABIN temperature selector on the P5-17 panel to the W position.

f) If the temperature in the control cabin increases, then you corrected the fault.

g) Remove conditioned air from the airplane. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

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OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999 (Continued)

4) If the filter is not dirty, then do these steps:

a) Replace the zone temperature sensor fan for the control cabin zone.

These are the tasks:

- Cabin Temperature Sensor Fan - Removal, AMM TASK 21-61-11-000-801
- Cabin Temperature Sensor Fan - Installation, AMM TASK 21-61-11-400-801

b) Supply conditioned air to the control cabin with both packs. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

c) Make sure the CONT CABIN temperature selector on the P5-17 panel is in the AUTO position.

d) Let the air conditioning packs operate for a few minutes to allow the temperature of the air from the packs to stabilize.

e) Put the CONT CABIN temperature selector on the P5-17 panel to the W position.

f) If the temperature in the control cabin increases, then you corrected the fault.

g) Remove conditioned air from the airplane. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(5) Do this wiring check (WDM 21-61-11, WDM 21-61-12):

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

(a) Disconnect the Electrical Connector D3860A from the ZTC No. 2, M1443.

1) Measure the resistance between these pins on the Electrical Connector D3860A:

ZONE TEMPERATURE CONTROLLER NO2 D3860A	**ZONE TEMPERATURE CONTROLLER NO2 D3860A**
pin A5	pin B5

(b) Disconnect the Electrical Connector D3858A from the ZTC No. 1, M1442.

1) Measure the resistance between these pins on the Electrical Connector D3858A:

ZONE TEMPERATURE CONTROLLER NO1 D3858A	**ZONE TEMPERATURE CONTROLLER NO1 D3858A**
pin A5	pin B5

OMR 820-826

(c) Disconnect the Electrical Connector D15816B from the R PTC, M2952.

1) Measure the resistance between these pins on the Electrical Connector D15816B:

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OMR 820-826 (Continued)

RIGHT PACK TEMPERATURE CONTROLLER D15816B	**RIGHT PACK TEMPERATURE CONTROLLER D15816B**
pin E9	pin E10

(d) Disconnect the Electrical Connector D15814B from the L PTC, M2951.

1) Measure the resistance between these pins on the Electrical Connector D15814B:

LEFT PACK TEMPERATURE CONTROLLER D15814B	**LEFT PACK TEMPERATURE CONTROLLER D15814B**
pin E9	pin E10

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(e) Compare the two resistance values from the CCDTS - Primary, T478 and the CCDTS - Backup, T477.

1) If the resistance values are different, replace the CCDTS - Primary, T478. These are the tasks:

- Duct Temperature Limit Sensor Removal, AMM TASK 21-61-04-000-803-002
- Duct Temperature Limit Sensor Installation, AMM TASK 21-61-04-400-803-002

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

2) If the resistance values are the same, replace the ZTC No. 2, M1443. These are the tasks:

- Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801
- Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801

OMR 820-826

3) If the resistance values are the same, replace the R PTC, M2952. These are the tasks:

- Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801
- Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801

——— END OF TASK ———

OMR 701-799

862. Passenger Cabin AUTO Temperature Control Does Not Operate - Fault Isolation

A. Description

(1) The passenger cabin automatic temperature control does not operate.

B. Possible Causes

(1) Temperature control system failure

(2) Passenger cabin zone temperature sensor fan, M1719

(3) Control cabin zone temperature sensor filter

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799 (Continued)

(4) Separation of sidewall riser ducts from transition duct.

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	2	C00258	AIR CONDITIONING TEMP CONTROL AUTO R

D. Related Data

(1) (SSM 21-61-22)

(2) (WDM 21-61-22)

E. Initial Evaluation

(1) Supply conditioned air to the passenger cabin with both packs. To supply conditioned air, do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) Make sure the passenger cabin temperature selector on the P5-17 panel is in the AUTO NORMAL position.

(3) Examine the portion of the passenger cabin sidewall panels on both sides of the aircraft near the floor that are adjacent to the forward wing-to-body fairing as follows:

(a) Make sure that the sidewall panels do not appear to be expanded inboard in the area just above the floor level and that the panels appear to have the same contour as the adjacent panels.

(b) Make sure that there are no whistling noises that could indicate the presence of leaking ducts.

(c) If either of the two conditions above are present, then do the Fault Isolation Procedure below.

(4) Let the air conditioning packs operate for a few minutes to allow the temperature of the air from the packs to stabilize.

(5) Put the passenger cabin temperature selector on the P5-17 panel to the AUTO WARM position.

(6) If the temperature in the passenger cabin does not increase, then do the Fault Isolation Procedure below.

(7) If the temperature in the passenger cabin increases, then there was an intermittent fault.

(a) Remove conditioned air from the airplane. To remove the conditioned air, do this task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804

F. Fault Isolation Procedure

(1) Do this check of the passenger cabin zone temperature sensor filter and fan:

(a) Hold a 3 inch (7.6 cm) by 4 inch (10 cm) sheet of lightweight paper over the air inlet grille of the temperature sensor assembly for the passenger cabin.

(b) If the fan does not hold the paper in its position, then do these steps:

1) Remove the passenger cabin zone temperature sensor filter. To remove the filter, do this task: Cabin Temperature Sensor Assembly Filter Removal, AMM TASK 21-61-06-000-801.

2) Examine the filter for dirt or other blockage.

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OMR 701-799 (Continued)

3) If the filter is dirty, then do these steps:
   a) Install a new filter. This is the task: Cabin Temperature Sensor Assembly Filter Installation, AMM TASK 21-61-06-400-801.
   b) Supply conditioned air to the passenger cabin with both packs. To supply conditioned air, do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.
   c) Make sure the passenger cabin temperature selector on the P5-17 panel is in the AUTO NORMAL position.
   d) Let the air conditioning packs operate for a few minutes to allow the temperature of the air from the packs to stabilize.
   e) Put the passenger cabin temperature selector on the P5-17 panel to the AUTO WARM position.
   f) If the temperature in the passenger cabin increases, then you corrected the fault.
   g) Remove conditioned air from the airplane. To remove the conditioned air, do this task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804

4) If the filter is not dirty, then do these steps:
   a) Replace the zone temperature sensor fan for the passenger cabin zone.

      These are the tasks:

      Cabin Temperature Sensor Fan - Removal, AMM TASK 21-61-11-000-801,

      Cabin Temperature Sensor Fan - Installation, AMM TASK 21-61-11-400-801.
   b) Supply conditioned air to the passenger cabin with both packs. To supply conditioned air, do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.
   c) Make sure the passenger cabin temperature selector on the P5-17 panel is in the AUTO NORMAL position.
   d) Let the air conditioning packs operate for a few minutes to allow the temperature of the air from the packs to stabilize.
   e) Put the passenger cabin temperature selector on the P5-17 panel to the AUTO WARM position.
   f) If the temperature in the passenger cabin increases, then you corrected the fault.
   g) Remove conditioned air from the airplane. To remove the conditioned air, do this task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804

(c) If the fan holds the paper in its position, then continue.
   1) Remove conditioned air from the airplane. To remove the conditioned air, do this task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804

(2) Do the BITE test of the cabin temperature controller, M345. To do the BITE test, do this task: Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.

(a) If the BITE test shows a fault, then do these steps:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799 (Continued)

1) Do the corrective action for the maintenance message that shows.
2) Supply conditioned air to the passenger cabin with both packs. To supply conditioned air, do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.
3) Make sure the passenger cabin temperature selector on the P5-17 panel is in the AUTO NORMAL position.
4) Let the air conditioning packs operate for a few minutes to allow the temperature of the air from the packs to stabilize.
5) Put the passenger cabin temperature selector on the P5-17 panel to the AUTO WARM position.
6) If the temperature in the passenger cabin increases, then you corrected the fault.
7) Remove conditioned air from the airplane. To remove the conditioned air, do this task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804

(3) If the Initial Evaluation indicated damage to the portion of the passenger cabin sidewall panels near the floor in the station location near the forward wing-to-body fairing, then do the steps that follow:

(a) Remove the applicable sidewall panel(s) as follows: Do this task: Sidewall Panel - Removal, AMM TASK 25-21-46-000-801.

(b) Inspect the sidewall risers and repair, if required, as follows: Do these tasks: Sidewall Riser Ducts/Transition Duct Inspection, AMM TASK 21-23-02-212-801 and Sidewall Riser Ducts/Transition Duct Repair, AMM TASK 21-23-02-360-801.

——— END OF TASK ———

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

863. Aft Cabin AUTO Temperature Control Does Not Operate - Fault Isolation

A. Description

(1) The Aft Cabin Automatic Temperature Control does not operate.

B. Possible Causes

(1) Pack/Zone Temperature Control System failure

(2) Trim Air Pressure Regulating and Shutoff Valve (TAPRSOV), V146

OMR 901-999

(3) Aft Cabin Temperature Sensor Module, M2848

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(4) Aft Cabin Duct Temperature Limit Sensor, T536

OMR 807

(5) Main Cargo Temperature Sensor Module, M3282

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OMR 802, 805, 809, 810, 812, 813

(6) Aft Cabin Zone Temperature Sensor Fan, M1446

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(7) Aft Cabin Zone Temperature Sensor Filter

(8) Separation of Sidewall Riser Ducts from Transition Duct.

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	3	C01170	A/C ZONE TEMP VALVE/FAN CONT AFT PASS
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

D. Related Data

(1) WDM 21-51-12

(2) WDM 21-61-14

(3) SSM 21-51-12

(4) SSM 21-61-14

E. Initial Evaluation

(1) Supply conditioned air to the Passenger Compartment with the two packs. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) On the Panel, P5-17 make sure that the AFT CAB Temperature Selector is in the AUTO position.

(3) Examine the portion of the Passenger Cabin Sidewall Panels on the two sides of the aircraft near the floor that are adjacent to the Forward Wing-to-Body Fairing as follows:

(a) Make sure that the sidewall panels do not appear to be expanded inboard in the area just above the floor level and that the panels appear to have the same contour as the adjacent panels.

(b) Make sure that there are no whistling noises that could indicate the presence of leaking ducts.

(c) If either of the two conditions above is present, then do the Fault Isolation Procedure below.

(4) Let the Air Conditioning Packs operate for a few minutes to allow the temperature of the air from the packs to stabilize.

(5) On the Panel, P5-17 put the AFT CAB Temperature Selector to the W position.

(a) If the temperature in the Aft Cabin increases, then there was an intermittent fault.

1) Remove the conditioned air from the airplane. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(b) If the temperature in the Aft Cabin does not increase, then do the Fault Isolation Procedure below.

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F. Fault Isolation Procedure

(1) Do the BITE Test of the Left Pack/Zone Temperature Controller, M1442. This is the task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the BITE Test shows a fault, then do these steps:

1) Do the corrective action for the maintenance message that shows.

2) Supply conditioned air to the Passenger Compartment with the two packs. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

3) On the Panel, P5-17 make sure that the AFT CAB Temperature Selector is in the AUTO position.

4) Let the Air Conditioning Packs operate for a few minutes to allow the temperature of the air from the packs to stabilize.

5) On the Panel, P5-17 put the AFT CAB Temperature Selector to the W position.

a) If the temperature in the Aft Cabin increases, then you corrected the problem.

<1> Remove the conditioned air from the airplane. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(b) If the BITE Test does not show a fault, then continue.

(2) Replace the TAPRSOV. These are the tasks:

- Trim Air Pressure Regulating and Shutoff Valve Removal, AMM TASK 21-62-01-000-801
- Trim Air Pressure Regulating and Shutoff Valve Installation, AMM TASK 21-62-01-400-801.

(3) Do this test of the TAPRSOV, V146:

(a) Close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L
D	9	C01172	AIR CONDITIONING TRIM AIR PRESS

(b) Do this task: Supply Pressure to the Pneumatic System with the APU, AMM TASK 36-00-00-860-803.

(c) Do these steps on the Air Conditioning Panel, P5-10:

1) Set the L PACK and R PACK Switches to AUTO.

2) Set the BLEED 1 and BLEED 2 Switches to ON.

3) Set the APU BLEED Switch to ON.

(d) Do these steps on the Temperature Control Panel, P5-17:

1) Set the CONT CAB Temperature Selector to midway between C and AUTO.

2) Set the AFT CAB Temperature Selector to midway between C and AUTO.

3) Set the FWD CAB Temperature Selector to midway between C and AUTO.

4) Set the TRIM AIR Switch to ON.

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5) Set the AIR TEMP Selector Switch to the AFT CAB position for SUPPLY DUCT Temperature Indications.

(e) Operate the packs until the Temperature Indication for the Aft Cabin Supply Duct does not change.

(f) On the Temperature Control Panel, P5-17 set the AIR TEMP Selector Switch to each of the three SUPPLY DUCT positions (CONT CAB, FWD and AFT).

(g) Make a record of the temperatures shown on the indicator for each of the three positions.

(h) Set the AFT CAB Temperature Selector Switch to W (full warm).

(i) Make sure that the Supply Duct Temperature for the Aft Passenger Cabin Zone is higher than the Supply Duct Temperatures for the Forward Passenger and Control Cabin Zones.

(j) If the Supply Duct Temperature for the Aft Passenger Cabin Zone is higher than the Supply Duct Temperatures for the Forward Passenger and Control Cabin Zones, then you corrected the problem.

NOTE: The Supply Duct Temperature for the Aft Passenger Cabin Zone should be at least 10 °F warmer than the Supply Duct Temperatures for the Forward Passenger and Control Cabin Zones.

(k) If the Supply Duct Temperature for the Aft Passenger Cabin Zone is not higher than the Supply Duct Temperatures for the Forward Passenger and Control Cabin Zones, then continue.

(4) Do a resistance check of the Aft Cabin Duct Temperature Limit Sensor, T536. This is the Task: Duct Temperature Limit Sensor Test, AMM TASK 21-61-04-000-805-002.

(a) If you replaced the Aft Cabin Duct Temperature Limit Sensor, T536 and the observed fault is gone, then you have corrected the fault.

(b) If the resistance check of the of the Aft Cabin Duct Temperature Limit Sensor, T536 is satisfactory, then continue.

(5) Do this check of the Aft Cabin Zone Temperature Sensor Filter and Fan:

(a) Hold a 3 in. (7.6 cm) by 4 in. (10 cm) sheet of lightweight paper over the Air Inlet Grille of the Temperature Sensor Assembly for the Aft Cabin.

(b) If the fan does not hold the paper in its position, then do these steps:

1) Remove the Aft Cabin Zone Temperature Sensor Filter. This is the task: Cabin Temperature Sensor Assembly Filter Removal, AMM TASK 21-61-06-000-801.

2) Examine the filter for dirt or other blockage.

3) If the filter is dirty, then do these steps:

a) Install a new filter. This is the task: Cabin Temperature Sensor Assembly Filter Installation, AMM TASK 21-61-06-400-801.

b) Supply conditioned air to the Passenger Compartment with the two packs. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

c) On the Panel, P5-17 make sure that the AFT CAB Temperature Selector is in the AUTO position.

d) Let the Air Conditioning Packs operate for a few minutes to allow the temperature of the air from the packs to stabilize.

e) On the Panel, P5-17 put the AFT CAB Temperature Selector to the W position.

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OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999 (Continued)

<1> If the temperature in the Aft Cabin increases, then you corrected the problem.

<a> Remove the conditioned air from the airplane. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

4) If the filter is not dirty, then do these steps:

a) Replace the Zone Temperature Sensor Fan for the Aft Cabin Zone. These are the tasks:

- Cabin Temperature Sensor Fan - Removal, AMM TASK 21-61-11-000-801
- Cabin Temperature Sensor Fan - Installation, AMM TASK 21-61-11-400-801

b) Supply conditioned air to the Passenger Compartment with the two packs. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

c) On the Panel, P5-17 make sure that the AFT CAB Temperature Selector is in the AUTO position.

d) Let the Air Conditioning Packs operate for a few minutes to allow the temperature of the air from the packs to stabilize.

e) On the Panel, P5-17 put the AFT CAB Temperature Selector to the W position.

<1> If the temperature in the Aft Cabin increases, then you corrected the problem.

<a> Remove the conditioned air from the airplane. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(c) If the fan holds the paper in its position, then continue.

1) Remove the conditioned air from the airplane. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(6) If the Initial Evaluation indicated damage to the portion of the Passenger Cabin Sidewall Panels near the floor in the station location near the Forward Wing-to-Body Fairing, then do the steps that follow:

(a) Remove the applicable Sidewall Panel(s).

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

1) This is the task: Sidewall Panel - Removal, AMM TASK 25-21-46-000-801.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(b) Examine the Sidewall Risers and repair as necessary. These are the applicable tasks:

- Sidewall Riser Ducts/Transition Duct Inspection, AMM TASK 21-23-02-212-801
- Sidewall Riser Ducts/Transition Duct Repair, AMM TASK 21-23-02-360-801 or Sidewall Riser Ducts/Transition Duct Repair, AMM TASK 21-23-02-360-802

——— END OF TASK ———

864. Forward Cabin AUTO Temperature Control Does Not Operate - Fault Isolation

A. Description

(1) The Fwd Cabin Automatic Temperature Control does not operate.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999 (Continued)

B. Possible Causes

(1) Pack/Zone Temperature Control System failure

(2) TAPRSOV, V146

(3) Forward Cabin Duct Temperature Limit Sensor, T533

OMR 807

(4) Supernumerary Temperature Sensor Module, M3283

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(5) Fwd Cabin Zone Temperature Sensor Fan, M1445

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(6) Fwd Cabin Zone Temperature Sensor Filter

(7) Separation of Sidewall Riser Ducts from Transition Duct

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	2	C01167	A/C ZONE TEMP VALVE/FAN CONT FWD PASS
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

D. Related Data

(1) WDM 21-51-22

(2) WDM 21-61-24

(3) SSM 21-51-22

(4) SSM 21-61-24

E. Initial Evaluation

(1) Supply conditioned air to the Passenger Compartment with the two packs. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) On the P5-17 Panel, make sure that the FWD CAB Temperature Selector is in the AUTO position.

(3) Examine the portion of the Passenger Cabin Sidewall Panels on the two sides of the aircraft near the floor that are adjacent to the Forward Wing-to-Body Fairing as follows:

(a) Make sure that the sidewall panels do not appear to be expanded inboard in the area just above the floor level and that the panels appear to have the same contour as the adjacent panels.

(b) Make sure that there are no whistling noises that could indicate the presence of leaking ducts.

(c) If either of the two conditions above is present, then do the Fault Isolation Procedure below.

(4) Let the Air Conditioning Packs operate for a few minutes to allow the temperature of the air from the packs to stabilize.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(5) On the P5-17 Panel, put the FWD CAB Temperature Selector to the W position.

(a) If the temperature in the Forward Cabin increases, then there was an intermittent fault.

1) Remove the conditioned air from the airplane. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(b) If the temperature in the Fwd Cabin does not increase, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Do the BITE Test of the Right Pack/Zone Temperature Controller, M1443. This is the task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the BITE Test shows a fault, then do these steps:

1) Do the corrective action for the maintenance message that shows.

2) Supply conditioned air to the passenger compartment with both packs. To supply conditioned air, do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

3) On the P5-17 Panel, make sure that the FWD CAB Temperature Selector is in the AUTO position.

4) Let the Air Conditioning Packs operate for a few minutes to allow the temperature of the air from the packs to stabilize.

5) On the P5-17 Panel, put the FWD CAB Temperature Selector to the W position.

a) If the temperature in the Fwd Cabin increases, then you corrected the problem.

<1> Remove the conditioned air from the airplane. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(b) If the BITE test does not show a fault, then continue.

(2) Replace the TAPRSOV. These are the tasks:

- Trim Air Pressure Regulating and Shutoff Valve Removal, AMM TASK 21-62-01-000-801
- Trim Air Pressure Regulating and Shutoff Valve Installation, AMM TASK 21-62-01-400-801.

(3) Do this test of the TAPRSOV, V146:

(a) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L
D	9	C01172	AIR CONDITIONING TRIM AIR PRESS

(b) Do this task: Supply Pressure to the Pneumatic System with the APU, AMM TASK 36-00-00-860-803.

(c) Do these steps on the P5-10 Air Conditioning Panel:

1) Set the L PACK and R PACK Switches to AUTO.

2) Set the BLEED 1 and BLEED 2 Switches to ON.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

3) Set the APU BLEED Switch to ON.

(d) Do these steps on the P5-17 air conditioning panel:

1) Set the CONT CAB Temperature Selector to a position midway between C and AUTO.

2) Set the FWD CAB Temperature Selector to a position midway between C and AUTO.

3) Set the AFT CAB Temperature Selector to a position midway between C and AUTO.

4) Set the TRIM AIR Switch to ON.

5) Set the AIR TEMP Selector Switch to the FWD position for SUPPLY DUCT temperature indications.

6) Set the AIR TEMP Selector Switch to each of the three SUPPLY DUCT positions (CONT CAB, FWD CAB and AFT CAB).

7) Make a record of the temperatures shown on the indicator for each of the three positions.

8) Set the FWD CAB Temperature Selector Switch to W (full warm).

9) Make sure that the Supply Duct Temperature for the Forward Cabin Zone is higher than the Supply Duct Temperatures for the Control and Aft Cabin Zones.

10) If the Supply Duct Temperature for the Forward Passenger Cabin Zone is higher than the Supply Duct Temperatures for the Control and Aft Passenger Cabin Zones, then you corrected the problem.

NOTE: The Supply Duct Temperature for the Forward Passenger Cabin Zone should be at least 10 °F warmer than the Supply Duct Temperatures for the Control and Aft Passenger Cabin Zones.

11) If the Supply Duct Temperature for the Forward Passenger Cabin Zone is not higher than the Supply Duct Temperatures for the Control and Aft Passenger Cabin Zones, then continue.

(4) Do a resistance check of the Fwd Cabin Duct Temperature Limit Sensor, T533. This is the Task: Duct Temperature Limit Sensor Test, AMM TASK 21-61-04-000-805-002.

(a) If you replaced the Fwd Cabin Duct Temperature Limit Sensor, T533 and the observed fault is gone, then you have corrected the fault.

(b) If the resistance check of the of the Fwd Cabin Duct Temperature Limit Sensor, T533 is satisfactory, then continue.

(5) Do this check of the Forward Cabin Zone Temperature Sensor Filter and Fan:

(a) Hold a 3 in. (7.6 cm) by 4 in. (10 cm) sheet of lightweight paper over the Air Inlet Grille of the Temperature Sensor Assembly for the Fwd Cabin.

(b) If the fan does not hold the paper in its position, then do these steps:

1) Remove the Forward Cabin Zone Temperature Sensor Filter. This is the task: Cabin Temperature Sensor Assembly Filter Removal, AMM TASK 21-61-06-000-801.

2) Examine the filter for dirt or other blockage.

3) If the filter is dirty, then do these steps:

a) Install a new filter. This is the task: Cabin Temperature Sensor Assembly Filter Installation, AMM TASK 21-61-06-400-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

b) Supply conditioned air to the Passenger Compartment with the two packs. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

c) On the P5-17 Panel, make sure that the FWD CAB Temperature Selector is in the AUTO position.

d) Let the Air Conditioning Packs operate for a few minutes to allow the temperature of the air from the packs to stabilize.

e) On the P5-17 Panel, put the FWD CAB Temperature Selector to the W position.

<1> If the temperature in the Fwd Cabin increases, then you corrected the problem.

<a> Remove the conditioned air from the airplane. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

4) If the filter is not dirty, then do these steps:

a) Replace the Zone Temperature Sensor Fan for the Forward Cabin Zone. These are the tasks:

- Cabin Temperature Sensor Fan - Removal, AMM TASK 21-61-11-000-801
- Cabin Temperature Sensor Fan - Installation, AMM TASK 21-61-11-400-801

b) Supply conditioned air to the Passenger Compartment with the two packs. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

c) On the P5-17 Panel, make sure that the FWD CAB Temperature Selector is in the AUTO position.

d) Let the Air Conditioning Packs operate for a few minutes to allow the temperature of the air from the packs to stabilize.

e) On the P5-17 Panel, put the FWD CAB Temperature Selector to the W position.

<1> If the temperature in the Fwd Cabin increases, then you corrected the problem.

<a> Remove the conditioned air from the airplane. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(c) If the fan holds the paper in its position, then continue.

1) Remove the conditioned air from the airplane. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(6) If the Initial Evaluation indicated damage to the portion of the Passenger Cabin Sidewall Panels near the floor in the station location near the Forward Wing-to-Body Fairing, then do the steps that follow:

(a) Remove the applicable Sidewall Panel(s).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

1) This is the task: Sidewall Panel - Removal, AMM TASK 25-21-46-000-801.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(b) Examine the Sidewall Risers and repair as necessary. These are the applicable tasks:

- Sidewall Riser Ducts/Transition Duct Inspection, AMM TASK 21-23-02-212-801
- Sidewall Riser Ducts/Transition Duct Repair, AMM TASK 21-23-02-360-801 or Sidewall Riser Ducts/Transition Duct Repair, AMM TASK 21-23-02-360-802

——— END OF TASK ———

OMR 701-799

865. TEMP CONTROL BOX - NO GO Red Light Indication - Fault Isolation

A. Description

(1) This task is for these maintenance messages:

(a) POWER CHECK - NO GO Red Light Indication (BITE Test Switch Position 2)

(b) OPEN VALVE DRIVE CHECK - NO GO Red Light Indication (BITE Test Switch Position 3)

(c) DEADBAND CHECK - NO GO Red Light Indication (BITE Test Switch Position 4)

(d) CLOSE VALVE DRIVE CHECK - NO GO Red Light Indication (BITE Test Switch Position 5)

B. Possible Causes

(1) Cabin Temperature Controller, M325

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	2	C00268	AIR CONDITIONING TEMP CONTROL AUTO L
B	2	C00258	AIR CONDITIONING TEMP CONTROL AUTO R

D. Related Data

(1) WDM 21-61-21

(2) SSM 21-61-21

E. Initial Evaluation

(1) Do this task: Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.

(a) If the NO-GO Red Indicator Light for BITE Switch position 2 to 5 is not ON, then there was an intermittent fault.

(b) If the NO-GO Red Indicator Light for one or more BITE Switch position 2 to 5 is ON, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Replace the CTC, M325. These are the tasks:

- Cabin Temperature Controller (CTC) Removal, AMM TASK 21-61-01-000-801
- Cabin Temperature Controller (CTC) Installation, AMM TASK 21-61-01-400-801

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OMR 701-799 (Continued)

(a) Do the Cabin Temperature Controller BITE Procedure, 21-61 TASK 801.

1) If the BITE Test is satisfactory, then you corrected the problem.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

801. Pack Temperature Control - BITE Procedure

A. General

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

(1) This task does a BITE Test of the Left (Right) Pack/Zone Temperature Controller (PZTC), M1442 (M1443).

(21-62 TASK SUPPORT Figure 301 or 21-62 TASK SUPPORT Figure 302)

(a) The BITE Test does the checks that follow:

1) PZTC Self Test
2) Zone Temperature Selector Test
3) Cabin Temperature Sensor Test
4) Duct Temperature Limit Sensor Test
5) Zone Trim Air Modulating Valve Test
6) Ram Air Control Temperature Sensor Test
7) Ram Air Inlet Actuator Test

OMR 815, 816, 818, 819, 901-999

8) SRADA (Smart Ram Air Door Actuator) Inlet and Exit Actuators Test

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

9) Pack Temperature Sensor Test
10) Temperature Control Valve Test
11) Standby Temperature Control Valve Test
12) Mix Manifold Temperature Sensor Test

OMR 820-826

(2) This task does a BITE Test of the Left (Right) Pack Flow Temperature Controller (PFTC), M2951 (M2952).

(21-62 TASK SUPPORT Figure 303 or 21-62 TASK SUPPORT Figure 304)

(a) The BITE Test does a check on these components:

1) PFTC
2) Flight Deck Temperature Sensor
3) Flight Deck Temperature Sensor Backup
4) Zone Temperature Sensor
5) Standby Zone Sensor
6) Flight Deck Duct Sensor
7) Flight Deck Duct Sensor Backup
8) Pack Temperature Sensor
9) Standby Pack Temperature Sensor
10) Mix Manifold Temperature Sensor
11) Flight Deck Selector
12) Zone Selector
13) Standby Selector

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 820-826 (Continued)

14) Pack Inlet Pressure Sensor
15) Pack Flow Sensor
16) Flow Control Valve
17) Temperature Control Valve
18) Flight Deck Trim Valve
19) Zone Trim Valve
20) Standby Temperature Control Valve
21) SRADA Inlet Actuator
22) SRADA Exit Actuator
23) SRADA Ram Air Sensor

(b) There are two identical Pack Flow Temperature Controllers (PFTC)s, located on the E3-3 Rack.

(c) If the PFTC shows FAULT on the Front Panel Display, the controller is detecting a fault.

NOTE: Faults could also be recorded in the FAULT HISTORY Mode of the PFTC.

(d) The BITE Procedure must be done on both PFTCs.

(e) The steps that follow give a summary of the function of the buttons on the front of the PFTC:

1) ON/OFF button: activates or de-activates the BITE functions on the PFTC.
2) MENU button: shows the BITE Menus.
   a) If you push the MENU button, you will go up to the previously displayed MENU.
3) YES button: used to respond YES to questions on the Display Screen.
4) NO button: used to respond NO to questions on the Display Screen.
5) UP Arrow button: used to scroll up through the menu or the results of a BITE Test.
6) DOWN Arrow button: used to scroll down through the menu or the results of a BITE Test.

(f) The PFTC BITE has five modes of operation as shown below:

1) Manual ON/OFF
2) EXISTING FAULTS
3) FAULT HISTORY
4) GROUND TESTS
5) OTHER FUNCTIONS

(g) The steps that follow give a summary of the MAIN MENU selections:

1) Manual ON/OFF
   a) The ON/OFF button supplies power to the Front Panel Displays.
2) EXISTING FAULTS
   a) The EXISTING FAULTS/MAIN MENU selection shows a list of faults that the PFTC detects at the present time.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 820-826 (Continued)

b) Maintenance action is necessary to correct the maintenance messages that show in EXISTING FAULTS.

3) FAULT HISTORY

a) The FAULT HISTORY state is activated when you push the DOWN Arrow button or NO button when the Front Panel Display shows EXISTING FAULTS or when you scroll UP/DOWN from the Front Panel Display MENU button.

b) Fault Memory is read and FLIGHT LEG Fault records show in reverse chronological order. The last fault on the last flight leg shows first and the first fault on the first flight leg shows last.

<1> The last fault on the last flight leg is displayed first and the first fault on the first flight leg is displayed last.

4) GROUND TESTS

a) GROUND TESTS is activated when you push the DOWN Arrow button or the NO button when the Front Panel Display shos FAULT HISTORY or when you scroll UP/DOWN from the Front Panel Display MENU button.

b) The GROUND TESTS menus provide the option to run a Ground Test with or without the FLOW DRIFT CHECK. There is also a Ground Test for the DISPLAY.

c) During the GROUND TESTS state, the BITE operator responds to the instructions on the display by pressing the Front Panel buttons.

5) OTHER FUNCTIONS

a) You can activate the OTHER FUNCTIONS option from the MAIN MENU.

b) The OTHER FUNCTIONS option has two modes of operation as shown below:

<1> SYSTEM CONFIG

<2> I/O MONITOR

<a> You can use these menus to monitor various PFTC Input and Output states/values

NOTE: The I/O MONITOR Menus do not update in real-time. For example, if you change the CONT CAB Switch on the Cabin Temperature Panel, P5-17, from AUTO to OFF, when you monitor the I/O MONITOR, DISCRETE INPUT Menu, the state of the switch does not change automatically in the menu. To see the changed switch state, go back to the Main Menu and then come back to the I/O MONITOR, DISCRETE INPUT Menu.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

B. BITE Procedure

(21-62 TASK SUPPORT Figure 301 or 21-62 TASK SUPPORT Figure 302 or 21-62 TASK SUPPORT Figure 303 or 21-62 TASK SUPPORT Figure 304)

(1) Make sure that there is no PACK/ZONE TEMP Light illuminated on the Panel, P5.

(a) If the PACK/ZONE TEMP Light is illuminated, press the TRIP RESET Switch on the Panel, P5 or MASTER CAUTION on the Glareshield Panel to reset the light.

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OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

(2) Do the BITE Procedure for the Pack/Zone Temperature Controller (PZTC) as follows:

(a) Put these switches on the P5-10 Air Conditioning Panel to the indicated positions below:

1) Put the BLEED 1, BLEED 2, and BLEED APU Switches to the OFF position.

a) Make sure that the L (R) Duct Pressure Indications on the DUCT PRESS Indication Gauge on the P5-10 Panel are Zero.

2) Put the L PACK and R PACK Switches to the AUTO position.

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

3) Put the L RECIRC FAN and R RECIRC FAN Switches to the AUTO position.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

4) Put the ISOLATION VALVE Switch to the OPEN position.

(b) Put these switches on the P5-17 Cabin Temperature Panel to the indicated positions below:

1) Put the TRIM AIR Switch to the ON position.

2) Put the CONT CAB, FWD CAB, and AFT CAB Temperature Selectors to the AUTO position.

(c) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

(d) Get access to the Zone Temperature Controllers located on the bottom shelf (E3-3) of the E3 Rack in the Electronics Equipment Compartment.

OMR 802, 805, 807, 809, 810, 812, 813

(e) Make sure that these lights are OFF for the Left and Right Pack/Zone Temperature Controllers:

1) CONTROLLER FAULT - PACK/ZONE

2) CONTROLLER FAULT - RAM AIR

3) CONTROLLER FAULT - STANDBY PACK

OMR 815, 816, 818, 819, 901-999

(f) Make sure these Lights are OFF for the Left and Right Pack/Zone Temperature Controllers:

1) CONTROLLER FAULT - PACK/ZONE

2) CONTROLLER FAULT - STANDBY PACK

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

(g) Do the steps that follow on the Left and Right Pack/Zone Temperature Controllers:

1) Push and hold the PRESS/TEST Switch on the controller.

a) Make sure that all the lights on the controller come ON.

b) When the PRESS/TEST is released, make sure that all lights go OFF.

2) Push the BIT Switch on the controller.

NOTE: The BIT Switch will show if failures occurred during the last flight.

a) The light above the BIT Switch will come ON when the switch is fully pushed.

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b) Wait for the Green GO Light to come ON.

c) If a Red light comes ON for a component, the component does not operate correctly.

<1> Make a record of that component.

3) Push the PREV FLT Switch on the controller. The light above the PREV FLT Switch comes ON when you fully push the switch.

NOTE: The PREV FLT Switch will show all the failures that occurred during the last nine flights.

a) Wait for the Green GO Light to come ON.

b) If a Red Light comes ON for a component, the component did not operate correctly on a previous flight.

<1> Make a record of that component.

WARNING MOVE ALL PERSONS AND EQUIPMENT AWAY FROM THE RAM AIR INLET DOOR. WHEN YOU PUSH THE VERIFY SWITCH, THE RAM AIR COMPONENTS WILL MOVE AND CAN CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

4) Push the VERIFY Switch on the controller.

a) The light above the VERIFY Switch comes ON when you fully push the switch.

NOTE: The VERIFY Switch will initiate a real time test of the Pack and Zone Temperature Control Components.

b) Wait for the Green GO Light to come ON.

c) If a Red light comes ON for a component, the component does not operate correctly.

NOTE: If there are no Trim Air Valve (TAV) faults during operation and no TAV faults until the real time test, then do the BITE again to clear the fault. It is not necessary to replace the TAV.

<1> Make a record of that component.

(h) If a Red Light comes ON, then refer to the Table at the end of this task (or the 21-MAINT MSG INDEX at the beginning of this Chapter) to find the applicable Fault Isolation Manual (FIM) Tasks for the maintenance messages that show.

(i) If a Green GO Light comes ON, then do this step to clear the controller's Fault Memory:

1) Push the RESET Switch when the VERIFY and GO Lights are ON.

(j) After the BITE Procedure is complete, on the P5-10 Air Conditioning Panel and P5-17 Cabin Temperature Panel, put these switches to the indicated positions below:

1) Put the L PACK and R PACK Switches to the OFF position.

2) Put all other switches back to their original configuration.

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OMR 820-826

(3) Do the BITE Procedure for the Pack/Flow Temperature Controller (PFTC) as follows:

(a) Make sure that the CAB/UTIL Switch on the P5-13 ELEC METER, BATTERY AND GALLEY POWER Module is in the ON position.

(b) On the P5-10 Air Conditioning Panel, put these switches to the positions shown below:

1) BLEED 1, BLEED 2 and BLEED APU Switches to OFF

2) Make sure that the L (R) Duct Pressure indications on the DUCT PRESS Indication Gauge on the P5-10 Panel are zero.

3) L PACK and R PACK to AUTO

4) ISOLATION VALVE to CLOSE

5) L RECIRC FAN and R RECIRC FAN Switches to OFF

(c) On the P5-17 Temperature Control Panel, put these switches to the positions shown below:

1) CONT CAB, FWD CAB and AFT CAB Switches to AUTO

2) TRIM AIR Switch to ON

(d) Get access to the Pack Flow Temperature Controllers in the E/E Equipment Bay:

1) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

C. BITE Procedure - EXISTING FAULTS

(1) Make sure that the PFTC Front Panel Display shows EXISTING FAULTS.

(2) Push the YES button on the front panel.

(3) The PFTC Front Panel Display shows the first maintenance message.

NOTE: The text of the fault will be the same as one of the maintenance messages shown in the Table at the end of this task. These existing faults are referred to as maintenance messages.

(4) Find and record the maintenance messages in EXISTING FAULTS as follows:

(a) Make a record of the first maintenance message.

(b) Continue to push the DOWN Arrow button on the PFTC Front Panel to identify and record all maintenance messages present until the PFTC Front Panel Display shows END OF LIST.

(5) Push the MENU button on the PFTC Front Panel.

(6) Refer to the Table at the end of this task (or the 21-MAINT MSG INDEX at the beginning of this Chapter) to find the applicable FIM Tasks for the maintenance messages that show.

D. BITE Procedure - FAULT HISTORY

(1) Refer to the PFTC Latched Maintenance Message index and PFTC Non-latched Maintenance Message index to determine if the maintenance messages that show in the fault history are Latched or Non-latched.

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OMR 820-826 (Continued)

(a) Refer to the Table at the end of this task (or the 21-MAINT MSG INDEX at the beginning of this Chapter) to find the applicable FIM Tasks for the Latched Maintenance Messages that show.

NOTE: It is required to do the applicable FIM Task for Latched Maintenance Messages that show in the Fault History.

1) Make a record of the applicable FIM Task that will be performed.

NOTE: While PFTC stores fault history for the most recent 64 flight legs, it is recommended that operators only review the faults recorded in the last (most recent) flight leg. Refer to FIM tasks for the Maintenance Messages recorded during the last (most recent) flight leg. It is not required to troubleshoot the same Latched Maintenance Message twice.

NOTE: For aircraft equipped with PFTC P/Ns 51090248-002/-003: If a Pack Off (i.e. un-commanded eFCV closure) condition is found via ACMS/AHM reporting or QAR/FDR data review, operators can review the faults recorded in the last (most recent) flight leg of Fault History to identify if a Latched Maintenance Message was stored (in addition to EXISTING FAULTS). Refer to the FIM tasks for the Maintenance Messages recorded during the last (most recent) flight leg.

NOTE: For aircraft equipped with PFTC P/N 51090248-005, operators are only required to review the Fault History for the last (most recent) flight leg recorded (in addition to EXISTING FAULTS) in correlation to the last Pack Light observed during Master Caution Recall.

NOTE: After you correct a problem related to a Latched Maintenance Message, you can clear the Maintenance Message as follows:

- Cycle the power to the PFTC or
- Perform the BITE ground test or
- Erase the EXISTING FAULTS from the PFTC Front Panel Menus.

(b) Refer to the Table at the end of this task (or the 21-MAINT MSG INDEX at the beginning of this Chapter) to find the applicable FIM Tasks for the Non-latched Maintenance Messages that show.

NOTE: It is not required to do the applicable FIM task for Non-latched Maintenance Messages that show the Fault History.

(2) Make sure that the PFTC Front Panel Display shows FAULT HISTORY.

(3) Push the YES button on the PFTC Front Panel.

(a) The PFTC Front Panel Display shows FLIGHT LEG XX.

NOTE: XX is the last flight leg that faults were recorded in history.

(4) If no FLIGHT LEGS show on the PFTC Front Panel, this indicates that there are no faults recorded in FAULT HISTORY.

(5) Push the YES button on the PFTC Front Panel.

(6) The PFTC Front Panel Display shows the first maintenance message in the FLIGHT LEG XX list.

NOTE: The text of the fault will be the same as one of the maintenance messages listed in the Table at the end of this task. These faults are referred to as maintenance messages.

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OMR 820-826 (Continued)

(7) Find and record the maintenance messages in the last (most recent) flight leg until the PFTC front panel display shows BOTTOM OF LIST for two seconds, then shows the maintenance message FLIGHT LEG XX list.

(8) Push the MENU button on the PFTC Front Panel until the display show FLIGHT LEG XX.

(9) Push the DOWN Arrow button as required on the PFTC Front Panel to look for the next flight leg that recorded faults in FAULT HISTORY.

NOTE: Unless maintenance messages in previous flight legs were never examined/resolved, it is only required to address the maintenance messages recorded during the last (most recent) flight leg.

(10) Do the above steps until all flight legs with maintenance messages in FAULT HISTORY have been examined and recorded.

(11) If the PFTC Front Panel Display is not at a BITE MAIN MENU selection, push the MENU button until you are in the MAIN MENU.

E. BITE Procedure - GROUND TESTS

(1) Make sure that the Front Panel Display on the Left (Right) PFTC, M2951 (M2952) shows GROUND TESTS.

(2) Push the YES button on the PFTC Front Panel.

(a) Make sure that the display shows BITE INSTRUCTION.

(3) Push the DOWN Arrow button.

(a) Make sure that the display shows 1: SET FOLLOWING AT FLIGHT DECK.

(4) Push the DOWN Arrow button.

(a) Make sure that the display shows ALL BLEED SWITCH OFF.

(b) If not already done, set the BLEED 1, BLEED 2 and APU BLEED Switches on the P5-10 Air Conditioning Panel to OFF.

(5) Push the DOWN Arrow button.

(a) Make sure that the display shows ALL ZONE TMP SEL AUTO.

(b) If not already done, set the CONT CAB, FWD CAB and AFT CAB Temperature Selectors on the P5-17 Temperature Control Panel to AUTO.

(6) Push the DOWN Arrow button.

(a) Make sure that the display shows LH RH PACK SWITCH AUTO.

(b) If not already done, set the L PACK and R PACK Switches on the P5-10 Air Conditioning Panel to AUTO.

(7) Push the DOWN Arrow button.

(a) Make sure that the display shows TRIM AIR SWITCH ON.

(b) If not already done, set the TRIM AIR Switch on the P5-17 Temperature Control Panel to ON.

(8) Push the DOWN Arrow button.

(a) Make sure that the display shows 2: WHEN PROMPTED SEL YES TO TEST.

(9) Push the DOWN Arrow button.

(a) Make sure that the display shows 3: FOR RESULTS FOLLOW PROMPT.

(10) Push the DOWN Arrow button.

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OMR 820-826 (Continued)

(a) Make sure that the display shows 4: AFTER TST SET PACK SWITCHES OFF.

OMR 820-826; AIRPLANES WITH PFTC P/N 51090248-002/-003

(11) Push the DOWN Arrow button.

(12) To omit the flow sensor drift check, do these steps:

(a) Make sure that the panel display shows FLOW DRIFT CK?.

(b) Push the NO button.

(c) Make sure that the panel display shows START GROUND TEST?.

(d) Push the YES button.

(e) Make sure that the panel display shows GROUND TEST IN PROGRESS FOR UP TO 150SEC.

(13) To do the flow sensor drift check, do these steps:

(a) Make sure that the panel display shows FLOW DRIFT CK?.

(b) Push the YES button.

(c) Make sure that the panel display shows START GROUND TEST?.

(d) Push the YES button.

(e) Make sure that the panel display shows GROUND TEST IN PROGRESS FOR UP TO 270SEC.

OMR 820-826; AIRPLANES WITH PFTC P/N 51090248-005

(14) Push the DOWN Arrow button.

(15) To bypass the flow sensor drift check, do these steps:

(a) Make sure that the panel display shows BYPASS FLOW DRIFT CHECK?.

(b) Push the YES button.

(c) Make sure that the panel display shows START GROUND TEST?.

(d) Push the YES button.

(e) Make sure that the panel display shows GROUND TEST IN PROGRESS FOR UP TO 150SEC.

(16) To do the flow sensor drift check, do these steps:

(a) Make sure that the panel display shows BYPASS FLOW DRIFT CHECK?.

(b) Push the NO button.

(c) Make sure that the panel display shows ENGINES/APU OFF AT LEAST 20MIN?.

1) Make sure that the engines and APU have been off for at least 20 minutes.

(d) Push the YES/DOWN button.

(e) Make sure that the panel display shows START GROUND TEST?.

(f) Push the YES button.

(g) Make sure that the panel display shows GROUND TEST IN PROGRESS FOR UP TO 330SEC.

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OMR 820-826 (Continued)

(17) After time has elapsed, make sure that the PFTC Front Panel Display shows GND TST COMPLETE PASS.

(18) After 5 seconds, the display shows GROUND TEST RESULTS?.

NOTE: If the panel display showed FAIL during the Ground Test BITE, you can push the YES button to go to the Existing Faults section to correct the problem.

(19) Make sure that the display shows NO FAULTS.

(20) Push the ON/OFF button on the PFTC Front Panel.

F. PFTC Latched Maintenance Message Index

(1) Use the below to find Latched Maintenance Messages.

Latched Maintenance Messages

LATCHED MAINTENANCE MESSAGES
21–51001 PFTC CNTRL INOP
21–51002 PFTC CNTRL INOP
21–51003 PFTC CNTRL INOP
21–51004 PFTC CNTRL INOP
21–51005 PFTC CNTRL INOP
21–51006 PFTC CNTRL INOP
21–51007 PFTC CNTRL INOP
21–51008 PFTC CNTRL INOP
21–51009 PFTC CNTRL INOP
21–51010 PFTC CNTRL INOP
21–51017 PFTC CNTRL INOP
21–51018 PFTC CNTRL INOP
21–51019 PFTC CNTRL INOP
21–51034 PFTC CNTRL INOP
21–51035 PFTC CNTRL INOP
21–51037 PFTC CNTRL INOP
21–51040 PFTC CNTRL INOP
21–51041 PFTC CNTRL INOP
21–51042 PFTC CNTRL INOP
21–51044 PFTC CNTRL INOP
21–51066 PFTC CNTRL INOP
21–51067 PFTC CNTRL INOP
21–51068 PFTC CNTRL INOP
21–51069 PFTC CNTRL INOP
21–51070 PFTC CNTRL INOP

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OMR 820-826 (Continued)

Latched Maintenance Messages (Continued)

LATCHED MAINTENANCE MESSAGES
21–51073 PFTC CNTRL INOP
21–51074 PFTC CNTRL INOP
21–51075 PFTC CNTRL INOP
21–51078 PFTC CNTRL INOP
21–51079 PFTC CNTRL INOP
21–51080 PFTC CNTRL INOP
21–51083 PFTC CNTRL INOP
21–51084 PFTC CNTRL INOP
21–51085 PFTC CNTRL INOP
21–51086 PFTC CNTRL CNTRL
21–51087 PFTC CNTRL CNTRL
21–51110 PFTC CNTRL INOP
21–51112 PFTC CNTRL INOP
21–51114 PFTC CNTRL INOP
21–51116 PFTC CNTRL INOP
21–51118 PFTC CNTRL INOP
21–51120 PFTC CNTRL INOP
21–51122 PFTC CNTRL INOP
21–51124 PFTC CNTRL INOP
21–51126 PFTC CNTRL INOP
21–51128 PFTC CNTRL INOP
21–51233 PACK IN PRESSURE
21–51239 PACK FLOW SENSOR
21–51246 PACK FLOW SENSOR
21–51247 PACK FLOW SENSOR
21–51248 PACK FLOW SENSOR
21–51249 PACK FLOW SENSOR
21–51250 FLOW CTRL VALVE
21–51251 FLOW CTRL VALVE
21–51252 FLOW CTRL VALVE
21–51255 FLOW CTRL VALVE
21–51260 TEMP CTRL VALVE
21–51261 TEMP CTRL VALVE

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OMR 820-826 (Continued)

Latched Maintenance Messages (Continued)

LATCHED MAINTENANCE MESSAGES
21–51263 TEMP CTRL VALVE
21–51264 TEMP CTRL VALVE
21–51265 TEMP CTRL VALVE
21–51268 F/D TRIM VALVE
21–51269 F/D TRIM VALVE
21–51271 F/D TRIM VALVE
21–51272 F/D TRIM VALVE
21–51273 F/D TRIM VALVE
21–51276 ZONE TRIM VALVE
21–51277 ZONE TRIM VALVE
21–51279 ZONE TRIM VALVE
21–51280 ZONE TRIM VALVE
21–51281 ZONE TRIM VALVE
21–51300 STBY TCV VALVE
21–51301 STBY TCV VALVE
21–51310 F/D TEMP SENSOR
21–51311 F/D SELECTOR
21–51312 F/D DUCT SENSOR
21–51313 PACK TEMP SENSOR
21–51314 MIXMAN TEMP SNSR
21–51409 SRADA INLET
21–51410 SRADA EXIT
21–51411 SRADA RAM SNSR
21–51500 PFTC CNTRL INOP
21–51501 PFTC CNTRL INOP
21–51502 PFTC CNTRL INOP
21–51503 PFTC CNTRL INOP
21–51504 PFTC CNTRL INOP
21–51505 PFTC CNTRL INOP
21–51506 PFTC CNTRL INOP
21–51507 PFTC CNTRL INOP
21–51508 F/D TRIM VALVE
21–51509 ZONE TRIM VALVE

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OMR 820-826 (Continued)

Latched Maintenance Messages (Continued)

LATCHED MAINTENANCE MESSAGES
21–51510 TEMP CTRL VALVE

G. PFTC Non-latched Maintenance Message Index

(1) Use the below to find Non-latched Maintenance Messages.

Non-latched Maintenance Messages

NON-LATCHED MAINTENANCE MESSAGES
21–51030 DISPLAY
21–51036 PFTC CNTRL INOP
21–51038 PFTC CNTRL INOP
21–51043 PFTC CNTRL INOP
21–51046 STBY 28V PWR OFF
21–51053 PFTC CNTRL INOP
21–51056 PFTC CNTRL INOP
21–51061 PFTC CNTRL INOP
21–51090 PFTC STBY CNTRL
21–51091 PFTC STBY CNTRL
21–51092 PFTC STBY CNTRL
21–51093 PFTC STBY CNTRL
21–51097 PFTC STBY CNTRL
21–51100 PFTC STBY CNTRL
21–51101 PFTC STBY CNTRL
21–51111 PFTC CNTRL INOP
21–51113 PFTC CNTRL INOP
21–51115 PFTC CNTRL INOP
21–51117 PFTC CNTRL INOP
21–51119 PFTC CNTRL INOP
21–51121 PFTC CNTRL INOP
21–51123 PFTC CNTRL INOP
21–51125 PFTC CNTRL INOP
21–51127 PFTC CNTRL INOP
21–51129 PFTC CNTRL INOP
21–51131 ARINC 429 INVLD
21–51132 ARINC 429 INVLD
21–51133 ARINC 429 INVLD

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OMR 820-826 (Continued)

Non-latched Maintenance Messages (Continued)

NON-LATCHED MAINTENANCE MESSAGES
21–51134 ARINC 429 INVLD
21–51135 ARINC 429 INVLD
21–51136 ARINC 429 INVLD
21–51143 ARINC 429 INVLD
21–51144 ARINC 429 INVLD
21–51145 ARINC 429 INVLD
21–51146 ARINC 429 INVLD
21–51147 ARINC 429 INVLD
21–51148 ARINC 429 INVLD
21–51149 ARINC 429 INVLD
21–51150 ARINC 429 INVLD
21–51151 ARINC 429 INVLD
21–51152 ARINC 429 INVLD
21–51153 ARINC 429 INVLD
21–51154 ARINC 429 INVLD
21–51190 F/D TEMP SENSOR
21–51191 F/D TEMP SENSOR
21–51193 ZONE TEMP SENSOR
21–51194 ZONE TEMP SENSOR
21–51196 STBY ZONE SENSOR
21–51197 STBY ZONE SENSOR
21–51199 F/D DUCT SENSOR
21–51200 F/D DUCT SENSOR
21–51202 ZONE DUCT SENSOR
21–51203 ZONE DUCT SENSOR
21–51205 PACK TEMP SENSOR
21–51206 PACK TEMP SENSOR
21–51208 STBY PACK TEMP
21–51209 STBY PACK TEMP
21–51211 MIXMAN TEMP SNSR
21–51212 MIXMAN TEMP SNSR
21–51217 F/D SELECTOR
21–51218 F/D SELECTOR

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OMR 820-826 (Continued)

Non-latched Maintenance Messages (Continued)

NON-LATCHED MAINTENANCE MESSAGES
21–51220 ZONE SELECTOR
21–51221 ZONE SELECTOR
21–51224 STBY SELECTOR
21–51225 STBY SELECTOR
21–51230 PACK IN PRESSURE
21–51231 PACK IN PRESSURE
21–51232 PACK IN PRESSURE
21–51240 PACK FLOW SENSOR
21–51253 FLOW CTRL VALVE
21–51254 FLOW CTRL VALVE
21–51266 TEMP CTRL VALVE
21–51267 TEMP CTRL VALVE
21–51274 F/D TRIM VALVE
21–51275 F/D TRIM VALVE
21–51282 ZONE TRIM VALVE
21–51283 ZONE TRIM VALVE
21–51331 DISPLAY
21–51332 PFTC CNTRL INOP
21–51333 PFTC CNTRL INOP
21–51400 SRADA CNTRL
21–51401 SRADA INLET
21–51402 SRDA EXIT
21–51403 SRDA RAM SNSR
21–51405 SRDA INLET
21–51406 SRDA EXIT
21–51407 SRDA RAM SNSR
21–51408 SRDA CNTRL

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

H. Pack Temperature Control Fault Code Index

(1) Refer to the table below to find the applicable FIM Task for the Maintenance Messages.

LRU/SYSTEM	MAINTENANCE MESSAGE	GO TO FIM TASK
PACK/ZN CON - L	BKUP ZONE CABIN SENSOR	21-62 TASK 815
PACK/ZN CON - L	BKUP ZONE SELECTOR	21-62 TASK 809

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LRU/SYSTEM	MAINTENANCE MESSAGE	GO TO FIM TASK
PACK/ZN CON - L	CONTROLLER FAULT - PACK/ZONE	21-62 TASK 802
PACK/ZN CON - L	CONTROLLER FAULT - RAM AIR	21-62 TASK 802
PACK/ZN CON - L	CONTROLLER FAULT - STANDBY	21-62 TASK 802
PACK/ZN CON - L	FLT DECK CABIN SENSOR	21-62 TASK 811
PACK/ZN CON - L	FLT DECK DUCT SENSOR	21-62 TASK 813
PACK/ZN CON - L	FLT DECK SELECTOR	21-62 TASK 805
PACK/ZN CON - L	FLT DECK TRIM VALVE	21-62 TASK 833
PACK/ZN CON - L	RAM AIR ACT	21-62 TASK 837
PACK/ZN CON - L	RAM AIR SNSR	21-62 TASK 821
PACK/ZN CON - L	SRADA EXIT	21-62 TASK 853
PACK/ZN CON - L	SRADA INLET	21-62 TASK 851
PACK/ZN CON - L	STBY OTHER PACK - PACK SNSR	21-62 TASK 824
PACK/ZN CON - L	STBY OTHER PACK - TCV	21-62 TASK 829
PACK/ZN CON - L	THIS PACK - MIX MANF SENSOR	21-62 TASK 827
PACK/ZN CON - L	THIS PACK - PACK SNSR	21-62 TASK 825
PACK/ZN CON - L	THIS PACK - TCV	21-62 TASK 830
PACK/ZN CON - L	ZONE CABIN SENSOR	21-62 TASK 817
PACK/ZN CON - L	ZONE DUCT SENSOR	21-62 TASK 819
PACK/ZN CON - L	ZONE SELECTOR	21-62 TASK 808
PACK/ZN CON - L	ZONE TRIM VALVE	21-62 TASK 835
PACK/ZN CON - R	BKUP ZONE CABIN SENSOR	21-62 TASK 816
PACK/ZN CON - R	BKUP ZONE SELECTOR	21-62 TASK 807
PACK/ZN CON - R	CONTROLLER FAULT - PACK/ZONE	21-62 TASK 803
PACK/ZN CON - R	CONTROLLER FAULT - RAM AIR	21-62 TASK 803
PACK/ZN CON - R	CONTROLLER FAULT - STANDBY	21-62 TASK 803
PACK/ZN CON - R	FLT DECK CABIN SENSOR	21-62 TASK 810
PACK/ZN CON - R	FLT DECK DUCT SENSOR	21-62 TASK 812
PACK/ZN CON - R	FLT DECK SELECTOR	21-62 TASK 804
PACK/ZN CON - R	FLT DECK TRIM VALVE	21-62 TASK 832
PACK/ZN CON - R	RAM AIR ACT	21-62 TASK 836
PACK/ZN CON - R	RAM AIR SNSR	21-62 TASK 820
PACK/ZN CON - R	SRADA EXIT	21-62 TASK 854
PACK/ZN CON - R	SRADA INLET	21-62 TASK 852
PACK/ZN CON - R	STBY OTHER PACK - PACK SNSR	21-62 TASK 823
PACK/ZN CON - R	STBY OTHER PACK - TCV	21-62 TASK 828

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LRU/SYSTEM	MAINTENANCE MESSAGE	GO TO FIM TASK
PACK/ZN CON - R	THIS PACK - MIX MANF SENSOR	21-62 TASK 826
PACK/ZN CON - R	THIS PACK - PACK SNSR	21-62 TASK 822
PACK/ZN CON - R	THIS PACK - TCV	21-62 TASK 831
PACK/ZN CON - R	ZONE CABIN SENSOR	21-62 TASK 814
PACK/ZN CON - R	ZONE DUCT SENSOR	21-62 TASK 818
PACK/ZN CON - R	ZONE SELECTOR	21-62 TASK 806
PACK/ZN CON - R	ZONE TRIM VALVE	21-62 TASK 834
PFTC	21-51001 PFTC CNTRL INOP	21-62 TASK 858
PFTC	21-51002 PFTC CNTRL INOP	21-62 TASK 858
PFTC	21-51003 PFTC CNTRL INOP	21-62 TASK 858
PFTC	21-51004 PFTC CNTRL INOP	21-62 TASK 858
PFTC	21-51005 PFTC CNTRL INOP	21-62 TASK 858
PFTC	21-51006 PFTC CNTRL INOP	21-62 TASK 858
PFTC	21-51007 PFTC CNTRL INOP	21-62 TASK 858
PFTC	21-51008 PFTC CNTRL INOP	21-62 TASK 858
PFTC	21-51009 PFTC CNTRL INOP	21-62 TASK 858
PFTC	21-51010 PFTC CNTRL INOP	21-62 TASK 858
PFTC	21-51017 PFTC CNTRL INOP	21-62 TASK 858
PFTC	21-51018 PFTC CNTRL INOP	21-62 TASK 858
PFTC	21-51019 PFTC CNTRL INOP	21-62 TASK 858
PFTC	21-51030 DISPLAY	21-62 TASK 858
PFTC	21-51034 PFTC CNTRL INOP	21-62 TASK 858
PFTC	21-51035 PFTC CNTRL INOP	21-62 TASK 858
PFTC	21-51036 PFTC CNTRL INOP	21-62 TASK 858
PFTC	21-51037 PFTC CNTRL INOP	21-62 TASK 858
PFTC	21-51038 PFTC CNTRL INOP	21-62 TASK 858
PFTC	21-51041 PFTC CNTRL INOP	21-62 TASK 858
PFTC	21-51042 PFTC CNTRL INOP	21-62 TASK 858
PFTC	21-51043 PFTC CNTRL INOP	21-62 TASK 858
PFTC	21-51044 PFTC CNTRL INOP	21-62 TASK 858
PFTC	21-51045 PFTC CNTRL INOP	21-62 TASK 858
PFTC	21-51046 STBY 28V PWR OFF	21-62 TASK 858
PFTC	21-51053 PFTC CNTRL INOP	21-62 TASK 862
PFTC	21-51056 PFTC CNTRL INOP	21-62 TASK 863
PFTC	21-51061 PFTC CNTRL INOP	21-62 TASK 859

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LRU/SYSTEM	MAINTENANCE MESSAGE	GO TO FIM TASK
PFTC	21-51066 PFTC CNTRL INOP	21-62 TASK 860
PFTC	21-51067 PFTC CNTRL INOP	21-62 TASK 860
PFTC	21-51068 PFTC CNTRL INOP	21-62 TASK 860
PFTC	21-51069 PFTC CNTRL INOP	21-62 TASK 857
PFTC	21-51070 PFTC CNTRL INOP	21-62 TASK 857
PFTC	21-51073 PFTC CNTRL INOP	21-62 TASK 876
PFTC	21-51074 PFTC CNTRL INOP	21-62 TASK 876
PFTC	21-51075 PFTC CNTRL INOP	21-62 TASK 876
PFTC	21-51078 PFTC CNTRL INOP	21-62 TASK 877
PFTC	21-51079 PFTC CNTRL INOP	21-62 TASK 877
PFTC	21-51080 PFTC CNTRL INOP	21-62 TASK 877
PFTC	21-51083 PFTC CNTRL INOP	21-62 TASK 874
PFTC	21-51084 PFTC CNTRL INOP	21-62 TASK 874
PFTC	21-51085 PFTC CNTRL INOP	21-62 TASK 874
PFTC	21-51086 PFTC STBY CNTRL	21-62 TASK 875
PFTC	21-51087 PFTC STBY CNTRL	21-62 TASK 875
PFTC	21-51090 PFTC STBY CNTRL	21-62 TASK 858
PFTC	21-51091 PFTC STBY CNTRL	21-62 TASK 858
PFTC	21-51092 PFTC STBY CNTRL	21-62 TASK 858
PFTC	21-51093 PFTC STBY CNTRL	21-62 TASK 858
PFTC	21-51097 PFTC STBY CNTRL	21-62 TASK 864
PFTC	21-51100 PFTC STBY CNTRL	21-62 TASK 868
PFTC	21-51101 PFTC STBY CNTRL	21-62 TASK 868
PFTC	21-51110 PFTC CNTRL INOP	21-62 TASK 887
PFTC	21-51111 PFTC CNTRL INOP	21-62 TASK 887
PFTC	21-51112 PFTC CNTRL INOP	21-62 TASK 888
PFTC	21-51113 PFTC CNTRL INOP	21-62 TASK 888
PFTC	21-51114 PFTC CNTRL INOP	21-62 TASK 889
PFTC	21-51115 PFTC CNTRL INOP	21-62 TASK 889
PFTC	21-51116 PFTC CNTRL INOP	21-62 TASK 890
PFTC	21-51117 PFTC CNTRL INOP	21-62 TASK 890
PFTC	21-51118 PFTC CNTRL INOP	21-62 TASK 891
PFTC	21-51119 PFTC CNTRL INOP	21-62 TASK 891
PFTC	21-51120 PFTC CNTRL INOP	21-62 TASK 892
PFTC	21-51121 PFTC CNTRL INOP	21-62 TASK 892

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LRU/SYSTEM	MAINTENANCE MESSAGE	GO TO FIM TASK
PFTC	21-51122 PFTC CNTRL INOP	21-62 TASK 893
PFTC	21-51123 PFTC CNTRL INOP	21-62 TASK 893
PFTC	21-51124 PFTC CNTRL INOP	21-62 TASK 894
PFTC	21-51125 PFTC CNTRL INOP	21-62 TASK 894
PFTC	21-51126 PFTC CNTRL INOP	21-62 TASK 895
PFTC	21-51127 PFTC CNTRL INOP	21-62 TASK 895
PFTC	21-51128 PFTC CNTRL INOP	21-62 TASK 896
PFTC	21-51129 PFTC CNTRL INOP	21-62 TASK 896
PFTC	21-51131 ARINC 429 INVLD	21-62 TASK 883
PFTC	21-51132 ARINC 429 INVLD	21-62 TASK 884
PFTC	21-51133 ARINC 429 INVLD	21-62 TASK 881
PFTC	21-51134 ARINC 429 INVLD	21-62 TASK 882
PFTC	21-51135 ARINC 429 INVLD	21-62 TASK 885
PFTC	21-51136 ARINC 429 INVLD	21-62 TASK 886
PFTC	21-51143 ARINC 429 INVLD	21-62 TASK 883
PFTC	21-51144 ARINC 429 INVLD	21-62 TASK 883
PFTC	21-51145 ARINC 429 INVLD	21-62 TASK 884
PFTC	21-51146 ARINC 429 INVLD	21-62 TASK 884
PFTC	21-51147 ARINC 429 INVLD	21-62 TASK 881
PFTC	21-51148 ARINC 429 INVLD	21-62 TASK 881
PFTC	21-51149 ARINC 429 INVLD	21-62 TASK 882
PFTC	21-51150 ARINC 429 INVLD	21-62 TASK 882
PFTC	21-51151 ARINC 429 INVLD	21-62 TASK 885
PFTC	21-51152 ARINC 429 INVLD	21-62 TASK 885
PFTC	21-51153 ARINC 429 INVLD	21-62 TASK 886
PFTC	21-51154 ARINC 429 INVLD	21-62 TASK 886
PFTC	21-51190 F/D TEMP SENSOR	21-62 TASK 865
PFTC	21-51191 F/D TEMP SENSOR	21-62 TASK 865
PFTC	21-51193 ZONE TEMP SENSOR	21-62 TASK 867
PFTC	21-51194 ZONE TEMP SENSOR	21-62 TASK 867
PFTC	21-51196 STBY ZONE SENSOR	21-62 TASK 868
PFTC	21-51197 STBY ZONE SENSOR	21-62 TASK 868
PFTC	21-51199 F/D DUCT SENSOR	21-62 TASK 866
PFTC	21-51200 F/D DUCT SENSOR	21-62 TASK 866
PFTC	21-51202 ZONE DUCT SENSOR	21-62 TASK 869

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LRU/SYSTEM	MAINTENANCE MESSAGE	GO TO FIM TASK
PFTC	21-51203 ZONE DUCT SENSOR	21-62 TASK 869
PFTC	21-51205 PACK TEMP SENSOR	21-62 TASK 871
PFTC	21-51206 PACK TEMP SENSOR	21-62 TASK 871
PFTC	21-51208 STBY PACK TEMP	21-62 TASK 872
PFTC	21-51209 STBY PACK TEMP	21-62 TASK 872
PFTC	21-51211 MIXMAN TEMP SNSR	21-62 TASK 873
PFTC	21-51212 MIXMAN TEMP SNSR	21-62 TASK 873
PFTC	21-51217 F/D SELECTOR	21-62 TASK 862
PFTC	21-51218 F/D SELECTOR	21-62 TASK 862
PFTC	21-51220 ZONE SELECTOR	21-62 TASK 863
PFTC	21-51221 ZONE SELECTOR	21-62 TASK 863
PFTC	21-51224 STBY SELECTOR	21-62 TASK 864
PFTC	21-51225 STBY SELECTOR	21-62 TASK 864
PFTC	21-51230 PACK IN PRESSURE	21-62 TASK 859
PFTC	21-51231 PACK IN PRESSURE	21-62 TASK 859
PFTC	21-51232 PACK IN PRESSURE	21-62 TASK 859
PFTC	21-51233 PACK IN PRESSURE	21-62 TASK 859
PFTC	21-51239 PACK FLOW SENSOR	21-62 TASK 860
PFTC	21-51240 PACK FLOW SENSOR	21-62 TASK 860
PFTC	21-51246 PACK FLOW SENSOR	21-62 TASK 860
PFTC	21-51247 PACK FLOW SENSOR	21-62 TASK 860
PFTC	21-51248 PACK FLOW SENSOR	21-62 TASK 860
PFTC	21-51249 PACK FLOW SENSOR	21-62 TASK 860
PFTC	21-51250 FLOW CTRL VALVE	21-62 TASK 857
PFTC	21-51251 FLOW CTRL VALVE	21-62 TASK 857
PFTC	21-51252 FLOW CTRL VALVE	21-62 TASK 857
PFTC	21-51253 FLOW CTRL VALVE	21-62 TASK 857
PFTC	21-51254 FLOW CTRL VALVE	21-62 TASK 857
PFTC	21-51255 FLOW CTRL VALVE	21-62 TASK 902
PFTC	21-51256 FLOW CTRL VALVE	21-62 TASK 861
PFTC	21-51257 FLOW CTRL VALVE	21-62 TASK 861
PFTC	21-51260 TEMP CTRL VALVE	21-62 TASK 874
PFTC	21-51261 TEMP CTRL VALVE	21-62 TASK 874
PFTC	21-51263 TEMP CTRL VALVE	21-62 TASK 874
PFTC	21-51264 TEMP CTRL VALVE	21-62 TASK 874

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LRU/SYSTEM	MAINTENANCE MESSAGE	GO TO FIM TASK
PFTC	21-51265 TEMP CTRL VALVE	21-62 TASK 874
PFTC	21-51266 TEMP CTRL VALVE	21-62 TASK 874
PFTC	21-51267 TEMP CTRL VALVE	21-62 TASK 874
PFTC	21-51268 F/D TRIM VALVE	21-62 TASK 876
PFTC	21-51269 F/D TRIM VALVE	21-62 TASK 876
PFTC	21-51271 F/D TRIM VALVE	21-62 TASK 876
PFTC	21-51272 F/D TRIM VALVE	21-62 TASK 876
PFTC	21-51273 F/D TRIM VALVE	21-62 TASK 876
PFTC	21-51274 F/D TRIM VALVE	21-62 TASK 876
PFTC	21-51275 F/D TRIM VALVE	21-62 TASK 876
PFTC	21-51276 ZONE TRIM VALVE	21-62 TASK 877
PFTC	21-51277 ZONE TRIM VALVE	21-62 TASK 877
PFTC	21-51279 ZONE TRIM VALVE	21-62 TASK 877
PFTC	21-51280 ZONE TRIM VALVE	21-62 TASK 877
PFTC	21-51281 ZONE TRIM VALVE	21-62 TASK 877
PFTC	21-51282 ZONE TRIM VALVE	21-62 TASK 877
PFTC	21-51283 ZONE TRIM VALVE	21-62 TASK 877
PFTC	21-51300 STBY TCV VALVE	21-62 TASK 875
PFTC	21-51301 STBY TCV VALVE	21-62 TASK 875
PFTC	21-51310 F/D TEMP SENSOR	21-62 TASK 865
PFTC	21-51311 F/D SELECTOR	21-62 TASK 862
PFTC	21-51312 F/D DUCT SENSOR	21-62 TASK 866
PFTC	21-51313 PACK TEMP SENSOR	21-62 TASK 871
PFTC	21-51314 MIXMAN TEMP SNSR	21-62 TASK 873
PFTC	21-51331 DISPLAY	21-62 TASK 858
PFTC	21-51332 PFTC CNTRL INOP	21-62 TASK 858
PFTC	21-51333 PFTC CNTRL INOP	21-62 TASK 858
PFTC	21-51400 SRADA CNTRL	21-62 TASK 878
PFTC	21-51401 SRADA INLET	21-62 TASK 878
PFTC	21-51402 SRADA EXIT	21-62 TASK 879
PFTC	21-51403 SRADA RAM SNSR	21-62 TASK 870
PFTC	21-51405 SRADA INLET	21-62 TASK 878
PFTC	21-51406 SRADA EXIT	21-62 TASK 879
PFTC	21-51407 SRADA RAM SNSR	21-62 TASK 870
PFTC	21-51408 SRADA CNTRL	21-62 TASK 880

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LRU/SYSTEM	MAINTENANCE MESSAGE	GO TO FIM TASK
PFTC	21-51409 SRADA INLET	21-62 TASK 878
PFTC	21-51410 SRADA EXIT	21-62 TASK 879
PFTC	21-51411 SRADA RAM SNSR	21-62 TASK 870
PFTC	21-51500 PFTC CNTRL INOP	21-62 TASK 860
PFTC	21-51501 PFTC CNTRL INOP	21-62 TASK 857
PFTC	21-51502 PFTC CNTRL INOP	21-62 TASK 860
PFTC	21-51503 PFTC CNTRL INOP	21-62 TASK 857
PFTC	21-51504 PFTC CNTRL INOP	21-62 TASK 857
PFTC	21-51505 PFTC CNTRL INOP	21-62 TASK 876
PFTC	21-51506 PFTC CNTRL INOP	21-62 TASK 877
PFTC	21-51507 PFTC CNTRL INOP	21-62 TASK 874
PFTC	21-51508 F/D TRIM VALVE	21-62 TASK 876
PFTC	21-51509 ZONE TRIM VALVE	21-62 TASK 877
PFTC	21-51510 TEMP CTRL VALVE	21-62 TASK 874

——— END OF TASK ———

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

802. Left Pack/Zone Temperature Controller Fault - Fault Isolation

A. Description

(1) This task is for these maintenance messages:

(a) CONTROLLER FAULT - PACK/ZONE

OMR 802, 805, 807, 809, 810, 812, 813

(b) CONTROLLER FAULT - RAM AIR

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

(c) CONTROLLER FAULT - STANDBY PACK

B. Possible Causes

(1) Pack/Zone temperature controller (PZTC), M1442 (No. 1)

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

D. Related Data

(1) (SSM 21-51-31)

(2) (WDM 21-51-12)

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OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999 (Continued)

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(2) If the maintenance message does not show, then there was an intermittent fault.

(3) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Replace the left pack/zone temperature controller, M1442.

These are the tasks:

Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801,

Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.

(a) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

1) If the maintenance message does not show, then you corrected the fault.

——— END OF TASK ———

803. Right Pack/Zone Temperature Controller Fault - Fault Isolation

A. Description

(1) This task is for these maintenance messages:

(a) CONTROLLER FAULT - PACK/ZONE

OMR 802, 805, 807, 809, 810, 812, 813

(b) CONTROLLER FAULT - RAM AIR

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

(c) CONTROLLER FAULT - STANDBY PACK

B. Possible Causes

(1) Pack/Zone temperature controller (PZTC), M1443 (No. 2)

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

D. Related Data

(1) (SSM 21-51-31)

(2) (WDM 21-51-22)

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(2) If the maintenance message does not show, then there was an intermittent fault.

(3) If the maintenance message shows, then do the Fault Isolation Procedure below.

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OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999 (Continued)

F. Fault Isolation Procedure

(1) Replace the right pack/zone temperature controller, M1443.

These are the tasks:

Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801,

Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.

(a) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

1) If the maintenance message does not show, then you corrected the fault.

——— END OF TASK ———

804. Control Cabin Temperature Selector (Primary) - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) FLT DECK SELECTOR

(2) This fault is for the Pack/Zone Temperature Controller (PZTC) No. 2.

B. Possible Causes

(1) Control Cabin Temperature Selector Switch, S2

(2) Wiring

(3) Pack/Zone Temperature Controller (PZTC) No. 2, M1443

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

D. Related Data

(1) SSM 21-51-31

(2) SSM 21-61-11

(3) WDM 21-51-22

(4) WDM 21-61-11

E. Initial Evaluation

(1) Do the Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then there was an intermittent fault.

(b) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Replace the Control Cabin Temperature Selector Switch, S2. These are the tasks:

- Cabin Temperature Selector Removal, AMM TASK 21-61-07-000-803-002
- Cabin Temperature Selector Installation, AMM TASK 21-61-07-400-803-002

(a) Do the Repair Confirmation at the end of this task.

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(2) Do this check of the wiring (WDM 21-61-11):

(a) Lower the Pilot Overhead Panel, P5.

(b) Disconnect the connector from the Control Cabin Temperature Selector Switch, S2.

(c) Remove the PZTC No. 2, M1443. This is the task: Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801.

CONT CAB TEMP SEL SW, S2 S2		ZONE TEMP CONT NO 2, M1443 D3860A
pin 1	. . . . .	pin C2
pin 2	. . . . .	pin B2
pin 3	. . . . .	pin A2

(e) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-connect the connector to the Control Cabin Temperature Selector Switch, S2.

3) Re-install the PZTC No. 2, M1443. This is the task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.

4) Do the Repair Confirmation at the end of this task.

(f) If you do not find a problem with the wiring, then do this step and continue:

1) Re-connect the connector to the Control Cabin Temperature Selector Switch, S2.

(3) Replace the PZTC No. 2, M1443. These are the tasks:

- Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801
- Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801

(a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the problem.

(b) If the maintenance message still shows, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

805. Control Cabin Temperature Selector (Backup) - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) FLT DECK SELECTOR

(2) This fault is for the PZTC No. 1.

B. Possible Causes

(1) Control Cabin Temperature Selector Switch, S2

(2) Wiring

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(3) Pack/Zone Temperature Controller (PZTC) No. 1, M1442

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

D. Related Data

(1) SSM 21-51-31

(2) SSM 21-61-12

(3) WDM 21-51-12

(4) WDM 21-61-12

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then there was an intermittent fault.

(b) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Replace the Control Cabin Temperature Selector Switch, S2. These are the tasks:

- Cabin Temperature Selector Removal, AMM TASK 21-61-07-000-803-002
- Cabin Temperature Selector Installation, AMM TASK 21-61-07-400-803-002

(a) Do the Repair Confirmation at the end of this task.

(2) Do this check of the wiring (WDM 21-61-12):

(a) Lower the Pilot Overhead Panel, P5.

(b) Disconnect the connector from the Control Cabin Temperature Selector Switch, S2.

(c) Remove the PZTC No. 1, M1442. This is the task: Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801.

CONT CAB TEMP SEL SW, S2		**ZONE TEMP CONT NO 1, M1442**
S2		**D3858A**
pin 4	. . . . .	pin C2
pin 5	. . . . .	pin B2
pin 6	. . . . .	pin A2

(e) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-connect the connector to the Control Cabin Temperature Selector Switch, S2.

3) Re-install the PZTC No. 1, M1442. This is the task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.

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4) Do the Repair Confirmation at the end of this task.

(f) If you do not find a problem with the wiring, then do this step and continue:

1) Re-connect the connector to the Control Cabin Temperature Selector Switch, S2.

(3) Replace the PZTC No. 1, M1442. These are the tasks:

- Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801
- Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801

(a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the problem.

(b) If the maintenance message still shows, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

806. Forward Cabin Temperature Selector (Primary) - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) ZONE SELECTOR

(2) This fault is for the PZTC No. 2.

B. Possible Causes

(1) Forward Cabin Temperature Selector Switch, S3

(2) Wiring

(3) Pack/Zone Temperature Controller (PZTC) No. 2, M1443

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

D. Related Data

(1) SSM 21-51-31

(2) SSM 21-61-13

(3) WDM 21-51-22

(4) WDM 21-61-13

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801 (21-62 TASK SUPPORT Figure 305).

(2) Do the Pack/Zone Temperature Controller Reset Procedure as follows:

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(a) Push the VERIFY Switch on the controller.

1) If a Red Light comes ON, then do the Fault Isolation Procedure below.

2) If the Green GO Light comes ON, push the RESET Switch while the VERIFY and GO Lights are ON. No further action is necessary.

F. Fault Isolation Procedure

(1) Replace the Forward Cabin Temperature Selector Switch, S3. These are the tasks:

- Cabin Temperature Selector Removal, AMM TASK 21-61-07-000-803-002
- Cabin Temperature Selector Installation, AMM TASK 21-61-07-400-803-002

(a) Do the Repair Confirmation at the end of this task.

(2) Do this check of the wiring (WDM 21-61-13):

(a) Lower the Pilot Overhead Panel, P5.

(b) Disconnect the connector from the Forward Cabin Temperature Selector Switch, S3.

(c) Remove the PZTC No. 2, M1443. This is the task: Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801.

FWD CABIN TEMP SEL SW, S3		**ZONE TEMP CONT NO 2, M1443**
S3		**D3860A**
pin 1	. . . . .	pin C3
pin 2	. . . . .	pin B3
pin 3	. . . . .	pin A3

1) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-connect the connector to the Forward Cabin Temperature Selector Switch, S3.

c) Re-install the PZTC No. 2, M1443. This is the task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.

d) Do the Repair Confirmation at the end of this task.

2) If you do not find a problem with the wiring, then do this step and continue:

a) Re-connect the connector to the Forward Cabin Temperature Selector Switch, S3.

(3) Replace the PZTC No. 2, M1443. These are the tasks:

- Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801
- Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801

(a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the problem.

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(b) If the maintenance message still shows, then continue the Fault Isolation Procedure at the subsequent step.

END OF TASK

807. Aft Cabin Temperature Selector (Backup) - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) BKUP ZONE SELECTOR

(2) This fault is for the PZTC No. 2.

B. Possible Causes

(1) Aft Cabin Temperature Selector Switch, S4

(2) Wiring

(3) Pack/Zone Temperature Controller (PZTC) No. 2, M1443

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

D. Related Data

(1) SSM 21-51-31

(2) SSM 21-61-14

(3) WDM 21-51-22

(4) WDM 21-61-14

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then there was an intermittent fault.

(b) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Replace the Aft Cabin Temperature Selector Switch, S4. These are the tasks:

- Cabin Temperature Selector Removal, AMM TASK 21-61-07-000-803-002
- Cabin Temperature Selector Installation, AMM TASK 21-61-07-400-803-002

(a) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

1) Do the Repair Confirmation at the end of this task.

(2) Do this check of the wiring (WDM 21-61-14):

(a) Lower the Pilot Overhead Panel, P5.

(b) Disconnect the connector from the Aft Cabin Temperature Selector Switch, S4.

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(c) Remove the PZTC No. 2, M1443. This is the task: Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801.

AFT CABIN TEMP SEL SW, S4 S4		ZONE TEMP CONT NO 2, M1443 D3860A
pin 1	. . . . .	pin C11
pin 2	. . . . .	pin B11
pin 3	. . . . .	pin B12

1) If you find a problem with the wiring, then do these steps:
   a) Repair the wiring.
   b) Re-connect the connector to the Aft Cabin Temperature Selector Switch, S4.
   c) Re-install the PZTC No. 2, M1443. This is the task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.
   d) Do the Repair Confirmation at the end of this task.
2) If you do not find a problem with the wiring, then do this step and continue:
   a) Re-connect the connector to the Aft Cabin Temperature Selector Switch, S4.

(3) Replace the PZTC No. 2, M1443. These are the tasks:
- Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801
- Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801

(a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.
   (a) If the maintenance message does not show, then you corrected the problem.
   (b) If the maintenance message still shows, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

808. Aft Cabin Temperature Selector (Primary) - Fault Isolation

A. Description

(1) This task is for this maintenance message:
   (a) ZONE SELECTOR

(2) This fault is for the PZTC No. 1.

B. Possible Causes

(1) Aft Cabin Temperature Selector Switch, S4
(2) Wiring
(3) Pack/Zone Temperature Controller (PZTC) No. 1, M1442

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OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999 (Continued)

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

D. Related Data

(1) SSM 21-51-31

(2) SSM 21-61-14

(3) WDM 21-51-12

(4) WDM 21-61-14

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801 (21-62 TASK SUPPORT Figure 305).

(2) Do the Pack/Zone Temperature Controller Reset Procedure as follows:

(a) Push the VERIFY Switch on the controller.

1) If a Red Light comes ON, then do the Fault Isolation Procedure below.

2) If the Green GO Light comes ON, push the RESET Switch while the VERIFY and GO Lights are ON. No further action is necessary.

F. Fault Isolation Procedure

(1) Replace the Aft Cabin Temperature Selector Switch, S4. These are the tasks:

- Cabin Temperature Selector Removal, AMM TASK 21-61-07-000-803-002
- Cabin Temperature Selector Installation, AMM TASK 21-61-07-400-803-002

(a) Do the Repair Confirmation at the end of this task.

(2) Do this check of the wiring (WDM 21-61-14):

(a) Lower the Pilot Overhead Panel, P5.

(b) Disconnect the connector from the Aft Cabin Temperature Selector Switch, S4.

(c) Remove the PZTC No. 1, M1442. This is the task: Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801.

AFT CAB TEMP SEL SW, S4 S4		**ZONE TEMP CONT NO 1, M1442 D3858A**
pin 4	. . . . .	pin C3
pin 5	. . . . .	pin B3
pin 6	. . . . .	pin A3

1) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

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b) Re-connect the connector to the Aft Cabin Temperature Selector Switch, S4.

c) Re-install the PZTC No. 1, M1442. This is the task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.

d) Do the Repair Confirmation at the end of this task.

2) If you do not find a problem with the wiring, then do this step and continue:

a) Re-connect the connector to the Aft Cabin Temperature Selector Switch, S4.

(3) Replace the PZTC No. 1, M1442. These are the tasks:

- Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801
- Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801

(a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the problem.

(b) If the maintenance message still shows, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

809. Forward Cabin Temperature Selector (Backup) - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) BKUP ZONE SELECTOR

(2) This fault is for the PZTC No. 1.

B. Possible Causes

(1) Forward Cabin Temperature Selector Switch, S3

(2) Wiring

(3) Pack/Zone Temperature Controller (PZTC) No. 1, M1442

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

D. Related Data

(1) SSM 21-51-31

(2) SSM 21-61-13

(3) WDM 21-51-12

(4) WDM 21-61-13

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OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999 (Continued)

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then there was an intermittent fault.

(b) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Replace the Forward Cabin Temperature Selector Switch, S3. These are the tasks:

- Cabin Temperature Selector Removal, AMM TASK 21-61-07-000-803-002
- Cabin Temperature Selector Installation, AMM TASK 21-61-07-400-803-002

(a) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

1) Do the Repair Confirmation at the end of this task.

(2) Do this check of the wiring (WDM 21-61-13):

(a) Lower the Pilot Overhead Panel, P5.

(b) Disconnect the connector from the Forward Cabin Temperature Selector Switch, S3.

(c) Remove the PZTC No. 1, M1442. This is the task: Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801.

FWD CABIN TEMP SEL SW, S3		**ZONE TEMP CONT NO 1, M1442**
S3		**D3858A**
pin 4	. . . . .	pin C11
pin 5	. . . . .	pin B11
pin 6	. . . . .	pin B12

1) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-connect the connector to the Forward Cabin Temperature Selector Switch, S3.

c) Re-install the PZTC No. 1, M1442. This is the task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.

d) Do the Repair Confirmation at the end of this task.

2) If you do not find a problem with the wiring, then do this step and continue:

a) Re-connect the connector to the Forward Cabin Temperature Selector Switch, S3.

(3) Replace the PZTC No. 1, M1442. These are the tasks:

- Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801
- Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801

(a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the problem.

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(b) If the maintenance message still shows, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

810. Flight Deck Cabin Temperature Sensor (Primary) - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) FLT DECK CABIN SENSOR

(2) The right pack/zone temperature controller has detected a short or an open circuit from the control cabin temperature sensor.

B. Possible Causes

(1) Flight deck cabin temperature sensor, M1718

(2) Wiring

(3) Pack/Zone temperature controller (PZTC), M1443 (No. 2).

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

D. Related Data

(1) (SSM 21-61-11)

(2) (WDM 21-61-11)

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(2) If the maintenance message does not show, then there was an intermittent fault.

(3) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Do this check of the control cabin temperature sensor, M1718:

(a) Lower the access panel for the control cabin temperature sensor that is aft of the overhead P5 panel.

(b) Disconnect electrical connector D10958 from the control cabin temperature sensor.

(c) Do a check for an open circuit or a short between pins 1 and 2 of the control cabin temperature sensor.

(d) If there is a short or an open circuit between pins 1 and 2, then do these steps:

1) Replace the control cabin temperature sensor, M1718.

These are the tasks:

Cabin Temperature Sensor Removal, AMM TASK 21-61-10-000-802,

Cabin Temperature Sensor Installation, AMM TASK 21-61-10-400-802.

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2) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

3) If the maintenance message does not show, then you corrected the fault.

(e) If there is not a short or an open circuit between pins 1 and 2, then continue.

(2) Do this check of the wiring between the control cabin temperature sensor and the right pack/zone temperature controller, M1443:

(a) Remove the right pack/zone temperature controller. To remove the controller, do this task: Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801.

(b) Do a wiring check between these pins of electrical connector D10958 of the control cabin temperature sensor and connector D3860A of the pack/zone temperature controller:

D10958		D3860A
pin 1	. . . . .	pin C4
pin 2	. . . . .	pin C5

(c) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-connect the electrical connector to the control cabin temperature sensor.

3) Re-install the right pack/zone temperature controller. To install the controller, do this task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.

4) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

5) If the maintenance message does not show, then you corrected the fault.

(d) If you do not find a problem with the wiring, then continue.

1) Re-connect the electrical connector to the control cabin temperature sensor.

(3) Replace the right pack/zone temperature controller, M1443. These are the tasks:

- Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801
- Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801

(a) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

1) If the maintenance message does not show, then you corrected the fault.

——— END OF TASK ———

811. Flight Deck Cabin Temperature Sensor (Backup) - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) FLT DECK CABIN SENSOR

(2) The left pack/zone temperature controller has detected a short or an open circuit from the control cabin temperature sensor.

B. Possible Causes

(1) Flight deck cabin temperature sensor, M1718

(2) Wiring

(3) Pack/Zone temperature controller (PZTC), M1442 (No. 1).

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OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999 (Continued)

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

D. Related Data

(1) (SSM 21-61-12)

(2) (WDM 21-61-12)

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(2) If the maintenance message does not show, then there was an intermittent fault.

(3) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Do this check of the control cabin temperature sensor, M1718:

(a) Lower the access panel for the control cabin temperature sensor that is aft of the overhead P5 panel.

(b) Disconnect electrical connector D10958 from the control cabin temperature sensor.

(c) Do a check for an open circuit or a short between pins 3 and 4 of the control cabin temperature sensor.

(d) If there is a short or an open circuit between pins 3 and 4, then do these steps:

1) Replace the control cabin temperature sensor, M1718.

These are the tasks:

Cabin Temperature Sensor Removal, AMM TASK 21-61-10-000-802,

Cabin Temperature Sensor Installation, AMM TASK 21-61-10-400-802.

2) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

3) If the maintenance message does not show, then you corrected the fault.

(e) If there is not a short or an open circuit between pins 3 and 4, then continue.

(2) Do this check of the wiring between the control cabin temperature sensor and the left pack/zone temperature controller, M1442:

(a) Remove the left pack/zone temperature controller. To remove the controller, do this task: Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801.

(b) Do a wiring check between these pins of electrical connector D10958 of the control cabin temperature sensor and connector D3858A of the pack/zone temperature controller:

D10958		D3858A
pin 3	. . . . .	pin C4
pin 4	. . . . .	pin C5

(c) If you find a problem with the wiring, then do these steps:

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1) Repair the wiring.
2) Re-connect the electrical connector to the control cabin temperature sensor.
3) Re-install the left pack/zone temperature controller. To install the controller, do this task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.
4) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.
5) If the maintenance message does not show, then you corrected the fault.

(d) If you do not find a problem with the wiring, then continue.
1) Re-connect the electrical connector to the control cabin temperature sensor.

(3) Replace the left pack/zone temperature controller, M1442. These are the tasks:
- Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801
- Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801

(a) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.
1) If the maintenance message does not show, then you corrected the fault.

END OF TASK

812. Flight Deck Duct Temperature Sensor (Primary) - Fault Isolation

A. Description

(1) This task is for this maintenance message:
(a) FLT DECK DUCT SENSOR

(2) The right pack/zone temperature controller has detected a short or an open circuit from the control cabin duct temperature sensor.

B. Possible Causes

(1) Control cabin duct temperature sensor, T478
(2) Wiring
(3) Pack/Zone temperature controller (PZTC), M1443 (No. 2).

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

D. Related Data

(1) (SSM 21-61-11)
(2) (WDM 21-61-11)

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.
(2) If the maintenance message does not show, then there was an intermittent fault.
(3) If the maintenance message shows, then do the Fault Isolation Procedure below.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Fault Isolation Procedure

(1) Do this check of the control cabin duct temperature sensor, T478:

(a) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

(b) Disconnect electrical connector D3884 from the control cabin duct temperature sensor.

(c) Do a check for an open circuit or a short between pins 1 and 2 of the control cabin duct temperature sensor.

(d) If there is a short or an open circuit between pins 1 and 2, then do these steps:

1) Replace the control cabin duct temperature sensor (Primary), T478.

These are the tasks:

Duct Temperature Limit Sensor Removal, AMM TASK 21-61-04-000-803-002,

Duct Temperature Limit Sensor Installation, AMM TASK 21-61-04-400-803-002.

2) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

3) If the maintenance message does not show, then you corrected the fault.

(e) If there is not a short or an open circuit between pins 1 and 2, then continue.

(2) Do this check of the wiring between the control cabin temperature sensor and the right pack/zone temperature controller, M1443:

(a) Remove the right pack/zone temperature controller. To remove the controller, do this task: Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801.

(b) Do a wiring check between these pins of electrical connector D3884 of the control cabin duct temperature sensor and connector D3860A of the pack/zone temperature controller:

D3884		D3860A
pin 1	. . . . .	pin B5
pin 2	. . . . .	pin A5

(c) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-connect the electrical connector to the control cabin temperature sensor.

3) Re-install the right pack/zone temperature controller. To install the controller, do this task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.

4) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

5) If the maintenance message does not show, then you corrected the fault.

(d) If you do not find a problem with the wiring, then continue.

1) Re-connect the electrical connector to the control cabin temperature sensor.

(3) Replace the right pack/zone temperature controller, M1443. These are the tasks:

- Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801
- Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

1) If the maintenance message does not show, then you corrected the fault.

——— END OF TASK ———

813. Flight Deck Duct Temperature Sensor (Backup) - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) FLT DECK DUCT SENSOR

(2) The left pack/zone temperature controller has detected a short or an open circuit from the control cabin duct temperature sensor.

B. Possible Causes

(1) Control cabin duct temperature sensor, T477

(2) Wiring

(3) PZTC, M1442 (No. 1).

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

D. Related Data

(1) (SSM 21-61-12)

(2) (WDM 21-61-12)

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(2) If the maintenance message does not show, then there was an intermittent fault.

(3) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Do this check of the control cabin duct temperature sensor, T477:

(a) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

(b) Disconnect electrical connector D3882 from the control cabin duct temperature sensor.

(c) Do a check for an open circuit or a short between pins 1 and 2 of the control cabin duct temperature sensor.

(d) If there is a short or an open circuit between pins 1 and 2, then do these steps:

1) Replace the control cabin duct temperature sensor (Backup), T477.

These are the tasks:

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OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999 (Continued)

Duct Temperature Limit Sensor Removal, AMM TASK 21-61-04-000-803-002,
Duct Temperature Limit Sensor Installation, AMM TASK 21-61-04-400-803-002.

2) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

3) If the maintenance message does not show, then you corrected the fault.

(e) If there is not a short or an open circuit between pins 1 and 2, then continue.

(2) Do this check of the wiring between the control cabin temperature sensor and the left pack/zone temperature controller, M1442:

(a) Remove the left pack/zone temperature controller. To remove the controller, do this task: Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801.

(b) Do a wiring check between these pins of electrical connector D3882 of the control cabin duct temperature sensor and connector D3858A of the pack/zone temperature controller:

D3882		D3858A
pin 1	. . . . .	pin B5
pin 2	. . . . .	pin A5

(c) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-connect the electrical connector to the control cabin temperature sensor.

3) Re-install the left pack/zone temperature controller. To install the controller, do this task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.

4) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

5) If the maintenance message does not show, then you corrected the fault.

(d) If you do not find a problem with the wiring, then continue.

1) Re-connect the electrical connector to the control cabin temperature sensor.

(3) Replace the left pack/zone temperature controller, M1442. These are the tasks:

- Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801
- Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801

(a) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

1) If the maintenance message does not show, then you corrected the fault.

——— END OF TASK ———

814. ZONE CABIN SENSOR (Primary-Right PZTC) - Fault Isolation

A. Description

(1) This task is for the ZONE CABIN SENSOR maintenance message reported by the Right Pack/Zone Temperature Controller.

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

(a) The Right Zone/Pack Temperature Controller has detected a Short or an Open Circuit from the Primary Forward Cabin Temperature Sensor.

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OMR 807

(b) The Right Zone/Pack Temperature Controller has detected a Short or an Open Circuit from the Primary Supernumerary Temperature Sensor.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

B. Possible Causes

OMR 802, 805, 807, 809, 810, 812, 813

(1) Forward Cabin Temperature Sensor Module, M1445

OMR 815, 816, 818, 819, 901-999

(2) Forward Cabin Temperature Sensor Module, M2847

OMR 807

(3) Supernumerary Temperature Sensor Module, M3283

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

(4) Wiring

(5) Right Pack/Zone Temperature Controller (PZTC), M1443

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

D. Related Data

(1) WDM 21-61-13

(2) SSM 21-61-13

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then there was an intermittent fault.

(b) If the maintenance message shows, then do the Fault Isolation Procedure below.

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

F. Fault Isolation Procedure

OMR 802, 805, 809, 810, 812, 813

(1) Do this check of the Forward Cabin Temperature Sensor Module, M1445:

(a) Lower the Passenger Service Unit (PSU) that is adjacent to the Forward Cabin Zone Temperature Sensor Module.

(b) Remove the trim panel that is adjacent to the PSU and beneath the Electrical Connector for the Zone Temperature Sensor Module.

(c) Disconnect Electrical Connector, D3908 from the Forward Cabin Temperature Sensor Module (WDM 21-61-13).

(d) Do a check for an Open Circuit or a Short between pins 1 and 2 of the Forward Cabin Temperature Sensor Module.

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OMR 802, 805, 809, 810, 812, 813 (Continued)

(e) If there is a Short or an Open Circuit between pins 1 and 2, then do these steps:

1) Replace the Forward Cabin Temperature Sensor Module, M1445. These are the tasks:

- Cabin Temperature Sensor Removal, AMM TASK 21-61-10-000-802
- Cabin Temperature Sensor Installation, AMM TASK 21-61-10-400-802

2) Do the Repair Confirmation at the end of this task.

(f) If there is not a Short or an Open Circuit between pins 1 and 2, then continue.

(2) Do this check of the wiring.

(a) Remove the Right Pack/Zone Temperature Controller, M1443. This is the task: Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801.

(b) Examine the wiring between the Forward Cabin Temperature Sensor Module, M1445, and the Pack/Zone Temperature Controller, M1443, as follows (WDM 21-61-13):

TEMP SNSR MODULE D3906		**RIGHT PZTC D3860A**
pin 1	. . . . .	pin B4
pin 2	. . . . .	pin A4

(c) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-connect the Electrical Connector to the Forward Cabin Temperature Sensor Module.

3) Re-install the Right Zone Temperature Controller, M1443. This is the task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.

4) Do the Repair Confirmation at the end of this task.

(d) If you do not find a problem with the wiring, then continue.

1) Re-connect the Electrical Connector to the Forward Cabin Temperature Sensor Module.

2) Install the trim panel adjacent to the PSU.

3) Close (lift) the PSU.

(3) Replace the Right Pack/Zone Temperature Controller, M1443. These are the tasks:

- Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801
- Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801

(a) Do the Repair Confirmation at the end of this task.

OMR 815, 816, 818, 819, 901-999

(4) Do this check of the Forward Cabin Temperature Sensor Module, M2847:

(a) Lower the PSU that is adjacent to the Forward Cabin Zone Temperature Sensor Module.

(b) Remove the trim panel that is adjacent to the PSU and beneath the Electrical Connector for the Zone Temperature Sensor Module.

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OMR 815, 816, 818, 819, 901-999 (Continued)

(c) Disconnect Electrical Connector, D15016 from the Forward Cabin Temperature Sensor Module (WDM 21-61-13).

(d) Do a check for an Open Circuit or a Short between pins 1 and 2 of the Forward Cabin Temperature Sensor Module.

(e) If there is a Short or an Open Circuit between pins 1 and 2, then do these steps:

1) Replace the Forward Cabin Temperature Sensor Module, M2847. These are the tasks:

- Cabin Temperature Sensor Removal, AMM TASK 21-61-10-000-802
- Cabin Temperature Sensor Installation, AMM TASK 21-61-10-400-802

a) Do the Repair Confirmation at the end of this task.

(f) If there is not a Short or an Open Circuit between pins 1 and 2, then continue.

(5) Do this check of the wiring.

(a) Remove the Right Pack/Zone Temperature Controller. This is the task: Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801.

(b) Examine the wiring between the Forward Cabin Temperature Sensor Module, M2847 and the Right Pack Temperature Controller Connector, M1443 as follows (WDM 21-61-13):

TEMP SNSR MODULE D15016	**RIGHT PZTC D3860A**
pin 1	pin B4
pin 2	pin A4

(c) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-connect the Electrical Connector to the Forward Cabin Temperature Sensor Module.

3) Re-install the Right Pack Temperature Controller, M1443. This is the task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.

4) Do the Repair Confirmation at the end of this task.

(d) If you do not find a problem with the wiring, then continue.

1) Re-connect the Electrical Connector to the Forward Cabin Temperature Sensor Module.

2) Install the trim panel adjacent to the PSU.

3) Close (lift) the PSU.

(6) Replace the Right Zone Temperature Controller, M1443. These are the tasks:

- Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801
- Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801

(a) Do the Repair Confirmation at the end of this task.

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OMR 807

G. Fault Isolation Procedure

(1) Do this check of the Supernumerary Temperature Sensor Module, M3283:

(a) Lower the PSU that is adjacent to the Supernumerary Temperature Sensor Module.

(b) Remove the trim panel that is adjacent to the PSU and beneath the Electrical Connector for the Zone Temperature Sensor Module.

(c) Disconnect Electrical Connector, D16874 from the Supernumerary Temperature Module.

(d) Do a check for an Open Circuit or a Short between pins 1 and 2 of the Supernumerary Temperature Module.

(e) If there is a Short or an Open Circuit between pins 1 and 2, then do these steps:

1) Replace the Supernumerary Temperature Sensor Module, M3283. These are the tasks:

- Cabin Temperature Sensor Removal, AMM TASK 21-61-10-000-802
- Cabin Temperature Sensor Installation, AMM TASK 21-61-10-400-802

2) Do the Repair Confirmation at the end of this task.

(f) If there is not a Short or an Open Circuit between pins 1 and 2, then continue.

(2) Do this check of the wiring.

(a) Remove the Right Pack/Zone Temperature Controller, M1443. This is the task: Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801.

(b) Do a Wiring Check between the Supernumerary Temperature Sensor Module, M3283, and the Right Pack/Zone Temperature Controller, M1443 as follows (WDM 21-61-13):

TEMP SNSR MODULE D16874	**RIGHT PZTC D3860A**
pin 1	pin B4
pin 2	pin A4

(c) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-connect the Electrical Connector to the Supernumerary Temperature Sensor Module.

3) Re-install the Right Pack/Zone Temperature Controller, M1443. This is the task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.

4) Do the Repair Confirmation at the end of this task.

(d) If you do not find a problem with the wiring, then continue.

1) Re-connect the Electrical Connector to the Supernumerary Temperature Sensor Module.

2) Install the trim panel adjacent to the PSU.

3) Close (lift) the PSU.

(3) Replace the Right Pack/Zone Temperature Controller, M1443. These are the tasks:

- Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801
- Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801

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OMR 807 (Continued)

(a) Do the Repair Confirmation at the end of this task.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

H. Repair Confirmation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the problem.

(b) If the maintenance message still shows, continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

815. BKUP ZONE CABIN SENSOR (Backup-Left PZTC) - Fault Isolation

A. Description

(1) This task is for the BKUP ZONE CABIN SENSOR maintenance message reported by the Left Pack/Zone Temperature Controller.

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

(a) The Left Pack/Zone Temperature Controller has detected a Short or an Open Circuit from the Backup Forward Cabin Temperature Sensor.

OMR 807

(b) The Left Pack/Zone Temperature Controller has detected a Short or an Open Circuit from the Backup Supernumerary Temperature Sensor.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

B. Possible Causes

OMR 802, 805, 807, 809, 810, 812, 813

(1) Forward Cabin Temperature Sensor Module, M1445

OMR 815, 816, 818, 819, 901-999

(2) Forward Cabin Temperature Sensor Module, M2847

OMR 807

(3) Supernumerary Temperature Sensor Module, M3283

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

(4) Wiring

(5) Left Pack/Zone Temperature Controller (PZTC), M1442

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

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OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999 (Continued)

D. Related Data

(1) WDM 21-61-13

(2) SSM 21-61-13

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then there was an intermittent fault.

(b) If the maintenance message shows, then do the Fault Isolation Procedure below.

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

F. Fault Isolation Procedure

OMR 802, 805, 809, 810, 812, 813

(1) Do this check of the Forward Cabin Temperature Sensor Module, M1445:

(a) Lower the PSU that is adjacent to the Forward Cabin Zone Temperature Sensor Module.

(b) Remove the trim panel that is adjacent to the PSU and beneath the Electrical Connector for the Zone Temperature Sensor Module.

(c) Disconnect Electrical Connector, D3908 from the Forward Cabin Temperature Sensor Module (WDM 21-61-13).

(d) Do a check for an Open Circuit or a Short between pins 3 and 4 of the Forward Cabin Temperature Sensor Module.

(e) If there is a Short or an Open Circuit between pins 3 and 4, then do these steps:

1) Replace the Forward Cabin Temperature Sensor Module, M1445. These are the tasks:

- Cabin Temperature Sensor Removal, AMM TASK 21-61-10-000-802
- Cabin Temperature Sensor Installation, AMM TASK 21-61-10-400-802

2) Do the Repair Confirmation at the end of this task.

(f) If there is not a Short or an Open Circuit between pins 3 and 4, then continue.

(2) Do this check of the wiring.

(a) Remove the Left Pack/Zone Temperature Controller, M1442. This is the task: Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801.

(b) Examine the wiring between the Forward Cabin Temperature Sensor Module, M1445, and the Left Pack/Zone Temperature Controller, M1442, as follows (WDM 21-61-13):

TEMP SNSR MODULE D3906		LEFT PZTC D3858A
pin 3	. . . . .	pin D14
pin 4	. . . . .	pin D13

(c) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-connect the Electrical Connector to the Forward Cabin Temperature Sensor Module.

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OMR 802, 805, 809, 810, 812, 813 (Continued)

3) Re-install the Left Zone Temperature Controller, M1442. This is the task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.

4) Do the Repair Confirmation at the end of this task.

(d) If you do not find a problem with the wiring, then continue.

1) Re-connect the Electrical Connector to the Forward Cabin Temperature Sensor Module.

2) Install the trim panel adjacent to the PSU.

3) Close (lift) the PSU.

(3) Replace the Left Pack/Zone Temperature Controller, M1442. These are the tasks:

- Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801
- Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801

(a) Do the Repair Confirmation at the end of this task.

OMR 815, 816, 818, 819, 901-999

(4) Do this check of the Forward Cabin Temperature Sensor Module, M2847:

(a) Lower the PSU that is adjacent to the Forward Cabin Zone Temperature Sensor Module.

(b) Remove the trim panel that is adjacent to the PSU and beneath the Electrical Connector for the Zone Temperature Sensor Module.

(c) Disconnect Electrical Connector, D15016 from the Forward Cabin Temperature Sensor Module (WDM 21-61-13).

(d) Do a check for an Open Circuit or a Short between pins 3 and 4 of the Forward Cabin Temperature Sensor Module.

(e) If there is a Short or an Open Circuit between pins 3 and 4, then do these steps:

1) Replace the Forward Cabin Temperature Sensor Module, M2847. These are the tasks:

- Cabin Temperature Sensor Removal, AMM TASK 21-61-10-000-802
- Cabin Temperature Sensor Installation, AMM TASK 21-61-10-400-802

a) Do the Repair Confirmation at the end of this task.

(f) If there is not a Short or an Open Circuit between pins 3 and 4, then continue.

(5) Do this check of the wiring.

(a) Remove the Left Pack/Zone Temperature Controller. This is the task: Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801.

(b) Examine the wiring between the Forward Cabin Temperature Sensor Module, M2847 and the Left Pack Temperature Controller Connector, M1442 as follows (WDM 21-61-13):

TEMP SNSR MODULE D15016	**LEFT PZTC D3858A**
pin 3	pin D14
pin 4	pin D13

(c) If you find a problem with the wiring, then do these steps:

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OMR 815, 816, 818, 819, 901-999 (Continued)

1) Repair the wiring.
2) Re-connect the Electrical Connector to the Forward Cabin Temperature Sensor Module.
3) Re-install the Left Pack Temperature Controller, M1442. This is the task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.
4) Do the Repair Confirmation at the end of this task.

(d) If you do not find a problem with the wiring, then continue.

1) Re-connect the Electrical Connector to the Forward Cabin Temperature Sensor Module.
2) Install the trim panel adjacent to the PSU.
3) Close (lift) the PSU.

(6) Replace the Left Zone Temperature Controller, M1442. These are the tasks:

- Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801
- Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801

(a) Do the Repair Confirmation at the end of this task.

OMR 807

G. Fault Isolation Procedure

(1) Do this check of the Supernumerary Temperature Sensor Module, M3283:

(a) Lower the PSU that is adjacent to the Supernumerary Temperature Sensor Module.

(b) Remove the trim panel that is adjacent to the PSU and beneath the Electrical Connector for the Zone Temperature Sensor Module.

(c) Disconnect Electrical Connector, D16874 from the Supernumerary Temperature Sensor Module.

(d) Do a check for an Open Circuit or a Short between pins 3 and 4 of the Supernumerary Temperature Sensor Module.

(e) If there is a Short or an Open Circuit between pins 3 and 4, then do these steps:

1) Replace the Supernumerary Temperature Sensor Module, M3283. These are the tasks:

- Cabin Temperature Sensor Removal, AMM TASK 21-61-10-000-802
- Cabin Temperature Sensor Installation, AMM TASK 21-61-10-400-802

2) Do the Repair Confirmation at the end of this task.

(f) If there is not a Short or an Open Circuit between pins 3 and 4, then continue.

(2) Do this check of the wiring.

(a) Remove the Left Pack/Zone Temperature Controller, M1443. This is the task: Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801.

(b) Do a Wiring Check between the Supernumerary Temperature Sensor Module, M3283, and the Left Pack/Zone Temperature Controller, M1443 as follows (WDM 21-61-13):

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OMR 807 (Continued)

TEMP SNSR MODULE D16874	LEFT PZTC D3858A
pin 3	pin D14
pin 4	pin D13

(c) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-connect the Electrical Connector to the Supernumerary Temperature Sensor Module.

3) Re-install the Left Pack/Zone Temperature Controller, M1443. This is the task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.

4) Do the Repair Confirmation at the end of this task.

(d) If you do not find a problem with the wiring, then continue.

1) Re-connect the Electrical Connector to the Supernumerary Temperature Sensor Module.

2) Install the trim panel adjacent to the PSU.

3) Close (lift) the PSU.

(3) Replace the Left Pack/Zone Temperature Controller, M1442. These are the tasks:

- Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801
- Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801

(a) Do the Repair Confirmation at the end of this task.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

H. Repair Confirmation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the problem.

(b) If the maintenance message still shows, continue the Fault Isolation Procedure at the subsequent step.

END OF TASK

816. BKUP ZONE CABIN SENSOR (Backup-Right PZTC) - Fault Isolation

A. Description

(1) This task is for the BKUP ZONE CABIN SENSOR maintenance message reported by the Right Pack/Zone Temperature Controller.

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

(a) The Right Pack/Zone Temperature Controller has detected a Short or an Open Circuit from the Backup Aft Cabin Temperature Sensor.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 807

(b) The Right Zone/Pack Temperature Controller has detected a Short or an Open Circuit from the Backup Main Cargo Temperature Sensor.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

B. Possible Causes

OMR 802, 805, 809, 810, 812, 813

(1) Aft Cabin Temperature Sensor Module, M1446

OMR 901-999

(2) Aft Cabin Temperature Sensor Module, M2848

OMR 807

(3) Main Cargo Temperature Sensor Module, M3282

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

(4) Wiring

(5) Right Pack/Zone Temperature Controller (PZTC), M1443

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

D. Related Data

(1) WDM 21-61-14

(2) SSM 21-61-14

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then there was an intermittent fault.

(b) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

OMR 802, 805, 809, 810, 812, 813

(1) Do this check of the Aft Cabin Temperature Sensor Module, M1446:

(a) Lower the PSU that is adjacent to the Aft Cabin Zone Temperature Sensor Module.

(b) Remove the trim panel that is adjacent to the PSU and beneath the Electrical Connector for the Aft Cabin Temperature Sensor Module.

(c) Disconnect Electrical Connector, D3910 from the Aft Cabin Temperature Sensor Module (WDM 21-61-14).

(d) Do a check for an Open Circuit or a Short between pins 3 and 4 of the Aft Cabin Temperature Sensor Module.

(e) If there is a Short or an Open Circuit between pins 3 and 4, then do these steps:

1) Replace the Aft Cabin Temperature Sensor Module, M1446. These are the tasks:

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OMR 802, 805, 809, 810, 812, 813 (Continued)

- Cabin Temperature Sensor Removal, AMM TASK 21-61-10-000-802
- Cabin Temperature Sensor Installation, AMM TASK 21-61-10-400-802

2) Do the Repair Confirmation at the end of this task.

(f) If there is not a Short or an Open Circuit between pins 3 and 4, then continue.

(2) Do this check of the wiring.

(a) Remove the Right Pack/Zone Temperature Controller, M1443. This is the task: Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801.

(b) Examine the wiring between the Aft Cabin Temperature Sensor Module, M1446, and the Right Pack/Zone Temperature Controller, M1443, as follows (WDM 21-61-14):

TEMP SNSR MODULE D3910		RIGHT PZTC D3860A
pin 3	. . . . .	pin D14
pin 4	. . . . .	pin D13

(c) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-connect the Electrical Connector to the Aft Cabin Temperature Sensor Module.

3) Re-install the Right Zone Temperature Controller, M1443. This is the task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.

4) Do the Repair Confirmation at the end of this task.

(d) If you do not find a problem with the wiring, then continue.

1) Re-connect the Electrical Connector to the Aft Cabin Temperature Sensor Module.

2) Install the trim panel adjacent to the PSU.

3) Close (lift) the PSU.

(3) Replace the Right Pack/Zone Temperature Controller, M1443. These are the tasks:

- Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801
- Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801

(a) Do the Repair Confirmation at the end of this task.

OMR 901-999

(4) Do this check of the Aft Cabin Temperature Sensor Module, M2848:

(a) Lower the PSU that is adjacent to the Aft Cabin Zone Temperature Sensor Module.

(b) Remove the trim panel that is adjacent to the PSU and beneath the Electrical Connector for the Aft Zone Temperature Sensor Module.

(c) Disconnect Electrical Connector, D15022 from the Aft Cabin Temperature Sensor Module (WDM 21-61-14).

(d) Do a check for an Open Circuit or a Short between pins 3 and 4 of the Aft Cabin Temperature Sensor Module.

(e) If there is a Short or an Open Circuit between pins 3 and 4, then do these steps:

1) Replace the Aft Cabin Temperature Sensor Module, M2848. These are the tasks:

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OMR 901-999 (Continued)

- Cabin Temperature Sensor Removal, AMM TASK 21-61-10-000-802
- Cabin Temperature Sensor Installation, AMM TASK 21-61-10-400-802

a) Do the Repair Confirmation at the end of this task.

(f) If there is not a Short or an Open Circuit between pins 3 and 4, then continue.

(5) Do this check of the wiring.

(a) Remove the Right Pack/Zone Temperature Controller. This is the task: Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801.

(b) Examine the wiring between the Aft Cabin Temperature Sensor Module, M2848 and the Right Pack Temperature Controller, M1443 as follows (WDM 21-61-14):

TEMP SNSR MODULE D15022	**RIGHT PZTC D3860A**
pin 3	pin D14
pin 4	pin D13

(c) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.
2) Re-connect the Electrical Connector to the Aft Cabin Temperature Sensor Module.
3) Re-install the Right Pack Temperature Controller, M1443. This is the task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.
4) Do the Repair Confirmation at the end of this task.

(d) If you do not find a problem with the wiring, then continue.

1) Re-connect the Electrical Connector to the Aft Cabin Temperature Sensor Module.
2) Install the trim panel adjacent to the PSU.
3) Close (lift) the PSU.

(6) Replace the Right Zone Temperature Controller, M1443. These are the tasks:

- Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801
- Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801

(a) Do the Repair Confirmation at the end of this task.

OMR 807

G. Fault Isolation Procedure

(1) Do this check of the Main Cargo Temperature Sensor Module, M3282:

(a) Lower the PSU that is adjacent to the Main Cargo Temperature Sensor Module.

(b) Remove the trim panel that is adjacent to the PSU and beneath the Electrical Connector for the Zone Temperature Sensor Module.

(c) Disconnect Electrical Connector, D16870 from the Main Cargo Temperature Module (WDM 21-61-14).

(d) Do a check for an Open Circuit or a Short between pins 3 and 4 of the Main Cargo Temperature Module.

(e) If there is a Short or an Open Circuit between pins 3 and 4, then do these steps:

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OMR 807 (Continued)

1) Replace the Main Cargo Temperature Sensor Module, M3282. These are the tasks:
   - Cabin Temperature Sensor Removal, AMM TASK 21-61-10-000-802
   - Cabin Temperature Sensor Installation, AMM TASK 21-61-10-400-802
2) Do the Repair Confirmation at the end of this task.

(f) If there is not a Short or an Open Circuit between pins 3 and 4, then continue.

(2) Do this check of the wiring.

(a) Remove the Right Pack/Zone Temperature Controller, M1443. This is the task: Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801.

(b) Do a Wiring Check between the Main Cargo Temperature Sensor Module, M3282, and the Right Pack/Zone Temperature Controller, M1443 as follows (WDM 21-61-14):

TEMP SNSR MODULE D16870	RIGHT PZTC D3860A
pin 3	pin D14
pin 4	pin D13

(c) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.
2) Re-connect the Electrical Connector to the Main Cargo Temperature Sensor Module.
3) Re-install the Right Pack/Zone Temperature Controller, M1443. This is the task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.
4) Do the Repair Confirmation at the end of this task.

(d) If you do not find a problem with the wiring, then continue.

1) Re-connect the Electrical Connector to the Main Cargo Temperature Sensor Module.
2) Install the trim panel adjacent to the PSU.
3) Close (lift) the PSU.

(3) Replace the Right Pack/Zone Temperature Controller, M1443. These are the tasks:
- Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801
- Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801

(a) Do the Repair Confirmation at the end of this task.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

H. Repair Confirmation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the problem.

(b) If the maintenance message still shows, continue the Fault Isolation Procedure at the subsequent step.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

817. ZONE CABIN SENSOR (Primary-Left PZTC) - Fault Isolation

A. Description

(1) This task is for the ZONE CABIN SENSOR maintenance message reported by the Left Pack/Zone Temperature Controller.

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

(a) The Left Pack/Zone Temperature Controller has detected a Short or an Open Circuit from the Primary Aft Cabin Temperature Sensor.

OMR 807

(b) The Left Zone/Pack Temperature Controller has detected a Short or an Open Circuit from the Primary Main Cargo Temperature Sensor.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

B. Possible Causes

OMR 802, 805, 807, 809, 810, 812, 813

(1) Aft Cabin Temperature Sensor Module, M1446

OMR 815, 816, 818, 819, 901-999

(2) Aft Cabin Temperature Sensor Module, M2848

OMR 807

(3) Main Cargo Temperature Sensor Module, M3282

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

(4) Wiring

(5) Left Pack/Zone Temperature Controller (PZTC), M1442

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

D. Related Data

(1) WDM 21-61-14

(2) SSM 21-61-14

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then there was an intermittent fault.

(b) If the maintenance message shows, then do the Fault Isolation Procedure below.

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OMR 802, 805, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

F. Fault Isolation Procedure

OMR 802, 805, 809, 810, 812, 813

(1) Do this check of the Aft Cabin Temperature Sensor Module, M1446:

(a) Lower the PSU that is adjacent to the Aft Cabin Zone Temperature Sensor Module.

(b) Remove the trim panel that is adjacent to the PSU and beneath the Electrical Connector for the Aft Cabin Temperature Sensor Module.

(c) Disconnect Electrical Connector, D3910 from the Aft Cabin Temperature Sensor Module (WDM 21-61-14).

(d) Do a check for an Open Circuit or a Short between pins 1 and 2 of the Aft Cabin Temperature Sensor Module.

(e) If there is a Short or an Open Circuit between pins 1 and 2, then do these steps:

1) Replace the Aft Cabin Temperature Sensor Module, M1446. These are the tasks:

- Cabin Temperature Sensor Removal, AMM TASK 21-61-10-000-802
- Cabin Temperature Sensor Installation, AMM TASK 21-61-10-400-802

2) Do the Repair Confirmation at the end of this task.

(f) If there is not a Short or an Open Circuit between pins 1 and 2, then continue.

(2) Do this check of the wiring.

(a) Remove the Left Pack/Zone Temperature Controller, M1442. This is the task: Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801.

(b) Examine the wiring between the Aft Cabin Temperature Sensor Module, M1446, and the Left Pack/Zone Temperature Controller, M1442, as follows (WDM 21-61-14):

TEMP SNSR MODULE D3910		LEFT PZTC D3858A
pin 1	. . . . .	pin A4
pin 2	. . . . .	pin B4

(c) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-connect the Electrical Connector to the Aft Cabin Temperature Sensor Module.

3) Re-install the Left Zone Temperature Controller, M1442. This is the task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.

4) Do the Repair Confirmation at the end of this task.

(d) If you do not find a problem with the wiring, then continue.

1) Re-connect the Electrical Connector to the Aft Cabin Temperature Sensor Module.

2) Install the trim panel adjacent to the PSU.

3) Close (lift) the PSU.

(3) Replace the Left Pack/Zone Temperature Controller, M1442. These are the tasks:

- Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801
- Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801

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OMR 802, 805, 809, 810, 812, 813 (Continued)

(a) Do the Repair Confirmation at the end of this task.

OMR 815, 816, 818, 819, 901-999

(4) Do this check of the Aft Cabin Temperature Sensor Module, M2848:

(a) Lower the PSU that is adjacent to the Aft Cabin Zone Temperature Sensor Module.

(b) Remove the trim panel that is adjacent to the PSU and beneath the Electrical Connector for the Aft Zone Temperature Sensor Module.

(c) Disconnect Electrical Connector, D15022 from the Aft Cabin Temperature Sensor Module (WDM 21-61-14).

(d) Do a check for an Open Circuit or a Short between pins 1 and 2 of the Aft Cabin Temperature Sensor Module.

(e) If there is a Short or an Open Circuit between pins 1 and 2, then do these steps:

1) Replace the Aft Cabin Temperature Sensor Module, M2848. These are the tasks:

- Cabin Temperature Sensor Removal, AMM TASK 21-61-10-000-802
- Cabin Temperature Sensor Installation, AMM TASK 21-61-10-400-802

a) Do the Repair Confirmation at the end of this task.

(f) If there is not a Short or an Open Circuit between pins 1 and 2, then continue.

(5) Do this check of the wiring.

(a) Remove the Right Pack/Zone Temperature Controller. This is the task: Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801.

(b) Examine the wiring between the Aft Cabin Temperature Sensor Module, M2848 and the Right Pack Temperature Controller, M1443 as follows (WDM 21-61-14):

TEMP SNSR MODULE D15022	**LEFT PZTC D3858A**
pin 1	pin A4
pin 2	pin B4

(c) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-connect the Electrical Connector to the Aft Cabin Temperature Sensor Module.

3) Re-install the Right Pack Temperature Controller, M1443. This is the task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.

4) Do the Repair Confirmation at the end of this task.

(d) If you do not find a problem with the wiring, then continue.

1) Re-connect the Electrical Connector to the Aft Cabin Temperature Sensor Module.

2) Install the trim panel adjacent to the PSU.

3) Close (lift) the PSU.

(6) Replace the Right Zone Temperature Controller, M1443. These are the tasks:

- Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801
- Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801

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OMR 815, 816, 818, 819, 901-999 (Continued)

(a) Do the Repair Confirmation at the end of this task.

OMR 807

G. Fault Isolation Procedure

(1) Do this check of the Main Cargo Temperature Sensor Module, M3282:

(a) Lower the PSU that is adjacent to the Main Cargo Temperature Sensor Module.

(b) Remove the trim panel that is adjacent to the PSU and beneath the Electrical Connector for the Zone Temperature Sensor Module .

(c) Disconnect Electrical Connector, D16870 from the Main Cargo Temperature Module (WDM 21-61-14).

(d) Do a check for an Open Circuit or a Short between pins 1 and 2 of the Main Cargo Temperature Module.

(e) If there is a Short or an Open Circuit between pins 1 and 2, then do these steps:

1) Replace the Main Cargo Temperature Sensor Module, M3282. These are the tasks:

- Cabin Temperature Sensor Removal, AMM TASK 21-61-10-000-802
- Cabin Temperature Sensor Installation, AMM TASK 21-61-10-400-802

2) Do the Repair Confirmation at the end of this task.

(f) If there is not a Short or an Open Circuit between pins 1 and 2, then continue.

(2) Do this check of the wiring.

(a) Remove the Left Pack/Zone Temperature Controller, M1443. This is the task: Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801.

(b) Do a Wiring Check between the Main Cargo Temperature Sensor Module, M3282, and the Right Pack/Zone Temperature Controller, M1443 as follows (WDM 21-61-13):

TEMP SNSR MODULE D16870		LEFT PZTC D3858A
pin 1	. . . . .	pin A4
pin 2	. . . . .	pin B4

(c) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-connect the Electrical Connector to the Main Cargo Temperature Sensor Module.

3) Re-install the Right Pack/Zone Temperature Controller, M1443. This is the task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.

4) Do the Repair Confirmation at the end of this task.

(d) If you do not find a problem with the wiring, then continue.

1) Re-connect the Electrical Connector to the Main Cargo Temperature Sensor Module.

2) Install the trim panel adjacent to the PSU.

3) Close (lift) the PSU.

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OMR 807 (Continued)

(3) Replace the Right Pack/Zone Temperature Controller, M1443. These are the tasks:

- Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801
- Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801

(a) Do the Repair Confirmation at the end of this task.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

H. Repair Confirmation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the problem.

(b) If the maintenance message still shows, continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

818. Forward Cabin Duct Temperature Sensor - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) ZONE DUCT SENSOR

(2) The right pack/zone temperature controller has detected a short or an open circuit from the forward cabin duct temperature sensor.

B. Possible Causes

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

(1) Forward cabin duct temperature sensor, T533

OMR 807

(2) Forward cabin duct temperature sensor, T551

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

(3) Wiring

(4) Pack/Zone temperature controller (PZTC), M1443 (No. 2).

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

D. Related Data

(1) (SSM 21-61-13)

(2) (WDM 21-61-13)

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(2) If the maintenance message does not show, then there was an intermittent fault.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999 (Continued)

(3) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

(1) Do this check of the forward cabin duct temperature sensor, T533:

(a) Remove the left ceiling panel that is just forward of the overwing escape hatches. To remove the panel, do this task: Ceiling Panel - Removal, AMM TASK 25-21-45-000-801 or Main Ceiling Panel - Removal, AMM TASK 25-21-45-000-803-001.

(b) Disconnect electrical connector D11830 from the forward cabin duct temperature sensor.

(c) Do a check for an open circuit or a short between pins 1 and 2 of the forward cabin duct temperature sensor.

(d) If there is a short or an open circuit between pins 1 and 2, then do these steps:

1) Replace the forward cabin duct temperature sensor, T533.

These are the tasks:

Duct Temperature Limit Sensor Removal, AMM TASK 21-61-04-000-803-002,

Duct Temperature Limit Sensor Installation, AMM TASK 21-61-04-400-803-002.

2) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

3) If the maintenance message does not show, then you corrected the fault.

(e) If there is not a short or an open circuit between pins 1 and 2, then continue.

OMR 807

(2) Do this check of the forward cabin duct temperature sensor, T551:

(a) Disconnect electrical connector D16844 from the forward cabin duct temperature sensor.

(b) Do a check for an open circuit or a short between pins 1 and 2 of the forward cabin duct temperature sensor.

(c) If there is a short or an open circuit between pins 1 and 2, then do these steps:

1) Replace the forward cabin duct temperature sensor, T551.

These are the tasks:

Duct Temperature Limit Sensor Removal, AMM TASK 21-61-04-000-803-002,

Duct Temperature Limit Sensor Installation, AMM TASK 21-61-04-400-803-002.

2) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

3) If the maintenance message does not show, then you corrected the fault.

(d) If there is not a short or an open circuit between pins 1 and 2, then continue.

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

(3) Do this check of the wiring between the forward cabin duct temperature sensor and the right pack/zone temperature controller, M1443:

(a) Remove the right pack/zone temperature controller. To remove the controller, do this task: Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801.

(b) Do a wiring check between these pins of electrical connector D11830 of the forward cabin duct temperature sensor and connector D3860A of the pack/zone temperature controller:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D11830		D3860A
pin 1	. . . . .	pin C6
pin 2	. . . . .	pin C7

(c) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.
2) Connect the electrical connector to the forward cabin duct temperature sensor.
3) Install the ceiling panel. To install the panel, do this task: Ceiling Panel - Installation, AMM TASK 25-21-45-400-801 or Main Ceiling Panel - Installation, AMM TASK 25-21-45-400-803-001.
4) Install the right pack/zone temperature controller. To install the controller, do this task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.
5) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.
6) If the maintenance message does not show, then you corrected the fault.

(d) If you do not find a problem with the wiring, then continue.

1) Connect the electrical connector to the forward cabin duct temperature sensor.
2) Install the ceiling panel. To install the panel, do this task: Ceiling Panel - Installation, AMM TASK 25-21-45-400-801 or Main Ceiling Panel - Installation, AMM TASK 25-21-45-400-803-001.

OMR 807

(4) Do this check of the wiring between the forward cabin duct temperature sensor and the right pack/zone temperature controller, M1443:

(a) Remove the right pack/zone temperature controller. To remove the controller, do this task: Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801.

(b) Do a wiring check between these pins of electrical connector D16844 of the forward cabin duct temperature sensor and connector D3860A of the pack/zone temperature controller:

D16844		D3860A
pin 1	. . . . .	pin C6
pin 2	. . . . .	pin C7

(c) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.
2) Connect the electrical connector to the forward cabin duct temperature sensor.
3) Install the right pack/zone temperature controller. To install the controller, do this task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.
4) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.
5) If the maintenance message does not show, then you corrected the fault.

(d) If you do not find a problem with the wiring, then continue.

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OMR 807 (Continued)

1) Connect the electrical connector to the forward cabin duct temperature sensor.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

(5) Replace the right pack/zone temperature controller, M1443. These are the tasks:

- Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801
- Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801

(a) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

1) If the maintenance message does not show, then you corrected the fault.

——— END OF TASK ———

819. Aft Cabin Duct Temperature Sensor - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) ZONE DUCT SENSOR

(2) The left pack/zone temperature controller has detected a short or an open circuit from the aft cabin duct temperature sensor.

B. Possible Causes

(1) Aft cabin duct temperature sensor, T536

(2) Wiring

(3) Pack/Zone temperature controller (PZTC), M1442 (No. 1).

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

D. Related Data

(1) (SSM 21-61-14)

(2) (WDM 21-61-14)

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(2) If the maintenance message does not show, then there was an intermittent fault.

(3) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Do this check of the aft cabin duct temperature sensor, T536:

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OMR 802, 805, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

(a) Remove the right ceiling panel that is adjacent to the overwing escape hatches. To remove the panel, do this task: Ceiling Panel - Removal, AMM TASK 25-21-45-000-801 or Main Ceiling Panel - Removal, AMM TASK 25-21-45-000-803-001.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

(b) Disconnect electrical connector D11840 from the aft cabin duct temperature sensor.

(c) Do a check for an open circuit or a short between pins 1 and 2 of the forward cabin duct temperature sensor.

(d) If there is a short or an open circuit between pins 1 and 2, then do these steps:

1) Replace the aft cabin duct temperature sensor, T536.

These are the tasks:

Duct Temperature Limit Sensor Removal, AMM TASK 21-61-04-000-803-002,

Duct Temperature Limit Sensor Installation, AMM TASK 21-61-04-400-803-002.

2) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

3) If the maintenance message does not show, then you corrected the fault.

(e) If there is not a short or an open circuit between pins 1 and 2, then continue.

(2) Do this check of the wiring between the aft cabin duct temperature sensor and the left pack/zone temperature controller, M1442:

(a) Remove the left pack/zone temperature controller. To remove the controller, do this task: Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801.

(b) Do a wiring check between these pins of electrical connector D11840 of the aft cabin duct temperature sensor and connector D3858A of the pack/zone temperature controller:

D11840	D3858A
pin 1	pin C6
pin 2	pin C7

(c) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Connect the electrical connector to the aft cabin duct temperature sensor.

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

3) Install the ceiling panel. To install the panel, do this task: Ceiling Panel - Installation, AMM TASK 25-21-45-400-801 or Main Ceiling Panel - Installation, AMM TASK 25-21-45-400-803-001.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

4) Install the left pack/zone temperature controller. To install the controller, do this task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.

5) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

6) If the maintenance message does not show, then you corrected the fault.

(d) If you do not find a problem with the wiring, then continue.

1) Connect the electrical connector to the aft cabin duct temperature sensor.

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OMR 802, 805, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

2) Install the ceiling panel. To install the panel, do this task: Ceiling Panel - Installation, AMM TASK 25-21-45-400-801 or Main Ceiling Panel - Installation, AMM TASK 25-21-45-400-803-001.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

(3) Replace the left pack/zone temperature controller, M1442. These are the tasks:

- Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801
- Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801

(a) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

1) If the maintenance message does not show, then you corrected the fault.

——— END OF TASK ———

820. Right Ram Air Control Temperature Sensor - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) RAM AIR SNSR

(2) The right pack/zone temperature controller has detected a short or an open circuit from the right ram air control temperature sensor.

B. Possible Causes

(1) Right ram air control temperature sensor, T482

(2) Wiring

(3) Pack/Zone temperature controller (PZTC), M1443 (No. 2).

OMR 815, 816, 818, 819, 901-999

(4) Right ram air inlet actuator, M2797

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

OMR 815, 816, 818, 819, 901-999

D. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

E. Related Data

(1) (SSM 21-51-24)

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OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999 (Continued)

(2) (WDM 21-51-24)

F. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(2) If the maintenance message does not show, then there was an intermittent fault.

(3) If the maintenance message shows, then do the Fault Isolation Procedure below.

G. Fault Isolation Procedure

OMR 802, 805, 807, 809, 810, 812, 813

(1) Do this check of the right ram air control temperature sensor, T482:

(a) Open this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

(b) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

(c) Disconnect electrical connector D3918 from the right ram air control temperature sensor.

(d) Do a check for an open circuit or a short between pins 1 and 2 of the right ram air control temperature sensor.

(e) If there is a short or an open circuit between pins 1 and 2, then do these steps:

1) Replace the right ram air control temperature sensor, T482.

These are the tasks:

Ram Air Control Temperature Sensor - Removal, AMM TASK 21-51-50-000-801,

Ram Air Control Temperature Sensor - Installation, AMM TASK 21-51-50-400-801.

2) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

3) If the maintenance message does not show, then you corrected the fault.

(f) If there is not a short or an open circuit between pins 1 and 2, then continue.

(2) Do this check of the wiring between the right ram air control temperature sensor and the right pack/zone temperature controller, M1443:

(a) Remove the right pack/zone temperature controller. To remove the controller, do this task: Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801.

(b) Do a wiring check between these pins of electrical connector D3918 of the right ram air control temperature sensor and connector D3860A of the pack/zone temperature controller:

D3918		D3860A
pin 1	. . . . .	pin A13
pin 2	. . . . .	pin B13

(c) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-connect the electrical connector to the right ram air control temperature sensor.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 802, 805, 807, 809, 810, 812, 813 (Continued)

3) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

4) Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

5) Re-install the right pack/zone temperature controller. To install the controller, do this task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.

6) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

7) If the maintenance message does not show, then you corrected the fault.

(d) If you do not find a problem with the wiring, then continue.

1) Re-connect the electrical connector to the right ram air control temperature sensor.

2) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

3) Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

OMR 815, 816, 818, 819, 901-999

(3) Do this check of the wiring between the right ram air control temperature sensor and the right ram air inlet actuator, M2797:

(a) Get access to the right ram air inlet actuator, M2797, as follows:

1) Open these access panels:

Number	Name/Location
191FR	Forward Wing To Body Fairing Panel - Mid Fairing, Above Ram Air Inlet
191GR	Ram Air Actuator Panel - Forward

(b) Disconnect electrical connector D14394 from the right ram air inlet actuator, M2797.

(c) Do a wiring check between these pins of electrical connector D3918 of the right ram air control temperature sensor and electrical connector D14394 of the right ram air inlet actuator, M2797:

M2797		**T482**
D14394		**D3918**
Pin 17	. . . . .	Pin 1
Pin 18	. . . . .	Pin 2

(d) If you find a problem with the wiring, then do these steps:

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OMR 815, 816, 818, 819, 901-999 (Continued)

1) Repair the wiring.
2) Reconnect electrical connector D3918 to the right ram air sensor, T482.
3) Reconnect electrical connector D14394 to the right ram air inlet actuator, M2797.
4) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.
5) If the maintenance message does not show, then you corrected the fault.
6) Close these access panels:

Number	Name/Location
191FR	Forward Wing To Body Fairing Panel - Mid Fairing, Above Ram Air Inlet
191GR	Ram Air Actuator Panel - Forward

7) If the maintenance message still shows, then continue.

(e) If you do not find a problem with the wiring, then continue.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

(4) Replace the right pack/zone temperature controller, M1443.

These are the tasks:

Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801,

Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.

(a) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.
1) If the maintenance message does not show, then you corrected the fault.

OMR 815, 816, 818, 819, 901-999

(5) Replace the right ram air inlet actuator, M2797, as follows:
(a) Do these tasks: Ram Air Inlet Actuator Removal, AMM TASK 21-51-21-000-801 and Ram Air Inlet Actuator Installation, AMM TASK 21-51-21-400-801.
(b) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.
(c) If the maintenance message does not show, then you corrected the fault.

END OF TASK

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

821. Left Ram Air Control Temperature Sensor - Fault Isolation

A. Description

(1) This task is for this maintenance message:
(a) RAM AIR SNSR
(2) The Left Pack/Zone Temperature Controller has detected a short or an open circuit from the Left Ram Air Control Temperature Sensor.

B. Possible Causes

(1) Left Ram Air Control Temperature Sensor, T481
(2) Wiring
(3) Left PZTC, M1442.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 815, 816, 818, 819, 901-999

(4) Left Ram Air Inlet Actuator, M2795

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

OMR 815, 816, 818, 819, 901-999

(2) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

D. Related Data

(1) SSM 21-51-14

(2) WDM 21-51-14

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then there was an intermittent fault.

(b) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

OMR 802, 805, 807, 809, 810, 812, 813

(1) Do this check of the Left Ram Air Control Temperature Sensor, T481:

(a) Open this access panel:

Number	Name/Location
192CL	ECS Access Door

(b) Disconnect electrical connector D3916 from the Left Ram Air Control Temperature Sensor, T481.

(c) Do a check for an open circuit or a short between pins 1 and 2 of the Left Ram Air Control Temperature Sensor, T481.

(d) If there is a short or an open circuit between pins 1 and 2, then do these steps:

1) Replace the Left Ram Air Control Temperature Sensor, T481. These are the tasks:

- Ram Air Control Temperature Sensor - Removal, AMM TASK 21-51-50-000-801
- Ram Air Control Temperature Sensor - Installation, AMM TASK 21-51-50-400-801

2) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

3) If the maintenance message does not show, then you corrected the problem.

(e) If there is not a short or an open circuit between pins 1 and 2, then continue.

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OMR 802, 805, 807, 809, 810, 812, 813 (Continued)

(2) Do this check of the wiring between the Left Ram Air Control Temperature Sensor, T481 and the Left PZTC, M1442:

(a) Remove the Left PZTC, M1442. This is the task: Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801.

(b) Do a wiring check between these pins of electrical connector D3916 of the Left Ram Air Control Temperature Sensor, T481 and connector D3858A of the Left PZTC, M1442:

D3916		D3858A
pin 1	. . . . .	pin A13
pin 2	. . . . .	pin B13

(c) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-connect the electrical connector to the Left Ram Air Control Temperature Sensor, T481.

3) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

4) Re-install the Left PZTC, M1442. This is the task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.

5) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

6) If the maintenance message does not show, then you corrected the problem.

(d) If you do not find a problem with the wiring, then continue.

1) Re-connect the electrical connector to the Left Ram Air Control Temperature Sensor, T481.

2) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

(3) Replace the Left PZTC, M1442. These are the tasks:

- Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801
- Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801

(a) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

1) If the maintenance message does not show, then you corrected the problem.

OMR 815, 816, 818, 819, 901-999

(4) Do this check of the wiring between the Left Ram Air Control Temperature Sensor, T481 and the Left Ram Air Inlet Actuator, M2795:

(a) Get access to the Left Ram Air Inlet Actuator, M2795, as follows:

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OMR 815, 816, 818, 819, 901-999 (Continued)

1) Open these access panels:

Number	Name/Location
191FL	Forward Wing To Body Fairing Panel - Mid Fairing, Above Ram Air Inlet
191GL	Ram Air Actuator Panel - Forward

(b) Disconnect electrical connector D14386 from the Left Ram Air Inlet Actuator, M2795.

(c) Disconnect electrical connector D3916 from the Left Ram Air Control Temperature Sensor, T481.

(d) Do a wiring check between these pins of electrical connector D3916 of the Left Ram Air Control Temperature Sensor, T481 and electrical connector D14386 of the Left Ram Air Inlet Actuator, M2795:

M2795		T481
D14386		D3916
Pin 17	. . . . .	Pin 1
Pin 18	. . . . .	Pin 2

(e) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Reconnect electrical connector D3916 to the left ram air sensor, T481.

3) Reconnect electrical connector D14386 to the Left Ram Air Inlet Actuator, M2795.

4) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

5) If the maintenance message does not show, then you corrected the problem.

6) Close these access panels:

Number	Name/Location
191FL	Forward Wing To Body Fairing Panel - Mid Fairing, Above Ram Air Inlet
191GL	Ram Air Actuator Panel - Forward

7) If the maintenance message still shows, then continue.

(5) Replace the Left Ram Air Inlet Actuator, M2795. These are the tasks:

- Ram Air Inlet Actuator Removal, AMM TASK 21-51-21-000-801
- Ram Air Inlet Actuator Installation, AMM TASK 21-51-21-400-801

(a) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(b) If the maintenance message does not show, then you corrected the problem.

——— END OF TASK ———

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

822. Right Pack Temperature Sensor (Primary) - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) THIS PACK - PACK SNSR

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(2) The right pack/zone temperature controller has detected a short or an open circuit from the right pack temperature sensor.

B. Possible Causes

(1) Right pack temperature sensor, T474

(2) Wiring

(3) Pack/Zone temperature controller (PZTC), M1443 (No. 2).

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

D. Related Data

(1) (SSM 21-51-22)

(2) (WDM 21-51-22)

(3) (SSM 21-51-23)

(4) (WDM 21-51-23)

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(2) If the maintenance message does not show, then there was an intermittent fault.

(3) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Do this check of the right pack temperature sensor, T474:

(a) Open this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

(b) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

(c) Disconnect electrical connector D3876 from the right pack temperature sensor.

(d) Do a check for an open circuit or a short between pins 1 and 2 of the right pack temperature sensor.

(e) If there is a short or an open circuit between pins 1 and 2, then do these steps:

1) Replace the right pack temperature sensor, T474.

These are the tasks:

Pack Temperature Sensor Removal, AMM TASK 21-51-52-000-801,

Pack Temperature Sensor Installation, AMM TASK 21-51-52-400-801.

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2) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

3) If the maintenance message does not show, then you corrected the fault.

(f) If there is not a short or an open circuit between pins 1 and 2, then continue.

(2) Do this check of the wiring between the right pack temperature sensor and the right pack/zone temperature controller, M1443:

(a) Remove the right pack/zone temperature controller. To remove the controller, do this task: Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801.

(b) Do a wiring check between these pins of electrical connector D3876 of the right pack temperature sensor and connector D3860A of the pack/zone temperature controller:

D3876		D3860A
pin 1	. . . . .	pin A6
pin 2	. . . . .	pin B6

(c) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-connect the electrical connector to the right pack temperature sensor.

3) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

4) Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

5) Re-install the right pack/zone temperature controller. To install the controller, do this task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.

6) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

7) If the maintenance message does not show, then you corrected the fault.

(d) If you do not find a problem with the wiring, then continue.

1) Re-connect the electrical connector to the right pack temperature sensor.

2) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

3) Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

(3) Replace the right pack/zone temperature controller, M1443. These are the tasks:

- Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801
- Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801

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(a) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

1) If the maintenance message does not show, then you corrected the fault.

——— END OF TASK ———

823. Left Pack Temperature Sensor (Backup) - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) STBY OTHER PACK - PACK SNSR

(2) The right pack/zone temperature controller has detected a short or an open circuit from the left pack temperature sensor.

B. Possible Causes

(1) Left pack temperature sensor, T471

(2) Wiring

(3) Pack/Zone temperature controller (PZTC), M1443 (No. 2).

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

D. Related Data

(1) (SSM 21-51-12)

(2) (WDM 21-51-12)

(3) (SSM 21-51-13)

(4) (WDM 21-51-13)

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(2) If the maintenance message does not show, then there was an intermittent fault.

(3) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Do this check of the left pack temperature sensor, T471:

(a) Open this access panel:

Number	Name/Location
192CL	ECS Access Door

(b) Disconnect electrical connector D3866 from the left pack temperature sensor.

(c) Do a check for an open circuit or a short between pins 3 and 4 of the left pack temperature sensor.

(d) If there is a short or an open circuit between pins 3 and 4, then do these steps:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Replace the left pack temperature sensor, T471.

These are the tasks:

Pack Temperature Sensor Removal, AMM TASK 21-51-52-000-801,

Pack Temperature Sensor Installation, AMM TASK 21-51-52-400-801.

2) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

3) If the maintenance message does not show, then you corrected the fault.

(e) If there is not a short or an open circuit between pins 3 and 4, then continue.

(2) Do this check of the wiring between the left pack temperature sensor and the right pack/zone temperature controller, M1443:

(a) Remove the right pack/zone temperature controller. To remove the controller, do this task: Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801.

(b) Do a wiring check between these pins of electrical connector D3866 of the left pack temperature sensor and connector D3860A of the pack/zone temperature controller:

D3866		D3860A
pin 3	. . . . .	pin D11
pin 4	. . . . .	pin D12

(c) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-connect the electrical connector to the left pack temperature sensor.

3) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

4) Re-install the right pack/zone temperature controller. To install the controller, do this task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.

5) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

6) If the maintenance message does not show, then you corrected the fault.

(d) If you do not find a problem with the wiring, then continue.

1) Re-connect the electrical connector to the left pack temperature sensor.

2) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

(3) Replace the right pack/zone temperature controller, M1443. These are the tasks:

- Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801
- Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801

(a) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

1) If the maintenance message does not show, then you corrected the fault.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

824. Right Pack Temperature Sensor (Backup) - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) STBY OTHER PACK - PACK SNSR

(2) The left pack/zone temperature controller has detected a short or an open circuit from the right pack temperature sensor.

B. Possible Causes

(1) Right pack temperature sensor, T474

(2) Wiring

(3) Pack/Zone temperature controller (PZTC), M1442 (No. 1).

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

D. Related Data

(1) (SSM 21-51-22)

(2) (WDM 21-51-22)

(3) (SSM 21-51-23)

(4) (WDM 21-51-23)

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(2) If the maintenance message does not show, then there was an intermittent fault.

(3) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Do this check of the right pack temperature sensor, T474:

(a) Open this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

(b) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

(c) Disconnect electrical connector D3876 from the right pack temperature sensor.

(d) Do a check for an open circuit or a short between pins 3 and 4 of the right pack temperature sensor.

(e) If there is a short or an open circuit between pins 3 and 4, then do these steps:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Replace the right pack temperature sensor, T474.

These are the tasks:

Pack Temperature Sensor Removal, AMM TASK 21-51-52-000-801,

Pack Temperature Sensor Installation, AMM TASK 21-51-52-400-801.

2) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

3) If the maintenance message does not show, then you corrected the fault.

(f) If there is not a short or an open circuit between pins 3 and 4, then continue.

(2) Do this check of the wiring between the right pack temperature sensor and the left pack/zone temperature controller, M1442:

(a) Remove the left pack/zone temperature controller. To remove the controller, do this task: Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801.

(b) Do a wiring check between these pins of electrical connector D3876 of the right pack temperature sensor and connector D3858A of the pack/zone temperature controller:

D3876		D3858A
pin 3	. . . . .	pin D11
pin 4	. . . . .	pin D12

(c) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-connect the electrical connector to the right pack temperature sensor.

3) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

4) Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

5) Re-install the left pack/zone temperature controller. To install the controller, do this task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.

6) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

7) If the maintenance message does not show, then you corrected the fault.

(d) If you do not find a problem with the wiring, then continue.

1) Re-connect the electrical connector to the right pack temperature sensor.

2) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

3) Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

(3) Replace the left pack/zone temperature controller, M1442. These are the tasks:

- Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801
- Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801

(a) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

1) If the maintenance message does not show, then you corrected the fault.

——— END OF TASK ———

825. Left Pack Temperature Sensor (Primary) - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) THIS PACK - PACK SNSR

(2) The left pack/zone temperature controller has detected a short or an open circuit from the left pack temperature sensor.

B. Possible Causes

(1) Left pack temperature sensor, T471

(2) Wiring

(3) Pack/Zone temperature controller (PZTC), M1442 (No. 1).

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

D. Related Data

(1) (SSM 21-51-12)

(2) (WDM 21-51-12)

(3) (SSM 21-51-13)

(4) (WDM 21-51-13)

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(2) If the maintenance message does not show, then there was an intermittent fault.

(3) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Do this check of the left pack temperature sensor, T471:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999 (Continued)

(a) Open this access panel:

Number	Name/Location
192CL	ECS Access Door

(b) Disconnect electrical connector D3866 from the left pack temperature sensor.

(c) Do a check for an open circuit or a short between pins 1 and 2 of the left pack temperature sensor.

(d) If there is a short or an open circuit between pins 1 and 2, then do these steps:

1) Replace the left pack temperature sensor, T471.

These are the tasks:

Pack Temperature Sensor Removal, AMM TASK 21-51-52-000-801,

Pack Temperature Sensor Installation, AMM TASK 21-51-52-400-801.

2) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

3) If the maintenance message does not show, then you corrected the fault.

(e) If there is not a short or an open circuit between pins 1 and 2, then continue.

(2) Do this check of the wiring between the left pack temperature sensor and the left pack/zone temperature controller, M1442:

(a) Remove the left pack/zone temperature controller. To remove the controller, do this task: Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801.

(b) Do a wiring check between these pins of electrical connector D3866 of the left pack temperature sensor and connector D3858A of the pack/zone temperature controller:

D3866		D3858A
pin 1	. . . . .	pin A6
pin 2	. . . . .	pin B6

(c) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-connect the electrical connector to the left pack temperature sensor.

3) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

4) Re-install the left pack/zone temperature controller. To install the controller, do this task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.

5) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

6) If the maintenance message does not show, then you corrected the fault.

(d) If you do not find a problem with the wiring, then continue.

1) Re-connect the electrical connector to the left pack temperature sensor.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

(3) Replace the left pack/zone temperature controller, M1442. These are the tasks:

- Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801
- Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801

(a) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

1) If the maintenance message does not show, then you corrected the fault.

——— END OF TASK ———

826. Right Mix Manifold Temperature Sensor - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) THIS PACK - MIX MANF SENSOR

(2) The right pack/zone temperature controller has detected a short or an open circuit from the right mix manifold temperature sensor.

B. Possible Causes

(1) Right mix manifold temperature sensor, T475

(2) Wiring

(3) Pack/Zone temperature controller (PZTC), M1443 (No. 2).

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

D. Related Data

(1) (SSM 21-51-22)

(2) (WDM 21-51-22)

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(2) If the maintenance message does not show, then there was an intermittent fault.

(3) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Do this check of the right mix manifold temperature sensor, T475:

(a) Open the forward cargo door.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Remove the aft center bulkhead liner in the forward cargo compartment. To remove the liner, do this task: Forward Cargo Compartment Aft Bulkhead Liner - Removal, AMM TASK 25-52-17-000-801.

(c) Disconnect electrical connector D3878 from the right mix manifold temperature sensor.

(d) Do a check for an open circuit or a short between pins 1 and 2 of the right mix manifold temperature sensor.

(e) If there is a short or an open circuit between pins 1 and 2, then do these steps:

1) Replace the right mix manifold temperature sensor, T475.

These are the tasks:

Mix Manifold Temperature Sensor Removal, AMM TASK 21-51-51-000-803,

Mix Manifold Temperature Sensor Installation, AMM TASK 21-51-51-400-802.

2) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

3) If the maintenance message does not show, then you corrected the fault.

(f) If there is not a short or an open circuit between pins 1 and 2, then continue.

(2) Do this check of the wiring between the right mix manifold temperature sensor and the right pack/zone temperature controller, M1443:

(a) Remove the right pack/zone temperature controller. To remove the controller, do this task: Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801.

(b) Do a wiring check between these pins of electrical connector D3878 of the right mix manifold temperature sensor and connector D3860A of the pack/zone temperature controller:

D3878		D3860A
pin 1	. . . . .	pin B7
pin 2	. . . . .	pin A7

(c) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-connect the electrical connector to the right mix manifold temperature sensor.

3) Install the aft center bulkhead liner in the forward cargo compartment. To install the liner, do this task: Forward Cargo Compartment Aft Bulkhead Liner - Installation, AMM TASK 25-52-17-400-801.

4) Close the forward cargo door.

5) Re-install the right pack/zone temperature controller. To install the controller, do this task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.

6) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

7) If the maintenance message does not show, then you corrected the fault.

(d) If you do not find a problem with the wiring, then continue.

1) Re-connect the electrical connector to the right mix manifold temperature sensor.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999 (Continued)

2) Install the aft center bulkhead liner in the forward cargo compartment. To install the liner, do this task: Forward Cargo Compartment Aft Bulkhead Liner - Installation, AMM TASK 25-52-17-400-801.

3) Close the forward cargo door.

(3) Replace the right pack/zone temperature controller, M1443. These are the tasks:

- Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801
- Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801

(a) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

1) If the maintenance message does not show, then you corrected the fault.

——— END OF TASK ———

827. Left Mix Manifold Temperature Sensor - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) THIS PACK - MIX MANF SENSOR

(2) The left pack/zone temperature controller has detected a short or an open circuit from the left mix manifold temperature sensor.

OMR 807

(3) For the 737-800BCF, it is possible to cause repetitive left MIX MANF SENSOR faults. It is a nuisance fault because of bad air mixture in the mix manifold because there are no recirculation fans. If the fault occurs again after you complete the fault isolation procedure, you can reset the fault. If the fault is reset and does not show again, you can release the airplane back into service with no more maintenance work necessary. This procedure to reset the fault can be done again and again when necessary.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

B. Possible Causes

(1) Left mix manifold temperature sensor, T472

(2) Wiring

(3) PZTC, M1442 (No. 1).

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

D. Related Data

(1) SSM 21-51-12

(2) WDM 21-51-12

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Initial Evaluation

(1) Do the Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(2) If the maintenance message does not show, then there was an intermittent fault.

(3) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Do this check of the left mix manifold temperature sensor, T472:

(a) Open the forward cargo door.

(b) Remove the aft center bulkhead liner in the forward cargo compartment. To remove the liner, do this task: Forward Cargo Compartment Aft Bulkhead Liner - Removal, AMM TASK 25-52-17-000-801.

(c) Disconnect electrical connector D3868 from the left mix manifold temperature sensor.

(d) Do a check for an open circuit or a short between pins 1 and 2 of the left mix manifold temperature sensor.

(e) If there is a short or an open circuit between pins 1 and 2, then do these steps:

1) Replace the left mix manifold temperature sensor, T472.

These are the tasks:

Mix Manifold Temperature Sensor Removal, AMM TASK 21-51-51-000-803,

Mix Manifold Temperature Sensor Installation, AMM TASK 21-51-51-400-802.

2) Do the Repair Confirmation at the end of this task.

(f) If there is not a short or an open circuit between pins 1 and 2, then continue.

(2) Do this check of the wiring between the left mix manifold temperature sensor and the left pack/zone temperature controller, M1442:

(a) Remove the left pack/zone temperature controller. To remove the controller, do this task: Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801.

(b) Do a wiring check between these pins of electrical connector D3868 of the left mix manifold temperature sensor and connector D3858A of the pack/zone temperature controller:

D3868		D3858A
pin 1	. . . . .	pin B7
pin 2	. . . . .	pin A7

(c) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-connect the electrical connector to the left mix manifold temperature sensor.

3) Install the aft center bulkhead liner in the forward cargo compartment. To install the liner, do this task: Forward Cargo Compartment Aft Bulkhead Liner - Installation, AMM TASK 25-52-17-400-801.

4) Close the forward cargo door.

5) Re-install the left pack/zone temperature controller. To install the controller, do this task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999 (Continued)

6) Do the Repair Confirmation at the end of this task.

(d) If you do not find a problem with the wiring, then continue.

1) Re-connect the electrical connector to the left mix manifold temperature sensor.

2) Install the aft center bulkhead liner in the forward cargo compartment. To install the liner, do this task: Forward Cargo Compartment Aft Bulkhead Liner - Installation, AMM TASK 25-52-17-400-801.

3) Close the forward cargo door.

(3) Replace the left pack/zone temperature controller, M1442. These are the tasks:

- Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801
- Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801

(a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do the Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the fault.

(b) If the maintenance message shows, then continue the Fault Isolation Procedure at the subsequent step.

OMR 807

NOTE: For the 737-800BCF, it is possible to cause repetitive left MIX MANF SENSOR faults. It is a nuisance fault because of bad air mixture in the mix manifold because there are no recirculation fans.

If the fault repeats after completing the Fault Isolation Procedure, the fault can be reset.

If the fault occurs again after you complete the fault isolation procedure, you can reset the fault.

If the fault is reset and does not show again, you can release the airplane back into service with no more maintenance work necessary.

This procedure to reset the fault can be done again and again when necessary.

——— END OF TASK ———

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

828. Left Standby Temperature Control Valve - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) STANDBY OTHER PACK - TCV

(2) The right pack/zone temperature controller has detected a fault from the left standby temperature control valve.

B. Possible Causes

(1) Left standby temperature control valve, V140

(2) Wiring

(3) Pack/Zone temperature controller (PZTC), M1443 (No. 2).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999 (Continued)

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

D. Related Data

(1) (SSM 21-51-13)

(2) (WDM 21-51-13)

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(2) If the maintenance message does not show, then there was an intermittent fault.

(3) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Replace the left standby temperature control valve, V140.

These are the tasks:

Standby Temperature Control Valve Removal, AMM TASK 21-51-11-000-801,

Standby Temperature Control Valve Installation, AMM TASK 21-51-11-400-801.

(a) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(b) If the maintenance message does not show, then you corrected the fault.

(c) If the maintenace message still shows, then continue.

(2) Do this check of the wiring between the left standby temperature control valve and the right pack/zone temperature controller, M1443:

(a) Remove the right pack/zone temperature controller. To remove the controller, do this task: Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801.

(b) Do a wiring check between these pins of electrical connector D3864 of the left standby temperature control valve and connector D3860A of the pack/zone temperature controller:

D3864		D3860A
pin 3	. . . . .	pin A15
pin 1	. . . . .	pin B15

(c) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-connect the electrical connector to the left standby temperature control valve.

3) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

4) Re-install the right pack/zone temperature controller. To install the controller, do this task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.

5) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

6) If the maintenance message does not show, then you corrected the fault.

(d) If you do not find a problem with the wiring, then continue.

1) Re-connect the electrical connector to the left standby temperature control valve.

2) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

(3) Replace the right pack/zone temperature controller, M1443. These are the tasks:

- Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801
- Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801

(a) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

1) If the maintenance message does not show, then you corrected the fault.

——— END OF TASK ———

829. Right Standby Temperature Control Valve - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) STANDBY OTHER PACK - TCV

(2) The left pack/zone temperature controller has detected a fault from the right standby temperature control valve.

B. Possible Causes

(1) Right standby temperature control valve, V142

(2) Wiring

(3) Pack/Zone temperature controller (PZTC), M1442 (No. 1).

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

D. Related Data

(1) (SSM 21-51-23)

(2) (WDM 21-51-23)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(2) If the maintenance message does not show, then there was an intermittent fault.

(3) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Replace the right standby temperature control valve, V142.

These are the tasks:

Standby Temperature Control Valve Removal, AMM TASK 21-51-11-000-801,

Standby Temperature Control Valve Installation, AMM TASK 21-51-11-400-801.

(a) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(b) If the maintenance message does not show, then you corrected the fault.

(c) If the maintenace message still shows, then continue.

(2) Do this check of the wiring between the right standby temperature control valve and the left pack/zone temperature controller, M1442:

(a) Remove the left pack/zone temperature controller. To remove the controller, do this task: Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801.

(b) Do a wiring check between these pins of electrical connector D3874 of the right standby temperature control valve and connector D3858A of the pack/zone temperature controller:

D3874		D3858A
pin 3	. . . . .	pin A15
pin 1	. . . . .	pin B15

(c) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-connect the electrical connector to the right standby temperature control valve.

3) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

4) Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

5) Re-install the left pack/zone temperature controller. To install the controller, do this task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.

6) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

7) If the maintenance message does not show, then you corrected the fault.

(d) If you do not find a problem with the wiring, then continue.

1) Re-connect the electrical connector to the right standby temperature control valve.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999 (Continued)

2) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

3) Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

(3) Replace the left pack/zone temperature controller, M1442. These are the tasks:

- Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801
- Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801

(a) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

1) If the maintenance message does not show, then you corrected the fault.

——— END OF TASK ———

830. Left Temperature Control Valve - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) THIS PACK - TCV

(2) The Left PZTC 1, has detected a fault in the Left Temperature Control Valve (TCV).

B. Possible Causes

(1) Left TCV, V139

(2) Air Conditioning Relay 1, M324

(3) Left PZTC 1, M1442

(4) Wiring

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	1	C01157	A/C TEMP CONT VALVE CLOSE LEFT
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
D	3	C01613	AIR COND PACK CONT VALVES L ALT

D. Related Data

(1) WDM 21-51-12

(2) SSM 21-51-12

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Initial Evaluation

(1) Make sure that the PACK Light is not illuminated on the Panel, P5.

(a) If the PACK Light is illuminated, press the TRIP RESET Switch on the Panel, P5 or MASTER CAUTION on the Glareshield Panel to reset the light.

NOTE: If the TRIP RESET does not extinguish the PACK light, there may be a nuisance Temperature Control Valve (TCV) fault after the BITE procedure is performed.

(2) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801 (Refer to the applicable sheet in 21-62 TASK SUPPORT Figure 305).

(a) If the Green GO Light comes ON, push the RESET Switch when the VERIFY and GO Lights are ON. No further action is necessary.

(b) If a Red Light comes ON when you push the VERIFY Switch on the controller, the fault is presently active. You can do this Circuit Breaker RESET Procedure to give the Left TCV, V139 an opportunity to get “unstuck” and clear the fault:

1) Open these circuit breakers, wait a minimum of two minutes, then close the circuit breakers.

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	1	C01157	A/C TEMP CONT VALVE CLOSE LEFT
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

a) Push the VERIFY Switch on the controller.

<1> If the Green GO Light comes ON, push the RESET Switch when the VERIFY and GO Lights are ON. No further action is necessary.

<2> If a Red Light comes ON, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Open this access panel:

Number	Name/Location
192CL	ECS Access Door

(2) Do a check of the electrical connector of the Left TCV, V139 (WDM 21-51-12).

(a) Disconnect connector D3862 from the Left TCV, V139 and do a visual inspection for bent pins, pushed-in pins, uncentered sockets, corrosion, moisture, or damages.

1) If you find a problem during the inspection, then do these steps:

a) Clean or repair the connector.

b) Re-connect the connector to the Left TCV, V139.

c) Do the Repair Confirmation at the end of this task.

2) If you did not find a problem during the inspection, then do these steps:

a) Re-connect the connector to the Left TCV, V139.

b) Do the Repair Confirmation at the end of this task.

(3) Do a voltage check of the Left TCV, V139 (WDM 21-51-12).

(a) Disconnect connector D3862 from the Left TCV, V139.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Do a check for 115V AC between pins 11 and 3 of the connector D3862.

(c) If you find 115V AC, replace the Left TCV, V139. These are the tasks:

- Temperature Control Valve Removal, AMM TASK 21-51-10-000-801
- Temperature Control Valve Installation, AMM TASK 21-51-10-400-801

1) Do the Repair Confirmation at the end of this task.

(d) If you do not find 115V AC, examine the wiring as follows:

LCC-R	LEFT TCV
D40444P	D3862
pin 13	pin 11

LEFT TCV	
D3862	
pin 3	AC GND

1) Repair the problems that you find.

2) Do the Repair Confirmation at the end of this task.

(4) Remove the Air Conditioning Relay 1, M324. This is the task: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801.

(5) Do a check for 28V DC at pin 22 of connector D458A (WDM 21-51-12).

(a) If you find 28V DC, install a new Air Conditioning Relay 1, M324. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

1) Do the Repair Confirmation at the end of this task.

(b) If you do not find 28V DC, examine the wiring as follows:

LCC-R	A/C RLY-1
CB	D458A
C1613	pin 22

1) Repair the problems that you find.

2) Do the Repair Confirmation at the end of this task.

(6) Do these wiring checks between the Air Conditioning Relay 1, M324, Left PZTC 1, M1442, and the Left TCV V139 (WDM 21-51-12):

(a) Disconnect connector D3862 from the Left TCV, V139.

(b) Remove the Left PZTC 1, M1442. This is the task: Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801.

(c) Remove the Air Conditioning Relay 1, M324. This is the task: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801.

(d) Examine the wiring as follows:

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A/C RLY-1 D458A	LEFT TCV D3862
pin 64	pin 1
pin 63	pin 2

PZTC-1 D3858B	LEFT TCV D3862
pin D10	pin 4
pin D9	pin 12

A/C RLY-1 D458A	PZTC-1 D3858B
pin 58	pin C2
pin 65	pin D13
pin 66	pin D12

LEFT TCV D3862	
pin 3	AC GND
pin 5	DC GND
pin 13	ST GND

1) If you find a problem with the wiring, then do these steps:
   a) Repair the wiring.
   b) Re-connect connector D3862 to the Left TCV, V139.
   c) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

   d) Re-install the Left PZTC 1, M1442. This is the task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.
   e) Re-install the Air Conditioning Relay 1, M324. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.
   f) Do the Repair Confirmation at the end of this task.
2) If you do not find a problem with the wiring, do these steps and then continue.
   a) Re-connect connector D3862 to the Left TCV, V139.
   b) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

   c) Re-install the Air Conditioning Relay 1, M324. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

(7) Replace the Left PZTC 1, M1442. These are the tasks:

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- Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801
- Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801

(a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the problem.

(b) If the maintenance message still shows, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

831. Right Temperature Control Valve - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) THIS PACK - TCV

(2) The Right Pack/Zone Temperature Controller, (PZTC)-2 has detected a fault in the Right TCV.

B. Possible Causes

(1) Right TCV, V141

(2) Air Conditioning Relay-2, M1455

(3) Wiring

(4) Pack/Zone Temperature Controller (PZTC)-2, M1443

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	1	C01160	A/C TEMP CONT VALVE CLOSE RIGHT
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC
D	4	C01614	AIR COND PACK CONT VALVES R ALT

D. Related Data

(1) WDM 21-51-22

(2) SSM 21-51-22

E. Initial Evaluation

(1) Make sure that the PACK Light is not illuminated on the Panel, P5.

(a) If the PACK Light is illuminated, press the TRIP RESET Switch on the Panel, P5 or MASTER CAUTION on the Glareshield Panel to reset the light.

NOTE: If the TRIP RESET does not extinguish the PACK light, there may be a nuisance Temperature Control Valve (TCV) fault after the BITE procedure is performed.

(2) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801 (Refer to the applicable sheet in 21-62 TASK SUPPORT Figure 305).

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OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999 (Continued)

(a) If the Green GO Light comes ON, push the RESET Switch when the VERIFY and GO Lights are ON. No further action is necessary.

(b) If a Red Light comes ON when you push the VERIFY Switch on the controller, the fault is presently active. You can do this Circuit Breaker RESET Procedure to give the Right TCV an opportunity to get “unstuck” and clear the fault:

1) Open these circuit breakers, wait a minimum of two minutes, then close the circuit breakers.

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	1	C01160	A/C TEMP CONT VALVE CLOSE RIGHT
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

a) Push the VERIFY Switch on the controller.

<1> If the Green GO Light comes ON, push the RESET Switch when the VERIFY and GO Lights are ON. No further action is necessary.

<2> If a Red Light comes ON, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Open these access panels:

Number	Name/Location
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

(2) Do a check of the electrical connector of the TCV (WDM 21-51-22).

(a) Disconnect connector D3872 from the right TCV and do a visual inspection for bent pins, pushed-in pins, uncentered sockets, corrosion, moisture, or damages.

1) If you find a problem during the inspection, then do these steps:

a) Clean or repair the connector.

b) Re-connect the connector to the right TCV.

c) Do the Repair Confirmation at the end of this task.

2) If you did not find a problem during the inspection, then do these steps:

a) Re-connect the connector to the right TCV.

b) Do the Repair Confirmation at the end of this task.

(3) Disconnect connector D3872 from the Right TCV (WDM 21-51-22).

(4) Do a check for 115V AC between pins 11 and 3 of the connector.

(a) If you find 115V AC, replace the Right TCV, V141. These are the tasks:

- Temperature Control Valve Removal, AMM TASK 21-51-10-000-801
- Temperature Control Valve Installation, AMM TASK 21-51-10-400-801

1) Do the Repair Confirmation at the end of this task.

(b) If you do not find 115V AC, examine the wiring as follows:

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LCC-R	RIGHT TCV
D41832P	D3872
pin 1	pin 11

RIGHT TCV	
D3872	
pin 3	AC GND

1) Repair the problems that you find.
2) Do the Repair Confirmation at the end of this task.

(5) Remove the Air Conditioning Relay-2, M1455. This is the task: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801.

(6) Do a check for 28V DC at pin 22 of connector D10002A (WDM 21-51-22).

(a) If you find 28V DC, install a new Air Conditioning Relay-2, M1455. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

1) Do the Repair Confirmation at the end of this task.

(b) If you do not find 28V DC, examine the wiring as follows:

LCC-R	A/C RLY-2
CB	D10002A
C1614	pin 22

1) Repair the problems that you find.
2) Do the Repair Confirmation at the end of this task.

(7) Do these wiring checks between the Air Conditioning Relay-2 M1455, PZTC-2, M1443, and the Right TCV, V141 (WDM 21-51-22):

(a) Disconnect connector D3872 from the Right TCV, V141.

(b) Remove the Pack/Zone Temperature Controller-2. This is the task: Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801.

(c) Remove the Air Conditioning Relay-2, M1455. This is the task: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801.

(d) Examine the wiring as follows:

A/C RLY-2	RIGHT TCV
D10002A	D3872
pin 64	pin 1
pin 63	pin 2

PZTC-2	RIGHT TCV
D3860B	D3872
pin D10	pin 4
pin D9	pin 12

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A/C RLY-2 D10002A	PZTC-2 D3860B
pin 58	pin C2
pin 65	pin D13
pin 66	pin D12

RIGHT TCV D3872	
pin 3	AC GND
pin 5	DC GND
pin 13	ST GND

1) If you find a problem with the wiring, then do these steps:
   a) Repair the wiring.
   b) Re-connect connector D3872 to the Right TCV.
   c) Close these access panels:

Number	Name/Location
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

   d) Re-install the PZTC-2. This is the task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.
   e) Re-install the Air Conditioning Relay-2, M1455. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.
   f) Do the Repair Confirmation at the end of this task.
2) If you do not find a problem with the wiring, do these steps and then continue.
   a) Re-connect connector D3872 to the Right TCV.
   b) Close these access panels:

Number	Name/Location
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

   c) Re-install the Air Conditioning Relay-2, M1455. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

(8) Replace PZTC-2, M1443. These are the tasks:
- Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801
- Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801

(a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.
   (a) If the maintenance message does not show, then you corrected the problem.

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(b) If the maintenance message still shows, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

832. Control Cabin Trim Air Valve (Primary) - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) FLT DECK TRIM VALVE

(2) The Right Pack/Zone Temperature Controller (PZTC)-2 (Primary Controller) has detected a fault in the Control Cabin TAV.

B. Possible Causes

(1) Control Cabin TAV, V143

(2) Air Conditioning Relay-1, M324

(3) Wiring

(4) Pack/Zone Temperature Controller (PZTC)-2, M1443

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	3	C01163	A/C ZONE TEMP VALVE/FAN CONT FLT DECK
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC
C	4	C01164	A/C ZONE TEMP DUCT OVHT FLT DECK

D. Related Data

(1) WDM 21-61-11

(2) SSM 21-61-11

E. Initial Evaluation

(1) Make sure that there is no ZONE TEMP Light illuminated on the Panel, P5.

(a) If the ZONE TEMP Light is illuminated, reset the light by pressing TRIP RESET on the Panel, P5, or MASTER CAUTION on the Glareshield Panel to reset the light.

NOTE: If the TRIP RESET switch is not pressed, there may be a nuisance Trim Air Valve (TAV) fault after the BITE procedure is performed.

(2) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801 (Refer to the applicable sheet in 21-62 TASK SUPPORT Figure 305).

(a) If the Green GO Light comes ON, push the RESET Switch when the VERIFY and GO Lights are ON. No further action is necessary.

(b) If a Red Light comes ON when you push the VERIFY Switch on the controller, the fault is presently active. You can do this Circuit Breaker RESET Procedure to give the Control Cabin TAV an opportunity to get “unstuck” and clear the fault:

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OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999 (Continued)

1) Open these circuit breakers, wait a minimum of two minutes, then close the circuit breakers.

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	3	C01163	A/C ZONE TEMP VALVE/FAN CONT FLT DECK
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

a) Push the VERIFY Switch on the controller.

<1> If the Green GO Light comes ON, push the RESET Switch when the VERIFY and GO Lights are ON. No further action is necessary.

<2> If a Red Light comes ON, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Open this access panel:

Number	Name/Location
192CL	ECS Access Door

(2) Do a check of the electrical connector of the TCV (WDM 21-61-11).

(a) Disconnect connector D3886 from the Control Cabin TCV and do a visual inspection for bent pins, pushed-in pins, uncentered sockets, corrosion, moisture, or damages.

1) If you find a problem during the inspection, then do these steps:

a) Clean or repair the connector.

b) Re-connect the connector to the Control Cabin TCV.

c) Do the Repair Confirmation at the end of this task.

2) If you did not find a problem during the inspection, then do these steps:

a) Re-connect the connector to the Control Cabin TCV.

b) Do the Repair Confirmation at the end of this task.

(3) Disconnect connector D3886 from the Control Cabin TAV, V143 (WDM 21-61-11).

(4) Do a check for 115V AC between pins 11 and 3 of the connector.

(a) If you find 115V AC, replace the Control Cabin TAV, V143. These are the tasks:

- Zone Trim Air Modulating Valve Removal, AMM TASK 21-61-21-000-801
- Zone Trim Air Modulating Valve Installation, AMM TASK 21-61-21-400-801

1) Do the Repair Confirmation at the end of this task.

(b) If you do not find 115V AC, examine the wiring as follows:

LCC-R	CONT CAB TAV
D40532P	D3886
pin 20	pin 11

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CONT CAB TAV	
D3872	
pin 3	AC GND

1) Repair the problems that you find.
2) Do the Repair Confirmation at the end of this task.

(5) Remove the Air Conditioning Relay-1, M324. This is the task: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801.

(6) Do a check for 28V DC at pin 19 of connector D458B (WDM 21-61-11).

(a) If you find 28V DC, install a new Air Conditioning Relay-1, M324. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

1) Do the Repair Confirmation at the end of this task.

(b) If you do not find 28V DC, examine the wiring as follows (WDM 21-61-11):

LCC-R	A/C RLY-1
D40444P	D458B
pin 3	pin 19

1) Repair the problems that you find.
2) Do the Repair Confirmation at the end of this task.

(7) Do these checks of the wiring between the Control Cabin TAV V143, PZTC-2 M1443 and the Air Conditioning Relay-1 M324 (WDM 21-61-11):

(a) Disconnect connector D3886 from the Control Cabin TAV, V143.

(b) Remove the PFTZ-2. This is the task:Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801.

(c) Remove the Air Conditioning Relay-1, M324. This is the task: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801.

(d) Examine the wiring as follows:

A/C RLY-1	CONT CAB TAV
D458A	D3886
pin 41	pin 1
pin 42	pin 2
pin 28	pin 10

PZTC-1	CONT CAB TAV
D3860B	D3886
pin B9	pin 12
pin B10	pin 4

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A/C RLY-1 D458A	PZTC-2 D3860B
pin 37	pin B12
pin 25	pin B13
pin 16	pin D2

CONT CAB TAV D3886	
pin 3	AC GND
pin 5	DC GND
pin 13	ST GND

1) If you find a problem with the wiring, then do these steps:
   a) Repair the wiring.
   b) Re-connect connector D3886 to the Control Cabin TAV, V143.
   c) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

   d) Re-install the PZTC-2. This is the task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.
   e) Re-install the Air Conditioning Relay-1, M324. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.
   f) Do the Repair Confirmation at the end of this task.
2) If you do not find a problem with the wiring, do these steps and then continue.
   a) Re-connect connector D3886 to the Control Cabin TAV, V143.
   b) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

   c) Re-install the Air Conditioning Relay-1, M324. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

(8) Replace the PZTC-2, M1443. These are the tasks:
- Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801
- Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801

(a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.
   (a) If the maintenance message does not show, then you corrected the problem.
   (b) If the maintenance message still shows, then continue the Fault Isolation Procedure at the subsequent step.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

833. Control Cabin Trim Air Valve (Backup) - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) FLT DECK TRIM VALVE

(2) The Left Pack/Zone Temperature Controller (PZTC)-1, M1442 (Backup Controller) has detected a fault in the Control Cabin TAV.

B. Possible Causes

(1) Control Cabin TAV, V143

(2) Air Conditioning Relay-1, M324

(3) Air Conditioning Relay-2, M1455

(4) Wiring

(5) Pack/Zone Temperature Controller (PZTC)-1, M1442

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	3	C01163	A/C ZONE TEMP VALVE/FAN CONT FLT DECK

D. Related Data

(1) WDM 21-61-11

(2) WDM 21-61-12

(3) SSM 21-61-11

(4) SSM 21-61-12

E. Initial Evaluation

(1) Make sure that there is no ZONE TEMP Light illuminated on the Panel, P5.

(a) If the ZONE TEMP Light is illuminated, reset the light by pressing TRIP RESET on the Panel, P5, or MASTER CAUTION on the Glareshield Panel to reset the light.

NOTE: If the TRIP RESET switch is not pressed, there may be a nuisance Trim Air Valve (TAV) fault after the BITE procedure is performed.

(2) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801 (Refer to the applicable sheet in 21-62 TASK SUPPORT Figure 305).

(a) If the Green GO Light comes ON, push the RESET Switch when the VERIFY and GO Lights are ON. No further action is necessary.

(b) If a Red Light comes ON when you push the VERIFY Switch on the controller, the fault is presently active. You can do this Circuit Breaker RESET Procedure to give the Control Cabin TAV an opportunity to get “unstuck” and clear the fault:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999 (Continued)

1) Open these circuit breakers, wait a minimum of two minutes, then close the circuit breakers.

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	3	C01163	A/C ZONE TEMP VALVE/FAN CONT FLT DECK

a) Push the VERIFY Switch on the controller.

<1> If the Green GO Light comes ON, push the RESET Switch when the VERIFY and GO Lights are ON. No further action is necessary.

<2> If a Red Light comes ON, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Open this access panel:

Number	Name/Location
192CL	ECS Access Door

(2) Do a check of the electrical connector of the TCV (WDM 21-61-11).

(a) Disconnect connector D3886 from the Control Cabin TCV and do a visual inspection for bent pins, pushed-in pins, uncentered sockets, corrosion, moisture, or damages.

1) If you find a problem during the inspection, then do these steps:

a) Clean or repair the connector.

b) Re-connect the connector to the Control Cabin TCV.

c) Do the Repair Confirmation at the end of this task.

2) If you did not find a problem during the inspection, then do these steps:

a) Re-connect the connector to the Control Cabin TCV.

b) Do the Repair Confirmation at the end of this task.

(3) Disconnect connector D3886 from the Control Cabin TAV, V143 (WDM 21-61-11).

(4) Do a check for 115V AC between pins 11 and 3 of the connector.

(a) If you find 115V AC, replace the Control Cabin TAV, V143. These are the tasks:

- Zone Trim Air Modulating Valve Removal, AMM TASK 21-61-21-000-801
- Zone Trim Air Modulating Valve Installation, AMM TASK 21-61-21-400-801

1) Do the Repair Confirmation at the end of this task.

(b) If you do not find 115V AC, examine the wiring as follows:

LCC-R	CONT CAB TAV
D40532P	D3886
pin 20	pin 11

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CONT CAB TAV
D3872
pin 3 . . . . . AC GND

1) Repair the problems that you find.
2) Do the Repair Confirmation at the end of this task.

(5) Remove the Air Conditioning Relay-1, M324. This is the task: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801.

(6) Do a check for 28V DC at pin 19 of connector D458B (WDM 21-61-11).

(a) If you find 28V DC, install a new Air Conditioning Relay-1, M324. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

1) Do the Repair Confirmation at the end of this task.

(b) If you do not find 28V DC, examine the wiring as follows (WDM 21-61-11):

LCC-R	**A/C RLY-1**
D40444P	**D458B**
pin 3	pin 19

1) Repair the problems that you find.
2) Do the Repair Confirmation at the end of this task.

(7) Replace the Air Conditioning Relay-2, M1455. These are the tasks:

- Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801
- Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801

(a) Do the Repair Confirmation at the end of this task.

(8) Do these checks of the wiring between the Control Cabin TAV V143, PZTC-1 M1442 and the Air Conditioning Relays-1, M324 and -2, M1455:

(a) Remove the PZTC-1. This is the task: Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801.

(b) Remove the Air Conditioning Relay-1, M324. This is the task: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801.

(c) Remove the Air Conditioning Relay-2, M1455. This is the task: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801.

(d) Examine the wiring between the Air Conditioning Relay-2, M1455 and PZTC-1, M1442 as follows (WDM 21-61-12):

A/C RELAY-2	**PZTC-1**
D10002A	**D3858B**
pin 25	pin B13
pin 37	pin B12
pin 16	pin D2

(e) Examine the wiring as follows (WDM 21-61-11):

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A/C RLY-1	**CONT CAB TAV**
D458A	**D3886**
pin 41	pin 1
pin 42	pin 2
pin 28	pin 10

PZTC-1	**CONT CAB TAV**
D3860B	**D3886**
pin B9	pin 12
pin B10	pin 4

A/C RELAY-1	**A/C RELAY-2**
D458A	**D10002B**
pin 28	pin 41

A/C RELAY-1	**A/C RELAY-2**
D458B	**D10002A**
pin 41	pin 27
pin 42	pin 26

CONT CAB TAV	
D3886	
pin 3	AC GND
pin 5	DC GND
pin 13	ST GND

1) If you find a problem with the wiring, then do these steps:
   a) Repair the wiring.
   b) Re-connect connector D3886 to the Control Cabin TAV, V143.
   c) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

   d) Re-install the PZTC-1. This is the task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.
   e) Re-install the Air Conditioning Relay-1, M324. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.
   f) Re-install the Air Conditioning Relay-2, M1455. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.
   g) Do the Repair Confirmation at the end of this task.
2) If you do not find a problem with the wiring, do these steps and then continue.
   a) Re-connect connector D3886 to the Control Cabin TAV, V143.

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b) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

c) Re-install the Air Conditioning Relay-1, M324. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

d) Re-install the Air Conditioning Relay-2, M1455. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

(9) Replace the PZTC-2, M1442. These are the tasks:

- Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801
- Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801

(a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the problem.

(b) If the maintenance message still shows, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

834. Forward Cabin Trim Air Valve - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) ZONE TRIM VALVE

(2) The Pack/Zone Temperature Controller (PZTC)-2 has detected a fault in the Forward Cabin TAV.

B. Possible Causes

(1) Forward Cabin TAV, V144

(2) Air Conditioning Relay-2, M1455

(3) Wiring

(4) Pack/Zone Temperature Controller (PZTC)-2, M1443

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	2	C01167	A/C ZONE TEMP VALVE/FAN CONT FWD PASS
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

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D. Related Data

(1) WDM 21-61-13

(2) SSM 21-61-13

E. Initial Evaluation

(1) Make sure that there is no ZONE TEMP Light illuminated on the Panel, P5.

(a) If the ZONE TEMP Light is illuminated, reset the light by pressing TRIP RESET on the Panel, P5, or MASTER CAUTION on the Glareshield Panel to reset the light.

NOTE: If the TRIP RESET switch is not pressed, there may be a nuisance Trim Air Valve (TAV) fault after the BITE procedure is performed.

(2) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801 (Refer to the applicable sheet in 21-62 TASK SUPPORT Figure 305).

(a) If the Green GO Light comes ON, push the RESET Switch when the VERIFY and GO Lights are ON. No further action is necessary.

(b) If a Red Light comes ON when you push the VERIFY Switch on the controller, the fault is presently active. You can do this Circuit Breaker RESET Procedure to give the Forward Cabin TAV an opportunity to get “unstuck” and clear the fault:

1) Open these circuit breakers, wait a minimum of two minutes, then close the circuit breakers.

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	2	C01167	A/C ZONE TEMP VALVE/FAN CONT FWD PASS
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

a) Push the VERIFY Switch on the controller.

<1> If the Green GO Light comes ON, push the RESET Switch when the VERIFY and GO Lights are ON. No further action is necessary.

<2> If a Red Light comes ON, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Open these access panels:

Number	Name/Location
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

(2) Do a check of the electrical connector of the TAV (WDM 21-61-13).

(a) Disconnect connector D3888 from the Fwd Cabin TAV and do a visual inspection for bent pins, pushed-in pins, uncentered sockets, corrosion, moisture, or damages.

1) If you find a problem during the inspection, then do these steps:

a) Clean or repair the connector.

b) Re-connect the connector to the Fwd Cabin TAV.

c) Do the Repair Confirmation at the end of this task.

2) If you did not find a problem during the inspection, then do these steps:

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a) Re-connect the connector to the Fwd Cabin TAV.

b) Do the Repair Confirmation at the end of this task.

(3) Disconnect connector D3888 from the Fwb Cabin TAV, V144 (WDM 21-61-13).

(4) Do a check for 115V AC between pins 11 and 3 of the connector.

(a) If you find 115V AC, replace the Fwd Cabin TAV, V144. These are the tasks:

- Zone Trim Air Modulating Valve Removal, AMM TASK 21-61-21-000-801
- Zone Trim Air Modulating Valve Installation, AMM TASK 21-61-21-400-801

1) Do the Repair Confirmation at the end of this task.

(b) If you do not find 115V AC, examine the wiring as follows:

LCC-R	**FWD TAV**
D41832P	**D3888**
pin 6	pin 11

FWD TAV	
D3888	
pin 3	AC GND

1) Repair the problems that you find.

2) Do the Repair Confirmation at the end of this task.

(5) Remove the Air Conditioning Relay-2, M1455. This is the task: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801.

(6) Do a check for 28V DC at pin 20 of connector D10002B (WDM 21-61-13).

(a) If you find 28V DC, install a new Air Conditioning Relay-2, M1455. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

1) Do the Repair Confirmation at the end of this task.

(b) If you do not find 28V DC, examine the wiring as follows (WDM 21-61-13):

LCC-R	**A/C RLY-2**
D41832P	**D10002B**
pin 15	pin 20

1) Repair the problems that you find.

2) Do the Repair Confirmation at the end of this task.

(7) Do these checks of the wiring between the Fwd Cabin TAV V144, the PZTC-2 M1443 and the Air Conditioning Relay-2 M1455 (WDM 21-61-13):

(a) Remove the Pack/Zone Temperature Controller-2. This is the task:Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801.

(b) Remove the Air Conditioning Relay-2, M1455. This is the task: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801.

(c) Examine the wiring as follows:

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A/C RLY-2 D10002A	FWD CAB TAV D3888
pin 52	pin 1
pin 51	pin 2
pin 67	pin 10

PZTC-2 D3860B	FWD CAB TAV D3888
pin C9	pin 12
pin C10	pin 4

A/C RLY-2 D10002A	PZTC-2 D3860B
pin 53	pin C12
pin 54	pin C13
pin 11	pin B2

FWD CAB TAV D3888	
pin 3	AC GND
pin 5	DC GND
pin 13	ST GND

1) If you find a problem with the wiring, then do these steps:
   a) Repair the wiring.
   b) Re-connect connector D3888 to the Fwd Cabin TAV, V144.
   c) Close these access panels:

Number	Name/Location
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

   d) Re-install the PZTC-2. This is the task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.
   e) Re-install the Air Conditioning Relay-2, M1455. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.
   f) Do the Repair Confirmation at the end of this task.
2) If you do not find a problem with the wiring, do these steps and then continue.
   a) Re-connect connector D3888 to the Fwd Cabin TAV, V144.
   b) Close these access panels:

Number	Name/Location
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

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c) Re-install the Air Conditioning Relay-2, M1455. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

(8) Replace the PZTC-2, M1443. These are the tasks:

- Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801
- Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801

(a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the problem.

(b) If the maintenance message still shows, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

835. Aft Cabin Trim Air Valve - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) ZONE TRIM VALVE

(2) The Pack/Zone Temperature Controller (PZTC)-1 has detected a fault in the Aft Cabin TAV.

B. Possible Causes

(1) Aft Cabin TAV, V145

(2) Air Conditioning Relay-1, M324

(3) Wiring

(4) Pack/Zone Temperature Controller (PZTC)-1, M1442

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	3	C01170	A/C ZONE TEMP VALVE/FAN CONT AFT PASS
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

D. Related Data

(1) WDM 21-61-14

(2) SSM 21-61-14

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Initial Evaluation

(1) Make sure that there is no ZONE TEMP Light illuminated on the Panel, P5.

(a) If the ZONE TEMP Light is illuminated, reset the light by pressing TRIP RESET on the Panel, P5, or MASTER CAUTION on the Glareshield Panel to reset the light.

NOTE: If the TRIP RESET switch is not pressed, there may be a nuisance Trim Air Valve (TAV) fault after the BITE procedure is performed.

(2) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801 (Refer to the applicable sheet in 21-62 TASK SUPPORT Figure 305).

(a) If the Green GO Light comes ON, push the RESET Switch when the VERIFY and GO Lights are ON. No further action is necessary.

(b) If a Red Light comes ON when you push the VERIFY Switch on the controller, the fault is presently active. You can do this Circuit Breaker RESET Procedure to give the Aft Cabin TAV an opportunity to get “unstuck” and clear the fault:

1) Open these circuit breakers, wait a minimum of two minutes, then close the circuit breakers.

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	3	C01170	A/C ZONE TEMP VALVE/FAN CONT AFT PASS
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

a) Push the VERIFY Switch on the controller.

<1> If the Green GO Light comes ON, push the RESET Switch when the VERIFY and GO Lights are ON. No further action is necessary.

<2> If a Red Light comes ON, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Open these access panels:

Number	Name/Location
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

(2) Do a check of the electrical connector of the TAV (WDM 21-61-14).

(a) Disconnect connector D3890 from the Aft Cabin TAV and do a visual inspection for bent pins, pushed-in pins, uncentered sockets, corrosion, moisture, or damages.

1) If you find a problem during the inspection, then do these steps:

a) Clean or repair the connector.

b) Re-connect the connector to the Aft Cabin TAV.

c) Do the Repair Confirmation at the end of this task.

2) If you did not find a problem during the inspection, then do these steps:

a) Re-connect the connector to the Aft Cabin TAV.

b) Do the Repair Confirmation at the end of this task.

(3) Disconnect connector D3890 from the Aft Cabin TAV, V145 (WDM 21-61-14).

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(4) Do a check for 115V AC between pins 11 and 3 of the connector.

(a) If you find 115V AC, replace the Aft Cabin TAV, V145. These are the tasks:

- Zone Trim Air Modulating Valve Removal, AMM TASK 21-61-21-000-801
- Zone Trim Air Modulating Valve Installation, AMM TASK 21-61-21-400-801

1) Do the Repair Confirmation at the end of this task.

(b) If you do not find 115V AC, examine the wiring as follows:

LCC-R	**AFT CAB TAV**
D41832P	**D3890**
pin 8	pin 11

FWD TAV	
D3888	
pin 3	AC GND

1) Repair the problems that you find.

2) Do the Repair Confirmation at the end of this task.

(5) Remove the Air Conditioning Relay-1, M324. This is the task: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801.

(6) Do a check for 28V DC at pin 20 of connector D458B (WDM 21-61-14).

(a) If you find 28V DC, install a new Air Conditioning Relay-1, M324. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

1) Do the Repair Confirmation at the end of this task.

(b) If you do not find 28V DC, examine the wiring as follows (WDM 21-61-14):

LCC-R	**A/C RLY-1**
D41832P	**D458B**
pin 14	pin 20

1) Repair the problems that you find.

2) Do the Repair Confirmation at the end of this task.

(7) Do these checks of the wiring between the Aft Cabin TAV V145, the PZTC-1 M1442 and the Air Conditioning Relay-1 M324 (WDM 21-61-14):

(a) Disconnect connector D3890 from the Aft Cabin TAV, V145.

(b) Remove the Pack/Zone Temperature Controller-1. This is the task:Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801.

(c) Remove the Air Conditioning Relay-1, M324. This is the task: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801.

(d) Examine the wiring as follows (WDM 21-61-14):

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A/C RLY-1 D458A	AFT CAB TAV D3890
pin 52	pin 1
pin 51	pin 2
pin 67	pin 10

PZTC-1 D3858B	AFT CAB TAV D3890
pin C9	pin 12
pin C10	pin 4

A/C RLY-1 D458A	PZTC-1 D3858B
pin 53	pin C12
pin 54	pin C13
pin 11	pin B2

AFT CAB TAV D3890	
pin 3	AC GND
pin 5	DC GND
pin 13	ST GND

1) If you find a problem with the wiring, then do these steps:
   a) Repair the wiring.
   b) Re-connect connector D3890 to the Aft Cabin TAV, V145.
   c) Close these access panels:

Number	Name/Location
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

   d) Re-install the PZTC-1. This is the task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.
   e) Re-install the Air Conditioning Relay-1, M324. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.
   f) Do the Repair Confirmation at the end of this task.
2) If you do not find a problem with the wiring, do these steps and then continue.
   a) Re-connect connector D3890 to the Aft Cabin TAV, V145.
   b) Close these access panels:

Number	Name/Location
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

   c) Re-install the Air Conditioning Relay-1, M324. This is the task: Air Conditioning

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Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

(8) Replace the PZTC-1, M1442. These are the tasks:

- Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801
- Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801

(a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the problem.

(b) If the maintenance message still shows, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

OMR 802, 805, 807, 809, 810, 812, 813

836. Right Pack Ram Air Actuator - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) RAM AIR ACT

(2) The Right PZTC, M1443 (No. 2) has detected a fault from the Right Pack Ram Air Inlet Actuator, M327.

B. Possible Causes

(1) Right Pack Ram Air Inlet Actuator, M327

(2) Air Conditioning Relays No. 2, M1455

(3) Right PZTC, M1443 (No. 2).

(4) Wiring

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

D. Related Data

(1) SSM 21-51-24

(2) WDM 21-51-24

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(2) If the maintenance message does not show, then there was an intermittent fault.

(3) If the maintenance message shows, then do the Fault Isolation Procedure below.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 802, 805, 807, 809, 810, 812, 813 (Continued)

F. Fault Isolation Procedure

(1) Replace the Right Pack Ram Air Inlet Actuator, M327. These are the tasks:

- Ram Air Inlet Actuator Removal, AMM TASK 21-51-21-000-801
- Ram Air Inlet Actuator Installation, AMM TASK 21-51-21-400-801

(a) Do the Repair Confirmation at the end of this task.

(2) Replace the Air Conditioning Relays No. 2, M1455. These are the tasks:

- Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801
- Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801

(a) Do the Repair Confirmation at the end of this task.

(3) Do these checks of the wiring between the Right Pack Ram Air Inlet Actuator, M327, the Right Pack/Zone Temperature Controller, M1443, and the Air Conditioning Relays No. 2, M1455:

(a) Remove the Right Pack/Zone Temperature Controller, M1443. This is the task: Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801.

(b) Remove the Air Conditioning Relays No. 2, M1455. This is the task: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801.

(c) Do a wiring check between these pins of electrical connector D3860B of the Right Pack/Zone Temperature Controller, M1443 and connector D10002A of the Air Conditioning Relays No. 2, M1455:

D10002A	D3860B
pin 61	pin D8
pin 47	pin D6
pin 12	pin A13
pin 13	pin A12

(d) Do a wiring check between these pins of electrical connector D968 of the Right Pack Ram Air Inlet Actuator, M327 and connector D10002A of the Air Conditioning Relays No. 2, M1455:

D10002A	D968
pin 36	pin 1
pin 24	pin 6
pin 24	pin 5
pin 23	pin 10

(e) Make sure pins 8 and 11 of connector D968 for the Right Pack Ram Air Inlet Actuator, M327 go to ground.

(f) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-connect the electrical connector to the Right Pack Ram Air Inlet Actuator, M327.

3) Close this access panel:

Number	Name/Location
191GR	Ram Air Actuator Panel - Forward

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 802, 805, 807, 809, 810, 812, 813 (Continued)

4) Re-install the Right Pack/Zone Temperature Controller, M1443. This is the task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.

5) Re-install the Air Conditioning Relays No. 2, M1455. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

6) Do the Repair Confirmation at the end of this task.

(g) If you do not find a problem with the wiring, then do these steps and continue.

1) Re-connect the electrical connector to the Right Pack Ram Air Inlet Actuator, M327.

2) Close this access panel:

Number	Name/Location
191GR	Ram Air Actuator Panel - Forward

3) Re-install the Air Conditioning Relays No. 2, M1455. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

(4) Install a new Right Pack/Zone Temperature Controller, M1443. These are the tasks:

- Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801
- Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801

(a) Do the Repair Confirmation at the end of this task.

(5) Do this task: Ram Air Inlet Door Linkage Inspection, AMM TASK 21-51-22-200-802.

(a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the problem.

(b) If the maintenance message still shows, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

837. Left Pack Ram Air Actuator - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) RAM AIR ACT

(2) The Left PZTC, M1442 (No. 1) has detected a fault from the Left Pack Ram Air Inlet Actuator, M326.

B. Possible Causes

(1) Left Pack Ram Air Inlet Actuator, M326

(2) Air Conditioning Relays No. 1, M324

(3) Left PZTC, M1442 (No. 1).

(4) Wiring

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 802, 805, 807, 809, 810, 812, 813 (Continued)

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

D. Related Data

(1) SSM 21-51-14

(2) WDM 21-51-14

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(2) If the maintenance message does not show, then there was an intermittent fault.

(3) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Replace the Left Pack Ram Air Inlet Actuator, M326. These are the tasks:

- Ram Air Inlet Actuator Removal, AMM TASK 21-51-21-000-801
- Ram Air Inlet Actuator Installation, AMM TASK 21-51-21-400-801

(a) Do the Repair Confirmation at the end of this task.

(2) Replace the Air Conditioning Accessory Unit (ACAU) No. 1, M324. These are the tasks:

- Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801
- Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801

(a) Do the Repair Confirmation at the end of this task.

(3) Do these checks of the wiring between the Left Pack Ram Air Inlet Actuator, M326, the Left PZTC, M1442, and the Air Conditioning Relays No. 1, M324:

(a) Remove the Left PZTC, M1442. This is the task: Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801.

(b) Remove the Air Conditioning Relays No. 1, M324. This is the task: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801.

(c) Do a wiring check between these pins of electrical connector D3858B of the Left PZTC, M1442 and connector D458A of the Air Conditioning Relays No. 1, M324:

D458A	D3858B
pin 61	pin D8
pin 47	pin D6
pin 12	pin A13
pin 13	pin A12

(d) Do a wiring check between these pins of electrical connector D966 of the Left Pack Ram Air Inlet Actuator, M326 and connector D458A of the Air Conditioning Relays No. 1, M324:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 802, 805, 807, 809, 810, 812, 813 (Continued)

D458A	D966
pin 36	pin 1
pin 24	pin 6
pin 24	pin 5
pin 23	pin 10

(e) Make sure pins 8 and 11 of connector D966 for the Left Pack Ram Air Inlet Actuator, M326 go to ground.

(f) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-connect the electrical connector to the Left Pack Ram Air Inlet Actuator, M326.

3) Close this access panel:

Number	Name/Location
191GL	Ram Air Actuator Panel - Forward

4) Re-install the Left PZTC, M1442. This is the task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.

5) Re-install the Air Conditioning Relays No. 1, M324. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

6) Do the Repair Confirmation at the end of this task.

(g) If you do not find a problem with the wiring, then do these steps and continue:

1) Re-connect the electrical connector to the Left Pack Ram Air Inlet Actuator, M326.

2) Close this access panel:

Number	Name/Location
191GL	Ram Air Actuator Panel - Forward

3) Re-install the Air Conditioning Relays No. 1, M324. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

(4) Install a new Left PZTC, M1442. These are the tasks:

- Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801
- Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801

(a) Do the Repair Confirmation at the end of this task.

(5) Do this task: Ram Air Inlet Door Linkage Inspection, AMM TASK 21-51-22-200-802.

(a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the problem.

(b) If the maintenance message still shows, then continue the Fault Isolation Procedure at the subsequent step.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

838. Trim Air Valve Does Not Operate - Fault Isolation

A. Description

(1) The Trim Air Pressure Regulating and Shutoff Valve (PRSOV) does not operate.

(2) The Trim Air PRSOV opens to let hot Bleed Air to the three Zone Trim Air Modulating Valves.

(a) When Temperature Control is not possible for all three Cabin Zones, then, the Trim Air PRSOV is the usual cause.

B. Possible Causes

(1) Trim Air PRSOV, V146

(2) Trim Air Switch, S5

(3) Wiring

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	9	C01172	AIR CONDITIONING TRIM AIR PRESS

D. Related Data

(1) WDM 21-61-15

(2) SSM 21-61-15

E. Initial Evaluation

(1) Put the TRIM AIR Switch to OFF.

(2) Supply Pneumatic Power. This is the task: Supply Pressure to the Pneumatic System (Selection), AMM TASK 36-00-00-860-801.

(3) Operate the Air Conditioning Packs. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(4) To get access to the Trim Air PRSOV, open these panels in this sequence:

Open this access panel:

Number	Name/Location
192CR	ECS Access Door

Open this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

(5) Put the TRIM AIR Switch to ON.

(6) On the P5-17 Cabin Temperature Panel, put the CONT CAB, FWD CAB and AFT CAB Temperature Selectors to AUTO (12:00) position.

(a) Make sure that the Visual Position Indicator on the Trim Air PRSOV moves in the direction of the open position.

(7) On the P5-17 Cabin Temperature Panel, put the FWD CAB Switch to the fully warm (W) position.

(a) Make sure that the Visual Position Indicator on the Trim Air PRSOV moves closer to the fully open position.

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(8) If the Trim Air PRSOV operates correctly, then there was an intermittent fault.

(9) If the Trim Air PRSOV does not operate correctly, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Make sure that the TRIM AIR Switch is in the ON position.

(2) Make sure that this circuit breaker is closed:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	9	C01172	AIR CONDITIONING TRIM AIR PRESS

(3) Do a check for 28 VDC between pins 1 and 3 (Ground) of the Trim Air PRSOV connector, D3914 (WDM 21-61-15):

(a) If you find 28V DC, replace the Trim Air PRSOV. These are the tasks:

- Trim Air Pressure Regulating and Shutoff Valve Removal, AMM TASK 21-62-01-000-801
- Trim Air Pressure Regulating and Shutoff Valve Installation, AMM TASK 21-62-01-400-801

1) Do the Repair Confirmation at the end of this task.

(b) If you do not find 28V DC, do these steps:

1) Open this circuit breaker and install safety lock:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	9	C01172	AIR CONDITIONING TRIM AIR PRESS

2) Disconnect connector D3914 from the Trim Air PRSOV, V146.

3) Disconnect connector D3998 from the Trim Air Switch, S5.

4) Examine the wiring between the Trim Air PRSOV, V146 and the TRIM AIR Switch, S5 as follows (WDM 21-61-15):

TRIM AIR PRSOV D3914		**TRIM AIR SW D3998**
pin 1	. . . . .	pin 16

TRIM AIR PRSOV D3914		
pin 3	. . . . .	GD6524-DC
pin 2	. . . . .	GD2358-ST

a) If you find a problem in the wiring, repair it as necessary.

<1> Connect connector D3914 to the Trim Air PRSOV, V146.

<2> Connect connector D3998 to the Trim Air Switch, S5.

<3> Do the Repair Confirmation at the end of this task.

b) If you do not find a problem in the wiring, then connect connector D3914 to the Trim Air PRSOV, V146 and continue.

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(4) Do this check of the Trim Air Switch, S5:

(a) Lower the P5 Forward Overhead Panel to get access to the Trim Air Switch electrical connections.

(b) Do these continuity checks with the TRIM AIR Switch set in the indicated positions:

S5 TERMINAL	S5 TERMINAL	SWITCH POSITION
term 5	term 6	OFF
term 1	term 2	ON

1) If the continuity checks are not as specified above, then replace the Trim Air Switch, S5.

a) Close the P5 Overhead Panel.

b) Connect connector D3998 to the Trim Air Switch, S5.

c) Do the Repair Confirmation at the end of this task.

2) If the continuity checks are correct, then continue.

(5) Remove the safety lock and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	9	C01172	AIR CONDITIONING TRIM AIR PRESS

(6) Do a check for 28V DC at pin 13 of the Trim Air Switch connector, D3998 (WDM 21-61-15):

(a) If you find 28V DC, replace the Trim Air Switch, S5.

1) Connect connector D3998 to the Trim Air Switch, S5.

2) Close the P5 Overhead Panel.

3) Do the Repair Confirmation at the end of this task.

(b) If you do not find 28V DC, do this check of the Power Wiring (WDM 21-61-15):

1) Open this circuit breaker and install safety lock:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	9	C01172	AIR CONDITIONING TRIM AIR PRESS

2) Disconnect connector D40536P at the Load Control Center-Right (LCC-R).

3) Examine the wiring from the Load Control Center-Right (LCC-R) to the TRIM AIR Switch, S5 as follows (WDM 21-61-15):

LCC-R D40536P	TRIM AIR SW D3998
pin 24	pin 13

a) Repair the problems that you find.

b) Connect connector D3998 to the Trim Air Switch, S5.

c) Close the P5 Overhead Panel.

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d) Remove the safety lock and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	9	C01172	AIR CONDITIONING TRIM AIR PRESS

4) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Supply Pneumatic Power. This is the task: Supply Pressure to the Pneumatic System (Selection), AMM TASK 36-00-00-860-801

(2) Operate the Air Conditioning Packs. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(3) To get access to the Trim Air PRSOV, open these panels in this sequence:

Open this access panel:

Number	Name/Location
192CR	ECS Access Door

Open this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

(4) Put the TRIM AIR Switch to ON.

(5) On the P5-17 Cabin Temperature Panel, put the CONT CAB, FWD CAB and AFT CAB Temperature Selectors to AUTO (12:00) position.

(a) Make sure that the Visual Position Indicator on the Trim Air PRSOV moves in the direction of the open position.

(6) On the P5-17 Cabin Temperature Panel, put the FWD CAB Switch to the fully warm (W) position.

(a) Make sure that the Visual Position Indicator on the Trim Air PRSOV moves in the direction of the open position.

(7) If the Visual Position Indicator on the Trim Air PRSOV moves in the direction of the open position, then you corrected the problem. Do these steps to return the airplane to its usual condition:

(a) On the P5-17 Cabin Temperature Panel, put the CONT CAB, FWD CAB and AFT CAB Temperature Selectors to AUTO (12:00) position.

(b) Shutdown the Air Conditioning Packs. This is the task: Remove Conditioned Air from the Airplane, AMM TASK 21-00-00-800-802.

(c) Remove the Pneumatic Power if it is not necessary. This is the task: Remove Pressure from the Pneumatic System, AMM TASK 36-00-00-860-806

(d) Close these panels in the specified sequence:

Close this access panel:

Number	Name/Location
192CR	ECS Access Door

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Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

(8) If the Trim Air PRSOV does not operate correctly, then continue the Fault Isolation Procedure below at the subsequent step..

——— END OF TASK ———

903. ZONE TEMP Light(s) Come ON During MASTER CAUTION Recall

A. Description

(1) One, Two, or ALL Three ZONE TEMP Lights on the Temperature Control Panel, P5-17, intermittently come ON during MASTER CAUTION Recall.

NOTE: Trim Air Valve maintenance messages can show when you do the BITE Test of the Temperature Controllers.

(2) The MASTER CAUTION Recall is required during Airplane Departure and Landing Approach.

B. Possible Causes

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818, 819, 901-999

(1) Fault(s) in L (R) PZTC, M1442 (M1443).

OMR 820-826

(2) Fault(s) in L (R) Pack Flow Temperature Controller (PFTC), M2951 (M2952).

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

C. Initial Evaluation

OMR 820-826

(1) Do these steps to clear the fault:

NOTE: PACK and/or ZONE TEMP Lights can come ON, if one or all of the Temperature Selectors (CONT CAB, FWD CAB, AFT CAB) are set to OFF.

(a) On the Cabin Temperature Panel, P5-17, make sure that all three Temperature Selectors (CONT CAB, FWD CAB, AFT CAB) are not set to OFF.

1) If a selector is set to OFF, move it from the OFF position.

(b) On the Front Panel Display of the applicable Pack Flow Temperature Controller, M2951 (M2952), clear the EXISTING FAULTS (21-62 TASK SUPPORT Figure 303 or 21-62 TASK SUPPORT Figure 304)

(c) On the Glareshield Panel, P7, push the Master Caution Light to reset it.

(d) Wait Thirty (30) seconds.

(e) Make sure that the ZONE TEMP Lights stay OFF.

(f) Look at the EXISTING FAULTS again.

1) If the maintenance message does not show, then there was an intermittent fault.

2) If a maintenance message shows or if a ZONE TEMP Light stays ON, then continue.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(2) On the P7 Glareshield Panel, push and release the Left or the Right Annunciator.

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(3) If one or more ZONE TEMP Light on the Temperature Control Panel intermittently comes ON, make a record and then do the Fault Isolation Procedure below.

D. Fault Isolation Procedure

(1) Do the Pack Temperature Control - BITE Procedure, 21-62 TASK 801 on the Left and the Right Temperature Controllers.

(a) If maintenance messages show, refer to the table at the end of the BITE Task to find the applicable FIM Tasks for the messages that show.

(2) Do the Pack Temperature Control - BITE Procedure, 21-62 TASK 801 again to make sure that you corrected the problem.

(3) On the P7 Glareshield Panel, push and release the Left or the Right Annunciator.

(4) Make sure that none of the ZONE TEMP Lights on the Temperature Control Panel, P5-17, come ON.

——— END OF TASK ———

904. ZONE TEMP Light(s) Come ON Without MASTER CAUTION Recall

A. Description

(1) ZONE TEMP Light(s) come ON without MASTER CAUTION Recall.

(2) ZONE TEMP Light(s) can come ON prior to the first flight of the day after aircraft parked overnight without power.

B. Possible Causes

(1) Control Cabin (Forward Cabin, Aft Cabin) Duct Overheat Switch, S931 (S1076, S1077).

(2) Wiring

(3) Temperature Control Panel, P5-17

(4) Loss of 28V DC Power to the Temperature Control Panel, P5-17

(5) Air Conditioning Panel, P5-10

C. Initial Evaluation

(1) Push the TRIP RESET switch on the air conditioning module on the Panel, P5.

NOTE: If the TRIP RESET switch is not pressed, there may be a nuisance Trim Air Valve (TAV) fault after the BITE procedure is performed.

(2) If the ZONE TEMP Light(s) came ON before the first flight of the day after the aircraft was parked overnight without power, do the Pack Temperature Control - BITE Procedure, 21-62 TASK 801 on the Left and the Right Temperature Controllers.

(a) If maintenance messages do not show, then no more action is required.

(b) If maintenance messages show, refer to the table at the end of the Built-In-Test Equipment (BITE) Task to find the applicable FIM Tasks for the messages that show.

D. Fault Isolation Procedure

(1) For the Control Cabin CONT CAB ZONE TEMP Light illumination only - Do the Pack Temperature Control - BITE Procedure, 21-62 TASK 801 on the Left and the Right Temperature Controllers.

(a) If maintenance messages show, refer to the table at the end of the BITE Task to find the applicable FIM Tasks for the messages that show.

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1) If you find a fault at the primary and backup flight compartment temperature control, then do the applicable FIM Task Procedures to correct the problem. There is no need to continue at the Fault Isolation Procedure below.

(b) If maintenance messages do not show, then continue.

(2) Do the Operational Test of the Control Cabin (Fwd Cabin, Aft Cabin) Duct Overheat Switch, S931 (S1076, S1077). This is the task: Duct Overheat Switch - Operational Test, AMM TASK 21-61-00-700-811-002 or Duct Overheat Switch - Operational Test, AMM TASK 21-61-00-710-804-002.

(a) If the ZONE TEMP Light did not come ON between 180°F (82°C) and 200°F (93°C), then replace the that Duct Overheat Switch. These are the tasks:

- Duct Overheat Switch Removal, AMM TASK 21-61-05-000-801.
- Duct Overheat Switch Installation, AMM TASK 21-61-05-400-801.

(b) If the ZONE TEMP Light came ON between 180°F (82°C) and 200°F (93°C), then continue.

(3) Do this check of the wiring:

(a) Disconnect these connectors from these Duct Overheat Switches:

- Connector D3892 from Control Cabin Duct Overheat Switch, S931.
- Connector D11834 from Forward Cabin Duct Overheat Switch, S1076.
- Connector D11836 from Aft Cabin Duct Overheat Switch, S1077.

(b) Remove the L (R) ACAU, M324 (M1455). This is the task: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801.

(c) Do a wiring check between the Overheat Switch and the ACAU. (WDM 21-61-11, WDM 21-61-13 and WDM 21-61-14):

Control Cabin Duct Overheat Switch

S931	**M324**
D3892	**D458B**
Pin 1	Pin 61
Pin 2	Ground

Forward Cabin Duct Overheat Switch

S1076	**M1455**
D11834	**D10002B**
Pin 1	Pin 48
Pin 2	Ground

Aft Cabin Duct Overheat Switch

S1077	**M324**
D11836	**D458B**
Pin 1	Pin 48
Pin 2	Ground

1) If you find a problem with the wiring, then repair the wiring.

2) If you do not find a problem with the wiring, then continue.

3) Reconnect the connectors to the duct overheat switches.

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4) Reinstall the L (R) air conditioning accessory unit. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

(4) Replace the Temperature Control Panel, P5-17. These are the tasks:
- Temperature Control Module - Removal, AMM TASK 21-61-15-000-801
- Temperature Control Module - Installation, AMM TASK 21-61-15-400-801.

(5) Do a check for 28V DC power at the Temperature Control Panel, P5-17 as follows:

(a) For the Control Cabin, disconnect connector D3996 from the Temperature Control Panel P5-17 (WDM 21-61-11).

1) Do a Voltage check for the 28V DC at pin 7 of connector D3996.

a) If you do not find 28V DC, then repair the wiring.

2) Connect the connector D3996 to the Temperature Control Panel P5-17.

(b) For The Forward Cabin, disconnect connector D3998 from the Temperature Control Panel P5-17 (WDM 21-61-13).

1) Do a Voltage check for the 28V DC at pin 7 of connector D3998.

a) If you do not find 28V DC, then repair the wiring.

(c) For the Aft Cabin, disconnect connector D3996 from the Temperature Control Panel P5-17 (WDM 21-61-14).

1) Do a Voltage check for the 28V DC at pin 20 of connector D3996.

a) If you do not find 28V DC, then repair the wiring.

(6) Replace the Air Conditioning Panel, P5-10. These are the tasks:
- Air Conditioning Module Removal, AMM TASK 21-51-65-000-801
- Air Conditioning Module Installation, AMM TASK 21-51-65-400-801

——— END OF TASK ———

905. Cabin Temperatures are Uncontrollable

A. Description

(1) Unable to control the cabin temperatures in the control cabin, forward cabin, or aft cabin.

B. Possible Causes

(1) Dirty filter screen of the Temperature Sensor Module(s):
- Control Cabin temperature Sensor Module, M1718.
- Forward Cabin temperature Sensor Module, M1445.
- Aft Cabin temperature Sensor Module, M1446.

(2) The Control Cabin (Forward Cabin, Aft Cabin) Sensor Fan.

(3) Separation of flight deck distribution ducts.

(4) Cabin Duct Temperature Sensors:
- Control Cabin Primary (Backup) Duct Temperature Sensors, T478 (T477).
- Forward Duct Temperature Sensor, T533.
- Aft Duct Temperature Sensor, T536.

(5) Control Cabin (Forward Cabin, Aft Cabin) Temperature Sensor

(6) Control Cabin (Forward Cabin, Aft Cabin) Temperature Selector.

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C. Fault Isolation Procedure

(1) Do this check of the Control Cabin (Forward Cabin, Aft Cabin) Zone Temperature Sensor filter and fan:

(a) Put the L(R) PACK switch on the P5-10 Air Conditioning Panel to the AUTO position.

(b) Hold a 3 in. (8 cm) by 4 in. (10 cm) sheet of lightweight paper over the air inlet grille of the temperature sensor assembly for the applicable cabin.

(c) If the fan does not hold the paper in its position, then do these steps:

1) Put the L(R) PACK switch on the P5-10 Air Conditioning Panel to the OFF position.

2) Remove the control cabin zone temperature sensor filter. This is the task: Cabin Temperature Sensor Assembly Filter Removal, AMM TASK 21-61-06-000-801.

3) Examine the filter for dirt or other blockages.

4) If the filter is dirty, then do these steps:

a) Install a new filter. This is the task: Cabin Temperature Sensor Assembly Filter Installation, AMM TASK 21-61-06-400-801.

b) Supply conditioned air to the control cabin with both packs. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

c) Make sure that the temperature selector on the P5-17 panel is in the AUTO NORMAL position.

d) Let the air conditioning packs operate for a few minutes to allow the temperature of the air from the packs to stabilize.

e) Put the temperature selector on the P5-17 panel to the AUTOWARM position.

f) If the temperature in the control cabin increases, then you corrected the fault.

g) Remove conditioned air from the airplane. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

5) If the filter is not dirty, then do this step:

a) Replace the zone temperature sensor fan for the applicable zone. These are the tasks:

- Cabin Temperature Sensor Fan - Removal, AMM TASK 21-61-11-000-801
- Cabin Temperature Sensor Fan - Installation, AMM TASK 21-61-11-400-801

(2) If unable to control the temperature in the Control Cabin, perform a visual inspection on the air distribution ducting from the mix manifold to the flight deck to check for duct separation.

(a) If duct separation is found, reconnect the ducts.

(b) If no duct separation is found, then continue.

(3) Replace the Control Cabin Primary (Backup, Forward Cabin, Aft Cabin) Duct Temperature Sensors. These are the tasks:

- Duct Temperature Limit Sensor Removal, AMM TASK 21-61-04-000-803-002
- Duct Temperature Limit Sensor Installation, AMM TASK 21-61-04-400-803-002

(4) Replace the Control Cabin (Forward Cabin, Aft Cabin) Zone Temperature Sensors. These are the tasks:

- Cabin Temperature Sensor Removal, AMM TASK 21-61-10-000-802
- Cabin Temperature Sensor Installation, AMM TASK 21-61-10-400-802

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(5) Replace the Control Cabin/Forward Cabin/Aft Cabin Temperature Selector. These are the tasks:

- Cabin Temperature Selector Removal, AMM TASK 21-61-07-000-803-002
- Cabin Temperature Selector Installation, AMM TASK 21-61-07-400-803-002

——— END OF TASK ———

849. Cannot Cool Cabins with All Temperature Selectors in Full COLD - Fault Isolation

A. Description

(1) The Flight Deck and or FWD and Aft Compartments cannot be adequately cooled on the Ground and or In-Flight with the CONT CAB, FWD CAB, and AFT CAB Temperature Selectors all set to the full COLD (C) Positions.

(a) This causes uncontrolled hot Cabins to occur In-Flight and on the Ground.

(2) This fault can also be shown when the TRIM AIR Switch is set to the ON Position, there are no faults on the Controllers, and one Pack Outlet Temperature is much higher than the others.

(a) You can observe this on the Ground when the AIR TEMP is selected with the Switch on the Cabin Temperature Control Panel, P5-17 for each L (R) PACK.

(3) The recommended Temperatures on the P5-17 Panel are: 50°F (10°C) (CONT CAB), 68°F (20°C) (FWD CAB/AFT CAB), and 86°F (30°C) (L/R PACK).

B. Possible Causes

(1) L (R) Standby Temperature Control Valve (STCV), V140 (V142)

(2) Primary Heat Exchanger (PHE)/Secondary Heat Exchanger (SHE) and Plenum/Diffuser Assembly

(3) Air Cycle Machine (ACM)

(4) Mix Manifold Temperature Sensor, T472 or T475

(5) Pack Temperature Sensor, T471 or T474

(6) Trim Air System

(7) Temperature Control System

(8) Air Distribution System

C. Initial Evaluation

(1) Do the BITE test of the PZTC or PFTC. This is the task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

NOTE: Do this BITE test before you begin each check. After you complete a check, fix or clear the BITE faults. Do this for each check.

(a) If the BITE test indicates faults, then correct each problem found by doing the Fault Isolation Procedure.

(b) Do the Repair Confirmation at the end of this task.

(2) If the BITE test does not indicate any faults, then do the ground pack run test to trend the temperatures on the P5-17 panel. This is the task: Ground Pack Run Test - APU AUTO Flow Mode, AMM TASK 21-51-00-700-807.

NOTE: The recommended temperatures on the P5-17 Panel are: 50°F (10°C) (CONT CAB), 68°F (20°C) (FWD CAB/AFT CAB), and 86°F (30°C) (L/R PACK).

(a) Do a check for leakages from the heat exchangers, the heat exchanger plenums, and the condenser.

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1) If you notice a leak, then do the Fault Isolation Procedure - Leakage Check.

(b) If the L(R) PACK temperature on the P5-17 panel is 86°F (30°C) or higher with the L(R) STCV open, then do the Fault Isolation Procedure - STCV Open.

(c) If the L(R) PACK temperature on the P5-17 panel is 86°F (30°C) or higher with both the L(R) STCV and the L(R) TCV closed, then do the Fault Isolation Procedure - STCV Closed.

(d) If the L(R) PACK temperature on the P5-17 panel is lower than 86°F (30°C), and CONT CAB Supply Duct is greater than 50°F (10°C), or FWD/AFT Supply Duct is greater than 68°F (20°C), then do the Fault Isolation Procedure - Temperature Control and Air Distribution System.

(e) Do a check to see if there is ice on the Conditioned Air Check Valve.

NOTE: Ice is an indicator that the pack is performing well.

1) If there is ice on the Conditioned Air Check Valve, then do the Fault Isolation Procedure - Temperature Control and Air Distribution System.

D. Fault Isolation Procedure - Leakage Check

(1) If there are any leakages when performing the ground pack run test, then replace the applicable part:

(a) Heat Exchanger and/or Heat Exchanger Plenum replacement:

- Heat Exchanger and Plenum/Diffuser Assembly - Removal, AMM TASK 21-51-03-000-805-002
- Heat Exchanger and Plenum/Diffuser Assembly - Installation, AMM TASK 21-51-03-400-804-002

1) Do the Repair Confirmation at the end of this task.

(b) Condenser replacement:

- Condenser Removal, AMM TASK 21-51-12-000-801
- Condenser Installation, AMM TASK 21-51-12-400-802

1) Do the Repair Confirmation at the end of this task.

E. Fault Isolation Procedure - Standby Temperature Control Valve (STCV) Open

(1) Do a check of the L (R) STCV:

NOTE: This check will make sure the STCV operates and the associated Sense Lines are not leaking or blocked. This check will also isolate a cracked ACM Turbine Scroll from a STCV problem.

(a) Open these doors to get access to the air conditioning bays:

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door

(b) Make sure that the Sense Lines connected to the L (R) STCV are not cross connected.

1) Do a check of the Conditioned Air Check Valve Sense Line and Condenser Low-Pressure (LP) Sense Lines from the Conditioned Air Check Valve and Condenser to the STCV for leakage or cracks.

2) Do a check of the ACM Turbine Outlet and Condenser High-Pressure (HP) Sense Lines from the ACM/Condenser to STCV for blockage.

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3) If you find any problems with the Sense Lines then correct the issue and do the Repair Confirmation at the end of this task.

4) If you do not find any problems with the Sense Lines then continue.

(c) At the ACM Turbine Outlet, remove the Sense Line to the L (R) STCV, V140 (V142).

NOTE: This check determines if the L (R) STCV is serviceable or the ACM Turbine Scroll is cracked which will cause the STCV to OPEN.

(d) Use the Auxiliary Power Unit (APU) to supply Pneumatic Power to the Airplane. This is the task: Supply Pressure to the Pneumatic System with the APU, AMM TASK 36-00-00-860-803.

1) On the P5-10 Air Conditioning Panel, set the L (R) PACK Switches to AUTO.

2) On the P5-17 Temperature Control Panel, set the CONT CAB, FWD CAB and AFT CAB Temperature selectors to the full COLD (C) Position.

3) On the P5-17 Temperature Control Panel, set the TRIM AIR to the ON Position.

(e) Do a check of the L (R) STCV Position, if L (R) STCV, V140 (V142) remains CLOSED, replace the ACM. These are the tasks:

- Air Cycle Machine Removal, AMM TASK 21-51-04-000-802-002
- Air Cycle Machine Installation, AMM TASK 21-51-04-400-802-002

1) Do the Repair Confirmation at the end of this task.

(f) If the L (R) STCV is OPEN, then disconnect the remaining Sense Lines.

(g) If the L (R) STCV does not close, then replace the STCV, V140 (V142). These are the tasks:

- Standby Temperature Control Valve Removal, AMM TASK 21-51-11-000-801
- Standby Temperature Control Valve Installation, AMM TASK 21-51-11-400-801

(h) Connect the Sense Lines to the L (R) STCV, V140 (V142).

(i) If the L (R) STCV closes, then do the Repair Confirmation at the end of this task.

F. Fault Isolation Procedure - Standby Temperature Control Valve (STCV) Closed



WARNING

DO NOT TOUCH THE DUCTS WHEN THEY ARE HOT. WHEN THE DUCTS ARE HOT, THEY CAN CAUSE INJURY TO PERSONNEL.

(1) If the L (R) STCV is in the fully CLOSED Position, feel near the downstream Duct of the STCV for false indication of the Butterfly Position.

NOTE: The Position Indicator on the L (R) STCV will show if the STCV is in the OPEN or CLOSED Position. If the butterfly on the STCV is broken or disconnected from the shaft, the indicator may show the STCV is CLOSED while it remains OPEN.

(a) If the downstream Duct of the L (R) STCV is hot, then replace the L (R) STCV, V140 (V142). These are the tasks:

- Standby Temperature Control Valve Removal, AMM TASK 21-51-11-000-801
- Standby Temperature Control Valve Installation, AMM TASK 21-51-11-400-801

1) Do the Repair Confirmation at the end of this task.

(b) If the downstream Duct of the L (R) STCV is not hot, then continue.

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(2) Make sure that the Fan Bypass Check Valve Door is installed and it operates correctly. This is the task: Fan Bypass Check Valve Door - Operational Check, AMM TASK 21-51-03-710-801.

(a) If you find any problems with the Fan Bypass Check Valve Door, then correct the issue and do the Repair Confirmation at the end of this task.

(b) If you did not find any problems with the Fan Bypass Check Valve Door, then continue.

(3) Do one of these tasks to clean the Heat Exchangers:

NOTE: Cleaning the heat exchanger using backflush equipment or with soap-water-air is more effective for cleaning dust, dirt, or small particles baked onto the surfaces. Cleaning the heat exchanger with nitrogen gas/compressed air is more effective for blowing out larger particles and blocked heat exchangers. For a more effective clean, Boeing recommends to first blow out debris with nitrogen gas/compressed air and then to clean with soap-water-air. The choice of task is dependent on the kind of contamination and local environment conditions.

- Heat Exchanger and Plenum/Diffuser Assembly Cleaning Using Backflush Equipment, AMM TASK 21-51-03-160-807
- Heat Exchanger - Blowing Out Debris/Blockages with Nitrogen Gas/Compressed Air, AMM TASK 21-51-03-160-811
- Heat Exchanger - Cleaning with Soap-Water-Air, AMM TASK 21-51-03-160-809

(a) Do the Repair Confirmation at the end of this task.

(4) Replace the Heat Exchangers, these are the tasks:

- Heat Exchanger and Plenum/Diffuser Assembly - Removal, AMM TASK 21-51-03-000-805-002
- Heat Exchanger and Plenum/Diffuser Assembly - Installation, AMM TASK 21-51-03-400-804-002

(a) Do a check of the Heat Exchanger Plenums for any white spots, discoloration, delamination, or soft spots.

1) If you find any damage, then also replace the Heat Exchanger Plenums.

(b) Do the Repair Confirmation at the end of this task.

(5) Do a check of the ACM operation:

NOTE: With the APU as the Pneumatic Source, this check will make sure the ACM starts with the highest Duct Pressure available.

(a) On the P5-10 Air Conditioning Panel, set the BLEED 1 and BLEED 2 Switches to the OFF Position.

(b) On the P5-17 Temperature Control Panel, set the TRIM AIR Switch to OFF Position.

(c) Do this task: Supply Pressure to the Pneumatic System with the APU, AMM TASK 36-00-00-860-803.

(d) Operate the applicable Air Conditioning Pack. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(e) Put the Zone Temperature Selectors to full COLD (C).

(f) Do a check of the Airflow at the Exhaust Duct Outlet for the applicable Pack Ram Air System.

1) If there is little or no Airflow at the Exhaust Duct Outlet, then do these steps:

a) Remove the Pack Air Supply. This is the task: Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

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b) Remove the Mix Muff down stream of the ACM.

c) Make sure that you can turn the ACM rotating components in the direction of normal rotation as follows:

> **CAUTION** TURN THE ROTATING COMPONENTS OF THE ACM ONLY IN THE DIRECTION THAT THEY TURN WHEN IN OPERATION. IF YOU TURN THE ACM ROTATING COMPONENTS IN THE INCORRECT DIRECTION, YOU CAN CAUSE DAMAGE TO THE FOIL AIR BEARINGS.

<1> If you turn the ACM Rotation Components at the Turbine Wheel, apply a maximum of 20 in-lb (2.3 N·m) of Torque to the Nut on the Turbine Wheel and turn the Rotation Components in the Clockwise direction.

<2> If you turn the ACM Rotation Components at the Axial Impeller, apply a maximum of 20 in-lb (2.3 N·m) to the Nut on the Axial Impeller and turn the Rotation Components in the Counterclockwise direction.

NOTE: This would require the removal of the Secondary Plenum/Diffuser Assembly which requires more effort than removal of the High Pressure Water Separator (HPWS) Mix Muff.

<a> Do a check of the impeller blades for any missing, broken, or cracked blades. If the impeller blades are missing, broken, or cracked, then replace the ACM. These are the tasks:

- Air Cycle Machine Removal, AMM TASK 21-51-04-000-802-002
- Air Cycle Machine Installation, AMM TASK 21-51-04-400-802-002

I. Do the Repair Confirmation at the end of this task.

d) If the ACM does not rotate with 20 in-lb (2.3 N·m) of Torque, then replace the ACM. These are the tasks:

- Air Cycle Machine Removal, AMM TASK 21-51-04-000-802-002
- Air Cycle Machine Installation, AMM TASK 21-51-04-400-802-002

<1> Do the Repair Confirmation at the end of this task.

e) If the ACM rotates with 20 in-lb (2.3 N·m) of Torque or less, then do the steps that follow:

<1> Install the Mix Muff. These are the tasks:

- High Pressure Water Separator Mix Muff Removal, AMM TASK 21-51-17-000-801
- High Pressure Water Separator Mix Muff Installation, AMM TASK 21-51-17-400-801

<2> Do the Repair Confirmation at the end of this task.

2) If the Airflow at the Exhaust Duct Outlet is satisfactory, and you can control the Temperature in the Cabin, then it was an intermittent fault.

3) If the Airflow at the Exhaust Duct Outlet is satisfactory, and you cannot control the Temperature in the Cabin, then continue.

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(6) Do this test of the Resistance of each Mix Manifold Sensor to make sure the Resistance has not drifted:

(a) Do these steps for each Mix Manifold Sensor:

1) Remove the Mix Manifold Sensors. This is the task: Mix Manifold Temperature Sensor Removal, AMM TASK 21-51-51-000-803.

2) Examine the Sensors to determine if they are dirty or contaminated.

3) If the Sensors are dirty or contaminated, do the following steps:

a) To remove dirt or contamination. This is the task: Mix Manifold Temperature Sensor Cleaning, AMM TASK 21-51-51-110-801.

4) Install the Mix Manifold Sensors. This is the task: Mix Manifold Temperature Sensor Installation, AMM TASK 21-51-51-400-802.

(b) Do this task: Mix Manifold Temperature Sensor Test, AMM TASK 21-51-51-000-802.

(c) Replace the Mix Manifold Sensor if it is out of tolerance. These are the tasks:

- Mix Manifold Temperature Sensor Removal, AMM TASK 21-51-51-000-803
- Mix Manifold Temperature Sensor Installation, AMM TASK 21-51-51-400-802

(d) Do the Repair Confirmation at the end of this task.

(7) Do this test of the Resistance of each Pack Temperature Sensor to make sure the Resistance has not drifted:

(a) Do a check of the Resistances of the Pack Temperature Sensors T471 and T474 as follows:

1) Do this task: Pack Temperature Sensor Test, AMM TASK 21-51-52-720-801.

(b) Replace the Pack Temperature Sensor if it is out of tolerance. These are the tasks:

- Pack Temperature Sensor Removal, AMM TASK 21-51-52-000-801
- Pack Temperature Sensor Installation, AMM TASK 21-51-52-400-801

(c) Do the Repair Confirmation at the end of this task.

G. Fault Isolation Procedure - Temperature Control and Air Distribution System

(1) Do this check of the zone temperature control. This is the task: Cabin Temperatures are Uncontrollable, 21-62 TASK 905.

(a) Do the Repair Confirmation at the end of the task.

(2) Do this test of the Trim Air Pressure Regulating and Shutoff Valve (TAPRSOV). This is the task: Trim Air Pressure Regulating and Shutoff Valve - Operational Test, AMM TASK 21-62-00-000-801.

(a) If there were any issues during the test, replace the TAPRSOV. These are the tasks:

- Trim Air Pressure Regulating and Shutoff Valve Removal, AMM TASK 21-62-01-000-801
- Trim Air Pressure Regulating and Shutoff Valve Installation, AMM TASK 21-62-01-400-801

1) Do the Repair Confirmation at the end of the task.

(3) If the passenger cabin remains warm, do this task to examine the sidewall riser ducts and transition ducts: Sidewall Riser Ducts/Transition Duct Inspection, AMM TASK 21-23-02-212-801.

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(a) If the inspection shows any issues, do this task to restore the sidewall riser ducts or transition ducts: Sidewall Riser Ducts/Transition Duct Repair, AMM TASK 21-23-02-360-801.

1) Do the Repair Confirmation at the end of the task.

(4) Do a check of the AS1591 flex hoses for any signs of damage or delamination.

(a) If damage or delamination is found, replace the flex hoses.

1) Do the Repair Confirmation at the end of the task.

(5) Do a check of overhead distribution ducts for debris and blockages.

(a) If debris or blockages were found, clean the ducts. This is the task: Overhead Distribution Duct Cleaning, AMM TASK 21-23-01-100-801.

1) Do the Repair Confirmation at the end of the task.

H. Repair Confirmation

(1) Do the ground pack run test to trend the temperatures on the P5-17 panel. This is the task: Ground Pack Run Test - APU AUTO Flow Mode, AMM TASK 21-51-00-700-807.

NOTE: The recommended temperatures on the P5-17 Panel are: 50°F (10°C) (CONT CAB), 68°F (20°C) (FWD CAB/AFT CAB), and 86°F (30°C) (L/R PACK).

(2) If the Zone Temperatures on the P5-17 Panel are as stated above, then you have corrected the problem.

(3) If the Cabin Temperatures are still uncontrollably high, then continue to the Fault Isolation Procedure at the subsequent step.

END OF TASK

850. Divergent Pack Out Duct Temperatures - Fault Isolation

A. Description

(1) The flight deck or forward/aft passenger cabin is consistently warmer than the other temperature zones with the packs in AUTO, the TRIM AIR switch at ON and the temperature control switches (CONT CAB, FWD CAB and AFT CAB) all set to the same setting.

(2) You can observe the divergent temperatures by selecting the AIR TEMP selector switch on the P5-17 temperature control panel for left and right pack duct temperature indications.

B. Possible Causes

(1) The mix manifold sensor, T472 or T475 (drifted)

(2) The pack temperature sensor, T471 or T474 (drifted)

C. Initial Evaluation

(1) Supply pressure to the pneumatic system (AMM TASK 36-00-00-860-801).

(2) Set the L and R PACK switches on the P5-10 air conditioning panel to the AUTO position.

(3) Set the TRIM AIR switch on the P5-17 temperature control panel to the ON position.

(4) Set the CONT CAB, FWD CAB and AFT CAB temperature selection switches on the P5-17 temperature control panel to the AUTO position.

(5) Allow the packs to operate until the temperatures stabilize.

(6) Set the AIR TEMP selector on the P5-17 temperature control panel to SUPPLY DUCT CONT CAB and note the temperature on the temperature indicator.

(7) Set the AIR TEMP selector on the P5-17 temperature control panel to SUPPLY DUCT FWD and note the temperature on the temperature indicator.

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(8) Set the AIR TEMP selector on the P5-17 temperature control panel to SUPPLY DUCT AFT and note the temperature on the temperature indicator.

(9) If the supply duct temperatures are almost the same, then there was an intermittent fault.

(10) If there is a large difference between any of the supply duct temperatures, then do the Fault Isolation Procedure - Divergent Pack Out Duct Temperatures.

D. Fault Isolation Procedure

(1) Remove conditioned air from the airplane. To remove conditioned air, do this task: AMM TASK 21-00-00-800-802.

(2) If the high temperature was noted in the duct for the flight deck, do these steps:

(a) Do a test of the mix manifold temperature sensor, T475. To test the mix manifold temperature sensor, T475, do this task: AMM TASK 21-51-51-000-802.

(b) If you replaced the mix manifold temperature sensor, do the Repair Confirmation at the end of this task.

(c) If the Repair Confirmation was satisfactory, then you corrected the fault.

(d) If the Repair Confirmation was not satisfactory or you did not find any fault with the mix manifold sensor, T475, then continue.

(3) If the high temperature was noted in the duct for the aft passenger compartment, do these steps:

(a) Do a test of the mix manifold temperature sensor, T472. To test the mix manifold temperature sensor, T472, do this task: AMM TASK 21-51-51-000-802.

(b) If you replaced the mix manifold temperature sensor, do the Repair Confirmation at the end of this task.

(c) If the Repair Confirmation was satisfactory, then you corrected the fault.

(d) If the Repair Confirmation was not satisfactory or you did not find any fault with the mix manifold sensor, T472, then continue.

(4) If the test of the mix manifold temperature sensor was satisfactory, then continue.

E. Repair Confirmation

(1) Supply conditioned air to the airplane. To supply the conditioned air, do this task:AMM TASK 21-00-00-800-803.

(2) Set the L and R PACK switches to AUTO.

(3) Make sure the TRIM AIR switch is set to ON.

(4) Set the CONT CAB, FWD CAB AND AFT CAB temperature selector switches on the P5-17 temperature control panel to AUTO.

(5) Set the AIR TEMP selector on the P5-17 temperature control panel to SUPPLY DUCT CONT CAB and note the temperature on the temperature indicator.

(6) Set the AIR TEMP selector on the P5-17 temperature control panel to SUPPLY DUCT FWD and note the temperature on the temperature indicator.

(7) Set the AIR TEMP selector on the P5-17 temperature control panel to SUPPLY DUCT AFT and note the temperature on the temperature indicator.

(8) Make sure that the supply duct temperatures are within a few degrees of each other.

(9) If the supply duct temperatures are not within a few degrees of each other, then continue the fault isolation procedure from the last step accomplished.

END OF TASK

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OMR 815, 816, 818, 819, 901-999

851. Left Ram Air Inlet Actuator - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) SRADA INLET

(2) The Left Pack/Zone Temperature Controller has detected a fault from the Left Ram Air Inlet Actuator.

(3) The Red INLET ACTUATOR FAULT Light on the Left Ram Air inlet Actuator, should be ON.

B. Possible Causes

(1) Left Ram Air Inlet Actuator, M2795

(2) Air Conditioning Relay-1, M324

(3) Wiring

(4) Left Pack/Zone Temperature Controller, M1442

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	8	C00555	AIR CONDITIONING RAM AIR MOD CONT L
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L

D. Related Data

(1) WDM 21-51-14

(2) SSM 21-51-14

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801. (Reference the applicable sheet in 21-62 TASK SUPPORT Figure 306)

(2) Cycle 28V DC for the Left SRADA to erase the SRADA Internal Fault Memory and allow actuator movement during the subsequent steps as follows:

(a) Open these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

(b) Close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT

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(c) Wait for a minimum of 10 seconds, then close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

(3) Do the PZTC Reset Procedure as follows:

(a) Push the VERIFY Switch on the PZTC Controller.

1) If a Red light comes ON, then do the Fault Isolation Procedure below.

2) If the Green GO Light comes ON, push the RESET Switch while the VERIFY and GO Lights are ON. No further action is necessary.

F. Fault Isolation Procedure

(1) Do these steps to get access to the ram air inlet actuator, M2795:

(a) Open these access panels:

Number	Name/Location
191FL	Forward Wing To Body Fairing Panel - Mid Fairing, Above Ram Air Inlet
191GL	Ram Air Actuator Panel - Forward

(2) Do a check for 28V DC Power to Ram Air Inlet Actuator, M2795, as follows:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT

(b) Disconnect electrical connector D14386 from the ram air inlet actuator, M2795.

(c) Close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT

(d) Do a check for 28V DC Power on the ship's wiring side of connector D14386 between pins 4 and 16 (ground).

(e) If 28V DC Power is not present, do these steps:

1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT

2) Do these wiring checks:

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OMR 815, 816, 818, 819, 901-999 (Continued)

P6-4 panel	**M2795 actuator**
C/B C01950	**D14386**
Load side	Pin 4

3) Repair the problems that you find.

4) Do the Repair Confirmation at the end of this task:

(f) If 28V DC power is present at pin 4 of connector D14386, then continue.

(3) If 28 V DC Power is present between pins 4 and 16 of connector D14386, then do the following:(Figure 302 (Sheet 1))

(a) Push and release either the Left MASTER CAUTION Annunciator or the Right MASTER CAUTION annunciator on the P7 Glareshield Panel.

(b) If the AIR COND Light on the MASTER CAUTION panel and the P5 PANEL PACK Light are not illuminated then you corrected the fault.

(c) If the AIR COND Light on the MASTER CAUTION panel or the P5 PANEL PACK Light are illuminated then do the steps that follow on the Left Pack/Zone Temperature Controller:

1) Push the BIT switch on the controller.

NOTE: The BIT Switch will show if failures occurred during the last flight.

a) The light above the BIT switch will come ON when the switch is fully pushed.

b) Wait for the Green GO Light to come ON.

c) If a Red Light comes ON for a component, the component does not operate correctly.

<1> Make a record of that component.

2) Push the PREV FLT Switch on the controller.

NOTE: The PREV FLT Switch will show all the failures that occurred during the last nine flights.

a) The light above the PREV FLT Switch will come ON when the switch is fully pushed.

b) Wait for the Green GO Light to come ON.

c) If a Red Light comes ON for a component, the component did not operate correctly during a previous flight.

<1> Make a record of that component.

WARNING MOVE ALL PERSONS AND EQUIPMENT AWAY FROM THE RAM AIR INLET DOOR. WHEN YOU PUSH THE VERIFY SWITCH, THE RAM AIR COMPONENTS WILL MOVE AND CAN CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

3) Push the VERIFY Switch on the controller to start a BIT Test.

NOTE: The VERIFY Switch will initiate a real time test of the Pack/Zone Temperature Controller components.

a) The light above the VERIFY Switch will come ON when the switch is fully pushed.

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b) Wait for the Green GO Light to come ON.

c) If a Red Light comes ON for a component, the component does not operate correctly.

<1> Make a record of that component.

4) If no Red Light on the Left Pack/Zone Temperature Controller comes ON then do these steps:

a) Push the RESET Switch on the controller.

b) Push the PRESS/TEST Switch on the controller.

c) You corrected the fault.

5) If a Red Light on the Left Pack/Zone Temperature Controller comes ON, but the SRADA Red Lights are not ON then refer to the Table at the end of Pack Temperature Control - BITE Procedure, 21-62 TASK 801 to find the applicable FIM Tasks for the maintenance messages that show, then continue.

6) If a Red Light on the Left Pack/Zone Temperature Controller comes ON, and the SRADA Red Lights are ON, and if this is not the 1st troubleshooting attempt then refer to the Table at the end of Pack Temperature Control - BITE Procedure, 21-62 TASK 801 to find the applicable FIM Tasks for the maintenance messages that show, then continue.

7) If a Red Light on the Left Pack/Zone Temperature Controller comes ON, and the SRADA Red Lights are ON and this is the 1st troubleshooting attempt, then do these steps:

a) Compare the Red Lights on the Left Pack/Zone Temperature Controller and the Red Lights on the Exit SRADA. If lights are not matched then do these steps:

<1> Examine the wiring as follows, then continue:

<a> Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

<b> Disconnect electrical connector D14386 from the Ram Air Inlet Actuator, M2795.

<c> Examine the face of the connectors for evidence of bent sockets.

<d> Remove the connector backshell to examine the connector grommets and inserted wires, and slightly pull on each wire to determine if each socket is properly seated and locked.

<e> Repair the problems that you find (SWPM 20-61-11), then re-install electrical connector D14386 to the Ram Air Inlet Actuator, M2795.

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OMR 815, 816, 818, 819, 901-999 (Continued)

<f> Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

b) Compare the Red Lights on the Left Pack/Zone Temperature Controller and the Red Lights on the Exit SRADA. If lights are matched then do these steps:

<1> Refer to the Table at the end of Pack Temperature Control - BITE Procedure, 21-62 TASK 801 to find the applicable FIM Tasks for the maintenance messages that show.

<2> Cycle 28V DC for the SRADA as follows:

<a> Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

<b> Wait for a minimum of 10 seconds and then close the following circuit breakers.

NOTE: Opening the above circuit breakers for 10 seconds minimum and then closing will clear the SRADA internal fault memory and allow actuator movement during subsequent steps.

<c> Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

<3> Do a 2nd troubleshooting attempt, starting from step F.(3)(c)1) above.

(4) If 28V DC Power is present between pins 4 and 16 of connector D14386, then replace the Ram Air Inlet Actuator, M2795. These are the tasks:

- Ram Air Inlet Actuator Removal, AMM TASK 21-51-21-000-801

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- Ram Air Inlet Actuator Installation, AMM TASK 21-51-21-400-801

(a) Do the Repair Confirmation at the end of this task:

(5) Make sure that the Trailing Edge Flaps are extended.

(6) Do these wiring checks to make sure that the signals from the M2061 Proximity Switch Electronics Unit and the S1051 Trailing Edge Flaps Up Switch to the actuator are satisfactory:

(a) Connect a Multimeter between pin 1 of electrical connector D14386 and a structural ground.

1) Make sure that there is continuity.

(b) Connect a Multimeter between pin 9 of electrical connector D14386 and a structural ground.

1) Make sure that there is continuity.

(c) Repair the problems that you find.

(d) Do the Repair Confirmation at the end of this task:

(7) Replace the Air Conditioning Accessory Unit, ACAU No.1, M324. These are the tasks:

- Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801
- Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801

(a) Do the Repair Confirmation at the end of this task:

(8) Do these checks of the wiring between the Left Ram Air Inlet Actuator, M2795, the Left Pack/Zone Temperature Controller, M1442 and the ACAU No.1 (M324):

(a) Get access to the Ram Air Inlet Actuator.

(b) Remove the Left Pack/Zone Temperature Controller, M1442. To remove the controller, do this task: Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801.

(c) Remove the Air Conditioning Accessory Unit, ACAU No.1, M324. Do this task: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801.

(d) Do a wiring check between these pins of electrical connector C3858B of the left pack/zone temperature controller and connector D458A of the ACAU No.1:

M324		**M1442**
D458A		**D3858B**
Pin 61	. . . . .	Pin D8
Pin 47	. . . . .	Pin D6

1) Repair the problems that you find.

2) Do the Repair Confirmation at the end of this task:

(e) Do a wiring check between these pins of electrical connector D3858B of the left pack/zone temperature controller and connector D14386 of the left ram air inlet actuator:

M1442		**M2795**
D3858B		**D14386**
Pin D6	. . . . .	Pin 19
Pin A13	. . . . .	Pin 10

(f) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

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OMR 815, 816, 818, 819, 901-999 (Continued)

2) Install the Left Pack/Zone Temperature Controller, M1442. This is the task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.

3) Install the Left ACAU, No.1 (M324). This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

4) Do the Repair Confirmation at the end of this task:

(g) If you do not find a problem with the wiring, then do these steps and continue.

1) Re-connect the electrical connector D14386 to the Ram Air Inlet Actuator, M2795.

2) Install the Left ACAU, No.1 (M324). This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

(9) Install a new Left Pack/Zone Temperature Controller, M1442. This is the task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.

(10) Do the Repair Confirmation at the end of this task.

(11) Do this task: Ram Air Inlet Door Linkage Inspection, AMM TASK 21-51-22-200-802.

(a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Reconnect all electrical connectors that were disconnected.

(2) Re-install all components that were removed.

(3) Close all circuit breakers that were opened.

NOTE: The SRADA circuit breakers must be pushed in at least 10 seconds before the PZTC circuit breakers are pushed in as part of the circuit breaker reset sequence.

(4) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the problem. Do the applicable steps that follow to return the system to a serviceable condition:

1) Make sure that the Green NO LRU FAULT Light is ON.

2) Make sure that the Red INLET ACTUATOR FAULT, EXIT ACTUATOR FAULT and RAM AIR SENSOR FAULT Lights are OFF.

3) Re-install all access panels that were removed.

(b) If the maintenance message still shows, continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

852. Right Ram Air Inlet Actuator - Fault Isolation

A. Description

(1) This task is for this maintenance message.

(a) SRADA INLET

(2) The right pack/zone temperature controller has detected a fault from the right ram air inlet actuator, M2797.

B. Possible Causes

(1) Right ram air inlet actuator

(2) Air Conditioning Relays No.2, M1455

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OMR 815, 816, 818, 819, 901-999 (Continued)

(3) Wiring

(4) Right pack/zone temperature controller, M1443

C. Circuit Breakers

(1) These are the primary circuit breakers related with the fault:

These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	8	C01178	AIR CONDITIONING RAM AIR MOD CONT R
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R

D. Related Data

(1) SSM 21-51-24

(2) WDM 21-51-24

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801. (Reference the applicable sheet in 21-62 TASK SUPPORT Figure 306)

(2) Cycle 28V DC for the Right SRADA to erase the SRADA Internal Fault Memory and allow actuator movement during the subsequent steps as follows:

(a) Open these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

(b) Close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT

(c) Wait for a minimum of 10 seconds, then close these circuit breakers.

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

(3) Do the PZTC Reset procedure as follows:

(a) Push the VERIFY Switch on the controller.

1) If a Red light comes ON, then do the Fault Isolation Procedure below.

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OMR 815, 816, 818, 819, 901-999 (Continued)

2) If the Green GO Light comes ON, push the RESET switch while the VERIFY and GO Lights are ON. No further action is necessary.

F. Fault Isolation Procedure

(1) Do these steps to get access to the ram air actuator, M2797:

(a) Open these access panels:

Number	Name/Location
191FR	Forward Wing To Body Fairing Panel - Mid Fairing, Above Ram Air Inlet
191GR	Ram Air Actuator Panel - Forward

(2) Do a check for 28V DC power to ram air inlet actuator, M2797, as follows:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT

(b) Disconnect electrical connector D14394 from the ram air inlet actuator, M2797.

(c) Close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT

(d) Do a check for 28V dc power on the ship's wiring side of connector D14394 between pins 4 and 16 (ground).

(e) If 28V DC power is not present, do these steps:

1) Open this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT

2) Do these wiring checks:

P6-4 panel	**M2797 actuator**
C/B 01951	**D14394**
Load side	Pin 4

3) Repair any problems that you find.

4) Do the Repair Confirmation at the end of this task:

a) If the Repair Confirmation is successful, then you corrected the fault.

b) If the Repair Confirmation is not successful, then continue.

(f) If 28V DC is present at pin 4 of connector D14394, then continue.

(3) If 28 V DC power is present between pins 4 and 16 of connector D14394, then do the following:(Figure 302 (Sheet 1))

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OMR 815, 816, 818, 819, 901-999 (Continued)

(a) Push and release either the left MASTER CAUTION annunciator or the right MASTER CAUTION annunciator on the P7 glareshield panel.

(b) If the AIR COND light on the MASTER CAUTION panel and the P5 PANEL PACK light are not illuminated then you corrected the fault.

(c) If the AIR COND light on the MASTER CAUTION panel or the P5 PANEL PACK light are illuminated then do the following steps on the right pack/zone temperature controller:

1) Push the BIT switch on the controller.

NOTE: The BIT switch will show if failures occurred during the last flight.

a) The light above the BIT switch will come ON when the switch is fully pushed.

b) Wait for the green GO Light to come ON.

c) If a red light comes ON for a component, the component is not operating correctly.

<1> Make a record of that component.

2) Push the PREV FLT switch on the controller.

NOTE: The PREV FLT switch will show all the failures that occurred during the last nine flights.

a) The light above the PREV FLT switch will come ON when the switch is fully pushed.

b) Wait for the green GO light to come ON.

c) If a red light comes ON for a component, the component did not operate correctly during a previous flight.

<1> Make a record of that component.

WARNING MOVE ALL PERSONS AND EQUIPMENT AWAY FROM THE RAM AIR INLET DOOR. WHEN YOU PUSH THE VERIFY SWITCH, THE RAM AIR COMPONENTS WILL MOVE AND CAN CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

3) Push the VERIFY switch on the controller to commence initiated BIT test.

NOTE: The VERIFY switch will initiate a real time test of the pack/zone temperature controller components.

a) The light above the VERIFY switch will come ON when the switch is fully pushed.

b) Wait for the green GO light to come ON.

c) If a red light comes ON for a component, the component is not operating correctly.

<1> Make a record of that component.

4) If no red light on the right pack/zone temperature controller comes ON then do the following:

a) Push the RESET switch on the controller.

b) Push the PRESS/TEST switch on the controller.

c) c) You corrected the fault.

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OMR 815, 816, 818, 819, 901-999 (Continued)

5) If a red light on the right pack/zone temperature controller comes ON, but the SRADA red lights are not ON then refer to the BITE Message Index Table at the end of FIM 21-62 TASK 801 to find the applicable Fault Isolation Manual (FIM) Tasks for the maintenance messages that show, then continue.

6) If a red light on the right pack/zone temperature controller comes ON, and the SRADA red lights are ON, and if this is not the 1st troubleshooting attempt then refer to the BITE Message Index Table at the end of FIM 21-62 TASK 801 to find the applicable Fault Isolation Manual (FIM) Tasks for the maintenance messages that show, then continue.

7) If a red light on the right pack/zone temperature controller comes ON, and the SRADA red lights are ON and this is the 1st troubleshooting then do the following:

a) Compare the red lights on the right pack/zone temperature controller and the red lights on the Exit SRADA. If lights are not matched then do the following:

<1> Investigate for a wiring issue by doing the following steps, then continue:

<a> Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

<b> Disconnect electrical connector D14394 from the ram air inlet actuator, M2797.

<c> Examine the face of the connectors for evidence of bent sockets.

<d> Remove the connector backshell to examine the connector grommets and inserted wires, and slightly pull on each wire to determine if each socket is properly seated and locked.

<e> Repair any problems that you find (reference SWPM 20-61-11), then re-install electrical connector D14394 to the ram air inlet actuator, M2797.

<f> Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

b) Compare the red lights on the right pack/zone temperature controller and the red lights on the Exit SRADA. If lights are matched then do the following:

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OMR 815, 816, 818, 819, 901-999 (Continued)

<1> Refer to the BITE Message Index Table at the end of FIM 21-62 TASK 801 to find the applicable Fault Isolation Manual (FIM) Tasks for the maintenance messages that show.

<2> Cycle 28VDC for the SRADA by the following steps:

<a> Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

<b> Wait for a minimum of 10 seconds and then close the following circuit breakers.

NOTE: Opening the above circuit breakers for 10 seconds minimum and then closing will clear the SRADA internal fault memory and allow actuator movement during subsequent steps.

<c> Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

<3> Do a 2nd troubleshooting attempt, starting from step F.(3)(c)1) above.

(4) If 28 V DC power is present between pins 4 and 16 of connector D14394, then replace the ram air inlet actuator (M2797) as follows:

(a) To replace the actuator, do these tasks:

- Ram Air Inlet Actuator Removal, AMM TASK 21-51-21-000-801
- Ram Air Inlet Actuator Installation, AMM TASK 21-51-21-400-801

(b) Do the Repair Confirmation at the end of this task.

1) If the maintenance message does not show, then you corrected the fault.

2) If the maintenance message still shows, then continue.

(5) Make sure that the trailing edge flaps are extended.

(6) Do these wiring checks to make sure that the signals from the M2061 Proximity Switch Electronics Unit and the M1746 Flap/Slat Electronics Unit to the actuator are satisfactory:

(a) Connect a multimeter between pin 1 of electrical connector D14394 and a structural ground.

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1) Make sure that there is continuity.

(b) Connect a multimeter between pin 9 of electrical connector D14394 and a structural ground.

1) Make sure that there is continuity.

(c) Repair any problems that you find.

(d) Do the Repair Confirmation at the end of this task:

1) If the Repair Confirmation is successful, then you corrected the fault.

2) If the Repair Confirmation is not successful, then continue.

(7) Do a wiring check to make sure that the signals from the M2061 Proximity Switch Electronics Unit (PSEU) and the M1455 A/C Relay No. 2 are satisfactory.

(a) Disconnect connector DM2061 from the PSEU.

(b) Disconnect connector D10002A from the A/C Relay No.2.

(c) Do these wiring checks:

M2061		**M1455**
D10988		**D10002A**
Pin 27	. . . . .	Pin 35

(d) Repair any problems that you find.

(e) Do the Repair Confirmation at the end of this task:

1) If the Repair Confirmation is successful, then you corrected the fault.

2) If the Repair Confirmation is not successful, then continue.

(8) Replace the air conditioning accessory unit (ACAU) No.2, M1455. Do these tasks:

- Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801
- Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801

(a) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(b) If the maintenance message does not show, then you corrected the fault.

(c) If the maintenance message still shows, then continue.

(9) Do these checks of the wiring between the right ram air inlet actuator (M2797), the right pack/zone temperature controller (M1443) and ACAU No.2 (M1455):

(a) Get access to the right ram air inlet actuator.

(b) Remove the right pack/zone temperature controller (M1443). To remove the controller, do this task: Pack/Zone Temperature Controller - Removal, AMM TASK 21-61-20-000-801.

(c) Remove the air conditioning accessory unit No.2 (M1455). Do this task: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801.

(d) Do a wiring check between these pins of electrical connector D3860B of the right pack/zone temperature controller and connector D10002A of ACAU No.2:

M1455		**M1443**
D10002A		**D3860B**
Pin 61	. . . . .	Pin D8
Pin 47	. . . . .	Pin D6

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Repair any problems that you find.
2) Do the Repair Confirmation at the end of this task:
   a) If the Repair Confirmation was successful, then you corrected the fault.
   b) If the Repair Confirmation was not successful, then continue.

(e) Do a wiring check between these pins of electrical connector D3860B of the right pack/zone temperature controller and connector D14394 of the right ram air inlet actuator M2797.

M1443	**M2797**
D3860B	**D14394**
Pin D6	Pin 19
Pin A13	Pin 10

(f) If you find a problem with the wiring, then do these steps:
1) Repair the wiring.
2) Re-connect the electrical connector D14394 to the right ram air inlet actuator, M2797.
3) Install the right pack/zone temperature controller (M1443). To install the controller, do this task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.
4) Install the right ACAU No.2 (M1455). To install the ACAU, do this task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.
5) Do the Repair Confirmation at the end of this task:
   a) If the Repair Confirmation was successful, then you corrected the fault.
   b) If the Repair Confirmation was not successful, then continue.

(g) If you do not find a problem with the wiring, then do these steps and continue:
1) Re-connect the electrical connector D14394 to the ram air inlet actuator M2797.
2) Install the right ACAU No.2 (M1455). To install the ACAU, do this task Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

(10) Install a new pack/zone temperature controller (M1443). To install the controller, do this task: Pack/Zone Temperature Controller - Installation, AMM TASK 21-61-20-400-801.

(11) Do this task: Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.
   (a) If the maintenance message does not show, then you corrected the fault.

(12) Do this task: Ram Air Inlet Door Linkage Inspection, AMM TASK 21-51-22-200-802.
   (a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Reconnect all electrical connectors that were disconnected.
(2) Re-install all components that were removed.
(3) Close all circuit breakers that were opened.

NOTE: The SRADA circuit breakers must be pushed in at least 10 seconds before the PZTC circuit breakers are pushed in as part of the circuit breaker reset sequence.

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OMR 815, 816, 818, 819, 901-999 (Continued)

(4) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(5) If the Repair Confirmation is not satisfactory, return to the step in the Fault Isolation procedure that you were at prior to performing the Repair Confirmation and continue the Fault Isolation procedure.

(a) Make sure that the green NO LRU FAULT light is on.

(b) Make sure that the red INLET ACTUATOR FAULT, EXIT ACTUATOR FAULT and RAM AIR SENSOR FAULT lights are off.

(c) Re-install all access panels that were removed.

(6) If the Repair Confirmation is satisfactory, make sure that you do all the applicable steps that follow to return the system to a serviceable condition:

(a) Re-install all access panels that were removed.

(b) Make sure that the green NO LRU FAULT light is on.

(c) Make sure that the red INLET ACTUATOR FAULT, EXIT ACTUATOR FAULT and RAM AIR SENSOR FAULT lights are off.

(d) Re-install all access panels that were removed.

——— END OF TASK ———

853. Left Ram Air Exit Actuator - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) SRADA EXIT

(2) The left pack/zone temperature controller has detected a fault from the left ram air exit actuator, M2796.

(3) The red EXIT ACTUATOR FAULT light on the left ram air exit actuator, M2796, should be on.

B. Possible Causes

(1) Left ram air exit actuator, M2796

(2) Left ram air inlet actuator, M2795

(3) Wiring

C. Circuit Breakers

(1) These are the circuit breakers related to the fault:

These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	8	C00555	AIR CONDITIONING RAM AIR MOD CONT L
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 815, 816, 818, 819, 901-999 (Continued)

D. Related Data

(1) SSM 21-51-14

(2) WDM 21-51-14

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801. (Reference the applicable sheet in 21-62 TASK SUPPORT Figure 306)

(2) Cycle 28V DC for the Left Smart Ram Air Door Actuator (SRADA) to erase the SRADA Internal Fault Memory and allow actuator movement during the subsequent steps as follows:

(a) Open these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

(b) Close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT

(c) Wait for a minimum of 10 seconds, then close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

(3) Do the PZTC Reset Procedure as follows:

(a) Push the VERIFY Switch on the PZTC.

1) If a Red light comes ON, then do the Fault Isolation Procedure below.

2) If the Green GO Light comes ON, push the RESET Switch while the VERIFY and GO Lights are ON. No further action is necessary.

F. Fault Isolation Procedure

(1) Get access to the ram air exit actuator, M2796, as follows:

(a) Open these access panels:

Number	Name/Location
192CL	ECS Access Door
192JL	Air Conditioning Panel - Aft

(2) Do this check for 28V DC power to the ram air exit actuator, M2796, as follows:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 815, 816, 818, 819, 901-999 (Continued)

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT

(b) Disconnect electrical connector D14390 from the actuator.

(c) Close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT

(d) Do a check for 28V DC between pins 4 and 16 of connector D14390.

(e) If 28V DC is not present between pins 4 and 16 of connector D14390, do these steps:

1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT

2) Do these wiring checks:

P6-4 panel **Circuit breaker** **C01950**	**M2796** **Connector** **D14390**
Load side	Pin 4

M2796 **D14390**	**Structure ground** **Ground**
Pin 16	GD656-DC

(f) Repair any problems that you find.

(g) Do the Repair Confirmation at the end of this task:

1) If the Repair Confirmation is successful, then you corrected the fault.

2) If the Repair Confirmation is not successful, then continue.

(3) If 28V DC is present, replace the left ram air exit actuator. Do these tasks:

- Ram Air Exhaust Actuator - Removal, AMM TASK 21-51-27-000-801
- Ram Air Exhaust Actuator - Installation, AMM TASK 21-51-27-400-801

(a) Do the Repair Confirmation at the end of this task:

1) If the Repair Confirmation is successful, then you corrected the fault.

2) If the Repair Confirmation is not successful, then continue.

(4) Get access to the ram air inlet actuator, M2795, as follows:

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OMR 815, 816, 818, 819, 901-999 (Continued)

(a) Open these access panels:

Number	Name/Location
191FL	Forward Wing To Body Fairing Panel - Mid Fairing, Above Ram Air Inlet
191GL	Ram Air Actuator Panel - Forward

(b) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT

(5) Disconnect electrical connector D14388 from the ram air inlet actuator, M2795, and electrical connector D14392 from the ram air exit actuator.

(6) Do these wiring checks:

M2795 D14388		M2796 D14392
Pin 1	. . . . .	Pin 1
Pin 2	. . . . .	Pin 2
Pin 3	. . . . .	Pin 3

(7) Repair any problems you find.

(8) Do the Repair Confirmation at the end of this task:

(a) If the Repair Confirmation is successful, then you corrected the fault.

(b) If the Repair Confirmation is not successful, then continue.

(9) Replace the ram air inlet actuator, M2795, as follows:

(a) To replace the actuator, do these tasks:

- Ram Air Inlet Actuator Removal, AMM TASK 21-51-21-000-801
- Ram Air Inlet Actuator Installation, AMM TASK 21-51-21-400-801

(10) Do the Repair Confirmation at the end of this task:

G. Repair Confirmation

(1) Reconnect all electrical connectors that were disconnected.

(2) Re-install all components that were removed.

(3) Close all circuit breakers that were opened.

NOTE: The SRADA circuit breakers must be pushed in at least 10 seconds before the PZTC circuit breakers are pushed in as part of the circuit breaker reset sequence.

(4) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(5) If the Repair Confirmation is not satisfactory, return to the step in the Fault Isolation procedure that you were at prior to performing the Repair Confirmation and continue the Fault Isolation procedure.

(6) If the Repair Confirmation is satisfactory, make sure that you do all the applicable steps that follow to return the system to a serviceable condition:

(a) Make sure that the green NO LRU FAULT light is on.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 815, 816, 818, 819, 901-999 (Continued)

(b) Make sure that the red INLET ACTUATOR FAULT, EXIT ACTUATOR FAULT and RAM AIR SENSOR FAULT lights are off.

(c) Re-install all access panels that were removed.

——— END OF TASK ———

854. Right Ram Air Exit Actuator - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) SRADA EXIT

(2) The right pack/zone temperature controller has detected a fault from the right ram air exit actuator, M2798.

(3) The red EXIT ACTUATOR FAULT light on the right ram air exit actuator, M2798, should be on.

B. Possible Causes

(1) Right ram air exit actuator, M2798

(2) Right ram air inlet actuator, M2797

(3) Wiring

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	8	C01178	AIR CONDITIONING RAM AIR MOD CONT R
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R

D. Related Data

(1) SSM 21-51-24

(2) WDM 21-51-24

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801. (Reference the applicable sheet in 21-62 TASK SUPPORT Figure 306)

(2) Cycle 28V DC for the Right SRADA to erase the SRADA Internal Fault Memory and allow actuator movement during the subsequent steps as follows:

(a) Open these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 815, 816, 818, 819, 901-999 (Continued)

(b) Close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT

(c) Wait for a minimum of 10 seconds, then close these circuit breakers.

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

(3) Do the PZTC Reset procedure as follows:

(a) Push the VERIFY Switch on the controller.

1) If a Red light comes ON, then do the Fault Isolation Procedure below.

2) If the Green GO Light comes ON, push the RESET switch while the VERIFY and GO Lights are ON. No further action is necessary.

F. Fault Isolation Procedure

(1) Get access to the ram air exit actuator, M2798, as follows:

(a) Open these access panels:

Number	Name/Location
192CR	ECS Access Door
192JR	Air Conditioning Panel - Aft

(2) Do this check for 28V DC power to the ram air exit actuator, M2798, as follows:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT

(b) Disconnect electrical connector D14398 from the actuator.

(c) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT

(d) Do a check for 28V DC between pins 4 and 16 of connector D14398.

(e) If 28V DC is not present between pins 4 and 16 of connector D14398, do these steps:

1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT

2) Do these wiring checks:

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OMR 815, 816, 818, 819, 901-999 (Continued)

P6-4 panel **Circuit breaker** **C01951**	**M2798** **Connector** **D14398**
Load side	Pin 4

M2798 **D14398**	**Structure ground** **Ground**
Pin 16	GD716-DC

(f) Repair any problems that you find.

(g) Do the Repair Confirmation at the end of this task:

1) If the Repair Confirmation is successful, then you corrected the fault.

2) If the Repair Confirmation is not successful, then continue.

(3) If 28V DC is present, replace the right ram air exit actuator, M2798. Do these tasks:

- Ram Air Exhaust Actuator - Removal, AMM TASK 21-51-27-000-801
- Ram Air Exhaust Actuator - Installation, AMM TASK 21-51-27-400-801

(a) Do the Repair Confirmation at the end of this task.

1) If the Repair Confirmation is successful, then you corrected the fault.

2) If the Repair Confirmation is not successful, then continue.

(4) Get access to the ram air inlet actuator, M2797, as follows:

(a) Open these access panels:

Number	Name/Location
191FL	Forward Wing To Body Fairing Panel - Mid Fairing, Above Ram Air Inlet
191GL	Ram Air Actuator Panel - Forward

(b) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT

(5) Disconnect electrical connector D14396 from the ram air inlet actuator, M2797, and electrical connector D14400 from the ram air exit actuator.

(6) Do these wiring checks:

M2797 **D14396**	**M2798** **D14400**
Pin 1	Pin 1
Pin 2	Pin 2
Pin 3	Pin 3

(7) Repair any problems that you find.

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OMR 815, 816, 818, 819, 901-999 (Continued)

(8) Do the Repair Confirmation at the end of this task:

(a) If the Repair Confirmation is successful, then you corrected the fault.

(b) If the Repair Confirmation is not successful, then continue.

(9) Replace the ram air inlet actuator, M2797, as follows:

(a) To replace the actuator, do these tasks: Ram Air Inlet Actuator Removal, AMM TASK 21-51-21-000-801 and Ram Air Inlet Actuator Installation, AMM TASK 21-51-21-400-801.

(10) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Reconnect all electrical connectors that were disconnected.

(2) Re-install all components that were removed.

(3) Close all circuit breakers that were opened.

NOTE: The SRADA circuit breakers must be pushed in at least 10 seconds before the PZTC circuit breakers are pushed in as part of the circuit breaker reset sequence.

(4) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(5) If the Repair Confirmation is not satisfactory, return to the step in the Fault Isolation procedure that you were at prior to performing the Repair Confirmation and continue the Fault Isolation procedure.

(6) If the Repair Confirmation is satisfactory, make sure that you do all the applicable steps that follow to return the system to a serviceable condition:

(a) Make sure that the green NO LRU FAULT light is on.

(b) Make sure that the red INLET ACTUATOR FAULT, EXIT ACTUATOR FAULT and RAM AIR SENSOR FAULT lights are off.

(c) Re-install all access panels that were removed.

——— END OF TASK ———

OMR 820-826

857. Flow Control Valve - Fault Isolation

A. Description

(1) This task is for these Latched Maintenance Messages:

NOTE: After you correct a problem related to a Latched Maintenance Message, you can clear the Maintenance Message as follows:

- Cycle the power to the PFTC or
- Perform the BITE ground test or
- Erase the EXISTING FAULTS from the PFTC Front Panel Menus.

(a) 21-51069 PFTC CNTRL INOP

1) Flow Control Valve (FCV) Driver High Fault

(b) 21-51070 PFTC CNTRL INOP

1) FCV Driver Low Fault

(c) 21-51250 FLOW CTRL VALVE

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OMR 820-826 (Continued)

1) Flow Control Valve Torque Motor Open Fault

(d) 21-51251 FLOW CTRL VALVE

1) Flow Control Valve Torque Motor Low Fault

(e) 21-51252 FLOW CTRL VALVE

1) Flow Control Valve Torque Motor Short Fault

(f) 21-51501 PFTC CNTRL INOP

1) FCV Shutoff Drive Fault

(g) 21-51503 PFTC CNTRL INOP

1) FCV Hardware Drive Tolerance Fault

(h) 21-51504 PFTC CNTRL INOP

1) Standby Hardware Drive Tolerance Fault

(2) This task is also for these Non-Latched Maintenance Messages:

(a) 21-51253 FLOW CTRL VALVE

1) Flow Control Valve Fail Open Position

(b) 21-51254 FLOW CTRL VALVE

1) Flow Control Valve Fail Close Position

(3) There is a problem with the Left (Right) Flow Control and Shutoff Valve (FCSOV).

B. Possible Causes

(1) Left (Right) FCSOV, V184 (V185)

(2) Wiring

(3) Left (Right) PFTC, M2951 (M2952)

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L

D. Related Data

(1) WDM 21-51-11

(2) WDM 21-51-21

(3) SSM 21-51-11

(4) SSM 21-51-21

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message shows in the PFTC Existing Faults, or it is a latched message that shows in the most recent flight leg, then do the Fault Isolation Procedure below.

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OMR 820-826 (Continued)

(b) If the maintenance message does not show in the PFTC Existing Faults, or as a latched message in the most recent flight leg, then no additional fault isolation steps are required.

F. Fault Isolation Procedure - Left Flow Control and Shutoff Valve, V184

(1) Do this check for power to the Left FCSOV, V184 (WDM 21-51-11):

(a) Disconnect electrical connector D15830 from the Left FCSOV, V184.

(b) Do a check for 28V DC from pin 1 of electrical connector D15830 to ground GD5524-DC.

1) If 28V DC is present, replace the Left FCSOV, V184. These are the tasks:

- Left Flow Control and Shutoff Valve Removal, AMM TASK 21-51-01-000-806-003
- Left Flow Control and Shutoff Valve Installation, AMM TASK 21-51-01-400-807-003

a) Do the Repair Confirmation at the end of this task.

2) If 28V DC is not present, do a continuity check of the wiring between electrical connector D15830, pin 1 and the load side of the AIR CONDITIONING PACK CONT VALVES L circuit breaker, C00262 (WDM 21-51-11).

a) If you find a wiring problem, repair the wiring.

b) Re-connect electrical connector D15830.

c) Do the Repair Confirmation at the end of this task.

(2) Replace the Left PFTC, M2951. These are the tasks:

- Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801
- Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801

(a) Do the Repair Confirmation at the end of this task.

G. Fault Isolation Procedure - Right Flow Control and Shutoff Valve, V185

(1) Do this check for power to the Right FCSOV, V185 (WDM 21-51-21):

(a) Disconnect electrical connector D15834 from the Right FCSOV, V185.

(b) Do a check for 28V DC from pin 1 of electrical connector D15834 to ground GD6524-DC.

1) If 28V DC is present, replace the Right FCSOV, V185. These are the tasks:

- Right Flow Control and Shutoff Valve Removal, AMM TASK 21-51-01-000-807-003
- Right Flow Control and Shutoff Valve Installation, AMM TASK 21-51-01-400-806-003

a) Do the Repair Confirmation at the end of this task.

2) If 28V DC is not present, do a continuity check of the wiring between electrical connector D15834, pin 1 and the load side of the AIR CONDITIONING PACK CONT VALVES R circuit breaker, C00263 (WDM 21-51-21).

a) If you find a wiring problem, repair the wiring.

b) Re-connect electrical connector D15834.

c) Do the Repair Confirmation at the end of this task.

(2) Replace the Right PFTC, M2952. These are the tasks:

- Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801
- Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801

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OMR 820-826 (Continued)

(a) Do the Repair Confirmation at the end of this task.

H. Repair Confirmation

(1) For the PFTC, do the "BITE Procedure" and the "BITE Procedure - EXISTING FAULTS" sections of this task:

- Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the problem.

(b) If the maintenance message still shows, continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

858. PFTC Internal Fault - Fault Isolation

A. Description

(1) This task is for these Latched Maintenance Messages:

NOTE: After you correct a problem related to a Latched Maintenance Message, you can clear the Maintenance Message as follows:

- Cycle the power to the PFTC or
- Perform the BITE ground test or
- Erase the EXISTING FAULTS from the PFTC Front Panel Menus.

(a) 21-51001, 21-51002, 21-51003, 21-51004, 21-51005, 21-51006, 21-51007, 21-51008, 21-51009, 21-51010, 21-51017, 21-51018, 21-51019, 21-51034, 21-51035, 21-51037, 21-51041, 21-51042 and 21-51044

- PFTC CNTRL INOP

OMR 820-826; AIRPLANES WITH PFTC P/N 51090248-002/-003

(b) 21-51001

- PFTC CNTRL INOP

OMR 820-826; AIRPLANES WITH PFTC P/N 51090248-005

(c) 21-51040

- PFTC CNTRL INOP

OMR 820-826

(2) This task is also for these Non-Latched Maintenance Messages:

(a) 21-51030, 21-51331

- DISPLAY

(b) 21-51036, 21-51038, 21-51043, 21-51045. 21-51332, 21-51333

- PFTC CNTRL INOP

(c) 21-51046

- STBY 28V PWR OFF

(d) 21-51090, 21-51091, 21-51092, 21-51093

- PFTC STBY CNTRL

(3) There is an internal problem in the Left (Right) Pack Flow and Temperature Controller (PFTC).

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OMR 820-826 (Continued)

B. Possible Causes

(1) Left PFTC, M2951

(2) Right PFTC, M2952

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

D. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message shows in the PFTC Existing Faults, or it is a latched message that shows in the most recent flight leg, then do the Fault Isolation Procedure below.

(b) If the maintenance message does not show in the PFTC Existing Faults, or as a latched message in the most recent flight leg, then no additional fault isolation steps are required.

E. Fault Isolation Procedure

(1) Replace the applicable Left (Right) Pack Flow Temperature Controller (PFTC), M2951 (M2952). These are the tasks:

- Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.
- Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

(2) For the PFTC, do the "BITE Procedure" and the "BITE Procedure - EXISTING FAULTS" sections of this task:

- Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the fault.

——— END OF TASK ———

859. Pack Inlet Pressure - Fault Isolation

A. Description

(1) This task is for these Maintenance Messages:

(a) 21-51061 PFTC CNTRL INOP

1) Pack Inlet Pressure Sensor Interface Fault

(b) 21-51230 PACK IN PRESSURE

1) Pack Inlet Pressure Sensor High Fault

(c) 21-51231 PACK IN PRESSURE

1) Pack Inlet Pressure Sensor Low Fault

(d) 21-51232 PACK IN PRESSURE

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OMR 820-826 (Continued)

1) Pack Inlet Pressure Sensor Drift Fault

(e) 21-51233 PACK IN PRESSURE

1) Pack Inlet Pressure Sensor Shorted Load Fault

2) This is a latched message.

NOTE: After you correct a problem related to a Latched Maintenance Message, you can clear the Maintenance Message as follows:

- Cycle the power to the PFTC or
- Perform the BITE ground test or
- Erase the EXISTING FAULTS from the PFTC Front Panel Menus.

(2) There is a problem with the Left (Right) Pack Inlet Pressure Sensor

B. Possible Causes

(1) Left Pack Inlet Pressure Sensor, M2955

(2) Right Pack Inlet Pressure Sensor, M2956

(3) Wiring

(4) L (R) PFTC, M2951 (M2952)

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

D. Related Data

(1) WDM 21-51-12

(2) WDM 21-51-22

(3) SSM 21-51-12

(4) SSM 21-51-22

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message shows in the PFTC Existing Faults, or it is a latched message that shows in the most recent flight leg, then do the Fault Isolation Procedure below.

(b) If the maintenance message does not show in the PFTC Existing Faults, or as a latched message in the most recent flight leg, then no additional fault isolation steps are required.

F. Fault Isolation Procedure

(1) Replace the applicable Left (Right) Pack Inlet Pressure Sensor, M2955 (M2956). These are the tasks:

- Pack Inlet Pressure Sensor Removal, AMM TASK 21-51-45-000-801

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OMR 820-826 (Continued)

- Pack Inlet Pressure Sensor Installation, AMM TASK 21-51-45-400-801

(a) Do the Repair Confirmation at the end of this task.

(2) If the maintenance message is for the Left Pack Inlet Pressure Sensor, do this wiring check (WDM 21-51-12):

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

(b) Remove the Left Pack Flow Temperature Controller. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

(c) Disconnect electrical connector D15826 from the Left Pack Inlet Pressure Sensor.

(d) Do a wiring check as follows:

L PFTC D15814B		L PACK PRESSURE SENSOR D15826
pin A8	. . . . .	pin 1
pin A9	. . . . .	pin 3
ground	. . . . .	pin 5

1) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Reconnect electrical connector D15826 to the Left Pack Inlet Pressure Sensor.

c) Re-install the L PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

d) Do the Repair Confirmation at the end of this task.

2) If you do not find a problem with the wiring, then do these steps:

a) Reconnect electrical connector D15826 to the Left Pack Inlet Pressure Sensor.

b) Replace the L PFTC, M2951. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

(3) If the maintenance message is for the Right Pack Inlet Pressure Sensor, do this wiring check (WDM 21-51-22):

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

(b) Remove the Right Pack Flow Temperature Controller. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

(c) Disconnect electrical connector D15828 from the Right Pack Inlet Pressure Sensor.

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OMR 820-826 (Continued)

(d) Do a wiring check as follows:

R PFTC D15816B		R PACK PRESSURE SENSOR D15828
pin A8	. . . . .	pin 1
pin A9	. . . . .	pin 3
ground	. . . . .	pin 5

1) If you find a problem with the wiring, then do these steps:
   a) Repair the wiring.
   b) Reconnect electrical connector D15828 to the Right Pack Inlet Pressure Sensor.
   c) Re-install the Right PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.
   d) Do the Repair Confirmation at the end of this task.
2) If you do not find a problem with the wiring, then do these steps:
   a) Reconnect electrical connector D15828 to the Right Pack Inlet Pressure Sensor.
   b) Replace the Right PFTC, M2952. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

G. Repair Confirmation

(1) For the PFTC, do the "BITE Procedure" and the "BITE Procedure - EXISTING FAULTS" sections of this task:

• Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the fault.

(b) If the maintenance message still shows, continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

860. PACK FLOW SENSOR - Fault Isolation

A. Description

(1) This task is for these Latched Maintenance Messages:

NOTE: After you correct a problem related to a Latched Maintenance Message, you can clear the Maintenance Message as follows:

• Cycle the power to the PFTC or

• Perform the BITE ground test or

• Erase the EXISTING FAULTS from the PFTC Front Panel Menus.

(a) 21-51066 PFTC CNTRL INOP

1) Pack Flow Sensor Heater Driver High Interface Fault

(b) 21-51067 PFTC CNTRL INOP

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OMR 820-826 (Continued)

1) Pack Flow Sensor Heater Driver Low Interface Fault

(c) 21-51068 PFTC CNTRL INOP

1) Pack Flow Sensor Heater Driver Drift Interface Fault

(d) 21-51239 PACK FLOW SENSOR

1) Pack Flow Sensor Drift Fault

(e) 21-51246 PACK FLOW SENSOR

1) Pack Flow Sensor Heater Open Fault

(f) 21-51247 PACK FLOW SENSOR

1) Pack Flow Sensor Heater Low Fault

(g) 21-51248 PACK FLOW SENSOR

1) Pack Flow Sensor Heater Short Fault

(h) 21-51249 PACK FLOW SENSOR

1) Pack Flow Sensor High Fault

(i) 21-51500 PFTC CNTRL INOP

1) Pack Flow Sensor Shutoff Drive Fault

(j) 21-51502 PFTC CNTRL INOP

1) Pack Flow Sensor Hardware Drive Tolerance Fault

(2) This task is also for this Non-Latched Maintenance Message:

(a) 21-51240 PACK FLOW SENSOR

1) Pack Flow Sensor Element-Calibration Resistor Fault

(3) There is a problem with the Left (Right) Pack Flow Sensor.

B. Possible Causes

(1) Left Pack Flow Sensor, M2953

(2) Right Pack Flow Sensor, M2954

(3) Wiring

(4) L (R) PFTC, M2951 (M2952)

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

D. Related Data

(1) SSM 21-51-12

(2) SSM 21-51-22

(3) WDM 21-51-12

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OMR 820-826 (Continued)

(4) WDM 21-51-22

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show in the PFTC Existing Faults, or as a latched message in the most recent flight leg, then no additional fault isolation steps are required.

(b) If the maintenance message shows in the PFTC Existing Faults, or it is a latched message that shows in the most recent flight leg, then do the Fault Isolation Procedure below.

1) Wait 20 minutes and repeat the Pack Temperature Control - BITE Procedure, 21-62 TASK 801 - GROUND TESTS.

NOTE: When the Pack Flow Sensor is heated to High Temperature Bleed Conditions, the two Flow Sensor Temperature Sensing Elements may require additional time to cool down to Ambient Temperature. If this occurs, it is recommended that operators do the PFTC Ground Test after approximately 20 minutes to make sure that there is no Pack Flow Sensor DRIFT fault. This will let the Flow Sensor Temperature Sensing Elements to cool and equalize.

a) If the maintenance message does not show in the PFTC Existing Faults, or as a latched message in the most recent flight leg, then no additional fault isolation steps are required.

b) If the maintenance message shows in the PFTC Existing Faults, or it is a latched message that shows in the most recent flight leg, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Replace the applicable Left (Right) Pack Flow Sensor, M2953 (M2954). These are the tasks:

- Pack Flow Sensor Removal, AMM TASK 21-51-43-000-801
- Pack Flow Sensor Installation, AMM TASK 21-51-43-400-801

(a) Do the Repair Confirmation at the end of this task.

(2) For a fault on the Left Pack, do these steps:

(a) Remove the Left PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

(b) Disconnect electrical connector D15822 from the Left Pack Flow Sensor.

(c) Do a wiring check between the Left PFTC and the Left Pack Flow Sensor as follows (WDM 21-51-12):

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OMR 820-826 (Continued)

L PFTC D15814B	L Pack Flow Sensor D15822
pin A11	pin 7
pin B10	pin 5
pin A12	pin 9
pin B11	pin 11
pin B12	pin 8
pin D1	pin 2
pin D2	pin 4
ground	pin 12

1) If you find a problem with the wiring, then do these steps:
   a) Repair the wiring.
   b) Reconnect electrical connector D15822 to the Left Pack Flow Sensor.
   c) Re-install the L PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.
   d) Do the Repair Confirmation at the end of this task.
2) If you do not find a problem with the wiring, then do these steps:
   a) Reconnect electrical connector D15822 to the Left Pack Flow Sensor.
   b) Replace the L PFTC, M2951. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

(3) For a fault on the Right Pack Flow Sensor, do these steps:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

(b) Remove the Right PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

(c) Disconnect electrical connector D15824 from the Right Pack Flow Sensor.

(d) Do this a wiring check between the Right PFTC and the Right Pack Flow Sensor as follows (WDM 21-51-22):

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OMR 820-826 (Continued)

R PFTC D15816B		R Pack Flow Sensor D15824
pin A11	. . . . .	pin 7
pin B10	. . . . .	pin 5
pin A12	. . . . .	pin 9
pin B11	. . . . .	pin 11
pin B12	. . . . .	pin 8
pin D1	. . . . .	pin 2
pin D2	. . . . .	pin 4
ground	. . . . .	pin 12

1) If you find a problem with the wiring, then do these steps:
   a) Repair the wiring.
   b) Reconnect electrical connector D15824 to the Right Pack Flow Sensor.
   c) Re-install the R PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.
   d) Do the Repair Confirmation at the end of this task.
2) If you do not find a problem with the wiring, then do these steps:
   a) Reconnect electrical connector D15824 to the Right Pack Flow Sensor.
   b) Replace the R PFTC, M2952. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

G. Repair Confirmation

(1) For the PFTC, do the "BITE Procedure" and the "BITE Procedure - EXISTING FAULTS" sections of this task:

• Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the fault.

(b) If the maintenance message still shows, continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

OMR 820-826; AIRPLANES WITH PFTC P/N 51090248-005

861. FLOW CTRL VALVE - Fault Isolation

A. Description

(1) This task is for these maintenance messages:

(a) 21-51256 FLOW CTRL VALVE

1) The Flow Control Valve (FCV) failed in the CLOSED position.

2) This message can show during these conditions:

• Unpressurized Take-off

• Unpressurized Landing

• No Engine Bleed Takeoff

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- No Engine Bleed Landing

3) The Pack Flow Temperature Controller (PFTC) cannot sense if upstream Bleed Pressure is available, and it sets the message.

(b) 21-51257 FLOW CTRL VALVE

1) The Flow Control Valve (FCV) failed in the OPEN position.

(2) These maintenance messages are not latched.

B. Fault Isolation Procedure

(1) These maintenance messages are for information and tracking in FAULT HISTORY only. Notify Boeing if one of these messages shows. No Fault Isolation is necessary.

——— **END OF TASK** ———

OMR 820-826

862. F/D SELECTOR - Fault Isolation

A. Description

(1) This task is for these Maintenance Messages:

(a) 21-51053 PFTC CNTRL INOP

1) Flight Deck Selector Temperature Interface Fault

(b) 21-51217 F/D SELECTOR

1) Flight Deck Selector Open Fault

(c) 21-51218 F/D SELECTOR

1) Flight Deck Selector Short Fault

(d) 21-51311 F/D SELECTOR

1) Flight Deck Selector Non Recoverable Fault

2) This is a latched message.

NOTE: After you correct a problem related to a Latched Maintenance Message, you can clear the Maintenance Message as follows:

- Cycle the power to the PFTC or
- Perform the BITE ground test or
- Erase the EXISTING FAULTS from the PFTC Front Panel Menus.

(2) These Maintenance Messages can show on the Left or the Right Pack Flow Temperature Controller (PFTC).

B. Possible Causes

(1) Control Cabin Temperature Selector Switch, S2

(2) Wiring

(3) L (R) PFTC, M2951 (M2952)

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC

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OMR 820-826 (Continued)

(Continued)

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	1	C01160	A/C TEMP CONT VALVE CLOSE RIGHT
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L

D. Related Data

(1) WDM 21-61-11

(2) WDM 21-61-12

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

OMR 820-826; AIRPLANES WITH PFTC P/N 51090248-002/-003

NOTE: Faults can occur on Pack Flow Temperature Controller (PFTC), if one or all of the Temperature Selectors (CONT CAB, FWD CAB, AFT CAB) are set to OFF.

OMR 820-826

(a) If the maintenance message does not show in the PFTC Existing Faults, or as a latched message in the most recent flight leg, then no additional fault isolation steps are required.

OMR 820-826; AIRPLANES WITH PFTC P/N 51090248-005

(b) If the maintenance message shows in the PFTC Existing Faults, or it is a latched message that shows in the most recent flight leg, then do the Fault Isolation Procedure below.

OMR 820-826; AIRPLANES WITH PFTC P/N 51090248-002/-003

(c) If the maintenance message shows in the PFTC Existing Faults, then do these steps to clear the fault:

1) On the P5-17 Cabin Temperature Panel, make sure that all three Temperature Selectors (CONT CAB, FWD CAB, AFT CAB) are not set to OFF.

a) If a Selector is set to OFF, move it away from the OFF position.

2) Erase the EXISTING FAULTS on the Front Panel Display of the applicable PFTC, M2951 (M2952) (21-62 TASK SUPPORT Figure 303 or 21-62 TASK SUPPORT Figure 304).

3) On the Glareshield Panel, P7, push the MASTER CAUTION Light to reset it.

4) Wait for thirty (30) seconds.

5) Make sure that the LH PACK, RH PACK, and ZONE TEMP Lights stay OFF.

6) Look at the EXISTING FAULTS again.

a) If the maintenance message does not show in the PFTC Existing Faults, or as a latched message in the most recent flight leg, then no additional fault isolation steps are required.

b) If the maintenance message still shows in the PFTC Existing Faults, then do the Fault Isolation Procedure below.

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OMR 820-826; AIRPLANES WITH PFTC P/N 51090248-002/-003 (Continued)

(d) If the maintenance message is a latched message that shows in the most recent flight leg, then do the Fault Isolation Procedure below.

OMR 820-826

F. Fault Isolation Procedure

(1) Replace the Control Cabin Temperature Selector Switch, S2. These are the tasks:

- Cabin Temperature Selector Removal, AMM TASK 21-61-07-000-803-002
- Cabin Temperature Selector Installation, AMM TASK 21-61-07-400-803-002

(a) Do the Repair Confirmation at the end of this task.

(2) If the maintenance message shows on the Left PFTC, then do these steps:

(a) Do this check of the wiring between the Control Cabin Temperature Panel, P5-17, and the Left PFTC, M2951 (WDM 21-61-12):

1) Lower the P5 Overhead Panel.

2) Disconnect connector D3996 from the Cabin Temperature Panel, P5-17.

3) Remove the Left PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

4) Do a wiring check as follows:

CONT CAB D3996		**L PFTC D15814B**
pin 4	. . . . .	pin D10
pin 5	. . . . .	pin D12
pin 6	. . . . .	pin D11

a) If you find a problem with the wiring, then do these steps:

<1> Repair the wiring.

<2> Re-connect connector D3996 to the Cabin Temperature Panel, P5-17.

<3> Re-install the Left PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

<4> Do the Repair Confirmation at the end of this task.

b) If you do not find a problem with the wiring, then do these steps:

<1> Re-connect connector D3996 to the Cabin Temperature Panel, P5-17.

<2> Replace the Left PFTC, M2951. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

(3) If the maintenance message shows on the Right PFTC, then do these steps:

(a) Do this check of the wiring between the Control Cabin Temperature Panel, P5-17, and the Right PFTC, M2952 (WDM 21-61-11):

1) Lower the P5 Overhead Panel.

2) Disconnect connector D3996 from the Cabin Temperature Panel, P5-17.

3) Remove the Right PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

4) Do a wiring check as follows:

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OMR 820-826 (Continued)

CONT CAB D3996	R PFTC D15816B
pin 9	pin D10
pin 10	pin D12
pin 11	pin D11

a) If you find a problem with the wiring, then do these steps:
   <1> Repair the wiring.
   <2> Re-connect connector D3996 to the Cabin Temperature Panel, P5-17.
   <3> Re-install the Right PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.
   <4> Do the Repair Confirmation at the end of this task.

b) If you do not find a problem with the wiring, then do these steps:
   <1> Re-connect connector D3996 to the Cabin Temperature Panel, P5-17.
   <2> Replace the Right PFTC, M2952. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

G. Repair Confirmation

(1) For the PFTC, do the "BITE Procedure" and the "BITE Procedure - EXISTING FAULTS" sections of this task:

• Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the fault.

(b) If the maintenance message still shows, continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

863. ZONE SELECTOR - Fault Isolation

A. Description

(1) This task is for these Maintenance Messages:
   (a) 21-51056 PFTC CNTRL INOP
      1) Zone Selector Interface Fault
   (b) 21-51220 ZONE SELECTOR
      1) Zone Selector Open Fault
   (c) 21-51221 ZONE SELECTOR
      1) Zone Selector Short Fault

(2) These Maintenance Messages can show on either the Left or the Right Pack Flow Temperature Controller (PFTC).

B. Possible Causes

(1) For maintenance messages that show on the Left PFTC, M2951:
   (a) Control Cabin Temperature Selector Switch, S4
   (b) Wiring

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OMR 820-826 (Continued)

(c) Left PFTC, M2951

(d) AFT Attendant Control Panel, M2786 (if applicable)

(2) For maintenance messages that show on the Right PFTC, M2952:

(a) Control Cabin Temperature Selector Switch, S3

(b) Wiring

(c) Right PFTC, M2952

(d) FWD Attendant Control Panel, M2785 (if applicable)

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	1	C01160	A/C TEMP CONT VALVE CLOSE RIGHT
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L

D. Related Data

(1) WDM 21-61-13

(2) WDM 21-61-14

E. Initial Evaluation

(1) Do the Pack Temperature Control - BITE Procedure, 21-62 TASK 801 - EXISTING FAULTS and GROUND TESTS (21-62 TASK SUPPORT Figure 307 or 21-62 TASK SUPPORT Figure 308).

OMR 820-826; AIRPLANES WITH PFTC P/N 51090248-002/-003

NOTE: Faults can occur on Pack Flow Temperature Controller (PFTC), if one or all of the Temperature Selectors (CONT CAB, FWD CAB, AFT CAB) are set to OFF.

OMR 820-826

(a) If the maintenance message does not show, then there was an intermittent fault. No further action is necessary.

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(b) If the maintenance message shows, then do the Fault Isolation Procedure below.

OMR 820-826; AIRPLANES WITH PFTC P/N 51090248-002/-003

(c) If the maintenance message still shows, then do these steps to clear the fault:

1) On the P5-17 Cabin Temperature Panel, make sure that all three Temperature Selectors (CONT CAB, FWD CAB, AFT CAB) are not set to OFF.

a) If a Selector is set to OFF, move it away from the OFF position.

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OMR 820-826; AIRPLANES WITH PFTC P/N 51090248-002/-003 (Continued)

2) Erase the EXISTING FAULTS on the Front Panel Display of the applicable PFTC, M2951 (M2952) (21-62 TASK SUPPORT Figure 303 or 21-62 TASK SUPPORT Figure 304).
3) On the Glareshield Panel, P7, push the MASTER CAUTION Light to reset it.
4) Wait for thirty (30) seconds.
5) Make sure that the LH PACK, RH PACK, and ZONE TEMP Lights stay OFF.
6) Look at the EXISTING FAULTS again.
   a) If the maintenance message does not show, then there was an intermittent problem. No further action is necessary.
   b) If the maintenance message still shows, then do the Fault Isolation Procedure below.

OMR 820-826

F. Fault Isolation Procedure

(1) If the maintenance message shows on the Left PFTC, then do these steps:
   (a) Replace the AFT Cabin Temperature Selector Switch, S4. These are the tasks:
   - Cabin Temperature Selector Removal, AMM TASK 21-61-07-000-803-002,
   - Cabin Temperature Selector Installation, AMM TASK 21-61-07-400-803-002.

   1) Do the Repair Confirmation at the end of this task.

   (b) Do this check of the wiring between the Cabin Temperature Panel, P5-17, and the Left PFTC, M2951 (WDM 21-61-14):
      1) Lower the P5 Overhead Panel.
      2) Disconnect connector D3996 from the Cabin Temperature Panel, P5-17.
      3) Remove the Left PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.
      4) Do a wiring check as follows:

AFT CAB	**L PFTC**
D3996	**D15814B**
pin 14	pin C10
pin 15	pin C12
pin 16	pin C11

         a) If you find a problem with the wiring, then do these steps:
            <1> Repair the wiring.
            <2> Re-connect connector D3996 to the Cabin Temperature Panel, P5-17.
            <3> Re-install the Left PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.
            <4> Do the Repair Confirmation at the end of this task.
         b) If you do not find a problem with the wiring, then do these steps:
            <1> Re-connect connector D3996 to the Cabin Temperature Panel, P5-17.

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OMR 820-826 (Continued)

<2> Replace the Left PFTC, M2951. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

(c) (If applicable) Make sure that there are no faults on the AFT Attendant Control Panel, M2786 - 23-42 TASK 816.

(2) If the maintenance message shows on the Right PFTC, then do these steps:

(a) Replace the FWD Cabin Temperature Selector Switch, S3. These are the tasks:

- Cabin Temperature Selector Removal, AMM TASK 21-61-07-000-803-002
- Cabin Temperature Selector Installation, AMM TASK 21-61-07-400-803-002

1) Do the Repair Confirmation at the end of this task.

(b) Do this check of the wiring between the Cabin Temperature Panel, P5-17, and the Right PFTC, M2952 (WDM 21-61-13):

1) Lower the P5 Overhead Panel.

2) Disconnect connector D3998 from the Cabin Temperature Panel, P5-17.

3) Remove the Right PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

4) Do a wiring check as follows:

FWD CAB D3998		R PFTC D15816B
pin 9	. . . . .	pin C10
pin 10	. . . . .	pin C12
pin 11	. . . . .	pin C11

a) If you find a problem with the wiring, then do these steps:

<1> Repair the wiring.

<2> Re-connect connector D3998 to the Cabin Temperature Panel, P5-17.

<3> Re-install the Right PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

<4> Do the Repair Confirmation at the end of this task.

b) If you do not find a problem with the wiring, then do these steps:

<1> Re-connect connector D3998 to the Cabin Temperature Panel, P5-17.

<2> Replace the Right PFTC, M2952. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

(c) (If applicable) Make sure that there are no faults on the FWD Attendant Control Panel, M2785 - 23-42 TASK 816.

G. Repair Confirmation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801 - GROUND TESTS.

(a) If the maintenance message does not show, then you corrected the problem. No further action is necessary.

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OMR 820-826 (Continued)

(b) If the maintenance message shows, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

864. STBY SELECTOR - Fault Isolation

A. Description

(1) This task is for these Maintenance Messages:

(a) 21-51097 PFTC STBY CNTRL

1) Backup Zone Selector Interface Fault

(b) 21-51224 STBY SELECTOR

1) Backup Zone Selector Open Fault

(c) 21-51225 STBY SELECTOR

1) Backup Zone Selector Short Fault

(2) These Maintenance Messages can show on the Left or the Right Pack Flow Temperature Controller (PFTC).

B. Possible Causes

(1) For maintenance messages that show on the Left PFTC, M2951:

(a) Control Cabin Temperature Selector Switch, S3

(b) Wiring

(c) Left PFTC, M2951

(2) For maintenance messages that show on the Right PFTC, M2952:

(a) Control Cabin Temperature Selector Switch, S4

(b) Wiring

(c) Right PFTC, M2952

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	1	C01160	A/C TEMP CONT VALVE CLOSE RIGHT
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L

D. Related Data

(1) WDM 21-61-13

(2) WDM 21-61-14

E. Initial Evaluation

(1) Do the Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

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OMR 820-826; AIRPLANES WITH PFTC P/N 51090248-002/-003

NOTE: Faults can occur on Pack Flow Temperature Controller (PFTC), if one or all of the Temperature Selectors (CONT CAB, FWD CAB, AFT CAB) are set to OFF.

OMR 820-826

(a) If the maintenance message does not show, then there was an intermittent fault.

OMR 820-826; AIRPLANES WITH PFTC P/N 51090248-005

(b) If the maintenance message shows, then do the Fault Isolation Procedure below.

OMR 820-826; AIRPLANES WITH PFTC P/N 51090248-002/-003

(c) If the maintenance message shows, then do these steps to clear the fault:

1) On the P5-17 Cabin Temperature Panel, make sure that all three Temperature Selectors (CONT CAB, FWD CAB, AFT CAB) are not set to OFF.
   a) If a Selector is set to OFF, move it away from the OFF position.
2) Erase the EXISTING FAULTS on the Front Panel Display of the applicable PFTC, M2951 (M2952) (21-62 TASK SUPPORT Figure 303 or 21-62 TASK SUPPORT Figure 304).
3) On the Glareshield Panel, P7, push the MASTER CAUTION Light to reset it.
4) Wait for thirty (30) seconds.
5) Make sure that the LH PACK, RH PACK, and ZONE TEMP Lights stay OFF.
6) Look at the EXISTING FAULTS again.
   a) If the maintenance message does not show, then there was an intermittent problem.
   b) If the maintenance message still shows, then do the Fault Isolation Procedure below.

OMR 820-826

F. Fault Isolation Procedure

(1) If the maintenance message shows on the Left PFTC, then do these steps:

(a) Replace the FWD Cabin Temperature Selector Switch, S3. These are the tasks:

- Cabin Temperature Selector Removal, AMM TASK 21-61-07-000-803-002
- Cabin Temperature Selector Installation, AMM TASK 21-61-07-400-803-002

1) Do the Repair Confirmation at the end of this task.

(b) Do this check of the wiring between the Cabin Temperature Panel, P5-17, and the Left PFTC, M2951 (WDM 21-61-13):

1) Lower the P5 Overhead Panel.
2) Disconnect connector D3998 from the Cabin Temperature Panel, P5-17.
3) Remove the Left PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.
4) Do a wiring check as follows:

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OMR 820-826 (Continued)

FWD CAB D3998		L PFTC D15814C
pin 4	. . . . .	pin 23
pin 5	. . . . .	pin 18
pin 6	. . . . .	pin 17

a) If you find a problem with the wiring, then do these steps:

<1> Repair the wiring.

<2> Re-connect connector D3998 to the Cabin Temperature Panel, P5-17.

<3> Re-install the Left PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

<4> Do the Repair Confirmation at the end of this task.

b) If you do not find a problem with the wiring, then do these steps:

<1> Re-connect connector D3998 to the Cabin Temperature Panel, P5-17.

<2> Replace the Left PFTC, M2951. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

(2) If the maintenance message shows on the Right PFTC, then do these steps:

(a) Replace the AFT Cabin Temperature Selector Switch, S4. These are the tasks:

- Cabin Temperature Selector Removal, AMM TASK 21-61-07-000-803-002
- Cabin Temperature Selector Installation, AMM TASK 21-61-07-400-803-002

1) Do the Repair Confirmation at the end of this task.

(b) Do this check of the wiring between the Cabin Temperature Panel, P5-17, and the Right PFTC, M2952 (WDM 21-61-14):

1) Lower the P5 Overhead Panel.

2) Disconnect connector D3996 from the Cabin Temperature Panel, P5-17.

3) Remove the Right PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

4) Do a wiring check as follows:

AFT CAB D3996		R PFTC D15816C
pin 21	. . . . .	pin 23
pin 22	. . . . .	pin 18
pin 23	. . . . .	pin 17

a) If you find a problem with the wiring, then do these steps:

<1> Repair the wiring.

<2> Re-connect connector D3996 to the Cabin Temperature Panel, P5-17.

<3> Re-install the Right PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

<4> Do the Repair Confirmation at the end of this task.

b) If you do not find a problem with the wiring, then do these steps:

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OMR 820-826 (Continued)

<1> Re-connect connector D3996 to the Cabin Temperature Panel, P5-17.

<2> Replace the Right PFTC, M2952. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

G. Repair Confirmation

(1) Do the Pack Temperature Control - BITE Procedure, 21-62 TASK 801

(a) If the maintenance message does not show, then you corrected the problem.

(b) If the maintenance message shows, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

865. F/D TEMP SENSOR - Fault Isolation

A. Description

(1) This task is for these Maintenance Messages:

(a) 21-51190 F/D TEMP SENSOR

1) Flight Deck Temperature Sensor Open Fault

(b) 21-51191 F/D TEMP SENSOR

1) Flight Deck Temperature Sensor Short Fault

(c) 21-51310 F/D TEMP SENSOR

1) Flight Deck Temperature Sensor Non Recoverable Fault

2) This is a latched message.

NOTE: After you correct a problem related to a Latched Maintenance Message, you can clear the Maintenance Message as follows:

- Cycle the power to the PFTC or
- Perform the BITE ground test or
- Erase the EXISTING FAULTS from the PFTC Front Panel Menus.

(2) These Maintenance Messages can show on either the Left or the Right Pack Flow Temperature Controller (PFTC).

B. Possible Causes

(1) Control Cabin Temperature Sensor

(2) Wiring

(3) L (R) PFTC, M2951 (M2952)

C. Circuit Breakers

(1) These are the primary circuit breakers related to this fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 820-826 (Continued)

D. Related Data

(1) WDM 21-61-11

(2) WDM 21-61-12

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message shows in the PFTC Existing Faults, or it is a latched message that shows in the most recent flight leg, then do the Fault Isolation Procedure below.

(b) If the maintenance message does not show in the PFTC Existing Faults, or as a latched message in the most recent flight leg, then no additional fault isolation steps are required.

F. Fault Isolation Procedure

(1) If the maintenance message shows on the Left PFTC, then do these steps:

(a) Do this check of the Control Cabin Temperature Sensor Module, M1718:

1) Lower the access panel for the Control Cabin Temperature Sensor that is aft of the Overhead P5 Panel.

2) Disconnect electrical connector D10958 from the Control Cabin Temperature Sensor (WDM 21-61-12) .

3) Do a check for an open circuit or a short between pins 3 and 4 of the Control Cabin Temperature Sensor.

4) If there is a short or an open circuit between pins 3 and 4, then do these steps:

a) Replace the Control Cabin Temperature Sensor. These are the tasks:

- Cabin Temperature Sensor Removal, AMM TASK 21-61-10-000-802
- Cabin Temperature Sensor Installation, AMM TASK 21-61-10-400-802

<1> Do the Repair Confirmation at the end of this task.

5) If there is not a short or an open circuit between pins 3 and 4, then continue.

(b) Do this check of the wiring between the Control Cabin Temperature Sensor and the Left PFTC, M2951 (WDM 21-61-12):

1) Remove the Left PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

2) Do a wiring check as follows:

TEMP SENSOR D10958	**L PFTC D15814B**
pin 3	pin E11
pin 4	pin E12

3) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-connect the connector D10958 to the Control Cabin Temperature Sensor.

c) Re-install the Left PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

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d) Do the Repair Confirmation at the end of this task.

4) If you do not find a problem with the wiring, then do these steps:

a) Re-connect the connector D10958 to Control Cabin Temperature Sensor.

b) Replace the Left PFTC, M2951. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

(2) If the maintenance message shows on the Right PFTC, then do these steps:

(a) Do this check of the Control Cabin Temperature Sensor Module, M1718:

1) Lower the access panel for the Control Cabin Temperature Sensor that is aft of the Overhead P5 Panel.

2) Disconnect electrical connector D10958 from the Control Cabin Temperature Sensor WDM 21-61-11).

3) Do a check for an open circuit or a short between pins 1 and 2 of the Control Cabin Temperature Sensor.

4) If there is a short or an open circuit between pins 1 and 2, then do these steps:

a) Replace the Control Cabin Temperature Sensor. These are the tasks:

- Cabin Temperature Sensor Removal, AMM TASK 21-61-10-000-802
- Cabin Temperature Sensor Installation, AMM TASK 21-61-10-400-802

<1> Do the Repair Confirmation at the end of this task.

5) If there is not a short or an open circuit between pins 1 and 2, then continue.

(b) Do this check of the wiring between the Control Cabin Temperature Sensor Module, M1718, and the Right PFTC, M2952 (WDM 21-61-11):

1) Remove the Right PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

2) Do a wiring check as follows:

TEMP SENSOR D10958	**R PFTC D15816B**
pin 1	pin E11
pin 2	pin E12

3) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-connect the connector D10958 to Control Cabin Temperature Sensor.

c) Re-install the Right PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

d) Do the Repair Confirmation at the end of this task.

4) If you do not find a problem with the wiring, then do these steps:

a) Re-connect the connector D10958 to the Control Cabin Temperature Sensor.

b) Replace the Right PFTC, M2952. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801

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G. Repair Confirmation

(1) For the PFTC, do the "BITE Procedure" and the "BITE Procedure - EXISTING FAULTS" sections of this task:

- Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the fault.

(b) If the maintenance message still shows, continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

866. F/D DUCT SENSOR - Fault Isolation

A. Description

(1) This task is for these Maintenance Messages:

(a) 21-51199 F/D DUCT SENSOR

1) Flight Deck Duct Temperature Sensor Open Fault

(b) 21-51200 F/D DUCT SENSOR

1) Flight Deck Duct Temperature Sensor Short Fault

(c) 21-51312 F/D DUCT SENSOR

1) Flight Deck Duct Temperature Sensor Non Recoverable Fault

2) This is a latched message.

NOTE: After you correct a problem related to a Latched Maintenance Message, you can clear the Maintenance Message as follows:

- Cycle the power to the PFTC or
- Perform the BITE ground test or
- Erase the EXISTING FAULTS from the PFTC Front Panel Menus.

(2) These Maintenance Messages can show on either the Left or the Right Pack Flow Temperature Controller (PFTC).

B. Possible Causes

(1) For maintenance messages that show on the Left PFTC, M2951:

(a) Control Cabin Duct Temperature Sensor Backup, T477

(b) Wiring

(c) Left PFTC, M2951

(2) For maintenance messages that show on the Right PFTC, M2952:

(a) Control Cabin Duct Temperature Sensor, T478

(b) Wiring

(c) Right PFTC, M2952

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C. Circuit Breakers

(1) These are the primary circuit breakers related to this fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

D. Related Data

(1) WDM 21-61-11

(2) WDM 21-61-12

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message shows in the PFTC Existing Faults, or it is a latched message that shows in the most recent flight leg, then do the Fault Isolation Procedure below.

(b) If the maintenance message does not show in the PFTC Existing Faults, or as a latched message in the most recent flight leg, then no additional fault isolation steps are required.

F. Fault Isolation Procedure

(1) If the maintenance message shows on the Left PFTC, then do these steps:

(a) Do this check of the Control Cabin Duct Temperature Sensor Backup, T477:

1) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

2) Disconnect electrical connector D3882 from the Control Cabin Duct Temperature Sensor Backup.

3) Do a check for an open circuit or a short between pins 1 and 2 of the Control Cabin Temperature Sensor Backup.

4) If there is a short or an open circuit between pins 1 and 2, then do these steps:

a) Replace the Control Cabin Duct Temperature Sensor Backup, T477. These are the tasks:

- Duct Temperature Limit Sensor Removal, AMM TASK 21-61-04-000-803-002
- Duct Temperature Limit Sensor Installation, AMM TASK 21-61-04-400-803-002

<1> Do the Repair Confirmation at the end of this task.

5) If there is not a short or an open circuit between pins 1 and 2, then continue.

(b) Do this check of the wiring between the Control Cabin Duct Temperature Sensor Backup, T477, and the Left PFTC, M2951 (WDM 21-61-12):

1) Remove the Left PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

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2) Do a wiring check as follows:

L PFTC D15814B	DUCT TEMP SENSOR D3882
pin E9	pin 1
pin E10	pin 2

3) If you find a problem with the wiring, then do these steps:
   a) Repair the wiring.
   b) Re-connect the connector D3882 to the Control Cabin Duct Temperature Sensor, T477.
   c) Re-install the Left PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.
   d) Do the Repair Confirmation at the end of this task.
4) If you do not find a problem with the wiring, then do these steps:
   a) Re-connect the connector D3882 to the Control Cabin Duct Temperature Sensor, T477.
   b) Replace the Left PFTC, M2951. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

(2) If the maintenance message shows on the Right PFTC, then do these steps:
   (a) Do this check of the Control Cabin Duct Temperature Sensor, T478:
      1) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

      2) Disconnect electrical connector D3884 from the Control Cabin Duct Temperature Sensor.
      3) Do a check for an open circuit or a short between pins 1 and 2 of the Control Cabin Duct Temperature Sensor.
      4) If there is a short or an open circuit between pins 1 and 2, then do these steps:
         a) Replace the Control Cabin Duct Temperature Sensor, T478. These are the tasks:
            - Duct Temperature Limit Sensor Removal, AMM TASK 21-61-04-000-803-002
            - Duct Temperature Limit Sensor Installation, AMM TASK 21-61-04-400-803-002

            <1> Do the Repair Confirmation at the end of this task.
      5) If there is not a short or an open circuit between pins 1 and 2, then continue.
   (b) Do this check of the wiring between the Control Cabin Duct Temperature Sensor, T478, and the Right PFTC, M2952 (WDM 21-61-11):
      1) Remove the Right PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.
      2) Do a wiring check as follows:

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R PFTC D15816B		DUCT TEMP SENSOR D3884
pin E9	. . . . .	pin 1
pin E10	. . . . .	pin 2

3) If you find a problem with the wiring, then do these steps:
   a) Repair the wiring.
   b) Re-connect the connector D3884 to the Control Cabin Duct Temperature Sensor, T478.
   c) Re-install the Right PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.
   d) Do the Repair Confirmation at the end of this task.
4) If you do not find a problem with the wiring, then do these steps:
   a) Re-connect the connector D3884 to the Control Cabin Duct Temperature Sensor, T478.
   b) Replace the Right PFTC, M2952. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

G. Repair Confirmation

(1) For the PFTC, do the "BITE Procedure" and the "BITE Procedure - EXISTING FAULTS" sections of this task:

• Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the fault.

(b) If the maintenance message still shows, continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

867. ZONE TEMP SENSOR - Fault Isolation

A. Description

(1) This task is for these Maintenance Messages:
   (a) 21-51193 ZONE TEMP SENSOR
      1) Zone Temperature Sensor Open Fault
   (b) 21-51194 ZONE TEMP SENSOR
      1) Zone Temperature Sensor Short Fault

(2) These Maintenance Messages can show on either the Left or the Right Pack Flow Temperature Controller (PFTC).

B. Possible Causes

(1) For maintenance messages that show on the Left PFTC, M2951:
   (a) AFT Cabin Temperature Sensor
   (b) Wiring
   (c) Left PFTC, M2951

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OMR 820-826 (Continued)

(2) For maintenance messages that show on the Right PFTC, M2952:

(a) Forward Cabin Temperature Sensor

(b) Wiring

(c) Right PFTC, M2952

C. Circuit Breakers

(1) These are the primary circuit breakers related to this fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	3	C01170	A/C ZONE TEMP VALVE/FAN CONT AFT PASS
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	2	C01167	A/C ZONE TEMP VALVE/FAN CONT FWD PASS
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

D. Related Data

(1) WDM 21-61-13

(2) WDM 21-61-14

E. Initial Evaluation

(1) Do the Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then there was an intermittent fault.

(b) If the maintenance message still shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) If the maintenance message shows on the Left PFTC, then do these steps:

NOTE: The AFT Cabin Temperature Sensor is located behind the bullnose covers below the overhead stowage bins. The Sensor Assembly is on the right side of the airplane.

(a) Do this check of the AFT Cabin Temperature Sensor:

1) Lower the access panel for the AFT Cabin Temperature Sensor Assembly.

2) Disconnect electrical connector D15506 from the AFT Cabin Temperature Sensor.

3) Do a check for an open circuit or a short between pins 1 and 2 of the AFT Cabin Temperature Sensor.

4) If there is a short or an open circuit between pins 1 and 2, then do these steps:

a) Replace the AFT Cabin Temperature Sensor. These are the tasks:

- Cabin Temperature Sensor Removal, AMM TASK 21-61-10-000-802
- Cabin Temperature Sensor Installation, AMM TASK 21-61-10-400-802

<1> Do the Repair Confirmation at the end of this task.

5) If there is not a short or an open circuit between pins 1 and 2, then continue.

(b) Do this check of the wiring between the Left PFTC, M2951, and the AFT Cabin Temperature Sensor (WDM 21-61-14):

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1) Remove the Left PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

2) Do a wiring check as follows:

L PFTC D15814B		AFT TEMP SENSOR D15506
pin C8	. . . . .	pin 1
pin C9	. . . . .	pin 2

3) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-connect the connector D15506 to the AFT Cabin Temperature Sensor.

c) Re-install the Left PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

d) Do the Repair Confirmation at the end of this task.

4) If you do not find a problem with the wiring, then do these steps:

a) Re-connect the connector D15506 to AFT Cabin Temperature Sensor.

b) Replace the Left PFTC, M2951. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

(2) If the maintenance message shows on the Right PFTC, then do these steps:

(a) Do this check of the Forward Cabin Temperature Sensor:

NOTE: The Forward Cabin Temperature Sensor is located behind the bullnose covers below the Overhead Stowage Bins. The Sensor Assembly is on the right side of the airplane.

1) Lower the panel for the Forward Cabin Temperature Sensor Assembly.

2) Disconnect electrical connector D15016 from the Forward Cabin Temperature Sensor (WDM 21-61-13).

3) Do a check for an open circuit or a short between pins 1 and 2 of the Forward Cabin Temperature Sensor.

4) If there is a short or an open circuit between pins 1 and 2, then do these steps:

a) Replace the Forward Cabin Temperature Sensor. These are the tasks:

- Cabin Temperature Sensor Removal, AMM TASK 21-61-10-000-802
- Cabin Temperature Sensor Installation, AMM TASK 21-61-10-400-802

<1> Do the Repair Confirmation at the end of this task.

5) If there is not a short or an open circuit between pins 1 and 2, then continue.

(b) Do this check of the wiring between the Right PFTC, M2952 and the Forward Cabin Temperature Sensor (WDM 21-61-13):

1) Remove the Right PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

2) Do a wiring check as follows:

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R PFTC D15816B		FWD Temp Sensor D15016
pin C9	. . . . .	pin 1
pin C8	. . . . .	pin 2

3) If you find a problem with the wiring, then do these steps:
   a) Repair the wiring.
   b) Re-connect the connector D15016 to Forward Cabin Temperature Sensor.
   c) Re-install the Right PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.
   d) Do the Repair Confirmation at the end of this task.
4) If you do not find a problem with the wiring, then do these steps:
   a) Re-connect the connector D15016 to the Forward Cabin Temperature Sensor.
   b) Replace the Right PFTC, M2952. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

G. Repair Confirmation

(1) Do the Pack Temperature Control - BITE Procedure, 21-62 TASK 801.
   (a) If the maintenance message does not show, then you corrected the problem.
   (b) If the maintenance message shows, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

868. STBY ZONE SENSOR - Fault Isolation

A. Description

(1) This task is for these maintenance messages:
   (a) 21-51100 PFTC STBY CNTRL
      1) Backup Zone Temp Error Output Fault
   (b) 21-51101 PFTC STBY CNTRL
      1) Backup Zone Temp Error Input Fault
   (c) 21-51196 STBY ZONE SENSOR
      1) Backup Zone Temperature Sensor Open Fault
   (d) 21-51197 STBY ZONE SENSOR
      1) Backup Zone Temperature Sensor Short Fault
(2) These maintenance messages can show on either the Left or the Right PFTC.

B. Possible Causes

(1) Forward (AFT) Cabin Temperature Sensor
(2) Wiring
(3) Left (Right) PFTC, M2951 (M2952)

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C. Circuit Breakers

(1) These are the primary circuit breakers related to this fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	3	C01170	A/C ZONE TEMP VALVE/FAN CONT AFT PASS
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	2	C01167	A/C ZONE TEMP VALVE/FAN CONT FWD PASS
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

D. Related Data

(1) SSM 21-61-13

(2) SSM 21-61-14

(3) WDM 21-61-13

(4) WDM 21-61-14

E. Initial Evaluation

(1) Do the Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then there was an intermittent fault.

(b) If the maintenance message still shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) If the maintenance message shows on the Left (Right) PFTC, then do these steps:

NOTE: The Forward (AFT) Cabin Temperature Sensor is located directly above the passenger seats on the right side of the aisle. The sensor assembly is on a panel between the PSUs.

(a) Do this check of the Forward (AFT) Cabin Temperature Sensor:

1) Lower the access panel for the applicable Forward (AFT) Cabin Temperature Sensor.

2) Disconnect the applicable connector:

- Connector D15016 from the Forward Passenger Cabin Temperature Sensor Module, M2847
- Connector D15022 from the Aft Passenger Cabin Temperature Sensor Module, M2848

3) Do a check for an open circuit or a short as follows (WDM 21-61-13, WDM 21-61-14):

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MAINTENANCE MESSAGE ON LEFT PFTC	
FWD TEMP SENSOR MODULE, M2847 RECEPTACLE	**FWD TEMP SENSOR MODULE, M2847 RECEPTACLE**
pin 3	pin 4

MAINTENANCE MESSAGE ON RIGHT PFTC	
AFT TEMP SENSOR MODULE, M2848 RECEPTACLE	**AFT TEMP SENSOR MODULE, M2848 RECEPTACLE**
pin 3	pin 4

4) If there is a short or an open circuit, the do these steps:
   a) Replace the Forward (AFT) Cabin Temperature Sensor. These are the tasks:
      - Cabin Temperature Sensor Removal, AMM TASK 21-61-10-000-802
      - Cabin Temperature Sensor Installation, AMM TASK 21-61-10-400-802
   b) Do the Repair Confirmation at the end of this task.
5) If there is not a short or an open circuit, then continue.

(2) Do this check of the wiring between the Left (Right) PFTC and the Forward (AFT) Cabin Temperature Sensor Module:
   (a) Remove the applicable Left (Right) PFTC, M2951 (M2952). This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.
   (b) Do a wiring check as follows (WDM 21-61-13, WDM 21-61-14):

MAINTENANCE MESSAGE ON LEFT PFTC	
L PFTC, M2951 D15814C	**FWD TEMP SENSOR MODULE, M2847 D15016**
pin 4	pin 3
pin 10	pin 4

MAINTENANCE MESSAGE ON RIGHT PFTC	
R PFTC, M2952 D15816C	**AFT TEMP SENSOR MODULE, M2848 D15022**
pin 4	pin 3
pin 10	pin 4

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(c) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-connect connector to the Forward (AFT) Cabin Temperature Sensor Module.

3) Re-install the Left (Right) PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

4) Do the Repair Confirmation at the end of this task.

(d) If you do not find a problem with the wiring, then do these steps:

1) Re-connect connector to the Forward (AFT) Cabin Temperature Sensor Module.

2) Install a new Left (Right) PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

3) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do the Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the problem.

(b) If the maintenance message shows, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

869. ZONE DUCT SENSOR - Fault Isolation

A. Description

(1) This task is for these Maintenance Messages:

(a) 21-51202 ZONE DUCT SENSOR

1) Zone Duct Temperature Sensor Open Fault

(b) 21-51203 ZONE DUCT SENSOR

1) Zone Duct Temperature Sensor Short Fault

(2) These Maintenance Messages can show on either the Left or the Right Pack Flow Temperature Controller (PFTC).

B. Possible Causes

(1) For maintenance messages that show on the Left PFTC, M2951:

(a) AFT Cabin Duct Temperature Sensor Backup, T536

(b) Wiring

(c) Left PFTC, M2951

(2) For maintenance messages that show on the Right PFTC, M2952:

(a) Forward Cabin Duct Temperature Sensor, T533

(b) Wiring

(c) Right PFTC, M2952

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OMR 820-826 (Continued)

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

D. Related Data

(1) WDM 21-61-13

(2) WDM 21-61-14

E. Initial Evaluation

(1) Do the Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then there was an intermittent fault.

(b) If the maintenance message still shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) If the maintenance message shows on the Left PFTC, then do these steps:

(a) Do this check of the AFT Cabin Duct Temperature Sensor, T536:

1) Remove the Right Ceiling Panel that is adjacent to the Overwing Escape Hatches. This is the task: Main Ceiling Panel - Removal, AMM TASK 25-21-45-000-803-001.

2) Disconnect electrical connector D11840 from the AFT Cabin Duct Temperature Sensor (WDM 21-61-14).

3) Do a check for an open circuit or a short between pins 1 and 2 of the AFT Cabin Duct Temperature Sensor.

4) If you find a short or an open circuit between pins 1 and 2, then do these steps:

a) Replace the AFT Cabin Duct Temperature Sensor, T536. These are the tasks:

- Duct Temperature Limit Sensor Removal, AMM TASK 21-61-04-000-803-002
- Duct Temperature Limit Sensor Installation, AMM TASK 21-61-04-400-803-002

<1> Do the Repair Confirmation at the end of this task.

5) If you do not find a short or an open circuit between pins 1 and 2, then continue.

(b) Do this check of the wiring between the Left PFTC, M2951, and the AFT Cabin Duct Temperature Sensor, T536 (WDM 21-61-14):

1) Remove the Left PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

2) Do a wiring check as follows:

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L PFTC D15814B		AFT DUCT TEMP SENSOR D11840
pin B8	. . . . .	pin 1
pin B9	. . . . .	pin 2

3) If you find a problem with the wiring, then do these steps:
   a) Repair the wiring.
   b) Re-connect the connector D11840 to the AFT Cabin Duct Temperature Sensor, T536.
   c) Re-install the Left PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.
   d) Do the Repair Confirmation at the end of this task.
4) If you do not find a problem with the wiring, then do these steps:
   a) Re-connect the connector D11840 to the AFT Cabin Duct Temperature Sensor, T536.
   b) Replace the Left PFTC, M2951. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

(2) If the maintenance message shows on the Right PFTC, then do these steps:
   (a) Do this check of the Forward Cabin Duct Temperature Sensor, T533:
      1) Remove the Left Ceiling Panel that is just forward of the Overwing Escape Hatches. This is the task: Main Ceiling Panel - Removal, AMM TASK 25-21-45-000-803-001.
      2) Disconnect electrical connector D11830 from the Forward Cabin Duct Temperature Sensor.
      3) Do a check for an open circuit or a short between pins 1 and 2 of the Forward Cabin Duct Temperature Sensor.
      4) If you find a short or an open circuit between pins 1 and 2, then do these steps:
         a) Replace the Forward Cabin Duct Temperature Sensor, T533. These are the tasks:
            - Duct Temperature Limit Sensor Removal, AMM TASK 21-61-04-000-803-002
            - Duct Temperature Limit Sensor Installation, AMM TASK 21-61-04-400-803-002

            <1> Do the Repair Confirmation at the end of this task.
      5) If you do not find a short or an open circuit between pins 1 and 2, then continue.
   (b) Do this check of the wiring between the Forward Cabin Duct Temperature Sensor, T533, and the Right PFTC, M2952 (WDM 21-61-13):
      1) Remove the Right PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.
      2) Do a wiring check as follows:

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R PFTC D15816B		FWD DUCT TEMP SENSOR D11830
pin B8	. . . . .	pin 1
pin B9	. . . . .	pin 2

3) If you find a problem with the wiring, then do these steps:
   a) Repair the wiring.
   b) Re-connect the connector D11830 to the Forward Cabin Duct Temperature Sensor, T533.
   c) Re-install the Right PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.
   d) Do the Repair Confirmation at the end of this task.
4) If you do not find a problem with the wiring, then do these steps:
   a) Re-connect the connector D11830 to the Forward Cabin Duct Temperature Sensor, T533.
   b) Replace the Right PFTC, M2952. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

G. Repair Confirmation

(1) Do the Pack Temperature Control - BITE Procedure, 21-62 TASK 801.
   (a) If the maintenance message does not show, then you corrected the problem.
   (b) If the maintenance message shows, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

870. SRADA RAM SENSOR - Fault Isolation

A. Description

(1) This task is for these Maintenance Messages:
   (a) 21-51403 SRADA RAM SENSOR
      1) SRADA Ram Sensor Interface Fault
   (b) 21-51407 SRADA RAM SENSOR
      1) SRADA Ram Sensor CBIT Fault
   (c) 21-51411 SRADA RAM SENSOR
      1) SRADA Ram Sensor IBIT Fault
      2) This is a latched message.

         NOTE: After you correct a problem related to a Latched Maintenance Message, you can clear the Maintenance Message as follows:
         - Cycle the power to the PFTC or
         - Perform the BITE ground test or
         - Erase the EXISTING FAULTS from the PFTC Front Panel Menus.

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OMR 820-826 (Continued)

(2) These Maintenance Messages can show on either the Left or the Right Pack Flow Temperature Controller (PFTC).

B. Possible Causes

(1) For maintenance messages that show on the Left PFTC, M2951:

(a) Left RAM Air Control Temperature Sensor, T481

(b) Wiring

(c) Left PFTC, M2951

(d) Left Forward (Inlet) Ram Air Actuator, M2795

(2) For maintenance messages that show on the Right PFTC, M2952:

(a) Right RAM Air Control Temperature Sensor, T482

(b) Wiring

(c) Right PFTC, M2952

(d) Right Forward (Inlet) Ram Air Actuator, M2797

C. Circuit Breakers

(1) These are the primary circuit breakers related to this fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	8	C00555	AIR CONDITIONING RAM AIR MOD CONT L
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	8	C01178	AIR CONDITIONING RAM AIR MOD CONT R
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L

D. Related Data

(1) WDM 21-51-14

(2) WDM 21-51-24

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message shows in the PFTC Existing Faults, or it is a latched message that shows in the most recent flight leg, then do the Fault Isolation Procedure below.

(b) If the maintenance message does not show in the PFTC Existing Faults, or as a latched message in the most recent flight leg, then no additional fault isolation steps are required.

F. Fault Isolation Procedure

(1) If the maintenance message shows on the Left PFTC, then do these steps:

(a) Do this check of the Left Ram Air Control Temperature Sensor, T481:

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OMR 820-826 (Continued)

1) Open this access panel:

Number	Name/Location
192CL	ECS Access Door

2) Disconnect electrical connector D3916 from the Left Ram Air Control Temperature Sensor, T481 ((WDM 21-51-14)).

3) Do a check for an open circuit or a short between pins 1 and 2 of the Left Ram Air Control Temperature Sensor.

4) If there is a short or an open circuit between pins 1 and 2, then do these steps:

   a) Replace the Left Ram Air Control Temperature Sensor. These are the tasks:

   - Ram Air Control Temperature Sensor - Removal, AMM TASK 21-51-50-000-801
   - Ram Air Control Temperature Sensor - Installation, AMM TASK 21-51-50-400-801

   b) Do the Repair Confirmation at the end of this task.

5) If there is not a short or an open circuit between pins 1 and 2, then continue.

(b) Do this check of the wiring between the Left Ram Air Control Temperature Sensor, T481, and the Left Forward (Inlet) Ram Air Actuator, M2795 (WDM 21-51-14):

1) Get access to the Left Forward (Inlet) Ram Air Actuator, M2795, as follows:

   a) Open these access panels:

Number	Name/Location
191FL	Forward Wing To Body Fairing Panel - Mid Fairing, Above Ram Air Inlet
191GL	Ram Air Actuator Panel - Forward

2) Disconnect electrical connector D14394 from the Left Forward (Inlet) Ram Air Actuator, M2795.

3) Do a wiring check as follows:

L RAM AIR SENSOR D3916		L FWD RAM ACTUATOR D14386
pin 1	. . . . .	pin 17
pin 2	. . . . .	pin 18

4) If you find a problem with the wiring, then do these steps:

   a) Repair the wiring.

   b) Reconnect electrical connector D3916 to the Left Ram Air Sensor, T481.

   c) Reconnect electrical connector D14386 to the Left Forward (Inlet) Ram Air Actuator, M2795.

   d) Do the Repair Confirmation at the end of this task.

5) If you do not find a problem with the wiring, then replace the Left Forward (Inlet) Ram Air Actuator, M2795. These are the tasks:

   - Ram Air Inlet Actuator Removal, AMM TASK 21-51-21-000-801

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- Ram Air Inlet Actuator Installation, AMM TASK 21-51-21-400-801

a) Do the Repair Confirmation at the end of this task.

(c) Do this check of the wiring between the Left PFTC, M2951, and the Left Forward (Inlet) Ram Air Actuator, M2795:

1) Remove the Left PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

2) Disconnect electrical connector D14386 from the Left Forward (Inlet) Ram Air Actuator, M2795.

3) Do a wiring check as follows:

L PFTC D15814B		L FWD RAM ACTUATOR D14386
pin G7	. . . . .	pin 12

4) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-install the Left PFTC, M2951. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

c) Reconnect electrical connector D14386 to the Left Forward (Inlet) Ram Air Actuator, M2795.

d) Do the Repair Confirmation at the end of this task.

5) If you do not find a problem with the wiring, then replace the Left PFTC, M2951. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

a) Do the Repair Confirmation at the end of this task.

(2) If the maintenance message shows on the Right PFTC, then do these steps:

(a) Do this check of the Right Ram Air Control Temperature Sensor, T482:

1) Open this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

2) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

3) Disconnect electrical connector D3918 from the Right Ram Air Control Temperature Sensor, T482 (WDM 21-51-24).

4) Do a check for an open circuit or a short between pins 1 and 2 of the Right Ram Air Control Temperature Sensor.

5) If there is a short or an open circuit between pins 1 and 2, then do these steps:

a) Replace the Right Ram Air Control Temperature Sensor. These are the tasks:

- Ram Air Control Temperature Sensor - Removal, AMM TASK 21-51-50-000-801

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OMR 820-826 (Continued)

- Ram Air Control Temperature Sensor - Installation, AMM TASK 21-51-50-400-801

b) Do the Repair Confirmation at the end of this task.

6) If there is not a short or an open circuit between pins 1 and 2, then continue.

(b) Do this check of the wiring between the Right Ram Air Control Temperature Sensor, T482, and the Right Forward (Inlet) Ram Air Actuator, M2797 (WDM 21-51-24):

1) Get access to the Right Forward (Inlet) Ram Air Actuator, M2797, as follows:

a) Open these access panels:

Number	Name/Location
191FR	Forward Wing To Body Fairing Panel - Mid Fairing, Above Ram Air Inlet
191GR	Ram Air Actuator Panel - Forward

2) Disconnect electrical connector D14394 from the Right Forward (Inlet) Ram Air Actuator, M2797.

3) Do a wiring check as follows:

R RAM AIR SENSOR D3918		R FWD RAM ACTUATOR D14394
pin 1	. . . . .	pin 17
pin 2	. . . . .	pin 18

4) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Reconnect electrical connector D3918 to the Right Ram Air Sensor, T482.

c) Reconnect electrical connector D14394 to the Right Forward (Inlet) Ram Air Actuator, M2797.

d) Do the Repair Confirmation at the end of this task.

5) If you do not find a problem with the wiring, then replace the Right Forward (Inlet) Ram Air Actuator, M2797. These are the tasks:

- Ram Air Inlet Actuator Removal, AMM TASK 21-51-21-000-801
- Ram Air Inlet Actuator Installation, AMM TASK 21-51-21-400-801

a) Do the Repair Confirmation at the end of this task.

(c) Do this check of the wiring between the Right PFTC, M2952, and the Right Forward (Inlet) Ram Air Actuator, M2797 (WDM 21-51-24):

1) Remove the Right PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

2) Disconnect electrical connector D14394 from the Right Forward (Inlet) Ram Air Actuator, M2797.

3) Do a wiring check as follows:

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OMR 820-826 (Continued)

R PFTC	R FWD RAM ACTUATOR
D15816B	D14394
pin G7	pin 12

4) If you find a problem with the wiring, then do these steps:
   a) Repair the wiring.
   b) Re-install the Right PFTC, M2952. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.
   c) Reconnect electrical connector D14394 to the Right Forward (Inlet) Ram Air Actuator, M2797.
   d) Do the Repair Confirmation at the end of this task.
5) If you do not find a problem with the wiring, then replace the Right PFTC, M2952. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.
   a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) For the PFTC, do the "BITE Procedure" and the "BITE Procedure - EXISTING FAULTS" sections of this task:
- Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the fault.

(b) If the maintenance message still shows, continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

871. PACK TEMP SENSOR - Fault Isolation

A. Description

(1) This task is for these Maintenance Messages:
   (a) 21-51205 PACK TEMP SENSOR
      1) PACK Temperature Sensor Open Fault
   (b) 21-51206 PACK TEMP SENSOR
      1) PACK Temperature Sensor Short Fault
   (c) 21-51313 PACK TEMP SENSOR
      1) PACK Temperature Sensor Non Recoverable Fault
      2) This is a latched message.

NOTE: After you correct a problem related to a Latched Maintenance Message, you can clear the Maintenance Message as follows:
- Cycle the power to the PFTC or
- Perform the BITE ground test or
- Erase the EXISTING FAULTS from the PFTC Front Panel Menus.

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OMR 820-826 (Continued)

(2) These Maintenance Messages can show on either the Left or the Right Pack Flow Temperature Controller (PFTC).

B. Possible Causes

(1) For maintenance messages that show on the Left PFTC, M2951:

(a) Left Pack Temperature Sensor, T471

(b) Wiring

(c) Left PFTC, M2951

(2) For maintenance messages that show on the Right PFTC, M2952:

(a) Right Pack Temperature Sensor, T474

(b) Wiring

(c) Right PFTC, M2952

C. Circuit Breakers

(1) These are the primary circuit breakers related to this fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

D. Related Data

(1) WDM 21-51-12

(2) WDM 21-51-22

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message shows in the PFTC Existing Faults, or it is a latched message that shows in the most recent flight leg, then do the Fault Isolation Procedure below.

(b) If the maintenance message does not show in the PFTC Existing Faults, or as a latched message in the most recent flight leg, then no additional fault isolation steps are required.

F. Fault Isolation Procedure

(1) If the maintenance message shows on the Left PFTC, M2951, then do these steps:

(a) Do this check of the Left Pack Temperature Sensor, T471:

1) Open this access panel:

Number	Name/Location
192CL	ECS Access Door

2) Disconnect electrical connector D3866 from the Left Pack Temperature Sensor (WDM 21-51-12).

3) Do a check for an open circuit or a short between pins 1 and 2 of the Left Pack Temperature Sensor.

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OMR 820-826 (Continued)

4) If there is a short or an open circuit between pins 1 and 2, then do these steps:
   a) Replace the Left Pack Temperature Sensor, T471. These are the tasks:
      • Pack Temperature Sensor Removal, AMM TASK 21-51-52-000-801
      • Pack Temperature Sensor Installation, AMM TASK 21-51-52-400-801
   b) Do the Repair Confirmation at the end of this task.
5) If there is not a short or an open circuit between pins 1 and 2, then continue.

(b) Do this check of the wiring between the Left PFTC, M2951, and the Left Pack Temperature Sensor, T471 (WDM 21-51-12):

1) Remove the Left PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.
2) Do a wiring check as follows:

L PFTC D15814B	L PACK TEMP SENSOR D3866
pin J9	pin 1
pin J10	pin 2

   a) If you find a problem with the wiring, then do these steps:
      <1> Repair the wiring.
      <2> Re-connect connector D3866 to the Left Pack Temperature Sensor, T471.
      <3> Close this access panel:

Number	Name/Location
192CL	ECS Access Door

      <4> Re-install the Left PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.
      <5> Do the Repair Confirmation at the end of this task.
   b) If you do not find a problem with the wiring, then do these steps:
      <1> Re-connect connector D3866 to the Left Pack Temperature Sensor, T471.
      <2> Close this access panel:

Number	Name/Location
192CL	ECS Access Door

      <3> Replace the Left PFTC, M2951. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

(2) If the maintenance message shows on the Right PFTC, M2952, then do these steps:
   (a) Do this check of the Right Pack Temperature Sensor, T474:

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OMR 820-826 (Continued)

1) Open this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

2) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

3) Disconnect electrical connector D3876 from the Right Pack Temperature Sensor (WDM 21-51-22).

4) Do a check for an open circuit or a short between pins 1 and 2 of the Right Pack Temperature Sensor.

5) If there is a short or an open circuit between pins 1 and 2, then do these steps:

a) Replace the Right Pack Temperature Sensor, T474. These are the tasks:

- Pack Temperature Sensor Removal, AMM TASK 21-51-52-000-801
- Pack Temperature Sensor Installation, AMM TASK 21-51-52-400-801

b) Do the Repair Confirmation at the end of this task.

6) If there is not a short or an open circuit between pins 1 and 2, then continue.

(b) Do this check of the wiring between the Right PFTC, M2952, and the Right Pack Temperature Sensor, T474 (WDM 21-51-22):

1) Remove the Right PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

2) Do a wiring check as follows:

R PFTC D15816B	R PACK TEMP SENSOR D3876
pin J9	pin 1
pin J10	pin 2

a) If you find a problem with the wiring, then do these steps:

<1> Repair the wiring.

<2> Re-connect connector D3876 to the Right Pack Temperature Sensor, T474.

<3> Close this access panel:

Number	Name/Location
192CR	ECS Access Door

<4> Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

<5> Re-install the Right PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

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OMR 820-826 (Continued)

<6> Do the Repair Confirmation at the end of this task.

b) If you do not find a problem with the wiring, then do these steps:

<1> Re-connect connector D3876 to the Right Pack Temperature Sensor, T474.

<2> Close this access panel:

Number	Name/Location
192CR	ECS Access Door

<3> Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

<4> Replace the Right PFTC, M2952. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

G. Repair Confirmation

(1) For the PFTC, do the "BITE Procedure" and the "BITE Procedure - EXISTING FAULTS" sections of this task:

- Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the fault.

(b) If the maintenance message still shows, continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

872. STBY PACK TEMP SENSOR - Fault Isolation

A. Description

(1) This task is for these Maintenance Messages:

(a) 21-51208 STBY PACK TEMP

1) Standby Pack Temperature Sensor Open Fault

(b) 21-51209 STBY PACK TEMP

1) Standby Pack Temperature Sensor Short Fault

(2) These Maintenance Messages can show on either the Left or the Right Pack Flow Temperature Controller (PFTC).

B. Possible Causes

(1) For maintenance messages that show on the Left PFTC, M2951:

(a) Right Pack Temperature Sensor, T474

(b) Wiring

(c) Left PFTC, M2951

(2) For maintenance messages that show on the Right PFTC, M2952:

(a) Left Pack Temperature Sensor, T471

(b) Wiring

(c) Right PFTC, M2952

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OMR 820-826 (Continued)

C. Circuit Breakers

(1) These are the primary circuit breakers related to this fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

D. Related Data

(1) WDM 21-51-13

(2) WDM 21-51-23

E. Initial Evaluation

(1) Do the Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then there was an intermittent fault.

(b) If the maintenance message still shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) If the maintenance message shows on the Left PFTC, M2951, then do these steps:

(a) Do this check of the Right Pack Temperature Sensor, T474:

1) Open this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

2) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

3) Disconnect electrical connector D3876 from the Right Pack Temperature Sensor (WDM 21-51-12).

4) Do a check for an open circuit or a short between pins 1 and 2 of the Right Pack Temperature Sensor.

5) If there is a short or an open circuit between pins 1 and 2, then do these steps:

a) Replace the Right Pack Temperature Sensor, T474. These are the tasks:

- Pack Temperature Sensor Removal, AMM TASK 21-51-52-000-801
- Pack Temperature Sensor Installation, AMM TASK 21-51-52-400-801

b) Do the Repair Confirmation at the end of this task.

6) If there is not a short or an open circuit between pins 1 and 2, then continue.

(b) Do this check of the wiring between the Left PFTC, M2951, and the Right Pack Temperature Sensor, T474 (WDM 21-51-12):

1) Remove the Left PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

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OMR 820-826 (Continued)

2) Do a wiring check as follows:

L PFTC D15814C	R PACK TEMP SENSOR D3876
pin 22	pin 3
pin 16	pin 4

a) If you find a problem with the wiring, then do these steps:

<1> Repair the wiring.

<2> Re-connect connector D3876 to the Right Pack Temperature Sensor, T474.

<3> Close this access panel:

Number	Name/Location
192CR	ECS Access Door

<4> Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

<5> Re-install the Left PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

<6> Do the Repair Confirmation at the end of this task.

b) If you do not find a problem with the wiring, then do these steps:

<1> Re-connect connector D3876 to the Right Pack Temperature Sensor, T474.

<2> Close this access panel:

Number	Name/Location
192CR	ECS Access Door

<3> Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

<4> Replace the Left PFTC, M2951. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

(2) If the maintenance message shows on the Right PFTC, M2952, then do these steps:

(a) Do this check of the Left Pack Temperature Sensor, T471:

1) Open this access panel:

Number	Name/Location
192CL	ECS Access Door

2) Disconnect electrical connector D3866 from the Left Pack Temperature Sensor (WDM 21-51-13).

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OMR 820-826 (Continued)

3) Do a check for an open circuit or a short between pins 1 and 2 of the Right Pack Temperature Sensor.

4) If there is a short or an open circuit between pins 1 and 2, then do these steps:

a) Replace the Left Pack Temperature Sensor, T471. These are the tasks:

- Pack Temperature Sensor Removal, AMM TASK 21-51-52-000-801
- Pack Temperature Sensor Installation, AMM TASK 21-51-52-400-801

b) Do the Repair Confirmation at the end of this task.

5) If there is not a short or an open circuit between pins 1 and 2, then continue.

(b) Do this check of the wiring between the Right PFTC, M2952, and the Left Pack Temperature Sensor, T471 (WDM 21-51-13):

1) Remove the Right PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

2) Do a wiring check as follows:

R PFTC D15816C	L PACK TEMP SENSOR D3866
pin 22	pin 3
pin 16	pin 4

a) If you find a problem with the wiring, then do these steps:

<1> Repair the wiring.

<2> Re-connect connector D3866 to the Left Pack Temperature Sensor, T471.

<3> Close this access panel:

Number	Name/Location
192CL	ECS Access Door

<4> Re-install the Right PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

<5> Do the Repair Confirmation at the end of this task.

b) If you do not find a problem with the wiring, then do these steps:

<1> Re-connect connector D3866 to the Left Pack Temperature Sensor, T471.

<2> Close this access panel:

Number	Name/Location
192CL	ECS Access Door

<3> Replace the Right PFTC, M2952. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

G. Repair Confirmation

(1) Do the Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the problem.

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OMR 820-826 (Continued)

(b) If the maintenance message shows, then continue the Fault Isolation Procedure at the subsequent step.

END OF TASK

873. MIXMAN TEMP SENSOR - Fault Isolation

A. Description

(1) This task is for these Maintenance Messages:

(a) 21-51211 MIXMAN TEMP SENSOR

1) Mix Manifold Temperature Sensor Open Fault

(b) 21-51212 MIXMAN TEMP SENSOR

1) Mix Manifold Temperature Sensor Short Fault

(c) 21-51314 MIXMAN TEMP SENSOR

1) Mix Manifold Temperature Sensor Non Recoverable Fault

2) This is a latched message.

NOTE: After you correct a problem related to a Latched Maintenance Message, you can clear the Maintenance Message as follows:

- Cycle the power to the PFTC or
- Perform the BITE ground test or
- Erase the EXISTING FAULTS from the PFTC Front Panel Menus.

(2) These Maintenance Messages can show on either the Left or the Right Pack Flow Temperature Controller (PFTC).

B. Possible Causes

(1) For maintenance messages that show on the Left PFTC, M2951:

(a) Left Mix Manifold Temperature Sensor, T472

(b) Wiring

(c) Left PFTC, M2951

(2) For maintenance messages that show on the Right PFTC, M2952:

(a) Right Mix Manifold Temperature Sensor, T475

(b) Wiring

(c) Right PFTC, M2952

C. Circuit Breakers

(1) These are the primary circuit breakers related to this fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

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D. Related Data

(1) WDM 21-51-12

(2) WDM 21-51-22

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message shows in the PFTC Existing Faults, or it is a latched message that shows in the most recent flight leg, then do the Fault Isolation Procedure below.

(b) If the maintenance message does not show in the PFTC Existing Faults, or as a latched message in the most recent flight leg, then no additional fault isolation steps are required.

F. Fault Isolation Procedure

(1) If the maintenance message shows on the Left PFTC, M2951, then do these steps:

(a) Do this check of the Left Mix Manifold Temperature Sensor, T472:

1) Open the Forward Cargo Door.

2) Remove the Aft Center Bulkhead Liner in the Forward Cargo Compartment. This is the task: Forward Cargo Compartment Aft Bulkhead Liner - Removal, AMM TASK 25-52-17-000-801.

3) Disconnect electrical connector D3868 from the Left Mix Manifold Temperature Sensor (WDM 21-51-12).

4) Do a check for an open circuit or a short between pins 1 and 2 of the Left Mix Manifold Temperature Sensor.

5) If there is a short or an open circuit between pins 1 and 2, then do these steps:

a) Replace the Left Mix Manifold Temperature Sensor, T472. These are the tasks:

- Mix Manifold Temperature Sensor Removal, AMM TASK 21-51-51-000-803
- Mix Manifold Temperature Sensor Installation, AMM TASK 21-51-51-400-802

b) Do the Repair Confirmation at the end of this task.

6) If there is not a short or an open circuit between pins 1 and 2, then continue.

(b) Do this check of the wiring between the Left PFTC, M2951, and the Left Mix Manifold Temperature Sensor, T472 (WDM 21-51-12):

1) Remove the Left PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

2) Do a wiring check as follows:

L PFTC D15814B		L MIX MANIFOLD TEMP SENSOR D3868
pin F9	. . . . .	pin 1
pin F10	. . . . .	pin 2

a) If you find a problem with the wiring, then do these steps:

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<1> Repair the wiring.

<2> Re-connect connector D3868 to the Left Mix Manifold Temperature Sensor, T472.

<3> Install the Aft Center Bulkhead Liner in the Forward Cargo Compartment. This is the task: Forward Cargo Compartment Aft Bulkhead Liner - Installation, AMM TASK 25-52-17-400-801.

<4> Close the Forward Cargo Door.

<5> Re-install the Left PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

<6> Do the Repair Confirmation at the end of this task.

b) If you do not find a problem with the wiring, then do these steps:

<1> Re-connect connector D3868 to the Left Mix Manifold Temperature Sensor, T472.

<2> Install the Aft Center Bulkhead Liner in the Forward Cargo Compartment. This is the task: Forward Cargo Compartment Aft Bulkhead Liner - Installation, AMM TASK 25-52-17-400-801.

<3> Close the Forward Cargo Door.

<4> Replace the Left PFTC, M2951. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

(2) If the maintenance message shows on the Right PFTC, M2952, then do these steps:

(a) Do this check of the Right Mix Manifold Temperature Sensor, T475:

1) Open the Forward Cargo Door.

2) Remove the Aft Center Bulkhead Liner in the Forward Cargo Compartment. This is the task: Forward Cargo Compartment Aft Bulkhead Liner - Removal, AMM TASK 25-52-17-000-801.

3) Disconnect electrical connector D3878 from the Right Mix Manifold Temperature Sensor (WDM 21-51-22).

4) Do a check for an open circuit or a short between pins 1 and 2 of the Right Mix Manifold Temperature Sensor.

5) If there is a short or an open circuit between pins 1 and 2, then do these steps:

a) Replace the Right Mix Manifold Temperature Sensor, T475. These are the tasks:

- Mix Manifold Temperature Sensor Removal, AMM TASK 21-51-51-000-803
- Mix Manifold Temperature Sensor Installation, AMM TASK 21-51-51-400-802

b) Do the Repair Confirmation at the end of this task.

6) If there is not a short or an open circuit between pins 1 and 2, then continue.

(b) Do this check of the wiring between the Right PFTC, M2952, and the Right Mix Manifold Temperature Sensor, T475 (WDM 21-51-22):

1) Remove the Right PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

2) Do a wiring check as follows:

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R PFTC D15816B		R MIX MANIFOLD TEMP SENSOR D3878
pin F9	. . . . .	pin 1
pin F10	. . . . .	pin 2

a) If you find a problem with the wiring, then do these steps:

<1> Repair the wiring.

<2> Re-connect connector D3878 to the Right Mix Manifold Temperature Sensor, T475.

<3> Install the Aft Center Bulkhead Liner in the Forward Cargo Compartment. This is the task: Forward Cargo Compartment Aft Bulkhead Liner - Installation, AMM TASK 25-52-17-400-801.

<4> Close the Forward Cargo Door.

<5> Re-install the Right PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

<6> Do the Repair Confirmation at the end of this task.

b) If you do not find a problem with the wiring, then do these steps:

<1> Re-connect connector D3878 to the Right Mix Manifold Temperature Sensor, T475.

<2> Install the Aft Center Bulkhead Liner in the Forward Cargo Compartment. This is the task: Forward Cargo Compartment Aft Bulkhead Liner - Installation, AMM TASK 25-52-17-400-801.

<3> Close the Forward Cargo Door.

<4> Replace the Right PFTC, M2952. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

G. Repair Confirmation

(1) For the PFTC, do the "BITE Procedure" and the "BITE Procedure - EXISTING FAULTS" sections of this task:

- Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the fault.

(b) If the maintenance message still shows, continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

874. TEMP CTRL VALVE - Fault Isolation

A. Description

(1) This task is for these Latched Maintenance Messages:

NOTE: After you correct a problem related to a Latched Maintenance Message, you can clear the Maintenance Message as follows:

- Cycle the power to the PFTC or

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• Perform the BITE ground test or

• Erase the EXISTING FAULTS from the PFTC Front Panel Menus.

(a) 21-51083 PFTC CNTRL INOP

1) Pack TCV Driver ON Fault

(b) 21-51084 PFTC CNTRL INOP

1) Pack TCV Driver OFF Fault

(c) 21-51085 PFTC CNTRL INOP

1) Pack TCV Driver Fault

(d) 21-51260 TEMP CTRL VALVE

1) Temperature Control Valve Open Motor Open Fault

(e) 21-51261 TEMP CTRL VALVE

1) Temperature Control Valve Close Motor Open Fault

(f) 21-51263 TEMP CTRL VALVE

1) Temperature Control Valve Motor Short Fault

(g) 21-51264 TEMP CTRL VALVE

1) Temperature Control Valve Seized Fail To Open Fault

(h) 21-51265 TEMP CTRL VALVE

1) Temperature Control Valve Seized Fail To Close Fault

(i) 21-51507 PFTC CNTRL INOP

1) Pack Temperature Control Valve Hardware Fault

(j) 21-51510 TEMP CTRL VALVE

1) Pack Temperature Control Valve Motor Fault

(2) This task is also for these Non-Latched Maintenance Messages:

(a) 21-51266 TEMP CTRL VALVE

1) Temperature Control Valve Limit Switch Fault

(b) 21-51267 TEMP CTRL VALVE

1) Temperature Control Valve Fault

(3) These Maintenance Messages can show on either the Left or the Right Pack Flow Temperature Controller (PFTC).

B. Possible Causes

(1) For maintenance messages that show on the Left PFTC, M2951:

(a) Left TCV, V139

(b) ACAU-1, M324

(c) Wiring

(d) Left PFTC, M2951

(2) For maintenance messages that show on the Right PFTC, M2952:

(a) Right TCV, V141

(b) Air Conditioning Accessory Unit (ACAU-2), M1455

(c) Wiring

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(d) Right PFTC, M2952

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	1	C01157	A/C TEMP CONT VALVE CLOSE LEFT
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	1	C01160	A/C TEMP CONT VALVE CLOSE RIGHT
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

D. Related Data

(1) WDM 21-51-12

(2) WDM 21-51-22

E. Initial Evaluation

(1) Make sure that the PACK Light is not illuminated on the Panel, P5.

(a) If the PACK Light is illuminated, press the TRIP RESET Switch on the Panel, P5 or MASTER CAUTION on the Glareshield Panel to reset the light.

NOTE: If the TRIP RESET does not extinguish the PACK light, there may be a nuisance Temperature Control Valve (TCV) fault after the BITE procedure is performed.

(2) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message shows in the PFTC Existing Faults, or it is a latched message that shows in the most recent flight leg, then do the Fault Isolation Procedure below.

(b) If the maintenance message does not show in the PFTC Existing Faults, or as a latched message in the most recent flight leg, then no additional fault isolation steps are required.

F. Fault Isolation Procedure

(1) If the maintenance message shows on the Left PFTC, then do these steps:

(a) Do a check on the valve connector (WDM 21-51-12).

1) Disconnect connector D3862 from the left TCV and do a visual inspection for bent pins, pushed-in pins, uncentered sockets, corrosion, moisture, or damages.

a) If you find a problem with the wiring, then do these steps:

<1> Clean or repair the connector.

<2> Re-connect the connector to the left TCV.

<3> Do the Repair Confirmation at the end of this task.

b) If you did not find a problem with the wiring, then do these steps:

<1> Re-connect the connector to the left TCV.

<2> Do the Repair Confirmation at the end of this task.

(b) Replace the Left TCV, V139. These are the tasks:

- Temperature Control Valve Removal, AMM TASK 21-51-10-000-801

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- Temperature Control Valve Installation, AMM TASK 21-51-10-400-801

1) Do the Repair Confirmation at the end of this task.

(c) Replace the Air Conditioning Accessory Unit (ACAU-1), M324. These are the tasks:

- Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801
- Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801

1) Do the Repair Confirmation at the end of this task.

(d) Do this check of the wiring between the ACAU-1, M324, the Left TCV, V139, and the Left PFTC, M2951 (WDM 21-51-12):

1) Remove the ACAU-1. This is the task: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801.

2) Remove the Left TCV. This is the task: Temperature Control Valve Removal, AMM TASK 21-51-10-000-801.

3) Do a wiring check as follows:

ACAU-1	**L TCV**
D458A	**D3862**
pin 50	pin 10
pin 64	pin 1
pin 63	pin 2

5) Make sure pins 3 and 5 of connector D3862 for the Left TCV connect to ground.

6) Remove the Left PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

7) Do a wiring check as follows:

L PFTC	**L TCV**
D15814B	**D3862**
pin B5	pin 4
pin B6	pin 12

a) If you find a problem with the wiring, then do these steps:

<1> Repair the wiring.

<2> Re-install the Left TCV. This is the task: Temperature Control Valve Installation, AMM TASK 21-51-10-400-801.

<3> Re-install the Air Conditioning Accessory Unit (ACAU-1). This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

<4> Re-install the Left PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

<5> Do the Repair Confirmation at the end of this task.

b) If you do not find a problem with the wiring, then replace the Left PFTC, M2951. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

<1> Do the Repair Confirmation at the end of this task.

(2) If the maintenance message shows on the Right PFTC, then do these steps:

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(a) Do a check on the valve connector (WDM 21-51-22).

1) Disconnect connector D3872 from the right TCV and do a visual inspection for bent pins, pushed-in pins, uncentered sockets, corrosion, moisture, or damages.

a) If you find a problem with the wiring, then do these steps:

<1> Clean or repair the connector.

<2> Re-connect the connector to the right TCV.

<3> Do the Repair Confirmation at the end of this task.

b) If you did not find a problem with the wiring, then do these steps:

<1> Re-connect the connector to the right TCV.

<2> Do the Repair Confirmation at the end of this task.

(b) Replace the Right TCV, V141. These are the tasks:

- Temperature Control Valve Removal, AMM TASK 21-51-10-000-801
- Temperature Control Valve Installation, AMM TASK 21-51-10-400-801

1) Do the Repair Confirmation at the end of this task.

(c) Replace the Air Conditioning Accessory Unit (ACAU-2), M1455. These are the tasks:

- Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801
- Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801

1) Do the Repair Confirmation at the end of this task.

(d) Do this check of the wiring between the ACAU-2, M1455, the Right TCV, V141, and the Right PFTC, M2952 (WDM 21-51-22):

1) Remove the ACAU-2. This is the task: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801

2) Remove the Right TCV. This is the task: Temperature Control Valve Removal, AMM TASK 21-51-10-000-801.

3) Do a wiring check as follows:

ACAU-2	**R TCV**
D10002A	**D3872**
pin 50	pin 10
pin 64	pin 1
pin 63	pin 2

5) Make sure pins 3 and 5 of connector D3872 for the Right TCV connect to ground.

6) Remove the Right PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

7) Do a wiring check as follows:

R PFTC	**R TCV**
D15816B	**D3872**
pin B5	pin 4
pin B6	pin 12

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OMR 820-826 (Continued)

a) If you find a problem with the wiring, then do these steps:

<1> Repair the wiring.

<2> Re-install the Right TCV. This is the task: Temperature Control Valve Installation, AMM TASK 21-51-10-400-801.

<3> Re-install the Air Conditioning Accessory Unit (ACAU-2). This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

<4> Re-install the Right PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

<5> Do the Repair Confirmation at the end of this task.

b) If you do not find a problem with the wiring, then replace the Right PFTC, M2952. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

<1> Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) For the PFTC, do the "BITE Procedure" and the "BITE Procedure - EXISTING FAULTS" sections of this task:

• Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the fault.

(b) If the maintenance message still shows, continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

875. STBY TEMP CTRL VALVE - Fault Isolation

A. Description

(1) This task is for these Latched Maintenance Messages:

NOTE: After you correct a problem related to a Latched Maintenance Message, you can clear the Maintenance Message as follows:

• Cycle the power to the PFTC or

• Perform the BITE ground test or

• Erase the EXISTING FAULTS from the PFTC Front Panel Menus.

(a) 21-51086 PFTC STBY CNTRL

1) Standby TCV Driver High Fault

(b) 21-51087 PFTC STBY CNTRL

1) Standby TCV Driver Low Fault

(c) 21-51300 STBY TCV VALVE

1) Standby TCV Valve Motor Open Fault

(d) 21-51301 STBY TCV VALVE

1) Standby TCV Valve Motor Short Fault

(2) These Maintenance Messages can show on the Left or Right Pack Flow Temperature Controller (PFTC).

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OMR 820-826 (Continued)

B. Possible Causes

(1) For maintenance messages that show on the Left PFTC, M2951:

(a) Right Standby Temperature Control Valve, V142

(b) Wiring

(c) Left PFTC, M2951

(2) For maintenance messages that show on the Right PFTC, M2952:

(a) Left Standby Temperature Control Valve, V140

(b) Wiring

(c) Right PFTC, M2952

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

D. Related Data

(1) WDM 21-51-13

(2) WDM 21-51-23

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message shows in the PFTC Existing Faults, or it is a latched message that shows in the most recent flight leg, then do the Fault Isolation Procedure below.

(b) If the maintenance message does not show in the PFTC Existing Faults, or as a latched message in the most recent flight leg, then no additional fault isolation steps are required.

F. Fault Isolation Procedure

(1) If the maintenance message shows on the Left PFTC, then do these steps:

(a) Replace the Right Standby Temperature Control Valve, V142. These are the tasks:

- Standby Temperature Control Valve Removal, AMM TASK 21-51-11-000-801
- Standby Temperature Control Valve Installation, AMM TASK 21-51-11-400-801

1) Do the Repair Confirmation at the end of this task.

(b) Do this check of the wiring between the Left PFTC, M2951, and the Right Standby Temperature Control Valve, V142 (WDM 21-51-23):

1) Remove the Left PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

2) Disconnect connector D3874 from the Right Standby Temperature Control Valve.

3) Do a wiring check as follows:

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L PFTC	R STBY TCV
D15814C	**D3874**
pin 12	pin 1
pin 6	pin 3

4) If you find a problem with the wiring, then do these steps:
   a) Repair the wiring.
   b) Re-connect connector D3874 to the Right Standby Temperature Control Valve.
   c) Re-install the Left PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.
   d) Do the Repair Confirmation at the end of this task.
5) If you do not find a problem with the wiring, then do these steps:
   a) Re-connect connector D3874 to the Right Standby Temperature Control Valve.
   b) Replace the Left PFTC, M2951. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

(2) If the maintenance message shows on the Right PFTC, then do these steps:
   (a) Replace the Left Standby Temperature Control Valve, V140. These are the tasks:
   - Standby Temperature Control Valve Removal, AMM TASK 21-51-11-000-801
   - Standby Temperature Control Valve Installation, AMM TASK 21-51-11-400-801

   1) Do the Repair Confirmation at the end of this task.
   (b) Do this check of the wiring between the Right PFTC, M2952, and the Left Standby Temperature Control Valve, V140 (WDM 21-51-13):
   1) Remove the Right PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.
   2) Disconnect connector D3864 from the Left Standby Temperature Control Valve.
   3) Do a wiring check as follows:

R PFTC	L STBY TCV
D15816C	**D3864**
pin 12	pin 1
pin 6	pin 3

   4) If you find a problem with the wiring, then do these steps:
      a) Repair the wiring.
      b) Re-connect connector D3864 to the Left Standby Temperature Control Valve.
      c) Re-install the Right PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.
      d) Do the Repair Confirmation at the end of this task.
   5) If you do not find a problem with the wiring, then do these steps:
      a) Re-connect connector D3864 to the Left Standby Temperature Control Valve.
      b) Replace the Right PFTC, M2952. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

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OMR 820-826 (Continued)

G. Repair Confirmation

(1) For the PFTC, do the "BITE Procedure" and the "BITE Procedure - EXISTING FAULTS" sections of this task:

- Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the fault.

(b) If the maintenance message still shows, continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

876. FLIGHT DECK TRIM VALVE - Fault Isolation

A. Description

(1) This task is for these Latched Maintenance Messages:

NOTE: After you correct a problem related to a Latched Maintenance Message, you can clear the Maintenance Message as follows:

- Cycle the power to the PFTC or
- Perform the BITE ground test or
- Erase the EXISTING FAULTS from the PFTC Front Panel Menus.

(a) 21-51073 PFTC CNTRL INOP

1) Flight Deck Trim Air Valve (TAV) Driver ON Fault

(b) 21-51074 PFTC CNTRL INOP

1) Flight Deck TAV Driver OFF Fault

(c) 21-51075 PFTC CNTRL INOP

1) Flight Deck TAV Driver Interface Fault

(d) 21-51268 F/D TRIM VALVE

1) Flight Deck Trim Valve Open Motor Open Fault

(e) 21-51269 F/D TRIM VALVE

1) Flight Deck Trim Valve Close Motor Open Fault

(f) 21-51271 F/D TRIM VALVE

1) Flight Deck Trim Valve Motor Short Fault

(g) 21-51272 F/D TRIM VALVE

1) Flight Deck Trim Valve Seized Fail To Open Fault

(h) 21-51273 F/D TRIM VALVE

1) Flight Deck Trim Valve Seized Fail To Close Fault

(i) 21-51505 PFTC CNTRL INOP

1) Flight Deck TAV Hardware Fault

(j) 21-51508 F/D TRIM VALVE

1) Flight Deck TAV Motor Fault

(2) This task is also for these Non-Latched Maintenance Messages:

(a) 21-51274 F/D TRIM VALVE

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OMR 820-826 (Continued)

1) Flight Deck Trim Valve Limit Switch Fault

(b) 21-51275 F/D TRIM VALVE

1) Flight Deck Trim Valve Fault

(3) These Maintenance Messages can show on either the Left or the Right Pack Flow Temperature Controller (PFTC).

B. Possible Causes

(1) For maintenance messages that show on the Left PFTC, M2951:

(a) Control Cabin TAV, V143

(b) ACAU-1, M324

(c) ACAU-2, M1455

(d) Wiring

(e) Left PFTC, M2951

(2) For maintenance messages that show on the Right PFTC, M2952:

(a) Control Cabin TAV, V143

(b) Air Conditioning Accessory Unit (ACAU-1), M324

(c) Wiring

(d) Right PFTC, M2952

C. Circuit Breakers

(1) These are the primary circuit breakers related to this fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	3	C01163	A/C ZONE TEMP VALVE/FAN CONT FLT DECK
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

D. Related Data

(1) WDM 21-61-11

(2) WDM 21-61-12

(3) SSM 21-61-11

(4) SSM 21-61-12

E. Initial Evaluation

(1) Make sure that there is no ZONE TEMP Light illuminated on the Panel, P5.

(a) If the ZONE TEMP Light is illuminated, reset the light by pressing TRIP RESET on the Panel, P5, or MASTER CAUTION on the Glareshield Panel to reset the light.

NOTE: If the TRIP RESET switch is not pressed, there may be a nuisance Trim Air Valve (TAV) fault after the BITE procedure is performed.

(2) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

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(a) If the maintenance message does not show in the PFTC Existing Faults, or as a latched message in the most recent flight leg, then no additional fault isolation steps are required.

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(b) If the maintenance message shows in the PFTC Existing Faults, or it is a latched message that shows in the most recent flight leg, then do the Fault Isolation Procedure below.

OMR 820-826; AIRPLANES WITH PFTC P/N 51090248-002/-003

(c) If a F/D TEMP SENSOR (21-51190, 21-51191, 21-51310), F/D DUCT SENSOR (21-51199, 21-51200, 21-51312), or F/D SELECTOR (21-51053, 21-51217, 21-51218, 21-51311) maintenance message shows in the PFTC Existing Faults, first do the applicable FIM Task(s) for these messages.

1) If a FLIGHT DECK TRIM VALVE maintenance message still shows in the PFTC Existing Faults, do the Pack Temperature Control - BITE Procedure, 21-62 TASK 801 - GROUND TESTS.

a) If the maintenance message does not show in the PFTC Existing Faults, or as a latched message in the most recent flight leg, then no additional fault isolation steps are required.

b) If the maintenance message still shows in the PFTC Existing Faults, then the fault is presently active. You can do this Circuit Breaker RESET Procedure to give the Flight Deck Trim Valve an opportunity to get “unstuck” and clear the fault:

<1> Open this circuit breaker for the Flight Deck Trim Valve, wait a minimum of two minutes, then close this circuit breaker.

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	3	C01163	A/C ZONE TEMP VALVE/FAN CONT FLT DECK

<2> Do the Pack Temperature Control - BITE Procedure, 21-62 TASK 801 - GROUND TESTS.

<a> If the maintenance message does not show in the PFTC Existing Faults, or as a latched message in the most recent flight leg, then no additional fault isolation steps are required.

<b> If the maintenance message still shows in the PFTC Existing Faults, then do the Fault Isolation Procedure below.

2) If the maintenance message does not show in the PFTC Existing Faults, or as a latched message in the most recent flight leg, then no additional fault isolation steps are required.

(d) If the maintenance message shows in the PFTC Existing Faults, or it is a latched message that shows in the most recent flight leg, then do the Fault Isolation Procedure below.

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OMR 820-826 (Continued)

F. Fault Isolation Procedure

(1) If the maintenance message shows on the Left PFTC, then do these steps:

(a) Do a check on the valve connector (WDM 21-61-11).

1) Disconnect connector D3886 from the Control Cabin TCV and do a visual inspection for bent pins, pushed-in pins, uncentered sockets, corrosion, moisture, or damages.

a) If you find a problem with the wiring, then do these steps:

<1> Clean or repair the connector.

<2> Re-connect the connector to the Control Cabin TCV.

<3> Do the Repair Confirmation at the end of this task.

b) If you did not find a problem with the wiring, then do these steps:

<1> Re-connect the connector to the Control Cabin TCV.

<2> Do the Repair Confirmation at the end of this task.

(b) Replace the Control Cabin TAV, V143. These are the tasks:

- Zone Trim Air Modulating Valve Removal, AMM TASK 21-61-21-000-801
- Zone Trim Air Modulating Valve Installation, AMM TASK 21-61-21-400-801

1) Do the Repair Confirmation at the end of this task.

(c) Replace the Air Conditioning Accessory Unit (ACAU-1), M324. These are the tasks:

- Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801
- Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801

1) Do the Repair Confirmation at the end of this task.

(d) Replace the Air Conditioning Accessory Unit (ACAU-2), M1455. These are the tasks:

- Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801
- Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801

1) Do the Repair Confirmation at the end of this task.

(e) Do this check of the wiring between the ACAU-2, M1455 and the ACAU-1, M324 (WDM 21-61-11):

1) Remove the ACAU-1 and the ACAU-2. This is the task: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801.

2) Do a wiring check as follows:

ACAU-2	**ACAU-1**
D10002A	**D458B**
pin 26	pin 42
pin 27	pin 41

4) Remove the Left PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

5) Do a wiring check as follows (WDM 21-61-12):

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OMR 820-826 (Continued)

L PFTC	**ACAU-2**
D15814C	**D10002A**
pin 13	pin 25
pin 14	pin 37

6) Do a wiring check as follows (WDM 21-61-12):

L PFTC	**ACAU-2**
D15814B	**D10002A**
pin G1	pin 16

a) If you find a problem with the wiring, then do these steps:

<1> Repair the wiring.

<2> Re-install the Control Cabin TAV. This is the task: Zone Trim Air Modulating Valve Installation, AMM TASK 21-61-21-400-801.

<3> Re-install the ACAU-1 and the ACAU-2. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

<4> Re-install the Left PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

<5> Do the Repair Confirmation at the end of this task.

b) If you do not find a problem with the wiring, then replace the Left PFTC, M2952. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

<1> Do the Repair Confirmation at the end of this task.

(2) If the maintenance message shows on the Right PFTC, then do these steps:

(a) Do a check on the valve connector (WDM 21-61-11).

1) Disconnect connector D3886 from the Control Cabin TCV and do a visual inspection for bent pins, pushed-in pins, uncentered sockets, corrosion, moisture, or damages.

a) If you find a problem with the wiring, then do these steps:

<1> Clean or repair the connector.

<2> Re-connect the connector to the Control Cabin TCV.

<3> Do the Repair Confirmation at the end of this task.

b) If you did not find a problem with the wiring, then do these steps:

<1> Re-connect the connector to the Control Cabin TCV.

<2> Do the Repair Confirmation at the end of this task.

(b) Replace the Control Cabin TAV, V143. These are the tasks:

- Zone Trim Air Modulating Valve Removal, AMM TASK 21-61-21-000-801
- Zone Trim Air Modulating Valve Installation, AMM TASK 21-61-21-400-801

1) Do the Repair Confirmation at the end of this task.

(c) Replace the Air Conditioning Accessory Unit (ACAU-1), M324. These are the tasks:

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OMR 820-826 (Continued)

- Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801
- Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801

1) Do the Repair Confirmation at the end of this task.

(d) Do this check of the wiring between the ACAU-1, M324, the Control Cabin TAV, V143, and the Right PFTC, M2952 (WDM 21-61-11):

1) Remove the ACAU-1. This is the task: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801.

2) Remove the Control Cabin TAV. This is the task: Zone Trim Air Modulating Valve Removal, AMM TASK 21-61-21-000-801.

3) Do a wiring check as follows:

ACAU-1 D458A	**CONTROL CABIN TAV D3886**
pin 41	pin 1
pin 42	pin 2
pin 28	pin 10

5) Make sure pins 3 and 5 of connector D3886 for the Control Cabin TAV, V143, connect to ground.

6) Remove the Right PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

7) Do a wiring check as follows:

R PFTC D15816B	**CONTROL CABIN TAV D3886**
pin C6	pin 12
pin C5	pin 4

8) Do a wiring check as follows:

R PFTC D15816C	**ACAU-1 D458A**
pin 13	pin 25
pin 14	pin 37

a) If you find a problem with the wiring, then do these steps:

<1> Repair the wiring.

<2> Re-install the Control Cabin TAV. This is the task: Zone Trim Air Modulating Valve Installation, AMM TASK 21-61-21-400-801.

<3> Re-install the Air Conditioning Accessory Unit (ACAU-1). This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

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OMR 820-826 (Continued)

<4> Re-install the Right PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

<5> Do the Repair Confirmation at the end of this task.

b) If you do not find a problem with the wiring, then replace the Right PFTC, M2952. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

<1> Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) For the PFTC, do the "BITE Procedure" and the "BITE Procedure - EXISTING FAULTS" sections of this task:

- Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the fault.

(b) If the maintenance message still shows, continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

877. ZONE TRIM VALVE - Fault Isolation

A. Description

(1) This task is for these Latched Maintenance Messages:

NOTE: After you correct a problem related to a Latched Maintenance Message, you can clear the Maintenance Message as follows:

- Cycle the power to the PFTC or
- Perform the BITE ground test or
- Erase the EXISTING FAULTS from the PFTC Front Panel Menus.

(a) 21-51078 PFTC CNTRL INOP

1) Zone Trim Air Valve (TAV) Driver ON Fault

(b) 21-51079 PFTC CNTRL INOP

1) Zone TAV Driver OFF Fault

(c) 21-51080 PFTC CNTRL INOP

1) Zone TAV Driver Interface Fault

(d) 21-51276 ZONE TRIM VALVE

1) Zone Trim Valve Open Motor Open Fault

(e) 21-51277 ZONE TRIM VALVE

1) Zone Trim Valve Close Motor Open Fault

(f) 21-51279 ZONE TRIM VALVE

1) Zone Trim Valve Motor Short Fault

(g) 21-51280 ZONE TRIM VALVE

1) Zone Trim Valve Seized Fail To Open Fault

(h) 21-51281 ZONE TRIM VALVE

1) Zone Trim Valve Seized Fail To Close Fault

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OMR 820-826 (Continued)

(i) 21-51506 PFTC CNTRL INOP

1) Zone TAV Hardware Fault

(j) 21-51509 ZONE TRIM VALVE

1) Zone TAV Motor Fault

(2) This task is also for these Non-Latched Latched Maintenance Messages:

(a) 21-51282 ZONE TRIM VALVE

1) Zone Trim Valve Limit Switch Fault

(b) 21-51283 ZONE TRIM VALVE

1) Zone Trim Valve Fault

(3) These Maintenance Messages can show on either the Left or the Right Pack Flow Temperature Controller (PFTC).

B. Possible Causes

(1) For maintenance messages that show on the Left PFTC, M2951:

(a) AFT Cabin TAV, V145

(b) ACAU-1, M324

(c) Wiring

(d) Left PFTC, M2951

(2) For maintenance messages that show on the Right PFTC, M2952:

(a) Forward Cabin TAV, V144

(b) Air Conditioning Accessory Unit (ACAU-2), M1455

(c) Wiring

(d) Right PFTC, M2952

C. Circuit Breakers

(1) These are the primary circuit breakers related to this fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	3	C01163	A/C ZONE TEMP VALVE/FAN CONT FLT DECK
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

D. Related Data

(1) WDM 21-61-13

(2) WDM 21-61-14

(3) SSM 21-61-13

(4) SSM 21-61-14

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OMR 820-826 (Continued)

E. Initial Evaluation

(1) Make sure that there is no ZONE TEMP Light illuminated on the Panel, P5.

(a) If the ZONE TEMP Light is illuminated, reset the light by pressing TRIP RESET on the Panel, P5, or MASTER CAUTION on the Glareshield Panel to reset the light.

NOTE: If the TRIP RESET switch is not pressed, there may be a nuisance Trim Air Valve (TAV) fault after the BITE procedure is performed.

(2) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show in the PFTC Existing Faults, or as a latched message in the most recent flight leg, then no additional fault isolation steps are required.

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(b) If the maintenance message shows in the PFTC Existing Faults, or it is a latched message that shows in the most recent flight leg, then do the Fault Isolation Procedure below.

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(c) If a ZONE TEMP SENSOR (21-51193, 21-51194), ZONE DUCT SENSOR (21-51202, 21-51203), or ZONE SELECTOR (21-51056, 21-51220, 21-51221) maintenance message shows in the PFTC Existing Faults, first do the applicable FIM Task(s) for these messages.

1) If a ZONE TRIM VALVE maintenance message still shows in the PFTC Existing Faults, then do the Pack Temperature Control - BITE Procedure, 21-62 TASK 801 - GROUND TESTS.

a) If the maintenance message does not show in the PFTC Existing Faults, or as a latched message in the most recent flight leg, then no additional fault isolation steps are required.

b) If the maintenance message still shows in the PFTC Existing Faults, then the fault is presently active. You can do this Circuit Breaker RESET Procedure to give the TAV an opportunity to get “unstuck” and clear the fault:

<1> Open these circuit breakers for the Aft Cabin TAV, wait for a minimum of two minutes, then close the circuit breakers.

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	3	C01170	A/C ZONE TEMP VALVE/FAN CONT AFT PASS
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

Open these circuit breakers for the Fwd Cabin TAV, wait for a minimum of two minutes, then close the circuit breakers.

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	2	C01167	A/C ZONE TEMP VALVE/FAN CONT FWD PASS

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(Continued)

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

<2> Do the Pack Temperature Control - BITE Procedure, 21-62 TASK 801 - GROUND TESTS.

<a> If the maintenance message does not show in the PFTC Existing Faults, or as a latched message in the most recent flight leg, then no additional fault isolation steps are required.

<b> If the maintenance message still shows in the PFTC Existing Faults, then do the Fault Isolation Procedure below.

2) If the maintenance message does not show in the PFTC Existing Faults, or as a latched message in the most recent flight leg, then no additional fault isolation steps are required.

(d) If the maintenance message shows in the PFTC Existing Faults, or it is a latched message that shows in the most recent flight leg, then do the Fault Isolation Procedure below.

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F. Fault Isolation Procedure

(1) If the maintenance message shows on the Left PFTC, then do these steps:

(a) Do a check on the valve connector (WDM 21-61-14).

1) Disconnect connector D3890 from the AFT Cabin TCV and do a visual inspection for bent pins, pushed-in pins, uncentered sockets, corrosion, moisture, or damages.

a) If you find a problem with the wiring, then do these steps:

<1> Clean or repair the connector.

<2> Re-connect the connector to the AFT Cabin TCV.

<3> Do the Repair Confirmation at the end of this task.

b) If you did not find a problem with the wiring, then do these steps:

<1> Re-connect the connector to the AFT Cabin TCV.

<2> Do the Repair Confirmation at the end of this task.

(b) Replace the AFT Cabin TAV, V145. These are the tasks:

- Zone Trim Air Modulating Valve Removal, AMM TASK 21-61-21-000-801
- Zone Trim Air Modulating Valve Installation, AMM TASK 21-61-21-400-801

1) Do the Repair Confirmation at the end of this task.

(c) Replace the Air Conditioning Accessory Unit (ACAU-1), M324. These are the tasks:

- Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801
- Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801

1) Do the Repair Confirmation at the end of this task.

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OMR 820-826 (Continued)

(d) Do this check of the wiring between the ACAU-1, M324, the AFT Cabin TAV, V145, and the Left PFTC, M2951 (WDM 21-61-14, SSM 21-61-14):

1) Remove the ACAU-1. This is the task: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801.

2) Remove the AFT Cabin TAV. This is the task: Zone Trim Air Modulating Valve Removal, AMM TASK 21-61-21-000-801.

3) Do a wiring check as follows:

ACAU-1	**AFT CABIN TAV**
D458A	**D3890**
pin 52	pin 1
pin 51	pin 2
pin 67	pin 10

5) Make sure pins 3 and 5 of connector D3890 for the AFT Cabin TAV, V145, connect to ground.

6) Remove the Left PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

7) Do a wiring check as follows:

L PFTC	**AFT CABIN TAV**
D15814B	**D3890**
pin A6	pin 12
pin A5	pin 4

8) Do a wiring check as follows:

L PFTC	**ACAU-1**
D15814C	**D458A**
pin 1	pin 53
pin 7	pin 54

a) If you find a problem with the wiring, then do these steps:

<1> Repair the wiring.

<2> Re-install the AFT Cabin TAV. This is the task: Zone Trim Air Modulating Valve Installation, AMM TASK 21-61-21-400-801.

<3> Re-install the Air Conditioning Accessory Unit (ACAU-1). This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

<4> Re-install the Left PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

<5> Do the Repair Confirmation at the end of this task.

b) If you do not find a problem with the wiring, then replace the Left PFTC, M2951. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

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OMR 820-826 (Continued)

<1> Do the Repair Confirmation at the end of this task.

(2) If the maintenance message shows on the Right PFTC, then do these steps:

(a) Do a check on the valve connector (WDM 21-61-13).

1) Disconnect connector D3888 from the Forward Cabin TCV and do a visual inspection for bent pins, pushed-in pins, uncentered sockets, corrosion, moisture, or damages.

a) If you find a problem with the wiring, then do these steps:

<1> Clean or repair the connector.

<2> Re-connect the connector to the Forward Cabin TCV.

<3> Do the Repair Confirmation at the end of this task.

b) If you did not find a problem with the wiring, then do these steps:

<1> Re-connect the connector to the Forward Cabin TCV.

<2> Do the Repair Confirmation at the end of this task.

(b) Replace the Forward Cabin TAV, V144. These are the tasks:

- Zone Trim Air Modulating Valve Removal, AMM TASK 21-61-21-000-801
- Zone Trim Air Modulating Valve Installation, AMM TASK 21-61-21-400-801

1) Do the Repair Confirmation at the end of this task.

(c) Replace the Air Conditioning Accessory Unit (ACAU-2), M1455. These are the tasks:

- Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801
- Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801

1) Do the Repair Confirmation at the end of this task.

(d) Do this check of the wiring between the ACAU-2, M1455, the Forward Cabin TAV, V144, and the Right PFTC, M2952 (WDM 21-61-13, SSM 21-61-13):

1) Remove the ACAU-2. This is the task: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801.

2) Remove the Forward Cabin TAV. This is the task: Zone Trim Air Modulating Valve Removal, AMM TASK 21-61-21-000-801.

3) Do a wiring check as follows:

ACAU-2	**FWD CAB TAV**
D10002A	**D3888**
pin 52	pin 1
pin 51	pin 2
pin 67	pin 10

5) Make sure pins 3 and 5 of connector D3888 for the Forward Cabin TAV, V144, connect to ground.

6) Remove the Right PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

7) Do a wiring check as follows:

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OMR 820-826 (Continued)

R PFTC	**FWD CAB TAV**
D15816B	**D3888**
pin A6	pin 12
pin A5	pin 4

8) Do a wiring check as follows:

R PFTC	**ACAU-2**
D15816C	**D10002A**
pin 1	pin 53
pin 7	pin 54

a) If you find a problem with the wiring, then do these steps:

<1> Repair the wiring.

<2> Re-install the Forward Cabin TAV. This is the task: Zone Trim Air Modulating Valve Installation, AMM TASK 21-61-21-400-801.

<3> Re-install the Air Conditioning Accessory Unit (ACAU-2). This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

<4> Re-install the Right PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

<5> Do the Repair Confirmation at the end of this task.

b) If you do not find a problem with the wiring, then replace the Right PFTC, M2952. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

<1> Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) For the PFTC, do the "BITE Procedure" and the "BITE Procedure - EXISTING FAULTS" sections of this task:

- Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the fault.

(b) If the maintenance message still shows, continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

878. SRADA Inlet - Fault Isolation

A. Description

(1) This task is for these Maintenance Messages:

(a) 21-51400 SRADA CNTRL

1) SRADA Power Sense Interface Fault

(b) 21-51401 SRADA INLET

1) SRADA Inlet Interface Fault

(c) 21-51405 SRADA INLET

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OMR 820-826 (Continued)

1) SRADA Inlet Continuous BIT (CBIT) Fault

(d) 21-51409 SRADA INLET

1) SRADA Inlet Initiated BIT (IBIT) Fault

2) This is a latched message.

NOTE: After you correct a problem related to a Latched Maintenance Message, you can clear the Maintenance Message as follows:

- Cycle the power to the PFTC or
- Perform the BITE ground test or
- Erase the EXISTING FAULTS from the PFTC Front Panel Menus.

(2) These Maintenance Messages can show on either the Left or the Right Pack Flow Temperature Controller (PFTC).

B. Possible Causes

(1) For maintenance messages that show on the Left PFTC, M2951:

(a) For 21-51400 and 21-51401 maintenance messages

1) Left PFTC, M2951

2) Wiring

(b) For 21-51405 and 21-51409 maintenance messages

1) Left Forward (Inlet) Ram Air Actuator, M2795

2) Wiring

3) Left PFTC, M2951

(2) For maintenance messages that show on the Right PFTC, M2952:

(a) For 21-51400 and 21-51401 maintenance messages

1) Right PFTC, M2952

2) Wiring

(b) For 21-51405 and 21-51409 maintenance messages

1) Right Forward (Inlet) Ram Air Actuator, M2797

2) Wiring

3) Right PFTC, M2952

C. Circuit Breakers

(1) These are the primary circuit breakers related to this fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	8	C00555	AIR CONDITIONING RAM AIR MOD CONT L
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	8	C01178	AIR CONDITIONING RAM AIR MOD CONT R
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC

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(Continued)

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L

D. Related Data

(1) WDM 21-51-14

(2) WDM 21-51-24

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If maintenance message 21-51400 or 21-51401 shows in the PFTC Existing Faults, then do the Fault Isolation Procedure "Fault Isolation Procedure - Maintenance Messages 21-51400 and 21-51401" below.

(b) If the maintenance message is a latched message that shows in the most recent flight leg, then do the Fault Isolation Procedure "Fault Isolation Procedure - Maintenance Messages 21-51405 and 21-51409" below.

(c) If maintenance message 21-51405 or 21-51409 shows in the PFTC Existing Faults, power cycle 28V DC to the applicable SRADA as follows:

1) If the fault is on Left PFTC M2951, cycle power to the Left SRADA to erase the SRADA Internal Fault Memory and allow actuator movement during the subsequent steps as follows:

a) Open these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

b) Close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT

c) Wait for a minimum of 10 seconds, then close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

2) If the fault is on Right PFTC M2952, cycle power to the Right SRADA to erase the SRADA Internal Fault Memory and allow actuator movement during the subsequent steps as follows:

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a) Open these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

b) Close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT

c) Wait for a minimum of 10 seconds, then close these circuit breakers.

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

3) Do the Pack Temperature Control - BITE Procedure, 21-62 TASK 801 - GROUND TESTS.

a) If the maintenance message does not show in the PFTC Existing Faults, or as a latched message in the most recent flight leg, then no additional fault isolation steps are required.

b) If maintenance message 21-51405 or 21-51409 still shows in the PFTC Existing Faults, then do the Fault Isolation Procedure "Fault Isolation Procedure - Maintenance Messages 21-51405 and 21-51409" below.

(d) If the maintenance message does not show in the PFTC Existing Faults, or as a latched message in the most recent flight leg, then no additional fault isolation steps are required.

F. Fault Isolation Procedure - Maintenance Messages 21-51405 and 21-51409

(1) If the maintenance message shows on the Left PFTC, then do these steps:

(a) Do these steps to do a visual inspection of the Fault LEDs on the Left AFT (Exhaust, Exit) Ram Air Actuator, M2796:

1) Open these access panels:

Number	Name/Location
192CL	ECS Access Door
192JL	Air Conditioning Panel - Aft

2) If the FWD/INLET ACTUATOR LED is ON, on the Left AFT (Exhaust, Exit) Ram Air Actuator, M2796, then replace the Left Forward (Inlet) Ram Air Actuator, M2795. These are the tasks:

- Ram Air Inlet Actuator Removal, AMM TASK 21-51-21-000-801
- Ram Air Inlet Actuator Installation, AMM TASK 21-51-21-400-801

a) Do the Repair Confirmation at the end of this task.

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3) If the FWD/INLET ACTUATOR LED is not ON, on the Left AFT (Exhaust, Exit) Ram Air Actuator, then continue.

(b) Do these steps to check for 28V DC at the Left Forward (Inlet) Ram Air Actuator, M2795:

1) Open these access panels:

Number	Name/Location
191FL	Forward Wing To Body Fairing Panel - Mid Fairing, Above Ram Air Inlet
191GL	Ram Air Actuator Panel - Forward

2) Disconnect connector D14386 from the Left Forward (Inlet) Ram Air Actuator.

3) Check for 28V DC at pin 4 of connector D14386.

4) If you did not find 28V DC at pin 4 of connector D14386, then do these steps.

a) Repair the wiring.

b) Connect connector D14386 to the Left Forward (Inlet) Ram Air Actuator.

c) Do the Repair Confirmation at the end of this task.

5) If you find 28V DC at pin 4 of connector D14386, then continue.

(c) Do this check of the wiring between the Left PFTC, M2951, and the Left Forward (Inlet) Ram Air Actuator, M2795 (WDM 21-51-14):

1) Remove the Left PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

2) Disconnect electrical connector D14386 from the Left Forward (Inlet) Ram Air Actuator, M2795.

3) Do a wiring check as follows:

L PFTC D15814B		L FWD RAM ACTUATOR D14386
pin G3	. . . . .	pin 19
pin G5	. . . . .	pin 10
pin G6	. . . . .	pin 8
pin G7	. . . . .	pin 12
pin G8	. . . . .	pin 13

4) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-install the Left PFTC, M2951. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

c) Reconnect electrical connector D14386 to the Left Forward (Inlet) Ram Air Actuator, M2795.

5) Do the Repair Confirmation at the end of this task.

(d) Replace the Left PFTC, M2951. These are the tasks:

- Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801
- Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801

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(2) If the maintenance message shows on the Right PFTC, then do these steps:

(a) Do these steps to do a visual inspection of the Fault LEDs on the Right AFT (Exhaust, Exit) Ram Air Actuator, M2798:

1) Open these access panels:

Number	Name/Location
192CR	ECS Access Door
192JR	Air Conditioning Panel - Aft

2) If the FWD/INLET ACTUATOR LED is ON, on the Right AFT (Exhaust, Exit) Ram Air Actuator, M2798, then replace the Right Forward (Inlet) Ram Air Actuator, M2797. These are the tasks:

- Ram Air Inlet Actuator Removal, AMM TASK 21-51-21-000-801
- Ram Air Inlet Actuator Installation, AMM TASK 21-51-21-400-801

a) Do the Repair Confirmation at the end of this task.

3) If the FWD/INLET ACTUATOR LED is not ON, on the Right AFT (Exhaust, Exit) Ram Air Actuator, then continue.

(b) Do these steps to check for 28V DC at the Right Forward (Inlet) Ram Air Actuator, M2797:

1) Open these access panels:

Number	Name/Location
191FR	Forward Wing To Body Fairing Panel - Mid Fairing, Above Ram Air Inlet
191GR	Ram Air Actuator Panel - Forward

2) Disconnect connector D14394 from the Right Forward (Inlet) Ram Air Actuator.

3) Check for 28V DC at pin 4 of connector D14394.

4) If you did not find 28V DC at pin 4 of connector D14394, then do these steps.

a) Repair the wiring.

b) Connect connector D14394 to the Right Forward (Inlet) Ram Air Actuator.

c) Do the Repair Confirmation at the end of this task.

5) If you find 28V DC at pin 4 of connector D14394, then continue.

(c) Do this check of the wiring between the Right PFTC, M2952, and the Right Forward (Inlet) Ram Air Actuator, M2797 (WDM 21-51-24):

1) Remove the Right PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

2) Disconnect electrical connector D14394 from the Right Forward (Inlet) Ram Air Actuator, M2797.

3) Do a wiring check as follows:

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R PFTC D15816B		R FWD RAM ACTUATOR D14394
pin G3	. . . . .	pin 19
pin G5	. . . . .	pin 10
pin G6	. . . . .	pin 8
pin G7	. . . . .	pin 12
pin G8	. . . . .	pin 13

4) If you find a problem with the wiring, then do these steps:
   a) Repair the wiring.
   b) Re-install the Right PFTC, M2952. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.
   c) Reconnect electrical connector D14394 to the Right Forward (Inlet) Ram Air Actuator, M2797.
5) Do the Repair Confirmation at the end of this task.

(d) Replace the Right PFTC, M2952. These are the tasks:
- Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801
- Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801

(3) Do this task: Ram Air Inlet Door Linkage Inspection, AMM TASK 21-51-22-200-802.
   (a) Do the Repair Confirmation at the end of this task.

G. Fault Isolation Procedure - Maintenance Messages 21-51400 and 21-51401

(1) If the maintenance message shows on the Left PFTC, then do these steps:
   (a) Remove the Left PFTC, M2951, off the aircraft and swap in the Right PFTC, M2952, into the removed PFTC position.
      1) If the maintenance message is cleared, replace the original Left PFTC (that was removed).
      2) If the maintenance message still shows, do this check of the wiring between the Left PFTC, M2951, and the Left Forward (Inlet) Ram Air Actuator, M2795 (WDM 21-51-14):
         a) Remove the Left PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.
         b) Disconnect electrical connector D14386 from the Left Forward (Inlet) Ram Air Actuator, M2795.
         c) Do a wiring check as follows:

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L PFTC D15814B		L FWD RAM ACTUATOR D14386
pin G3	. . . . .	pin 19
pin G5	. . . . .	pin 10
pin G6	. . . . .	pin 8
pin G7	. . . . .	pin 12
pin G8	. . . . .	pin 13

d) If you find a problem with the wiring, then do these steps:

<1> Repair the wiring.

<2> Re-install the Left PFTC, M2951. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

<3> Reconnect electrical connector D14386 to the Left Forward (Inlet) Ram Air Actuator, M2795.

<4> Do the Repair Confirmation at the end of this task.

(2) If the maintenance message shows on the Right PFTC, then do these steps:

(a) Remove the Right PFTC, M2952, off the aircraft and swap in the Left PFTC, M2951, into the removed PFTC position.

1) If the maintenance message is cleared, replace the original Right PFTC (that was removed).

2) If the maintenance message still shows, do this check of the wiring between the Right PFTC, M2952, and the Right Forward (Inlet) Ram Air Actuator, M2797 (WDM 21-51-24):

a) Remove the Right PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

b) Disconnect electrical connector D14394 from the Right Forward (Inlet) Ram Air Actuator, M2797.

c) Do a wiring check as follows:

R PFTC D15816B		R FWD RAM ACTUATOR D14394
pin G3	. . . . .	pin 19
pin G5	. . . . .	pin 10
pin G6	. . . . .	pin 8
pin G7	. . . . .	pin 12
pin G8	. . . . .	pin 13

d) If you find a problem with the wiring, then do these steps:

<1> Repair the wiring.

<2> Re-install the Right PFTC, M2952. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

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<3> Reconnect electrical connector D14394 to the Right Forward (Inlet) Ram Air Actuator, M2797.

<4> Do the Repair Confirmation at the end of this task.

(3) Do this task: Ram Air Inlet Door Linkage Inspection, AMM TASK 21-51-22-200-802.

(a) Do the Repair Confirmation at the end of this task.

H. Repair Confirmation

(1) For the PFTC, do the "BITE Procedure" and the "BITE Procedure - EXISTING FAULTS" sections of this task:

• Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the fault.

(b) If the maintenance message still shows, continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

879. SRADA EXIT - Fault Isolation

A. Description

(1) This task is for these Maintenance Messages:

(a) 21-51402 SRADA EXIT

1) SRADA Exit Interface Fault

(b) 21-51406 SRADA EXIT

1) SRADA Exit Continuous BIT (CBIT) Fault

(c) 21-51410 SRADA EXIT

1) SRADA Exit Initiated BIT (IBIT) Fault

2) This is a latched message.

NOTE: After you correct a problem related to a Latched Maintenance Message, you can clear the Maintenance Message as follows:

• Cycle the power to the PFTC or

• Perform the BITE ground test or

• Erase the EXISTING FAULTS from the PFTC Front Panel Menus.

(2) These Maintenance Messages can show on either the Left or the Right Pack Flow Temperature Controller (PFTC).

B. Possible Causes

(1) For maintenance messages that show on the Left PFTC, M2951:

(a) For 21-51402 maintenance message

1) Left PFTC, M2951

2) Wiring

(b) For 21-51406 and 21-51410 maintenance messages

1) Left AFT (Exhaust, Exit) Ram Air Actuator, M2796

2) Left Forward (Inlet) Ram Air Actuator, M2795

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3) Wiring
4) Left PFTC, M2951

(2) For maintenance messages that show on the Right PFTC, M2952:

(a) For 21-51402 maintenance message

1) Right PFTC, M2952
2) Wiring

(b) For 21-51406 and 21-51410 maintenance messages

1) Right AFT (Exhaust, Exit) Ram Air Actuator, M2798
2) Right Forward (Inlet) Ram Air Actuator, M2797
3) Wiring
4) Right PFTC, M2952

C. Circuit Breakers

(1) These are the primary circuit breakers related to this fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	8	C00555	AIR CONDITIONING RAM AIR MOD CONT L
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	8	C01178	AIR CONDITIONING RAM AIR MOD CONT R
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L

D. Related Data

(1) WDM 21-51-14

(2) WDM 21-51-24

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If maintenance message 21-51402 shows in the PFTC Existing Faults, then do the Fault Isolation Procedure "Fault Isolation Procedure - Maintenance Messages 21-51402" below.

(b) If the maintenance message is a latched message that shows in the most recent flight leg, then do the Fault Isolation Procedure "Fault Isolation Procedure - Maintenance Messages 21-51406 and 21-51410" below.

(c) If maintenance message 21-51406 or 21-51410 shows in the PFTC Existing Faults, power cycle 28V DC to the applicable SRADA as follows:

1) If the fault is on Left PFTC M2951, cycle power to the Left SRADA to erase the SRADA Internal Fault Memory and allow actuator movement during the subsequent steps as follows:

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a) Open these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

b) Close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT

c) Wait for a minimum of 10 seconds, then close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

2) If the fault is on Right PFTC M2952, cycle power to the Right SRADA to erase the SRADA Internal Fault Memory and allow actuator movement during the subsequent steps as follows:

a) Open these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

b) Close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT

c) Wait for a minimum of 10 seconds, then close these circuit breakers.

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

3) Do the Pack Temperature Control - BITE Procedure, 21-62 TASK 801 - GROUND TESTS.

a) If the maintenance message does not show in the PFTC Existing Faults, or as a latched message in the most recent flight leg, then no additional fault isolation steps are required.

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b) If maintenance message 21-51406 or 21-51410 still shows in the PFTC Existing Faults, then do the Fault Isolation Procedure "Fault Isolation Procedure - Maintenance Messages 21-51406 and 21-51410" below.

(d) If the maintenance message does not show in the PFTC Existing Faults, or as a latched message in the most recent flight leg, then no additional fault isolation steps are required.

F. Fault Isolation Procedure - Maintenance Messages 21-51406 and 21-51410

(1) If the maintenance message shows on the Left PFTC, do these steps:

(a) Do these steps to do a visual inspection of the LEDs on the Left AFT (Exhaust, Exit) Ram Air Actuator, M2796:

1) Open these access panels:

Number	Name/Location
192CL	ECS Access Door
192JL	Air Conditioning Panel - Aft

2) If the AFT Actuator LED is ON, on the Left AFT (Exhaust, Exit) Ram Air Actuator, then replace the Left AFT (Exhaust, Exit) Ram Air Actuator, M2796. These are the tasks:

- Ram Air Exhaust Actuator - Removal, AMM TASK 21-51-27-000-801
- Ram Air Exhaust Actuator - Installation, AMM TASK 21-51-27-400-801

a) Do the Repair Confirmation at the end of this task.

b) If the AFT Actuator LED is still ON, on the Left AFT (Exhaust, Exit) Ram Air Actuator, then replace the Left Forward (Inlet) Ram Air Actuator, M2795. These are the tasks:

- Ram Air Inlet Actuator Removal, AMM TASK 21-51-21-000-801
- Ram Air Inlet Actuator Installation, AMM TASK 21-51-21-400-801

<1> Do the Repair Confirmation at the end of this task.

3) If no LEDs are ON, on the Left AFT (Exhaust, Exit) Ram Air Actuator, then do a voltage check at the Left AFT (Exhaust, Exit) Ram Air Actuator as follows (WDM 21-51-14):

a) Disconnect connector D14390 from the Left AFT (Exhaust, Exit) Ram Air Actuator.

b) Do a check for 28V DC at pin 4 of connector D14390.

c) If you did not find 28V DC at pin 4 of connector D14390, then do these steps.

<1> Repair the wiring.

<2> Connect electrical connector D14390 to the Left AFT (Exhaust, Exit) Ram Air Actuator.

<3> Do the Repair Confirmation at the end of this task.

d) If you find 28V DC at pin 4 of connector D14390, then replace the Left AFT (Exhaust, Exit) Ram Air Actuator, M2796. These are the tasks:

- Ram Air Exhaust Actuator - Removal, AMM TASK 21-51-27-000-801
- Ram Air Exhaust Actuator - Installation, AMM TASK 21-51-27-400-801

<1> Do the Repair Confirmation at the end of this task.

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(b) Do this check of the wiring between the Left Forward (Inlet) Ram Air Actuator, M2795, and the Left AFT (Exhaust, Exit) Ram Air Actuator, M2796 (WDM 21-51-14):

1) Open these access panels:

Number	Name/Location
191FL	Forward Wing To Body Fairing Panel - Mid Fairing, Above Ram Air Inlet
191GL	Ram Air Actuator Panel - Forward

2) Disconnect connectors D14386 and D14388 from the Left Forward (Inlet) Ram Air Actuator.

3) Disconnect connectors D14390 and D14392 from the Left AFT (Exhaust, Exit) Ram Air Actuator.

4) Do a wiring check as follows:

L FWD RAM ACTUATOR D14388		L AFT RAM ACTUATOR D14392
pin 1	. . . . .	pin 1
pin 2	. . . . .	pin 2
pin 3	. . . . .	pin 3

5) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Connect connectors D14386 and D14388 to the Left Forward (Inlet) Ram Air Actuator.

c) Connect connectors D14390 and D14392 to the Left AFT (Exhaust, Exit) Ram Air Actuator.

6) Do the Repair Confirmation at the end of this task.

7) Do this check of the wiring between the Left PFTC, M2951, and the Left Forward (Inlet) Ram Air Actuator, M2795 (WDM 21-51-14):

a) Remove the Left PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

b) Disconnect electrical connector D14386 from the Left Forward (Inlet) Ram Air Actuator, M2795.

c) Do a wiring check as follows:

L PFTC D15814B		L FWD RAM ACTUATOR D14386
pin G3	. . . . .	pin 19
pin G5	. . . . .	pin 10
pin G6	. . . . .	pin 8
pin G7	. . . . .	pin 12
pin G8	. . . . .	pin 13

d) If you find a problem with the wiring, then do these steps:

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<1> Repair the wiring.

<2> Re-install the Left PFTC, M2951. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

<3> Reconnect electrical connector D14386 to the Left Forward (Inlet) Ram Air Actuator, M2795.

e) Do the Repair Confirmation at the end of this task.

8) Replace the Left PFTC, M2951. These are the tasks:

- Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801
- Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801

(2) If the maintenance message shows on the Right PFTC, do these steps:

(a) Do these steps to do a visual inspection of the LEDs on the Right AFT (Exhaust, Exit) Ram Air Actuator, M2798:

1) Open these access panels:

Number	Name/Location
192CR	ECS Access Door
192JR	Air Conditioning Panel - Aft

2) If the AFT Actuator LED is ON, on the Right AFT (Exhaust, Exit) Ram Air Actuator, then replace the Right AFT (Exhaust, Exit) Ram Air Actuator, M2798. These are the tasks:

- Ram Air Exhaust Actuator - Removal, AMM TASK 21-51-27-000-801
- Ram Air Exhaust Actuator - Installation, AMM TASK 21-51-27-400-801

a) Do the Repair Confirmation at the end of this task.

b) If the AFT Actuator LED is still ON, on the Right AFT (Exhaust, Exit) Ram Air Actuator, then replace the Right Forward (Inlet) Ram Air Actuator, M2797. These are the tasks:

- Ram Air Inlet Actuator Removal, AMM TASK 21-51-21-000-801
- Ram Air Inlet Actuator Installation, AMM TASK 21-51-21-400-801

<1> Do the Repair Confirmation at the end of this task.

3) If no LEDs are ON, on the Right AFT (Exhaust, Exit) Ram Air Actuator, then do a voltage check at the Right AFT (Exhaust, Exit) Ram Air Actuator as follows (WDM 21-51-24):

a) Disconnect connector D14398 from the Right AFT (Exhaust, Exit) Ram Air Actuator.

b) Do a voltage check for 28V DC at pin 4 of connector D14398.

c) If you did not find 28V DC at pin 4 of connector D14398, then do these steps.

<1> Repair the wiring.

<2> Connect electrical connector D14398 to the Right AFT (Exhaust, Exit) Ram Air Actuator.

<3> Do the Repair Confirmation at the end of this task.

d) If you find 28V DC at pin 4 of connector D14398, then replace the Right AFT (Exhaust, Exit) Ram Air Actuator, M2798. These are the tasks:

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- Ram Air Exhaust Actuator - Removal, AMM TASK 21-51-27-000-801
- Ram Air Exhaust Actuator - Installation, AMM TASK 21-51-27-400-801

<1> Do the Repair Confirmation at the end of this task.

(b) Do this check of the wiring between the Right Forward (Inlet) Ram Air Actuator, M2797, and the Right AFT (Exhaust, Exit) Ram Air Actuator, M2798 (WDM 21-51-24):

1) Open these access panels:

Number	Name/Location
191FL	Forward Wing To Body Fairing Panel - Mid Fairing, Above Ram Air Inlet
191GL	Ram Air Actuator Panel - Forward

2) Disconnect connectors D14394 and D14396 from the Right Forward (Inlet) Ram Air Actuator.

3) Disconnect connectors D14398 and D14400 from the Right AFT (Exhaust, Exit) Ram Air Actuator.

4) Do a wiring check as follows:

R FWD RAM ACTUATOR D14396		R AFT RAM ACTUATOR D14400
pin 1	. . . . .	pin 1
pin 2	. . . . .	pin 2
pin 3	. . . . .	pin 3

5) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Connect connectors D14394 and D14396 to the Right Forward (Inlet) Ram Air Actuator.

c) Connect connectors D14398 and D14400 to the Right AFT (Exhaust, Exit) Ram Air Actuator.

6) Do the Repair Confirmation at the end of this task.

7) Do this check of the wiring between the Right PFTC, M2952, and the Right Forward (Inlet) Ram Air Actuator, M2797 (WDM 21-51-24):

a) Remove the Right PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

b) Disconnect electrical connector D14394 from the Right Forward (Inlet) Ram Air Actuator, M2797.

c) Do a wiring check as follows:

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OMR 820-826 (Continued)

R PFTC D15816B		R FWD RAM ACTUATOR D14394
pin G3	. . . . .	pin 19
pin G5	. . . . .	pin 10
pin G6	. . . . .	pin 8
pin G7	. . . . .	pin 12
pin G8	. . . . .	pin 13

d) If you find a problem with the wiring, then do these steps:

<1> Repair the wiring.

<2> Re-install the Right PFTC, M2952. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

<3> Reconnect electrical connector D14394 to the Right Forward (Inlet) Ram Air Actuator, M2797.

e) Do the Repair Confirmation at the end of this task.

8) Replace the Right PFTC, M2952. These are the tasks:

- Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801
- Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801

G. Fault Isolation Procedure - Maintenance Message 21-51402

(1) If the maintenance message shows on the Left PFTC, M2951, then do these steps:

(a) Remove the Left PFTC, M2951, off the aircraft and swap in the Right PFTC, M2952, into the removed PFTC position.

1) If the maintenance message is cleared, replace the original Left PFTC (that was removed).

2) If the maintenance message still shows, do this check of the wiring between the Left Forward (Inlet) Ram Air Actuator, M2795, and the Left AFT (Exhaust, Exit) Ram Air Actuator, M2796 (WDM 21-51-14):

a) Open these access panels:

Number	Name/Location
191FL	Forward Wing To Body Fairing Panel - Mid Fairing, Above Ram Air Inlet
191GL	Ram Air Actuator Panel - Forward

b) Disconnect connectors D14386 and D14388 from the Left Forward (Inlet) Ram Air Actuator.

c) Disconnect connectors D14390 and D14392 from the Left AFT (Exhaust, Exit) Ram Air Actuator.

d) Do a wiring check as follows:

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OMR 820-826 (Continued)

L FWD RAM ACTUATOR D14388		L AFT RAM ACTUATOR D14392
pin 1	. . . . .	pin 1
pin 2	. . . . .	pin 2
pin 3	. . . . .	pin 3

e) If you find a problem with the wiring, then do these steps:

<1> Repair the wiring.

<2> Connect connectors D14386 and D14388 to the Left Forward (Inlet) Ram Air Actuator.

<3> Connect connectors D14390 and D14392 to the Left AFT (Exhaust, Exit) Ram Air Actuator.

<4> Do the Repair Confirmation at the end of this task.

(2) If the maintenance message shows on the Right PFTC, M2952, then do these steps:

(a) Remove the Right PFTC, M2952, off the aircraft and swap in the Left PFTC, M2951, into the removed PFTC position.

1) If the maintenance message is cleared, replace the original Right PFTC (that was removed).

2) If the maintenance message still shows, do this check of the wiring between the Right Forward (Inlet) Ram Air Actuator, M2797, and the Right AFT (Exhaust, Exit) Ram Air Actuator, M2798 (WDM 21-51-24):

a) Open these access panels:

Number	Name/Location
191FL	Forward Wing To Body Fairing Panel - Mid Fairing, Above Ram Air Inlet
191GL	Ram Air Actuator Panel - Forward

b) Disconnect connectors D14394 and D14396 from the Right Forward (Inlet) Ram Air Actuator.

c) Disconnect connectors D14398 and D14400 from the Right AFT (Exhaust, Exit) Ram Air Actuator.

d) Do a wiring check as follows:

R FWD RAM ACTUATOR D14396		R AFT RAM ACTUATOR D14400
pin 1	. . . . .	pin 1
pin 2	. . . . .	pin 2
pin 3	. . . . .	pin 3

e) If you find a problem with the wiring, then do these steps:

<1> Repair the wiring.

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OMR 820-826 (Continued)

<2> Connect connectors D14394 and D14396 to the Right Forward (Inlet) Ram Air Actuator.

<3> Connect connectors D14398 and D14400 to the Right AFT (Exhaust, Exit) Ram Air Actuator.

<4> Do the Repair Confirmation at the end of this task.

H. Repair Confirmation

(1) For the PFTC, do the "BITE Procedure" and the "BITE Procedure - EXISTING FAULTS" sections of this task:

- Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the fault.

(b) If the maintenance message still shows, continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

880. PFTC SRADA Power Fault - Fault Isolation

A. Description

(1) This task is for Maintenance Message:

(a) 21-51408 SRADA CNTRL

1) SRADA Power Fault

(2) This Maintenance Message can show on either the Left or the Right Pack Flow Temperature Controller (PFTC).

B. Possible Causes

(1) For maintenance messages that show on the Left PFTC, M2951:

(a) Wiring

(b) Left Forward (Inlet) Ram Air Actuator, M2795

(c) Left PFTC, M2951

(2) For maintenance messages that show on the Right PFTC, M2952:

(a) Wiring

(b) Right Forward (Inlet) Ram Air Actuator, M2797

(c) Right PFTC, M2952

C. Circuit Breakers

(1) These are the primary circuit breakers related to this fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	8	C00555	AIR CONDITIONING RAM AIR MOD CONT L
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	8	C01178	AIR CONDITIONING RAM AIR MOD CONT R

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OMR 820-826 (Continued)

(Continued)

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L

D. Related Data

(1) WDM 21-51-14

(2) WDM 21-51-24

E. Initial Evaluation

(1) Do the Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then there was an intermittent fault.

(b) If the maintenance message still shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) If the maintenance message shows on the Left PFTC, then do these steps:

(a) Do a voltage check at the Left Forward (Inlet) Ram Air Actuator, M2795 as follows:

1) Open these access panels:

Number	Name/Location
191FL	Forward Wing To Body Fairing Panel - Mid Fairing, Above Ram Air Inlet
191GL	Ram Air Actuator Panel - Forward

2) Disconnect electrical connector D14386 from the Left Forward (Inlet) Ram Air Actuator, M2795 (WDM 21-51-14).

3) Do a check for 28V DC at pin 4 of connector D14386.

a) If you do not find 28V DC at pin 4 of connector D14386, then do these steps:

<1> Repair the wiring.

<2> Connect electrical connector D14386 to the Left Forward (Inlet) Ram Air Actuator.

<3> Do the Repair Confirmation at the end of this task.

b) If you find 28V DC at pin 4 of connector D14386, then continue.

(b) Do this check of the wiring between the Left PFTC, M2951, and the Left Forward (Inlet) Ram Air Actuator, M2795 (WDM 21-51-14):

1) Remove the Left PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

2) Do a wiring check as follows:

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OMR 820-826 (Continued)

L PFTC D15814B		L FWD RAM ACTUATOR D14386
pin G5	. . . . .	pin 10
pin G6	. . . . .	pin 8
pin G7	. . . . .	pin 12
pin G8	. . . . .	pin 13

3) If you find a problem with the wiring, then do these steps:
   a) Repair the wiring.
   b) Re-install the Left PFTC, M2951. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.
   c) Connect electrical connector D14386 to the Left Forward (Inlet) Ram Air Actuator.
   d) Do the Repair Confirmation at the end of this task.
4) If you do not find a problem with the wiring, then do these steps:
   a) Replace the Left Forward (Inlet) Ram Air Actuator, M2795. These are the tasks:
      • Ram Air Inlet Actuator Removal, AMM TASK 21-51-21-000-801
      • Ram Air Inlet Actuator Installation, AMM TASK 21-51-21-400-801
   b) Re-install the Left PFTC, M2951. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.
   c) Do the Repair Confirmation at the end of this task.

(c) Replace the Left PFTC, M2951. These are the tasks:
   • Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801
   • Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801

(2) If the maintenance message shows on the Right PFTC, then do these steps:

(a) Do a voltage check for 28V DC at the Right Forward (Inlet) Ram Air Actuator, M2797:

1) Open these access panels:

Number	Name/Location
191FR	Forward Wing To Body Fairing Panel - Mid Fairing, Above Ram Air Inlet
191GR	Ram Air Actuator Panel - Forward

2) Disconnect electrical connector D14394 from the Right Forward (Inlet) Ram Air Actuator, M2797 (WDM 21-51-24).
3) Do a check for 28V DC at pin 4 of connector D14394.
   a) If you do not find 28V DC at pin 4 of connector D14394, then do these steps:
      <1> Repair the wiring.
      <2> Connect electrical connector D14394 to the Right Forward (Inlet) Ram Air Actuator.
      <3> Do the Repair Confirmation at the end of this task.

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OMR 820-826 (Continued)

b) If you find 28V DC at pin 4 of connector D14394, then continue.

(b) Do this check of the wiring between the Right PFTC, M2952, and the Right Forward (Inlet) Ram Air Actuator, M2797 (WDM 21-51-24):

1) Remove the Right PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

2) Do a wiring check as follows:

R PFTC D15816B	R FWD RAM ACTUATOR D14394
pin G5	pin 10
pin G6	pin 8
pin G7	pin 12
pin G8	pin 13

3) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-install the Right PFTC, M2952. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

c) Connect electrical connector D14394 to the Right Forward (Inlet) Ram Air Actuator.

d) Do the Repair Confirmation at the end of this task.

4) If you do not find a problem with the wiring, then do these steps:

a) Replace the Right Forward (Inlet) Ram Air Actuator, M2797. These are the tasks:

- Ram Air Inlet Actuator Removal, AMM TASK 21-51-21-000-801
- Ram Air Inlet Actuator Installation, AMM TASK 21-51-21-400-801

b) Re-install the Right PFTC, M2952. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

c) Do the Repair Confirmation at the end of this task.

(c) Replace the Right PFTC, M2952. These are the tasks:

- Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801
- Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801

G. Repair Confirmation

(1) Do the Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the problem.

(b) If the maintenance message shows, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

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OMR 820-826 (Continued)

881. ADIRU1 ARINC - Fault Isolation

A. Description

(1) This task is for these Maintenance Messages:

(a) 21-51133 ARINC 429 INVLD

1) Left Air Data Inertial Reference Unit (ADIRU) 1 ARINC Data Invalid Fault

(b) 21-51147 ARINC 429 INVLD

1) ADIRU 1 (Left) Parity Error Fault

(c) 21-51148 ARINC 429 INVLD

1) ADIRU 1 (Left) No Activity Fault

(2) These Maintenance Messages can show on either the Left or the Right Pack Flow Temperature Controller (PFTC).

B. Possible Causes

(1) For maintenance messages that show on the Left PFTC, M2951:

(a) Left ADIRU, M1749

(b) Wiring

(c) Left PFTC, M2951

(d) Right PFTC, M2952

(2) For maintenance messages that show on the Right PFTC, M2952:

(a) Left ADIRU, M1749

(b) Wiring

(c) Right PFTC, M2952

(d) Left PFTC, M2951

C. Circuit Breakers

(1) These are the primary circuit breakers related to this fault:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
E	5	C01009	ADIRU LEFT DC
E	7	C01007	ADIRU LEFT AC

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	8	C00425	ADIRU LEFT EXC

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

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OMR 820-826 (Continued)

D. Related Data

(1) WDM 21-51-12

(2) WDM 21-51-22

(3) WDM 34-21-14

E. Initial Evaluation

(1) Do the Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then there was an intermittent fault.

(b) If the maintenance message still shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Replace the Left ADIRU, M1749. These are the tasks:

- Air Data Inertial Reference Unit Removal, AMM TASK 34-21-01-000-801
- Air Data Inertial Reference Unit - Installation, AMM TASK 34-21-01-400-801

(a) Do the Repair Confirmation at the end of this task.

(2) If the maintenance message shows on the Left PFTC, then do these steps:

(a) Do this check of the wiring between the Left PFTC, M2951, and the Left ADIRU, M1749 (WDM 21-51-12, WDM 34-21-14):

1) Remove the Left PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

2) Remove the Left ADIRU, M1749. This is the task: Air Data Inertial Reference Unit Removal, AMM TASK 34-21-01-000-801.

3) Do a wiring check as follows:

L PFTC		**L ADIRU**
D15814B		**D3687A**
pin G14	. . . . .	pin B12
pin G15	. . . . .	pin D12

4) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-install the Left PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

c) Re-install the Left ADIRU. This is the task: Air Data Inertial Reference Unit - Installation, AMM TASK 34-21-01-400-801.

d) Do the Repair Confirmation at the end of this task.

5) Replace the Left PFTC, M2951. These are the tasks:

- Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801
- Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801

a) Do the Repair Confirmation at the end of this task.

6) Replace the Right PFTC, M2952. These are the tasks:

- Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801

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- Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801

a) Do the Repair Confirmation at the end of this task.

(3) If the maintenance message shows on the Right PFTC, then do these steps:

(a) Do this check of the wiring between the Right PFTC, M2952, and the Left ADIRU, M1749 (WDM 21-51-22, WDM 34-21-14):

1) Remove the Right PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

2) Remove the Left ADIRU, M1749. This is the task: Air Data Inertial Reference Unit Removal, AMM TASK 34-21-01-000-801.

3) Do a wiring check as follows:

R PFTC		L ADIRU
D15816B		D3687A
pin G14	. . . . .	pin B12
pin G15	. . . . .	pin D12

4) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-install the Right PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

c) Re-install the Left ADIRU. This is the task: Air Data Inertial Reference Unit - Installation, AMM TASK 34-21-01-400-801.

d) Do the Repair Confirmation at the end of this task.

5) Replace the Right PFTC, M2952. These are the tasks:

- Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801
- Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801

a) Do the Repair Confirmation at the end of this task.

6) Replace the Left PFTC, M2951. These are the tasks:

- Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801
- Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801

a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do the Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the problem.

(b) If the maintenance message shows, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

882. ADIRU2 ARINC - Fault Isolation

A. Description

(1) This task is for these Maintenance Messages:

(a) 21-51134 ARINC 429 INVLD

1) Right Air Data Inertial Reference Unit (ADIRU) 2 ARINC Data Invalid Fault

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OMR 820-826 (Continued)

(b) 21-51149 ARINC 429 INVLD

1) ADIRU 2 (Right) Parity Error Fault

(c) 21-51150 ARINC 429 INVLD

1) ADIRU 2 (Right) No Activity Fault

(2) These Maintenance Messages can show on either the Left or the Right Pack Flow Temperature Controller (PFTC).

B. Possible Causes

(1) For maintenance messages that show on the Left PFTC, M2951:

(a) Right ADIRU, M1752

(b) Wiring

(c) Left PFTC, M2951

(d) Right PFTC, M2952

(2) For maintenance messages that show on the Right PFTC, M2952:

(a) Right ADIRU, M1752

(b) Wiring

(c) Right PFTC, M2952

(d) Left PFTC, M2951

C. Circuit Breakers

(1) These are the primary circuit breakers related to this fault:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	14	C01008	ADIRU RIGHT AC
C	15	C00426	ADIRU RIGHT EXC
C	17	C01010	ADIRU RIGHT DC

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

D. Related Data

(1) WDM 21-51-12

(2) WDM 21-51-22

(3) WDM 34-21-24

E. Initial Evaluation

(1) Do the Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then there was an intermittent fault.

(b) If the maintenance message still shows, then do the Fault Isolation Procedure below.

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OMR 820-826 (Continued)

F. Fault Isolation Procedure

(1) Replace the Right ADIRU, M1752. These are the tasks:

- Air Data Inertial Reference Unit Removal, AMM TASK 34-21-01-000-801
- Air Data Inertial Reference Unit - Installation, AMM TASK 34-21-01-400-801

(a) Do the Repair Confirmation at the end of this task.

(2) If the maintenance message shows on the Left PFTC, then do these steps:

(a) Do this check of the wiring between the Left PFTC, M2951, and the Right ADIRU, M1752 (WDM 21-51-12, WDM 34-21-24):

1) Remove the Left PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

2) Remove the Right ADIRU, M1752. This is the task: Air Data Inertial Reference Unit Removal, AMM TASK 34-21-01-000-801.

3) Do a wiring check as follows:

L PFTC		**R ADIRU**
D15814B		**D3693A**
pin D14	. . . . .	pin B12
pin D15	. . . . .	pin D12

4) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-install the Left PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

c) Re-install the Right ADIRU. This is the task: Air Data Inertial Reference Unit - Installation, AMM TASK 34-21-01-400-801.

d) Do the Repair Confirmation at the end of this task.

5) Replace the Left PFTC, M2951. These are the tasks:

- Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801
- Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801

a) Do the Repair Confirmation at the end of this task.

6) Replace the Right PFTC, M2952. These are the tasks:

- Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801
- Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801

a) Do the Repair Confirmation at the end of this task.

(3) If the maintenance message shows on the Right PFTC, then do these steps:

(a) Do this check of the wiring between the Right PFTC, M2952, and the Right ADIRU, M1752 (WDM 21-51-22, WDM 34-21-24):

1) Remove the Right PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

2) Remove the Right ADIRU, M1752. This is the task: Air Data Inertial Reference Unit Removal, AMM TASK 34-21-01-000-801.

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OMR 820-826 (Continued)

3) Do a wiring check as follows:

R PFTC D15816B		R ADIRU D3693A
pin D14	. . . . .	pin B12
pin D15	. . . . .	pin D12

4) If you find a problem with the wiring, then do these steps:
   a) Repair the wiring.
   b) Re-install the Right PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.
   c) Re-install the Right ADIRU. This is the task: Air Data Inertial Reference Unit - Installation, AMM TASK 34-21-01-400-801.
   d) Do the Repair Confirmation at the end of this task.
5) Replace the Right PFTC, M2952. These are the tasks:
   • Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801
   • Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801
   a) Do the Repair Confirmation at the end of this task.
6) Replace the Left PFTC, M2951. These are the tasks:
   • Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801
   • Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801
   a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do the Pack Temperature Control - BITE Procedure, 21-62 TASK 801.
   (a) If the maintenance message does not show, then you corrected the problem.
   (b) If the maintenance message shows, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

883. CPC1 ARINC - Fault Isolation

A. Description

(1) This task is for these Maintenance Messages:
   (a) 21-51131 ARINC 429 INVLD
      1) Cabin Pressure Controller (CPC) 1 ARINC Data Invalid Fault
   (b) 21-51143 ARINC 429 INVLD
      1) Cabin Pressure Controller (CPC) 1 Parity Error Fault
   (c) 21-51144 ARINC 429 INVLD
      1) Cabin Pressure Controller (CPC) 1 No Activity Fault
(2) These Maintenance Messages can show on either the Left or the Right Pack Flow Temperature Controller (PFTC).

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B. Possible Causes

(1) For maintenance messages that show on the Left PFTC, M2951:

(a) CPC 1, M1653

(b) Wiring

(c) Left PFTC, M2951

(d) Right PFTC, M2952

(2) For maintenance messages that show on the Right PFTC, M2952:

(a) CPC 1, M1653

(b) Wiring

(c) Right PFTC, M2952

(d) Left PFTC, M2951

C. Circuit Breakers

(1) These are the primary circuit breakers related to this fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC
F	3	C01270	PRESSURIZATION CONTROL AUTO 1

D. Related Data

(1) WDM 21-51-12

(2) WDM 21-51-22

(3) WDM 31-35-04

E. Initial Evaluation

(1) Do the Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then there was an intermittent fault.

(b) If the maintenance message still shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Replace the Cabin Pressure Controller, M1653. These are the tasks:

- Cabin Pressure Controller (CPC) Removal, AMM TASK 21-31-01-000-801
- Cabin Pressure Controller (CPC) Installation, AMM TASK 21-31-01-400-801

(a) Do the Repair Confirmation at the end of this task.

(2) If the maintenance message shows on the Left PFTC, then do these steps:

(a) Do this check of the wiring between the Left PFTC, M2951, and the Cabin Pressure Controller 1, M1653 (WDM 21-51-12, WDM 31-35-04):

1) Remove the Left PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

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2) Remove the Cabin Pressure Controller 1. This is the task: Cabin Pressure Controller (CPC) Removal, AMM TASK 21-31-01-000-801.

3) Do a wiring check as follows:

L PFTC		**CPC 1**
D15814B		**D10738A**
pin H14	. . . . .	pin A5
pin H15	. . . . .	pin A6

4) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-install the Left PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

c) Re-install CPC 1. This is the task: Cabin Pressure Controller (CPC) Installation, AMM TASK 21-31-01-400-801.

d) Do the Repair Confirmation at the end of this task.

5) Replace the Left PFTC, M2951. These are the tasks:

- Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801
- Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801

a) Do the Repair Confirmation at the end of this task.

6) Replace the Right PFTC, M2952. These are the tasks:

- Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801
- Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801

a) Do the Repair Confirmation at the end of this task.

(3) If the maintenance message shows on the Right PFTC, then do these steps:

(a) Do this check of the wiring between the Right PFTC, M2952, and the Cabin Pressure Controller 1, M1653 (WDM 21-51-22, WDM 31-35-04):

1) Remove the Right PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

2) Remove the Cabin Pressure Controller 1. This is the task: Cabin Pressure Controller (CPC) Removal, AMM TASK 21-31-01-000-801.

3) Do a wiring check as follows:

R PFTC		**CPC 1**
D15816B		**D10738A**
pin H14	. . . . .	pin A5
pin H15	. . . . .	pin A6

4) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-install the Right PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

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c) Re-install CPC 1. This is the task: Cabin Pressure Controller (CPC) Installation, AMM TASK 21-31-01-400-801.

d) Do the Repair Confirmation at the end of this task.

5) Replace the Right PFTC, M2952. These are the tasks:

- Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801
- Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801

a) Do the Repair Confirmation at the end of this task.

6) Replace the Left PFTC, M2951. These are the tasks:

- Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801
- Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801

a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do the Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the problem.

(b) If the maintenance message shows, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

884. CPC2 ARINC - Fault Isolation

A. Description

(1) This task is for these Maintenance Messages:

(a) 21-51132 ARINC 429 INVLD

1) Cabin Pressure Controller (CPC) 2 ARINC Data Invalid Fault

(b) 21-51145 ARINC 429 INVLD

1) Cabin Pressure Controller (CPC) 2 Parity Error Fault

(c) 21-51146 ARINC 429 INVLD

1) Cabin Pressure Controller (CPC) 2 No Activity Fault

(2) These Maintenance Messages can show on either the Left or the Right Pack Flow Temperature Controller (PFTC).

B. Possible Causes

(1) For maintenance messages that show on the Left PFTC, M2951:

(a) CPC 2, M1654

(b) Wiring

(c) Left PFTC, M2951

(d) Right PFTC, M2952

(2) For maintenance messages that show on the Right PFTC, M2952:

(a) CPC 2, M1654

(b) Wiring

(c) Right PFTC, M2952

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(d) Left PFTC, M2951

C. Circuit Breakers

(1) These are the primary circuit breakers related to this fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC
F	5	C01271	PRESSURIZATION CONTROL AUTO 2

D. Related Data

(1) WDM 21-51-12

(2) WDM 21-51-22

(3) WDM 31-35-04

E. Initial Evaluation

(1) Do the Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then there was an intermittent fault.

(b) If the maintenance message still shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Replace the Cabin Pressure Controller, M1654. These are the tasks:

- Cabin Pressure Controller (CPC) Removal, AMM TASK 21-31-01-000-801
- Cabin Pressure Controller (CPC) Installation, AMM TASK 21-31-01-400-801

(a) Do the Repair Confirmation at the end of this task.

(2) If the maintenance message shows on the Left PFTC, then do these steps:

(a) Do this check of the wiring between the Left PFTC, M2951, and the Cabin Pressure Controller 2, M1654 (WDM 21-51-12, WDM 31-35-04):

1) Remove the Left PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

2) Remove the Cabin Pressure Controller 2. This is the task: Cabin Pressure Controller (CPC) Removal, AMM TASK 21-31-01-000-801.

3) Do a wiring check as follows:

L PFTC D15814B		**CPC 2 D10740A**
pin C14	. . . . .	pin A5
pin C15	. . . . .	pin A6

4) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-install the Left PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

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c) Re-install CPC 2. This is the task: Cabin Pressure Controller (CPC) Installation, AMM TASK 21-31-01-400-801.

d) Do the Repair Confirmation at the end of this task.

5) Replace the Left PFTC, M2951. These are the tasks:

- Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801
- Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801

a) Do the Repair Confirmation at the end of this task.

6) Replace the Right PFTC, M2952. These are the tasks:

- Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801
- Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801

a) Do the Repair Confirmation at the end of this task.

(3) If the maintenance message shows on the Right PFTC, then do these steps:

(a) Do this check of the wiring between the Right PFTC, M2952, and the Cabin Pressure Controller 2, M1654 (WDM 21-51-22, WDM 31-35-04):

1) Remove the Right PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

2) Remove the Cabin Pressure Controller 2. This is the task: Cabin Pressure Controller (CPC) Removal, AMM TASK 21-31-01-000-801.

3) Do a wiring check as follows:

R PFTC		**CPC 2**
D15816B		**D10740A**
pin C14	. . . . .	pin A5
pin C15	. . . . .	pin A6

4) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-install the Right PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

c) Re-install CPC 2. This is the task: Cabin Pressure Controller (CPC) Installation, AMM TASK 21-31-01-400-801.

d) Do the Repair Confirmation at the end of this task.

5) Replace the Right PFTC, M2952. These are the tasks:

- Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801
- Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801

a) Do the Repair Confirmation at the end of this task.

6) Replace the Left PFTC, M2951. These are the tasks:

- Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801
- Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801

a) Do the Repair Confirmation at the end of this task.

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OMR 820-826 (Continued)

G. Repair Confirmation

(1) Do the Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the problem.

(b) If the maintenance message shows, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

885. COMM1 ARINC - Fault Isolation

A. Description

(1) This task is for these Maintenance Messages:

(a) 21-51135 ARINC 429 INVLD

1) Communications 1 ARINC Data Invalid Fault

(b) 21-51151 ARINC 429 INVLD

1) Communications 1 Parity Error Fault

(c) 21-51152 ARINC 429 INVLD

1) Communications 1 No Activity Fault

(2) These Maintenance Messages can show on either the Left or the Right Pack Flow Temperature Controller (PFTC).

B. Possible Causes

(1) For maintenance messages that show on the Left PFTC, M2951:

(a) Left PFTC, M2951

(b) Wiring

(c) Right PFTC, M2952

(2) For maintenance messages that show on the Right PFTC, M2952:

(a) Right PFTC, M2952

(b) Wiring

(c) Left PFTC, M2951

C. Circuit Breakers

(1) These are the primary circuit breakers related to this fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

D. Related Data

(1) WDM 21-51-12

(2) WDM 21-51-22

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OMR 820-826 (Continued)

E. Initial Evaluation

(1) Do the Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then there was an intermittent fault.

(b) If the maintenance message still shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) If the maintenance message shows on either the Right PFTC or the Left PFTC, then do these steps:

(a) Do this check of the wiring between the Right PFTC, M2952, and the Left PFTC, M2951 (WDM 21-51-12, WDM 21-51-22):

1) Remove the Right PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

2) Remove the Left PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

3) Do a wiring check as follows:

L PFTC D15814B	R PFTC D15816B
pin A14	pin B14
pin A15	pin B15
pin B14	pin A14
pin B15	pin A15

4) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-install the Right PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

c) Re-install the Left PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

d) Do the Repair Confirmation at the end of this task.

5) Replace the Right PFTC, M2952. These are the tasks:

- Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801
- Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801

a) Do the Repair Confirmation at the end of this task.

6) Replace the Left PFTC, M2951. These are the tasks:

- Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801
- Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801

a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do the Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the problem.

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OMR 820-826 (Continued)

(b) If the maintenance message shows, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

886. COMM2 ARINC - Fault Isolation

A. Description

(1) This task is for these Maintenance Messages:

(a) 21-51136 ARINC 429 INVLD

1) Communications 2 ARINC Data Invalid Fault

(b) 21-51153 ARINC 429 INVLD

1) Communications 2 Parity Error Fault

(c) 21-51154 ARINC 429 INVLD

1) Communications 2 No Activity Fault

(2) These Maintenance Messages can show on either the Left or the Right Pack Flow Temperature Controller (PFTC).

B. Possible Causes

(1) For maintenance messages that show on the Left PFTC, M2951:

(a) Wiring

(b) Right PFTC, M2952

(c) Left PFTC, M2951

(2) For maintenance messages that show on the Right PFTC, M2952:

(a) Wiring

(b) Right PFTC, M2952

(c) Left PFTC, M2951

C. Circuit Breakers

(1) These are the primary circuit breakers related to this fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

D. Related Data

(1) WDM 21-51-12

(2) WDM 21-51-22

E. Initial Evaluation

(1) Do the Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then there was an intermittent fault.

(b) If the maintenance message still shows, then do the Fault Isolation Procedure below.

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OMR 820-826 (Continued)

F. Fault Isolation Procedure

(1) If the maintenance message shows on either the Right PFTC or the Left PFTC, then do these steps:

(a) Do this check of the wiring between the Right PFTC, M2952, and the Left PFTC, M2951 (WDM 21-51-12, WDM 21-51-22):

1) Remove the Right PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

2) Remove the Left PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

3) Do a wiring check as follows:

L PFTC D15814B	**R PFTC D15816B**
pin F14	pin E14
pin F15	pin E15
pin E14	pin F14
pin E15	pin F15

4) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-install the Right PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

c) Re-install the Left PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

d) Do the Repair Confirmation at the end of this task.

5) Replace the Right PFTC, M2952. These are the tasks:

- Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801
- Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801

a) Do the Repair Confirmation at the end of this task.

6) Replace the Left PFTC, M2951. These are the tasks:

- Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801
- Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801

a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do the Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the problem.

(b) If the maintenance message shows, then continue the Fault Isolation Procedure at the subsequent step.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 820-826 (Continued)

887. PFTC Flight Deck INOP Discrete Output - Fault Isolation

A. Description

(1) This task is for this Latched Maintenance Message:

NOTE: After you correct a problem related to a Latched Maintenance Message, you can clear the Maintenance Message as follows:

- Cycle the power to the PFTC or
- Perform the BITE ground test or
- Erase the EXISTING FAULTS from the PFTC Front Panel Menus.

(a) 21-51110 PFTC CNTRL INOP

1) Flight Deck INOP Interface Fault

(2) This task is also for this Non-Latched Maintenance Message:

(a) 21-51111 PFTC CNTRL INOP

1) Flight Deck INOP Load Fault

(3) These Maintenance Messages can show on either the Left or the Right Pack Flow Temperature Controller (PFTC).

B. Possible Causes

(1) For maintenance messages that show on the Left PFTC, M2951:

(a) Wiring

(b) Left PFTC, M2951

(c) Air Conditioning Module, P5-10

(2) For maintenance messages that show on the Right PFTC, M2952:

(a) Wiring

(b) Right PFTC, M2952

(c) Air Conditioning Module, P5-10

C. Circuit Breakers

(1) These are the primary circuit breakers related to this fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

D. Related Data

(1) WDM 21-61-11

(2) WDM 21-61-12

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OMR 820-826 (Continued)

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message shows in the PFTC Existing Faults, or it is a latched message that shows in the most recent flight leg, then do the Fault Isolation Procedure below.

(b) If the maintenance message does not show in the PFTC Existing Faults, or as a latched message in the most recent flight leg, then no additional fault isolation steps are required.

F. Fault Isolation Procedure

(1) If the maintenance message shows on the Left PFTC, then do these steps:

(a) Do this check of the wiring between the Air Conditioning Module, P5-10, and the Left PFTC, M2951 (WDM 21-61-12):

1) Remove the Left PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

2) Remove the Air Conditioning Module. This is the task: Air Conditioning Module Removal, AMM TASK 21-51-65-000-801.

3) Do a wiring check as follows:

AIR CONDITIONING MODULE D646	**L PFTC D15814B**
pin 27	pin G2

4) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-install the Left PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

c) Re-install the Air Conditioning Module. This is the task: Air Conditioning Module Installation, AMM TASK 21-51-65-400-801.

d) Do the Repair Confirmation at the end of this task.

5) If you do not find a problem with the wiring, then do these steps:

a) Replace the Left PFTC, M2951. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

b) Re-install the Air Conditioning Module. This is the task: Air Conditioning Module Installation, AMM TASK 21-51-65-400-801.

c) Do the Repair Confirmation at the end of this task.

(b) Replace the Air Conditioning Module, P5-10. These are the tasks:

- Air Conditioning Module Removal, AMM TASK 21-51-65-000-801
- Air Conditioning Module Installation, AMM TASK 21-51-65-400-801

1) Do the Repair Confirmation at the end of this task.

(2) If the maintenance message shows on the Right PFTC, then do these steps:

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OMR 820-826 (Continued)

(a) Do this check of the wiring between the Air Conditioning Module, P5-10, and the Right PFTC, M2952 (WDM 21-61-11):

1) Remove the Right PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

2) Remove the Air Conditioning Module. This is the task: Air Conditioning Module Removal, AMM TASK 21-51-65-000-801.

3) Do a wiring check as follows:

AIR CONDITIONING MODULE D646	R PFTC D15816B
pin 40	pin G2

4) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-install the Right PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

c) Re-install the Air Conditioning Module. This is the task: Air Conditioning Module Installation, AMM TASK 21-51-65-400-801.

d) Do the Repair Confirmation at the end of this task.

5) If you do not find a problem with the wiring, then do these steps:

a) Replace the Right PFTC, M2952. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

b) Re-install the Air Conditioning Module. This is the task: Air Conditioning Module Installation, AMM TASK 21-51-65-400-801.

c) Do the Repair Confirmation at the end of this task.

(b) Replace the Air Conditioning Module, P5-10. These are the tasks:

- Air Conditioning Module Removal, AMM TASK 21-51-65-000-801
- Air Conditioning Module Installation, AMM TASK 21-51-65-400-801

1) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) For the PFTC, do the "BITE Procedure" and the "BITE Procedure - EXISTING FAULTS" sections of this task:

- Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the fault.

(b) If the maintenance message still shows, continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 820-826 (Continued)

888. PFTC Flight Deck Fault Discrete Output - Fault Isolation

A. Description

(1) This task is for this Latched Maintenance Message:

NOTE: After you correct a problem related to a Latched Maintenance Message, you can clear the Maintenance Message as follows:

- Cycle the power to the PFTC or
- Perform the BITE ground test or
- Erase the EXISTING FAULTS from the PFTC Front Panel Menus.

(a) 21-51112 PFTC CNTRL INOP

1) Flight Deck Fault Interface Fault

(2) This task is also for this Non-Latched Maintenance Message:

(a) 21-51113 PFTC CNTRL INOP

1) Flight Deck Fault Load Fault

(3) These Maintenance Messages can show on either the Left or the Right Pack Flow Temperature Controller (PFTC).

B. Possible Causes

(1) For maintenance messages that show on the Left PFTC, M2951:

(a) Wiring

(b) Left PFTC, M2951

(c) Air Conditioning Accessory Unit (ACAU-2), M1455

(d) Air Conditioning Module, P5-10

(2) For maintenance messages that show on the Right PFTC, M2952:

(a) Wiring

(b) Right PFTC, M2952

(c) ACAU-1, M324

C. Circuit Breakers

(1) These are the primary circuit breakers related to this fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

D. Related Data

(1) WDM 21-61-11

(2) WDM 21-61-12

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OMR 820-826 (Continued)

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message shows in the PFTC Existing Faults, or it is a latched message that shows in the most recent flight leg, then do the Fault Isolation Procedure below.

(b) If the maintenance message does not show in the PFTC Existing Faults, or as a latched message in the most recent flight leg, then no additional fault isolation steps are required.

F. Fault Isolation Procedure

(1) If the maintenance message shows on the Left PFTC, then do these steps:

(a) Do this check of the wiring between the ACAU-2, M1455, the Left PFTC, M2951, and the Air Conditioning Module, P5-10 (WDM 21-61-12):

1) Remove the Left PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

2) Remove the ACAU-2. This is the task: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801.

3) Do a wiring check as follows:

ACAU-2	**L PFTC**
D10002A	**D15814B**
pin 16	pin G1

4) Remove the Air Conditioning Module, P5-10. This is the task: Air Conditioning Module Removal, AMM TASK 21-51-65-000-801.

5) Do a wiring check as follows:

AIR CONDITIONING MODULE	**ACAU-2**
D646	**D10002A**
pin 48	pin 16

6) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-install the Left PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

c) Re-install the Air Conditioning Accessory Unit (ACAU-2). This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

d) Re-install the Air Conditioning Module. This is the task: Air Conditioning Module Installation, AMM TASK 21-51-65-400-801.

e) Do the Repair Confirmation at the end of this task.

7) If you do not find a problem with the wiring, then do these steps:

a) Replace the Left PFTC, M2951. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

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OMR 820-826 (Continued)

b) Re-install the ACAU-2. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

c) Re-install the Air Conditioning Module. This is the task: Air Conditioning Module Installation, AMM TASK 21-51-65-400-801.

d) Do the Repair Confirmation at the end of this task.

(b) Replace the ACAU-2. These are the tasks:

- Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801
- Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801

1) Do the Repair Confirmation at the end of this task.

(c) Replace the Air Conditioning Module, P5-10. These are the tasks:

- Air Conditioning Module Removal, AMM TASK 21-51-65-000-801
- Air Conditioning Module Installation, AMM TASK 21-51-65-400-801

1) Do the Repair Confirmation at the end of this task.

(2) If the maintenance message shows on the Right PFTC, then do these steps:

(a) Do this check of the wiring between the ACAU-1, M324, and the Right PFTC, M2952 (WDM 21-61-11):

1) Remove the Right PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

2) Remove the ACAU-1. This is the task: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801.

3) Do a wiring check as follows:

ACAU-1	**R PFTC**
D458A	**D15816B**
pin 16	pin G1

4) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-install the Right PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

c) Re-install the Air Conditioning Accessory Unit (ACAU-1). This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

d) Do the Repair Confirmation at the end of this task.

5) If you do not find a problem with the wiring, then do these steps:

a) Replace the Right PFTC, M2952. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

b) Re-install the ACAU-1. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

c) Do the Repair Confirmation at the end of this task.

(b) Replace the ACAU-1. These are the tasks:

- Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801

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- Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801

1) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) For the PFTC, do the "BITE Procedure" and the "BITE Procedure - EXISTING FAULTS" sections of this task:

- Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the fault.

(b) If the maintenance message still shows, continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

889. PFTC Zone Fault Discrete Output - Fault Isolation

A. Description

(1) This task is for this Latched Maintenance Message:

NOTE: After you correct a problem related to a Latched Maintenance Message, you can clear the Maintenance Message as follows:

- Cycle the power to the PFTC or
- Perform the BITE ground test or
- Erase the EXISTING FAULTS from the PFTC Front Panel Menus.

(a) 21-51114 PFTC CNTRL INOP

1) Zone Fault Interface Fault

(2) This task is also for this Non-Latched Maintenance Message:

(a) 21-51115 PFTC CNTRL INOP

1) Zone Fault Load Fault

(3) These Maintenance Messages can show on either the Left or the Right Pack Flow Temperature Controller (PFTC).

B. Possible Causes

(1) For maintenance messages that show on the Left PFTC, M2951:

(a) Wiring

(b) Left PFTC, M2951

(c) Air Conditioning Accessory Unit (ACAU-1), M324

(d) Air Conditioning Module, P5-10

(2) For maintenance messages that show on the Right PFTC, M2952:

(a) Wiring

(b) Right PFTC, M2952

(c) Air Conditioning Accessory Unit (ACAU-2), M1455

(d) Air Conditioning Module, P5-10

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OMR 820-826 (Continued)

C. Circuit Breakers

(1) These are the primary circuit breakers related to this fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

D. Related Data

(1) WDM 21-61-13

(2) WDM 21-61-14

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message shows in the PFTC Existing Faults, or it is a latched message that shows in the most recent flight leg, then do the Fault Isolation Procedure below.

(b) If the maintenance message does not show in the PFTC Existing Faults, or as a latched message in the most recent flight leg, then no additional fault isolation steps are required.

F. Fault Isolation Procedure

(1) If the maintenance message shows on the Left PFTC, then do these steps:

(a) Do this check of the wiring between the ACAU-1, M324, the Left PFTC, M2951, and the Air Conditioning Module, P5-10 (WDM 21-61-14):

1) Remove the Left PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

2) Remove the ACAU-1. This is the task: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801.

3) Do a wiring check as follows:

ACAU-1	**L PFTC**
D458A	**D15814B**
pin 11	pin H2

4) Remove the Air Conditioning Module, P5-10. This is the task: Air Conditioning Module Removal, AMM TASK 21-51-65-000-801.

5) Do a wiring check as follows:

AIR CONDITIONING MODULE	**ACAU-1**
D646	**D458A**
pin 51	pin 11

6) If you find a problem with the wiring, then do these steps:

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OMR 820-826 (Continued)

a) Repair the wiring.

b) Re-install the Left PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

c) Re-install the Air Conditioning Accessory Unit (ACAU-1). This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

d) Re-install the Air Conditioning Module. This is the task: Air Conditioning Module Installation, AMM TASK 21-51-65-400-801.

e) Do the Repair Confirmation at the end of this task.

7) If you do not find a problem with the wiring, then do these steps:

a) Replace the Left PFTC, M2951. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

b) Re-install the ACAU-1. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

c) Re-install the Air Conditioning Module. This is the task: Air Conditioning Module Installation, AMM TASK 21-51-65-400-801.

d) Do the Repair Confirmation at the end of this task.

(b) Replace the ACAU-1. These are the tasks:

- Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801
- Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801

1) Do the Repair Confirmation at the end of this task.

(c) Replace the Air Conditioning Module, P5-10. These are the tasks:

- Air Conditioning Module Removal, AMM TASK 21-51-65-000-801
- Air Conditioning Module Installation, AMM TASK 21-51-65-400-801

1) Do the Repair Confirmation at the end of this task.

(2) If the maintenance message shows on the Right PFTC, then do these steps:

(a) Do this check of the wiring between the ACAU-2, M1455, the Right PFTC, M2952, and the Air Conditioning Module, P5-10 (WDM 21-61-13):

1) Remove the Right PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

2) Remove the ACAU-2. This is the task: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801.

3) Do a wiring check as follows:

ACAU-2	**R PFTC**
D10002A	**D15816B**
pin 11	pin H2

4) Remove the Air Conditioning Module, P5-10. This is the task: Air Conditioning Module Removal, AMM TASK 21-51-65-000-801.

5) Do a wiring check as follows:

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OMR 820-826 (Continued)

AIR CONDITIONING MODULE	**ACAU-2**
D680	**D10002A**
pin 40	pin 11

6) If you find a problem with the wiring, then do these steps:
   a) Repair the wiring.
   b) Re-install the Right PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.
   c) Re-install the Air Conditioning Accessory Unit (ACAU-2). This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.
   d) Re-install the Air Conditioning Module. This is the task: Air Conditioning Module Installation, AMM TASK 21-51-65-400-801.
   e) Do the Repair Confirmation at the end of this task.

7) If you do not find a problem with the wiring, then do these steps:
   a) Replace the Right PFTC, M2952. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.
   b) Re-install the ACAU-2. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.
   c) Re-install the Air Conditioning Module. This is the task: Air Conditioning Module Installation, AMM TASK 21-51-65-400-801.
   d) Do the Repair Confirmation at the end of this task.

(b) Replace the ACAU-2. These are the tasks:
- Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801
- Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801

1) Do the Repair Confirmation at the end of this task.

(c) Replace the Air Conditioning Module, P5-10. These are the tasks:
- Air Conditioning Module Removal, AMM TASK 21-51-65-000-801
- Air Conditioning Module Installation, AMM TASK 21-51-65-400-801

1) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) For the PFTC, do the "BITE Procedure" and the "BITE Procedure - EXISTING FAULTS" sections of this task:
- Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the fault.

(b) If the maintenance message still shows, continue the Fault Isolation Procedure at the subsequent step.

END OF TASK

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OMR 820-826 (Continued)

890. PFTC Backup Zone Fault Discrete Output - Fault Isolation

A. Description

(1) This task is for this Latched Maintenance Message:

NOTE: After you correct a problem related to a Latched Maintenance Message, you can clear the Maintenance Message as follows:

- Cycle the power to the PFTC or
- Perform the BITE ground test or
- Erase the EXISTING FAULTS from the PFTC Front Panel Menus.

(a) 21-51116 PFTC CNTRL INOP

1) Backup Zone Fault Interface Fault

(2) This task is also for this Non-Latched Maintenance Message:

(a) 21-51117 PFTC CNTRL INOP

1) Backup Zone Fault Load Fault

(3) These Maintenance Messages can show on either the Left or the Right Pack Flow Temperature Controller (PFTC).

B. Possible Causes

(1) For maintenance messages that show on the Left PFTC, M2951:

(a) Wiring

(b) Left PFTC, M2951

(c) Air Conditioning Module, P5-10

(2) For maintenance messages that show on the Right PFTC, M2952:

(a) Wiring

(b) Right PFTC, M2952

(c) Air Conditioning Module, P5-10

C. Circuit Breakers

(1) These are the primary circuit breakers related to this fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

D. Related Data

(1) WDM 21-61-13

(2) WDM 21-61-14

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OMR 820-826 (Continued)

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message shows in the PFTC Existing Faults, or it is a latched message that shows in the most recent flight leg, then do the Fault Isolation Procedure below.

(b) If the maintenance message does not show in the PFTC Existing Faults, or as a latched message in the most recent flight leg, then no additional fault isolation steps are required.

F. Fault Isolation Procedure

(1) If the maintenance message shows on the Left PFTC, then do these steps:

(a) Do this check of the wiring between the Air Conditioning Module, P5-10, and the Left PFTC, M2951 (WDM 21-61-13):

1) Remove the Left PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

2) Remove the Air Conditioning Module, P5-10. This is the task: Air Conditioning Module Removal, AMM TASK 21-51-65-000-801.

3) Do a wiring check as follows:

AIR CONDITIONING MODULE D680		**L PFTC D15814B**
pin 5	. . . . .	pin H3

4) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-install the Left PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

c) Re-install the Air Conditioning Module. This is the task: Air Conditioning Module Installation, AMM TASK 21-51-65-400-801.

d) Do the Repair Confirmation at the end of this task.

5) If you do not find a problem with the wiring, then do these steps:

a) Replace the Left PFTC, M2951. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

b) Re-install the Air Conditioning Module. This is the task: Air Conditioning Module Installation, AMM TASK 21-51-65-400-801.

c) Do the Repair Confirmation at the end of this task.

(b) Replace the Air Conditioning Module, P5-10. These are the tasks:

- Air Conditioning Module Removal, AMM TASK 21-51-65-000-801
- Air Conditioning Module Installation, AMM TASK 21-51-65-400-801

1) Do the Repair Confirmation at the end of this task.

(2) If the maintenance message shows on the Right PFTC, then do these steps:

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OMR 820-826 (Continued)

(a) Do this check of the wiring between the Air Conditioning Module, P5-10, and the Right PFTC, M2952 (WDM 21-61-14):

1) Remove the Right PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

2) Remove the Air Conditioning Module, P5-10. This is the task: Air Conditioning Module Removal, AMM TASK 21-51-65-000-801.

3) Do a wiring check as follows:

AIR CONDITIONING MODULE D646	R PFTC D15816B
pin 16	pin H3

4) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-install the Right PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

c) Re-install the Air Conditioning Module. This is the task: Air Conditioning Module Installation, AMM TASK 21-51-65-400-801.

d) Do the Repair Confirmation at the end of this task.

5) If you do not find a problem with the wiring, then do these steps:

a) Replace the Right PFTC, M2952. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

b) Re-install the Air Conditioning Module. This is the task: Air Conditioning Module Installation, AMM TASK 21-51-65-400-801.

c) Do the Repair Confirmation at the end of this task.

(b) Replace the Air Conditioning Module, P5-10. These are the tasks:

- Air Conditioning Module Removal, AMM TASK 21-51-65-000-801
- Air Conditioning Module Installation, AMM TASK 21-51-65-400-801

1) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) For the PFTC, do the "BITE Procedure" and the "BITE Procedure - EXISTING FAULTS" sections of this task:

- Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the fault.

(b) If the maintenance message still shows, continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 820-826 (Continued)

891. PFTC Pack INOP Discrete Output - Fault Isolation

A. Description

(1) This task is for this Latched Maintenance Message:

NOTE: After you correct a problem related to a Latched Maintenance Message, you can clear the Maintenance Message as follows:

- Cycle the power to the PFTC or
- Perform the BITE ground test or
- Erase the EXISTING FAULTS from the PFTC Front Panel Menus.

(a) 21-51118 PFTC CNTRL INOP

1) Pack INOP Interface Fault

(2) This task is also for this Non-Latched Maintenance Message:

(a) 21-51119 PFTC CNTRL INOP

1) Pack INOP Load Fault

(3) These Maintenance Messages can show on either the Left or the Right Pack Flow Temperature Controller (PFTC).

B. Possible Causes

(1) For maintenance messages that show on the Left PFTC, M2951:

(a) Wiring

(b) Left PFTC, M2951

(c) Air Conditioning Accessory Unit (ACAU-1), M324

(d) Air Conditioning Module, P5-10

(2) For maintenance messages that show on the Right PFTC, M2952:

(a) Wiring

(b) Right PFTC, M2952

(c) Air Conditioning Accessory Unit (ACAU-2), M1455

(d) Air Conditioning Module, P5-10

C. Circuit Breakers

(1) These are the primary circuit breakers related to this fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

D. Related Data

(1) WDM 21-51-12

(2) WDM 21-51-22

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OMR 820-826 (Continued)

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message shows in the PFTC Existing Faults, or it is a latched message that shows in the most recent flight leg, then do the Fault Isolation Procedure below.

(b) If the maintenance message does not show in the PFTC Existing Faults, or as a latched message in the most recent flight leg, then no additional fault isolation steps are required.

F. Fault Isolation Procedure

(1) If the maintenance message shows on the Left PFTC, then do these steps:

(a) Do this check of the wiring between the ACAU-1, M324, the Left PFTC, M2951, and the Air Conditioning Module, P5-10 (WDM 21-51-12):

1) Remove the Left PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

2) Remove the ACAU-1. This is the task: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801.

3) Do a wiring check as follows:

ACAU-1	**L PFTC**
D458A	**D15814B**
pin 58	pin H1

4) Remove the Air Conditioning Module, P5-10. This is the task: Air Conditioning Module Removal, AMM TASK 21-51-65-000-801.

5) Do a wiring check as follows:

AIR CONDITIONING MODULE	**ACAU-1**
D646	**D458A**
pin 10	pin 58

6) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-install the Left PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

c) Re-install the Air Conditioning Accessory Unit (ACAU-1). This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

d) Re-install the Air Conditioning Module. This is the task: Air Conditioning Module Installation, AMM TASK 21-51-65-400-801.

e) Do the Repair Confirmation at the end of this task.

7) If you do not find a problem with the wiring, then do these steps:

a) Replace the Left PFTC, M2951. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

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OMR 820-826 (Continued)

b) Re-install the ACAU-1. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

c) Re-install the Air Conditioning Module. This is the task: Air Conditioning Module Installation, AMM TASK 21-51-65-400-801.

d) Do the Repair Confirmation at the end of this task.

(b) Replace the ACAU-1. These are the tasks:

- Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801
- Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801

1) Do the Repair Confirmation at the end of this task.

(c) Replace the Air Conditioning Module, P5-10. These are the tasks:

- Air Conditioning Module Removal, AMM TASK 21-51-65-000-801
- Air Conditioning Module Installation, AMM TASK 21-51-65-400-801

1) Do the Repair Confirmation at the end of this task.

(2) If the maintenance message shows on the Right PFTC, then do these steps:

(a) Do this check of the wiring between the ACAU-2, M1455, the Right PFTC, M2952, and the Air Conditioning Module, P5-10 (WDM 21-51-22):

1) Remove the Right PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

2) Remove the ACAU-2. This is the task: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801.

3) Do a wiring check as follows:

ACAU-2	**R PFTC**
D10002A	**D15816B**
pin 58	pin H1

4) Remove the Air Conditioning Module, P5-10. This is the task: Air Conditioning Module Removal, AMM TASK 21-51-65-000-801.

5) Do a wiring check as follows:

AIR CONDITIONING MODULE	**ACAU-2**
D680	**D10002A**
pin 1	pin 58

6) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-install the Right PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

c) Re-install the Air Conditioning Accessory Unit (ACAU-2). This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

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d) Re-install the Air Conditioning Module. This is the task: Air Conditioning Module Installation, AMM TASK 21-51-65-400-801.

e) Do the Repair Confirmation at the end of this task.

7) If you do not find a problem with the wiring, then do these steps:

a) Replace the Right PFTC, M2952. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

b) Re-install the ACAU-2. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

c) Re-install the Air Conditioning Module. This is the task: Air Conditioning Module Installation, AMM TASK 21-51-65-400-801.

d) Do the Repair Confirmation at the end of this task.

(b) Replace the ACAU-2. These are the tasks:

- Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801
- Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801

1) Do the Repair Confirmation at the end of this task.

(c) Replace the Air Conditioning Module, P5-10. These are the tasks:

- Air Conditioning Module Removal, AMM TASK 21-51-65-000-801
- Air Conditioning Module Installation, AMM TASK 21-51-65-400-801

1) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) For the PFTC, do the "BITE Procedure" and the "BITE Procedure - EXISTING FAULTS" sections of this task:

- Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the fault.

(b) If the maintenance message still shows, continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

892. PFTC Pack Fault Discrete Output - Fault Isolation

A. Description

(1) This task is for this Latched Maintenance Message:

NOTE: After you correct a problem related to a Latched Maintenance Message, you can clear the Maintenance Message as follows:

- Cycle the power to the PFTC or
- Perform the BITE ground test or
- Erase the EXISTING FAULTS from the PFTC Front Panel Menus.

(a) 21-51120 PFTC CNTRL INOP

1) Pack Fault Interface Fault

(2) This task is also for this Non-Latched Maintenance Message:

(a) 21-51121 PFTC CNTRL INOP

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OMR 820-826 (Continued)

1) Pack Fault Load Fault

(3) These Maintenance Messages can show on either the Left or the Right Pack Flow Temperature Controller (PFTC).

B. Possible Causes

(1) For maintenance messages that show on the Left PFTC, M2951:

(a) Wiring

(b) Left PFTC, M2951

(c) Air Conditioning Module, P5-10

(2) For maintenance messages that show on the Right PFTC, M2952:

(a) Wiring

(b) Right PFTC, M2952

(c) Air Conditioning Module, P5-10

C. Circuit Breakers

(1) These are the primary circuit breakers related to this fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

D. Related Data

(1) WDM 21-51-12

(2) WDM 21-51-22

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message shows in the PFTC Existing Faults, or it is a latched message that shows in the most recent flight leg, then do the Fault Isolation Procedure below.

(b) If the maintenance message does not show in the PFTC Existing Faults, or as a latched message in the most recent flight leg, then no additional fault isolation steps are required.

F. Fault Isolation Procedure

(1) If the maintenance message shows on the Left PFTC, then do these steps:

(a) Do this check of the wiring between the Air Conditioning Module, P5-10, and the Left PFTC, M2951 (WDM 21-61-12):

1) Remove the Left PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

2) Remove the Air Conditioning Module. This is the task: Air Conditioning Module Removal, AMM TASK 21-51-65-000-801.

3) Do a wiring check as follows:

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OMR 820-826 (Continued)

AIR CONDITIONING MODULE D646	L PFTC D15814B
pin 55	pin J1

4) If you find a problem with the wiring, then do these steps:
   a) Repair the wiring.
   b) Re-install the Left PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.
   c) Re-install the Air Conditioning Module. This is the task: Air Conditioning Module Installation, AMM TASK 21-51-65-400-801.
   d) Do the Repair Confirmation at the end of this task.
5) If you do not find a problem with the wiring, then do these steps:
   a) Replace the Left PFTC, M2951. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.
   b) Re-install the Air Conditioning Module. This is the task: Air Conditioning Module Installation, AMM TASK 21-51-65-400-801.
   c) Do the Repair Confirmation at the end of this task.

(b) Replace the Air Conditioning Module, P5-10. These are the tasks:
- Air Conditioning Module Removal, AMM TASK 21-51-65-000-801
- Air Conditioning Module Installation, AMM TASK 21-51-65-400-801

1) Do the Repair Confirmation at the end of this task.

(2) If the maintenance message shows on the Right PFTC, then do these steps:

(a) Do this check of the wiring between the Air Conditioning Module, P5-10, and the Right PFTC, M2952 (WDM 21-51-22):

1) Remove the Right PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.
2) Remove the Air Conditioning Module. This is the task: Air Conditioning Module Removal, AMM TASK 21-51-65-000-801.
3) Do a wiring check as follows:

AIR CONDITIONING MODULE D680	R PFTC D15816B
pin 55	pin J1

4) If you find a problem with the wiring, then do these steps:
   a) Repair the wiring.
   b) Re-install the Right PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

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OMR 820-826 (Continued)

c) Re-install the Air Conditioning Module. This is the task: Air Conditioning Module Installation, AMM TASK 21-51-65-400-801.

d) Do the Repair Confirmation at the end of this task.

5) If you do not find a problem with the wiring, then do these steps:

a) Replace the Right PFTC, M2952. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

b) Re-install the Air Conditioning Module. This is the task: Air Conditioning Module Installation, AMM TASK 21-51-65-400-801.

c) Do the Repair Confirmation at the end of this task.

(b) Replace the Air Conditioning Module, P5-10. These are the tasks:

- Air Conditioning Module Removal, AMM TASK 21-51-65-000-801
- Air Conditioning Module Installation, AMM TASK 21-51-65-400-801

1) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) For the PFTC, do the "BITE Procedure" and the "BITE Procedure - EXISTING FAULTS" sections of this task:

- Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the fault.

(b) If the maintenance message still shows, continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

893. PFTC Standby Pack Fault Discrete Output - Fault Isolation

A. Description

(1) This task is for this Latched Maintenance Message:

NOTE: After you correct a problem related to a Latched Maintenance Message, you can clear the Maintenance Message as follows:

- Cycle the power to the PFTC or
- Perform the BITE ground test or
- Erase the EXISTING FAULTS from the PFTC Front Panel Menus.

(a) 21-51122 PFTC CNTRL INOP

1) Standby Pack Fault Interface Fault

(2) This task is also for this Non-Latched Maintenance Message:

(a) 21-51123 PFTC CNTRL INOP

1) Standby Pack Fault Load Fault

(3) These Maintenance Messages can show on either the Left or the Right Pack Flow Temperature Controller (PFTC).

B. Possible Causes

(1) For maintenance messages that show on the Left PFTC, M2951:

(a) Wiring

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OMR 820-826 (Continued)

(b) Left PFTC, M2951

(c) Air Conditioning Module, P5-10

(2) For maintenance messages that show on the Right PFTC, M2952:

(a) Wiring

(b) Right PFTC, M2952

(c) Air Conditioning Module, P5-10

C. Circuit Breakers

(1) These are the primary circuit breakers related to this fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

D. Related Data

(1) WDM 21-51-12

(2) WDM 21-51-22

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message shows in the PFTC Existing Faults, or it is a latched message that shows in the most recent flight leg, then do the Fault Isolation Procedure below.

(b) If the maintenance message does not show in the PFTC Existing Faults, or as a latched message in the most recent flight leg, then no additional fault isolation steps are required.

F. Fault Isolation Procedure

(1) If the maintenance message shows on the Left PFTC, then do these steps:

(a) Do this check of the wiring between the Air Conditioning Module, P5-10, and the Left PFTC, M2951 (WDM 21-51-22):

1) Remove the Left PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

2) Remove the Air Conditioning Module. This is the task: Air Conditioning Module Removal, AMM TASK 21-51-65-000-801.

3) Do a wiring check as follows:

AIR CONDITIONING MODULE D680		**L PFTC D15814B**
pin 2	. . . . .	pin J2

4) If you find a problem with the wiring, then do these steps:

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OMR 820-826 (Continued)

a) Repair the wiring.

b) Re-install the Left PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

c) Re-install the Air Conditioning Module. This is the task: Air Conditioning Module Installation, AMM TASK 21-51-65-400-801.

d) Do the Repair Confirmation at the end of this task.

5) If you do not find a problem with the wiring, then do these steps:

a) Replace the Left PFTC, M2951. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

b) Re-install the Air Conditioning Module. This is the task: Air Conditioning Module Installation, AMM TASK 21-51-65-400-801.

c) Do the Repair Confirmation at the end of this task.

(b) Replace the Air Conditioning Module, P5-10. These are the tasks:

- Air Conditioning Module Removal, AMM TASK 21-51-65-000-801
- Air Conditioning Module Installation, AMM TASK 21-51-65-400-801

1) Do the Repair Confirmation at the end of this task.

(2) If the maintenance message shows on the Right PFTC, then do these steps:

(a) Do this check of the wiring between the Air Conditioning Module, P5-10, and the Right PFTC, M2952 (WDM 21-51-12):

1) Remove the Right PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

2) Remove the Air Conditioning Module. This is the task: Air Conditioning Module Removal, AMM TASK 21-51-65-000-801.

3) Do a wiring check as follows:

AIR CONDITIONING MODULE D646	**R PFTC D15816B**
pin 12	pin J2

4) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-install the Right PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

c) Re-install the Air Conditioning Module. This is the task: Air Conditioning Module Installation, AMM TASK 21-51-65-400-801.

d) Do the Repair Confirmation at the end of this task.

5) If you do not find a problem with the wiring, then do these steps:

a) Replace the Right PFTC, M2952. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

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b) Re-install the Air Conditioning Module. This is the task: Air Conditioning Module Installation, AMM TASK 21-51-65-400-801.

c) Do the Repair Confirmation at the end of this task.

(b) Replace the Air Conditioning Module, P5-10. These are the tasks:

- Air Conditioning Module Removal, AMM TASK 21-51-65-000-801
- Air Conditioning Module Installation, AMM TASK 21-51-65-400-801

1) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) For the PFTC, do the "BITE Procedure" and the "BITE Procedure - EXISTING FAULTS" sections of this task:

- Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the fault.

(b) If the maintenance message still shows, continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

894. PFTC Activate Standby Pack Discrete Output - Fault Isolation

A. Description

(1) This task is for this Latched Maintenance Message:

NOTE: After you correct a problem related to a Latched Maintenance Message, you can clear the Maintenance Message as follows:

- Cycle the power to the PFTC or
- Perform the BITE ground test or
- Erase the EXISTING FAULTS from the PFTC Front Panel Menus.

(a) 21-51124 PFTC CNTRL INOP

1) Activate Standby Pack Interface Fault

(2) This task is also for this Non-Latched Maintenance Message:

(a) 21-51125 PFTC CNTRL INOP

1) Activate Standby Pack Load Fault

(3) These Maintenance Messages can show on either the Left or the Right Pack Flow Temperature Controller (PFTC).

B. Possible Causes

(1) For maintenance messages that show on the Left PFTC, M2951:

(a) Wiring

(b) Left PFTC, M2951

(c) Right PFTC, M2952

(2) For maintenance messages that show on the Right PFTC, M2952:

(a) Wiring

(b) Right PFTC, M2952

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OMR 820-826 (Continued)

(c) Left PFTC, M2951

C. Circuit Breakers

(1) These are the primary circuit breakers related to this fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

D. Related Data

(1) WDM 21-51-13

(2) WDM 21-51-23

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message shows in the PFTC Existing Faults, or it is a latched message that shows in the most recent flight leg, then do the Fault Isolation Procedure below.

(b) If the maintenance message does not show in the PFTC Existing Faults, or as a latched message in the most recent flight leg, then no additional fault isolation steps are required.

F. Fault Isolation Procedure

(1) If the maintenance message shows on the Left PFTC, then do these steps:

(a) Do this check of the wiring between the Left PFTC, M2951, and the Right PFTC, M2952 (WDM 21-51-13):

1) Remove the Left PFTC and the Right PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

2) Do a wiring check as follows:

L PFTC D15814B	R PFTC D15816B
pin J3	pin J8

3) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-install the Left PFTC and the Right PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

c) Do the Repair Confirmation at the end of this task.

4) If you do not find a problem with the wiring, then do these steps:

a) Replace the Left PFTC, M2951. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

b) Re-install the Right PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

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OMR 820-826 (Continued)

c) Do the Repair Confirmation at the end of this task.

5) Replace the Right PFTC, M2952. These are the tasks:

- Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801
- Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801

a) Do the Repair Confirmation at the end of this task.

(2) If the maintenance message shows on the Right PFTC, then do these steps:

(a) Do this check of the wiring between the Right PFTC, M2952, and the Left PFTC, M2951 (WDM 21-51-23):

1) Remove the Right PFTC and the Left PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

2) Do a wiring check as follows:

R PFTC	L PFTC
D15816B	D15814B
pin J3	pin J8

3) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-install the Right PFTC and the Left PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

c) Do the Repair Confirmation at the end of this task.

4) If you do not find a problem with the wiring, then do these steps:

a) Replace the Right PFTC, M2952. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

b) Re-install the Left PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

c) Do the Repair Confirmation at the end of this task.

(b) Replace the Left PFTC, M2951. These are the tasks:

- Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801
- Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801

1) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) For the PFTC, do the "BITE Procedure" and the "BITE Procedure - EXISTING FAULTS" sections of this task:

- Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the fault.

(b) If the maintenance message still shows, continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 820-826 (Continued)

895. PFTC Controller INOP Discrete Output - Fault Isolation

A. Description

(1) This task is for this Latched Maintenance Message:

NOTE: After you correct a problem related to a Latched Maintenance Message, you can clear the Maintenance Message as follows:

- Cycle the power to the PFTC or
- Perform the BITE ground test or
- Erase the EXISTING FAULTS from the PFTC Front Panel Menus.

(a) 21-51126 PFTC CNTRL INOP

1) Controller INOP Interface Fault

(2) This task is also for this Non-Latched Maintenance Message:

(a) 21-51127 PFTC CNTRL INOP

1) Controller INOP Load Fault

(3) These Maintenance Messages can show on either the Left or the Right Pack Flow Temperature Controller (PFTC).

B. Possible Causes

(1) For maintenance messages that show on the Left PFTC, M2951:

(a) Wiring

(b) Left PFTC, M2951

(c) Right PFTC, M2952

(2) For maintenance messages that show on the Right PFTC, M2952:

(a) Wiring

(b) Right PFTC, M2952

(c) Left PFTC, M2951

C. Circuit Breakers

(1) These are the primary circuit breakers related to this fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

D. Related Data

(1) WDM 21-51-13

(2) WDM 21-51-23

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OMR 820-826 (Continued)

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message shows in the PFTC Existing Faults, or it is a latched message that shows in the most recent flight leg, then do the Fault Isolation Procedure below.

(b) If the maintenance message does not show in the PFTC Existing Faults, or as a latched message in the most recent flight leg, then no additional fault isolation steps are required.

F. Fault Isolation Procedure

(1) If the maintenance message shows on the Left PFTC, then do these steps:

(a) Do this check of the wiring between the Left PFTC, M2951, and the Right PFTC, M2952 (WDM 21-51-13):

1) Remove the Left PFTC and the Right PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

2) Do a wiring check as follows:

L PFTC		R PFTC
D15814B		D15816B
pin K3	. . . . .	pin K9

3) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-install the Left PFTC and the Right PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

c) Do the Repair Confirmation at the end of this task.

4) If you do not find a problem with the wiring, then do these steps:

a) Replace the Left PFTC, M2951. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

b) Re-install the Right PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

c) Do the Repair Confirmation at the end of this task.

5) Replace the Right PFTC, M2952. These are the tasks:

- Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801
- Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801

a) Do the Repair Confirmation at the end of this task.

(2) If the maintenance message shows on the Right PFTC, then do these steps:

(a) Do this check of the wiring between the Right PFTC, M2952, and the Left PFTC, M2951 (WDM 21-51-23):

1) Remove the Right PFTC and the Left PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

2) Do a wiring check as follows:

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OMR 820-826 (Continued)

R PFTC		L PFTC
D15816B		D15814B
pin K3	. . . . .	pin K9

3) If you find a problem with the wiring, then do these steps:
   a) Repair the wiring.
   b) Re-install the Right PFTC and the Left PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.
   c) Do the Repair Confirmation at the end of this task.
4) If you do not find a problem with the wiring, then do these steps:
   a) Replace the Right PFTC, M2952. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.
   b) Re-install the Left PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.
   c) Do the Repair Confirmation at the end of this task.

(b) Replace the Left PFTC, M2951. These are the tasks:
- Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801
- Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801

1) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) For the PFTC, do the "BITE Procedure" and the "BITE Procedure - EXISTING FAULTS" sections of this task:
- Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the fault.

(b) If the maintenance message still shows, continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

896. PFTC SRADA BITE Discrete Output - Fault Isolation

A. Description

(1) This task is for this Latched Maintenance Message:

NOTE: After you correct a problem related to a Latched Maintenance Message, you can clear the Maintenance Message as follows:
- Cycle the power to the PFTC or
- Perform the BITE ground test or
- Erase the EXISTING FAULTS from the PFTC Front Panel Menus.

(a) 21-51128 PFTC CNTRL INOP
   1) SRADA BITE Test Interface Fault

(2) This task is also for this Non-Latched Maintenance Message:

(a) 21-51129 PFTC CNTRL INOP

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OMR 820-826 (Continued)

1) SRADA BITE Load Fault

(3) These Maintenance Messages can show on either the Left or the Right Pack Flow Temperature Controller (PFTC).

B. Possible Causes

(1) For maintenance messages that show on the Left PFTC, M2951:

(a) Wiring

(b) Left PFTC, M2951

(c) Left Forward (Inlet) Ram Air Actuator, M2795

(d) Air Conditioning Accessory Unit (ACAU-1), M324

(2) For maintenance messages that show on the Right PFTC, M2952:

(a) Wiring

(b) Right PFTC, M2952

(c) Right Forward (Inlet) Ram Air Actuator, M2797

(d) ACAU-2, M1455

C. Circuit Breakers

(1) These are the primary circuit breakers related to this fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	8	C00555	AIR CONDITIONING RAM AIR MOD CONT L
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	8	C01178	AIR CONDITIONING RAM AIR MOD CONT R
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L

D. Related Data

(1) WDM 21-51-14

(2) WDM 21-51-24

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message shows in the PFTC Existing Faults, or it is a latched message that shows in the most recent flight leg, then do the Fault Isolation Procedure below.

(b) If the maintenance message does not show in the PFTC Existing Faults, or as a latched message in the most recent flight leg, then no additional fault isolation steps are required.

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OMR 820-826 (Continued)

F. Fault Isolation Procedure

(1) If the maintenance message shows on the Left PFTC, then do these steps:

(a) Do this check of the wiring between the Left PFTC, M2951, the ACAU-1, M324, and the Left Forward (Inlet) Ram Air Actuator, M2795 (WDM 21-51-14):

1) Remove the Left PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

2) Remove the ACAU-1. This is the task: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801.

3) Do a wiring check as follows:

L PFTC		**ACAU-1**
D15814B		**D458A**
pin G3	. . . . .	pin 47

4) Get access to the Left Forward (Inlet) Ram Air Actuator, M2795, as follows:

a) Open these access panels:

Number	**Name/Location**
191FL	Forward Wing To Body Fairing Panel - Mid Fairing, Above Ram Air Inlet
191GL	Ram Air Actuator Panel - Forward

5) Disconnect electrical connector D14386 from the Left Forward (Inlet) Ram Air Actuator.

6) Do a wiring check as follows:

		L RAM AIR INLET ACTUATOR
L PFTC		**D14386**
D15814B		
pin G3	. . . . .	pin 19

7) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-install the Left PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

c) Re-install the Air Conditioning Accessory Unit (ACAU-1). This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

d) Connect electrical connector D14386 to the Left Forward (Inlet) Ram Air Actuator.

e) Do the Repair Confirmation at the end of this task.

8) If you do not find a problem with the wiring, then do these steps:

a) Replace the Left PFTC, M2951. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

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OMR 820-826 (Continued)

b) Re-install the ACAU-1. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

c) Connect electrical connector D14386 to the Left Forward (Inlet) Ram Air Actuator.

d) Do the Repair Confirmation at the end of this task.

(b) Replace the Left Forward (Inlet) Ram Air Actuator, M2795. These are the tasks:

- Ram Air Inlet Actuator Removal, AMM TASK 21-51-21-000-801
- Ram Air Inlet Actuator Installation, AMM TASK 21-51-21-400-801

1) Do the Repair Confirmation at the end of this task.

(c) Replace the ACAU-1. These are the tasks:

- Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801
- Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801

1) Do the Repair Confirmation at the end of this task.

(2) If the maintenance message shows on the Right PFTC, then do these steps:

(a) Do this check of the wiring between the Right PFTC, M2952, the ACAU-2, M1455, and the Right Forward (Inlet) Ram Air Actuator, M2797 (WDM 21-51-24):

1) Remove the Right PFTC. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

2) Remove the ACAU-2. This is the task: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801.

3) Do a wiring check as follows:

R PFTC	**ACAU-2**
D15816B	**D10002A**
pin G3	pin 47

4) Get access to the Right Forward (Inlet) Ram Air Actuator, M2797, as follows:

a) Open these access panels:

Number	Name/Location
191FR	Forward Wing To Body Fairing Panel - Mid Fairing, Above Ram Air Inlet
191GR	Ram Air Actuator Panel - Forward

5) Disconnect electrical connector D14394 from the Right Forward (Inlet) Ram Air Actuator.

6) Do a wiring check as follows:

R PFTC	**R RAM AIR INLET ACTUATOR**
D15816B	**D14394**
pin G3	pin 19

7) If you find a problem with the wiring, then do these steps:

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OMR 820-826 (Continued)

a) Repair the wiring.

b) Re-install the Right PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

c) Re-install the Air Conditioning Accessory Unit (ACAU-2). This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

d) Connect electrical connector D14394 to the Right Forward (Inlet) Ram Air Actuator.

e) Do the Repair Confirmation at the end of this task.

8) If you do not find a problem with the wiring, then do these steps:

a) Replace the Right PFTC, M2952. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

b) Re-install the ACAU-2. This is the task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

c) Connect electrical connector D14394 to the Right Forward (Inlet) Ram Air Actuator.

d) Do the Repair Confirmation at the end of this task.

(b) Replace the Right Forward (Inlet) Ram Air Actuator, M2797. These are the tasks:

- Ram Air Inlet Actuator Removal, AMM TASK 21-51-21-000-801
- Ram Air Inlet Actuator Installation, AMM TASK 21-51-21-400-801

1) Do the Repair Confirmation at the end of this task.

(c) Replace the ACAU-2. These are the tasks:

- Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801
- Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801

1) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) For the PFTC, do the "BITE Procedure" and the "BITE Procedure - EXISTING FAULTS" sections of this task:

- Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the fault.

(b) If the maintenance message still shows, continue the Fault Isolation Procedure at the subsequent step.

OMR 802, 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

——— END OF TASK ———

901. Control Cabin Too Cold, No Temperature Control in AUTO - Fault Isolation

A. Description

(1) The Control Cabin Temperature is too cold with the CONT CAB Temperature Selector in the AUTO position.

(2) On the P5-17 Temperature Control Panel, you can put the AIR TEMP Selector to CONT CAB to observe the temperature in the Control Cabin.

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B. Possible Causes

(1) Control Cabin Duct Primary Temperature Sensor, T478

(2) Control Cabin Duct Backup Temperature Sensor, T477

(3) Control Cabin Temperature Sensor Module, M1718

(4) Pack Flow Control and Shutoff Valve, V18

(5) Trim Air PRSOV, V146

(6) Left Pack Temperature Sensor, T471

(7) Left Air Conditioning Accessory Unit (ACAU), M324

(8) Wiring

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	1	C01157	A/C TEMP CONT VALVE CLOSE LEFT
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	3	C01163	A/C ZONE TEMP VALVE/FAN CONT FLT DECK
C	4	C01164	A/C ZONE TEMP DUCT OVHT FLT DECK
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L

D. Related Data

(1) WDM 21-51-11

(2) WDM 21-61-11

(3) WDM 21-61-15

(4) WDM 21-61-12

(5) SSM 21-51-11

(6) SSM 21-61-11

(7) SSM 21-61-12

(8) SSM 21-61-15

E. Initial Evaluation

(1) Supply conditioned air to the Passenger Compartment with the two packs. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) On the P5-17 Temperature Control Panel, put the CONT CAB, FWD CAB, and AFT CAB Temperature Selectors to the AUTO (12 o'clock) position.

(3) On the P5-17 Panel, put the TRIM AIR Switch to ON.

(4) Operate the pack until the Pack Output Temperatures are stable.

(a) On the P5-17 Panel, put the AIR TEMP Selector Switch to CONT CAB, then PASS CAB FWD and PASS CAB AFT to show the temperatures.

1) The Control Cabin Temperature and the Forward Passenger Cabin Temperatures should be about the same (approximately 75°F (24°C)).

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(5) On the P5-17 Panel, put the CONT CABIN, FWD CAB, AFT CAB Temperature Selectors to the W (fully warm) position.

(a) Make sure that the temperatures increase and are approximately the same (85°F (29°C)).

(6) On the P5-17 Panel, put the CONT CABIN, FWD CAB, AFT CAB Temperature Selectors to the C (fully cold) position.

(a) Let the packs operate until the Cabin Temperatures are stable.

(b) Make sure that the Temperatures are about the same (65°F (18°C)).

(7) If the Control Cabin Temperature became warmer when the temperature selectors was in the W (fully warm) position, and colder when the Temperature Selector was in the C (fully cold) position, and it was about the same as the Forward Cabin Temperature in each of the two positions (W and C), then there was an intermittent fault.

(8) If the Control Cabin Temperature did not become warmer when the Temperature Selector was in the W (fully warm) position, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Do the Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If maintenance messages show, then refer to the table at the end of the BITE Task (or 21-MAINT MSG INDEX) to find the applicable FIM Tasks and do the corrective actions for the messages that show.

(2) Examine the Sidewall Panels in the Control Cabin area on the two sides of the aircraft.

- Make sure that the Sidewall Panels are not expanded inboard, in the area adjacent to the floor level, and that the panels have the same contour as the adjacent panels.
- Make sure that there are no noises that could be a duct leak.

(a) If one of the two conditions above is present, then examine the Sidewall Riser Ducts. This is the task: Sidewall Riser Ducts/Transition Duct Repair, AMM TASK 21-23-02-360-801 or Sidewall Riser Ducts/Transition Duct Repair, AMM TASK 21-23-02-360-802.:

1) Do the Repair Confirmation at the end of this task.

(b) If the Sidewall Panels are satisfactory, then continue.

(3) Do this check of the Trim Air PRSOV, V146.

(a) On the P5-17 Temperature Control Panel, make sure that the TRIM AIR Switch is ON.

(b) Disconnect the Trim Air PRSOV connector D3914 (WDM 21-61-15).

(c) Do a check for 28V DC at pin 1 of connector D3914.

1) If you find 28V DC, replace the Trim Air PRSOV, V146. These are the tasks:

- Trim Air Pressure Regulating and Shutoff Valve Removal, AMM TASK 21-62-01-000-801
- Trim Air Pressure Regulating and Shutoff Valve Installation, AMM TASK 21-62-01-400-801

a) Do the Repair Confirmation at the end of this task.

2) If you do not find 28V DC, at pin 1 of D3914, examine the wiring between the Trim Air PRSOV connector and the Load Control Center Right (LCC-R) as follows:

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LCC-R D40536P	TRIM AIR PRSOV D3914
pin 24	pin 1

TRIM AIR PRSOV D3914	
pin 3	GD6524-DC
pin 2	GD2358-ST

a) If you find problems with the wiring, repair as necessary.

<1> Do the Repair Confirmation at the end of this task.

b) If the wiring, is OK, continue.

(4) Do a check of the Control Cabin Duct Primary and Backup Temperature Sensors, T478 and T477.

(a) Do a Resistance Check for the two sensors. This is the task: Duct Temperature Limit Sensor Test, AMM TASK 21-61-04-000-805-002.

1) If the Resistances are not as specified, replace the applicable Control Cabin Duct Primary (Backup) Temperature Sensor, T478 (T477). These are the tasks:

- Duct Temperature Limit Sensor Removal, AMM TASK 21-61-04-000-803-002
- Duct Temperature Limit Sensor Installation, AMM TASK 21-61-04-400-803-002

a) Do the Repair Confirmation at the end of this task.

(5) Make sure that the Filter Screen of the Control Cabin Temperature Sensor Module, M1718 is free from unwanted material.

(a) If necessary, replace the filter. These are the tasks:

- Cabin Temperature Sensor Assembly Filter Removal, AMM TASK 21-61-06-000-801
- Cabin Temperature Sensor Assembly Filter Installation, AMM TASK 21-61-06-400-801

1) Do the Repair Confirmation at the end of this task.

(6) Do a check of the Air Conditioning Accessory Unit (ACAU) and the Pack Flow Control and Shutoff Valve, V18 as follows:

(a) At Terminal Block TB4102, disconnect the Diode R508 at the B3-side terminal.

(b) Do a wire check as follows (WDM 21-61-11):

TB4102 term	
term B3	GD518-DC

1) If you do not find continuity to Ground at term B3, repair the wiring as necessary.

a) Re-connect Diode R508.

b) Do the Repair Confirmation at the end of this task.

2) If you find continuity to Ground at term B3, continue.

(c) Disconnect connector D486 from the Pack Flow Control and Shutoff Valve, V18.

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(d) Do a check for 28V DC between pins 3 and 1 of connector D486.

1) If you find 28V DC, replace the Left Pack Flow Control and Shutoff Valve, V18. These are the tasks:

- Left Flow Control and Shutoff Valve - Removal, AMM TASK 21-51-01-000-802-002 or Left Flow Control and Shutoff Valve Removal, AMM TASK 21-51-01-000-806-003
- Right Flow Control and Shutoff Valve Installation, AMM TASK 21-51-01-400-806-003 or Right Flow Control and Shutoff Valve - Installation, AMM TASK 21-51-01-400-802-002

a) Do the Repair Confirmation at the end of this task.

2) If you do not find 28V DC, between pins 3 and 1 of D486, continue.

(e) Remove the ACAU. This is the task: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801.

(f) Disconnect these connectors:

- D486 from the Pack Flow Control and Shutoff Valve, V18
- D3886 from the Control Cabin Trim Modulating Valve, V143

1) Examine the wiring between the ACAU, M324 and the Pack Flow Control and Shutoff Valve, V18 as follows (WDM 21-51-11):

ACAU D458A	**L PACK FLOW CTRL AND S/O VLV D486**
pin 33	pin 3

L PACK FLOW CTRL AND S/O VLV D486	
pin 1	GND

2) Examine the wiring between the ACAU, M324 and the Control Cabin Trim Modulating Valve, V143 as follows (WDM 21-61-11):

ACAU D458B	**CONT CAB TRIM MOD VLV D3886**
pin 41	pin 1
pin 42	pin 2

CONT CAB TRIM AIR MOD VLV D3886	
pin 3	GND
pin 13	GND

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(g) If you find a problem in the wiring, repair it as necessary.

1) Do the Repair Confirmation at the end of this task.

(h) If the wiring, is OK, continue.

(7) Install a new Left Air Conditioning Accessory Unit (ACAU). This is the task:

- Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801
- Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801

(a) Do the Repair Confirmation at the end of this task.

(8) Replace the Temperature Control Module. These are the tasks:

- Temperature Control Module - Removal, AMM TASK 21-61-15-000-801
- Temperature Control Module - Installation, AMM TASK 21-61-15-400-801

(a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Supply conditioned air to the airplane. This is the task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(2) Set the L PACK and R PACK Switches to AUTO.

(3) Set the TRIM AIR Switch to ON.

(4) On the P5-17 Temperature Control Panel, set the CONT CAB, FWD CAB and AFT CAB Temperature Selectors to AUTO.

(5) On the P5-17 Temperature Control Panel, set the AIR TEMP Selector to CONT CAB and record the temperature from the Temperature Indicator.

(6) On the P5-17 Temperature Control Panel, set the AIR TEMP Selector to SUPPLY DUCT FWD and record the temperature from the Temperature Indicator.

(7) Make sure that the Supply Duct and Control Cabin Temperatures are within a few degrees of each other.

(a) If the Supply Duct and Control Cabin Temperatures are within a few degrees of each other, then you corrected the problem.

(b) If the Supply Duct and Control Cabin Temperatures are not within a few degrees of each other, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

OMR 820-826

902. Flow Control Valve Overpressure - Fault Isolation

A. Description

(1) This task is for this Latched Maintenance Message:

NOTE: After you correct a problem related to a Latched Maintenance Message, you can clear the Maintenance Message as follows:

- Cycle the power to the PFTC or
- Perform the BITE ground test or
- Erase the EXISTING FAULTS from the PFTC Front Panel Menus.

(a) 21-51255 FLOW CTRL VALVE

1) Flow Control Valve Overpressure

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OMR 820-826 (Continued)

B. Possible Causes

(1) Pack Pressure Sensor Sense Line

(2) Left Pack Inlet Pressure Sensor, M2955

(3) Right Pack Inlet Pressure Sensor, M2956

(4) Wiring

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

D. Related Data

(1) WDM 21-51-12

(2) WDM 21-51-22

(3) SSM 21-51-12

(4) SSM 21-51-22

E. Initial Evaluation

(1) Do this task: Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message shows in the PFTC Existing Faults, or it is a latched message that shows in the most recent flight leg, then do the Fault Isolation Procedure below.

(b) If the maintenance message does not show in the PFTC Existing Faults, or as a latched message in the most recent flight leg, then no additional fault isolation steps are required.

F. Fault Isolation Procedure - Left Flow Control and Shutoff Valve, V184

(1) Do the Pack Pressure Sensor Sense Line cleaning.

(a) Do this task: Pack Inlet Pressure Sensor Sense Line Cleaning, AMM TASK 21-51-45-100-801.

1) Do the Repair Confirmation at the end of this task.

(2) Replace the applicable Left (Right) Pack Inlet Pressure Sensor, M2955 (M2956). These are the tasks:

- Pack Inlet Pressure Sensor Removal, AMM TASK 21-51-45-000-801
- Pack Inlet Pressure Sensor Installation, AMM TASK 21-51-45-400-801.

(a) Do the Repair Confirmation at the end of this task.

(3) Do this wiring check for the Left Pack Inlet Pressure Sensor, M2955 (WDM 21-51-12):

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OMR 820-826 (Continued)

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

(b) Remove the Left Pack Flow Temperature Controller. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

(c) Disconnect electrical connector D15826 from the Left Pack Inlet Pressure Sensor.

(d) Do a wiring check as follows:

L PFTC D15814B		L PACK PRESSURE SENSOR D15826
pin A8	. . . . .	pin 1
pin A9	. . . . .	pin 3
ground	. . . . .	pin 5

1) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Reconnect electrical connector D15826 to the Left Pack Inlet Pressure Sensor.

c) Re-install the L PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.

d) Do the Repair Confirmation at the end of this task.

(4) Do this wiring check for the Right Pack Inlet Pressure Sensor, M2956 (WDM 21-51-22):

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

(b) Remove the Right Pack Flow Temperature Controller. This is the task: Pack Flow Temperature Controller - Removal, AMM TASK 21-61-17-020-801.

(c) Disconnect electrical connector D15828 from the Right Pack Inlet Pressure Sensor.

(d) Do a wiring check as follows:

R PFTC D15816B		R PACK PRESSURE SENSOR D15828
pin A8	. . . . .	pin 1
pin A9	. . . . .	pin 3
ground	. . . . .	pin 5

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OMR 820-826 (Continued)

1) If you find a problem with the wiring, then do these steps:
   a) Repair the wiring.
   b) Reconnect electrical connector D15828 to the Right Pack Inlet Pressure Sensor.
   c) Re-install the Right PFTC. This is the task: Pack Flow Temperature Controller - Installation, AMM TASK 21-61-17-400-801.
   d) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) For the PFTC, do the "BITE Procedure" and the "BITE Procedure - EXISTING FAULTS" sections of this task:

- Pack Temperature Control - BITE Procedure, 21-62 TASK 801.

(a) If the maintenance message does not show, then you corrected the fault.

(b) If the maintenance message still shows, continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

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BIT Check and Erase Faults

1. Push BIT - shows faults found during the last flight.
2. Push PREV FLT - shows faults that occurred during the 9 flights before the last flight.
3. Push VERIFY - this is a check that shows all faults that are current at this time.
4. Push RESET - removes all faults kept in memory. Push RESET only after the VERIFY button is pushed and the result is a green GO indication. Do not push RESET if faults show.
5. PRESS/TEST - does a check of all fault lights when pushed. Replace the controller if a fault light does not come on.

Fault Isolation

1. Always examine controller BIT to do fault isolation of a PACK or ZONE INOP indication in the flight deck.
2. Examine the overheat switches if you have a PACK or ZONE INOP indication and no controller BIT faults.
3. Remove controllers if a PACK/ZONE, RAM AIR, or STANDBY PACK controller fault shows. Remove controllers if a light does not come on when the PRESS/TEST button is pushed.
4. Do not remove a controller for a valve, sensor or selector fault.
5. The source of a LRU fault can be aircraft wires, the ACAU, or the LRU.
6. Always use VERIFY to do a check for existing faults.
7. TAV, TCV and ram air actuator faults in BIT storage will stay until the correct reset procedure is done. See step 4 of BIT Check and Erase Faults for correct reset procedure. All other faults will move from BIT to PREV FLT after the takeoff.
8. Erase all faults in memory before you change controllers. Push the BIT, PREV FLT, and VERIFY buttons. You erased the memory if a green GO lamp comes on after you push each of the three buttons.

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PACK/ZONE CABIN TEMPERATURE CONTROLLER, M1442, M1443 (LOCATED IN EE BAY - E3 RACK) 1

1 HONEYWELL P/N 622814-5
BOEING P/N 10-62091-17

2158157 S0000471722_V1

Pack/Zone Temperature Controller Troubleshooting Tips
Figure 301/21-62-00-990-801 (Sheet 1 of 2)

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PACK/ZONE CABIN TEMPERATURE CONTROLLER, M1442, M1443 (LOCATED IN EE BAY - E3 RACK) 1

BIT Check and Erase Faults

1. Push BIT - shows faults found during the last flight.
2. Push PREV FLT - shows faults that occurred during the 9 flights before the last flight.
3. Press VERIFY - this is a REAL TIME check (IBIT initiation) that displays all active faults.

3a. Replace faulted LRUs. If SRADA or Ram Temperature Sensor faults, cycle power to SRADA.

3b. Press VERIFY again to run IBIT.

4. Push RESET - removes all faults kept in memory. Push RESET only after the VERIFY button is pushed and the result is a green GO indication. Do not push RESET if faults show.
5. PRESS/TEST - does a check of all fault lights when pushed. Replace the controller if a fault light does not come on.

Fault Isolation

1. Always examine controller BIT to do fault isolation of a PACK or ZONE INOP indication in the flight deck.
2. Examine the overheat switches if you have a PACK or ZONE INOP indication and no controller BIT faults.
3. Remove controllers if a PACK/ZONE or STANDBY PACK controller fault shows. Remove controllers if a light does not come on when the PRESS/TEST button is pushed.
4. Do not remove a controller for a valve, sensor or selector fault.
5. The source of a LRU fault can be aircraft wires, the ACAU, or the LRU.
6. Always use VERIFY to do a check for existing faults.
7. TAV, TCV and SRADA Inlet or Exit faults in BIT storage will stay until the correct reset procedure is done. See step 4 of BIT Check and Erase Faults for correct reset procedure. All other faults will move from BIT to PREV FLT after the takeoff.
8. Erase all faults in memory before you change controllers. Push the BIT, PREV FLT, and VERIFY buttons. You erased the memory if a green GO lamp comes on after you push each of the three buttons.
9. If SRADA fault, compare exit SRADA light to PZTC light. If lights match, replace LRU. If not, cycle power and check wiring.

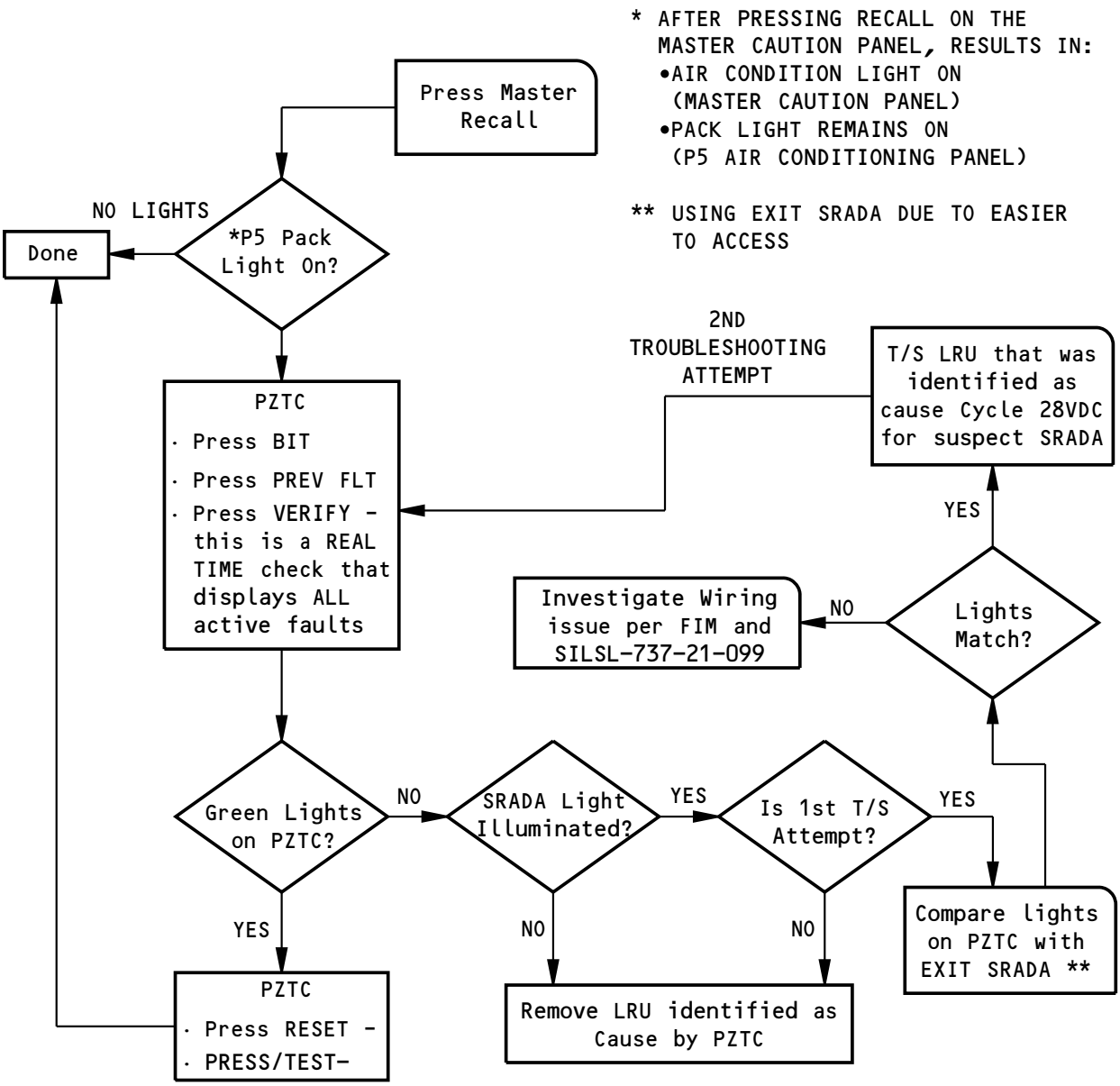
Honeywell Contact:
Gary Boudreau
Staff Engineer
Phone: 905-608-6105
Email: gary.boudreau@honeywell.com

1 HONEYWELL P/N 51090160-1
BOEING P/N 10-62091-19

2391562 S0000550410_V2

Pack/Zone Temperature Controller Troubleshooting Tips
Figure 301/21-62-00-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



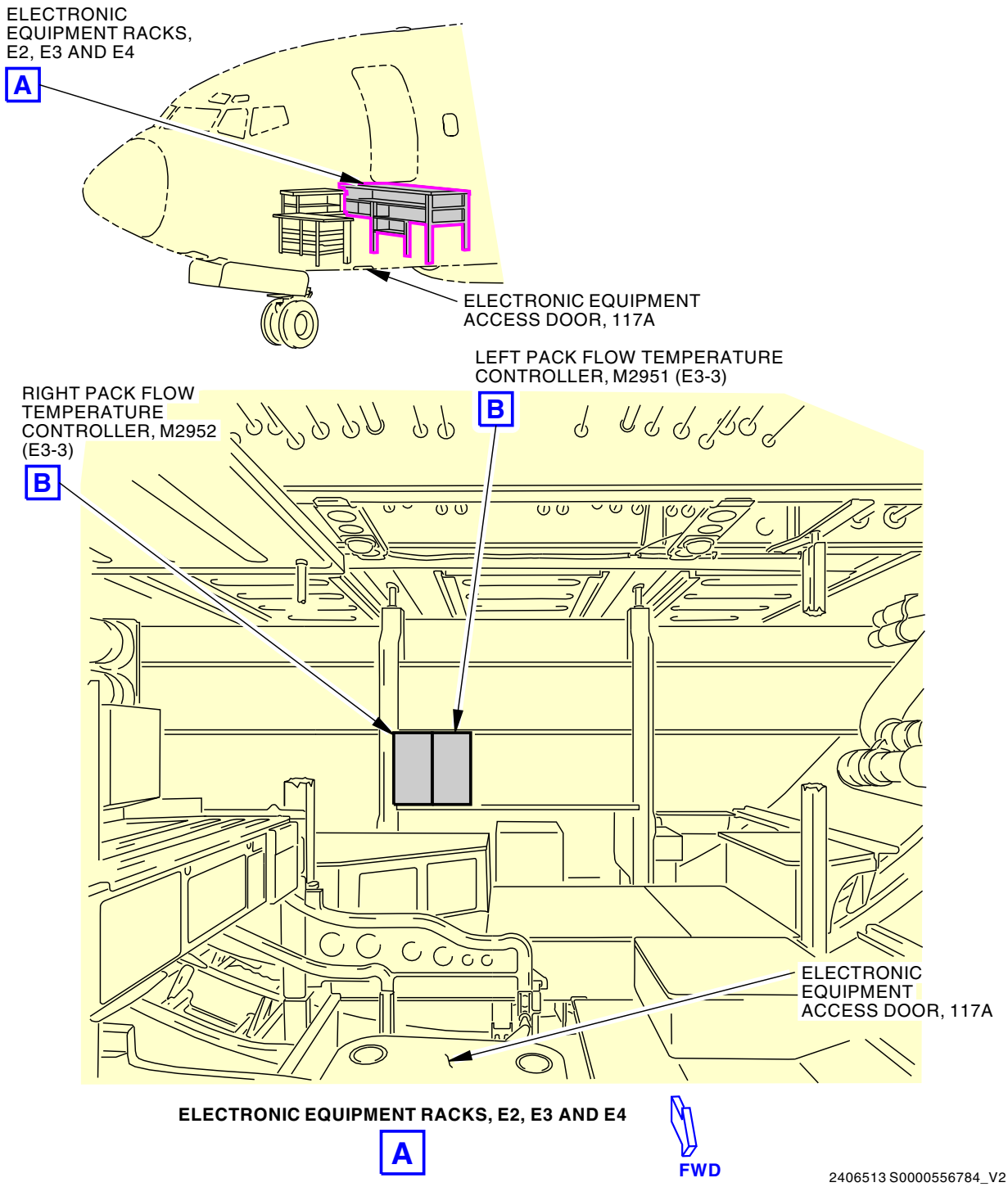
NOTE: FOR DOOR OPEN LIGHT DURING CRUISE, FOLLOW NORMAL FIM TROUBLESHOOTING PROCEDURE.

RECOMMENDED TROUBLE SHOOTING PROCEDURE FOR 737NG SRADA AND PACK ZONE TEMPERATURE CONTROLLER

2478252 S0000581413_V1

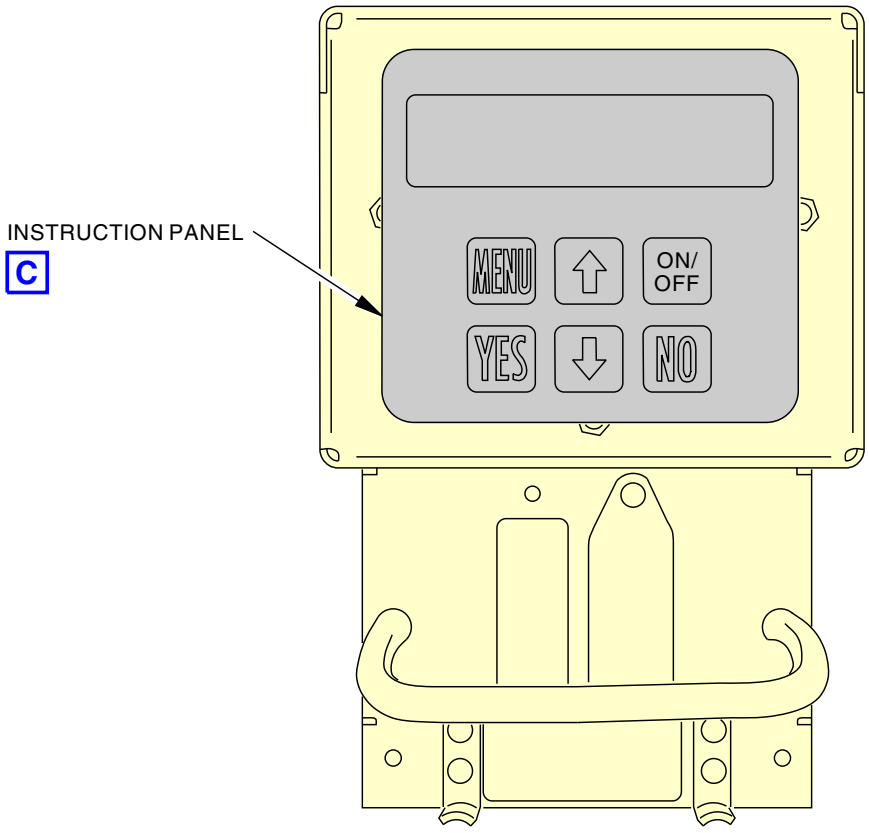
Checklist – Pack Zone Temperature Controller and SRADA
Figure 302/21-62-00-990-804

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Pack Flow Temperature Controller BITE Test
Figure 303/21-62-00-990-802 (Sheet 1 of 3)

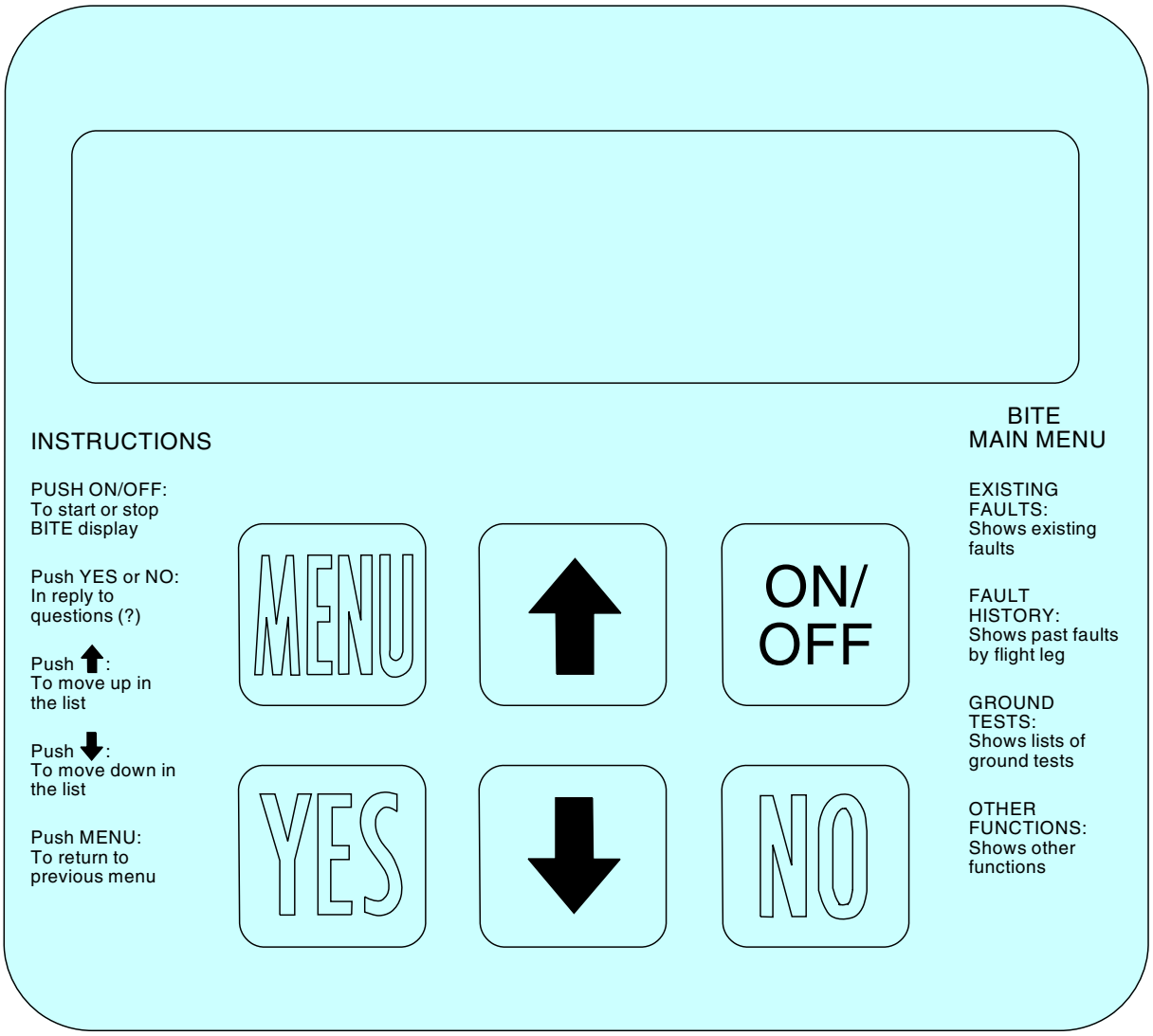
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2406564 S0000556786_V1

Pack Flow Temperature Controller BITE Test
Figure 303/21-62-00-990-802 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



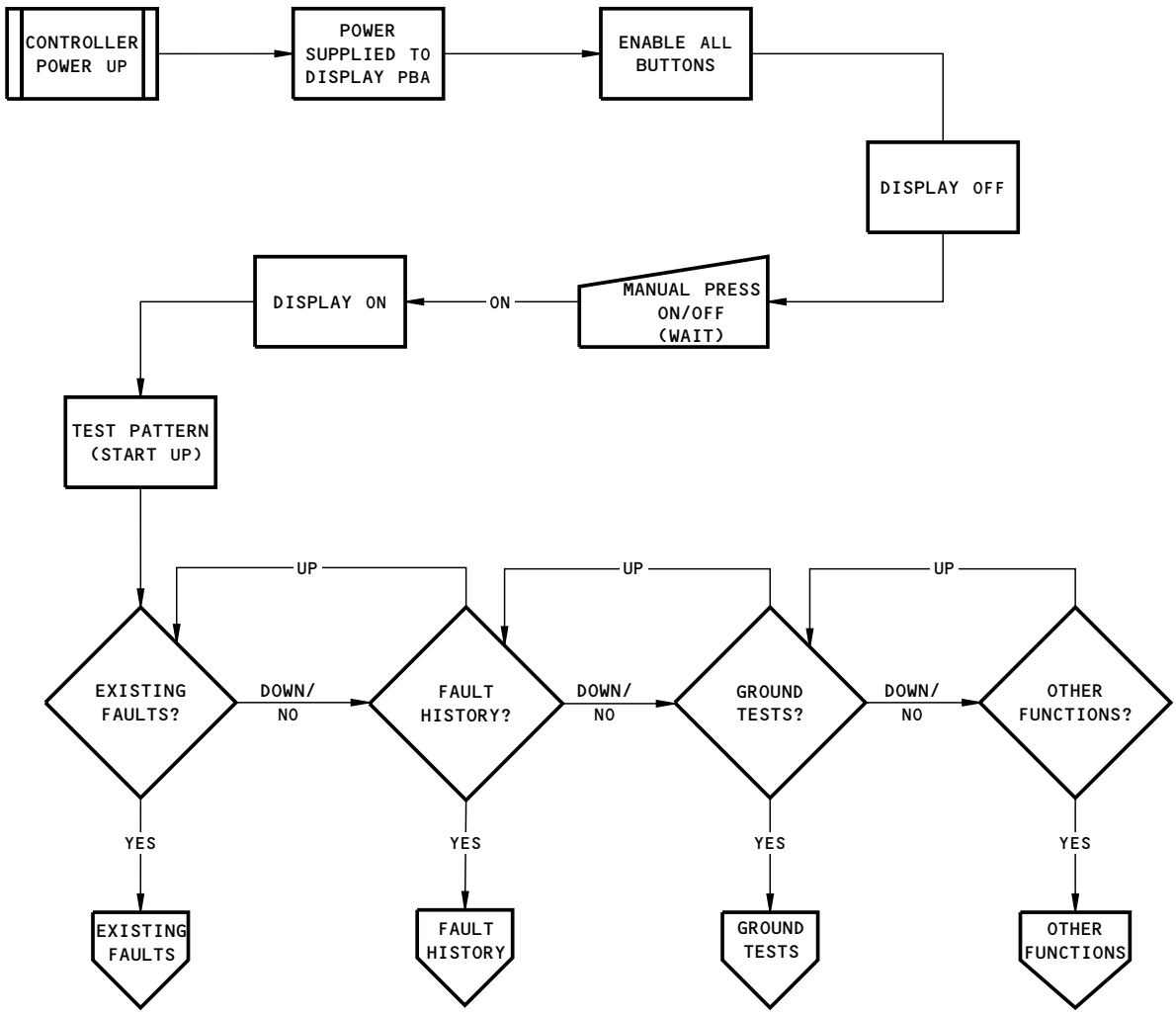
INSTRUCTION PANEL

C

2406579 S0000557050_V1

Pack Flow Temperature Controller BITE Test
Figure 303/21-62-00-990-802 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

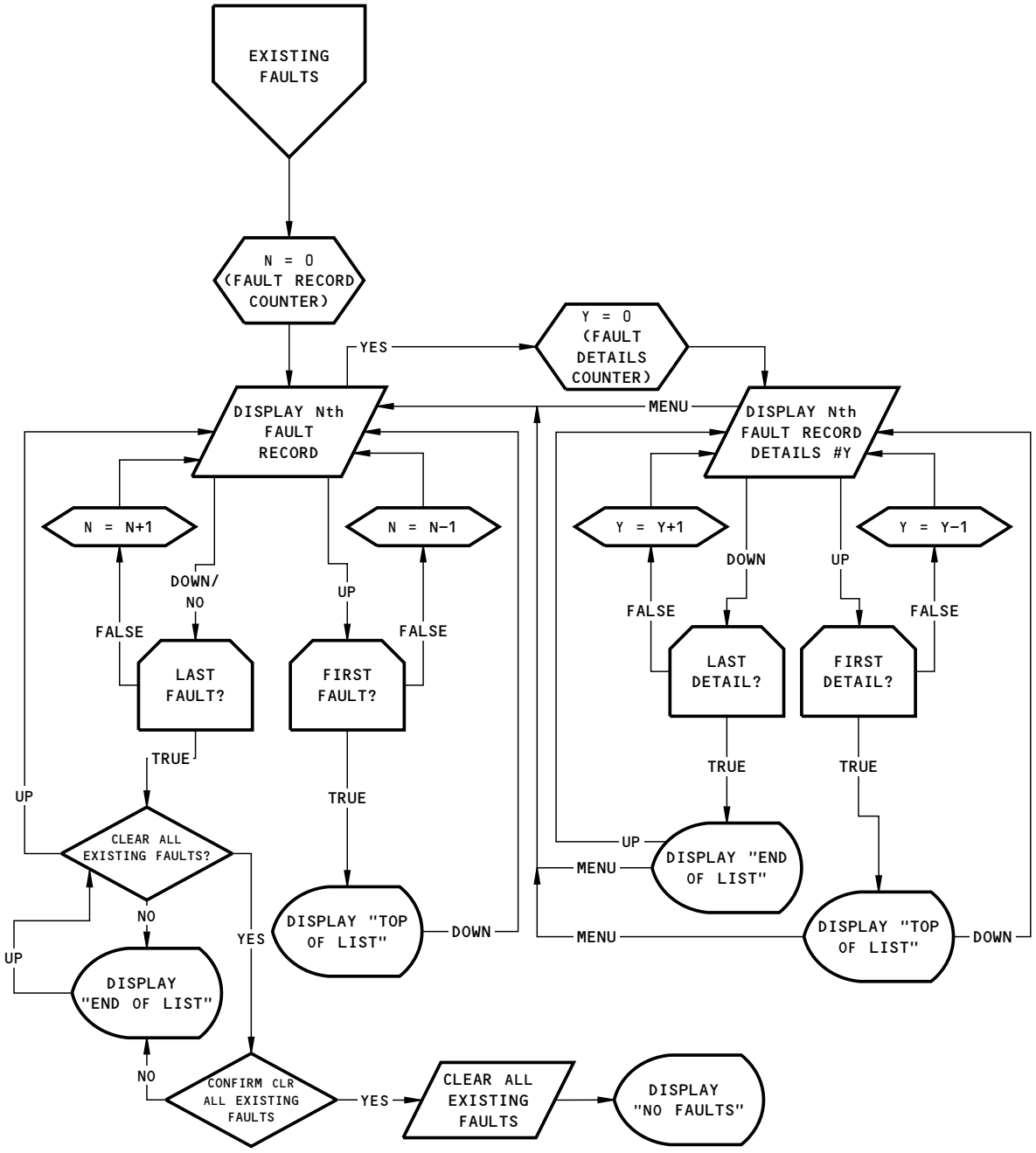


PACK FLOW TEMPERATURE CONTROLLER BITE PROCESS DIAGRAM

2407862 S0000557204_V1

Pack Flow Temperature Controller BITE Procedure
Figure 304/21-62-00-990-803 (Sheet 1 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



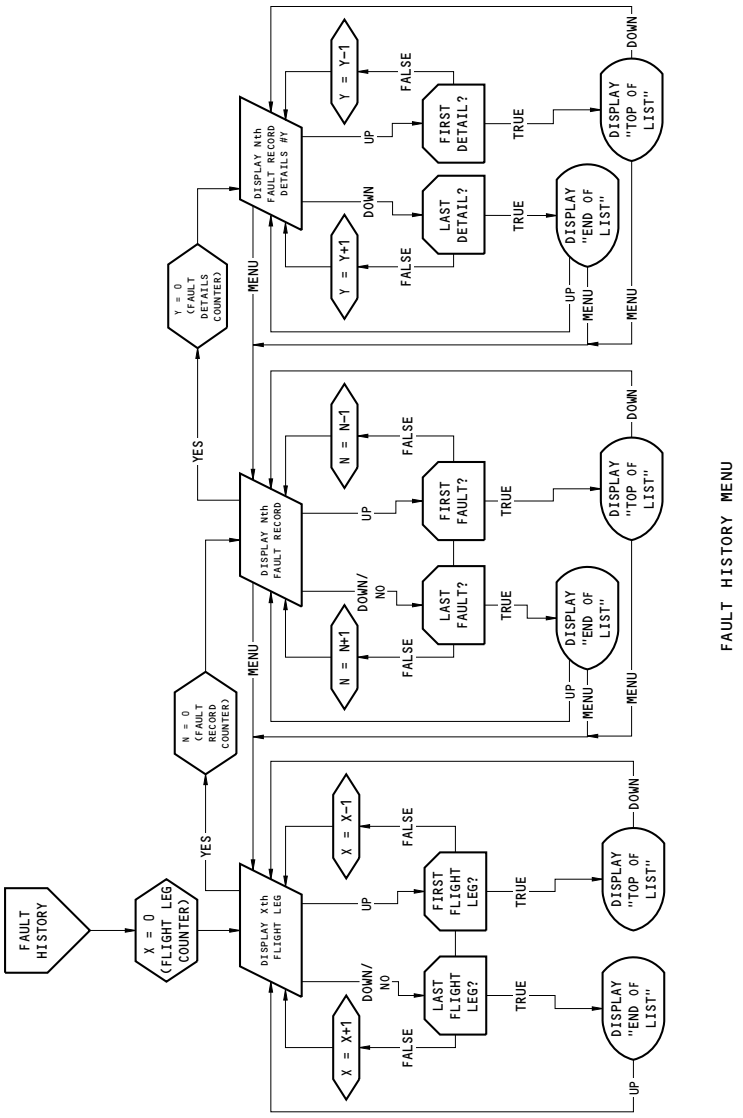
EXISTING FAULTS MENU

2407925 S0000557207_V1

Pack Flow Temperature Controller BITE Procedure
Figure 304/21-62-00-990-803 (Sheet 2 of 8)

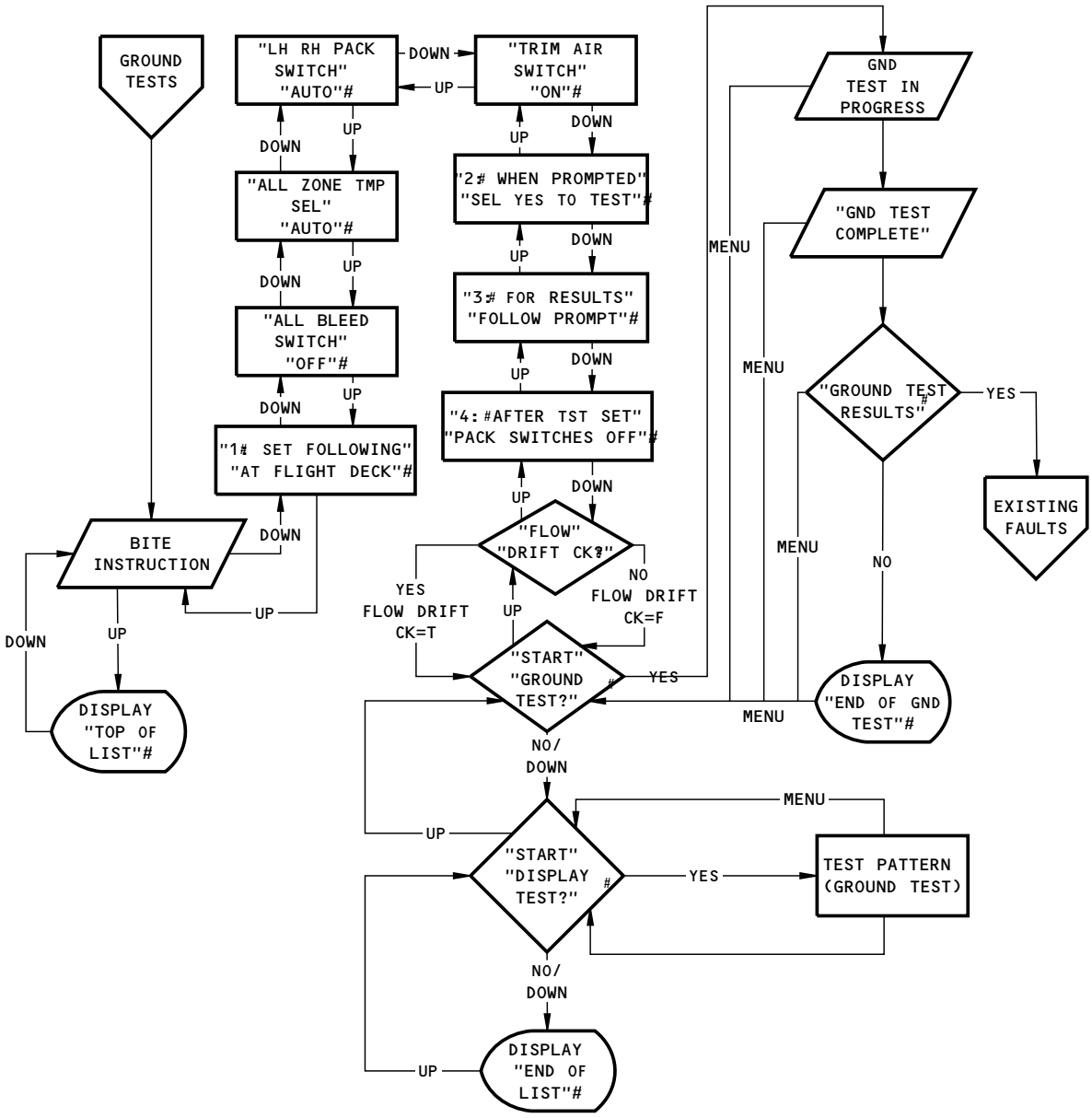
ECCN 9E991 BOEING PROPRIETARY - See title page for details

2407994 S0000557209_V1



Pack Flow Temperature Controller BITE Procedure
Figure 304/21-62-00-990-803 (Sheet 3 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

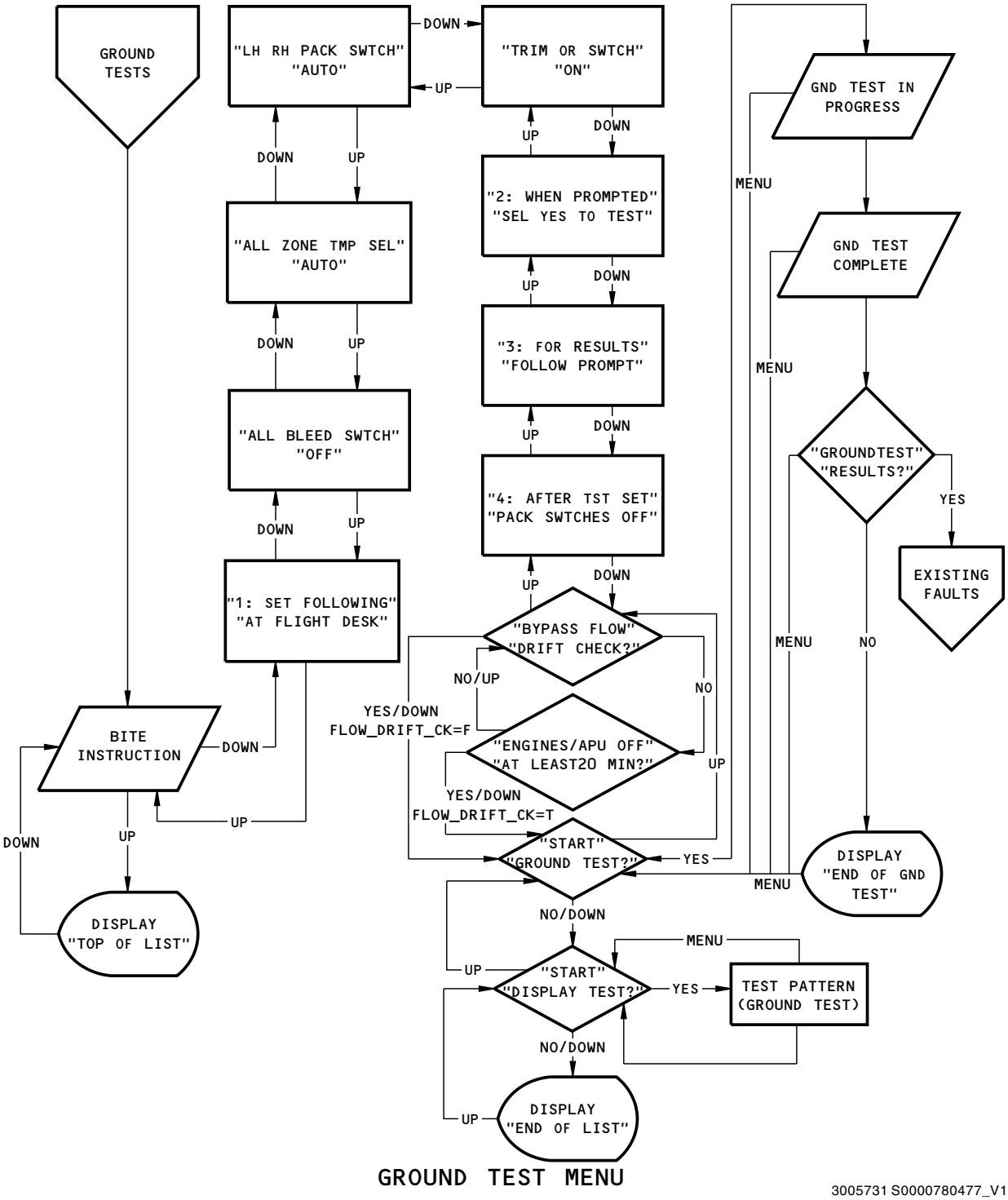


GROUND TEST MENU

2408194 S0000557211_V1

Pack Flow Temperature Controller BITE Procedure
Figure 304/21-62-00-990-803 (Sheet 4 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

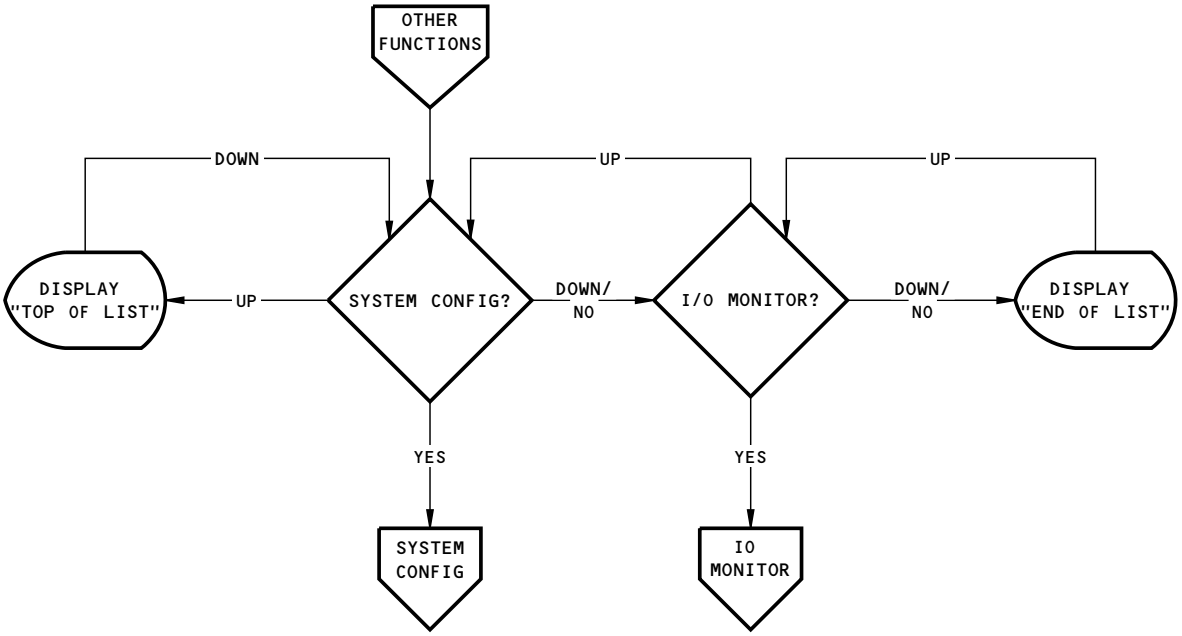


GROUND TEST MENU

3005731 S0000780477_V1

Pack Flow Temperature Controller BITE Procedure
Figure 304/21-62-00-990-803 (Sheet 5 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

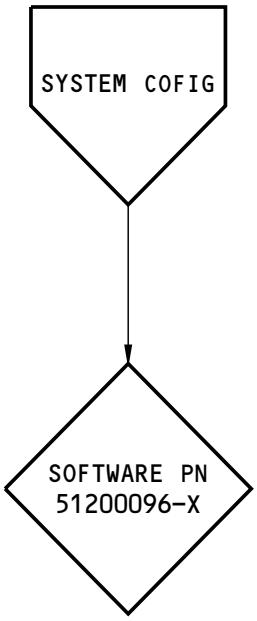


OTHER FUNCTIONS MENU

2408267 S0000557213_V1

Pack Flow Temperature Controller BITE Procedure
Figure 304/21-62-00-990-803 (Sheet 6 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

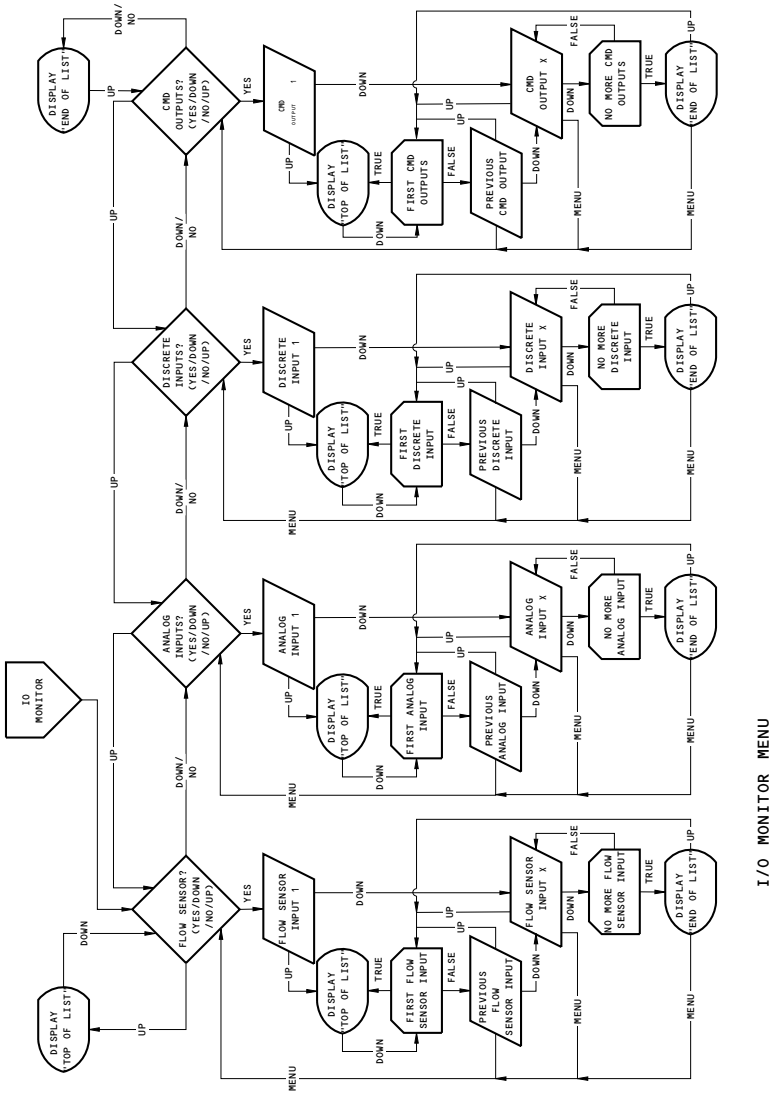


SYSTEM CONFIGURATION

2408286 S0000557215_V1

Pack Flow Temperature Controller BITE Procedure
Figure 304/21-62-00-990-803 (Sheet 7 of 8)

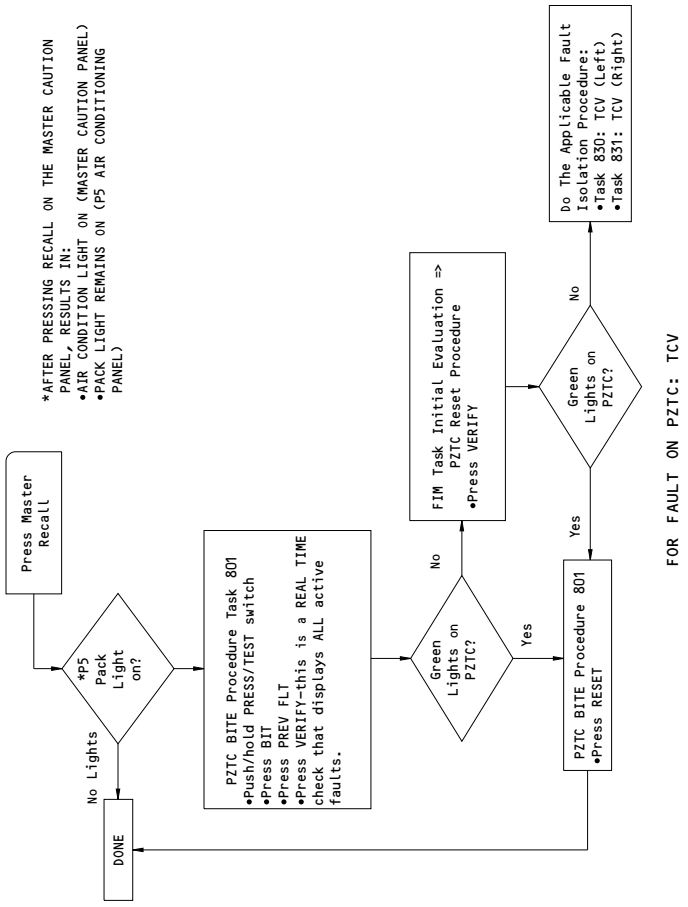
ECCN 9E991 BOEING PROPRIETARY - See title page for details



Pack Flow Temperature Controller BITE Procedure
Figure 304/21-62-00-990-803 (Sheet 8 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

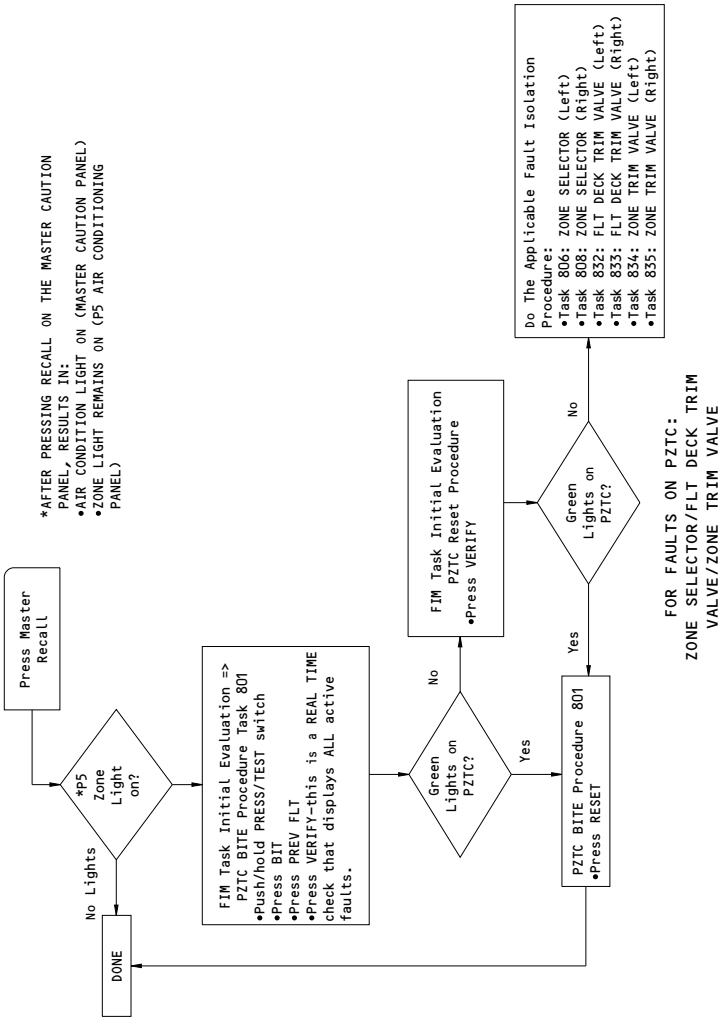
2571120 S0000616900_V2



Fault Lights on PZTC
Figure 305/21-62-00-990-805 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

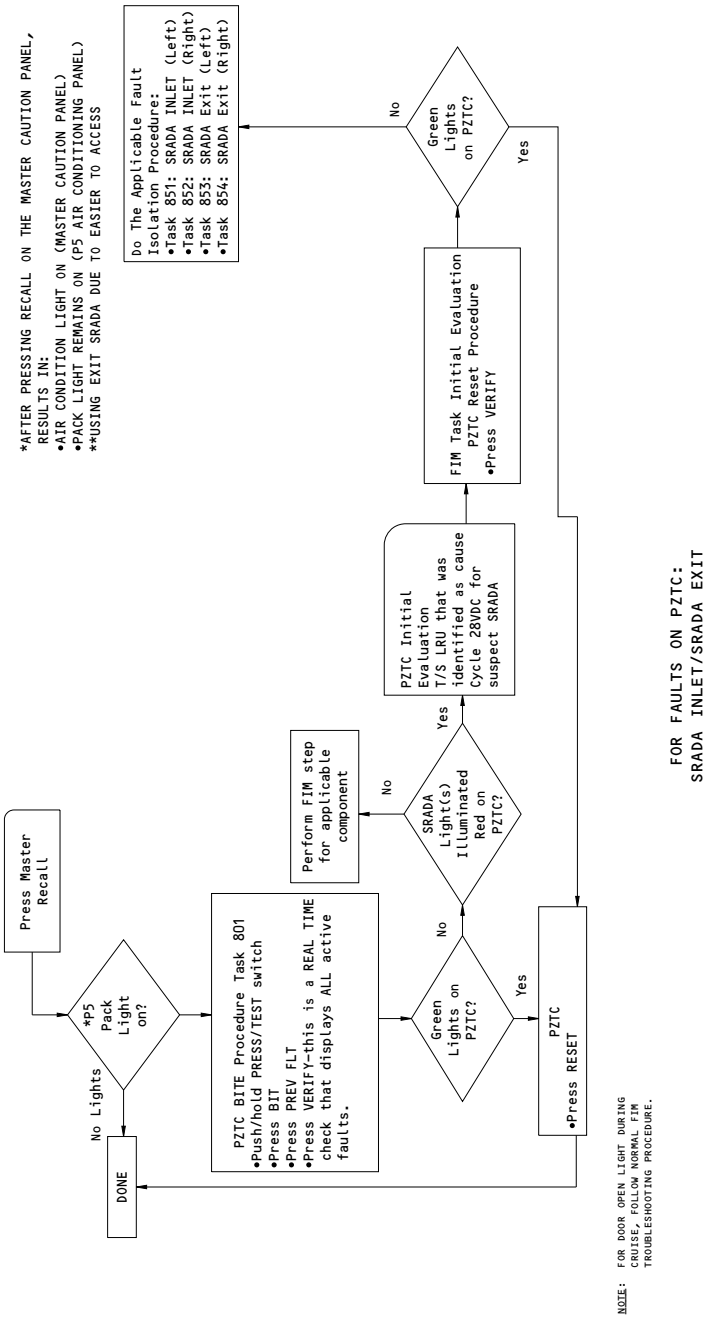
2571122 S0000617012_V1



Fault Lights on PZTC
Figure 305/21-62-00-990-805 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

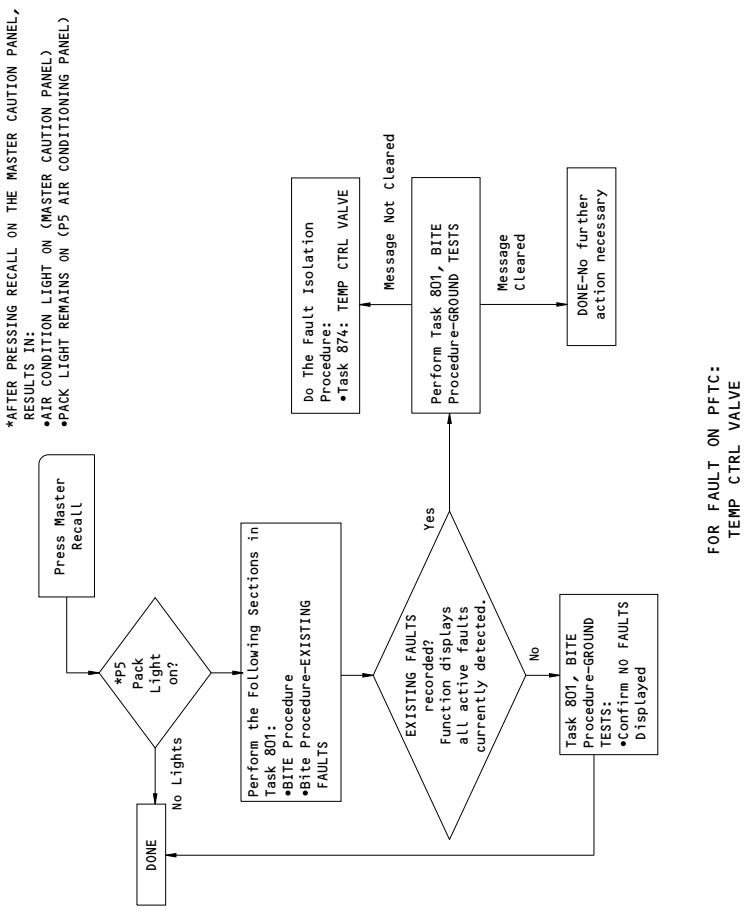
2571126 S0000616903_V1



SRADA and Pack Zone Temperature Controller
Figure 306/21-62-00-990-806

ECCN 9E991 BOEING PROPRIETARY - See title page for details

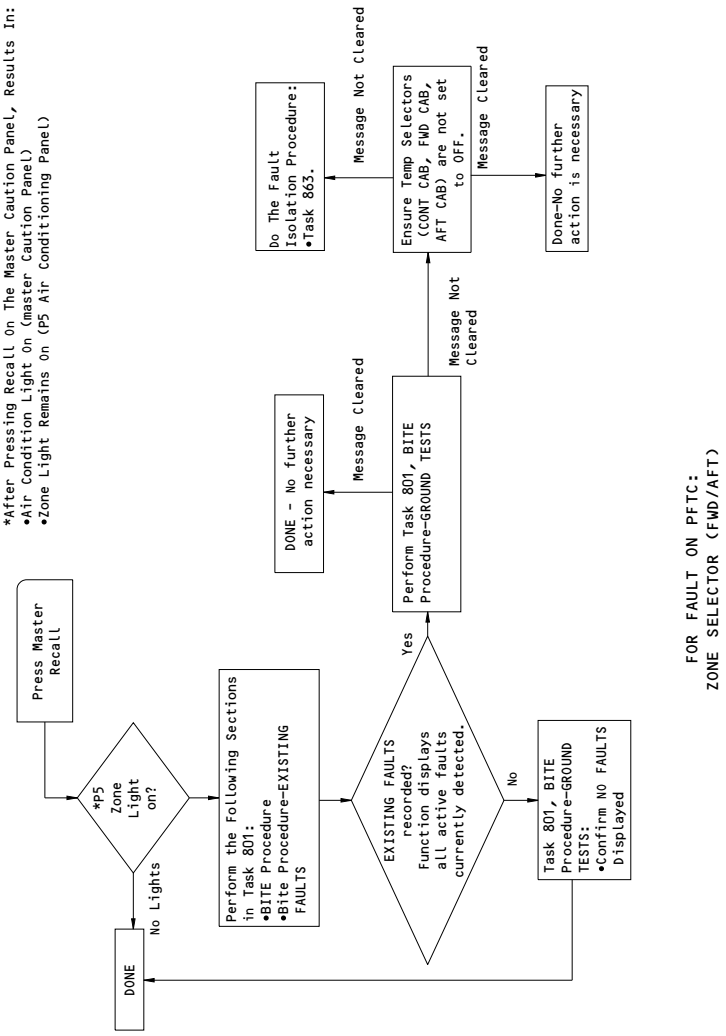
2571138 S0000616906_V1



Maintenance Messages on PFTC
Figure 307/21-62-00-990-807 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

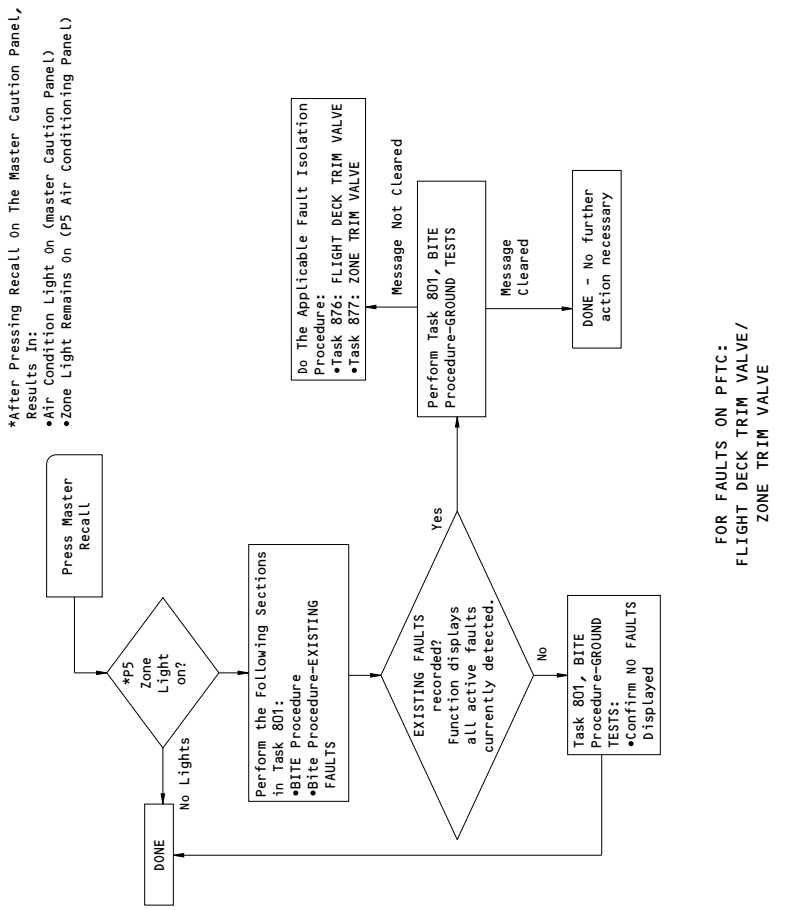
2571140 S0000616921_V2



Maintenance Messages on PFTC
Figure 307/21-62-00-990-807 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

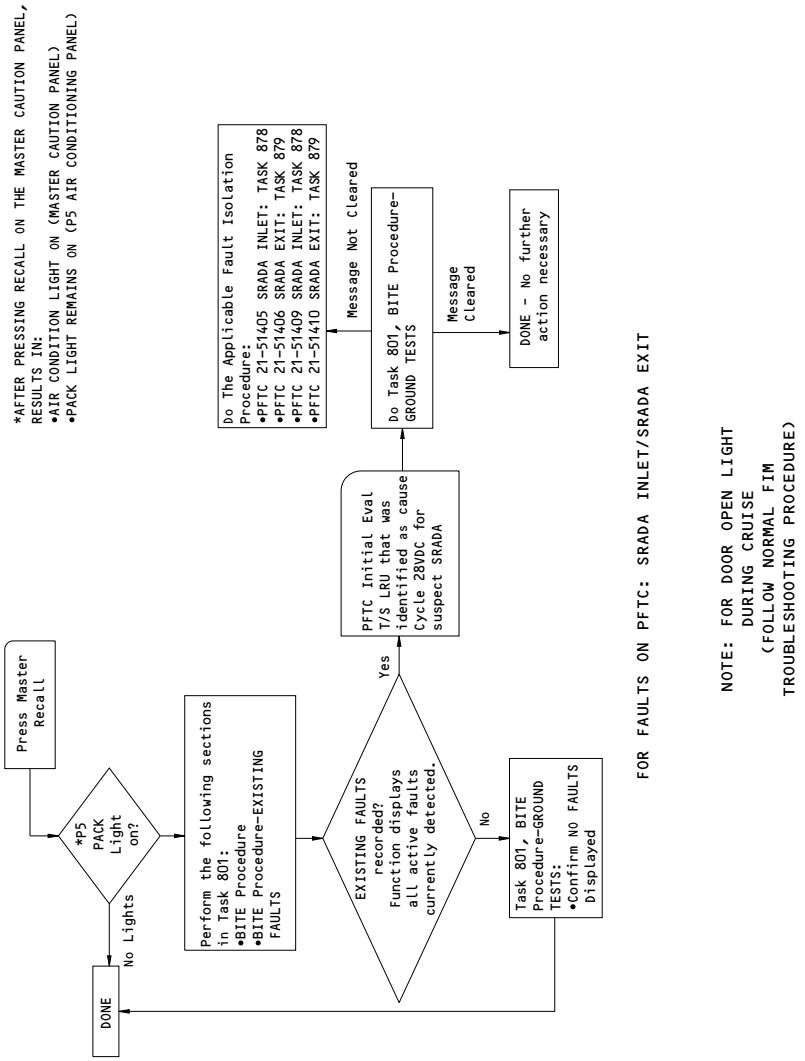
2571143 S0000617031_V1



Maintenance Messages on PFTC
Figure 307/21-62-00-990-807 (Sheet 3 of 4)

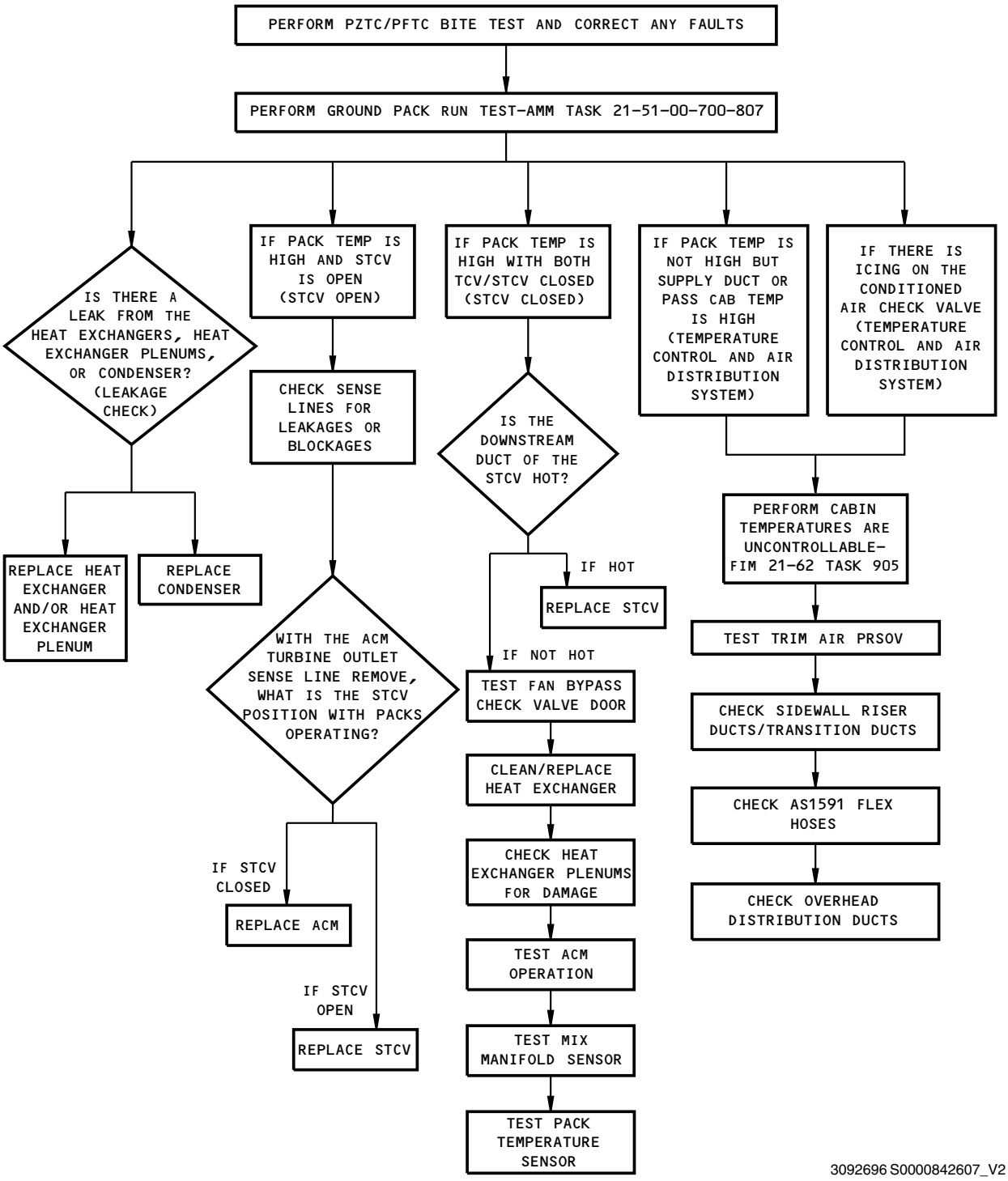
ECCN 9E991 BOEING PROPRIETARY - See title page for details

2572582 S0000617220_V2



Maintenance Messages on PFTC
Figure 307/21-62-00-990-807 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Cannot Cool Cabin
Figure 308/21-62-00-990-808

ECCN 9E991 BOEING PROPRIETARY - See title page for details

801. Pack Air Supply Cleaner System Does Not Operate - Fault Isolation

A. Description

(1) The Pack Air Supply Cleaner System for the air conditioning packs does not operate as expected.

B. Possible Causes

(1) Left Air Cleaner Purge Valve, V177

(2) Right Air Cleaner Purge Valve, V178

(3) Smoke Mode Interrupt Relay, R975

(4) Left Purge Valve Control Relay, R977

(5) Cargo Fire Alarm Interrupt Relay, R976

(6) Right Purge Valve Control Relay, R978

(7) Aux Smoke Control Relay, R1023

(8) Wiring

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	8	C00265	AIR CONDITIONING RAM AIR MOD CONT L
B	8	C00266	AIR CONDITIONING RAM AIR MOD CONT R
D	3	C01671	PNEUMATIC PURGE VALVE L
D	4	C01672	PNEUMATIC PURGE VALVE CONT
D	5	C01673	PNEUMATIC PURGE VALVE R

D. Related Data

(1) SSM 21-75-11

(2) WDM 21-75-11

(3) SSM 21-51-14

(4) SSM 21-51-24

E. Initial Evaluation

(1) The pack air supply cleaner receives discrete inputs from the ram air modulation control relays. Correct faults in the ram air inlet system before you continue.

(2) Do an operational test of the pack air supply cleaner system. To do the test, do this task: AMM TASK 21-75-00-700-801.

(a) If the operational test is not satisfactory for the air cleaner system, then do the Fault Isolation Procedure that follows.

(b) If the operational test is satisfactory, then there was an intermittent fault.

(c) If the purge valve operates but there is no airflow from the purge line, examine the purge line for blockage.

F. Fault Isolation Procedure - Pack Air Supply Cleaner System Does Not Operate

(1) Supply conditioned air to the airplane. Use the cooling pack with the pack air supply cleaner that does not operate. To supply conditioned air, do this task: Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(2) Make sure the AIR CLEANER shutoff switch on the aft overhead panel is set to ON.

(3) Do this check for 115V AC to the purge valve, V177 (V178):

(a) Disconnect connector D14002 (D14004) from the purge valve, V177 (V178).

(b) Do a check for nominal 115v ac at connector D14002 (D14004) between pins 3 and 1:

(c) If the voltage check is satisfactory, replace the purge valve. To replace the purge valve, do the tasks that follow:

Pack Supply Air Cleaner Purge Valve Removal, AMM TASK 21-75-11-000-801,

Pack Supply Air Cleaner Purge Valve Installation, AMM TASK 21-75-11-400-801.

1) Do the Repair Confirmation at the end of this task.

(d) If the voltage check is not satisfactory, then do the steps that follow.

1) Do a check of the wiring from the control relay, R977 (R978) to the purge valve, V177 (V178):

D14002		**D13998**
pin 1	. . . . .	pin A1
pin 2	. . . . .	pin A3

D14004		**D14000**
pin 1	. . . . .	pin A1
pin 2	. . . . .	pin A3

2) If you find a problem with the wiring, then do the steps that follow:

a) Repair the wiring.

b) Re-connect connector D14002 (D14004) to the purge valve.

c) Do the Repair Confirmation at the end of this task.

d) If the Repair Confirmation is not satisfactory, then continue.

3) If you do not find a problem with the wiring, re-connect connector D14002 (D14004) to the purge valve and continue.

(4) Do this check for 115v ac to the purge valve control relay R977 (R978):

(a) Remove the purge valve control relay R977 (R978) from the J4 junction box located in the .

(b) Do a check for nominal 115v ac at connector D13998 (D14000), pin A2.

(c) If the voltage check is satisfactory, then replace the purge valve control relay R977 (R978).

1) Do the Repair Confirmation at the end of this task.

2) If the Repair Confirmation is not satisfactory, then continue.

(d) If the voltage check is not satisfactory, then do the steps that follow:

1) Do a check of the wiring from the connector D13998 (D14000), pin A2 to the disconnected load side of the circuit breaker C1671 (C1673) in the P6-4 circuit breaker panel.

2) If you find a problem with the wiring, then do the steps that follow:

a) Repair the wiring.

b) Install the purge valve control relay R977 (R978).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

c) Do the Repair Confirmation at the end of this task.

d) If the Repair Confirmation is not satisfactory, then continue.

3) If you do not find a problem with the wiring, do a check for 28v dc from D13998 (D14000), pin X1 to structure ground.

4) If the voltage check is not satisfactory, then do these wiring checks:

a) Remove the cargo fire alarm interrupt relay R976 from the J4 junction box located in the compartment on the right side of the nose landing gear wheel well.

b) Do a check of the wiring from connector D13996, pin 13 to relay R977, pin X1 and from connector D13996, pin 3 to relay R978, pin X1.

c) If you find a problem with the wiring, then do the steps that follow:

<1> Repair the wiring.

<2> Install any removed relay.

<3> Reconnect any disconnected connectors.

<4> Do the Repair Confirmation at the end of this task.

<5> If the Repair Confirmation is not satisfactory, then continue.

(5) Do these checks of the cargo fire alarm interrupt relay R976 and the wiring:

(a) Remove the cargo fire alarm interrupt relay R976 from the the J4 junction box located in the compartment on the right side of the nose landing gear wheel well.

(b) Do a check for 28v dc at connector D13996, pin X1 to structure ground.

(c) If there is 28v dc at connector D13996, pin X1 to structure ground, then replace the cargo fire alarm interrupt relay R976.

(d) Do the Repair Confirmation at the end of this task.

(e) If the Repair Confirmation is not satisfactory, then continue.

(f) If the voltage check is not satisfactory, then do the steps that follow;

1) Do a check of the wiring from D13996, pin 6 to D01408, pin 2.

2) Do a check to make sure there is an open circuit at connector D13996, pin 12.

3) Do a check of the wiring from connector D13994, pin A1 to connector D13996, pin 1.

4) Do a check of the wiring from connector D13994, pin B1 to connector D13996, pin 5.

(g) If you find a problem with the wiring, then do the steps that follow:

1) Repair the wiring.

2) Install the cargo fire alarm interrupt relay R976.

3) Reconnect connector D13996.

4) Do the Repair Confirmation at the end of this task.

5) If the Repair Confirmation is not satisfactory, then continue.

(6) Do these checks of the wiring and the smoke mode interrupt relay R975:

(a) Do a check for 28v dc from connector D13994, pin X1 to a structure ground.

(b) If the voltage check is not satisfactory, then do the steps that follow:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Do a check of the wiring from connector D01408, pin 2 to connector D13994, pin X1.
2) Do a check of the wiring from connector D13994, pin X2 to connector D14206, pin A3.
3) If you find a problem with the wiring, then do the steps that follow:
   a) Repair the wiring.
   b) Replace relay R1023 if there was no ground at D13994, pin X2.
   c) Install the R975 relay.
   d) Connect all disconnected electrical connectors.
   e) Do the Repair Confirmation at the end of this task.
   f) If the Repair Confirmation is not satisfactory, then continue.
4) If you do not find a problem with the wiring, reconnect all disconnected connectors and continue.

(7) Do these checks of the wiring and the AIR CLEANER shutoff switch on the P5 aft overhead panel:
   (a) Disconnect electrical connector D01408 from the AIR CLEANER shutoff switch on the P5 aft overhead panel.
   (b) Do a check for 28v dc at connector D01408, pin 2.
   (c) If the voltage check is not satisfactory, then replace the AIR CLEANER shutoff switch.
      1) Do the Repair Confirmation at the end of this task.
      2) If the Repair Confirmation is not satisfactory, then continue.
   (d) Make sure that this circuit breaker is open and has safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	4	C01672	PNEUMATIC PURGE VALVE CONT

   (e) Do a wiring check between connector D01408, pin 1 and the load side of circuit breaker C01672.
   (f) If you find a problem with the wiring, then do the steps that follow:
      1) Repair the wiring.
      2) Reconnect the electrical connector D01408.
      3) Make sure that this circuit breaker is closed:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	4	C01672	PNEUMATIC PURGE VALVE CONT

      4) Do the Repair Confirmation at the end of this task.
      5) If the Repair Confirmation is not satisfactory, then continue.

(8) Do these checks for 28v dc from the ram air modulation control logic:
   (a) Remove smoke mode interrupt relay R975 from the the J4 junction box located in the compartment on the right side of the nose landing gear wheel well.
   (b) Do a check for 28v dc at connector D13994, pin A2 (pin B2).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) If the voltage check is not satisfactory, then do the steps that follow:

1) Remove the ACAU. To remove the ACAU, do this task: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801.

2) Do a check of the wiring from connector D458B (D454A), pin 30 to connector D13994, pin A2 (pin B2).

3) If you find a problem with the wiring, then do the steps that follow:

a) Repair the wiring.

b) Install the smoke mode interrupt relay R975 in the J4 junction box.

c) Install the ACAU. To install the ACAU, do this task: Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

d) Do the Repair Confirmation at the end of this task.

e) If the Repair Confirmation is not satisfactory, then continue.

4) If you do not find a problem with the wiring, replace the ACAU. To replace the ACAU, do these tasks: Air Conditioning Accessory Unit (ACAU) Removal, AMM TASK 21-51-02-000-801and Air Conditioning Accessory Unit (ACAU) Installation, AMM TASK 21-51-02-400-801.

G. Repair Confirmation

(1) Do an operational test of the pack air supply cleaner system. To do the test, do this task: AMM TASK 21-75-00-700-801.

(a) If the operational test is satisfactory, then you corrected the fault.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details